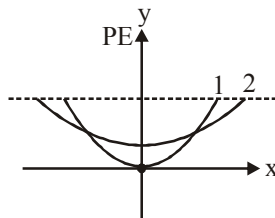


SECTION-I

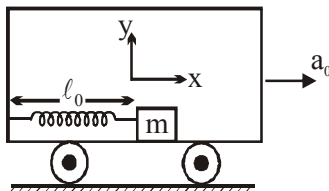
Single Correct Answer Type

13 Q. [3 M (-1)]

1. There are two potential energy (PE) displacement curves as shown for two different particles (of same mass) doing SHM about same mean position $x = 0$ on x-axis then we can say.



- (A) $\omega_1 > \omega_2$ (B) $\omega_1 < \omega_2$ (C) $\omega_1 = \omega_2$ (D) none
2. A spring block system is fixed in a train suddenly train starts moving with constant acceleration a_0 . Taking origin at initial position of block, what is equation of motion for block in reference frame of train. Force constant of spring is k . Neglect friction between block and train.

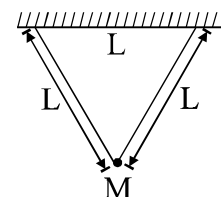


- (A) $-\frac{ma_0}{k} + \frac{ma_0}{k} \cos\left(\sqrt{\frac{k}{m}} t\right)$ (B) $\frac{ma_0}{k} - \frac{ma_0}{k} \cos\left(\sqrt{\frac{k}{m}} t\right)$
- (C) $-\frac{ma_0}{k} - \frac{ma_0}{k} \cos\left(\sqrt{\frac{k}{m}} t\right)$ (D) $\frac{ma_0}{k} + \frac{ma_0}{k} \cos\left(\sqrt{\frac{k}{m}} t\right)$
3. A particle is subjected to two simple harmonic motions, one along the x-axis and the other on a line making an angle of 45° with the x-axis. The two motions are given by $x = x_0 \sin \omega t$ and $s = s_0 \sin \omega t$:-
- (A) Path of the particle is straight line. (B) Path of the particle is parabola.
- (C) Path of the particle is ellipse (D) Path of the particle is circle
4. Pendulum A is a physical pendulum made from a thin rigid and uniform rod whose length is ℓ . One end of this rod is attached to the ceiling by a frictionless hinge so that rod is free to swing back and forth.

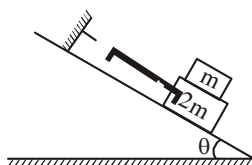
Pendulum B is a simple pendulum whose length is also ℓ . The ratio $\frac{T_A}{T_B}$ for small angular oscillations-

- (A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$
5. A man is swinging on a swing made of 2 ropes of equal length L and in direction perpendicular to the plane of paper. The time period of the small oscillations about the mean position is :

- (A) $2\pi \sqrt{\frac{L}{2g}}$ (B) $2\pi \sqrt{\frac{\sqrt{3}L}{2g}}$
- (C) $2\pi \sqrt{\frac{L}{2\sqrt{3}g}}$ (D) $\pi \sqrt{\frac{L}{g}}$

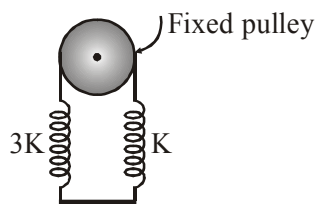


6. The coefficient of friction between block of mass m and $2m$ is $\mu = 2 \tan \theta$. There is no friction between block of mass $2m$ and inclined plane. The maximum amplitude of two block system for which there is no relative motion between both the blocks.



- (A) $\frac{2mg \sin \theta}{k}$ (B) $\frac{3mg \sin \theta}{k}$ (C) $\frac{6mg \sin \theta}{k}$ (D) $\frac{8mg \sin \theta}{k}$

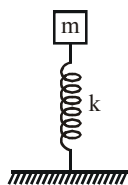
7. A bar of mass m is suspended horizontally on two vertical springs of spring constant k and $3k$. The bar bounces up and down while remaining horizontal. Find the time period of oscillation of the bar (Neglect mass of springs and friction everywhere).



- (A) $2\pi\sqrt{\frac{m}{k}}$ (B) $2\pi\sqrt{\frac{m}{3k}}$ (C) $\pi\sqrt{\frac{2m}{3k}}$ (D) $\pi\sqrt{\frac{3m}{4k}}$

8. A block of mass m is attached to a spring of force constant k whose other end is fixed to a horizontal surface. Initially the spring is in its natural length and the block is released from rest. If average force acting on the surface by the spring till the instant when the block has zero acceleration for the first time

in time T is F . Then find F ($T = \frac{\pi}{2}\sqrt{\frac{m}{k}}$) :-

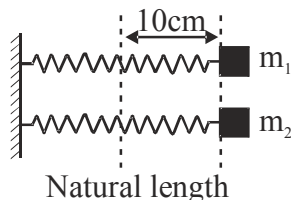


- (A) $\frac{mg}{\pi}$ (B) $\frac{2mg}{\pi}$ (C) $\frac{3mg}{\pi}$ (D) $\frac{4mg}{\pi}$

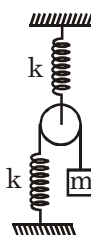
9. A particle of mass 10 gm is placed in a potential field given by $V = (50x^2 + 100) \text{ J/kg}$. The frequency of oscillation in cycle/sec is :-

- (A) $\frac{10}{\pi}$ (B) $\frac{5}{\pi}$ (C) $\frac{100}{\pi}$ (D) $\frac{50}{\pi}$

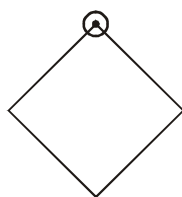
10. The drawing shows a top view of a frictionless horizontal surface, where there are two identical springs with particles of mass m_1 and m_2 attached to them. Each spring has a spring constant of 1200 N/m . The particles are pulled to the right and then released from the positions shown in the drawing. How much time passes before the particles are again side by side for the first time if $m_1 = 3.0 \text{ kg}$ and $m_2 = 27 \text{ kg}$?



- (A) $\frac{\pi}{40} \text{ sec}$ (B) $\frac{\pi}{20} \text{ sec}$ (C) $\frac{3\pi}{40} \text{ sec}$ (D) $\frac{\pi}{10} \text{ sec}$
11. Figure shows a system consisting of a massless pulley and two springs of equal constants k each, a block is attached with an ideal string as shown. If the block is slightly displaced vertically down from its equilibrium position and then released, the time period of vertical oscillations is :



- (A) $2\pi\sqrt{\frac{m}{k}}$ (B) $2\pi\sqrt{\frac{m}{2k}}$ (C) $2\pi\sqrt{\frac{2m}{k}}$ (D) $2\pi\sqrt{\frac{5m}{k}}$
12. A square frame of mass m and side ℓ is suspended as shown. Find the time period of small oscillations in vertical plane :-



- (A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{5\sqrt{2}}{6} \frac{\ell}{g}}$ (C) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$ (D) $2\pi\sqrt{\frac{5\sqrt{2}}{3} \frac{\ell}{g}}$
13. A particle starts from rest and performs SHM of amplitude A . Find the ratio of time taken by it from mean to $\frac{A\sqrt{3}}{2}$ to time taken by it from A to $\frac{A}{2}$:-
- (A) 1 (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{4}$

Multiple Correct Answer Type

2 Q. [4 M (-1)]

14. Acceleration of a particle which is at rest at $x = 0$ is $\vec{a} = (4 - 2x)\hat{j}$. Select the correct alternatives (s) :-
- (A) particle further comes to rest at $x = 4$ (B) particle oscillates about $x = 2$
 (C) maximum speed of particle is 4 units (D) all of the above
15. Tick the correct alternative (s)
- (A) The displacement of the particle varies with time as $x = 12 \sin \omega t - 16 \sin^3 \omega t$. The motion of particle is SHM with amplitude 4 units.
- (B) A particle oscillates according to equation $x = 7 \cos \frac{\pi t}{2}$ where t is in seconds. The point moves from the point of equilibrium to maximum displacement in 1 second
- (C) If a simple pendulum of length ℓ_0 has maximum angular displacement θ_0 , then the maximum speed of the bob is $2\sqrt{g\ell_0} \sin \frac{\theta_0}{2}$
- (D) None of the above

Linked Comprehension Type

(1 Para × 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 16 to 18

Two particles A and B are performing SHM along x and y -axis respectively with equal amplitude and frequency of 2 cm & 1Hz respectively. Equilibrium positions of the particles A and B are at the coordinates (3 cm, 0) and (0, 4 cm) respectively. At $t = 0$, B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin.

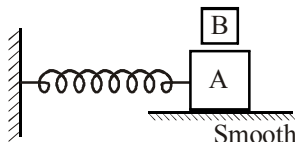
16. Equation of motion of particle A can be written as-
- (A) $x = (2 \text{ cm}) \cos 2\pi t$ (B) $x = (3 \text{ cm}) - (2 \text{ cm}) \cos 2\pi t$
 (C) $x = (2 \text{ cm}) \sin 2\pi t$ (D) $x = (3 \text{ cm}) - (2 \text{ cm}) \sin 2\pi t$
17. Equation of motion of particle B can be written as-
- (A) $y = (2 \text{ cm}) \cos 2\pi t$ (B) $y = (4 \text{ cm}) - (2 \text{ cm}) \cos 2\pi t$
 (C) $y = (2 \text{ cm}) \sin 2\pi t$ (D) $y = (4 \text{ cm}) - (2 \text{ cm}) \sin 2\pi t$
18. Minimum and maximum distance between A and B during the motion is-
- (A) $\sqrt{5}$ cm and $\sqrt{61}$ cm (B) 3 cm and 7 cm
 (C) 1 cm and 5 cm (D) 9 cm and 16 cm

SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

1. In the figure shown below block A is executing SHM on a smooth level ground. Another block B is kept gently on A in any of the positions mentioned in column-I. The block B sticks to A. Match with appropriate descriptions in column II.



Column I

- (A) Block B placed on the block A when A is at right extreme.
(B) Block B placed on the block A when A is at mean position.
(C) Block B placed on the block A when A is midway between extreme and mean.

Column II

- (P) Momentum of (A + B) can be assumed to be conserved at the time block A is placed on block B
(Q) Mechanical energy is conserved.
(R) Time period of SHM increases.
(S) Amplitude of SHM decreases.

2. Column I

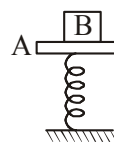
- (A) A bob B hanging from a string A of length 3m is projected to left with a speed of 10 m/s
(B) The plat form is pushed down by a distance $2x_0$ below mean and released x_0 is compression in mean position
(C) A spherical solid ball B is released on a perfectly rough spherical surface A as shown $\theta = \frac{\pi}{3}$
(D) Block B is released on a smooth track A as shown

Column II

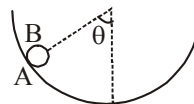
- (P) The acceleration of centre of mass of B at some time can be equal to g
(Q) The force exerted by A on B can be zero at some point



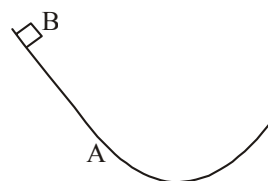
- (Q) The force exerted by A on B can be zero at some point



- (R) The speed of the body B varies sinusoidally with time when force exerted by A on B is not zero



- (S) The motion of B is oscillatory



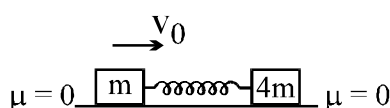
- (T) The motion of B is periodic

SECTION-I

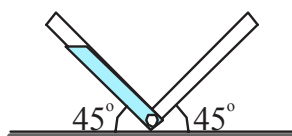
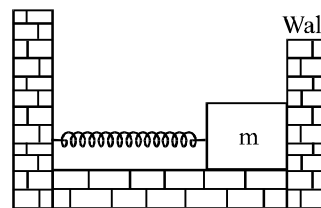
Single Correct Answer Type

6 Q. [3 M (–1)]

- Two particles P and Q describe simple harmonic motions of same period, same amplitude, along the same line about the same equilibrium position O. When P and Q are on opposite sides of O at the same distance from O they have the same speed of 1.2 m/s in the same direction, when their displacements are the same they have the same speed of 1.6 m/s in opposite directions. The maximum velocity in m/s of either particle is :
(A) 2.8 (B) 2.5 (C) 2.4 (D) 2
- Two blocks of masses m and $4m$ lie on a smooth horizontal surface connected with a spring in its natural length. Mass m is given velocity v_0 through an impulse as shown in the diagram. Which of the following is **not true** about subsequent motion



- Kinetic energy is maximum in ground frame and centre of mass (COM) frame simultaneously
 - Value of maximum kinetic energy & minimum kinetic energy is same in COM frame & ground frame
 - Minimum kinetic energy is zero in COM frame but non zero in ground frame
 - Maximum and minimum kinetic energies of m in ground frame is respectively $\frac{1}{2}mv_0^2$ and zero
- A particle is performing simple harmonic motion with time period T . At an instant its speed is 60% of its maximum value and is increasing. After an interval Δt its speed becomes 80% of its maximum value and is decreasing. The smallest value of Δt in terms of T will be :-
(A) $\frac{T}{4}$ (B) $\frac{T}{2}$ (C) $\frac{3T}{8}$ (D) $\frac{3T}{4}$
 - In the figure, the block of mass m , attached to the spring of stiffness k is in contact with the completely elastic wall, and the compression in the spring is ' e '. The spring is compressed further by ' e ' by displacing the block towards left and is then released. If the collision between the block and the wall is completely elastic then the time period of oscillations of the block will be:
(A) $\frac{2\pi}{3} \sqrt{\frac{m}{k}}$ (B) $2\pi \sqrt{\frac{m}{k}}$
(C) $\frac{\pi}{3} \sqrt{\frac{m}{k}}$ (D) $\frac{\pi}{6} \sqrt{\frac{m}{k}}$
 - A thin L-shaped glass tube is fixed in the vertical plane as shown. Initially, the left part of the tube contains a column of water of length d . A valve at the bottom of the tube prevents the water from moving. Suddenly, the valve is opened. Find the time water takes to move completely into the right part of the tube. Neglect viscous forces.



- $t = 2\pi \sqrt{\frac{d}{\sqrt{2}g}}$
- $t = 2\pi \sqrt{\frac{d}{2g}}$
- $t = \pi \sqrt{\frac{d}{2g}}$
- $t = \pi \sqrt{\frac{d}{\sqrt{2}g}}$

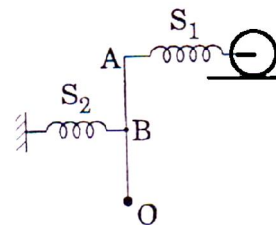
6. A solid sphere (m kg, R meter) placed on a horizontal rough plane, is attached to a spring S_1 , fixed to a light rod OA of length L meter, as shown in the **figure**. The other end O of the rod is hinged to rotate in the plane about vertical axis passing through it. A spring S_2 is fixed to the midpoint B of the rod. If the spring constants of both springs are $k \text{ Nm}^{-1}$ each, then the angular frequency of oscillations of the mass would be (in rad s^{-1})

(A) $\omega = \sqrt{\frac{k}{7m}}$

(B) $\omega = \sqrt{\frac{11k}{28m}}$

(C) $\omega = \sqrt{\frac{2k}{7m}}$

(D) $\omega = \sqrt{\frac{2k}{5m}}$



Multiple Correct Answer Type

4 Q. [4 M (-1)]

7. Two particles of mass $\frac{3M}{4}$ and M, are connected by a massless spring of free length L and force constant k. These masses are initially at rest L apart on a horizontal frictionless table. A particle of mass $\frac{M}{4}$ moving with speed v along the line joining the two connected masses, collides with and sticks to the particle of mass $\frac{3M}{4}$. Find the amplitude A and period T with which each of the masses vibrate

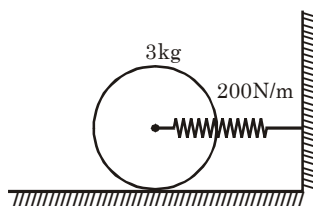
(A) $A = \frac{v}{8} \sqrt{\frac{M}{2k}}$

(B) $A = \frac{v}{4} \sqrt{\frac{M}{2k}}$

(C) $T = 2\pi \sqrt{\frac{M}{2k}}$

(D) $T = \pi \sqrt{\frac{M}{2k}}$

8. The potential energy U of a particle is given by $U = \{20 + (x-4)^2\} \text{ J}$. Total mechanical energy of the particle is 36 J. Select the correct alternative(s)
 (A) the particle oscillates about point $x=4 \text{ m}$
 (B) the amplitude of the particle is 4m
 (C) the kinetic energy of the particle at $x=2 \text{ m}$ is 12 J
 (D) the motion of the particle is periodic but not simple harmonic
9. A 20 gm particle is subjected to two simple harmonic motions $x_1 = 2 \sin 10t$, $x_2 = 4 \sin \left(10t + \frac{\pi}{3}\right)$ along same straight line. Where x_1 and x_2 are in metre and t is in sec.
 (A) the displacement of the particle at $t = 0$ will be $2\sqrt{3} \text{ m}$
 (B) maximum speed of the particle will be $20\sqrt{7} \text{ m/s}$
 (C) magnitude of maximum acceleration of the particle will be $200\sqrt{7} \text{ m/s}^2$.
 (D) Energy of the resultant motion will be 28 J.
10. The uniform solid cylinder rolls without slipping on the surface as shown. If the maximum compression in spring is 15cm, the possible values of friction force acting on the cylinder during its motion is :



(A) 4N

(B) 10N

(C) 12N

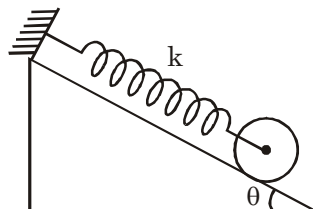
(D) 15N

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions No. 11 and 12

A disc of mass M is connected with ideal spring of force constant k is released from unstretched position of the spring, then the disc roll down the inclined plane. Friction is sufficient to prevent slipping.



11. What is the amplitude of and time period of oscillation respectively.

- (A) $\frac{mg \sin \theta}{k}, 2\pi\sqrt{\frac{m}{k}}$ (B) $\frac{mg \sin \theta}{k}, 2\pi\sqrt{\frac{3m}{2k}}$
(C) $\frac{2mg \sin \theta}{k}, 2\pi\sqrt{\frac{3m}{2k}}$ (D) $\frac{mg \sin \theta}{2k}, 2\pi\sqrt{\frac{m}{k}}$

12. Friction force at the mean position is

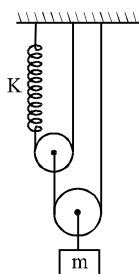
- (A) $\frac{1}{3} mg \cos \theta$ (B) $\frac{2}{3} mg \cos \theta$ (C) Zero (D) $mg \sin \theta$

SECTION-II

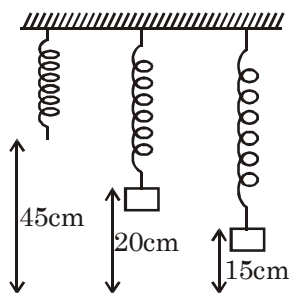
Numerical Answer Type Question (upto second decimal place)

2Q.[3(0)]

1. Find the natural angular frequency (in rad/s) of oscillation of the system as shown in figure. Pulleys are massless and frictionless. Spring and string are also massless. Take spring constant as 100 N/m and mass $m = 1$ kg



2. A massless spring is suspended from a hook at the top. A small mass of 180 gm is suspended from the bottom of the hook. At the equilibrium it is at 20 cm from ground. It is pulled down to distance 15 cm from ground and released. Find the time (in sec) at which it is first at 25 cm from the ground. Take $\pi^2 = 10$.

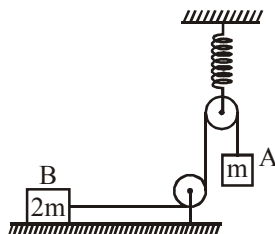


SECTION-III

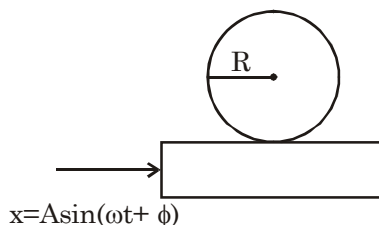
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

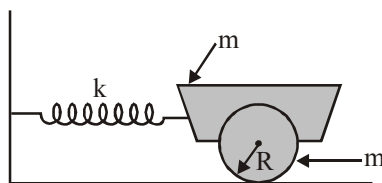
1. In the given figure, spring and pulleys are massless. Block A is performing SHM of amplitude 1m and time period π sec. If block B remains at rest, then minimum value of coefficient of friction between block B and surface will be μ . Fill 10μ in OMR sheet. ($g = 10 \text{ m/s}^2$)



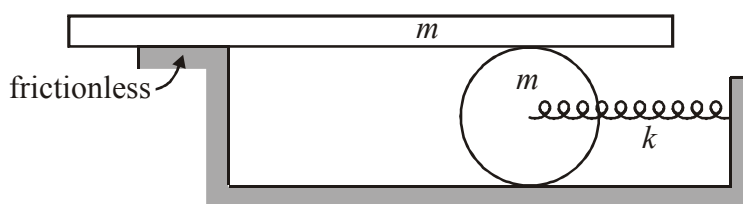
2. A solid sphere of mass m and radius R is resting on platform and sphere is free to rotate about its axis. Plate is given a displacement $x = A \sin(\omega t + \phi)$ along its length as shown in figure. There is no slipping between sphere and platform. If maximum torque acting on sphere is $\frac{2m\omega^2 AR}{\beta}$. Find β .



3. A cart consists of a body of mass m and two wheels, each of mass m and radius R . The cart is attached to a spring of constant k . The other end of the spring is fixed to a wall as shown in figure. If time period of oscillation is $\beta \times 2\pi \sqrt{\frac{m}{k}}$ then find the value of β

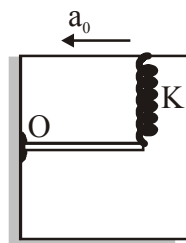


4. The axle of a uniform cylinder with mass m and radius R is connected to a spring with spring constant k , as shown in the Fig. A horizontal board with mass m rests on top of the cylinder, and the board also rests on top of a frictionless support near its left end. The system is slightly displaced from equilibrium. There is no slipping between the cylinder and the board, or between the cylinder and the ground. The angular frequency of the oscillatory motion is p . The value of $\frac{11p^2 m}{k}$ is .



5. A rod of mass m and length l is pivoted at a point O in a car whose acceleration towards left is a_0 . The rod is free to oscillate in vertical plane. In the equilibrium state the rod remains horizontal when other end is suspended by a spring of stiffness K . The time period of small oscillations of the rod is given by

$T = \frac{2\pi}{C\sqrt{3}}$. Find the value of C and fill in OMR sheet. Given value : $K = 20 \text{ N/m}$, $a_0 = 10 \text{ m/s}^2$, $m = 1 \text{ kg}$, $l = 1 \text{ m}$.

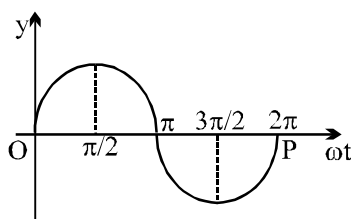


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

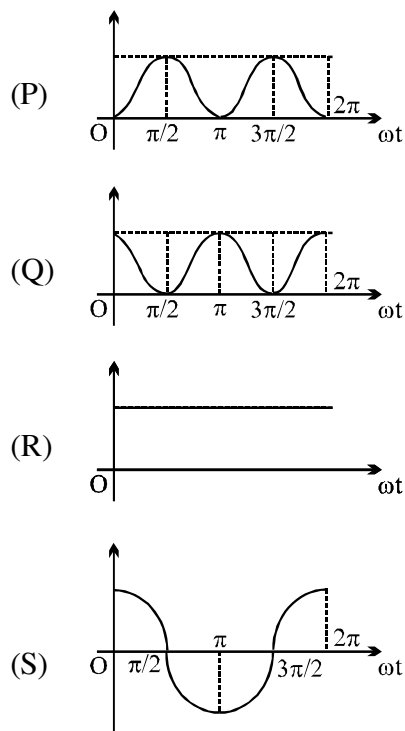
1. The graph plotted between phase angle (ϕ) and displacement of a particle from equilibrium position (y) is a sinusoidal curve as shown below. Then the best matching is



Column A

- (A) K.E. versus phase angle curve
(B) P.E. versus phase angle curve
(C) T.E. versus phase angle curve
(D) Velocity versus phase angle curve

Column B

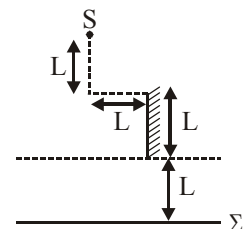
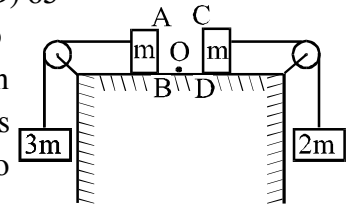
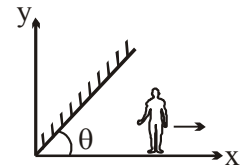


SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

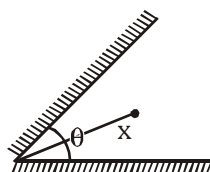
- A man is walking under an inclined mirror at a constant velocity V m/s along the X axis. If the mirror is inclined at an angle θ with the horizontal then what is the velocity of the image?
 (A) $V \sin \theta \hat{i} + V \cos \theta \hat{j}$ (B) $V \cos \theta \hat{i} + V \sin \theta \hat{j}$
 (C) $V \sin 2\theta \hat{i} + V \cos 2\theta \hat{j}$ (D) $V \cos 2\theta \hat{i} + V \sin 2\theta \hat{j}$
- A ray of light is incident on a plane mirror, along the direction given by, $\vec{A} = 2\hat{i} - 3\hat{j} + 4\hat{k}$. Find the unit vector along the reflected ray. Take normal to mirror along the direction of $\vec{B} = 3\hat{i} - 6\hat{j} + 2\hat{k}$
 (A) $\frac{-94\hat{i} + 237\hat{j} + 68\hat{k}}{49\sqrt{29}}$ (B) $\frac{-94\hat{i} + 68\hat{j} - 237\hat{k}}{49\sqrt{29}}$ (C) $\frac{3\hat{i} + 6\hat{j} - 2\hat{k}}{7}$ (D) None of these
- Sunbeam makes an angle $\theta = 40^\circ$ with the surface of the Earth. At what angle to the horizontal should we place a flat mirror so that a ray of sunlight after reflection falls on the bottom of a deep well?
 (A) 40° (B) 50° (C) 80° (D) 65°
- Two blocks each of mass m lie on a smooth table. They are attached to two other masses as shown in the figure. The pulleys and strings are light. An object O is kept at rest on the table. The sides AB & CD of the two blocks are made reflecting. The acceleration of two images formed in those two reflecting surfaces w.r.t. each other is:
 (A) $5g/6$ (B) $5g/3$ (C) $g/3$ (D) $17g/6$
- A point source of light is placed in front of a plane mirror as shown in the figure. Determine the length of reflected patch of light on the screen Σ :-
 (A) L (B) $2L$
 (C) $\frac{3L}{2}$ (D) $\frac{L}{2}$



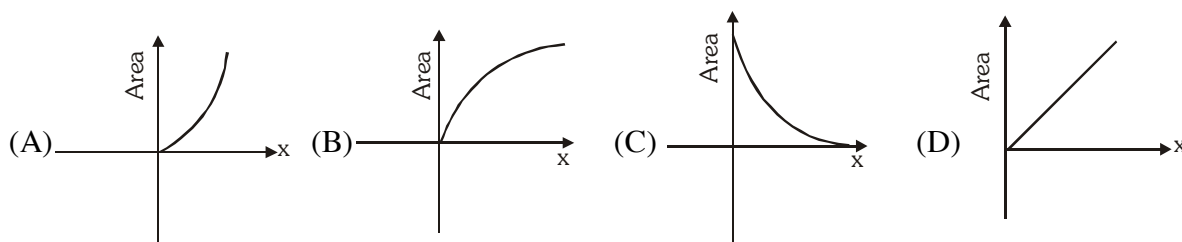
Linked Comprehension Type (2 Para $\times$ 3Q.) (1 Para $\times$ 2Q.) [3 M (-1)] (Single Correct Answer Type)

Paragraph for Questions 6 to 8

Whenever an object is placed on the bisector of two mirrors inclined at an angle θ , it forms several images which lie on a circle of radius equal to the distance between the vertex of mirror and the object



6. Draw graph between the area formed by joining all the images and object and the distance between the vertex and object



7. If the angle between the two mirror is 60° . Find the area of regular polygon bounded by the images and the object

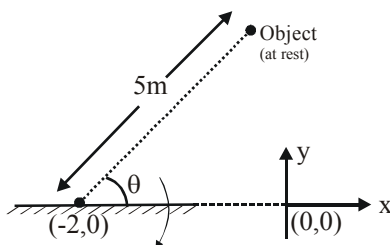
(A) $\frac{x^2(4\sqrt{3}+1)}{4}$ (B) $\sqrt{\left(\frac{4\sqrt{3}+1}{4}\right)}x$ (C) $\frac{3\sqrt{3}x^2}{2}$ (D) $\frac{5\sqrt{3}x^2}{4}$

8. If an object is placed between more than two mirror then the locus of a point going through all the images and object will be

- (A) Equation of sphere
(B) Family of concentric circle
(C) Family of a circle passing through one common point
(D) None of these

Paragraph for Questions no. 9 to 11

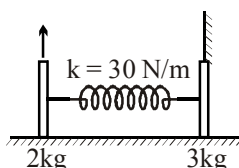
The figure shows time varying angle θ which is given by equation $\theta = (2 + 6t^2)$. The mirror is rotating about a horizontal axis in clockwise direction.



9. Locus of the image is :-
(A) $x^2 + y^2 = 5^2$ (B) $(x + 2)^2 + y^2 = 5^2$ (C) $(x - 2)^2 + y^2 = 5^2$ (D) $x^2 + (y - 2)^2 = 5^2$
10. Angular velocity of the image at $t = 2$ s :-
(A) 12 rad/s (B) 24 rad/s (C) 36 rad/s (D) 48 rad/s
11. Velocity of the image at $t = 1$ s :-
(A) 20 m/s (B) 40 m/s (C) 60 m/s (D) 120 m/s

Paragraph for Question no. 12 and 13

A mirror is mounted on a stand as shown. The mirror has negligible mass. On the other side, massless arrow is mounted. The natural length of the spring is 1m. The whole system is on frictionless surface. The two bodies are pulled out such that spring is extended by 10 cm and released.



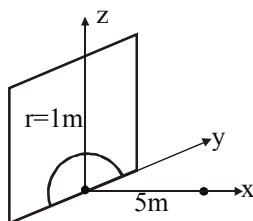
12. What is the minimum distance between arrow and its image during subsequent motion ?
(A) 1.6 m (B) 1.8 m (C) 2 m (D) 1.85 m
13. What is the maximum velocity of the image of the arrow as seen by a person on the ground ?
(A) 70 cm/s (B) 40 cm/s (C) 35 cm/s (D) 28 cm/s

SECTION-III

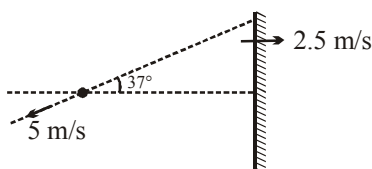
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

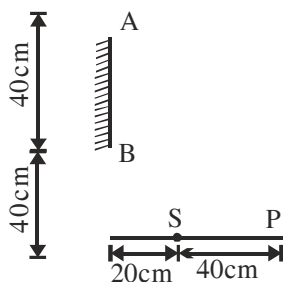
1. A plane mirror of semicircular shape (radius = 1m) is placed in y-z plane with its diameter along y-axis and centre at origin. An object is placed on the axis of mirror at point (5, 0, 0). At a distance of 10 m from plane mirror there is a wall also parallel to y-z plane. The ratio of area of spot of light on wall to that of mirror is



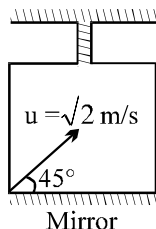
2. A plane mirror is placed such that it passes through points (0, 0, a), (0, a, 0) and (a, 0, 0). A light ray traveling along negative x-axis is incident on it. After reflection, this ray makes an angle $\cos^{-1}\left(\frac{1}{n}\right)$ with x axis. Find n.
3. A point object is moving with a speed of 5 m/s in front of a mirror moving with a speed of 2.5 m/s as shown in figure. The velocity of image of the object with respect to ground is $\sqrt{10x}$ m/s, then x is



4. In figure shown AB is a plane mirror of length 40cm placed at a height 40cm from ground. There is a light source S at a point on the ground. The minimum height and maximum height of a man (eye height) required to see the image of the source if he is standing at a point P on ground shown in figure are x and y respectively then find $3x + y$ (in meters).



5. An elevator at rest which is at 10th floor of a building is having a plane mirror fixed to its floor. A particle is projected with a speed $\sqrt{2}$ m/s and at 45° with the horizontal as shown in the figure. At the very instant of projection, the cable of the elevator breaks and the elevator starts falling freely. What will be the separation between the particle and its image 0.5 second after the instant of projection ?



SECTION-IV

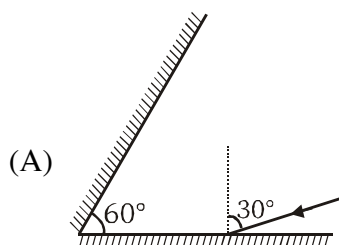
Matrix Match Type (4×4)

1 Q. [8 M (for each entry +2(0))]

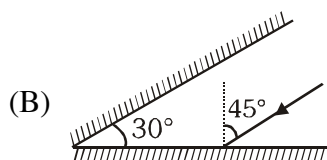
1. Different arrangement of two mirror with incident light are shown in column-I and in column II corresponding deviations are given.

Column I

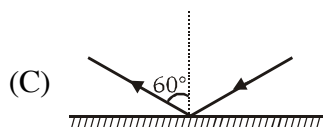
Column II



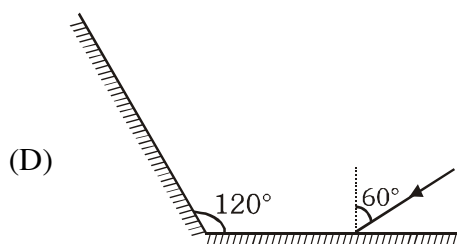
(P) Total deviation after two reflection is 240° clockwise



(Q) Total deviation after three reflection is 180°



(R) Total deviation after two reflection is 60° anticlockwise



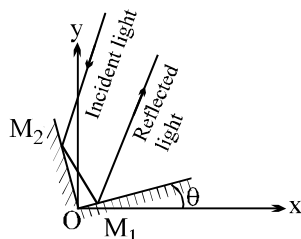
(S) Total deviation after one reflection is 60° clockwise

SECTION-I

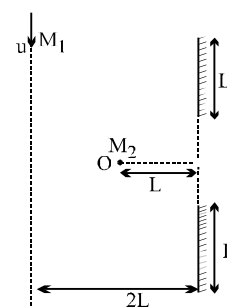
Single Correct Answer Type

7 Q. [3 M (-1)]

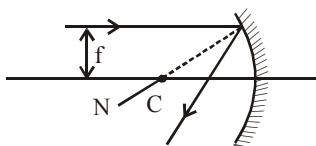
1. A light ray gets reflected from a pair of mutually perpendicular mirrors, not necessarily along axes. The intersection point of mirrors is at origin. The incident light is along $y = x + 2$. If the light ray strikes both mirrors in succession, then it may get reflected finally along the line:



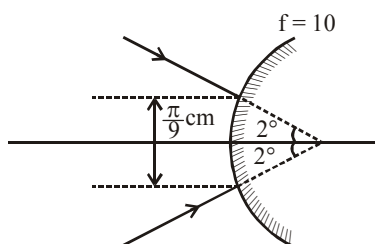
- (A) $y = 2x - 2$ (B) $y = -x + 2$ (C) $y = -x - 2$ (D) $y = x - 4$
2. Two plane mirrors of length L are separated by distance L and a man M_2 is standing at distance L from the connecting line of mirrors as shown in figure. A man M_1 is walking in a straight line at distance $2L$ parallel to mirrors at speed u , then man M_2 at O will be able to see image of M_1 for total time:



- (A) $\frac{4L}{u}$ (B) $\frac{3L}{u}$
(C) $\frac{6L}{u}$ (D) $\frac{9L}{u}$
3. A student holds a hand plane mirror to observe the back of her head while standing in front of and looking into a wall plane mirror. If she is standing 4 ft away from wall mirror in front of the wall mirror and she holds the hand mirror vertically 1 ft behind her head, she will see the back of her head approximately how far behind the wall mirror?
- (A) 6 ft (B) 5 ft (C) 4 ft (D) 3 ft
4. A light ray is coming parallel to principal axis, the distance between ray and axis is equal to focal length (as shown). Find the angle of deviation after reflection :-

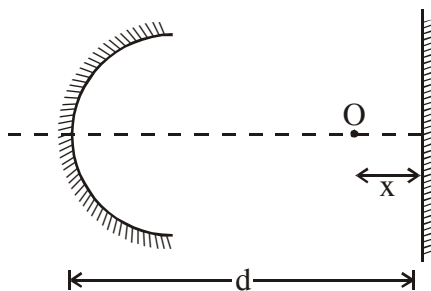


- (A) 60° (B) 90° (C) 120° (D) small deviation
5. A converging beam of light having angle of convergence 4° is incident upon a convex mirror as shown. Find the angle of convergence after reflection. Focal length of mirror is 10 cm.

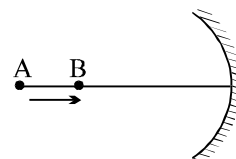


- (A) 0.5° (B) 1° (C) 1.5° (D) 2°

6. A plane mirror and a spherical mirror of focal length f is arranged as given. A point object O is placed between them, the value of x so that final image coincide the object ($d > 2f$) is :



- (A) $\frac{d}{2}$ (B) $\sqrt{d(d-2f)}$ (C) $\sqrt{2d(d-2f)}$ (D) $\sqrt{3d(d-2f)}$
7. A linear object AB is placed along the axis of a concave mirror. The object is moving towards the mirror with speed U . The speed of the image of the point A is $4U$ and the speed of the image of B is also $4U$. If the center of the line AB is at a distance L from the mirror then find out the length of the object.

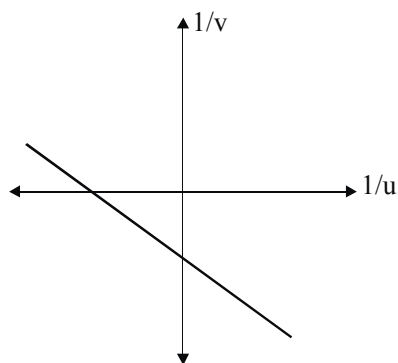


- (A) $3L/2$ (B) $5L/3$ (C) L (D) None

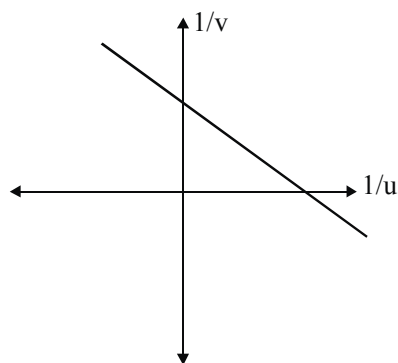
Multiple Correct Answer Type

4 Q. [4 M (-1)]

8. Choose the **CORRECT** statement(s).

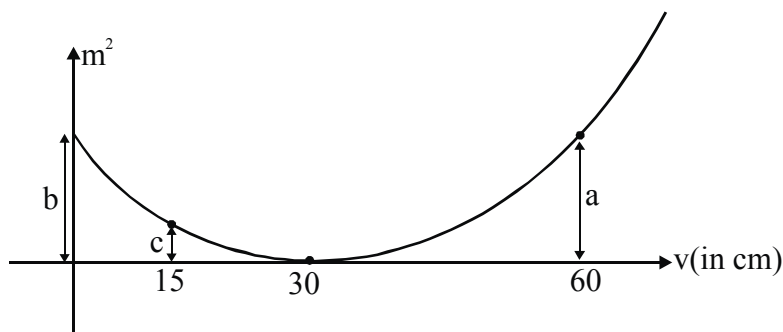


concave mirror



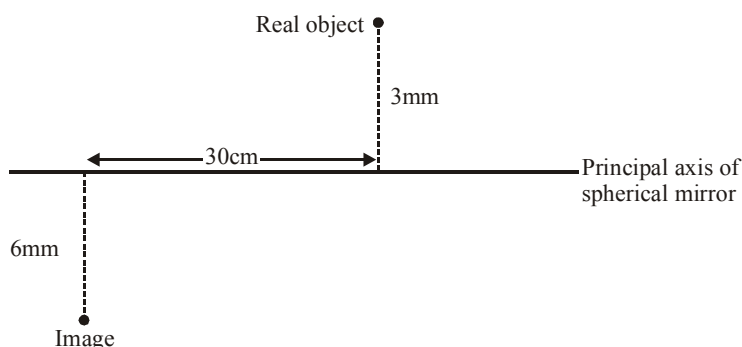
convex mirror

- (A) Concave mirror can not form erect magnified image of an object placed in front of it.
(B) Concave mirror can form inverted magnified image of an object placed in front of it.
(C) Convex mirror can not form real image of an object placed in front of it.
(D) Convex mirror can not form magnified image of an object placed in front of it.
9. For a concave mirror, graph of square of magnification and image distance from pole is given for real object. Choose the correct statement(s).

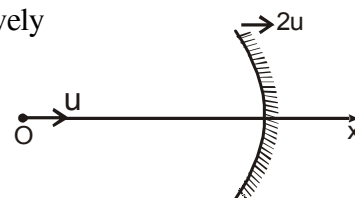


- (A) This graph is hyperbola. (B) length a must be equal to length b .
(C) radius of concave mirror is 60 cm (D) the length c will be half of length a .

10. A spherical mirror forms image of a real object as shown, choose the **CORRECT** option(s) :-



- (A) Mirror is concave of radius 40 cm
(B) Mirror of convex of radius 40 cm
(C) Mirror is 30 cm from object along principal axis
(D) Mirror is 60 cm from image along principal axis
11. An object & a concave mirror is moving with velocities $u\hat{i}$ & $2u\hat{i}$ respectively as shown in figure. The image formed by the concave mirror :-
- (A) will have speed greater than the speed of the object
(B) may have speed greater than the speed of the object
(C) will have speed greater than the speed of the mirror
(D) must move away from the mirror.

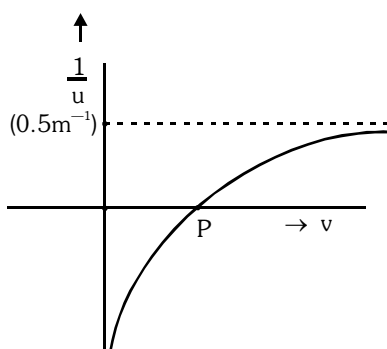


Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Questions no. 12 to 14

Sign convention is taken as +ve direction in the direction of light ray and the graph is drawn between $\frac{1}{u}$ and v for a spherical mirror.

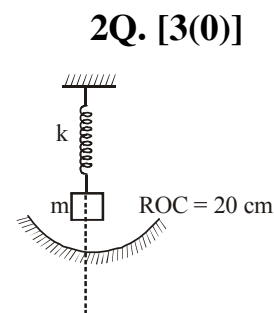


12. What is the focal length of the mirror?
(A) -50cm (B) -200 cm (C) +200 cm (D) +50 cm
13. The coordinates of point P are
(A) (50cm, 0 cm) (B) (100cm, 0 cm) (C) (200cm, 0cm) (D) data insufficient
14. Magnitude of slope of curve at point P
(A) 1 m^{-2} (B) 0.25 m^{-2} (C) 2 m^{-2} (D) data insufficient

SECTION-II

Numerical Answer Type Question
(upto second decimal place)

1. A block of mass 'm' is attached to a spring. Block is released from rest, when spring is in natural length. Block is along the principal axis of concave mirror and size of block is very small. When spring is in natural length then block is at a distance of 20 cm from the mirror, distance (in cm) in which the image formed by mirror oscillates is (mass of the block = 5×10^{-2} kg, spring constant $k = 20$ N/m.)
2. A plane mirror is placed 25 cm away from a concave spherical mirror perpendicularly to the principal axis of the concave mirror. What should be the distance (in cm) in front of concave mirror, where we place a candle if its images formed by the two mirror independently are at the same distances from the candle? The radius of the concave mirror is 40 cm. (Consider images formed by single reflection only.)



SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1. When an object is placed at a distance of 25 cm from a concave mirror the magnification is m . The object is moved 15 cm further away from the mirror with respect to the earlier position, and the magnification becomes m_2 . If $m_1/m_2 = 4$, then find the focal length (in dm) of the mirror (Assume image is real and m_1, m_2 are numerical values of transverse magnification of the image).
2. The distance between object and its erect but diminished image formed due to a spherical mirror is 3 times the distance between image and the focus while distance between object and focus is 4 x. The distance of object from mirror is nx. Find the value of n.
3. A handheld mirror contains a plane mirror on one of its side and a concave mirror of radius of curvature of 30 cm on its other side. You hold the plane mirror 10 cm from your face and notice a spot. By what factor will the image of the spot be enlarged if you turn the mirror, but still keep it at a distance of 10 cm from your face?

SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. Plane mirror are arranged parallel to each other as shown in column I and the number of images formed by the combination is in column II.

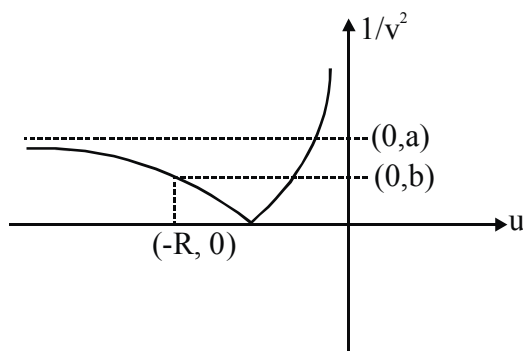
Column I	Column II
(A)	(P) 3
(B)	(Q) 2
(C)	(R) ∞
(D)	(S) 0

SECTION-I

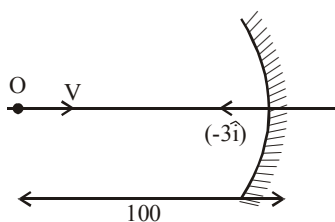
Single Correct Answer Type

7 Q. [3 M (-1)]

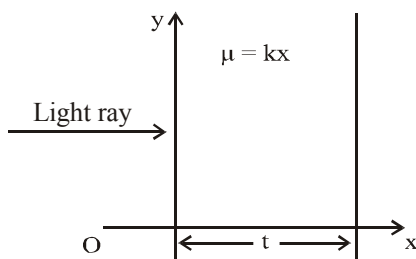
1. For a concave mirror, graph of $\frac{1}{v^2}$ and u for real object is drawn. What is the value of $\frac{b}{a}$



- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{8}$
2. Radius of curvature of a concave mirror varies as $R = \left| \frac{6}{t^2 - 5t + 6} \right| \text{ cm}$. Initially the object was at a distance 100 cm from the pole as shown. At $t = 0$ the object started moving with velocity $v = 2t\hat{i} \text{ (cm/s)}$. Find the speed of image at time $t = 3$ sec. Mirror is moving with constant velocity $(-3\hat{i}) \text{ (cm/s)}$.

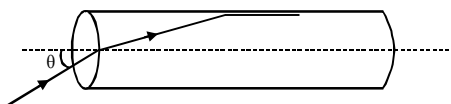


- (A) 12 cm/s (B) 20 cm/s (C) 5 cm/s (D) 24 cm/s
3. Refractive index of a transparent slab varies as $\mu = kx$ where x is the distance from origin. Time taken by the light to travel the slab of thickness t (as shown in the figure) :-

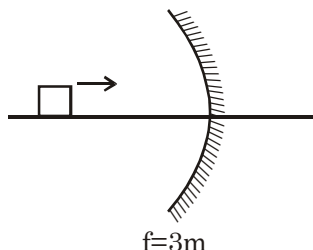


- (A) $\frac{t^2 k}{2c}$ (B) $\frac{t^2 k}{c}$ (C) $\frac{tk}{c}$ (D) $\frac{2tk}{2c}$

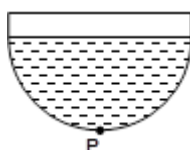
4. A transparent solid cylindrical rod has a refractive index of $\frac{4}{\sqrt{3}}$. It is surrounded by a medium of refractive index 2. A light ray is incident at the mid-point of one end of the rod as shown in the figure. The incident angle θ for which the light ray grazes along the wall of the rod is :-



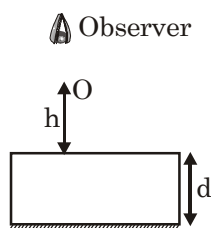
- (A) $\sin^{-1}\left(\frac{1}{2}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (C) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$
5. A block of mass 1 kg is moving on a rough horizontal surface along the principal axis of a concave mirror as shown. At $t = 0$, it is 15m away from the pole, moving with a velocity of 7m/s. At $t = 1$ sec, it's image is at $\frac{57}{13}$ m away from the pole on left hand side of the mirror. Where will the image be at $t = 3$ sec.



- (A) 5m to left of mirror (B) $\frac{123}{23}$ m to left of mirror
(C) $\frac{138}{23}$ m to left of mirror (D) 7.5m to left of mirror.
6. A hemispherical bowl of radius 10 cm is filled with liquid of refractive index $\mu = 4/3$. A glass plate of refractive index 1.5 is placed on the top of bowl. If for the observer above the plate the shift in position of a point P on the bottom is 3 cm find the thickness of glass plate.



- (A) 1.5 cm (B) 1 cm (C) 7 cm (D) 10 cm
7. A point luminous object (O) is at a distance h from front face of a glass slab of width d and of refractive index n . On the back face of slab is a reflecting plane mirror. An observer sees the image of object in mirror as shown in figure. Distance of image from front face as seen by observer will be :-

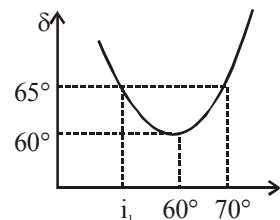


- (A) $h + \frac{2d}{n}$ (B) $2h + 2d$ (C) $h + d$ (D) $h + \frac{d}{n}$

Multiple Correct Answer Type

3 Q. [4 M (-1)]

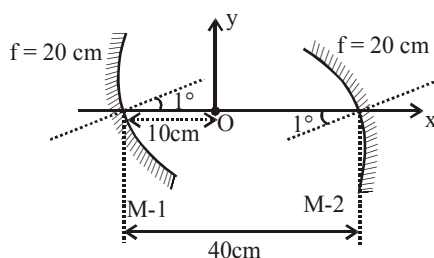
8. A concave spherical mirror has a radius of curvature of 50 cm. Find two positions of an object for which the image is four times as large as the object.
- (A) $\frac{75}{4}$ cm from mirror (B) 125 cm from mirror
- (C) $\frac{125}{4}$ cm from mirror (D) 32 cm from mirror
9. The image of an object kept at a distance 20 cm in front of a concave mirror is found to coincide with itself. If a glass slab ($\mu=1.5$) of thickness 3 cm is introduced between the mirror and the object, then in order that the final image again coincides with the object :
- (A) the mirror should be displaced away from the object
(B) the mirror should be displaced towards the object
(C) the magnitude of displacement is 1 cm
(D) the magnitude of displacement is 0.5 cm
10. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Choose the **CORRECT** statement(s).
- (A) The angle of prism is 60°
(B) The refractive index of the prism is $n = \sqrt{3}$
(C) For deviation to be 65° the angle of incidence $i_1 = 55^\circ$
(D) for minimum deviation, the angle of emergence $e = 60^\circ$.



Linked Comprehension Type (1 Para $\times$ 3Q.) (1 Para $\times$ 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Questions 11 to 13

In the diagram shown, principal axis of each concave mirror is inclined at 1° with respect to x-axis. Object O is placed at the origin of x-y coordinate system. The distance of object from mirror M-1 is 10cm. Separation between the two mirrors M-1 & M-2 is 40 cm. Focal length of each mirror is 20 cm. Then :-



11. The coordinates of image due to reflection from mirror M-2 is :-
- (A) $\left(-30\text{cm}, \frac{2\pi}{3}\text{cm}\right)$ (B) $\left(-30\text{cm}, -\frac{2\pi}{3}\text{cm}\right)$ (C) $\left(-30\text{cm}, \frac{\pi}{3}\text{cm}\right)$ (D) None of these
12. The coordinates of image due to reflection from mirror M-1 is :-
- (A) $\left(-20\text{cm}, \frac{\pi}{9}\text{cm}\right)$ (B) $\left(-20\text{cm}, -\frac{\pi}{9}\text{cm}\right)$ (C) $\left(-30\text{cm}, -\frac{2\pi}{9}\text{cm}\right)$ (D) None of these
13. Taking first reflection at mirror M-2 and then at mirror M-1, the coordinate of image after two reflection is :-
- (A) $\left(0, \frac{2\pi}{9}\text{cm}\right)$ (B) $\left(0, -\frac{2\pi}{9}\text{cm}\right)$ (C) $\left(0, \frac{\pi}{9}\text{cm}\right)$ (D) None of these

Paragraph for Questions 14 and 15

Rear view mirrors

Rear view mirrors in automobiles are generally convex. Suppose a car A moves with a constant speed of 40.0 kilometre per hour on a straight level road and is followed by another car B moving with the constant speed 60.0 kilometer per hour. At a given instant of time, we denote : [INPhO-2012]

x : distance of the car B from the mirror of car A,

y : distance of the car B from A as seen by the driver of A in the mirror,

v_x : speed of approach of B relative to A and

v_y : speed of approach of B as seen in the mirror of A.

14. Obtain an expression for v_y in terms of x , v_x and the radius of curvature R of the convex mirror used as the rear view mirrors in the car A.

(A) $v_y = \frac{3R^2}{(2x - 2R)^2} v_x$ (B) $v_y = \frac{R^2}{(2x + R)^2} v_x$ (C) $v_y = \frac{R^2}{(2x - R)^2} v_x$ (D) $v_y = \frac{R^2}{(x - R)^2} v_x$

15. If $R = 2.0$ m, what is the speed of approach of B in kilometre per hour as seen by the driver of A in the mirror for $x = 2.0$ m ?

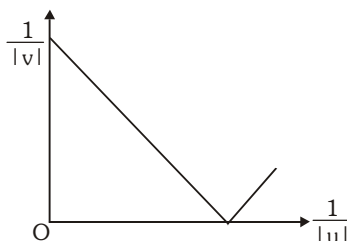
(A) 2.22 km/h (B) 3 km/h (C) 4.62 km/h (D) 6.87 km/h

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. A flat glass slab of thickness 6 cm and refractive index 1.5 is placed in front of a plane mirror. An observer is standing behind the glass slab and looking at the mirror. The actual distance of the observer from the mirror is 50 cm. The observer sees the image at a distance of d_1 cm from himself. Now the slab is removed and the observer sees his image in plane mirror at a distance of d_2 cm from himself. What is the value of $d_2 - d_1$?
2. Figure shows the variation of $\frac{1}{|v|}$ versus $\frac{1}{|u|}$ for a concave mirror for real object. The line $y = 3x$ cuts the graphs at an ordinate 1.5 m^{-1} . The radius of the mirror is



SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. In column-I, figure shows the optical (or principal) axis XX' , and position of real object O and image I . Match this with relevant parameters of spherical mirror in column-II

Column-I

Column-II

(A) $X \xrightarrow{\bullet O} \bullet I X'$

(P) concave mirror, virtual image

(B) $X \xrightarrow{\bullet O} \bullet I X'$

(Q) concave mirror, real image

(C) $X \xrightarrow{\bullet O} \bullet I X'$

(R) convex mirror, virtual image

(D) $X \xrightarrow{\bullet O} \bullet I X'$

(S) concave mirror, magnified image

SECTION-I

Single Correct Answer Type

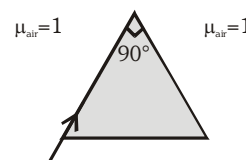
7 Q. [3 M (–1)]

- A pole 6m high stands amidst a lake having water to a depth of 2m. If the sun is at angle of inclination 37° , what will be the length of the shadow in water?
(A) 4.5 m (B) 6.8 m (C) 5.3 m (D) 6m
- A beam of light of width t is incident on an air-water boundary at an angle of incidence 45° . The width of the beam in water is (Refractive index of water $=\mu$)

- (A) $(\mu - 1)t$ (B) $\frac{\sqrt{2\mu^2 - 1}}{\sqrt{2}\mu} \cdot t$ (C) $\frac{\sqrt{2\mu^2 - 1}}{\mu} \cdot t$ (D) $\frac{\sqrt{\mu^2 - 1}}{\mu} \cdot t$

- Consider a right angled prism, if the ray enters at grazing incidence and just fails to emerge on the facing side. The value of refractive index of the prism is–

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$
(C) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\sin^{-1}(\sqrt{2})$

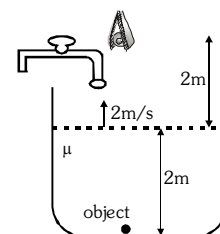


- The cross-section of a transparent prism having a refractive index of $\sqrt{2}$ is a right triangle with the angle of refraction of 90° and the acute angles of 30° and 60° . Find the angle of deflection of a ray incident normal to the steeper face of the prism.

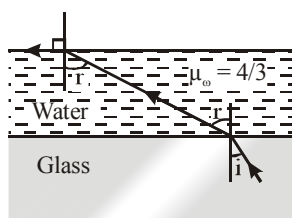
- (A) 45° (B) 90° (C) 180° (D) None of these

- In a container, water level is rising with a constant rate of 2m/s. The velocity (in m/s) of image as observed by eye is [given $\mu = 4/3$]

- (A) $\frac{1}{2}$ upward (B) $\frac{1}{2}$ down ward
(C) 2 upward (D) 2 down ward



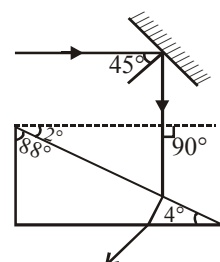
- A ray of light is incident at the glass-water interface at an angle i . It emerges finally parallel to the surface of water, then the value of μ_g would be :-



- (A) $(4/3) \sin i$ (B) $1/\sin i$ (C) $4/3$ (D) 1

- A ray of light strikes a plane mirror at an angle of incidence 45° as shown in the figure. After reflection, the ray passes through a prism of refractive index 1.5, whose apex angle is 4° . The angle through which the mirror should be rotated if the total deviation of the ray is to be 90° is :-

- (A) 1° clockwise (B) 1° anticlockwise
(C) 2° clockwise (D) 2° anticlockwise



Multiple Correct Answer Type

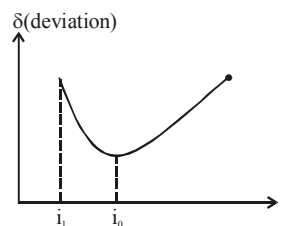
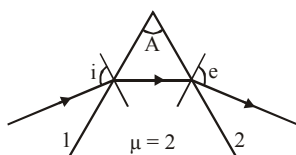
8. For the system shown in the figure, the image formed by the concave mirror

(A) may have speed greater than the speed of the object
(B) will have positive velocity.
(C) may have speed greater than the speed of the mirror
(D) sign of velocity will depend upon thickness of slab.

9. A ray of light from a denser medium strikes a rarer medium at an angle of incidence i . The reflected and refracted rays make an angle of 90° with each other. The angles of reflection and refraction are i and r . The critical angle is :-

(A) $\sin^{-1}(\cot r)$
(B) $\sin^{-1}(\tan i)$
(C) $\sin^{-1}(\tan r)$
(D) $\tan^{-1}(\sin i)$

10. A monochromatic light is incident on a prism as shown. Deviation suffered by ray varies with angle of incidence as shown. Assuming refraction takes place at 1 & 2. Choose the correct alternative(s).



(A) i_1 can be equal to zero if $A = 30^\circ$
(B) If angle of incidence is increased by 2° , then angle of emergence may decrease by less than 2° .
(C) If angle of incidence is increased by 2° , then angle of emergence may decrease by more than 2° .
(D) If angle of incidence is increased by 2° , then angle of emergence may increase by equal to 2° .

11. In the figure shown O, O_1 and O_2 are three observers respectively in three different medium. Choose the **CORRECT** option(s).

(A) According to O the apparent distance between

O_1 & O_2 is 3.5 m

(B) O observes O_1 to be 4 m away

(C) O observes O_2 to be 4 m away

(D) O_2 observes O_1 to be 2.1 m away

12. A ray of light strikes the surface AB of a rectangular prism at an angle of incidence equal to critical angle of prism-air interface as shown then (neglect normal reflections) :-

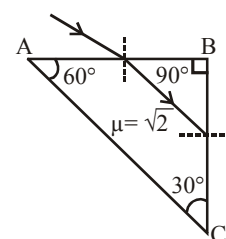
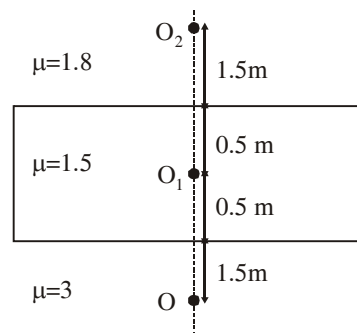
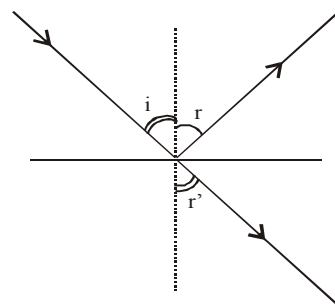
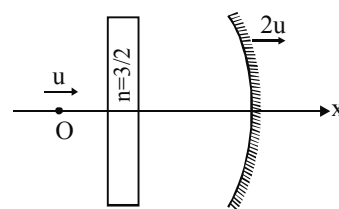
(A) Angle of deviation of emergent ray with respect to the incident ray will be 60° ACW.

(B) Ray after refraction from surface AB will be parallel to the base AC.

(C) Angle of incidence is same as angle of emergence.

(D) Total three refraction will take place.

5 Q. [4 M (-1)]



Linked Comprehension Type (Single Correct Answer Type)

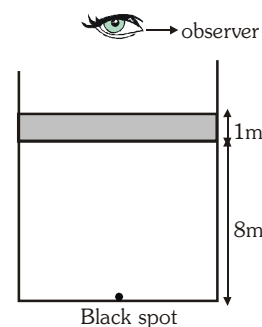
(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question 13 to 15

Shown below a system in which one mole of helium gas is filled in a vessel with a fixed piston made up of glass having refractive index $4/3$ (constant). The gas is supplied with heat at a constant rate of 1000 J/s . The refractive

index of He varies with temperature as $\mu = \frac{T}{300}$.

[Initially the gas is at room temperature of 300 K]

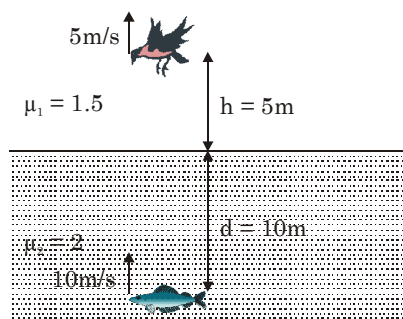


13. The magnitude of the velocity of the point black spot as seen by an observer outside the vessel when $T=400\text{K}$ is
(A) 1.2 m/s (B) 1.6 m/s (C) 2.4 m/s (D) 0.72 m/s
14. Maximum distance traversed by spot as seen by observer is
(A) 2m (B) 8m (C) 1m (D) 4m
15. If the gas is replaced by CO_2 , magnitude of velocity of black spot as observed by an observer outside the vessel when $T=400 \text{ K}$ is
(A) 1.2 m/s (B) 1.6 m/s (C) 2.4 m/s (D) 0.72 m/s

Matching List Type (4 × 4)

1Q.[3 M (-1)]

16. Diagram shows a bird B and a fish F moving in two different medium of refractive index 1.5 and 2 with speed 10 m/s and 5 m/s respectively. At the given instant bird is located at height 5m above the interface while fish is located at depth of 10m . List-I indicates various parameters while list-II gives its values. Match them correctly :-



List-I

- (P) Separation of fish as observed by bird
(Q) Separation of bird as observed by fish
(R) Speed of fish as observed by bird
(S) Speed of bird as observed by fish

Code :-

P	Q	R	S
(A) 3	2	1	4
(B) 2	1	4	3
(C) 4	1	2	3
(D) 1	2	3	4

List-II

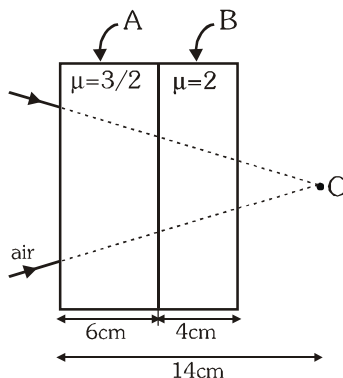
- (1) $\frac{50}{3}$ unit
(2) 12.5 unit
(3) $\frac{10}{3}$ unit
(4) 2.5 unit

SECTION-III

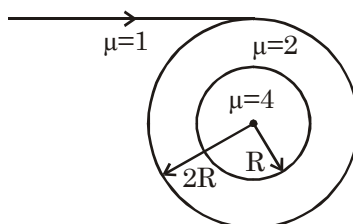
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

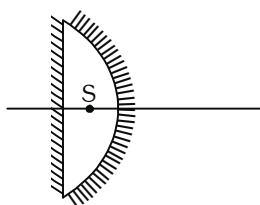
1. A convergent beam is incident on two slabs placed in contact as shown in figure. Where will the rays finally converge from the interface of B and air ?



2. Figure shows two spherical surface of radii R and $2R$ separating three transparent media of refractive index $\mu = 4$, $\mu = 2$ and $\mu = 1$. A ray of light travelling in medium $\mu = 1$ is incident on outer sphere tangentially. If the net deviation suffered by light ray in clockwise direction is β (in degree) then find value of $\beta/30$.



3. In the figure shown a thick plano convex lens is silvered at both the surfaces. A point source of light 'S' is inside the lens. The radius of curvature of the curved surface is $\frac{8}{3}$ cm and the distance of 'S' from the curved surface is 2 cm. If the images formed due to direct reflections from the plane and curved surface coincide then find the thickness (in cm) of the lens at the middle.

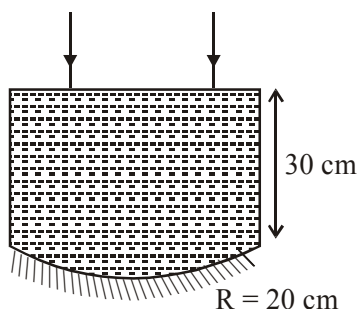


SECTION-I

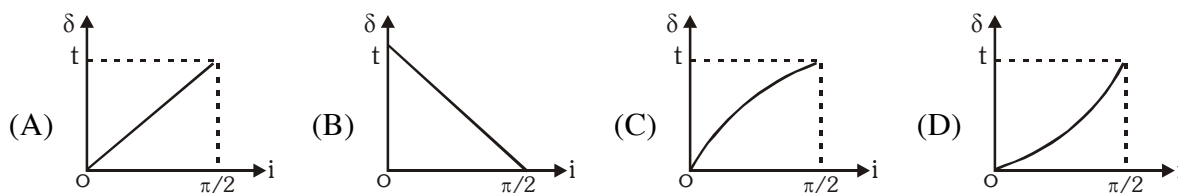
Single Correct Answer Type

7 Q. [3 M (–1)]

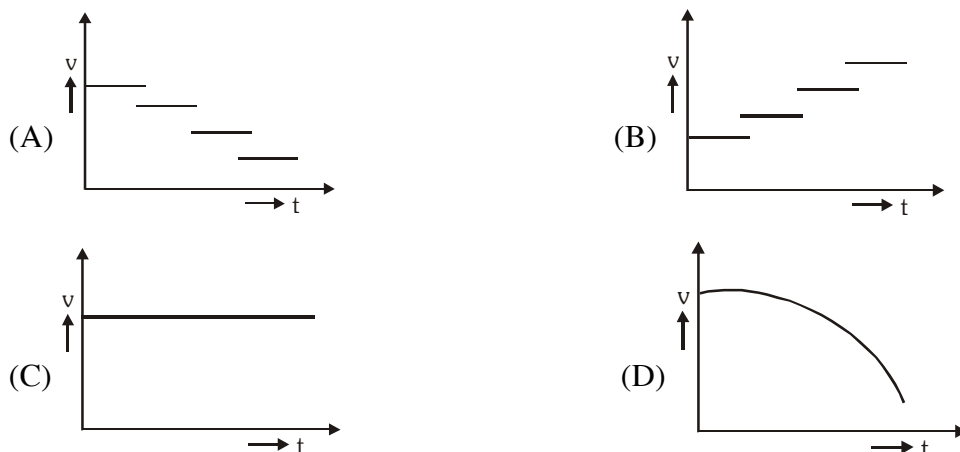
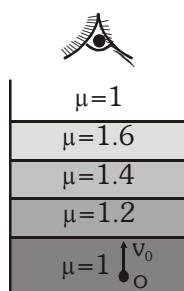
1. A vessel with a silvered concave bottom is filled with water of refractive index $4/3$. A parallel beam of light is incident on the water surface as shown in figure. What is position of final image from water surface:-



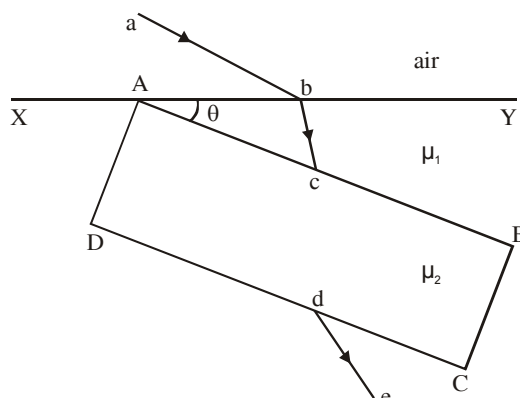
- (A) 15 cm (B) 20 cm (C) 10 cm (D) 7.5 cm
2. A ray of light passes through a rectangular slab of thickness t . The variation of lateral shift (δ) with angle of incidence (i) is



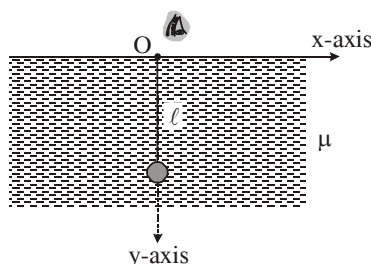
3. An object is moving with a constant velocity in a vessel containing layers of immiscible liquids of different refractive indices as shown in figure. Variation of velocity with time (until object comes out) is best represented by :



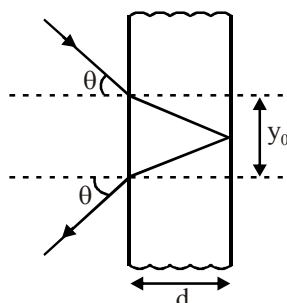
4. A ray of light ab passing through air, enters a liquid of refractive index μ_1 , at the boundary XY . In the liquid, the ray is shown as bc . The angle between ab and bc is δ (angle of deviation). The ray then passes through a rectangular slab $ABCD$ of refractive index μ_2 ($\mu_2 > \mu_1$), and emerges from the slab as ray de . The angle between XY and AB is θ . The angle between ab and de is-



- (A) δ (B) $\delta + \theta$ (C) $\delta + \sin^{-1}\left(\frac{\mu_1}{\mu_2}\right)$ (D) $\delta + \theta - \sin^{-1}\left(\frac{\mu_1}{\mu_2}\right)$
5. A pendulum of length ℓ is free to oscillate in vertical plane about point O in a medium of refractive index μ . An observer in air is viewing the bob of the pendulum directly from above. The pendulum is performing small oscillations about its equilibrium position. The equation of trajectory of bob as seen by observer is

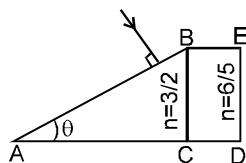


- (A) $x^2 + y^2 = \ell^2$ (B) $\frac{x^2}{(\ell/\mu)^2} + \frac{y^2}{\ell^2} = 1$ (C) $\frac{x^2}{\ell^2} + \frac{y^2}{(\ell/\mu)^2} = 1$ (D) $x^2 + y^2 = \left(\frac{\ell}{\mu}\right)^2$
6. A ray of light incident from air on a glass plate of refractive index n is partly reflected and partly refracted at the two surfaces of the glass. The displacement y_0 in the figure is



- (A) $\frac{2d \sin \theta}{\sqrt{n^2 - \sin^2 \theta}}$ (B) $\frac{2d \sin \theta}{\sqrt{\sin^2 \theta - \frac{1}{n^2}}}$ (C) $\frac{2d \sqrt{n^2 - \sin^2 \theta}}{\sin \theta}$ (D) None of these

7. In the figure ABC is the cross section of a right angled prism and BCDE is the cross section of a glass slab. The value of θ so that light incident normally on the face AB does not cross the face BC is [Given $\sin^{-1}(3/5) = 37^\circ$]



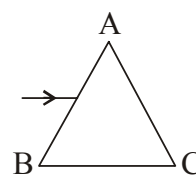
- (A) $\theta \leq 37^\circ$ (B) $\theta < 37^\circ$ (C) $\theta \leq 53^\circ$ (D) $\theta < 53^\circ$

Multiple Correct Answer Type

4 Q. [4 M (-1)]

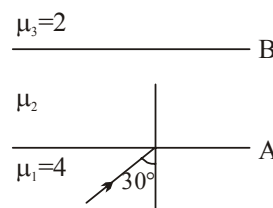
8. A ray of light is incident on an equilateral triangular prism parallel to its base as shown in the figure. The ray just fails to emerge from the face AC. If μ be the refractive index of the prism then the incorrect relation(s) is/are :

- (A) $2\sin^{-1}\left(\frac{1}{\mu}\right) = \frac{\pi}{3}$ (B) $\sin^{-1}\left(\frac{1}{\mu}\right) + \sin^{-1}\left(\frac{1}{2\mu}\right) = \frac{\pi}{6}$
(C) $\sin^{-1}\left(\frac{1}{\mu}\right) + \sin^{-1}\left(\frac{1}{2\mu}\right) = \frac{\pi}{3}$ (D) $\sin^{-1}\left(\frac{\mu}{2}\right) + \sin^{-1}\left(\frac{\mu}{4}\right) = \frac{\pi}{3}$

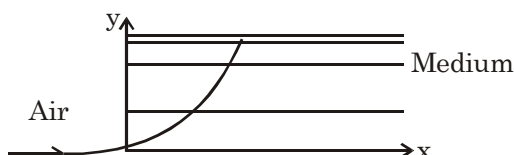


9. A light ray is incident on lower medium boundary at an angle 30° with the normal. Which of following statement is/are **TRUE**?

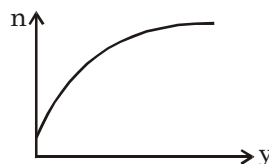
- (A) If $\mu_2 > 2$ then total deviation is 60°
(B) If $\mu_2 < 2$ then total deviation is 60°
(C) If $\mu_2 > 2$ then total deviation is 120°
(D) If $\mu_2 < 2$ then total deviation is 120°



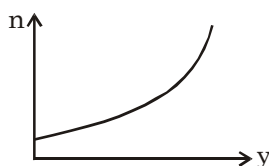
10. The refractive index of the medium within a certain region $x > 0, y > 0$, changes continuously with y . A thin light ray travelling in air in the x -direction strikes the medium at right angles and moves through the medium along a circular arc of radius R .



- (A) Refractive index of medium varies with y as

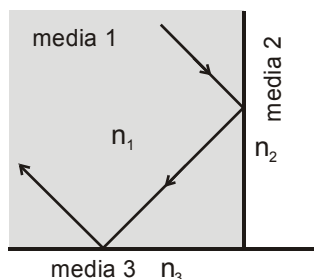


- (B) Refractive index of medium varies with y as



- (C) If refractive index of medium can increase upto a value $n = 2.5$, the maximum value of y is $\frac{3R}{5}$
(D) If refractive index of medium can increase upto a value $n = 2.5$, the maximum value of y is $5R$

11. In the diagram shown, light is incident on the interface between media 1 (refractive index n_1) and 2 (refractive index n_2) at angle slightly greater than the critical angle, and is totally reflected. The light is then also totally reflected at the interface between media 1 and 3 (refractive index n_3), after which it travels in a direction opposite to its initial direction. The media must have a refractive indices such that



(A) $n_1 < n_2 < n_3$

(B) $n_1^2 - n_3^2 > n_2^2$

(C) $n_1^2 - n_2^2 < n_3^2$

(D) $n_1^2 + n_2^2 > n_3^2$

Linked Comprehension Type
(Single Correct Answer Type)

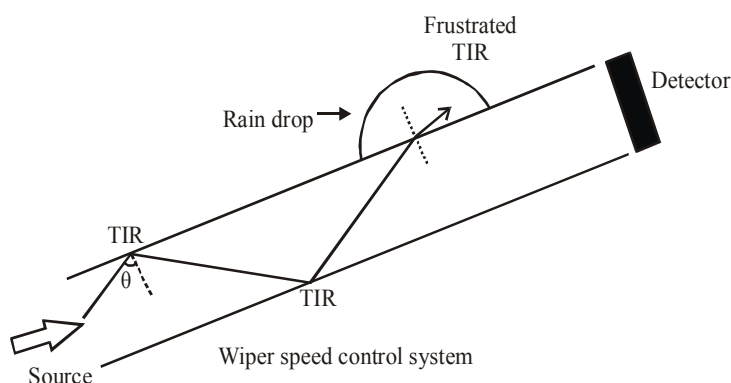
(2 Para × 3Q.) [3 M (-1)]

Paragraph for Question No. 12 to 14

Frustrated TIR (F-TIR) : In optics, when light rays traveling in a denser medium strike at medium boundary at an angle greater than critical angle, TIR occurs and a surface wave which is called **Evanescent wave** forms in rarer medium. An evanescent wave is a near-field **standing wave (waves which do not carry energy)** with an intensity that exhibits exponential decay with distance (less than wavelength) from the boundary at which the wave is formed.

Imagine that a beam of light traveling within a block of glass is internally reflected at a boundary. Presumably if you pressed another piece of glass against the first, the air-glass interface would be made to vanish and the beam would then propagate onward undisturbed. Furthermore, this transition from total to no reflection occurs gradually as the air film between them thinned out as explained above, when third medium with a higher refractive index (than the low-index second medium) is placed within less than several wavelengths distance from the interface between the first medium and the second medium, the evanescent wave will be different from the one under "ordinary conditions" and it will pass energy across the second into the third medium (evanescent wave coupling). This process is called "frustrated" total internal reflection (FTIR) and is very similar to quantum tunneling. An example of application of this principle is automatic **Wiper speed control** found in high end cars like Skoda, Audi, BMW etc.

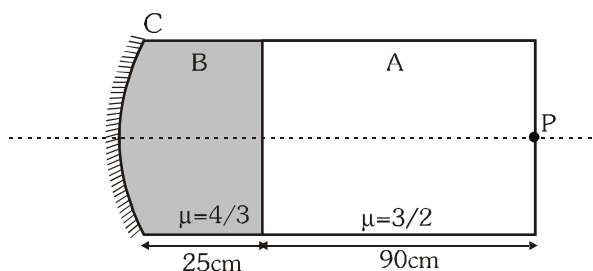
Figure shows an example of an optical system designed to detect the amount of water present on the windshield of a car to adjust the wiper speed. As shown in this figure, we can use the windshield as a waveguide to guide the light from a source located at one end (bottom of the windshield) to a detector located in the opposite end. The light suffers total-internal reflection (TIR) at the glass-air interface. However, when rain drops are present, some of the light will suffer frustrated TIR escaping outside the waveguide. Since we know the power of the light source, a given drop in power can be correlated to the amount of water present and used to adjust the wiper speed.



12. Choose the **CORRECT** statement.
 (A) The energy carried by evanescent wave is zero in both TIR and FTIR.
 (B) The energy carried by evanescent wave is non-zero in both TIR and FTIR.
 (C) The energy carried by evanescent wave is zero in TIR and non-zero in FTIR.
 (D) The energy carried by evanescent wave is non-zero in TIR and zero in FTIR.
13. In the following systems which should not be based on FTIR?
 (A) Allen student card attendance system (B) Thumb attendance system
 (C) Finger print scanner (D) Multi touch screen
14. For the wiper speed control system to work, the angle of incidence on the glass air interface is θ , then
 (A) $\theta < \sin^{-1} \left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}} \right)$ (B) $\theta > \sin^{-1} \left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}} \right)$
 (C) $\sin^{-1} \left(\frac{\mu_{\text{air}}}{\mu_{\text{glass}}} \right) < \theta < \sin^{-1} \left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}} \right)$ (D) $\theta > \sin^{-1} \left(\frac{\mu_{\text{air}}}{\mu_{\text{glass}}} \right)$

Paragraph for Questions no. 15 to 17

An object P is placed on the glass slab A, which is 90 cm thick with refractive index $3/2$. A container B of thickness 25 cm having water of refractive index $4/3$ is sandwiched between glass slab A and a concave mirror C of radius of curvature 70 cm.



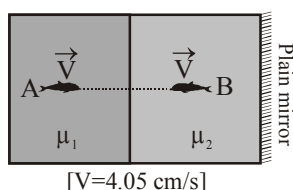
15. As a result of first refraction, find the distance of image from glass-water interface
 (A) 50 cm (B) 60 cm (C) 75 cm (D) 80 cm
16. Second image will be formed after reflection from the concave mirror, the distance of second image from the mirror will be
 (A) 57.50 cm (B) 52.50 cm (C) 27.5 cm (D) 32.5 cm
17. The distance of final image from the object P, is
 (A) 70.2 cm (B) 68.25 cm (C) 62.25 cm (D) 39.37 cm

SECTION-III

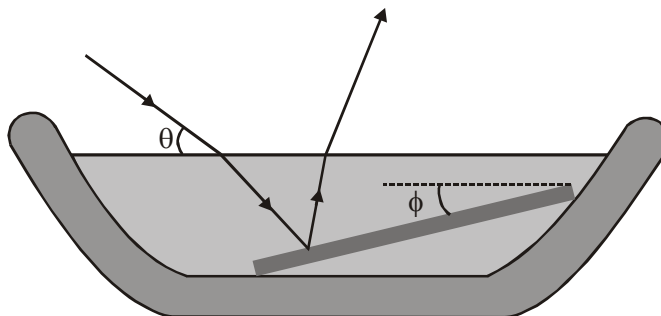
Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. An aquarium is bifurcated by a thin sheet of transparent material as shown in the figure. Each of the two portions contains different kinds of liquid (refractive indices $\mu_1 = 4/3$ and $\mu_2 = 27/20$ respectively). Two fishes A and B swim along each other with their line of approach perpendicular to the interface. One of the side walls is a plane mirror. The velocity of separation of the two images of the fish B that are being observed by the fish 'A' in cm/s is:



2. A small mirror is placed in a bowl of water, making an angle $\phi = 13.5^\circ$ with the water surface as shown in figure. A beam of light strikes the water at angle $\theta = \cos^{-1}\left(\frac{2}{3}\right)$ from its surface and subsequently reflects from the mirror. At what angle (in degree) from the water surface does the ray emerge from the water?



SECTION-IV

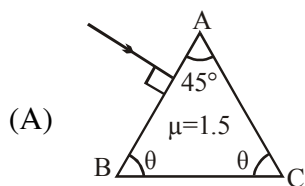
Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

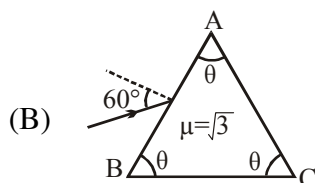
1. Match the following :-

Column-I

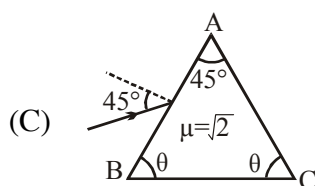
Column-II



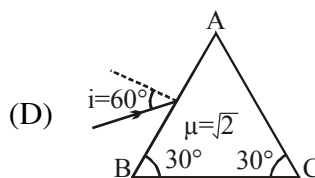
(P) At surface AC TIR will take place



(Q) At surface AC light will be refracted



(R) Ray refracted at AB will be parallel to base BC



(S) Ray refracted at AB will not be parallel to base BC

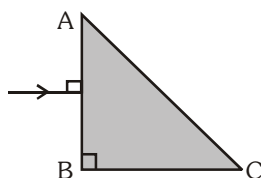
(T) None of these

SECTION-I

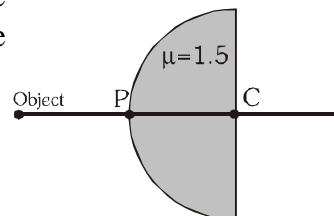
Single Correct Answer Type

7 Q. [3 M (–1)]

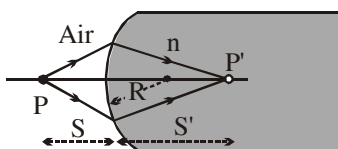
1. In the given figure ABC is a right angled isosceles prism kept in air. A ray of light is incident normally as shown. Refractive index of prism is varying with time t as $\mu = (1 + 0.4t)$, here t is in seconds. The angular velocity of the emergent ray at $t = 1$ second is :-



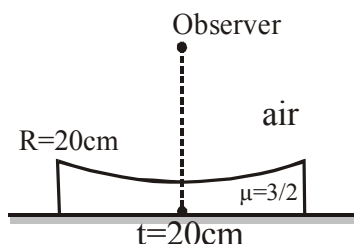
- (A) 1.2 rad/second (B) 1 rad/second (C) 2 rad/second (D) 1.5 rad/second
2. A point object is placed at a distance of $3R$ from point P on glass hemisphere of radius R . If refractive index of hemisphere is 1.5 then position of its image from C is :



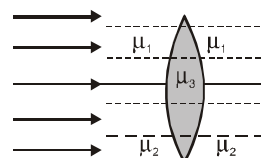
- (A) $\frac{15}{2}R$ to the right (B) $\frac{16}{3}R$ to the right
(C) $7.5R$ to the left (D) $\frac{8}{15}R$ to the left
3. Which of these actions will move the real image point P' farther from the boundary ?
(1) Decrease the index of refraction n . (2) Increase the distance S .
(3) Decrease the radius of curvature R .



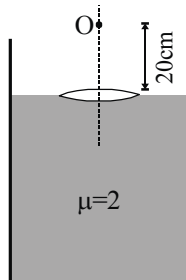
- (A) 1, 2, 3 (B) 1 only (C) 2 & 3 only (D) 2 only
4. A 20 cm thick planoconcave lens is placed on a paper on which flower is drawn. How far above its actual position does the flower appear to an observer looking down normally from top ?



- (A) 10 cm (B) 15 cm (C) 50 cm (D) None of these
5. A double convex lens, made of a material of refractive index μ_3 , is placed inside two liquids of refractive indices μ_1 and μ_2 , as shown $\mu_1 > \mu_3 > \mu_2$. A wide, parallel beam of light is incident on the lens from the left. The lens will give rise to :-
(A) a single convergent beam
(B) two different convergent beams
(C) two different divergent beams
(D) a convergent and a divergent beam



6. There is a small black dot at the centre C of a solid glass sphere of refractive index μ , when seen from outside, the dot will appear to be located
(A) away from the C for all values of μ .
(B) at C for all values of μ .
(C) at C for $\mu=1.5$, but away from C for $\mu \neq 1.5$
(D) at C for $2 < \mu < 1.5$
7. A point object O is placed at a distance of 20 cm in front of an equi-convex lens ($\mu_g = 1.5$) of focal length 10 cm. The lens is placed on a liquid of refractive index 2 as shown. Image will be formed at a distance h from lens. The value of h is :-

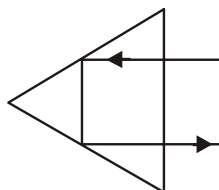


- (A) 5 cm (B) 10 cm (C) 20 cm (D) 40 cm

Multiple Correct Answer Type

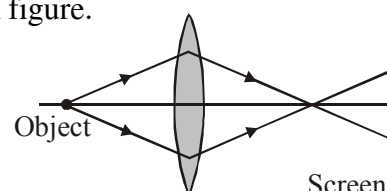
4 Q. [4 M (-1)]

8. A right angled isosceles prism is used in air ($\mu_{\text{air}} = 1$) for reflecting back a ray as shown. This is possible if refractive index of prism (μ) :

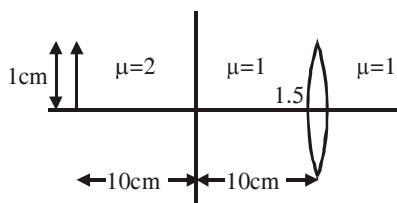


- (A) $\mu = 1.25$ (B) $\mu = 1.35$ (C) $\mu = 1.50$ (D) $\mu = 1.65$

9. A lens forms a blurred image of a point object on a screen as shown in figure. Which of the following can lead to a sharp image on the screen?

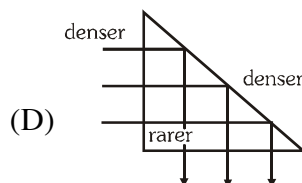
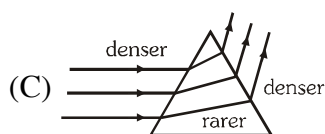
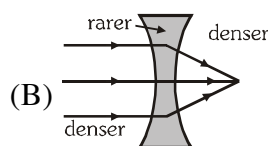
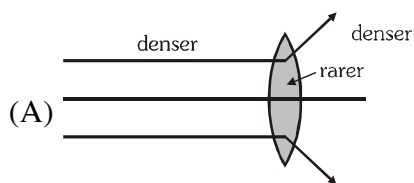


- (A) By moving the screen towards lens.
(B) By moving the object towards the lens.
(C) By using a lens of larger focal length.
(D) By using a lens of smaller focal length.
10. An object of length 1 cm is placed on a principle axis of an equiconvex lens of radii 5 cm. Refractive index of medium of the lens is 1.5. Distance between the lens and object is 20 cm. Space between the lens and object is filled with medium of two different refractive index 2 and 1, lens being placed in the medium of refractive index 1. Boundary of both medium is mid-way between the object and lens as shown in figure.



- (A) The image will be formed at distance of 7.5 cm from the optical centre
(B) The image will be formed at distance of $20/3$ cm from the optical centre
(C) The size of the image is 0.5 cm
(D) The size of the image is 0.4 cm

11. When light rays pass from an optically denser medium through an optically less denser medium as shown below, which is/are possible?

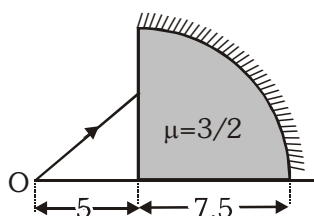


Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

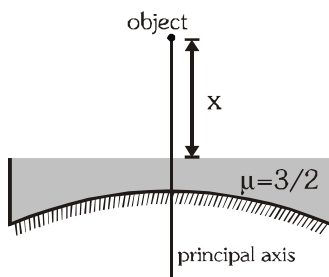
Paragraph for Questions no. 12 to 14

If a ray is incident normally on a mirror, the ray retraces its former path and final image of the object forms on the object itself.

12. Radius of curvature of spherical surface, for which image forms on the object itself will be

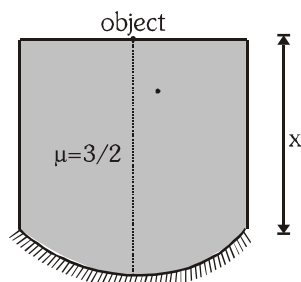


- (A) 10 cm (B) 15 cm (C) 20 cm (D) None of these
13. If amount of liquid in curved part of the mirror is very small, focal length of mirror is 10 cm, the value of x for which final image forms on the object itself



- (A) 10 cm (B) 15 cm (C) 20 cm (D) None of these

14. Radius of curvature of spherical surface is 20 cm. If image of the object forms on the object itself. The value of x is



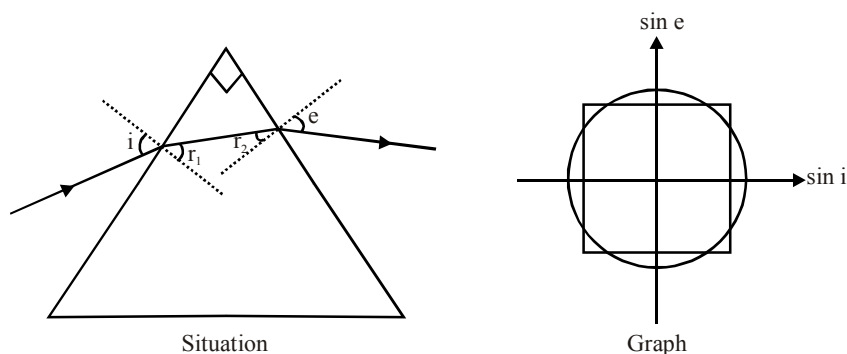
- (A) 10 cm (B) 15 cm (C) 20 cm (D) None of these

Paragraph for Question 15 and 16

Consider a right angled prism as shown. The light ray is incident on its left face at an angle i and emerges at an angle e . When we successively apply snell's law at both faces, we get

$$\sin^2 i + \sin^2 e = \mu^2$$

If we plot $\sin i$ on x-axis and $\sin e$ on y-axis, this is a equation of circle with radius μ . Study of this circle gives some interesting insight into behaviour of this prism. If we draw a square of side 2×2 as shown the parts of circle that lie outside the square represent the case of total internal reflection because at the point of intersection either $\sin i = 1$ or $\sin e = 1$. The negative values of $\sin i$ and $\sin e$ refer to the cases where the incident or the emergent ray is above the normal.



15. Point on the circle which corresponds to minimum deviation is obtained at :-

- (A) At an angle $\theta = \sin^{-1} \left(\frac{1}{\mu} \right)$ with positive $\sin i$ axis
(B) At an angle $\theta = \cos^{-1} \left(\frac{1}{\mu} \right)$ with positive $\sin i$ axis
(C) Intersection of circle with positive $\sin i$ axis
(D) At an angle $\theta = 45^\circ$ with positive $\sin i$ axis

16. For the given situation, which quadrant of graph is representing the situation which can be achieved.

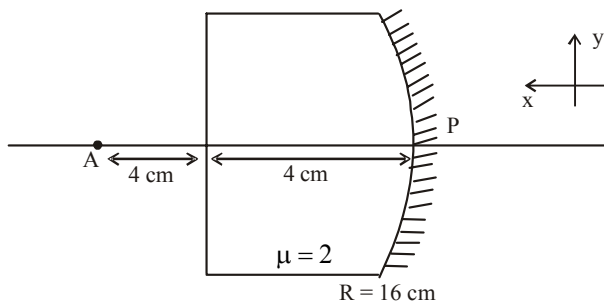
- (A) 1st quadrant (B) IIIrd quadrant (C) IVth quadrant (D) All of these

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

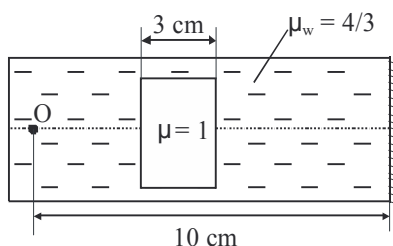
1. A curved thick glass surface is silvered at curved face & not silvered on plane surface. Object is placed at A as shown in figure. Considering P (pole of the silvered surface) as origin. If x - co-ordinate of final image is (2n) cm then find n.



2. Light falls on a prism of angle $\cos^{-1}\left(\frac{1}{8}\right)$. If the minimum value of refractive index of the prism-material,

so that light entering one face may not exit through the other is $\frac{k}{\sqrt{7}}$, then k = ?

3. A point object O is placed at a distance of 10 cm from a mirror in a medium containing water ($\mu_w = 4/3$). In between the object and mirror a 3 cm vacuum is created by some means. The light ray undergoes two refractions and then reflection at mirror and then again two refractions. Find the distance ' ℓ ' (in cm) of final image from the mirror. Fill ' $\ell/2$ ' in OMR sheet.

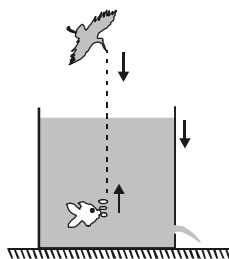


SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. A bird in air is diving vertically over a tank with speed 5 cm/s, base of tank is silvered. A fish in the tank is rising upward along the same line with speed 2 cm/s. Water level is falling at rate of 2 cm/s. [Take : $\mu_{\text{water}} = 4/3$]



Column I (cm/s)

- (A) Speed of the image of fish as seen by the bird directly
- (B) Speed of the image of fish formed after reflection in the mirror as seen by the bird
- (C) Speed of image of bird relative to the fish looking upwards
- (D) Speed of image of bird relative to the fish looking downwards in the mirror

Column II

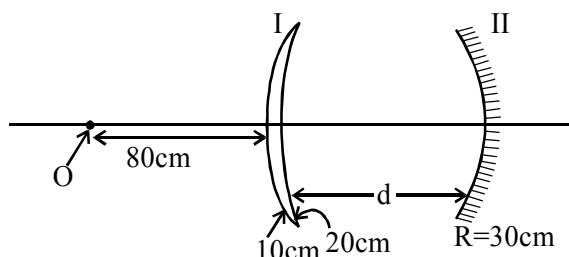
- (P) 8
- (Q) 6
- (R) 3
- (S) 4

SECTION-I

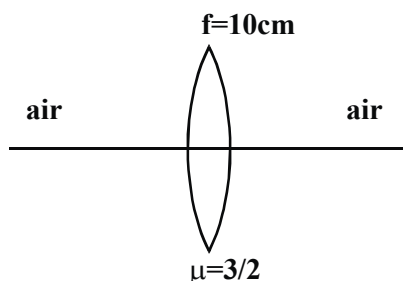
Single Correct Answer Type

5 Q. [3 M (-1)]

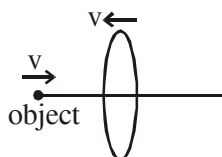
- When a thin convergent glass lens ($\mu_g = 1.5$) of power +5 D is immersed in a liquid of refractive index μ_ℓ it acts as a divergent lens of focal length 100 cm. Then μ_ℓ is :
(A) 4/3 (B) 5/3 (C) 5/4 (D) 6/5
- If final image formed after two refractions through the lens of refractive index 1.5 and one reflection from the mirror is forming at same point 'O' then d is equal to :-



- (A) 100 cm (B) 120 cm (C) 90 cm (D) 80 cm
- Given set-up which is shown in figure, converges parallel beam of light at point P_1 . If the surrounding medium of above set-up is replaced by transparent fluid of refractive index $\mu_m = 2$, then same parallel beam converge at point P_2 then distance P_1P_2 is



- (A) 70 cm (B) 20 cm (C) 30 cm (D) 10 cm
- In the diagram shown, the lens is moving towards the object with a velocity V m/s and the object is also moving towards the lens with the same speed. What is speed of the image with respect to earth when the object is at a distance $2f$ from the lens ? (f is the focal length) :-

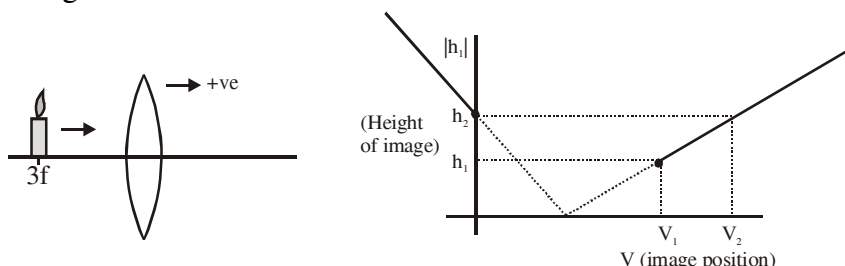


- (A) 2V (B) 4V (C) 3V (D) V
- A plano-convex lens, when silvered at its plane surface is equivalent to a concave mirror of focal length 28 cm. When its curved surface is silvered and the plane surface not silvered, it is equivalent to a concave mirror of focal length 10 cm, then the refractive index of the material of the lens is:-
(A) 9/14 (B) 14/9 (C) 17/9 (D) none

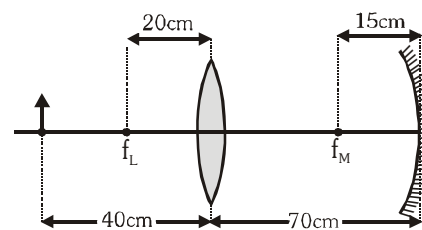
Multiple Correct Answer Type

3 Q. [4 M (-1)]

6. A candle of height h is being moved along the axis from $u = 3f$ towards a thin converging lens of focal length f . The adjacent graph represents the height of image as a function of the position of image w.r.t. lens as shown in figure.



- (A) The value of h_2 is h .
(B) The value of h_2 is $2h$.
(C) The value of h_1 is $h/2$.
(D) The ratio v_1/v_2 is $3/4$.
7. Mark the **CORRECT** statement(s) :-
(A) Laws of reflection are valid at each point on any reflecting surface whether plane or curved
(B) In normal situations the wavelength is several thousands times smaller than the usual obstacles or openings hence light can then be treated as light rays.
(C) When sunlight falls on small water droplets suspended in air during or after a rain, it suffers refraction, total internal reflection and dispersion.
(D) Dispersion takes place because the refractive index of medium for different wavelengths (colours) is different.
8. An upright object is placed at a distance in front of a converging lens equal to twice the focal length 20cm of the lens. On the other side there is a concave mirror of focal length 15cm separated from the lens by a distance of 70cm . Then select the correct statements from the following
(A) Magnification for the system is $-1/2$
(B) Magnification for the system is -1
(C) Final image by the system will real and at distance of 80cm from centre of curvature of spherical mirror
(D) Magnification for the system is $+1/2$

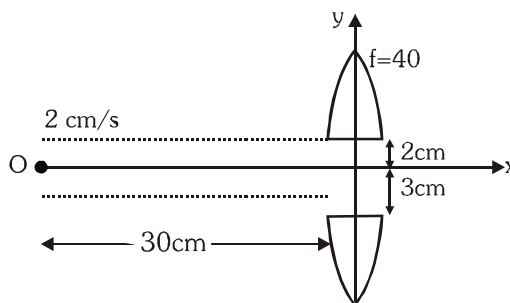


Linked Comprehension Type (Single Correct Answer Type)

(2 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question 9 to 11

An object is approaching two pieces of a lens halves are placed according to diagram.



9. x coordinate of images
(A) -120 (B) -30 (C) -40 (D) $+120$
10. y coordinates of image formed by upper and lower half
(A) $-6\text{ cm}, 9\text{ cm}$ (B) $-8\text{ cm}, -12\text{ cm}$ (C) $8\text{ cm}, -12\text{ cm}$ (D) $-8\text{ cm}, +12\text{ cm}$
11. Component of relative velocities of images parallel to x-axis.
(A) zero (B) 1 cm/s (C) 2 cm/s (D) 3 cm/s

Paragraph for Question 12 to 14

A plano convex lens has thickness 4 cm, when placed on a horizontal table with the curved surface in contact with it, the apparent depth of the bottom most point of the lens is found to be 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of the plane face of the lens is found to be $25/8$ cm.

12. Radius of curvature of curved surface is
(A) 25 cm (B) 50 cm (C) 79 cm (D) 100 cm
13. Focal length of the plano convex lens is
(A) 25 cm (B) 50 cm (C) 75 cm (D) 100 cm
14. Refractive index of the lens is
(A) $4/3$ (B) $8/5$ (C) $5/3$ (D) $8/7$

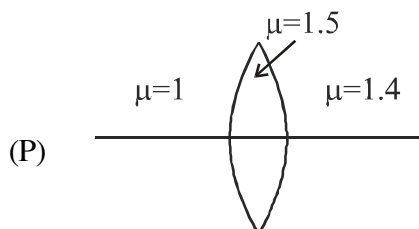
Matching List Type (4×4)

1Q. [3 M (-1)]

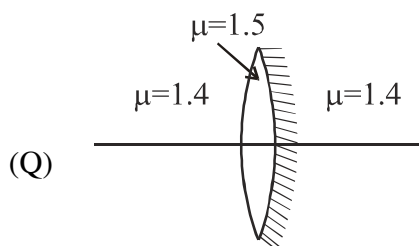
15. List-I gives different lens configurations. The radius of curvature of each curved surface is R. Parallel paraxial rays of light parallel to the axis of lens traversing through the lens get focused at distance f from the lens. List-II gives corresponding values of f. (μ represent refractive index of medium)

List-I

List-II

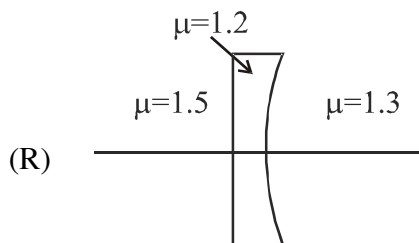


(1) 13 R



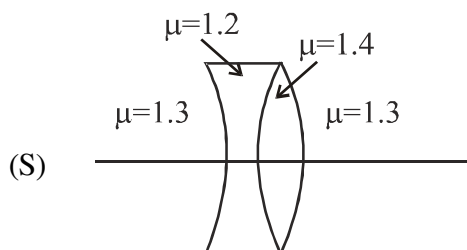
(2) $\frac{13R}{4}$

Lens is silvered



(3) $\frac{7R}{3}$

Plano concave lens



(4) $\frac{7R}{16}$

Codes

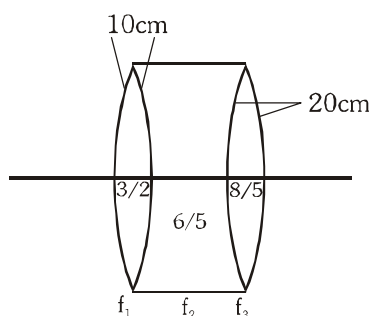
	P	Q	R	S		P	Q	R	S
(A)	4	1	3	2	(B)	2	3	4	1
(C)	1	2	4	3	(D)	3	4	1	2

SECTION-III

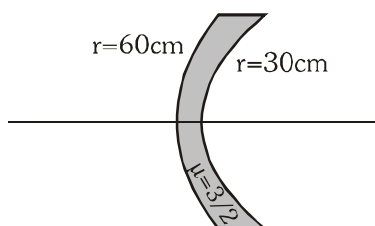
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

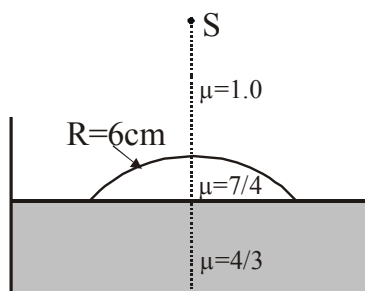
1. In the shown figure the focal length (in cm) of equivalent system is in the form of $\left(\frac{50x}{13}\right)$. Find the value of x.



2. The given lens is having radius of curvature 30 cm and 60 cm as shown in the figure respectively. The power of lens is $-5/\alpha$ dioptre. Find the value of α .



3. Water (with refractive index = $4/3$) in a tank is 18 cm deep. Oil of refractive index $7/4$ lies on water making a convex surface of radius of curvature ' $R = 6$ cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at ' x ' cm above the bottom of the tank. Then ' x ' is



SECTION-IV

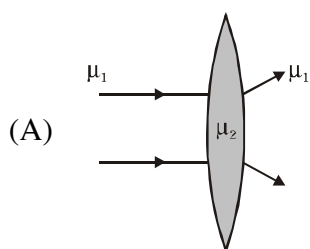
Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

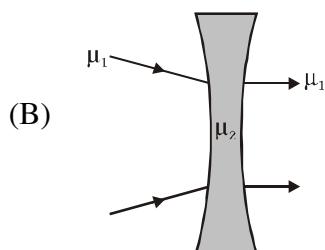
1. Column I shows incident ray, refracted ray associated with different lenses whereas column-II shows refractive index, of the lens surrounding medium and nature of optical instruments. Match the entries of column-I with column-II

Column I

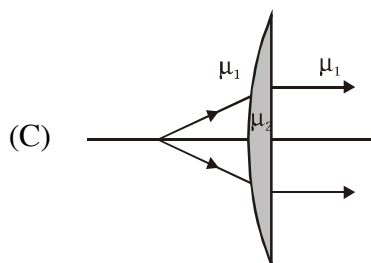
Column II



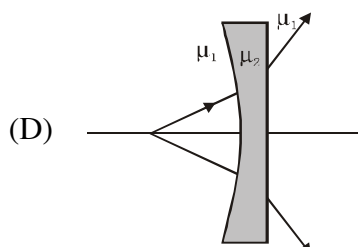
(P) $\mu_2 > \mu_1$



(Q) $\mu_2 < \mu_1$



(R) diverging



(S) converging

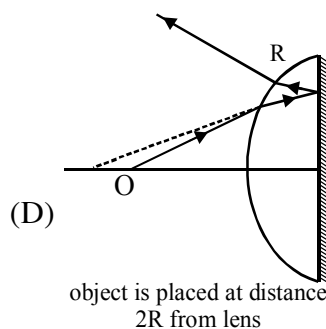
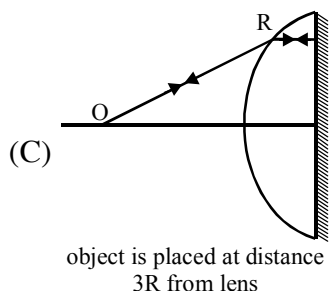
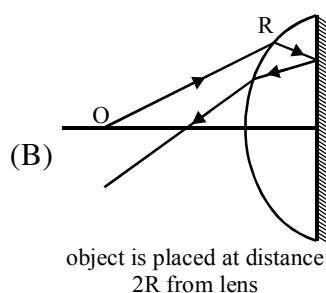
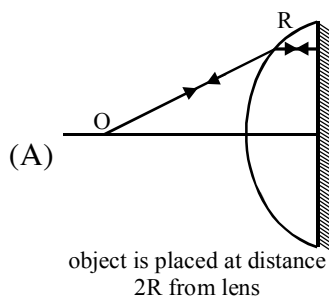
(T) None of these

SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

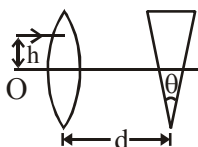
1. A thin plano-convex glass lens ($\mu = 1.5$) has its plane surface reflecting and R is the radius of curvature of curved part, then which of the following ray diagram is true for an object placed at O ?



2. Two thin symmetrical lenses of different nature have equal radii of curvature $R = 20$ cm. The lenses are put close together and immersed in water. The converging focal length of the system is 24 cm. Find the difference between refractive indices of the two lenses. (refractive index for water = $\frac{4}{3}$)

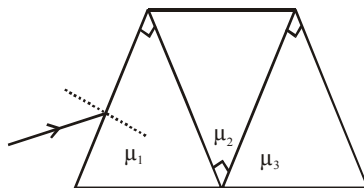
(A) $(\mu_1 - \mu_2) = \frac{1}{24}$ (B) $(\mu_1 - \mu_2) = \frac{1}{20}$ (C) $(\mu_1 - \mu_2) = \frac{5}{9}$ (D) $(\mu_1 - \mu_2) = \frac{4}{3}$

3. A ray of light parallel to the axis of converging lens (having focal length f) strikes it at a small distance 'h' from its optical centre. A thin prism having angle θ and refractive index μ is placed normal to the axis of lens at a distance 'd' from it. What should be the value of μ so that the ray emerges parallel to the lens axis?

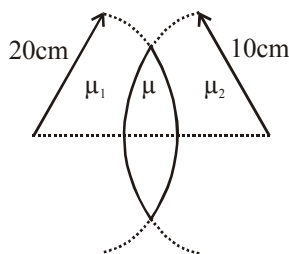


(A) $\frac{h}{f\theta}$ (B) $\frac{h}{f\theta} + 1$ (B) $\frac{h}{(d+f)\theta}$ (D) $\frac{h}{(d+f)\theta} + 1$

4. Three right-angled prisms of refractive indices μ_1 , μ_2 and μ_3 are joined together as shown in the figure. If the ray passes through the whole system undeviated then :-



- (A) $\mu_1^2 + \mu_2^2 + \mu_3^2 = 1$ (B) $\mu_1^2 + \mu_3^2 = 1 + \mu_2^2$ (C) $\mu_1^2 + \mu_2^2 = 1 + \mu_3^2$ (D) $\mu_1^2 = \mu_2^2 + \mu_3^2 + 1$
5. Radii of curvature of surfaces of a biconvex lens are 10 cm and 20 cm and refractive index of its material is $\mu = 1.5$. If the refractive indices of the medium on the two sides of the lens are μ_1 and μ_2 , then the parallel rays incident on the lens from left will focus at a distance of f_2 (distance being measured from the optical centre of the lens) where f_2 is equal to :-

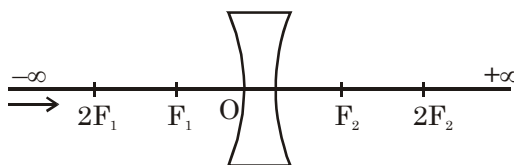


- (A) $\frac{20\mu_1}{4.5 - (\mu_1 + \mu_2)}$ (B) $\frac{20\mu_2}{4.5 - (\mu_1 + 2\mu_2)}$ (C) $\frac{20\mu_2}{4.5 - (2\mu_1 + \mu_2)}$ (D) $\frac{20\mu_1}{4.5 - (\mu_1 + 2\mu_2)}$

Multiple Correct Answer Type

2 Q. [4 M (-1)]

6. A object is moving on the optical axis of a diverging lens from $-\infty$ to O optical centre of the lens. Choose the **CORRECT** statement(s) about the image of particle :

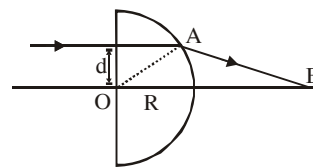


- (A) When object is moving between $(-\infty, 2F_1)$ its image will be virtual & diminished.
(B) When object is moving between $(2F_1, F_1)$ its image will be real and diminished.
(C) When object is moving between F_1 to O its image will be virtual & enlarge.
(D) For complete motion of object, image will lie on same side of the lens.
7. A real object is kept in front of a lens in air. The object is a linear extended object with its length perpendicular to the optical axis of lens. With reference to different cases of image formation by lens, choose the correct options :
- (A) If the image has a magnification -2.5 then image is real and power of lens is positive.
(B) If the magnification of the image is $+0.5$ then image is virtual and power of lens is negative
(C) If length of image is the same as that of object then image is real and power of lens is positive.
(D) If length of image is the same as that of object then image is virtual and power of lens is negative.

Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question 8 to 10

A semi cylinder made of a transparent plastic has a refraction index of $n = \sqrt{2}$ and a radius of R . There is a narrow incident light ray perpendicular to the flat side of the semi cylinder at d distance from the axis of symmetry.



8. What can the maximum value of d be so that the light ray can still leave the other side of the semi cylinder?
- (A) $d = \frac{R}{\sqrt{2}}$ (B) $d = \frac{R}{2\sqrt{2}}$ (C) $\frac{R}{2}$ (D) None of these
9. When the value of d chosen is such that TIR just takes place then time for which light remains inside the cylinder is
- (A) $4R/c$ (B) $\frac{4\sqrt{2}R}{c}$ (C) $\frac{2\sqrt{2}R}{c}$ (D) $2R/c$
10. Distance d is now varied, so that the ray always emerges from the other side of the semi cylinder. What is the range of OB ?
- (A) $2\sqrt{2}R > OB > R\sqrt{2}$ (B) $5\sqrt{2}R > OB > R$
(C) $\infty > OB > R\sqrt{2}$ (D) $\infty > OB > 2R\sqrt{2}$

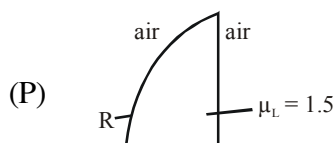
Matching List Type (4 × 4)

1Q. [3 M (-1)]

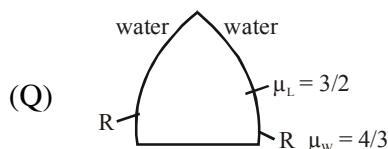
11. Using the lens maker formula, relate the focal length of thin lens in List-I with value in List-II.

List-I

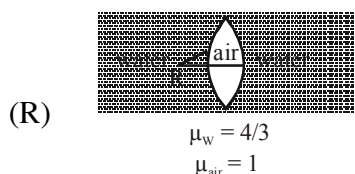
List-II



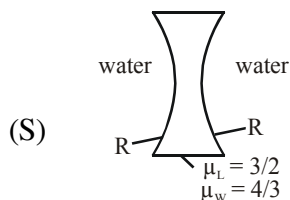
(1) $4R$



(2) $-2R$



(3) $-4R$



(4) $2R$

Codes :

	P	Q	R	S
(A)	1	3	4	2
(B)	4	2	1	3
(C)	3	2	1	4
(D)	4	1	2	3

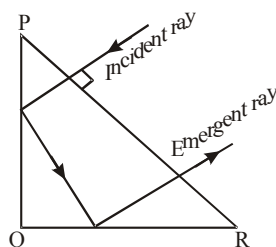
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

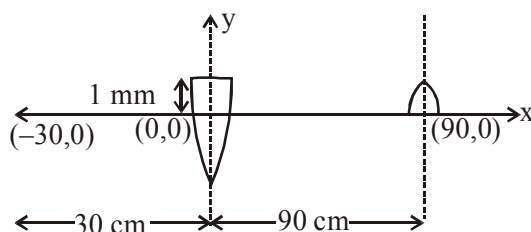
7 Q. [4 M (0)]

1. For a right angled prism PQR immersed in water, the incident and emergent rays are parallel as shown

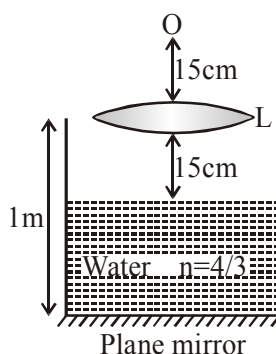
in figure. The minimum value of refractive index of the prism is $\frac{N\sqrt{2}}{3}$. Then find the value of N.



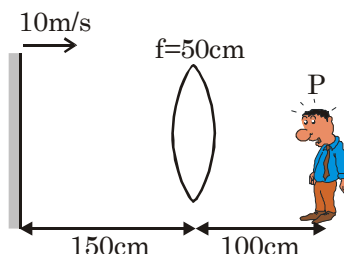
2. A thin convex lens having focal length 20 cm is cut into two parts 1 mm above the principle axis. The lower portion is placed with optical centre at origin and upper portion at (90, 0) as shown in the figure. A point object is placed at (-30, 0). Find the magnitude of y-coordinates (in mm) of the image. Assuming paraxial ray approximation to remain valid.



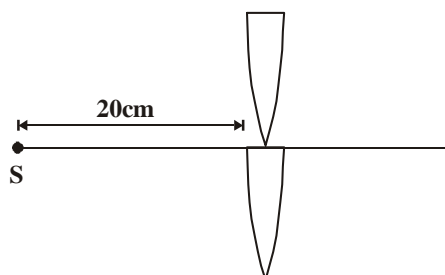
3. In the figure shown, O is a point object placed above the lens L of focal length 10 cm. The distance between the object O and its image formed in the plane mirror is $n \times 20$ cm find n.



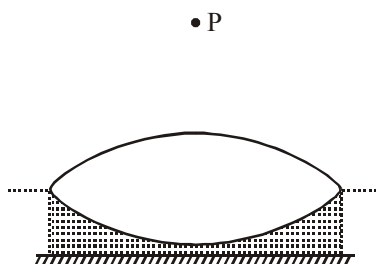
4. At the instant shown in the figure, find the magnitude of velocity 'v' (in m/s) of image of person as seen by himself. Fill $\frac{9v}{5}$ in OMR sheet.



5. A convex lens of focal length 12 cm and 4 mm aperture is cut in to two equal halves and are placed as shown in the figure. A point source S is placed on the principal axis of the lens at a distance 20 cm from the lens as shown in the figure. Find the separation (in mm) between the images formed.



6. A lens forms the image of sun on a screen 30 cm away. How far (in decimeter) from the first lens should a second lens of focal length 30 cm be placed (between first lens and screen) so that the screen has to be moved 8 cm towards the first lens for the new image to be in focus?
7. Figure shows an equiconvex lens of $\mu = 1.50$, in contact with a liquid layer on the top of a plane mirror. When a point object is placed at 45 cm from lens, it is found that image coincides with object. When liquid is removed and experiment is repeated, the new distance for object and image to coincide is found to be 30 cm. Then refractive index of liquid would be $\frac{n}{3}$. Find the value of n.



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Match the entries of column-I with the entries of column-II, which describes the angle of deviation of ray with angle of incidence.

Column-I

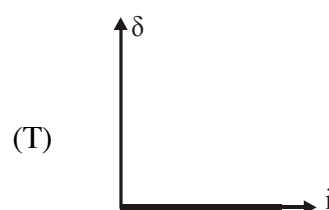
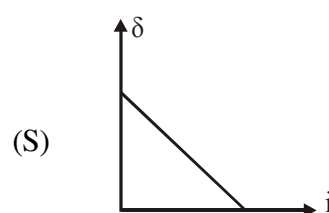
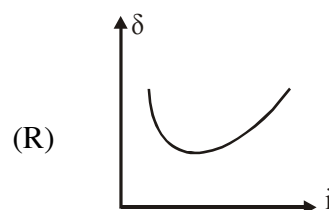
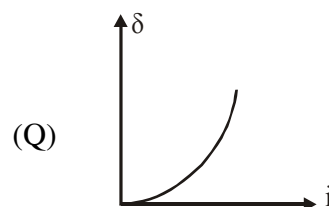
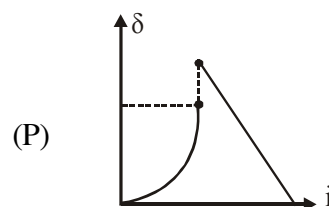
(A) A ray is falling on a plane smooth mirror

(B) A ray is going from a rarer to denser medium

(C) A ray is going from a denser to rarer medium

(D) A ray is falling parallel slab

Column-II

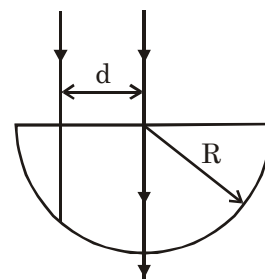


SECTION-I

Single Correct Answer Type

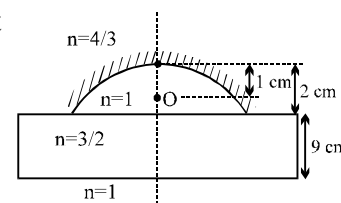
5 Q. [3 M (-1)]

1. As shown, a narrow beam of light is incident onto a semi-circular glass cylinder of radius R . Light can exit the cylinder when the beam is at the centre. When the beam is moved parallelly to a distance d from the central line, no light can exit the cylinder from its lower surface. Find the refractive index of the glass.

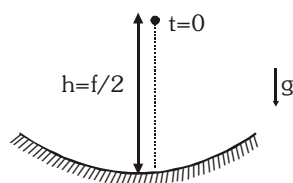


- (A) $\frac{R}{d}$ (B) $\frac{d}{R}$
(C) $\frac{R}{\sqrt{R^2 - d^2}}$ (D) $\frac{\sqrt{R^2 - d^2}}{R}$

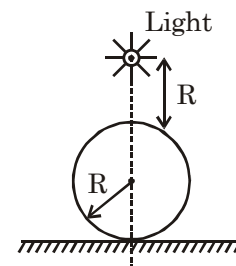
2. A concave mirror of focal length 2 cm is placed on a glass slab as shown in the figure. Then the image of object O formed due to reflection at mirror and then refraction by the slab :



- (A) will be virtual and will be at 2 cm from the pole of the concave mirror
(B) will be virtual and formed on the pole of the mirror
(C) will be real and on the object itself
(D) none of these
3. An object was placed upright 25 cm in front of a converging lens with a focal length of 20 cm. A concave mirror with a focal length of 15 cm was placed 120 cm behind the lens. Which of these describes the final image?
(A) real, enlarged (B) virtual, upright (C) virtual, inverted (D) inverted, diminished
4. A particle is dropped along the axis from a height $\frac{f}{2}$ on a concave mirror of focal length f as shown in figure. The maximum speed of image is :



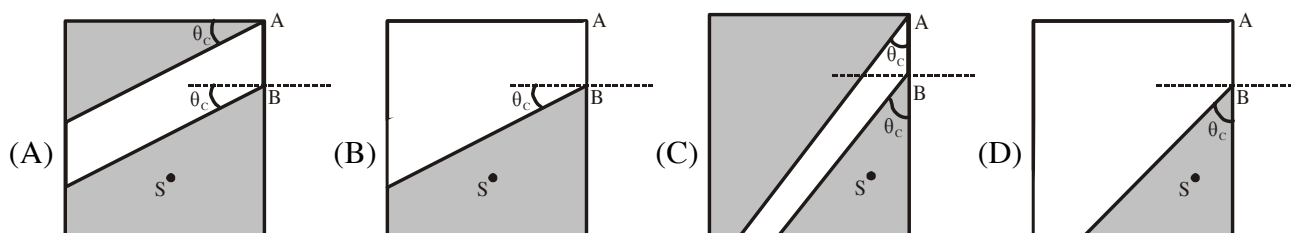
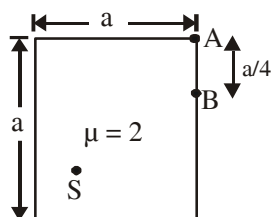
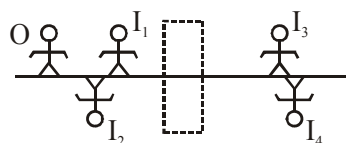
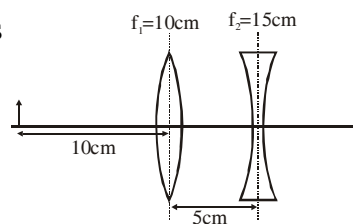
- (A) ∞ (B) $\frac{3}{4}\sqrt{3fg}$ (C) $\frac{3}{2}\sqrt{3fg}$ (D) None of these
5. An opaque sphere of radius R lies on a horizontal plane. A light source is placed above sphere as shown. Then
(i) area of shadow on the plane is $2\pi R^2$
(ii) area of shadow on the plane is $3\pi R^2$
(iii) if the sphere is just submerged in some liquid, area of the shadow on plane decreases
(iv) if the sphere is just submerged in some liquid, area of the shadow on plane increases
(A) (i), (ii) (B) (ii), (iii) (C) (i), (iii) (D) (ii), (iv)



Multiple Correct Answer Type

4 Q. [4 M (–1)]

6. A lens forms 3 times magnified image of an object on screen. When the distance between the screen and the lens is increased by 10 cm and object is also moved to get 5 times magnified image again on the screen. Then :-
 (A) Focal length of the lens is 5 cm
 (B) Focal length of the lens is –5 cm
 (C) The displacement of the object is $\frac{2}{3}$ cm towards the lens
 (D) The displacement of the object is $\frac{2}{3}$ cm away the lens
7. An object is placed nearly at focus of a converging lens of focal length 10 cm as shown in figure. A diverging lens of focal length of 15 cm is placed 5 cm behind the converging lens as shown in the figure.
 (A) Final image forms at the object
 (B) Image forms at a distance 30 cm from concave lens
 (C) Magnification of image is 1.5
 (D) Magnification can not be determined.
8. In figure, stick figure O stands in front of a thin, symmetric mirror that is mounted within the boxed region; the central axis through the mirror is shown. The four stick figures I_1 to I_4 suggest general locations and orientations for the images that might be produced by the mirror. (The figure are only sketched in; their height and their distance from the mirror is not drawn to scale.)
 (A) I_4 cannot be a possible image
 (B) I_1 cannot be a possible image.
 (C) If I_2 is an image, mirror must be concave only.
 (D) If I_3 is an image, mirror must be convex only.
9. Figure shows a transparent block of front sides 'a' and 'a'. The third dimension of the block is negligible. A point source S which can emit light in all directions can move inside the block. It is desired that no direct light of 'S' should pass through AB. The region in which S should be present to satisfy this condition shown by shaded region. Choose correct option.

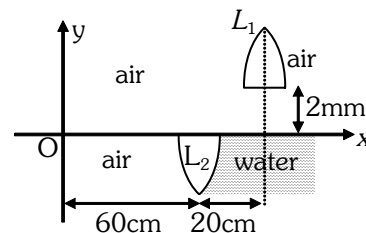


Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question 10 to 12

A point object O is placed at the origin of coordinate system. An equi-convex thin lens ($\mu_g = 1.5$) of focal length $f=20$ cm in air is placed so that its principal axis is along x-axis. Now the lens is cut at the middle (along the principal axis) and upper half is shifted along x-axis and y-axis by 20 cm and 2 mm respectively and right side of lower half is filled with water ($\mu_w = 4/3$) as shown in figure.



10. Total number of images formed by the combination will be
(A) 1 (B) 2 (C) 3 (D) 4
11. Coordinates of the image produced by the lens L_1 will be
(A) $\left(\frac{320}{3}\text{cm}, \frac{4}{3}\text{mm}\right)$ (B) $\left(\frac{160}{3}\text{cm}, \frac{8}{3}\text{mm}\right)$ (C) $\left(\frac{320}{3}\text{cm}, \frac{8}{3}\text{mm}\right)$ (D) $\left(\frac{160}{3}\text{cm}, \frac{4}{3}\text{mm}\right)$
12. Coordinates of the image produced by the lens L_2 will be
(A) 140cm, 0 (B) 140cm, 20 (C) 70cm, 0 (D) 140cm, 30

Matching list based comprehension Type (4 × 4 × 4) 1 Table × 3 Q. [3(-1)] Single option correct (Three Columns and Four Rows)

Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table.

In the below question u and v represents distance of object and image from the lens/mirror. Both are measured using usual conventional method i.e. distance in the direction of incident rays are positive and opposite to incident rays are negative. Column 2 gives the nature of optical element and column 3 gives the variation of image distance from lens/mirror as object distance changes. To answer the related questions we have to consider the part of graph from 1 to 2 as shown in column 3.

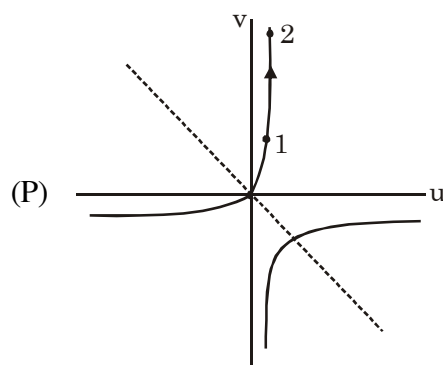
Column-1

Column-2

Column-3

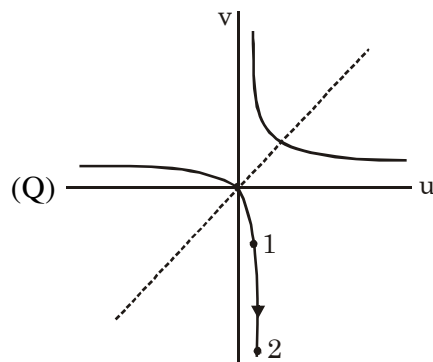
(I) $u < 0, v > 0$

(i) Convex mirror



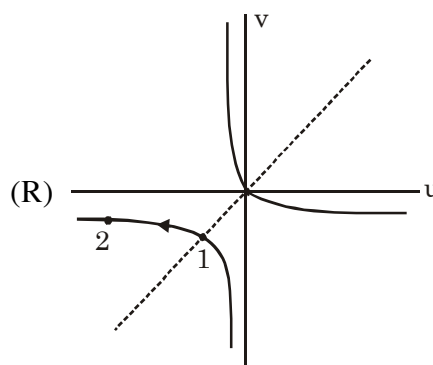
(II) $u > 0, v > 0$

(ii) Convex lens



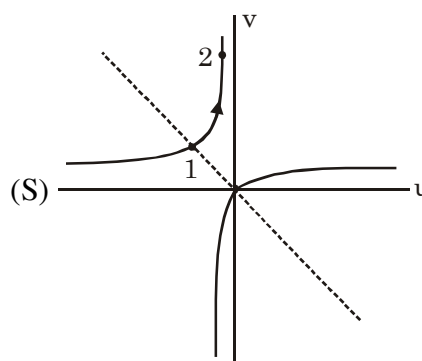
(III) $u < 0$ $v < 0$

(iii) Concave mirror



(IV) $u > 0$ $v < 0$

(iv) Concave lens



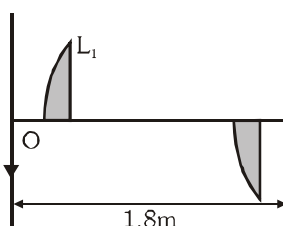
13. In which of the following case. Real image of virtual object is moving away from mirror/lens:-
 (A) (I) (iii) (R) (B) (II) (iv) (P) (C) (II) (ii) (P) (D) (III) (i) (S)
14. Which of the following represents real image of real object going towards mirror/lens :-
 (A) (IV) (ii) (S) (B) (II) (i) (R) (C) (III) (iii) (R) (D) (I) (iii) (P)
15. Which represents real image of real object going away from mirror/lens :-
 (A) (I) (ii) (S) (B) (III) (i) (P) (C) (III) (iv) (R) (D) (II) (iii) (Q)

SECTION-III

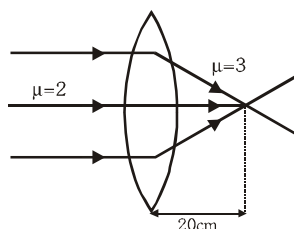
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

1. When a bright source is placed 30 cm in front of a lens there is an erect image 7.5 cm from the lens. There is also of faint inverted image 6 cm in front of the lens due to reflection from the front surface of the lens. When the lens is turned around this faint inverted image is seen 10 cm in front of the lens.
- Refractive index of the lens is $1 + \frac{x}{10}$. Find the value of x.
2. A thin plano-convex lens of focal length f is split into two halves, one of the halves is shifted along the optical axis. The separation between object and image plane is 1.8 m. The magnification of the image formed by one of the half lenses is 2. Find the focal length of the lens in decimetre. [1 decimeter = 10cm]

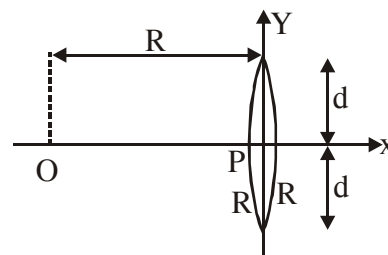


3. A convex lens is placed in such a way that the left side of lens has refractive index 2 and right of lens has refractive index 3. If parallel ray coming from left side focus at 20 cm from lens, the distance from lens where rays will focus if they are coming from right side is $\frac{20n}{3}$ cm. Find the value of n.



4. A biconvex thin lens of radius of curvature R is made up of variable refractive index $\mu = 2\left(1 + \frac{|y|}{d}\right)$. Assume very small aperture $2d \ll R$.

A point object O is placed at a distance $R = 7.5$ m on the principal axis from the lens (as shown). Due to variable refractive index of lens, there are infinite number of image on the principal axis. These image are spreaded over the length ℓ . Find the value of ℓ (in m).



SECTION-IV

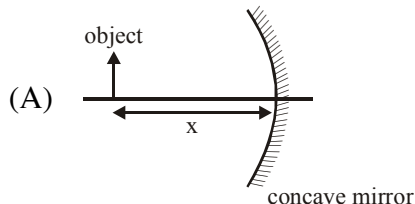
Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

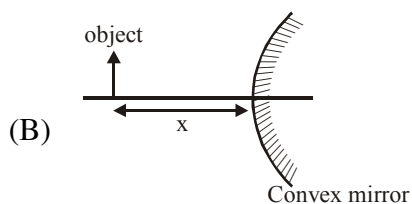
1. In column I, all the objects ($x \geq 0$) shown are real. In column II, nature of image is mentioned. Rays are paraxial, consider all the possibilities of nature of images then match column-I with column-II.

Column-I

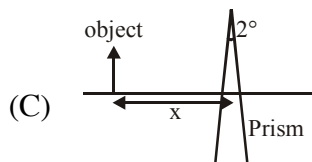
Column-II



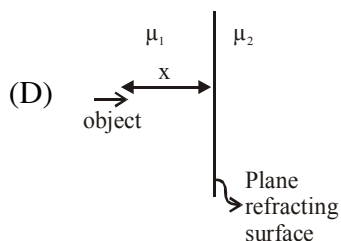
(P) Real



(Q) Virtual



(R) Magnified (Large)



(S) Diminished (Small)

(T) Same size

SECTION-I

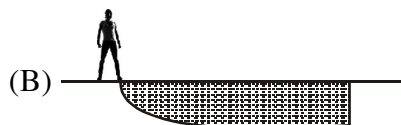
Single Correct Answer Type

6 Q. [3 M (–1)]

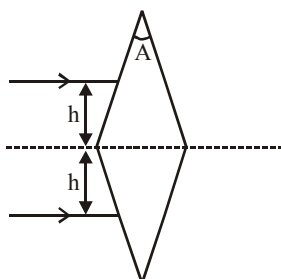
1. A concave mirror is placed on a horizontal surface and two thin uniform layers of different transparent liquids (which do not mix or interact) are formed on the reflecting surface. The refractive indices of the upper and lower liquids are μ_1 and μ_2 respectively. The bright point source at a height 'd' (d is very large in comparison to the thickness of the film) above the mirror coincides with its own final image. The radius of curvature of the reflecting surface therefore is :

- (A) $\frac{\mu_1 d}{\mu_2}$ (B) $\mu_1 \mu_2 d$ (C) $\mu_1 d$ (D) $\mu_2 d$

2. An observer stands at the edge of a swimming pool, as sketched in the figure below. This observer will perceive the pool as :-

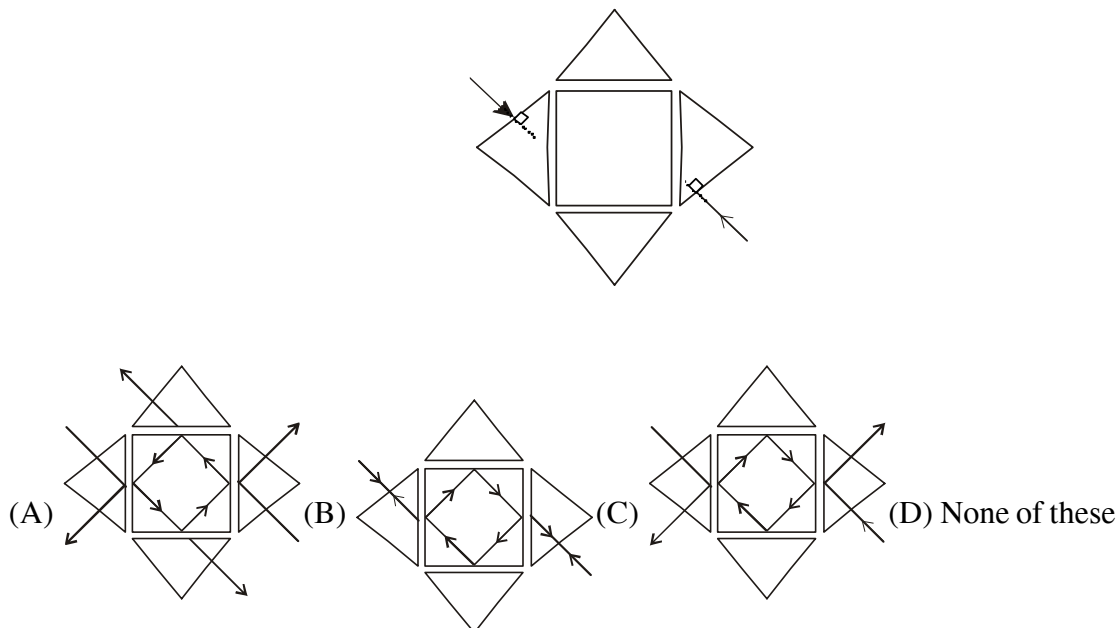


3. Two identical isosceles thin prisms of prism angle A and refractive index μ are placed with their bases touching each other. This system acts as a converging lens. What is the focal length of this system for parallel rays at a distance h from the base of the prism ?



- (A) $\frac{2h}{(\mu-1)A}$ (B) $\frac{h}{(\mu-1)A}$ (C) $\frac{h}{2(\mu-1)A}$ (D) None of these

4. Figure shows a glass cube surrounded by four glass prisms in very close proximity to its sides. The path that will be taken by the two rays incident normally on the sides of the prisms is:



5. Three point charge q , $-2q$ and $-2q$ are placed at the vertices of an equilateral triangle of side a . The minimum work done by some external force to increase their separation to $2a$ will be

- (A) $\frac{1}{4\pi\epsilon_0} \frac{2q^2}{a}$ (B) negative (C) zero (D) $\frac{1}{4\pi\epsilon_0} \frac{3q^2}{a}$

6. A charge of $6\mu\text{C}$ is fixed at the origin. Another identical charge is placed at a distance of 2 m from the fixed charge and is released. The charges have a mass of 0.1 kg each. Then choose the correct statement:-

- (A) The maximum speed achieved by the 2nd charge will be 1.8 m/sec
(B) The maximum speed achieved by the 2nd charge will depend on the direction of its motion
(C) As time passes, potential energy of the system increases.
(D) The 2nd charge will achieve a maximum velocity of $0.9\sqrt{2}$ m/s

Multiple Correct Answer Type

3 Q. [4 M (-1)]

7. Consider mercury (Hg) rotating about a vertical axis with uniform angular velocity ω filled in a cylindrical container. The liquid surface is curved. The figure shows a cross sectional view of the curved surface. Ignore surface tension and viscosity. [Hint : coordinates of focus for parabola $x^2 = 4ay$ is given by $(0, a)$] Mark the **CORRECT** statement(s) :

- (A) Steady state angle θ made by tangent to the surface at $P(x, y)$ with the

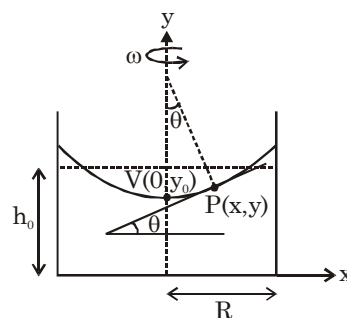
horizontal is given by $\tan^{-1}\left(\frac{\omega^2 x}{g}\right)$

- (B) Steady state angle θ made by tangent to the surface at $P(x, y)$ with the horizontal is given by

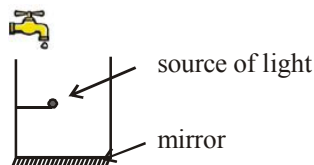
$\sin^{-1}\left(\frac{\omega^2 x}{g}\right)$

- (C) If the focal length of the mirror formed by shiny liquid surface is 20 cm then ω is 5 rad/s

- (D) If the focal length of the mirror formed by shiny liquid surface is 20 cm then ω is 10 rad/s



8. A small source of light is mounted inside a cylindrical container of height h . The bottom of the container is covered with a mirror. Initially, the container is empty. Then a clear liquid with the index of refraction n is slowly poured into the container. The level of liquid H rises steadily, reaching the top of the container in time T . Let h_1 be the distance of source of the light from the bottom of the container. Consider paraxial ray approximation.



Consider two cases in which the observer is in the air observing the image of the source in the mirror.

- (1) $H \leq h_1$ (2) $h_1 \leq H \leq h$

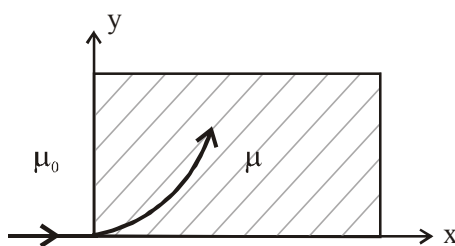
(A) The speed of the image of the source in case (1) during this process is $\frac{2h}{T} \left(1 - \frac{1}{n}\right)$.

(B) The speed of the image of the source in case (2) during this process is $\frac{h}{T} \left(1 - \frac{1}{n}\right)$.

(C) The speed of the image of the source in case (1) during this process is $\frac{h}{2T} \left(1 - \frac{1}{n}\right)$.

(D) The speed of the image of the source in case (2) during this process is $\frac{h}{2T} \left(\frac{1}{n}\right)$.

9. The refractive index of the medium with a certain region, $x > 0, y > 0$, changes with y . A thin light ray travelling in the x -direction in medium having refractive index $\mu_0 = 1$ strikes another medium of refractive index μ at right angles and moves through the medium along a circular arc of radius R as shown in the figure. The material with the greatest known refractive index is diamond, but even the refractive index of this material does not reach the value $\mu_{\max} = 2.5$. It is this limit that sets the maximum angular size of the arc the light ray can cover. Angular size of arc is the angle subtended by the arc at the centre.



(A) The variation of refractive index μ with y is given as $\mu = \frac{R}{R - y}$

(B) The unit vector in the direction of refracted light at $y = \frac{R}{2}$ is $\frac{1}{2} \hat{i} + \frac{\sqrt{3}}{2} \hat{j}$

(C) If the maximum angular size of the arc of light is $\theta_{\max}$ then $\sin \theta_{\max} = 2/5$

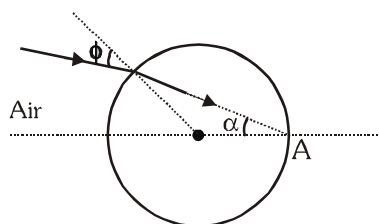
(D) If the maximum angular size of the arc of light is $\theta_{\max}$ then $\cos \theta_{\max} = 2/5$

Linked Comprehension Type (Single Correct Answer Type)

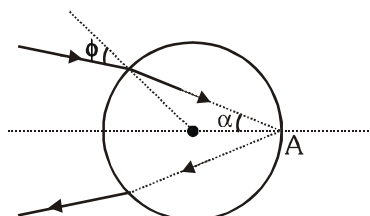
(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question 10 to 12

A ray of light enters a spherical drop of water of refractive index μ as shown in the figure.



10. Select the correct statement
 (A) Incident rays are partially reflected at point A
 (B) Incident rays are totally reflected at point A
 (C) Incident rays are totally transmitted through point A
 (D) None of these
11. An expression of the angle between incident ray and emergent ray (angle of deviation) as shown in the figure is



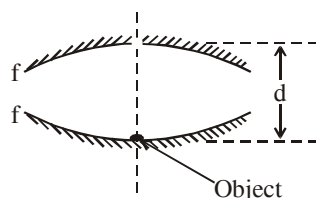
- (A) 0° (B) ϕ (C) $\alpha - \phi$ (D) $\pi - 4\alpha + 2\phi$
12. The angle ϕ for which minimum deviation is produced will be given by
 (A) $\cos^2 \phi = \frac{\mu^2 + 1}{3}$ (B) $\cos^2 \phi = \frac{\mu^2 - 1}{3}$ (C) $\sin^2 \phi = \frac{\mu^2 + 1}{3}$ (D) $\sin^2 \phi = \frac{\mu^2 - 1}{3}$

Linked Comprehension Type (Multiple Correct Answer Type)

(1 Para × 2 Q.) [4 M (-1)]

Paragraph for Questions 13 and 14

Two concave mirrors with equal focal length f , are placed one above other at a separation of d . Upper mirror has a small hole at its centre as given. A small object is placed at centre of lower mirror. Take first reflection at above mirror and second at lower and answer the given questions for these two reflections only.



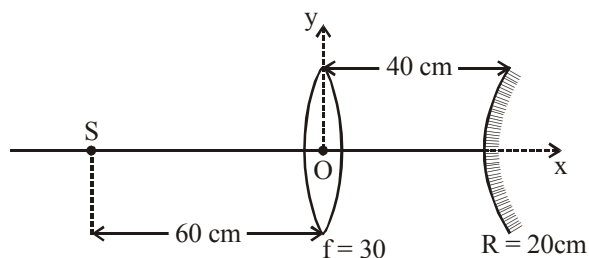
- For final image to be at the hole of the upper mirror
13. Value of d for given possibility
 (A) f (B) $2f$ (C) $3f$ (D) $\frac{f}{3}$
14. Nature of image
 (A) Erect with equal size (B) Erect with unequal size
 (C) inverted with equal size (D) inverted with unequal size

SECTION-III

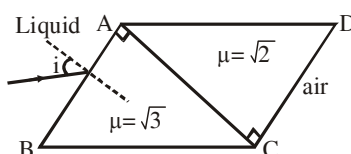
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

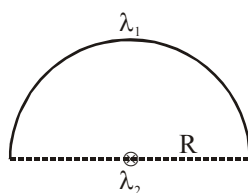
- In the shown arrangement a point source of light S is placed 60 cm left to a convex lens of focal length 30 cm and 2 cm above the common principal axis of the convex lens and a convex mirror of focal length 10 cm. Consider optical centre of the lens to be the origin, the x-coordinate of the final image of the source S for the viewer left of the lens, is equal to $-10K$ cm. Calculate the value of K.



- A man of height 2 m stands on a straight road on a hot day. The vertical temperature in the air results in a variation of refractive index with height y as $\mu = \mu_0 \sqrt{1+ay}$ where μ_0 is the refractive index of air near the road and $a = 2 \times 10^{-6}/\text{m}$. What is the actual length of the road (in km), man is able to see
- A ray of light from a liquid ($\mu = \sqrt{3}$) is incident on a system of two right angled prism of refractive indices $\sqrt{3}$ and $\sqrt{2}$ as shown. The ray suffers zero deviation when emerges into air from CD. The angle of incidence i is π/n . Find the value of n .



- An infinite long uniformly charged wire is kept along axis of a uniformly charged semi circular ring of radius R as shown. Linear charge density of ring and wire are respectively λ_1 and λ_2 . Net electrostatic force on wire due to ring is $\frac{4k\lambda_1\lambda_2}{R^{n-1}}$ value of n is



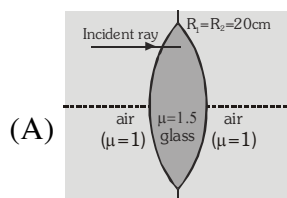
SECTION-IV

Matrix Match Type (4×5)

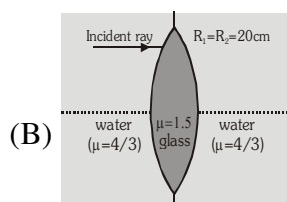
1. Column I (optical system)

2 Q. [8 M (for each entry +2(0))]

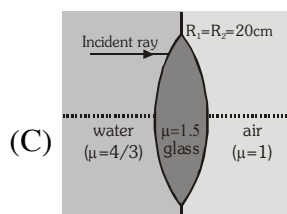
Column II (focal length)



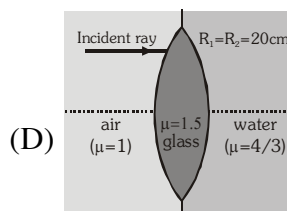
(P) 80 cm



(Q) 40 cm



(R) 30 cm



(S) 20 cm

2. The greatest thickness of a plano convex lens when viewed normally through the plane surface appears to be 3 cm & when viewed normally through the curved surface it appears to be 3.6 cm. Actual thickness of lens is 4.5 cm.

Column-I

Column-II

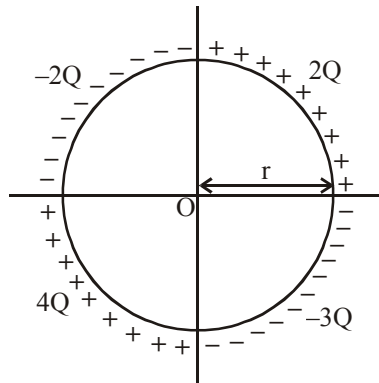
- | | |
|--|--------|
| (A) The refractive index of the material of the lens in multiple of 10^{-1} is | (P) 9 |
| (B) The radius of curvature of lens in cm | (Q) 3 |
| (C) The focal length if its plane surface is silvered in cm | (R) 16 |
| (D) The focal length if its curved surface is silvered in cm | (S) 15 |

SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

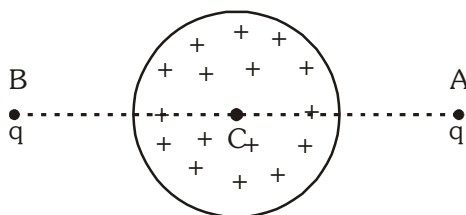
- Three point charge q , $-2q$ and $-2q$ are placed at the vertices of an equilateral triangle of side a . The minimum work done by some external force to increase their separation to $2a$ will be
 (A) $\frac{1}{4\pi\epsilon_0} \frac{2q^2}{a}$ (B) negative (C) zero (D) $\frac{1}{4\pi\epsilon_0} \frac{3q^2}{a}$
- A charge of $6\mu\text{C}$ is fixed at the origin. Another identical charge is placed at a distance of 2 m from the fixed charge and is released. The charges have a mass of 0.1 kg each. Then choose the correct statement:-
 (A) The maximum speed achieved by the 2nd charge will be 1.8 m/sec
 (B) The maximum speed achieved by the 2nd charge will depend on the direction of its motion
 (C) As time passes, potential energy of the system increases.
 (D) The 2nd charge will achieve a maximum velocity of $0.9\sqrt{2}$ m/s
- Charge distribution on a ring in x-y plane is shown in the figure. Then electric potential at origin is :



- (A) $\frac{Q}{4\pi\epsilon_0 r}$ (B) Zero (C) $\frac{Q}{2\pi\epsilon_0 r}$ (D) $\frac{3Q}{4\pi\epsilon_0 r}$
- A solid sphere of radius R_1 and volume charge density $\rho = \frac{\rho_0}{r}$ is enclosed by a hollow sphere of radius R_2 with negative surface charge density σ , such that the total charge in the system is zero, ρ_0 is a positive constant and r is the distance from the centre of the sphere. The ratio $\frac{R_2}{R_1}$ is
 (A) $\frac{\sigma}{\rho_0}$ (B) $\sqrt{2\sigma/\rho_0}$ (C) $\sqrt{\rho_0/2\sigma}$ (D) $\frac{\rho_0}{\sigma}$
- A thin dielectric rod of length l lies along the x-axis with one end at the origin and the other end at the point $(l, 0)$. It is charged uniformly along its length with a total charge Q . The potential at a point $(x, 0)$ when $x > l$ is :
 (A) $\frac{Q}{4\pi\epsilon_0 l}$ (B) $\frac{Q}{4\pi\epsilon_0 l} \log_e \left(\frac{x}{l} \right)$ (C) $\frac{Q}{4\pi\epsilon_0 l} \log_e \left(\frac{x}{x-l} \right)$ (D) $\frac{Q}{4\pi\epsilon_0 l} \log_e \left(\frac{x-l}{x} \right)$

6. A uniformly charged non-conducting sphere having charge Q and radius R is kept in gravity free space. Two identical charge particles having charge q are released from rest from 'A' and 'B' simultaneously. The speed of one of the charge particle when separation between them is very large, is (mass of one of the particle is m)

[C $\rightarrow$ centre of sphere and A,B and C in same line. $AC = BC = 2R$, $K = \frac{1}{4\pi\epsilon_0}$]



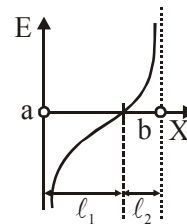
- (A) $\sqrt{\frac{Kq}{2mR} \left(\frac{q}{2} + 2Q \right)}$ (B) $\sqrt{\frac{Kq}{mR} \left(\frac{q}{2} + Q \right)}$ (C) $\sqrt{\frac{Kq^2}{2mR}}$ (D) $\sqrt{\frac{KqQ}{2mR}}$

Multiple Correct Answer Type

5 Q. [4 M (-1)]

7. Two point charges are located on the x-axis. The first is a charge $+Q$ at $x = -a$. The second is an unknown charge located at $x = +3a$. The net electric field these charges produce at the origin has a magnitude of $\frac{2kQ}{a^2}$. What are the possible value (s) of the unknown charge ?
- (A) $-9Q$ (B) $+9Q$ (C) $+27Q$ (D) $+8Q$
8. A ring carries a linear charge density on one half and the linear charge density of same magnitude but opposite sign on the other half.
- (A) the component of electric field along the axis of ring at all points on the axis is zero.
(B) the component of electric field along the axis of ring at point on the axis is zero only at the centre.
(C) the resultant field at the centre is zero.
(D) the electric field at all points on the axis of ring is perpendicular to axis.
9. Electric field in a region is given as $\vec{E} = (10 - 5x)\hat{i}$. A charge particle of mass 5kg and charge $Q (= 1\text{C})$ is situated at origin and free to move in given electric field. Then choose the correct options (Neglect gravity) :-
- (A) Motion of charge particle is Oscillatory
(B) Maximum displacement of charge particle from origin is 4 SI units
(C) Maximum velocity gain by charge particle is 2 SI units
(D) The position of charge particle, when velocity gained by particle is maximum, is 2 SI units
10. Charges Q_1 & Q_2 lie inside and outside respectively of a closed surface S . Let E be the field at any point on S and ϕ be the flux of E over S .
- (A) If Q_1 changes, both E & ϕ will change
(B) If Q_2 changes, E will change but ϕ will not change
(C) If $Q_1 = 0$ and $Q_2 \neq 0$ then $E \neq 0$ but $\phi = 0$
(D) If $Q_1 \neq 0$ and $Q_2 = 0$ then $E = 0$ but $\phi \neq 0$

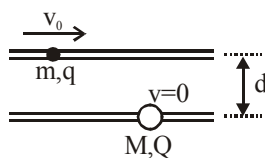
11. Two point charges are placed at a and b at a certain distance from each other. Assuming the field strength is positive in the direction coinciding with the positive direction of the x axis ($\ell_1 > \ell_2$) :-
- (A) Charge at a is positive and at b negative
(B) Magnitude of charge at a is greater than that of charge at b
(C) Both charges at a and b are negative
(D) Magnitude of charge at b is greater than that of charge at a



Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question no 12 and 13

Two small beads slide without friction, one on each of two long, horizontal, parallel, fixed rods set a distance d apart. The masses of the beads are m and M , and they carry respective charges of q and Q . Initially, the large mass M is at rest and the other one is far away approaching it at speed v_0 .



12. If the beads stop w.r.t. each other and at this time they are at minimum distance d , what is the velocity v_0 ?
- (A) $\sqrt{\frac{qQ(m+M)}{4\pi\epsilon_0 d(mM)}}$ (B) $\sqrt{\frac{qQ(m+M)}{2\pi\epsilon_0 d(mM)}}$ (C) $\sqrt{\frac{qQ(mM)}{2\pi\epsilon_0 d(m+M)}}$ (D) $\sqrt{\frac{qQ(mM)}{4\pi\epsilon_0 d(m+M)}}$
13. Consider system to be beads. Mark the **INCORRECT** statement :
- (A) Acceleration of centre of mass of system is zero
(B) Momentum of system is conserved
(C) Angular momentum of system about centre of mass of system is conserved
(D) Mechanical energy of system is constant

SECTION-III

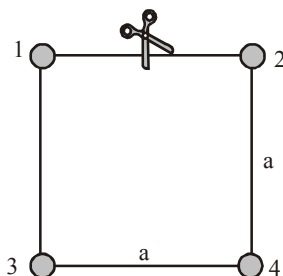
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

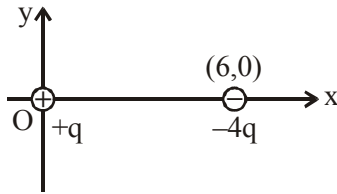
1. Consider three identical metal spheres A, B and C. Spheres A carries charge $+0.6C$ and sphere B carries charge $-0.3C$. Sphere C carries no charge. Spheres A and C are touched together and then separated. Sphere B is then touched to sphere A and separated from it. Finally the sphere C is touched to sphere B and separated from it. Final charge on sphere C is $\frac{x}{20}C$. Find x .

2. Four balls, each with mass m , are connected by four nonconducting strings to form a square with side a as shown in figure. The assembly is placed on a horizontal, nonconducting, frictionless surface. Balls 1 and 2 each have charge q , and balls 3 and 4 are uncharged. Find the maximum speed of balls 3 and 4

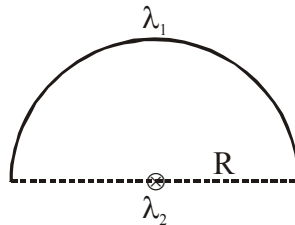
after the string connecting balls 1 and 2 is cut. If your answer is $\sqrt{\frac{q^2}{2N\pi\epsilon_0 am}}$. Fill the value of N



3. The ratio of magnitude of electric field due to $+q$ and $-4q$ at a point between them where net potential is zero on x-axis is $z : 1$, then z will be



4. An infinite long uniformly charged wire is kept along axis of a uniformly charged semi circular ring of radius R as shown. Linear charge density of ring and wire are respectively λ_1 and λ_2 . Net electrostatic force on wire due to ring is $\frac{4k\lambda_1\lambda_2}{R^{n-1}}$ value of n is



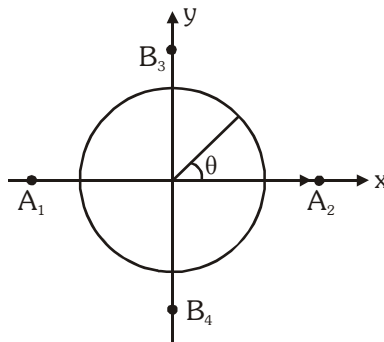
5. For a hypothetical electric field in x-direction, its potential is continuously decreasing and becoming half after every 20 cm. The rate of decreasing of potential with distance is also directly proportional to the potential itself. If the ratio of electric field and potential is $x \ln 2 (\text{m}^{-1})$. Find the value of x .

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. A circular non-conducting ring has charge density $\lambda = \lambda_0 \sin \theta$ as shown in figure.



Column I (Point)

- (A) A_1
(B) A_2
(C) B_3
(D) B_4

Column II (Direction of electric field)

- (P) Positive x-direction
(Q) Negative x-direction
(R) Positive y-direction
(S) Negative y-direction
(T) Positive z-direction

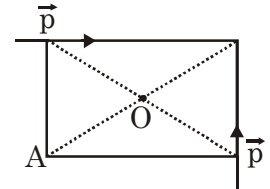
SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

1. Two short electric dipoles of dipole moment $\vec{p}$ each are placed at two corners of a square as shown in figure. What is ratio of magnitudes of electric field at two point O and A :-

- (A) 2 (B) $2\sqrt{2}$
(C) 1 (D) $\sqrt{2}$

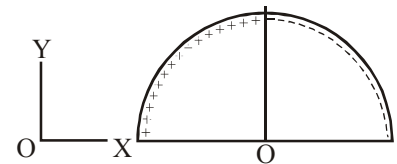


2. Three equal charges Q are placed at the vertices of an equilateral triangle. What should be the value of a charge so that when placed at the centroid, self energy of the system becomes zero. (excluding the self energies of point charges)

- (A) $\frac{-Q}{2}$ (B) $\frac{-Q}{3}$ (C) $\frac{-Q}{2\sqrt{3}}$ (D) $\frac{-Q}{\sqrt{3}}$

3. A wire, of length $L (= 20 \text{ cm})$, is bent into a semi-circular arc. If the two equal halves, of the arc, were each to be uniformly charged with charges $\pm Q$, [$|Q| = 10^3 \epsilon_0 \text{ Coulomb}$ where ϵ_0 is the permittivity (in SI units) of free space] the net electric field at the centre O of the semi-circular arc would be :-

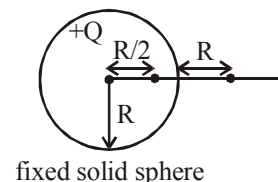
- (A) $(50 \times 10^3 \text{ N/C}) \hat{j}$ (B) $(50 \times 10^3 \text{ N/C}) \hat{i}$ (C) $(25 \times 10^3 \text{ N/C}) \hat{i}$ (D) $(25 \times 10^3 \text{ N/C}) \hat{j}$



4. Two planes of charge with no thickness, A and B, are parallel and vertical. The electric field in region-1 to the left of plane A has magnitude $\frac{3\sigma}{2\epsilon_0}$ and points to the left. The electric field in region-2 to the right of B has magnitude $\frac{3\sigma}{2\epsilon_0}$ and points to the right. The electric field in region-3 between the two planes has magnitude $\frac{\sigma}{2\epsilon_0}$ and points to the right. The surface charge density on planes A and B respectively is :-

- (A) $-\frac{\sigma}{2}, \sigma$ (B) $+\frac{\sigma}{2}, \sigma$ (C) $\sigma, -\frac{\sigma}{2}$ (D) $2\sigma, \sigma$

5. A point charge (+q) of mass m is released from a distance $\left(\frac{R}{2}\right)$ from the centre of a solid sphere of radius R and charge Q (uniformly charged in volume). The speed of the point charge when it reaches a distance R from the surface of the sphere (there is no interaction between sphere and point

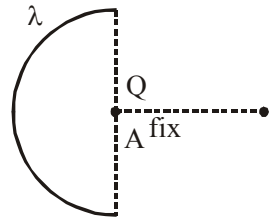


charge other than electrostatic). $k = \frac{1}{4\pi\epsilon_0}$

- (A) $\sqrt{\frac{7kQq}{16mR}}$ (B) $\sqrt{\frac{7kQq}{4mR}}$ (C) $\sqrt{\frac{5kQq}{16mR}}$ (D) $\sqrt{\frac{3kQq}{4mR}}$

6. A charge is kept fixed at point A as shown in the figure in a gravity free space. A semicircular ring of linear charge density λ , mass m & radius R is released from rest. Initially centre of semicircular ring coincides with the fixed charge. The initial acceleration of the semicircular ring is :-

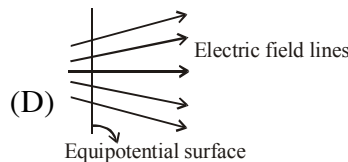
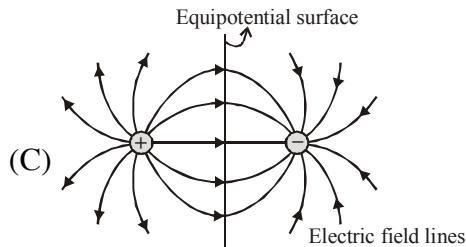
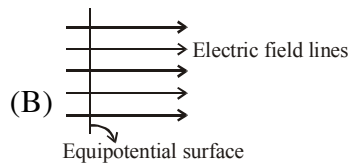
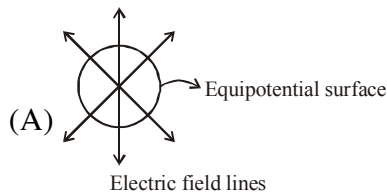
- (A) $\frac{2K\lambda Q}{mR}$ (B) $\frac{K\lambda Q}{2mR}$ (C) $\frac{K\lambda Q}{mR}$ (D) 0



5 Q. [4 M (-1)]

Multiple Correct Answer Type

7. If we move in the direction perpendicular to the electric field, work done by the electric field will be zero and surface thus generated is called equipotential surface. Choose the correct matching :-



8. Consider a non-conducting sphere of radius a carrying uniformly distributed charge q surrounded by a spherical shell of radius $2a$. The region between $a \leq r \leq 2a$ is filled with charge density $\rho = \frac{C}{r}$, where r is the distance from the centre. If the magnitude of electric field in the region $a \leq r \leq 2a$ is constant, then:

- (A) $C = \frac{q}{2\pi a^2}$
(B) Magnitude of charge in the volume $a \leq r \leq 2a$ will be $3q$.
(C) Electric field outside the spherical shell remains constant.
(D) $C = \frac{q}{\pi a^2}$

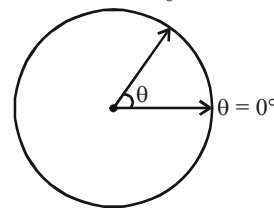
9. A non-conducting solid sphere of radius 30 cm and relative permittivity 1 has the volume charge density $\rho = \left(\frac{5}{2\pi} \mu\text{C}/\text{m}^3\right)\left(1 - \frac{r}{30\text{cm}}\right)$, where r (in cm) is the radial distance of any point in the space from the centre of the sphere. Then choose the correct statement(s).
(A) Density of electric field lines is maximum at $r = 25$ cm
(B) Density of electric field lines is maximum at $r = 20$ cm
(C) Electrical energy stored in the space $r = 30$ cm to $r \rightarrow \infty$, is same as that of a solid non-conducting sphere carrying charge of $0.0225 \mu\text{C}$ and radii 30 cm having uniform volume charge density in the same space $r = 30$ cm to $r \rightarrow \infty$.
(D) Variation of magnitude of electric field due to the sphere with respect to radial distance r is rectangular hyperbola in the region $r = 30$ to $r \rightarrow \infty$.

10. In a solid sphere of radius $R = 2m$ charge density varies as

$$\rho = \frac{A}{r^2} - \frac{B}{r} \quad 0 < r \leq R$$

where $A = 1 \text{ C/m}$ and $B = 1 \text{ C/m}^2$. For this sphere

- (A) Flux of electric field linked with a sphere of radius r ($0 < r \leq R$), increases as r increases
 (B) Flux of electric field linked with a sphere of radius r ($0 < r \leq R$), first increases and then decreases as r increases
 (C) Magnitude of electric field intensity continuously decreases as r increases from $r > 0$ to $r = R$.
 (D) Forces experienced by two point charges placed at distance $3m$ and $4m$ from centre of sphere are same.
11. The linear charge density on a non conducting ring of radius R is varying with θ as $\lambda = \lambda_0 \cos \theta$:-
- (A) Electric field at the center of the ring is zero
 (B) Electric field at center of ring is towards left
 (C) Electric field on a line perpendicular to the plane of the ring and passing through the center of the ring is zero
 (D) Electric field on a line perpendicular to the plane of the ring and passing through the center of the ring is non-zero.

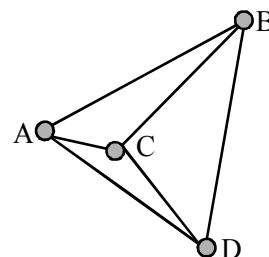


Linked Comprehension Type (1 Para × 3Q.)
(Single Correct Answer Type)

[3 M (-1)]

Paragraph for Question Nos. 12 to 14

Four charged particles (A, B, C, D), of mass m and charge q each are connected by light silk threads of length d forming a tetrahedron floating in outer space. The thread connecting particles A and B suddenly snaps.



12. Mark the **CORRECT** statement :-

- (A) Displacement of centre of mass of system of particles is $\frac{d}{\sqrt{3}}$
 (B) Angular momentum of system about CM is $\frac{4Md^2}{\sqrt{3}}\omega$, where ω is maximum angular velocity of particles about centre of mass.
 (C) When particles attain maximum speed direction of motion of particles C and D is opposite to that of A and B.
 (D) When particles attain maximum speed direction of motion of all the particles is in same direction

13. The potential energy of particle A and B when the maximum speed is attained by charged particles :-

- (A) $U = \frac{q^2}{4\pi\epsilon_0(d\sqrt{2})}$ (B) $U = \frac{q^2}{4\pi\epsilon_0(d\sqrt{3})}$
 (C) $U = \frac{q^2}{4\pi\epsilon_0 d} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right)$ (D) $U = \frac{q^2}{4\pi\epsilon_0 d} \left(\frac{\sqrt{3}-1}{\sqrt{3}} \right)$

14. Maximum speed of particles is :-

- (A) $v = \sqrt{\frac{q^2}{8\pi\epsilon_0 md} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right)}$ (B) $v = \sqrt{\frac{q^2}{8\pi\epsilon_0 md} \left(\frac{\sqrt{3}-1}{\sqrt{3}} \right)}$
 (C) $v = \sqrt{\frac{q^2}{8\pi\epsilon_0 md} \left(\frac{1}{\sqrt{2}} \right)}$ (D) $v = \sqrt{\frac{q^2}{8\pi\epsilon_0 md} \left(\frac{1}{\sqrt{3}} \right)}$

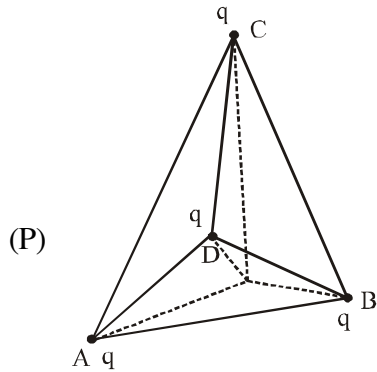
Matching List Type (4 × 4)

1Q.[3 M (-1)]

15. List-I shows different configurations of charge distribution. List-II gives the corresponding expressions of physical quantities mentioned in list-I :-

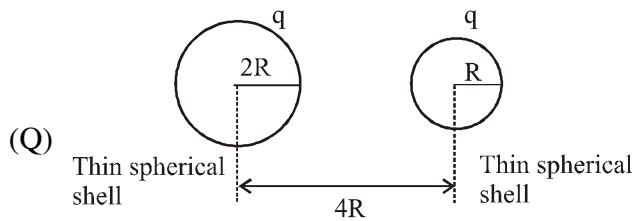
List-I

List-II



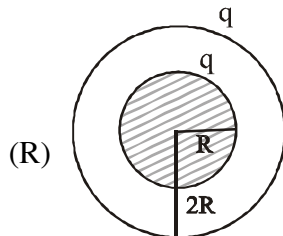
(1) $\frac{6kq^2}{R}$

Four point charges are kept at the vertices of a regular tetrahedron of side R. Total electrostatics energy of the configuration.



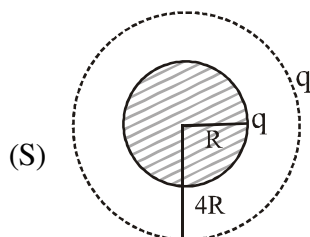
(2) $\frac{6kq^2}{R} \left(\frac{9}{40} \right)$

Two thin spherical uniformly charged shells having charge q are separated by distance 4R. Total electrostatics energy of the configuration.



(3) $\frac{6kq^2}{6R}$

A solid non-conducting sphere of radius R having uniform charge q is kept inside a thin spherical non-conducting shell of radius 2R having uniform charge q. Total electrostatics energy of the configuration.



(4) $\frac{6kq^2}{40R} (3)$

Work done in contracting the sphere having uniform charge q from radius 4R to R

Codes :

	P	Q	R	S
(A)	1	3	4	2
(B)	4	2	3	1
(C)	3	2	1	4
(D)	1	3	2	4

SECTION-II

Numerical Answer Type Question
(upto second decimal place)

1Q.[3(0)]

- In a 2-D region of x-y space, a uniform electric field exists, given by : $E = (2 \text{ V/m}) (3\hat{i} + 4\hat{j})$. A charged particle of mass 10^{-2} g and charge 10^{-5} C is fired from the origin with initial velocity 20 m/s , directed along the negative y-direction. Find the minimum speed (in m/s) of the particle during its subsequent motion. Neglect gravity.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

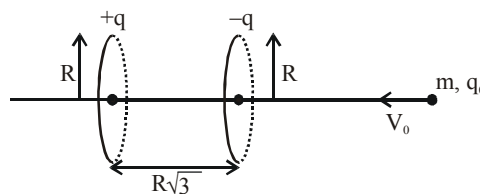
3 Q. [4 M (0)]

- Two point charges $2 \mu\text{C}$ and $8 \mu\text{C}$ are placed at position $x = 0$ and $x = 4 \text{ m}$ respectively. The distance of the neutral point from the smaller charge is found to be at $x = \alpha/\beta$. Find the value of $(\alpha + \beta)$.
- A positive charged particle q_0 , when launched with some velocity $\vec{v}_0$ in a uniform field is found to deviate by 60° in a certain time, such that its speed is halved. If $-ve$ charged particle $-2q_0$, is launched with the same velocity $\vec{v}_0$ in the same field then after same time its speed is v_2 . Mass of both particles is

same then $\frac{v_2^2}{v_0^2}$ would be :-

- Two uniformly charged co-axial, identical rings (Radius R) are fixed as shown. A point charge q_0 is projected with speed V_0 along common axis of rings from large distance. Minimum value of V_0 for

which the charge particle can just pass through both rings is given as $V_0 = \sqrt{\frac{kqq_0}{nmR}}$. Find $n \left(k = \frac{1}{4\pi\epsilon_0} \right)$.



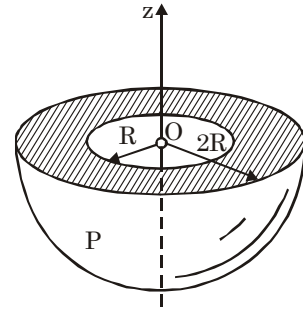
SECTION-I

Single Correct Answer Type

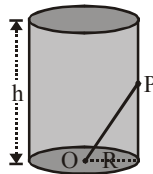
6 Q. [3 M (-1)]

1. A hemispherical bowl of inner radii R and outer radii $2R$ is charged with density $\rho = \rho_0 \frac{r}{R}$. Find the electric field at point O .

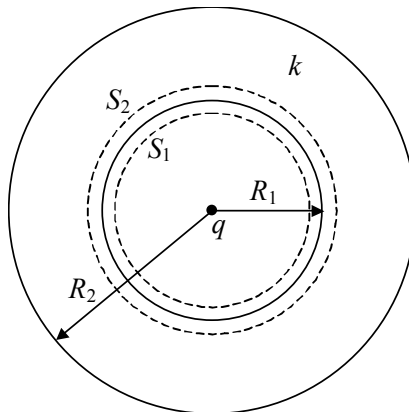
- (A) $\frac{2\rho_0 R}{3\epsilon_0}$ (B) $\frac{3\rho_0 R}{8\epsilon_0}$
(C) $\frac{\rho_0 R}{2\epsilon_0}$ (D) $\frac{3\rho_0 R}{2\epsilon_0}$



2. Electric field at point P is given by $\vec{E} = E_0 \vec{r}$. The total flux through the given cylinder of radius R and height h is :-



- (A) $E_0 \pi R^2 h$ (B) $2E_0 \pi R^2 h$ (C) $3E_0 \pi R^2 h$ (D) $4E_0 \pi R^2 h$
3. A positive charge q is kept at the center of a thick shell of inner radius R_1 and outer radius R_2 which is made up of conducting material. If ϕ_1 is flux through closed gaussian surface S_1 whose radius is just less than R_1 and ϕ_2 is flux through closed gaussian surface S_2 whose radius is just greater than R_1 then:-



- (A) $\phi_1 > \phi_2$ (B) $\phi_2 > \phi_1$ (C) $\phi_1 = \phi_2 = \frac{q}{\epsilon_0}$ (D) $\phi_1 = \phi_2 = \frac{kq}{\epsilon_0}$
4. Consider a solid insulating sphere of radius R with charge density varying as $\rho = \rho_0 r^2$ (ρ_0 is a constant and r is measure from centre). Consider two points A and B at distance x and y respectively ($x < R$, $y > R$) from the centre. If magnitudes of electric fields at points A and B are equal, then :-

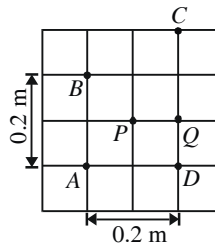
- (A) $x^2 y = R^3$ (B) $x^3 y^2 = R^5$ (C) $x^2 y^3 = R^5$ (D) $\frac{x^4}{y} = R^5$

5. Consider a uniform electric field $\vec{E} = 3 \times 10^3 \hat{i} \text{ N/C}$. What is the flux of this field through a square of 10 cm side length and normal to the plane makes α, β, γ from x-axis, y-axis and z-axis and $\sin \alpha = \frac{\sqrt{3}}{2}$, $\sin \beta = \frac{\sqrt{3}}{2}$, $\sin \gamma = \frac{1}{\sqrt{2}}$:-

$$\beta = \frac{\sqrt{3}}{2}, \sin \gamma = \frac{1}{\sqrt{2}} \quad \therefore$$

- (A) $30 \frac{\text{Nm}^2}{\text{C}}$ (B) $15 \frac{\text{Nm}^2}{\text{C}}$ (C) $\frac{45}{2} \frac{\text{Nm}^2}{\text{C}}$ (D) $20 \frac{\text{Nm}^2}{\text{C}}$

6. Points A, B, C, D, P and Q are shown in a region of uniform electric field. The potentials at some of the points are $V(A) = 0 \text{ V}$ & $V(P) = V(B) = V(D) = 4 \text{ V}$. The potential at C is :



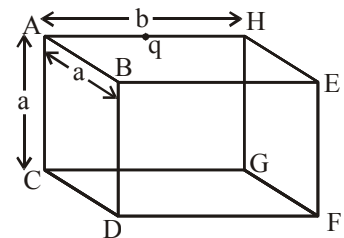
- (A) 10 V (B) -10 V (C) 8 V (D) -8 V

Multiple Correct Answer Type

5 Q. [4 M (-1)]

7. A cuboid of dimension $[a \times a \times b]$. Charge q is placed at the centre of edge having length 'b'. If flux through face 'ABCD' is $\frac{q}{32 \epsilon_0}$ then select the correct statement(s) :-

- (A) Flux through the entire cuboid is $\frac{q}{4 \epsilon_0}$
 (B) Flux through the face 'ABEH' is zero.
 (C) Flux through the face 'BEFD' is $\frac{3q}{32 \epsilon_0}$
 (D) Flux through the face 'BEFD' is $\frac{q}{16 \epsilon_0}$



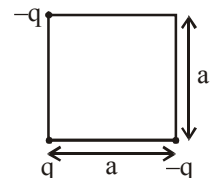
8. Three charges are kept at the corners of a square as shown. A 4th charge q is brought slowly at the 4th corner from infinity. Then choose the **CORRECT** statement(s) [Take reference potential energy at infinity to be zero & $k = \frac{1}{4\pi \epsilon_0}$]

(A) Work done by electrostatic field is $\frac{kq^2}{a} \left[\frac{1-2\sqrt{2}}{\sqrt{2}} \right] \text{ J}$

(B) Work done by external agent is $\frac{kq^2}{a} \left[\frac{1-2\sqrt{2}}{\sqrt{2}} \right] \text{ J}$

(C) Total potential energy of the system after the charge is placed at the corner is $\frac{kq^2}{a} [\sqrt{2} - 4]$

(D) For the system thus achieved centre of square will be a point of equilibrium.



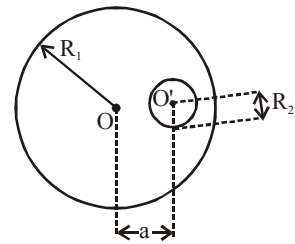
9. A sphere of radius R_1 has charge density ρ uniform with in its volume, except for a small spherical hollow region of radius R_2 located a distance "a" from centre (Take zero potential at infinity).

(A) Electric field at O' is $\frac{\rho \vec{a}}{3 \epsilon_0}$

(B) Electric field at O' is $\frac{\rho a^3}{3 \epsilon_0} \hat{a}$

(C) Electric potential at O' is $\frac{\rho}{6 \epsilon_0} [3(R_1^2 - R_2^2) - a^2]$

(D) Electric potential at any point in hollow region is constant



10. In a region having electric field, when we move along x-axis from $x = 2\text{m}$ to $x = 3\text{m}$, potential increases by 5 Volt. Which of the following is/are possible for the electric field?

(A) It can be uniform with magnitude 7 V/m.

(B) It can be uniform with magnitude 3 V/m.

(C) From $x = 2\text{m}$ to $x = 3\text{m}$ it has positive x component at every point.

(D) It can be non uniform.

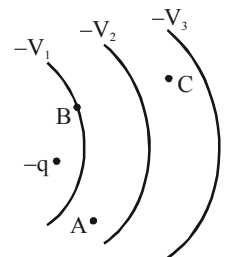
11. Consider three equipotential surfaces due to a charge $-q$. Another charge $+q$ is taken slowly along two paths from A to B and A to C, then :-

(A) Work done by electric field in moving it from A to B is +ve

(B) Work done by electric field in moving it from A to B is -ve

(C) Work done by an external agent in moving it from A to C = +ve

(D) Work done by an external agent in moving it from A to C = -ve



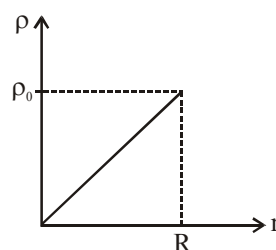
Linked Comprehension Type

(2 Para $\times$ 2Q.) [3 M (-1)]

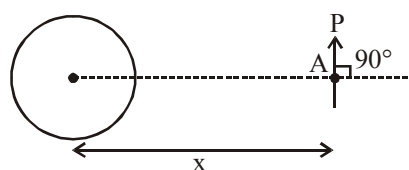
(Single Correct Answer Type)

Paragraph for Question no. 12 and 13

A non conducting sphere having relative permittivity unity ($\epsilon_r = 1$) has radius R and have a non uniform charge distribution. The volume charge density $\rho(r)$ varies with radial distance r as per the following graph. Here ρ_0 is a constant.



12. If a small dipole of dipole moment P is placed at point A which is at large distance x from centre of sphere as shown in figure. The force exerted by dipole on sphere would be :



(A) $\frac{\rho_0 P R^3}{4 \epsilon_0 x^3}$

(B) $\frac{\rho_0 P R^3}{2 \epsilon_0 x^3}$

(C) $\frac{\rho_0 P R^3}{8 \epsilon_0 x^3}$

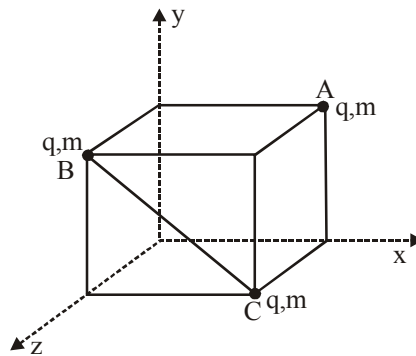
(D) $\frac{\rho_0 P R^3}{\epsilon_0 x^3}$

13. Consider an isolated sphere (as explained in paragraph). If $\vec{E}$ is electric field at any point because of sphere and if C is centre of sphere and S is surface of sphere find the value of $\int_S \vec{E} \cdot d\vec{r}$.

- (A) $\frac{-\rho_0 R^2}{12\epsilon_0}$ (B) $\frac{+\rho_0 R^2}{12\epsilon_0}$ (C) $\frac{-\rho_0 R^2}{4\epsilon_0}$ (D) $\frac{+\rho_0 R^2}{4\epsilon_0}$

Paragraph for Question no. 14 and 15

Three identical charge particles A, B and C are kept at three corners of cube of side length 'a'. B and C are also joined with massless rod. (Neglect gravitational force)



14. The magnitude of electrostatic force on particle 'A' is :
- (A) $\frac{q^2}{8\pi\epsilon_0 a^2} \sqrt{3}$ (B) $\frac{q^2}{8\pi\epsilon_0 a^2}$ (C) $\frac{q^2 \sqrt{3}}{4\pi\epsilon_0 a^2}$ (D) $\frac{q^2}{4\pi\epsilon_0 a^2}$
15. If system is released from rest, find kinetic energy of A, when A is very far from B and C :
- (A) $\frac{q^2}{4\pi\epsilon_0 a\sqrt{2}}$ (B) $\frac{q^2}{12\pi\epsilon_0 a\sqrt{2}}$ (C) $\frac{q^2}{3\pi\epsilon_0 a\sqrt{2}}$ (D) $\frac{3q^2}{7\pi\epsilon_0 a\sqrt{2}}$

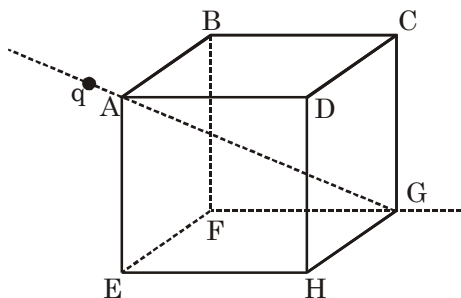
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

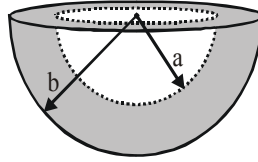
1. A charge q is kept just outside a cube on extension of diagonal GA. The magnitude of electric flux of side

ABCD is $\phi = \frac{\alpha q}{168 \epsilon_0}$. Find value of α .

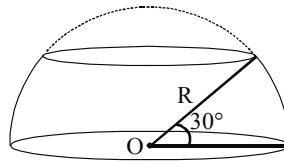


2. A uniformly charged hemisphere of radius b and charge density ρ has a hemispherical cavity of radius

$a \left(a = \frac{b}{2} \right)$ cut from its centre. If the potential at the centre of the cavity is $\frac{n\rho b^2}{16\epsilon_0}$ then $n = ?$



3. Figure shows the part of a hemisphere of radius (R) = 2m and surface charge density (σ) = $2\epsilon_0$ C/m². Calculate the electric potential (in volt) at centre O.



4. Point charges $-2q$, $-2q$ and $+q$ are put on the vertices of an equilateral triangle of side a . Find the work done by some external force in increasing the side of triangle to $2a$ (in J)
5. A spherical distribution of charge consists of uniform charge density ρ_1 from $r = 0$ to $r = a/2$ and

a uniform charge density ρ_2 from $r = a/2$ to $r = a$. If the electric potential at $r = a$ is $\frac{K\pi a^2}{n} [7\rho_2 + \rho_1]$.

Then find n .

SECTION-I

Single Correct Answer Type

1. A charge q_1 & q_2 are placed inside a spherical cavity made inside uncharged conducting sphere of radius R . Electric field at P due all induced charge on the sphere :-

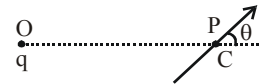
(A) $\frac{kq_1}{a^2} \hat{i} + \frac{kq_2}{b^2} \hat{j}$ (B) $-\frac{kq_1}{a^2} \hat{i} - \frac{kq_2}{b^2} \hat{j}$
(C) $\frac{kq_1}{r_1^2} \hat{i} + \frac{kq_2}{r_2^2} \hat{j}$ (D) $-\frac{kq_1}{r_1^2} \hat{i} - \frac{kq_2}{r_2^2} \hat{j}$

2. Consider a metallic sphere of radius R concentric with another hollow metallic sphere of inner radius $2R$ and outer radius $3R$ as shown in the figure. The inner sphere is given a charge $-Q$ and outer $+2Q$. The electrostatic potential energy of the system is :-

(A) $\frac{Q^2}{16\pi\epsilon_0 R}$ (B) $\frac{3Q^2}{16\pi\epsilon_0 R}$
(C) $\frac{5Q^2}{48\pi\epsilon_0 R}$ (D) $\frac{11Q^2}{48\pi\epsilon_0 R}$

3. A small dipole P and a fixed point charge $+q$ are kept in free space. Net electrostatic force on dipole due to charge q is F and torque on dipole about point O is τ then :-

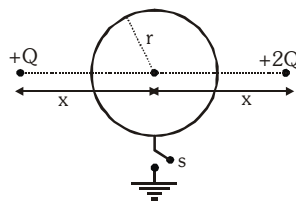
- (A) for every value of θ ; $F = 0$ and $\tau = 0$
(B) for every value of θ ; $F \neq 0$ and $\tau = 0$
(C) for every value of θ ; $F \neq 0$ and $\tau \neq 0$
(D) τ is zero or nonzero, depends on value of θ .



4. A bead of mass m and charge q is threaded on a smooth wire kept along x -axis. A second charge Q is fixed at the point $x = 0$, $y = -d$. Now bead is moved with constant speed v so that it is initially near $x = -\infty$ and it ends up near $x = +\infty$. Find the impulse exerted by the fixed charge on the bead. Neglect gravity.

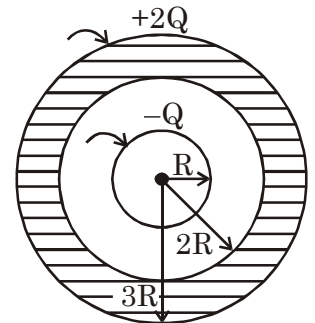
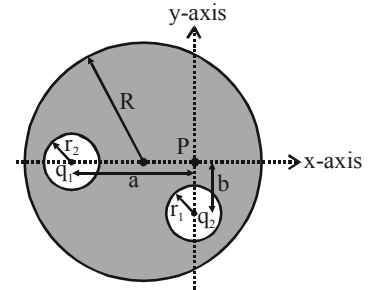
(A) $\frac{qQ}{2\pi\epsilon_0 vd}$ (B) $\frac{qQ}{4\pi\epsilon_0 vd}$ (C) $\frac{qQ}{\pi\epsilon_0 vd}$ (D) $\frac{2qQ}{\pi\epsilon_0 vd}$

5. Two particles having positive charges $+Q$ and $+2Q$ are fixed at equal distance x from centre of a conducting solid sphere having zero net charge and radius r as shown. Initially the switch S is open. After the switch S is closed, the net charge flowing out of sphere is

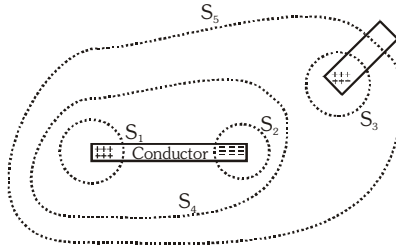


(A) $\frac{Qr}{x}$ (B) $-\frac{Qr}{x}$ (C) $\frac{3Qr}{x}$ (D) $-\frac{3Qr}{x}$

6 Q. [3 M (-1)]



6. A rod containing charge $+Q$ is brought near an initially uncharged isolated conducting rod as shown. Regions with total surface charge $+Q$ and $-Q$ are induced in the conductor as shown in the figure. The only regions where the net charge in this configuration is non-zero are indicated by the "+" and "-" signs. Let us denote the total flux of electric field outward through closed surface S_1 as ϕ_1 , through S_2 as ϕ_2 , etc. Which of the following is necessarily false.

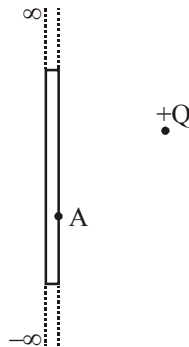


- (A) $\phi_1 > 0$ (B) $\phi_2 = \phi_1$ (C) $\phi_3 = \phi_1$ (D) $\phi_4 = 0$

Multiple Correct Answer Type

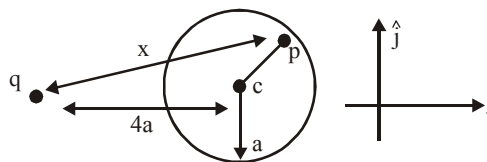
2 Q. [4 M (-1)]

7. A positive point charge is kept in front of an infinite large uncharged conducting plate as shown in figure. What must not be the direction of electric field at the point A on the surface of plate due to charges appearing on plate:-



- (A) $\longrightarrow$ (B) $\longleftarrow$ (C) $\swarrow$ (D) $\nwarrow$

8. There is a spherical conductor of radius a . A point charge is placed at a distance $4a$ from the centre of the spherical conductor as shown in figure.



- (A) The electric field at 'c' due to spherical conductor is $\frac{kq}{16a^2}(-\hat{i})$
- (B) Potential at P due to induced charge on spherical conductor is $\left(\frac{kq}{4a} - \frac{kq}{x}\right)$
- (C) Net electric field at 'P' is ZERO.
- (D) Now the conductor is earthed then charge on the spherical conductor is $\left(-\frac{q}{4}\right)$

Linked Comprehension Type
(Single Correct Answer Type)

(2 Para × 3Q.) [3 M (-1)]

Paragraph for Questions 9 to 11

In physics, Gauss's law, also known as Gauss's flux theorem, is a law relating the distribution of electric charge to the resulting electric field. It states that,

"The net electric flux through any closed surface is equal to $\frac{1}{\epsilon_0}$ time the net electric charge enclosed within that closed surface".

Mathematically it states that,

$$\oint \vec{E} \cdot \vec{ds} = \frac{Q_{\text{enclosed}}}{\epsilon_0} \rightarrow \text{Integral form of Gauss's law}$$

In differential form the law can be stated mathematically as,

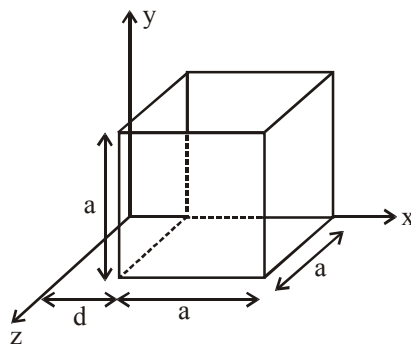
$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (\rho \text{ is charge density in the enclosed area})$$

$$\text{where } \vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \quad (\text{in cartesian coordinate})$$

and in spherical coordinate system, $\vec{\nabla}$ is written as,

$$\vec{\nabla} = \frac{d}{dr} \hat{r} \quad \text{for radial dependence only.}$$

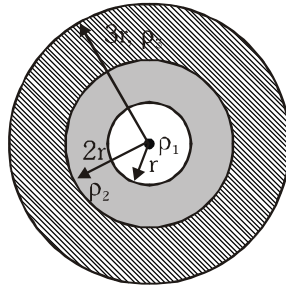
9. Consider a cube of side "a" placed such that its faces are parallel to x-y; y-z & z-x plane. (see figure) An electric field exists in the region as $y\hat{i} + x\hat{j} = \vec{E}$. The net charge enclosed in the cube will be :-



- (A) $2a^3\epsilon_0$ (B) zero (C) $a^3\epsilon_0$ (D) Information insufficient
10. In a region it was found that electric field varies radially as $E = \alpha r^3$, α is constant. Then in the region charge density linked with the field will be :
- (A) $5\alpha r^4\epsilon_0$ (B) $\alpha r\epsilon_0$ (C) $3\alpha r^2\epsilon_0$ (D) 0
11. Considering area vector to be positive along inward normal for a closed surface, flux linked to a charge Q enclosed in a sphere of same radius will be (assume flux to be defined in the same way as is defined in normal system)
- (A) $\frac{Q}{\epsilon_0}$ (B) $-\frac{Q}{\epsilon_0}$
- (C) zero (D) depends on radius of sphere

Paragraph for Questions 12 to 14

A solid sphere is made by 3 materials taken as shown. Total charge on system is Q . If magnitude of electric field at every boundary point is same then



12. Ratio of charges in material 1, material 2 and material 3.
 (A) 1 : 2 : 3 (B) 1 : 3 : 5 (C) 1 : 4 : 9 (D) 1 : 1 : 1
13. Charge density of material 1 will be
 (A) $\frac{Q}{12\pi r^3}$ (B) $\frac{3Q}{4\pi r^3}$ (C) $\frac{Q}{4\pi r^3}$ (D) $\frac{Q}{8\pi r^3}$
14. Electric field at distance $r/2$ from centre will be
 (A) $\frac{1}{56\pi\epsilon_0} \frac{Q}{r^2}$ (B) $\frac{1}{72\pi\epsilon_0} \frac{Q}{r^2}$ (C) $\frac{1}{36\pi\epsilon_0} \frac{Q}{r^2}$ (D) $\frac{Q}{r^2}$

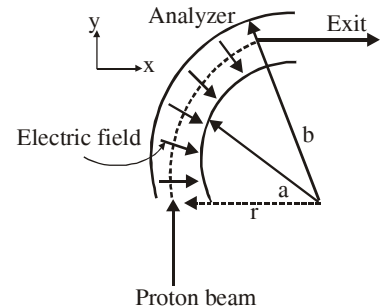
Linked Comprehension Type

(1 Para × 2 Q.) [4 M (-1)]

(Multiple Correct Answer Type)

Paragraph for Questions 15 & 16

Figure shows a schematic view of an electrostatic analyzer. It can sort out charged particles by speed and charge to mass ratio. Spacecraft uses such analyzers to characterize charged particles in interplanetary space. Two curved metal plates establish an electric field given by $E = E_0(b/r)$ where E_0 and b are positive constants. The field points towards the centre of curvature and r is distance from centre. There is no influence of gravity. Proton (charge $+e$; mass ' m ') enters along y -axis and exits along x -axis while moving along a circular path.



15. Mark the **CORRECT** option (s) :-

- (A) Speed with which proton is to be projected is given by $v = \sqrt{\frac{eE_0b}{m}}$
- (B) centripetal acceleration of electron is a_c is given by $a_c = \frac{e}{m} E_0 \left(\frac{b}{r} \right)$
- (C) Electric force on proton is given by $\frac{e}{2} E_0 \left(\frac{b}{r} \right)$
- (D) Electric force on proton is given by $2eE_0 \left(\frac{b}{r} \right)$

16. Mark the **CORRECT** option(s)

(A) Work done by electric field on proton is zero.

(B) If $v = \sqrt{\frac{2eE_0b}{m}}$ proton may strike outer surface of analyzer.

(C) If $v = \sqrt{\frac{2eE_0b}{m}}$ proton may strike inner surface of analyzer.

(D) If an electron is released with zero initial velocity from inner surface of analyzer, it will strike outer

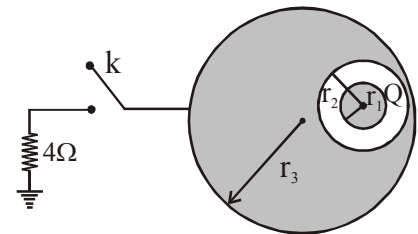
surface with velocity $v = \sqrt{\frac{2eE_0b}{m_e} \ln\left(\frac{b}{a}\right)}$, where m_e is mass of electron.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1. Charge $Q = 2\mu\text{C}$ is distributed on a small conducting sphere (radius $r_1 = 0.1\text{ m}$) as shown inside a spherical cavity of radius $r_2 = 0.5\text{ m}$ made inside a neutral solid conducting sphere of radius $r_3 = 2\text{ m}$ as shown in the figure. The outer surface is grounded using a resistance of 4Ω by closing key k . The heat dissipated on switching on the key is given by $\alpha\text{ mJ}$. Fill the value of α .



2. Electric dipole of moment $\vec{P} = P\hat{i}$ is kept at a point (x, y) in an electric field, $\vec{E} = 4xy^2\hat{i} + 4x^2y\hat{j}$. Find the forces on the dipole. If your answer is $n\text{ py } (y^2 + 4x^2)^{1/2}$ Then find n .
3. Two charges q and $-2q$ are placed at $(-3, 0)$ and $(3, 0)$ in x - y plane. The locus of the point in the plane of the charges, where the field potential is zero is $(x + \alpha)^2 + y^2 = 4\beta^2$. Find the value of $(\alpha + \beta)$.

SECTION-IV

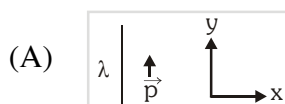
Matrix Match Type (4×4)

1 Q. [8 M (for each entry +2(0))]

1. A dipole with an electric dipole moment $\vec{p}$ is located at distance r from a long thread charged uniformly with a linear density λ . Orientation of dipole is given in column-I & electric force on the dipole is given in column-II.

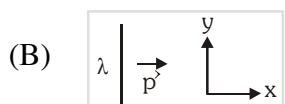
Column-I

Column-II



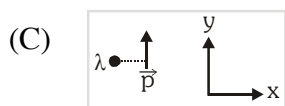
Along the thread

(P) $-\frac{\lambda p}{2\pi \epsilon_0 r^2} \hat{i}$



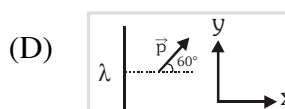
Along the radius vector

(Q) $-\frac{\lambda p}{4\pi \epsilon_0 r^2} \hat{i}$



At right angles to the thread and radius vector r

(R) Zero



At angle 60° from the radius vector in the plane

(S) $\frac{\lambda \vec{p}}{2\pi \epsilon_0 r^2}$

carrying wire & radius vector

SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

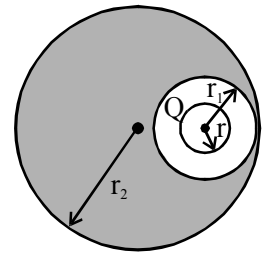
1. Charge Q is distributed on a small hollow conducting sphere (radius r) as shown inside a spherical cavity of radius r_1 made inside a solid conducting sphere of radius r_2 as shown in the figure. The total energy of the system is :-

(A) $kQ^2 \left(\frac{1}{2r_2} - \frac{1}{2r_1} + \frac{1}{2r} \right)$

(B) $kQ^2 \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{2r} \right)$

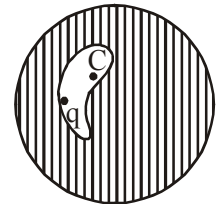
(C) $kQ^2 \left(\frac{1}{2r_1} + \frac{1}{4r_2} + \frac{1}{2r} \right)$

(D) $kQ^2 \left(\frac{1}{4r_1} + \frac{1}{2r_2} + \frac{1}{2r} \right)$



2. The figure shows a charge q placed inside a cavity in an uncharged conductor. Now if an external electric field is switched on then :

- (A) only induced charge on outer surface will redistribute
(B) only induced charge on inner surface will redistribute
(C) Both induced charge on outer and inner surface will redistribute
(D) force on charge q placed inside the cavity will change



3. There are two conducting concentric spherical thin shells of radius

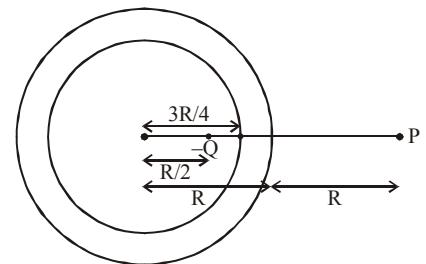
$\frac{3R}{4}$ and R . Thin shell of radius R is given a charge $4Q$. A point charge $-Q$ is also placed in the cavity at a distance $R/2$ from the centre as shown in figure. Find the electric potential at a distance $2R$ from the centre of concentric spherical shells along the line joining point charge :

(A) $\frac{1}{8\pi\epsilon_0} \frac{5Q}{R}$

(B) $-\frac{1}{8\pi\epsilon_0} \frac{5Q}{R}$

(C) $\frac{1}{8\pi\epsilon_0} \frac{3Q}{R}$

(D) $\frac{1}{2\pi\epsilon_0} \frac{Q}{3R}$



4. A solid uncharged conducting sphere has radius $3a$ contains a hollowed spherical region of radius $2a$. A point charge $+Q$ is placed at a position a from the common center of the spheres. What is the magnitude of the electric field at the position $r = 4a$ from the center of the spheres as

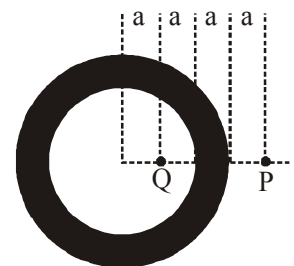
marked in the figure by P? $\left(k = \frac{1}{4\pi\epsilon_0} \right) :-$

(A) 0

(B) $\frac{kQ}{16a^2}$

(C) $\frac{3kQ}{16a^2}$

(D) $\frac{kQ}{9a^2}$



5. An isolated and charged spherical soap bubble has a radius ' r ' and the pressure inside is 1 atmosphere. If ' T ' is the surface tension of soap solution, then charge on the soap bubble is :-

(A) $2\sqrt{\frac{2rT}{\epsilon_0}}$

(B) $8\pi r\sqrt{2rT\epsilon_0}$

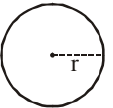
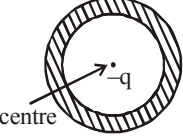
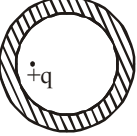
(C) $8\pi r\sqrt{rT\epsilon_0}$

(D) $8\pi r\sqrt{\frac{2rT}{\epsilon_0}}$

Multiple Correct Answer Type

3 Q. [4 M (-1)]

6. In which of the cases we will get uniform charge distribution of (+q) on external spherical surface. Given every object is a conductor :

- (A)  thin shell having net charge +q
- (B)  thick shell having net charge zero
- (C)  thick shell having net charge zero
- (D)  solid sphere having net charge +q

7. A long thin straight wire with linear charge density λ runs along axis of a thin hollow metal cylinder of radius R. The cylinder has a net linear charge density 2λ . Assume λ is positive. Mark correct options:-

(A) $\vec{E}(r > R) = \frac{3\lambda}{2\pi\epsilon_0} \frac{\hat{r}}{r}$

(B) $\vec{E}(r < R) = \frac{3\lambda}{2\pi\epsilon_0} \frac{\hat{r}}{r}$

(C) Linear charge density on inner surface of cylinder is $-\lambda$

(D) Linear charge density on outer surface of cylinder is 3λ

8. A point charge q is placed at P(0, 0, a). Then :-

(A) Flux through surface formed by joining points (0, 0, 0) ; (0, a, 0) ; (a, 0, 0) is greater than $\frac{q}{48\epsilon_0}$

(B) Flux through surface formed by joining points (0, a, 0) ; (a, 0, 0) ; (a, a, 0) is greater than $\frac{q}{24\epsilon_0}$

(C) Flux through surface formed by joining points (0, a, 0); (0, -a, 0); (-a, -a, 0); (-a, a, 0) is equal to $\frac{q}{12\epsilon_0}$

(D) Flux through surface formed by joining points (0, 0, 0) ; (0, 0, a) ; (0, a, a) ; (0, a, 0) is equal to $\frac{q}{48\epsilon_0}$

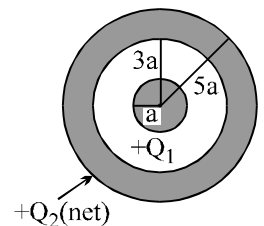
Linked Comprehension Type

(1 Para $\times$ 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question No. 9 to 11

A spherical insulator of radius a is concentric with a conducting spherical shell having inner radius 3a and outer radius 5a as shown in the figure. The charge $+Q_1$ is uniformly distributed throughout the volume of the insulator, and the net charge on the conductor is $+Q_2$.



9. What is the magnitude of the electric field a distance of 2a from the center of the insulator?

- (A) $(Q_1 + Q_2) / 4\pi\epsilon_0 a^2$ (B) $Q_1 / 16\pi\epsilon_0 a^2$ (C) $Q_1 / 4\pi\epsilon_0 a^2$ (D) None of these

10. What is the flux of the electric field through a Gaussian sphere of radius $4a$ that is concentric with both the insulator and conducting shell?
- (A) 0 (B) Q_1 / ϵ_0 (C) $(Q_1 + Q_2) / \epsilon_0$ (D) None of these
11. What is the surface charge density on the outer surface of the conductor?
- (A) $Q_2 / 100\pi a^2$ (B) $(Q_1 + Q_2) / 100\pi a^2$
(C) $3(Q_1 + Q_2) / 100\pi a^2$ (D) None of these

Matching list based comprehension Type ($4 \times 4 \times 4$)

1 Table $\times$ 3 Q. [3(-1)]

Single option correct (Three Columns and Four Rows)

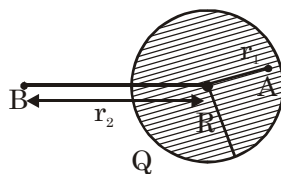
Answer Q.12, Q.13 and Q.14 by appropriately matching the information given in the three columns of the following table.

Column-I & II represents the magnitude of electric field at points A & B respectively. Column-III represents the magnitude of potential difference between the two points. (Take potential at infinity to be

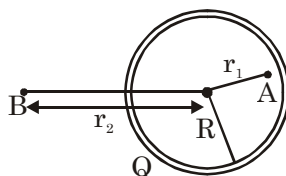
zero) $\left(k = \frac{1}{4\pi\epsilon_0} \right)$

Column-I Electric field at point A	Column-II Electric field at point B	Column-III Potential difference between two points
(I) $\frac{\rho r_1}{2\epsilon_0}$	(i) $\frac{\rho R^2}{4\epsilon_0 r_2}$	(P) $ V_A - V_B = \frac{kQ}{2R} \left(3 - \frac{r_1^2}{R^2} \right) - \frac{kQ}{r_2}$
(II) $\frac{kQr_1}{R^3}$	(ii) $\frac{\rho R^2}{2\epsilon_0 r_2}$	(Q) $ V_A - V_B = \frac{3kQ}{2R} - \frac{kQ}{r_2}$
(III) 0	(iii) $\frac{kQ}{r_2^2}$	(R) $ V_C - V_B = \frac{\rho R^2}{2\epsilon_0} \ln \left(\frac{r_3}{r_2} \right)$
(IV) $\frac{\rho r_1}{4\epsilon_0}$	(iv) $\frac{kQr_2}{R^3}$	(S) $ V_A - V_B = \frac{kQ}{R} - \frac{kQ}{r_2}$

12. Which of the following combination represents the situation due to a uniformly charged non-conducting solid sphere having total charge Q & radius R :-

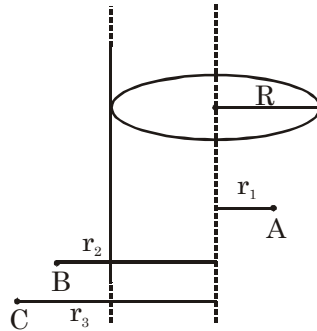


- (A) (II) (iii) (Q) (B) (III) (iii) (S) (C) (II) (iii) (P) (D) (III) (iv) (P)
13. Which of the following combination represents the situation due to a uniformly charged non-conducting hollow thin sphere having total charge Q & radius R .



- (A) (III) (iii) (S) (B) (II) (iii) (P) (C) (III) (iv) (Q) (D) (II) (iv) (S)

14. Which of the following combination represents the situation due to a uniformly charged infinite solid cylinder of radius R and charge density ρ .



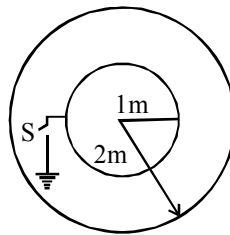
- (A) (III) (i) (P) (B) (I) (i) (R) (C) (IV) (ii) (P) (D) (I) (ii) (R)

SECTION-III

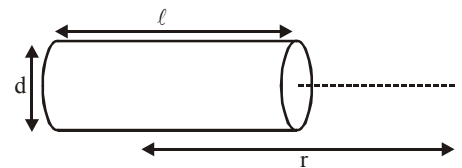
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

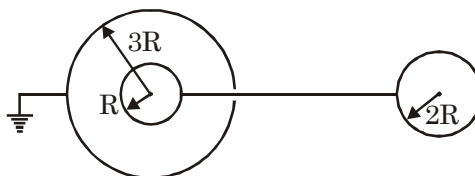
- An arrangement of source charges produces the electric potential $V = 5000x^2$ along the x -axis, where V is in volts and x is in meters. If a charge particle of mass 1g and charge 1nC is present in this field and its turning points are at $\pm 8.0\text{ cm}$? What is the particle's maximum speed (in mm/s).
- Figure shows two concentric spherical shell having radii 1 m and 2 m respectively. Outer spherical surface has charge $-3.2 \times 10^{-15}\text{ Coulomb}$. If inner surface is now earthed by closing the switch then $x \times 10^{+y}$ number of electron move towards earth, then $x + y$ will be ($x, y \in \mathbb{I}, 0 \leq x, y \leq 9$).



- Figure shows a cylindrical conducting rod of diameter d and length $\ell \gg d$ uniformly charged in vacuum such that electric field near its surface and far from its ends is E_0 . Electric field at $r \gg \ell$ on the axis of the cylinder is $\frac{3E_0 \ell d}{2Nr^2}$. Then find the value of N .



- A spherical conductor of radius 4cm carries a charge of $\frac{10}{9} \times 10^{-10}\text{ C}$. At what distance (in cm) from of sphere should a point charge of $\frac{40}{9} \times 10^{-10}\text{ C}$ should be placed so that the potential of the conductor becomes 75 V . Assume potential is zero at infinity.
- Two metal balls of radii R and $2R$ are connected to a long thin conducting wire and has total charge $Q = 21\text{ }\mu\text{C}$. Then the ball of radius R is placed inside a grounded metal sphere of radius $3R$ as shown. What charge (in μC) will flow through the thin wire in this process?



SECTION-I

Single Correct Answer Type

7 Q. [3 M (-1)]

1. Consider a thin conducting spherical shell of radius R having charge Q . A point charge q is placed at a distance $R/3$ from the center of the shell and another point charge $2q$ is placed at a distance $3R$ from the center of the shell. If the shell is grounded, find the charge flown through the grounding wire.

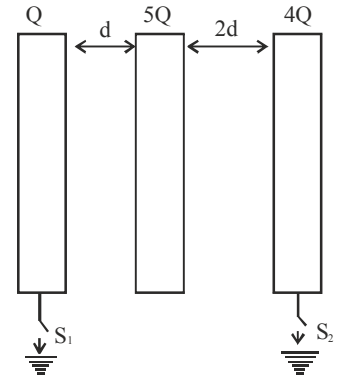
(A) $Q + q$ (B) $Q + 3q$ (C) $Q + \frac{2}{3}q$ (D) $Q + \frac{5}{3}q$

2. Three concentric spherical shells have radii a, b and c ($a < b < c$) and have surface charge densities $\sigma, -\sigma$ and σ respectively. If V_A, V_B and V_C denote the potentials of the three shells, then for $c = a + b$, we have :-

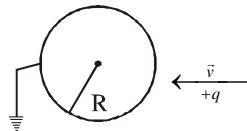
(A) $V_C = V_B = V_A$ (B) $V_C = V_A \neq V_B$ (C) $V_C = V_B \neq V_A$ (D) $V_C \neq V_B \neq V_A$

3. The metal plate on the left carries charge $+Q$, and on the right has charge $4Q$ respectively as shown. The central plate has initially charge equal to $5Q$. When both S_1 and S_2 switches are closed simultaneously then the charge flowed through S_1 upto steady state is :-

(A) Q (B) $\frac{5Q}{2}$
(C) $\frac{3Q}{2}$ (D) None of these

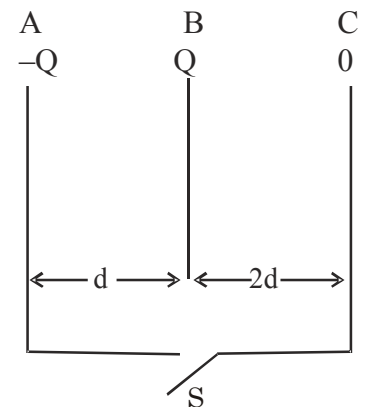


4. Consider a solid metallic sphere which is grounded. Consider a charge $+q$ which is moved towards the sphere with a constant velocity by an external agent as shown. Then

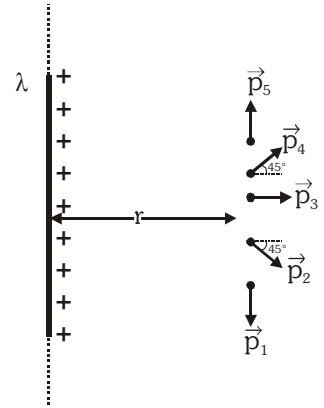


- (A) The charge distribution on the surface of the sphere will not change with time.
(B) The surface charge density of the sphere will increase with time.
(C) The surface charge density of the sphere will decrease with time.
(D) The potential of the sphere will decrease as the charge moves towards the sphere.
5. Three large conducting plates A, B & C are placed parallel to each other as shown in figure & having charges $-Q, +Q$ & 0 respectively. Long time after closing the switch s two particles having charge $-q$ mass m are projected normally with velocity v_A & v_C from plate B but in opposite direction towards A & C respectively. Then find the ratio of velocity (V_A/V_C) needed so that these two charge particles can reach plates A & C respectively.

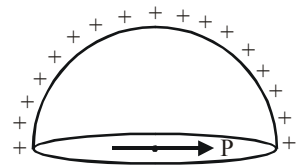
(A) $\frac{1}{1}$ (B) $\frac{\sqrt{2}}{1}$
(C) $\frac{2}{1}$ (D) $\frac{3}{1}$



6. Five identical small electric dipoles are placed at a distance r from an infinitely long straight wire carrying a uniform charge density λ . If they experience force of magnitude F_1, F_2, F_3, F_4 and F_5 respectively then
- (A) $F_1 = F_2 = F_3 = F_4 = F_5$
 (B) $F_3 < F_2 = F_4 < F_1 = F_5$
 (C) $F_3 > F_2 = F_4 > F_1 = F_5$
 (D) Can't be determined the relation between F_1, F_2, F_3, F_4 and F_5



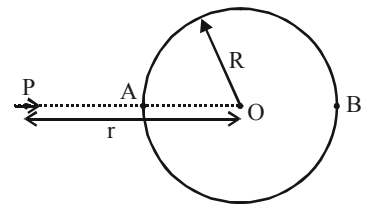
7. A dipole is kept in circular plane of a uniformly charged hemispherical shell. Center of dipole coincide with center of hemispherical shell and dipole is restricted to move in given circular plane. Identify its type of equilibrium if dipole is slightly rotate in this plane :-
- (A) Neutral
 (B) Unstable
 (C) Stable
 (D) Depends upon the direction of rotation



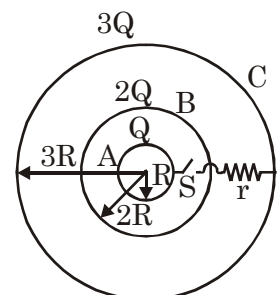
Multiple Correct Answer Type

7 Q. [4 M (-1)]

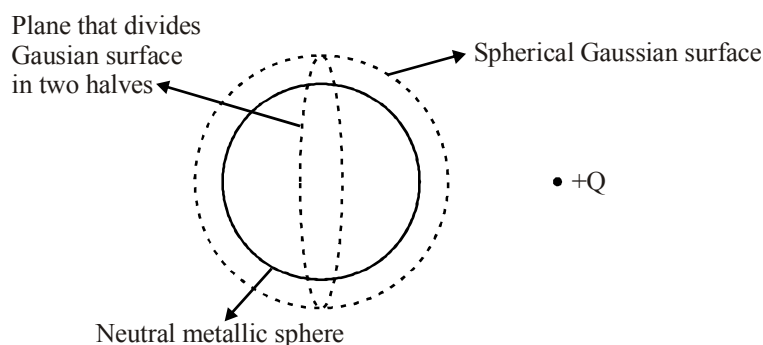
8. Imagine that a dipole is at the centre of a spherical non conducting surface. If magnitude of electric field at a certain point on the surface of sphere is 10 N/C, then which of the following cannot be the magnitude of electric field anywhere on the surface of sphere :-
- (A) 4 N/C
 (B) 8 N/C
 (C) 16 N/C
 (D) 32 N/C
9. An ideal dipole of dipole moment $\vec{p}$ is placed in front of an uncharged conducting sphere of radius R as shown.



- (A) The potential at point A is $\frac{K P}{(r - R)^2}$
 (B) The potential at point A is $\frac{K P}{r^2}$
 (C) The potential due to dipole at point B is $\frac{K P}{(r + R)^2}$
 (D) The potential due to dipole at point B is $\frac{K P}{r^2}$
10. Three thin concentric conducting spherical shells A, B, C of radius $R, 2R$ and $3R$ are given charges $Q, 2Q, 3Q$ respectively as shown. The inner most and outermost can be joined through switch 'S' which is initially opened. Choose the **INCORRECT** options :-
- (A) After closing switch S, potential of shell C does not change.
 (B) After closing switch S, charge on shell A becomes zero.
 (C) After closing switch S, potential of shell B does not change.
 (D) After closing switch S, charge on shell C does not change.



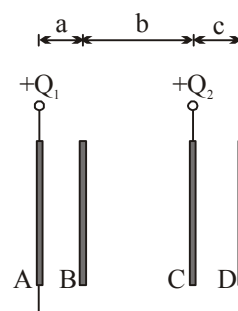
11. Figure shows a neutral metallic sphere with a point charge $+Q$ placed near its surface. Electrostatic equilibrium conditions exist on metallic sphere. Mark the **CORRECT** statement(s) :



- (A) Net flux through Gaussian surface due to charge Q is zero
 (B) Net flux through Gaussian surface due to charges appearing on the outer surface of metallic sphere must be zero
 (C) If point charge Q is displaced towards metallic sphere, magnitude of flux through right hemispherical part of Gaussian surface due to this charge increases.
 (D) If point charge Q is displaced towards metallic sphere, charge distribution on outer surface of sphere will change
12. A spherical soap bubble of radius R has uniformly distributed charge over its surface with surface charge density σ then $[T = \text{surface tension of the soap solution}]$

- (A) excess pressure inside the bubble is $\frac{4T}{R} - \frac{\sigma^2}{2\epsilon_0}$ (B) excess pressure inside the bubble is $\frac{4T}{R} + \frac{\sigma^2}{2\epsilon_0}$
 (C) excess pressure inside the bubble is $\frac{4T}{R}$ (D) electrostatic pressure is $\frac{\sigma^2}{2\epsilon_0}$

13. Figure shows an arrangement of four identical rectangular plates A, B, C and D each of area S . Ignore the separation between the plates in comparison to the plate dimensions.
- (A) Potential difference between plates A & B is independent of Q_1 .
 (B) Potential difference between plates C & D is independent of Q_1 .
 (C) Potential difference between plates A & B is independent of Q_2 .
 (D) Potential difference between plates C & D is independent of Q_2 .



14. Two concentric conducting shells have radii R and $2R$; charges q_A and q_B and potentials $2V$ and $\frac{3V}{2}$ respectively if reference for electric potential is taken at infinity. Now shell B is earthed and let charges on them become q'_A and q'_B . Then:
- (A) $\frac{q_A}{q_B} = \frac{1}{2}$
 (B) $\frac{q'_A}{q'_B} = 1$
 (C) Potential of A after earthing becomes $\frac{3V}{2}$
 (D) Potential difference $(V_A - V_B)$ between A and B after earthing becomes $V/2$

Linked Comprehension Type
(Single Correct Answer Type)

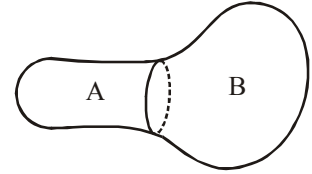
(1 Para × 2Q.)

[3 M (-1)]

Paragraph for Question no 15 and 16

Figure shows an arbitrary body through which electric charge is distributed in an arbitrary way. The body is divided in two region A and B. Using the result,

The electric force that the charge in region A exerts on the charge in region B is equal to the force that the charge on the entire body exerts on the charge in region B or otherwise solve next two problems.



15. Consider a uniformly charged spherical shell of radius R having total charge Q . Find electric force $\vec{F}$ exerted on each hemisphere by the charge on the other hemisphere.

(A) $\frac{Q^2}{4\pi\epsilon_0 R^2}$ (B) $\frac{Q^2}{16\pi\epsilon_0 R^2}$ (C) $\frac{Q^2}{8\pi\epsilon_0 R^2}$ (D) $\frac{Q^2}{32\pi\epsilon_0 R^2}$

16. Consider a uniformly charged solid dielectric sphere of total charge Q and radius R . The force exerted by one hemisphere on other hemisphere is :

(A) $\frac{3Q^2}{16\pi\epsilon_0 R^2}$ (B) $\frac{3Q^2}{64\pi\epsilon_0 R^2}$ (C) $\frac{Q^2}{9\pi\epsilon_0 R^2}$ (D) $\frac{Q^2}{36\pi\epsilon_0 R^2}$

Matching List Type (4 × 4)

1Q.[3 M (-1)]

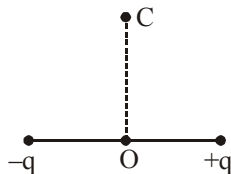
17. List-I represents certain charge configurations & list-II comments on their electric field at point C :

List-I

- (P) An electric dipole and point C is on the perpendicular bisector

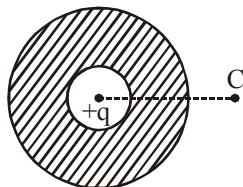
List-II

- (1) Electric field at C will be zero

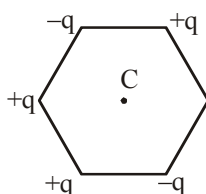


- (Q) A uniform hollow neutral conducting sphere & (2) Electric field at C is towards right positive charge is placed at centre of cavity.

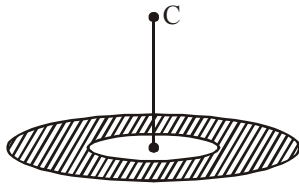
C is a point outside the sphere



- (R) A regular hexagon with 5 charges as shown. (3) Electric field at C is towards left C is centre of hexagon.



- (S) A uniformly positively charged disc with a cavity inside the disc. C is a point on its axis. (4) Electric field at C is towards upward



Code :

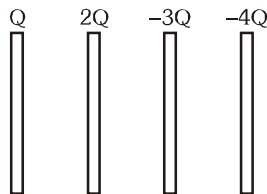
	P	Q	R	S
(A)	3	2	4	1
(B)	1	2	4	3
(C)	1	4	2	3
(D)	3	2	2	4

SECTION-III

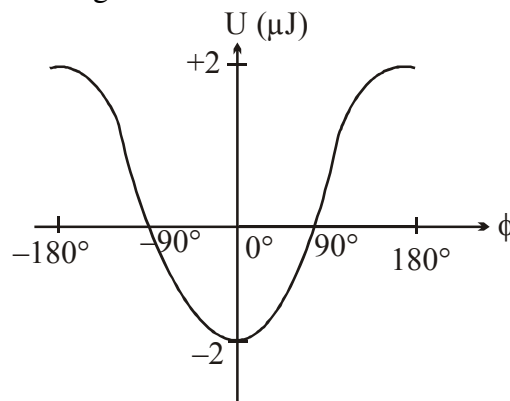
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1. Four very large metal plates are given charges as shown. The middle two are then connected through a conducting wire. The charge that will flow through the wire is nQ . Find the value of n .

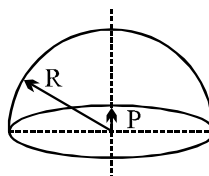


2. The graph shows the potential energy of an electric dipole that oscillates between $\pm 60^\circ$. What is dipole's kinetic energy (in μJ) when it is aligned with the field.



3. A dipole of dipole moment P , is kept at the centre, along the axis of the hemispherical shell as shown in the figure. The electric flux through the surface of the shell, if radius of the shell is R is given by

$$\frac{P}{N \epsilon_0 R} \cdot \text{Find } N.$$

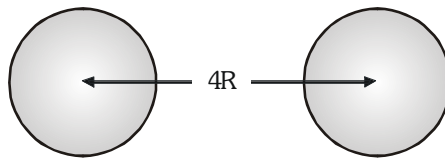


SECTION-I

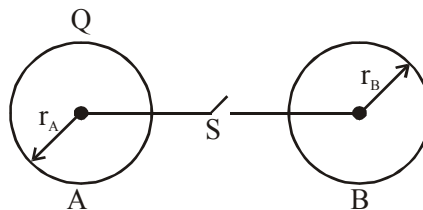
Single Correct Answer Type

6 Q. [3 M (-1)]

1. Two identical spheres each of radius R are kept at center-to-center spacing $4R$ as shown in the figure. They are charged and the electrostatic force of interaction between them is first calculated assuming them point like charges at their centers and the force is also measured experimentally. The calculated and measured forces are denoted by F_c and F_m respectively. (F_c and F_m denote magnitude of force)



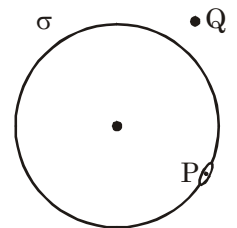
- (A) When they carry charges of the same sign $F_c > F_m$ and when they carry charges of opposite signs $F_c < F_m$ only when they are insulator.
 (B) When they carry charges of the same sign $F_c > F_m$ and when they carry charges of opposite signs $F_c < F_m$ only when they are conductor.
 (C) When they carry charges of the same sign $F_c < F_m$ and when they carry charges of opposite signs $F_c > F_m$ irrespective of their material.
 (D) When they carry charges of the same sign $F_c > F_m$ and when they carry charges of opposite signs $F_c < F_m$ irrespective of their material.
2. A conducting sphere A of radius r_A carries a charge Q . Another conducting sphere B of radius r_B is uncharged initially. Both the spheres are separated by a very large distance. Sphere A is connected with sphere B through a long conducting wire of zero resistance through a switch S as shown. As the current flow through the wire vanishes, switch S is opened, sphere A is again charged to a total charge Q and then switch S is again closed and the process is continued for infinite times. If q is the charge transferred to sphere B when the switch S is closed for the first time, then total charge on the sphere B after large number of such processes is :



- (A) Q (B) $\frac{qQ}{Q-q}$ (C) $Q \frac{r_A}{r_B}$ (D) None of these
3. P is a point at the centre of a tiny hole present on the surface of a conducting shell having charge density σ and Q is a point just outside the shell as shown in figure.

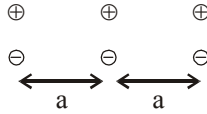
Two charge particles A (q, m) and B $\left(q, \frac{m}{2}\right)$ are released at $t = 0$ from Q & P

respectively. Find the ratio of magnitudes of their accelerations $\left(\frac{a_A}{a_B}\right)$.

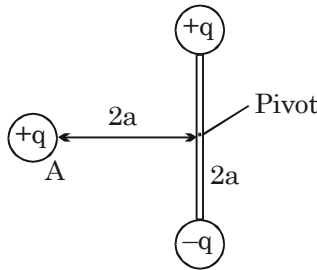


- (A) 2 (B) $\frac{1}{2}$ (C) 1 (D) 4

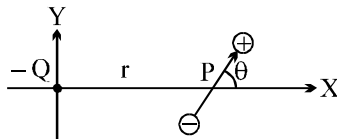
4. Three identical small electric dipoles are arranged parallel to each other at separation a as shown in figure. Their total interaction energy is U . Now one of the end dipole is gradually reversed. Work done by electric forces is :



- (A) $\frac{17U}{8}$ (B) $\frac{16U}{17}$ (C) $\frac{16U}{7}$ (D) $\frac{18U}{17}$
5. Figure shows a ball having a charge q fixed at a point A. Two identical balls of mass m having charge $+q$ & $-q$ are attached to the end of a light rod of length $2a$. The system is released from the situation shown in figure. Find the angular velocity of the rod when the rod turns through 90° .



- (A) $\frac{\sqrt{2}q}{3\pi\epsilon_0 ma^3}$ (B) $\frac{q}{\sqrt{3\pi\epsilon_0 ma^3}}$ (C) $\frac{q}{\sqrt{6\pi\epsilon_0 ma^3}}$ (D) $\frac{\sqrt{2}q}{4\pi\epsilon_0 ma^3}$
6. A point negative charge $-Q$ is placed at a distance r from a dipole with dipole moment P as shown in figure. The x component of force acting on the charge $-Q$ is -



- (A) $-\frac{PKQ}{r}\cos\theta\hat{i}$ (B) $\frac{PKQ}{r}\cos\theta\hat{i}$ (C) $-\frac{2PKQ}{r^3}\cos\theta\hat{i}$ (D) $\frac{2PKQ}{r^3}\cos\theta\hat{i}$

Multiple Correct Answer Type

4 Q. [4 M (-1)]

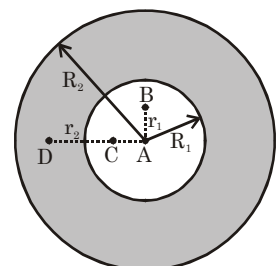
7. Consider a thick conducting shell of inner radius R_1 & outer radius R_2 as shown in the figure. Shell is neutral and a point charge Q is kept at point A i.e. centre of shell. Point B and D lie at distance r_1 and r_2 respectively from centre. Choose **CORRECT** option(s) :

(A) Potential at point D is $\frac{kQ}{R_2} + \frac{kQ}{r_2}$

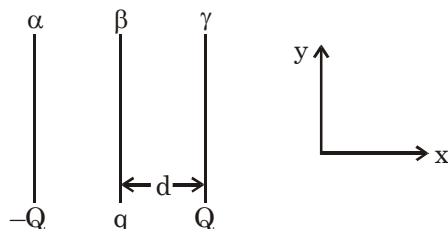
(B) Potential at point B is $\frac{kQ}{R_2} + \frac{kQ}{r_1} - \frac{kQ}{R_1}$

(C) Potential at D will change if point charge shifted to position C

(D) Potential at B will change if point charge is shifted to position C.



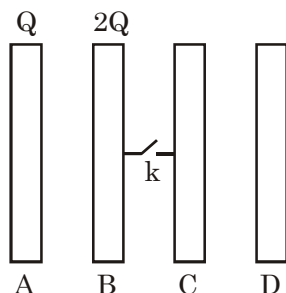
8. Two fixed, identical conducting plates (α & β), each of surface area S on one side are charged to $-Q$ and q , respectively, where $Q > q > 0$. A third identical plate (γ) with charge Q , free to move is located on the right side of the plates at a distance d (figure). The third plate is released and collides with the plate β . Assume the collision is elastic and the time of collision is sufficient to redistribute charge amongst β & γ .



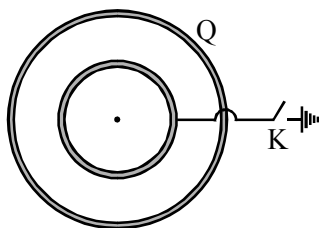
- (A) The electric field acting on the plate γ before collision is $\frac{q-Q}{S(2\epsilon_0)}$, to the right
 (B) The charges on β after the collision is $q_1 = (Q + q/2)$
 (C) The charges on γ after the collision is $q_2 = (q/2)$
 (D) The velocity of the plate γ after the collision and at a distance d from the plate β is

$$v = (Q - q/2) \left(\frac{d}{m\epsilon_0 S} \right)^{1/2}$$

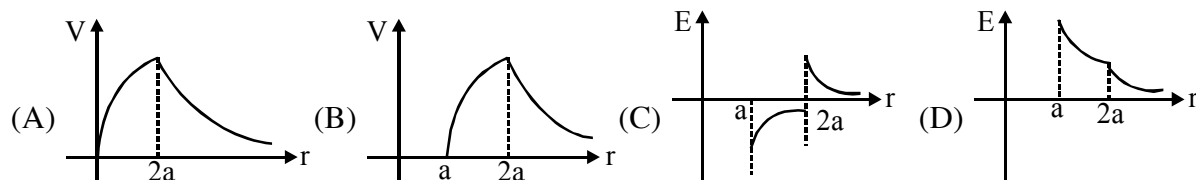
9. Initially two plates (A and B) are given charges as shown. Find the final charges on the plates after the key k is closed. All plates are conducting, parallel and of infinite length and breadth.



- (A) Final charges on plate A is Q
 (B) Final charge on plate B is 0
 (C) Final charge on plate C is $3Q/2$
 (D) Final charge on plate D is 0
10. In the given figure two concentric conducting spherical shells of radii a and $2a$ have charge zero and Q respectively. When the key K is closed :-



The $V = f(r)$ & $E = f(r)$ graph is (r is the distance from the centre) :-



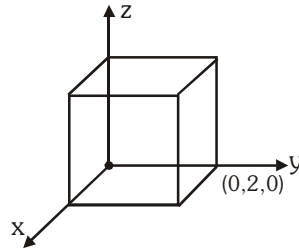
Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Questions no. 11 to 13

Electric field in a region is given by following equation $\vec{E} = \frac{K(\vec{r} - \vec{r}_0)}{|\vec{r} - \vec{r}_0|^2}$ where $\vec{r} = x\hat{i} + y\hat{j}$ and $\vec{r}_0 = \hat{i} + \hat{j}$

and an imaginary cube is made as shown in the diagram.



11. Find total flux through this cube due to the given electric field.
(A) $4\pi K$ (B) $6\pi K$ (C) $16K$ (D) $8K$
12. Find total charge enclosed in this imaginary shown cube.
(A) $4K\pi\epsilon_0$ (B) $8K\pi\epsilon_0$ (C) $6K\pi\epsilon_0$ (D) $2K\pi\epsilon_0$
13. If $\vec{r}_0 = (2\hat{i} + 2\hat{j})$ then find the value of total flux through the walls of the shown cube
(A) $2K\pi$ (B) $K\pi$ (C) $4K\pi$ (D) $4K$

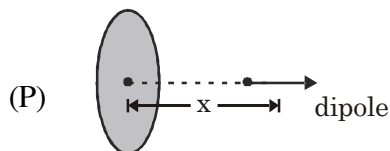
Matching List Type (4 × 4)

2 Q. [3 M (-1)]

14. List-I shows different charge distributions and short electric dipole at a distance x from the charge distributions. List-II gives the dependence of force acting on the dipole as of function of x .

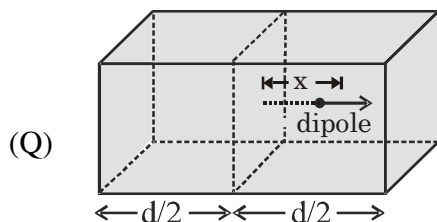
List-I

List-II



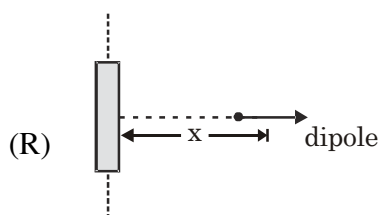
(1) x^{-2}

Uniformly charged thin infinite disc.



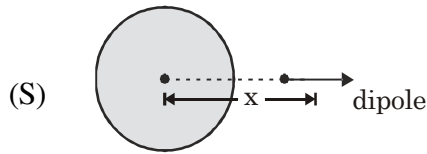
(2) x^{-3}

Uniformly charged thick infinite sheet.
Dipole is placed inside the sheet.



(3) x^0

Uniform infinite line of charge.



(4) $x^{-\infty}$

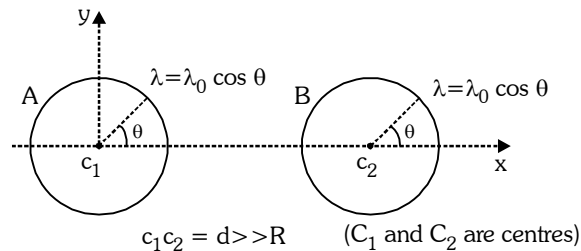
Uniformly charged sphere.

Dipole is placed outside the sphere.

Code

	P	Q	R	S
(A)	4	1	3	2
(B)	4	3	1	2
(C)	1	2	4	3
(D)	3	4	1	2

15. If the given two identical charged rings lie in xy plane both having linear charge density λ varies as per $\lambda = \lambda_0 \cos \theta$ ($\lambda_0 = \text{constant}$) where θ is measured from +x-axis. Radius for both the rings is R



List-I

- (P) Dipole moment of each ring is $x\lambda_0\pi R^2$ then x is
- (Q) Magnitude of work done by electric field in rotating ring B by 90° about x-axis is

$$\frac{xK\lambda_0^2\pi^2R^4}{2d^3} \text{ then x is}$$

- (R) Magnitude of work done by electric field in rotating ring B by 90° about z-axis is

$$\frac{xK\lambda_0^2\pi^2R^4}{2d^3} \text{ then x is}$$

- (S) Electric force between the two rings is $\frac{xK\lambda_0^2\pi^2R^4}{d^4}$ (4) 4

then x is

Codes :

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	4	3
(C)	2	3	1	4
(D)	3	4	2	1

List-II

(1) 0

(2) 1

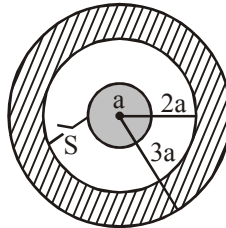
(3) 6

SECTION-III

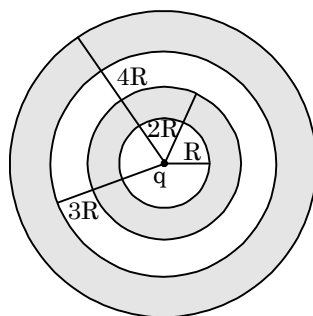
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

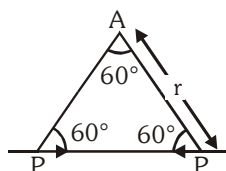
1. A solid conducting sphere of radius 'a' having charge q is surrounded by a concentric conducting spherical shell of inner radius 2a and outer radius 3a as shown in figure. If the amount of heat produced when switch is closed is $x \left(\frac{kq^2}{24a} \right)$. Fill the value of x.



2. The centres of two small metal balls, whose radii are equal, are initially at a distance of r. If both balls are given different magnitude and opposite charges, and then they are released, they begin to move towards each other. They touch each other and then they move away from each other again. When the distance between their centres is 4r, then the force exerted between them is one-twentieth of the initial force exerted between them. The initial charge of one of the balls was 2×10^{-7} C. If the initial charge of the other ball is $-x \times 10^{-8}$ C, find the value of x. (x is an integer, the sum of charges is positive)
3. A point charge particle kept at the center of two concentric hollow conducting sphere of inner radii R and 3R, outer radii 2R and 4R respectively. Amount of energy stored in electric field in the absence of conducting spheres is U_1 and in the presence of conducting spheres is U_2 . If $U_1 - U_2 = \frac{nq^2}{96\pi\epsilon_0 R}$. Then fill the value of n.



4. Potential at point A due to dipoles is $\frac{3KP}{\beta r^\gamma}$ then find the value of $\beta + \gamma$?



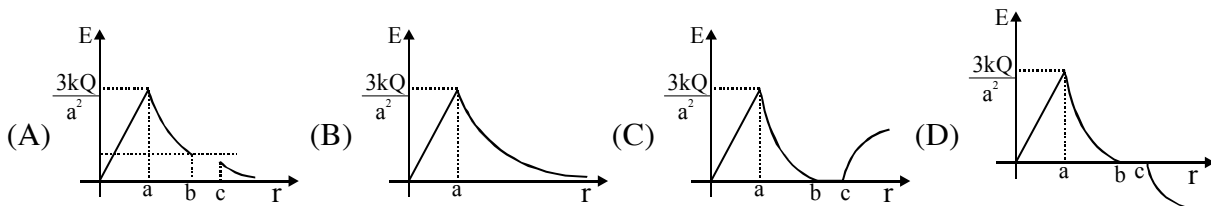
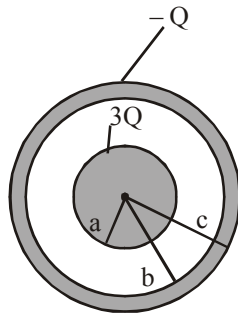
SECTION-I

Single Correct Answer Type

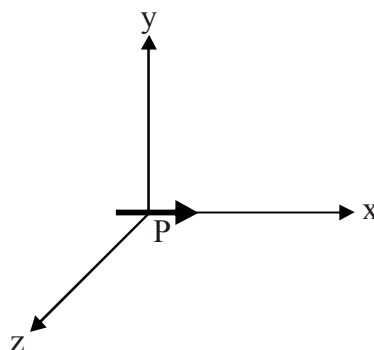
5 Q. [3 M (-1)]

1. A insulating sphere of radius 'a' carries a net positive charge $3Q$, uniformly distributed throughout its volume. Concentric with this sphere is a conducting spherical shell with inner radius 'b' and outer radius 'c', and having a net charge $-Q$, as shown in figure. Electric field varies with distance r from the centre

as $\left(K = \frac{1}{4\pi\epsilon_0} \right)$



2. A short electric dipole is oriented along x-direction at origin. At which of following point the electric field have no x component.

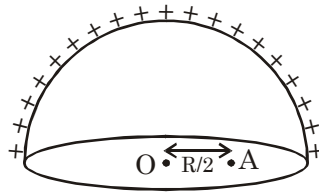
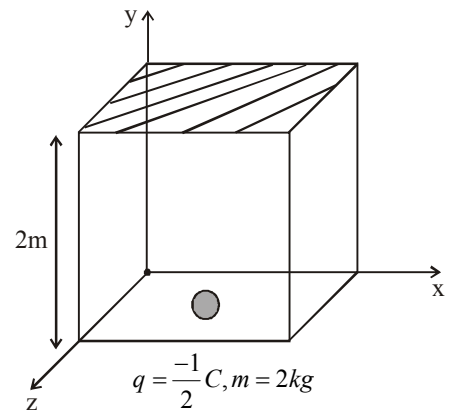


- (A) $(1, 1, 0)$ (B) $(\sqrt{2}, 1, 0)$ (C) $(1, \sqrt{2}, 0)$ (D) $(1, 0, 0)$

3. An infinitely small electric dipole is kept on the axis of a uniformly charged ring at distance $R/\sqrt{2}$ from the centre of the ring. The direction of the dipole moment is along the axis. The dipole moment is P , charge of the ring is Q and radius of the ring is R . The force on the dipole is nearly.

- (A) $\frac{4kPQ}{3\sqrt{3}R^2}$ (B) $\frac{4kPQ}{3\sqrt{3}R^3}$ (C) $\frac{2kPQ}{3\sqrt{3}R^3}$ (D) zero

4. Consider a gravity free container as shown. System is initially at rest and electric potential in the region is $V = (y^3 + 2)$ J/C. A ball of charge q and mass m is released from rest from base starts to move up due to electric field and collides with the shaded face as shown. If its speed just after collision is 1.5 m/s and time for which ball is in contact with shaded face is 0.1 sec, find external force required to hold the container fixed in its position during collision assuming ball exerts constant force on wall during entire span of collision.
- (A) 70 N
(B) 72 N
(C) 74 N
(D) 76 N
5. A hemispherical bowl of mass m is uniformly charged with charge density ' σ '. Electric potential due to charge distribution at a point 'A' is (which lies at centre of radius as shown).

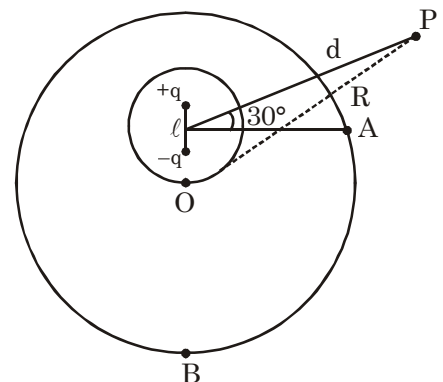


- (A) $\frac{\sigma R}{4\epsilon_0}$ (B) $\frac{\sigma R}{3\epsilon_0}$ (C) $\frac{\sigma R}{2\epsilon_0}$ (D) $\frac{\sigma R}{\epsilon_0}$

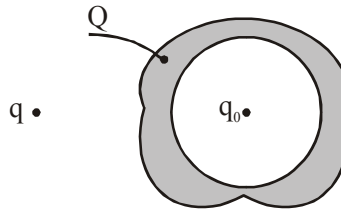
Multiple Correct Answer Type

7 Q. [4 M (-1)]

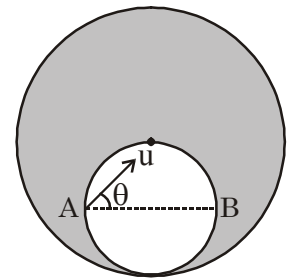
6. A non conducting sphere of radius R is given a volume charge density $\rho = kr$, where r is the distance of a point in the spherical volume from its centre and k is a positive constant. Then mark the **CORRECT** options :
- (A) Self energy of the sphere is given by dR^7 , where d is some constant.
- (B) Electrostatic force of interaction between the left and right hemispheres is $\frac{k^2 \pi R^6}{12 \cdot \epsilon_0}$.
- (C) Electric field at distance r from centre is $\frac{kr^2}{4 \epsilon_0}$ ($r \leq R$).
- (D) Electrostatic pressure at a distance r ($r \leq R$), from the centre is $P(r) = a(b-r^4)$, where a & b are some constants.
7. A cavity is taken out from a uniform conducting sphere. Inside the cavity a dipole is placed as shown in the figure. Find which of the following is/are true (Given : $q = 10^{-1} C$, $\ell = 0.1$ mm, $\theta = 30^\circ$, $d = 10$ cm, $R = 12$ cm)
- (A) Potential at points A & B are same
(B) Potential at points A & B are different
(C) Potential at point P is 0 volt
(D) Potential at point P is 2 volt



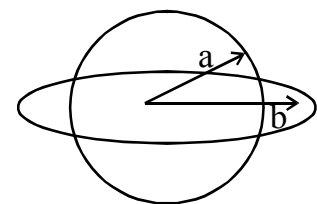
8. Consider a conductor of arbitrary shape with a spherical cavity inside it. A charge q_0 is placed inside cavity, a charge Q is given to conductor and a charge q is placed outside conductor. Mark the **CORRECT** statement (s) :-



- (A) If cavity is spherical and charge q_0 is placed asymmetrically inside it, it experiences a resultant force
 (B) Force due to charges on outer surface of conductor on q_0 is non zero.
 (C) Field of external charge q is cancelled off for all inside points by charges appearing on outer surface of conductor
 (D) Charges on inner surface of cavity cancel off field of charge q_0 for all points outside cavity
9. A thick shell with radius $2R$ has uniform density ρ C/m^3 . It has a spherical cavity of radius R as shown in the figure. A charge particle of mass m and charge Q is projected along the line AB at an angle θ in the cavity as shown. Select the correct statement (Assume particle doesn't come out of cavity) :

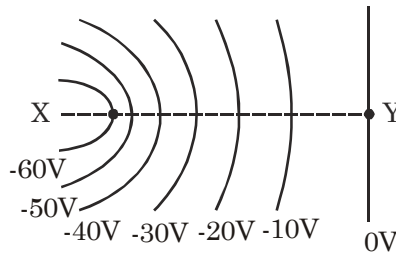


- (A) Particle crosses line AB at a distance of $\left[\frac{u^2 \sin 2\theta}{Q\rho R} \right] 3m \epsilon_0$ from point A .
 (B) Average velocity of particle during the time it crosses line AB is greater than $u \cos \theta$.
 (C) Final speed when it crosses line AB is equal to initial speed.
 (D) It will move in a parabolic path in the cavity.
10. An insulating ring of charge $2\pi\lambda b$ and radius b is concentric with a charged conducting solid sphere of charge Q and radius a ($b > a$) as shown in the figure. Then

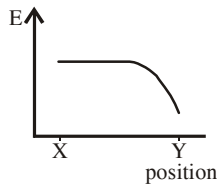


- (A) Potential of the sphere is $\frac{Q}{4\pi \epsilon_0 a} + \frac{\lambda}{2 \epsilon_0}$
 (B) If sphere is grounded then charge on the sphere will be $\frac{-a\lambda}{8\pi^2 \epsilon_0}$
 (C) Electric field inside the sphere is non-zero
 (D) The sphere is equipotential body.
11. A point charge $+Q$ is fixed at a point O . Another point charge $+q$ & mass m is projected from infinity towards $+Q$ with speed v_0 . Minimum distance of approach is r . If it is now given speed $2v_0$ at infinity then
- (A) Minimum distance of approach becomes $\frac{r}{2}$
 (B) Minimum distance of approach becomes $\frac{r}{4}$
 (C) Angular momentum conservation will hold true about O .
 (D) Mechanical energy will remain conserved throughout the motion.

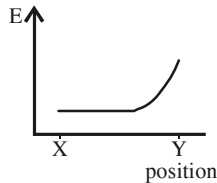
12. The diagram below shows some equipotential lines in the region of an electric field.



- (A) Graph which shows the magnitude E of the electric field strength along the line XY is



- (B) Graph which shows the magnitude E of the electric field strength along the line XY is



- (C) If an electron is released from rest (from point x) then its velocity increases & its acceleration first remains constant and then decreases
(D) If an electron is released from rest (from point x) then its velocity increases & its acceleration first remains constant and then increases

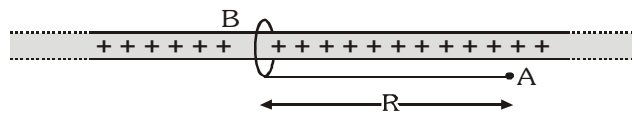
Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Questions no. 13 to 15

Figure shows a very long smooth non conducting rod of radius r , charged with uniform charge density λ , fixed horizontally.

A neutral and smooth ring B of mass M can slide freely on the rod which happens to just fit in it. A is a nonconducting particle having mass m and charge q , attached to the ring by means of a non-conducting and inextensible string of length R . If P is released from the position shown in figure then answer the following questions.



13. The loss in electrostatic potential energy when string becomes vertical, is:-

(A) $\frac{q\lambda}{4\pi\epsilon_0} \ln\left(1 + \frac{R}{r}\right)$ (B) $\frac{q\lambda}{2\pi\epsilon_0} \ln\left(1 + \frac{R}{r}\right)$ (C) $\frac{q\lambda}{2\pi\epsilon_0} \ln\left(1 - \frac{r}{R}\right)$ (D) $\frac{q\lambda}{2\pi\epsilon_0} \ln\left(1 + \frac{r}{R}\right)$

14. If $M = 2m$ and $R = 2r$ then the speed of particle when string becomes vertical, is:-

(A) $\sqrt{\frac{2}{3} \left(2gR + \frac{q\lambda}{\pi\epsilon_0 m} \ln 3 \right)}$ (B) $\frac{2}{3} \sqrt{2gR + \frac{q\lambda}{\pi\epsilon_0 m} \ln 3}$
(C) $\sqrt{2gR + \frac{q\lambda}{2\pi\epsilon_0 m} \ln 3}$ (D) None of these

15. If $M = 2m$ and $R = 2r$ then the tension in the string when the string is vertical, is:–

(A) $3mg + \frac{q\lambda}{6\pi\epsilon_0 r} [2 + 3\ln 3]$

(B) $mg + \frac{q\lambda}{12\pi\epsilon_0 r} [2 + 3\ln 3]$

(C) $4mg + \frac{q\lambda}{12\pi\epsilon_0 r} [2 + 9\ln 3]$

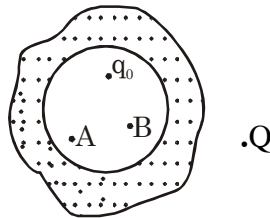
(D) None of these

Matching List Type (4 × 4)

1 Q. [3 M (–1)]

16. A point charge is placed inside a spherical cavity of a neutral conductor. Another point charge Q is placed outside the conductor as shown.

List – I gives assertion statements and List- II gives its reason statements. Match the assertions with proper reason



List -I

(P) Total charge induced on cavity wall is equal & opposite to q_0

(Q) As the point charge Q placed outside is pushed away from conductor, the potential difference ($V_A - V_B$) between two points A and B within the cavity of sphere remains constant.

(R) Electric field due to induced charge on the inner surface of cavity at a point outside the conductor is non zero

(S) Electric potential at a point outside the conductor due to induced charges on the inner surface of cavity is nonzero

List -II

(1) If cavity is surrounded by a Gaussian surface, where, all parts of Gaussian surface are located inside conductor then, $\int \vec{E} \cdot d\vec{A} = 0$ hence, total charge enclosed by Gaussian surface must be zero

(2) The resultant electric field due to charges on outer surface of conductor & outside the conductor is zero at all points inside the conductor

(3) The resultant electric field due to charge induced on inner surface of conductor cavity and charge q_0 placed inside cavity is zero at all points outside cavity.

(4) Net electric potential due to charge induced on the inner surface of cavity and the charge q_0 placed inside cavity is zero at all points outside cavity.

Codes :

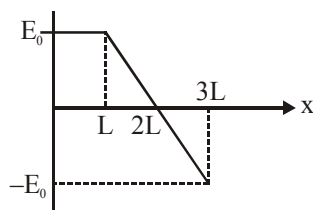
	P	Q	R	S
(A)	2	3	1	4
(B)	1	3	4	3
(C)	4	2	3	1
(D)	1	2	3	4

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

- For a hypothetical electric field in x-direction, its potential is continuously decreasing and becoming half after every 20 cm. The rate of decreasing of potential with distance is also directly proportional to the potential itself. If the ratio of electric field and potential is $x \ln 2 (\text{m}^{-1})$. Find the value of x.
- The electric potential varies in space according to the relation $V = 3x + 4y$. A particle of mass 10 kg and charge $+1 \mu\text{C}$ starts from rest from point (2, 3.2) under the influence of this field. Find the velocity (in mm/s) of the particle when it crosses the x-axis. Assume V and (x, y) are in S.I. units.
- Consider the E versus x graph. What is the minimum velocity that should be given to a point charge $-Q$ of mass M at $x = 3L$ so that it can reach the origin? Take $Q = 1 \text{ mC}$, mass of the charge as 1g, $E_0 = 5 \text{ N/C}$ and $L = 5\text{m}$.



SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

- The electric potential in a region of space is varies as $V = 4x + 3y - 5z^2$. All physical quantities are in SI units. Match the entries of column I with entries of column II.

Column I

- Magnitude of electric field at $(-34, 63, 1)$
- Equation of equipotential surface
- Magnitude of force experienced by $\sqrt{5}\text{C}$ charge located at origin.
- The potential difference between the point $(3, 4, 0)$ and $(6, 0, 0)$

Column II

- $4x + 3y = 5z^2 + 59$
- $7 = 10z$
- 0
- $5\sqrt{5}$
- $5z^2 - 4x - 3y = 59$

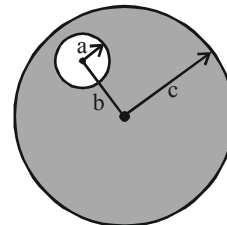
SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

1. A spherical conductor have a cavity of radius 'a'. If radius of conductor have radius 'c' and the distance between centre of conductor and center of cavity is 'b'. Charge given to conductor is Q. Then select the correct statement :-

- (A) Electric field in the cavity is non zero and uniform
(B) Electric field in the cavity is zero through out the volume
(C) Electric field intensity in the cavity depends on b
(D) Electric field intensity depends on radius of cavity 'a'



2. The potential (in volts) of a charge distribution is given by

$$V(z) = 30 - 5z^2 \text{ for } |z| \leq 1 \text{ m}$$

$$V(z) = 35 - 10|z| \text{ for } |z| \geq 1 \text{ m.}$$

$V(z)$ does not depend on x and y. If this potential is generated by a uniform charge per unit volume ρ_0 (in units of ϵ_0) which is spread over a certain region, then choose the correct statement.

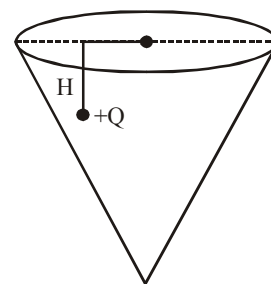
- (A) $\rho_0 = 20 \epsilon_0$ in the entire region
(B) $\rho_0 = 10 \epsilon_0$ for $|z| \leq 1 \text{ m}$ and $\rho_0 = 0$ elsewhere
(C) $\rho_0 = 40 \epsilon_0$ in the entire region
(D) $\rho_0 = 20 \epsilon_0$ for $|z| \leq 1 \text{ m}$ and $\rho_0 = 0$ elsewhere

3. A charge +Q is located somewhere inside a vertical cone such that the depth of the charge from the free surface of the cone is H. It is found that the flux

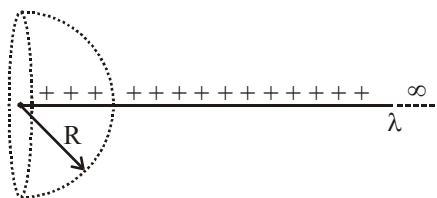
associated with the cone with the curved surface is $\frac{3Q}{5\epsilon_0}$. If the charge is raised

vertically through a height 2H, then the flux through the curved surface is

- (A) $\frac{3Q}{5\epsilon_0}$ (B) $\frac{2Q}{5\epsilon_0}$ (C) $\frac{4Q}{5\epsilon_0}$ (D) Zero



4. Consider an imaginary hemispherical surface. A semi-infinite wire of charge density λ is kept with one of its end coinciding with centre of hemisphere and wire is kept along the symmetric axis of the hemisphere as shown in the figure. The electric flux passing through the spherical surface of the hemisphere is :-

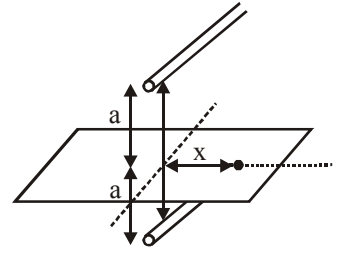


- (A) $\frac{\lambda R}{3\epsilon_0}$ (B) $\frac{\lambda R}{4\epsilon_0}$ (C) $\frac{3\lambda R}{4\epsilon_0}$ (D) $\frac{\lambda R}{2\epsilon_0}$

5. A sphere of radius 10 cm has a total charge Q distributed over its surface. Imagine this sphere to be resting on the open end of a cylinder of radius 8 cm and height 10 cm. Find the electric flux through the curved cylindrical surface.

- (A) $\frac{Q}{10\epsilon_0} (2\sqrt{5} - 3)$ (B) $\frac{Q}{5\epsilon_0}$ (C) $\frac{Q}{2\epsilon_0} \left(1 - \frac{2}{\sqrt{5}}\right)$ (D) None of these

6. Two infinite wires are charged uniformly with same charge density and kept parallel to each other at a distance of $2a$ to each other. In the symmetry plane passing through the mid point of line joining both the wire; there is a point at a distance x from the central point where electric field is maximum. Find x .



(A) $x = a$

(B) $x = \frac{a}{\sqrt{2}}$

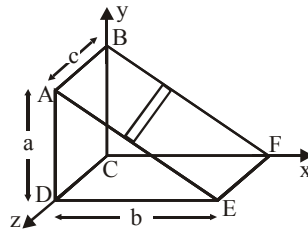
(C) $x = \frac{a}{2}$

(D) $\sqrt{2}a$

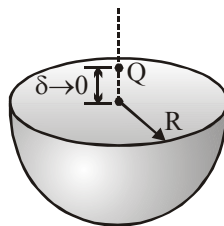
Multiple Correct Answer Type

5 Q. [4 M (-1)]

7. In the given hypothetical electric field $\vec{E} = [(d + x)\hat{i} - E_0\hat{j}] \text{ N/C}$, a hypothetical closed surface is taken as shown in figure :-



- (A) Net flux through plane ABCD is $-acd$ unit.
 (B) Net flux through plane CDEF is bcE_0 unit.
 (C) Net flux through plane ABEF is $\left[-bcE_0 + acd + \frac{acb}{2}\right]$ unit.
 (D) Net charge enclosed by the closed surface is $abc\epsilon_0$ unit.
8. A particle with charge Q is located immediately above the centre of the flat face of a hemisphere of radius R as shown in figure. Choose CORRECT statement(s) :- (Outward area is to be taken positive)



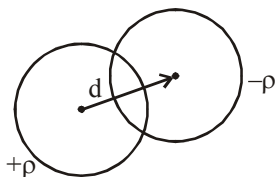
(A) Flux through the flat surface is $-\frac{Q}{2\epsilon_0}$.

(B) Flux through the curved surface is $\frac{Q}{2\epsilon_0}$.

(C) If charge Q is displaced up flux through flat surface is less than $\frac{Q}{2\epsilon_0}$ in magnitude

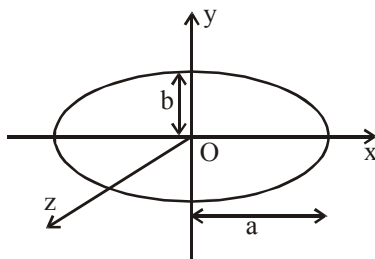
(D) If charge Q is displaced up flux through curved surface is less than $\frac{Q}{2\epsilon_0}$ in magnitude.

9. Figure shows two spheres, each of radius R and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed so that they partially overlap. Let the vector from the positive centre to the negative centre is $\vec{d}$. Then choose the **CORRECT** statement(s) :-

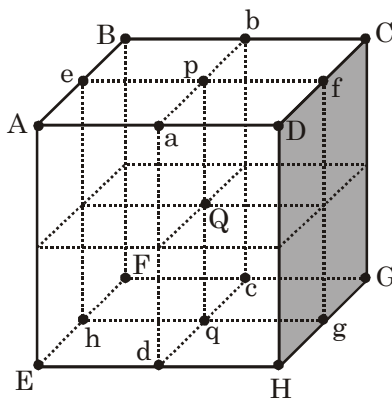


- (A) Electric field inside the overlapped region will be uniform
(B) Electric field inside the overlapped region is $\frac{\rho}{3\epsilon_0} \vec{d}$
(C) Electric field inside the overlapped region is zero
(D) Gauss's law can not be applied if only one of the spheres is taken as Gaussian surface.
10. An ellipsoidal gaussian surface with semi major axis a and minor axis b is shown in figure. An electric

field $\vec{E} = \frac{C(\hat{x}i + \hat{y}j + \hat{z}k)}{(x^2 + y^2 + z^2)^{3/2}}$ exist in space.



- (A) Net flux through constant gaussian surface is $4\pi C$
(B) Net charge enclosed by gaussian surface is $4\pi C\epsilon_0$
(C) If given surface is a conductor electric potential at all its inside points may be constant for given field function
(D) If given surface is a conductor electric field at all its inside points may be constant for given field function
11. The following figure shows a charge Q kept at the centre of a cube. Let ϕ represent the flux of field due to the charge Q , the correct options are :



(A) $\phi_{ABCD} = \frac{Q}{6\epsilon_0}$

(B) $\phi_{CBPF} = \frac{Q}{24\epsilon_0}$

(C) $\phi_{abcd} = 0$

(D) $\phi_{abcd} = \frac{Q}{2\epsilon_0}$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question nos. 12 to 14

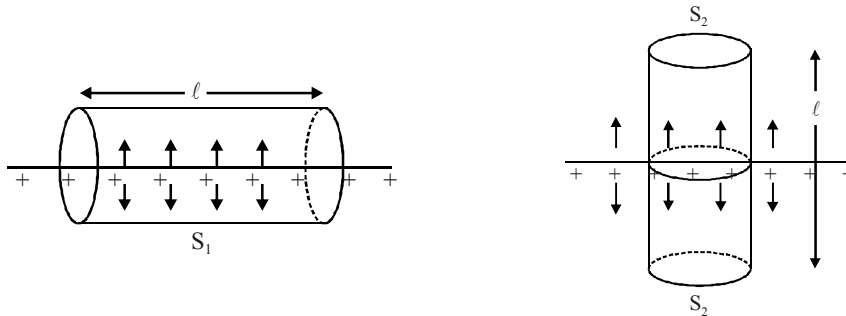
While dealing with a system of point charges, the principle of superposition is helpful in obtaining the electric field vector at any point. However, evaluating electric field at a point for continuous charge distribution in space often becomes a mathematically challenging job. The application of Gauss' law is sometimes useful in such cases. Gauss' law states that the total electric flux through a closed surface is

proportional to total electric charge enclosed within the surface with $\left(\frac{1}{\epsilon_0}\right)$ as constant of proportionality.

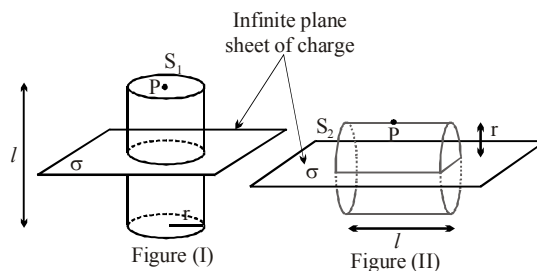
One such application is in finding electric field for an infinite linear distribution of charge, having λ as linear charge density.

Taking S_1 as the gaussian surface, $\phi_E = \int \vec{E} \cdot d\vec{s} = E(2\pi r\ell)$, $q_{\text{enclosed}} = \lambda\ell$. Hence, $E = \frac{\lambda}{2\pi\epsilon_0 r}$.

Mr. Becker rather decides to take an upright cylinder shown as surface S_2 . He applies Gauss' law on surface S_2 as ; $\phi_E = \int \vec{E} \cdot d\vec{s} = E(2\pi r^2)$, $q_{\text{enclosed}} = (2r\lambda)$. It gives $E = \frac{\lambda}{\pi r\epsilon_0}$. The difference in values of electric field calculated by using surfaces S_1 and S_2 is debatable.

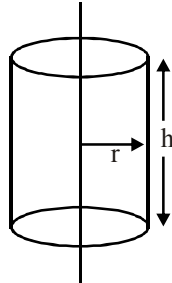


12. Which of the following correctly points out the error in Mr. Beckers analysis ?
 (A) Gaussian surface cannot cut the continuous distribution of charge
 (B) The area vectors at the two ends of S_2 must be in the same direction
 (C) $\oint \vec{E} \cdot d\vec{s} \neq E(2\pi r^2)$
 (D) $q_{\text{enclosed}} \neq 2r\lambda$
13. To calculate the electric field due to an infinite plane sheet of uniform charge density σ , We choose two different gaussian cylinders as shown in figure (I) & figure (II). The appropriate choice is figure (I) because



- (A) Gaussian surface cannot cut the continuous distribution of charge
 (B) The area vectors at the two ends of S_2 must be in the same direction
 (C) for S_2 , $\oint \vec{E} \cdot d\vec{s} \neq E(2\pi r l)$
 (D) $q_{\text{enclosed}} \neq 2r l \sigma$

14. Consider a finite wire having a uniform charge distribution over its length. It is symmetrically enclosed by coaxial cylindrical Gaussian surface as shown in figure. We apply Gauss's law to find electric field and for that purpose, we wish to evaluate $\oint \vec{E} \cdot d\vec{s}$. Which of the following is incorrect reason for our inability to evaluate the integral.



- (A) The flux will be due to the components of electric field normal to the wire only.
- (B) Electric field is varying in magnitude over the surface
- (C) The angle between electric field and area vector is not constant at all the points on the gaussian surface
- (D) The flux through the circular faces will be non zero

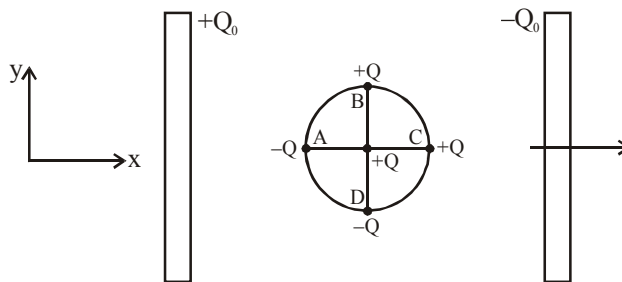
Linked Comprehension Type

(1 Para $\times$ 2 Q.) [4 M (-1)]

(Multiple Correct Answer Type)

Paragraph for Question no 15 and 16

Figure shows a massless wheel of radius R and massless spokes with five charges each of charge Q and mass m . System is placed in a field created by two large fixed plates having charges $+Q_0$ and $-Q_0$ respectively. Entire assembly lies in a smooth horizontal plane. Wheel is placed in horizontal plane and constrained to move in horizontal plane. Initially spokes are released along x & y axis as shown in figure. (A = surface area of plate)



15. Mark the **CORRECT** statement(s) :

- (A) Acceleration of centre of mass is $\frac{QQ_0}{5\epsilon_0 mA}$
- (B) Acceleration of centre of mass is $\frac{QQ_0}{10\epsilon_0 mA}$
- (C) Instantaneous angular acceleration of system is $\frac{QQ_0}{2\epsilon_0 AmR}$
- (D) Instantaneous angular acceleration of system is $\frac{QQ_0}{5\epsilon_0 AmR}$

16. Mark the **CORRECT** statement(s) :

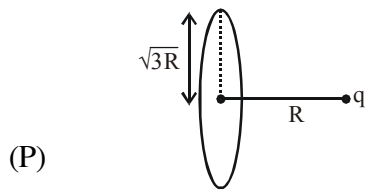
- (A) When released from rest system executes periodic motion in the reference frame-fixed to centre of mass.
- (B) When released from rest system executes simple harmonic motion in the reference frame-fixed to centre of mass.
- (C) Acceleration of point A immediately after release is $\frac{QQ_0}{5\epsilon_0 mA} \hat{i} + \frac{QQ_0}{2\epsilon_0 Am} \hat{j}$.
- (D) Acceleration of point A immediately after release is $\frac{QQ_0}{5\epsilon_0 mA} \hat{i} - \frac{QQ_0}{5\epsilon_0 Am} \hat{j}$.

Matching List Type (4 × 4)

1 Q. [3 M (-1)]

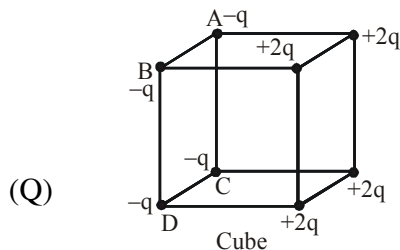
17. List-I

List-II



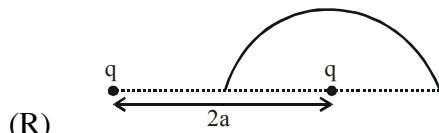
(1) 0

Flux through given circle



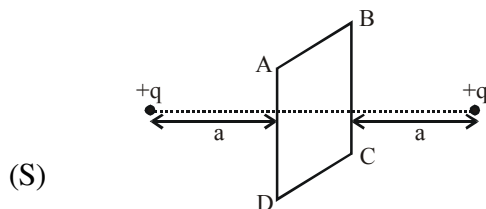
(2) $\frac{q}{2\epsilon_0}$

Net flux through surface ABCD



(3) $\frac{q}{3\epsilon_0}$

Net flux through given hemispherical surface



(4) $\frac{q}{4\epsilon_0}$

Net flux through surface ABCD

Code :-

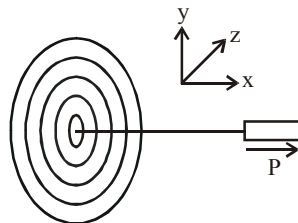
	P	Q	R	S
(A)	4	3	2	1
(B)	3	2	1	4
(C)	2	3	4	1
(D)	1	4	2	3

SECTION-III

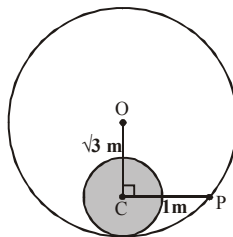
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

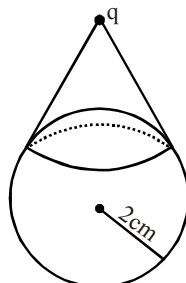
1. A long uniformly charged wire with linear charge density $\lambda = \epsilon_0$ coulomb/m is placed in the form of a spiral of outer radius $R = 1\text{m}$ and having a total $N = 160$ turns as shown in figure. A dipole of dipole moment $|\vec{p}| = 0.1\text{ Cm}$ is placed along x-axis at a $= \sqrt{3}\text{ m}$ point $(a, 0, 0)$. Find the magnitude of force (in N) experienced by dipole.



2. A uniformly charged sphere is placed inside a charged hollow sphere as shown in figure. O is the centre of hollow sphere and C is the centre of solid sphere. The magnitude of charge on both the spheres is $4\left(\sqrt{\frac{3}{7}}\right)\mu\text{C}$. The electric field at point 'P' which lie just outside the hollow sphere is given by $3\alpha\text{ kN/C}$. Find the value of α .



3. A point charge $q (= 4\text{C})$ is placed at a distance $2\sqrt{2}\text{ cm}$ from the centre of sphere having radius 2 cm as shown in figure. Flux through the surface of sphere enclosed by all the tangents to the sphere passing through the charge as shown in figure is $\frac{\sqrt{\alpha}(\sqrt{2}-1)}{\epsilon_0}$. Find the value of α .



SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

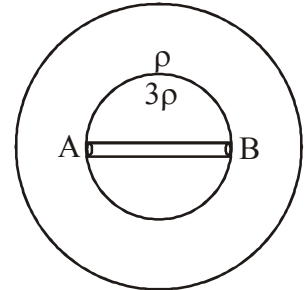
1. A planet of core density 3ρ and outer crust of density ρ has small tunnel in core. A small particle of mass m is released from end A then time required to reach end B :

(A) $\sqrt{\frac{\pi}{\rho G}}$

(B) $\frac{1}{2}\sqrt{\frac{\pi}{\rho G}}$

(C) $\pi\sqrt{\frac{1}{\rho G}}$

(D) $2\pi\sqrt{\frac{1}{\rho G}}$



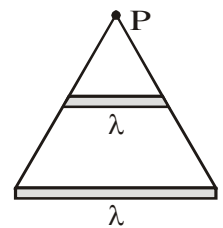
2. The two sticks shown in diagram have same linear mass density λ , and they subtend same angle at point P. Their distances from point P are ℓ and 2ℓ . Which stick creates a larger force on a point mass placed at point P?

(A) The top stick

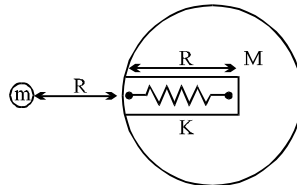
(B) The bottom stick

(C) They produce equal forces

(D) The relative magnitude of forces depends on ℓ



3. A small ball of mass 'm' is released at a height 'R' above the earth surface, as shown in the figure above. If the maximum depth of the ball to which it goes is $R/2$ inside the earth through a narrow groove before coming to rest momentarily. The groove contains an ideal spring of spring constant K and natural length R, find the value of K if R is radius of earth and M mass of earth



(A) $\frac{3GMm}{R^3}$

(B) $\frac{6GMm}{R^3}$

(C) $\frac{9GMm}{R^3}$

(D) $\frac{7GMm}{R^3}$

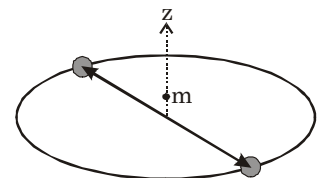
4. Two stars, each of mass M and separated by a distance, orbit about their centre of mass. A planetoid of mass m ($m \ll M$) moves along the axis of this system perpendicular to the orbital plane. Let T_p be the period of simple harmonic motion for the planetoid for small displacement from the center of mass along the z-axis and let T_s be the period of motion for the two stars. Determine the ratio $\frac{T_p}{T_s}$:-

(A) The ratio of $\frac{T_p}{T_s} = 2$

(B) The ratio of $\frac{T_p}{T_s} = \frac{1}{2}$

(C) The ratio of $\frac{T_p}{T_s} = \frac{1}{2\sqrt{2}}$

(D) The ratio of $\frac{T_p}{T_s} = \frac{1}{3}$

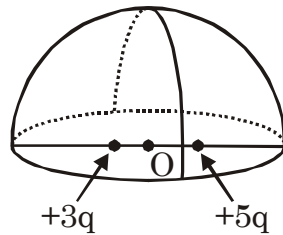


5. An electric field $\vec{E} = E_0\hat{i} + E_0\hat{j}$ exists in a space. Find the flux through a triangular loop with vertices at

$$\left(\frac{a}{2}, 0, 0\right), \left(\frac{a}{2}, 0, a\right) \text{ and } \left(\frac{a}{2}, a, \frac{a}{2}\right)$$

- (A) $\sqrt{2} E_0 a^2$ (B) $\frac{E_0 a^2}{2}$ (C) $\frac{\sqrt{3}}{2} E_0 a^2$ (D) $\frac{\sqrt{3}}{4} E_0 a^2$

6. The figure shows a hollow hemisphere of radius R in which two charges $3q$ and $5q$ are placed symmetrically about the centre O on the planar surface. The electric flux over the curved surface is :

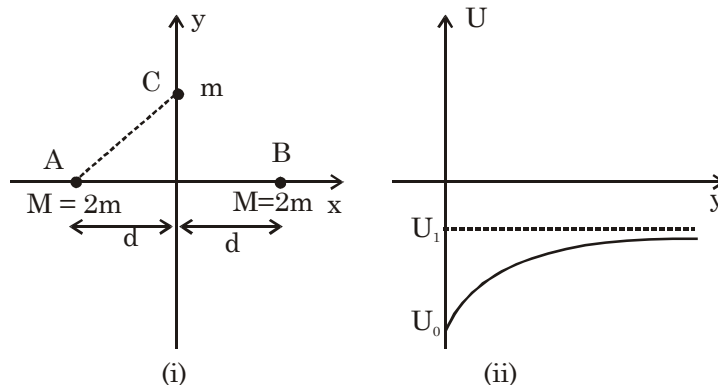


- (A) $\frac{15}{2} \frac{q}{\epsilon_0}$ (B) $\frac{4q}{\epsilon_0}$ (C) $\frac{q}{\epsilon_0}$ (D) $\frac{2q}{\epsilon_0}$

Multiple Correct Answer Type

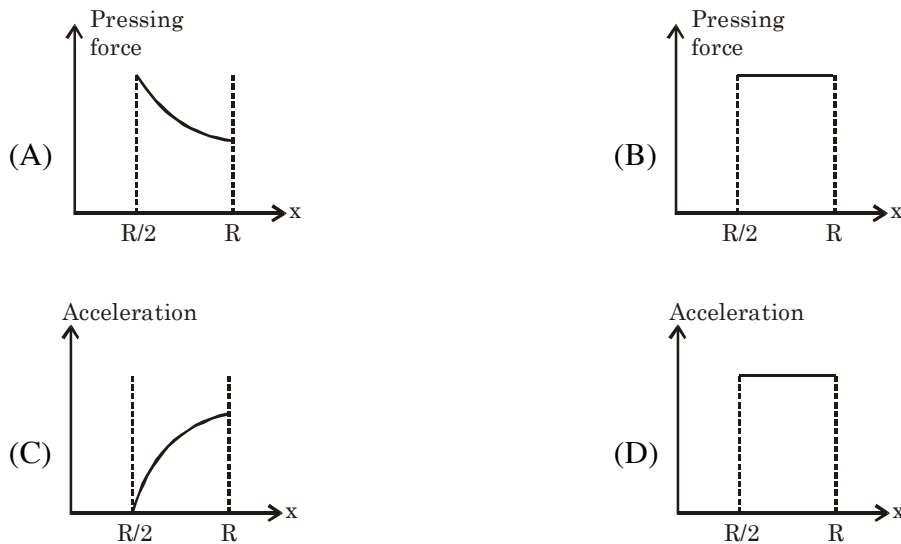
5 Q. [4 M (-1)]

7. Two point masses, each of mass M are fixed at points A and B respectively. A third point m is released from infinity, so that it can move along y -axis under the influence of mutual gravitational attraction on it due to point masses kept at A and B respectively as shown in the figure -1. Figure -2 represents the potential energy of system (includes m , M at A and M at B) with position of m at y -axis. (Neglect any forces other than gravity on particle C) (Given: $Gm^2/d = 12$ Joule, $m = 6$ kg). Choose the correct option(s) :-

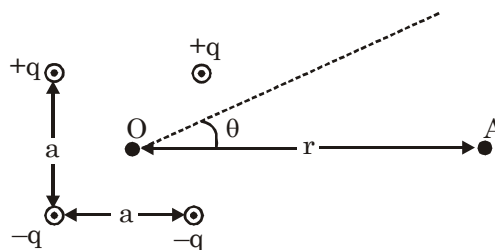


- (A) Point mass m will perform periodic motion
(B) Value of U_1 must be equal to -24 Joule if gravitational potential energy of two point mass system is taken to be zero when separation between them is infinite.
(C) Maximum speed of particle is 24 m/s
(D) Maximum speed of particle is 4 m/s

8. A smooth tunnel is dug along the chord of earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth. The pressing force by the particle on the wall and the acceleration of the particle varies with x (distance of particle from centre of earth) as :



9. A spherical planet of radius R has spherically symmetrical distribution of mass density, varying as square of the distance from the centre, from zero at centre to maximum value ρ_0 at its surface.
- (A) The value of escape velocity of a mass m at the surface of planet is $\sqrt{\frac{4\pi G \rho_0 R^2}{5}}$.
- (B) The value of acceleration due to gravity ' g ' varies inside the planet as cube of the distance from centre.
- (C) The value of escape velocity is same as the escape velocity from another planet of same total mass & radius but having uniform mass density.
- (D) The energy required to impart escape velocity to particles of masses ' m ' & ' $2m$ ', at the surface of planet, will be in ratio 1:2.
10. The electric quadrupole consists of two positive and two negative point charges of equal magnitude, q , located at the vertices of a square. Find electric field at A and B in terms of polar coordinates at A ($1000a, 0^\circ$) and B ($1000a, 45^\circ$),



- (A) $E_A \approx \frac{2kqa}{(1000a)^3}(-\hat{\theta})$
- (B) $E_B \approx \frac{kq\sqrt{2}a}{(1000a)^3}[2\hat{r} - \hat{\theta}]$
- (C) $E_A \approx \frac{2kqa}{(1000a)^3}(\hat{r})$
- (D) $E_B \approx \frac{kq\sqrt{2}a}{(1000a)^3}(\hat{r} + \hat{\theta})$

11. Mark the **CORRECT** statement(s) :

(A) The electric field at a height z above the center of a square sheet of side a carrying a uniform surface

charge σ is given by $\frac{\sigma}{2\epsilon_0} \left[\frac{2}{\pi} \tan^{-1} \sqrt{1 + \frac{a^2}{2z^2}} - 1 \right]$

(B) Potential difference between the top and the center of an inverted hemispherical bowl of radius R

carrying a uniform surface charge density σ is $\frac{R\sigma}{2\epsilon_0} (\sqrt{2} - 1)$

(C) A sphere of radius R carries a charge density $\rho(r) = kr$, k is some +ve constant, then energy of

configuration is $\frac{\pi k^2 R^7}{7\epsilon_0}$

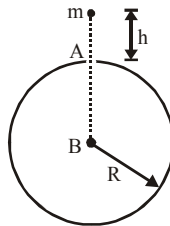
(D) A point charge q is inside a cavity in an uncharged conductor, then the force on q will be necessarily zero.

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Question no. 12 and 13

A hollow spherical planet has mass M & radius ' R '. A small particle of mass m is released from rest from a height h above a small hole in the planet. Assume that the mass density of planet is constant.



12. What time will the particle take to move from A to B?

(A) $R\sqrt{\frac{R(R+h)}{GMh}}$ (B) $R\sqrt{\frac{2R(R+h)}{GMh}}$ (C) $R\sqrt{\frac{R(R+h)}{2GMh}}$ (D) $2R\sqrt{\frac{R(R+h)}{GMh}}$

13. With what approximate speed will the particle reach B?

(A) $2\sqrt{\frac{GMh}{R(R+h)}}$ (B) $\sqrt{\frac{GMh}{2R(R+h)}}$ (C) $\sqrt{\frac{GMh}{R(R+h)}}$ (D) $\sqrt{\frac{2GMh}{R(R+h)}}$

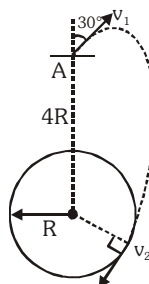
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

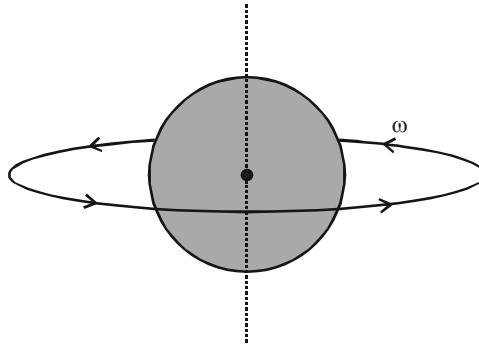
1. A particle is projected from point A, that is at a distance $4R$ from the centre of the Earth, with speed v_1 in a direction making 30° with the line joining the centre of the Earth and point A, as shown. Find the speed v_1 if particle passes grazing the surface of the earth. Consider gravitational interaction only between

these two. (use $\frac{GM}{R} = 6.4 \times 10^7 \text{ m}^2/\text{s}^2$) Express your answer in the form $(500\sqrt{2} X) \text{ m/s}$ and write the value of X .



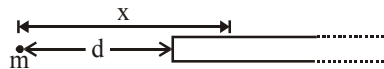
2. A ring of mass m and radius $3R$ is rotating with constant angular speed ω around a planet of mass M and radius R . Center of ring and planet coincide with each other. Tension in the ring is given as

$$T = \frac{GMm}{3n\pi R^2}. \text{ Find value of } n \left(\omega = \sqrt{\frac{GM}{9R^3}} \right)$$



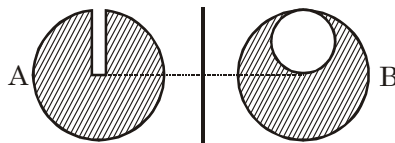
3. Find the gravitational force of interaction between the mass m and infinite rod of varying mass density λ such that $\lambda(x) = \frac{k}{x}$, where x is the distance from mass m . Given that mass m is placed at a distance d

from the end of the rod on its axis as shown in the figure. If force is $\frac{Gmk}{nd^2}$, then find the value of n .

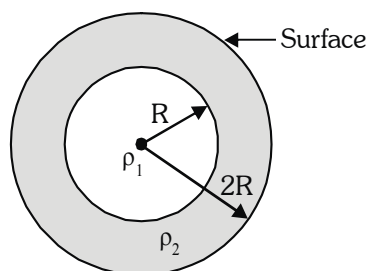


4. Nowadays ISRO is working on a space research program. In this program they discover spherical Asteroid made up titanium, a precious metal. They dig a tunnel from surface to centre to find depth of titanium. Titanium was uniformly distributed in the sphere. Now they want to mine this precious material out of the asteroid due to which cavity is formed as shown in figure B. If a piece of metal falls in the tunnel it takes time t_1 and when it falls in cavity it takes time t_2 to reach at the bottom. Ratio of time

$$\frac{t_1}{t_2} = \left(\frac{\pi}{n} \right). \text{ Find the value of 'n'.$$



5. A planet is made of two materials of density ρ_1 and ρ_2 as shown in figure. The acceleration due to gravity at surface of planet is same as a depth 'R'. The ratio of $\frac{\rho_1}{\rho_2}$ is $\frac{a}{b}$. Find value of $\frac{a^b + 1}{86}$ (a and b are lowest possible integers)



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. **Column-I**

- (A) Magnitude of angular momentum of system about O increases after the instant shown.

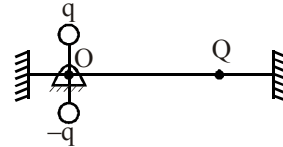
- (B) The net torque on body A is zero about O at the instant shown.

- (C) The net force on negative charge increases in magnitude after the instant shown.

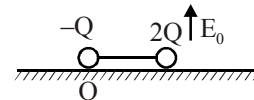
- (D) Mechanical energy of the system increases after the instant shown.

Column-II

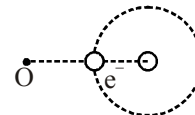
- (P) A dipole is hinged on a long smooth fixed horizontal wire. A charge Q (A) is threaded on wire and is free to move on the wire. The system comprises of the dipole and point charge. Gravity is absent.



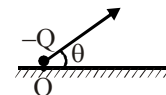
- (Q) A system of two charges $-Q$ and $2Q$ (A) of mass m each are connected by a light rod and kept on smooth level ground. An external uniform electric field $E_0 = mg/Q$ is applied as shown. Gravity is present.



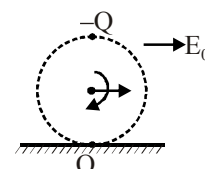
- (R) An electron (A) is revolving about a fixed proton in anticlockwise direction. The system comprises of the electron and proton. Gravity is absent.



- (S) System consists of a particle (A) of mass m and charge $-Q$. It is projected from level ground in presence of constant electric field $E = mg/Q$ in vertically upward direction. Gravity is present.



- (T) System consists of a uniform ring and particle (A) of mass m and charge $-Q$ on its circumference as shown. The ring is in pure rolling -on a sufficiently rough horizontal surface. Gravity is present.

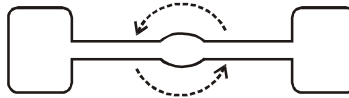


SECTION-I

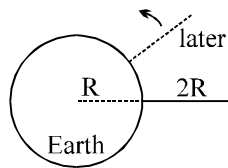
Single Correct Answer Type

6 Q. [3 M (-1)]

1. A space station consists of two living modules attached to a central hub on opposite sides of the hub by long corridors of equal length. Each living module contains N astronauts of equal mass. The mass of the space station is negligible compared to the mass of the astronauts, and the size of the central hub and living modules is negligible compared to the length of the corridors. At the beginning of the day, the space station is rotating so that the astronauts feel as if they are in gravitational field of strength g . Two astronauts, one from each module, climb into the central hub, and the remaining astronauts now feel a gravitational of strength g' . What is the ratio g'/g in terms of N ?



- (A) $\sqrt{\frac{N}{N-1}}$ (B) $\frac{N}{(N-1)}$ (C) $\sqrt{\frac{(N-1)}{N}}$ (D) $\left(\frac{N}{N-1}\right)^2$
2. Consider a very long stick of length $2R$, which extends from just above the surface of the earth, to a radius $3R$. If initial conditions have been set up so that this stick moves in a circular orbit while always pointing radially. What is the period of this orbit. g represents acceleration due to gravity on surface of earth.



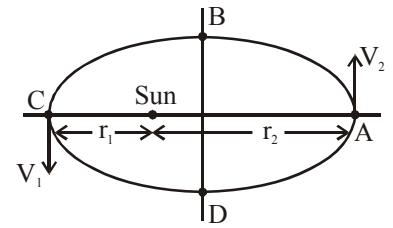
- (A) $6\pi\sqrt{\frac{R}{2g}}$ (B) $2\pi\sqrt{6}\sqrt{\frac{R}{g}}$ (C) $\pi\sqrt{6}\sqrt{\frac{R}{g}}$ (D) none of these
3. Two identical thin uniform rods of mass M and length L are placed in a line at separation of L . Find the gravitational force acting between them



- (A) $\frac{GM^2}{L^2} \ln\left(\frac{3}{4}\right)$ (B) $\frac{GM^2}{L^2} \ln\left(\frac{5}{3}\right)$ (C) $\frac{GM^2}{L^2} \ln 2$ (D) $\frac{GM^2}{L^2} \ln 3$
4. A smooth tunnel is dug along the radius of the earth that ends at the centre and a ball is released from the surface of earth along the tunnel. If the coefficient of restitution is 0.2 between the surface and ball then the distance travelled by the ball before second collision at the centre is :
- (A) $\frac{6}{5}R$ (B) $\frac{7}{5}R$ (C) $\frac{9}{5}R$ (D) $\frac{3}{2}R$
5. An artificial satellite is moving in a circular orbit around the earth with a speed equal to half the magnitude of the escape velocity from the surface of earth. If the satellite is stopped suddenly in its orbit and allowed to fall freely, onto the earth, find the speed (in km/sec) with which it hits the surface of the earth ($g = 10 \text{ m/s}^2$ and $R = 6400 \text{ km}$) :-
- (A) 8 km/sec (B) $2\sqrt{6}$ km/sec (C) 2 km/sec (D) 4 km/sec

6. A planet moves around the sun in an elliptical orbit as shown.

Eccentricity of ellipse is $\frac{1}{2}$. Time taken by planet to move from D to B ($D \rightarrow A \rightarrow B$) and B to D ($B \rightarrow C \rightarrow D$) are respectively T_{DAB} and T_{BCD}



(A) $\frac{T_{DAB}}{T_{BCD}} = \frac{\pi + 2}{\pi - 2}$

(B) $\frac{T_{DAB}}{T_{BCD}} = \frac{2\pi + 1}{2\pi - 1}$

(C) Velocity of planet of point B is $\sqrt{3}V_2$

(D) Velocity of planet of point B is $\sqrt{2}V_2$

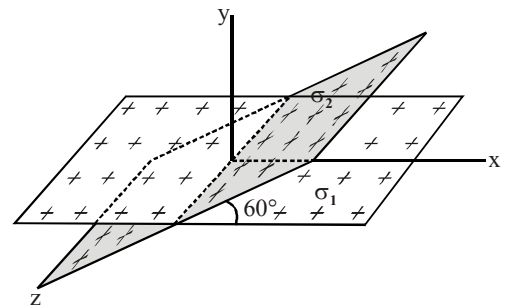
Multiple Correct Answer Type

4 Q. [4 M (-1)]

7. A ring carries a linear charge density on one half and the linear charge density of same magnitude but opposite sign on the other half.

- (A) the component of electric field along the axis of ring at all points on the axis is zero.
(B) the component of electric field along the axis of ring at point on the axis is zero only at the centre.
(C) the resultant field at the centre is zero.
(D) the electric field at all points on the axis of ring is perpendicular to axis.

8. An infinite plane in the xz plane carries a uniform surface charge density $\sigma_1 = 8.85 \text{ nC/m}^2$. A second infinite plane carrying a uniform charge density $\sigma_2 = 17.7 \text{ nC/m}^2$ intersects the xz plane at the z axis and makes an angle of 60° with the xz plane as shown in figure. The electric field in the xy plane.

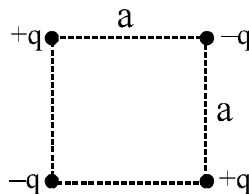


- (A) at $x = 6\text{m}$, $y = 2\text{m}$ is $500\sqrt{3} \text{ N/C}$
(B) at $x = -5\text{m}$, $y = 0$ is $500\sqrt{3} \text{ N/C}$
(C) at $x = 2\text{m}$, $y = 6\text{m}$ is $500\sqrt{7} \text{ N/C}$
(D) at $x = -1\text{m}$, $y = -1\text{m}$ is $500\sqrt{7} \text{ N/C}$

9. Two very thin, long, insulating rods, each carrying uniform linear charge density λ , lie in perpendicular directions at a distance d from each other. The force of repulsion between them is F . Which of the following statements is/are correct ?

- (A) $F \propto \lambda^2$ (B) $F \propto \frac{1}{d^2}$ (C) $F \propto \frac{1}{d}$ (D) F is independent of d .

10. In a system of two dipoles placed in the way as shown in figure:



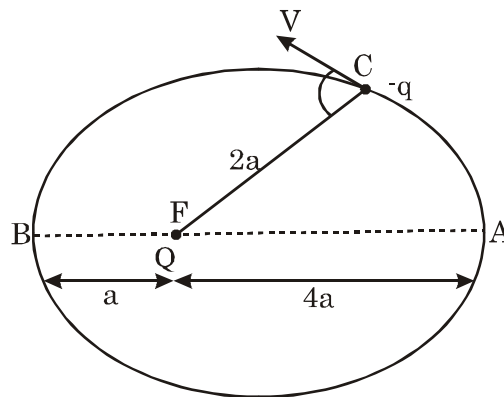
- (A) It is possible to consider a spherical surface of radius a and whose centre lies within the square shown, through which total flux is +ve.
(B) It is possible to consider a spherical surface of radius a and whose centre lies within the square shown through which total flux is -ve
(C) There are two points within the square at which EF is zero.
(D) It is possible to consider a spherical surface of radius a and whose centre lies within the square shown, through which total flux is zero.

Linked Comprehension Type
(Single Correct Answer Type)

(2 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 11 & 12

Consider a situation in which a negatively charged particle of mass M and charge $-q$ revolves in elliptical path around a fixed charge Q . The closest and farthest distance of the moving particle from fixed charge is a and $4a$ respectively. Point A and B are the two extreme points of the major axis of ellipse as shown in the figure. Charge Q is present at one of the Foci of the ellipse.



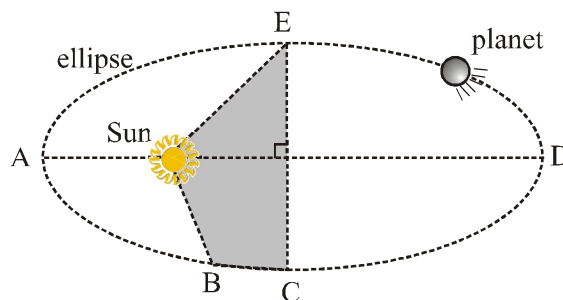
11. The speed (in SI units) of revolving particle when it reaches point A, will be? Use $\left(K = \frac{1}{4\pi\epsilon_0}\right)$ and

$$\left(\frac{KQq}{Ma} = 1000 \text{ S.I units}\right)$$

- (A) 10 (B) 20 (C) 30 (D) 40
12. The radius of curvature of path (in SI units) of revolving particle at point B, will be ?
Use $[a = 25 \text{ (S.I units)}]$
- (A) 40 (B) 30 (C) 20 (D) 10

Paragraph for Question no. 13 and 14

In the picture below, the planet orbits around the sun with a period of 40 months and takes 12 months to translate from the point D to point E and 1 month from point B to C. The area of the ellipse is A . Mass of the satellite is m .



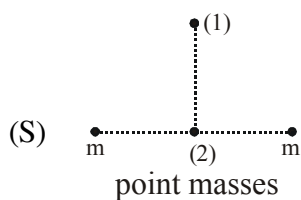
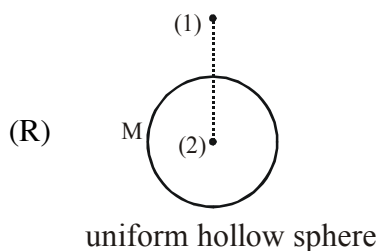
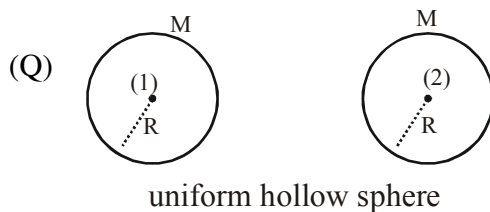
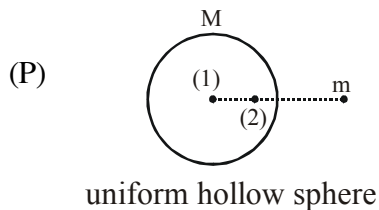
13. The area of the shaded region
- (A) $A/20$ (B) $A/8$ (C) $A/40$ (D) $A/4$
14. The angular momentum of planet per unit mass per unit area is :-
- (A) $5 \times 10^{-8} \text{ /s}$ (B) $16 \times 10^{-8} \text{ /s}$ (C) $2 \times 10^{-8} \text{ /s}$ (D) 0

Matching List Type (4×4)

2 Q. [3 M (-1)]

15. Assuming reference position for gravitational potential is at ∞ . Consider the situation of List-I and match them with the conditions in List-II.

List-I



List-II

- (1) Gravitational potential at position (1) & (2) is same
- (2) Gravitational potential at position (1) is less than at position (2)
- (3) magnitude of gravitational field at position (1) is less than at position (2)
- (4) magnitude of gravitational field at position (1) is more than at position (2)

Code :

	P	Q	R	S
(A)	3	1	2	4
(B)	3	2	4	4
(C)	3	1	4	4
(D)	2	1	3	4

16. In List-I certain situations are indicated where graph can be plotted. List-II lists the comments on the modulus of physical quantity on y-axis. If the modulus of physical quantity on x-axis is increasing, then match with List-II.

List-I

- (P) y-axis → Speed
x-axis → Distance from centre of earth
Situation : A particle projected with a speed

$$\text{of } \frac{3}{2} \sqrt{\frac{GM_{\text{earth}}}{R_{\text{earth}}}} \text{ from earth vertically}$$

- (Q) y-axis → Electric field
x-axis → Distance from centre
Situation : A uniformly charged solid sphere

- (R) y-axis → Temperature
x-axis → Pressure
Situation : Isochoric process

- (S) y-axis → Time period
x-axis → Mass
Situation : Simple pendulum

List-II

- (1) Graph first increase & then decrease
(2) Continuously decreasing graph
(3) Graph will be parallel to x-axis
(4) Continuously increasing graph

Code :

	P	Q	R	S
(A)	2	4	3	1
(B)	3	1	2	4
(C)	2	1	4	3
(D)	2	3	4	1

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. Binary stars of comparable masses m_1 and m_2 rotate under the influence of each other's gravity with angular velocity ω . If they are stopped suddenly in their motions, their relative velocity when they

collide with each other is $\left[\frac{2G(m_1 + m_2)}{(R_1 + R_2)} - 2(G^2(m_1 + m_2)^2 \omega^2)^\alpha \right]^\beta$ where R_1 and R_2 are radii of stars and

G is the universal gravitational constant. Write down the value of $\left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$

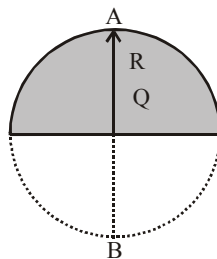
2. In some hypothetical space the gravitational field intensity is given by $\vec{g} = -2\vec{r}$, where $\vec{r}$ is position vector. If x-z plane is a rigid elastic plane and a particle of 1 kg mass is released from (1, 1, 0) position, the magnitude of change in momentum of the particle during its first eventual elastic collision with plane, is $n\sqrt{2}$ Nm. Find n .

SECTION-I

Single Correct Answer Type

6 Q. [3 M (–1)]

- A crew of scientists has built a new space station. The space station is shaped like a wheel of radius R , with essentially all its mass M at the rim. When the crew arrives, the station will be set rotating at a rate that causes an object at the rim to have radial acceleration g , thereby simulating Earth's surface gravity. This is accomplished by two small rockets, each with thrust T newtons, mounted on the station's rim. How long a time t does one need to fire the rockets to achieve the desired condition?
 (A) $t = \frac{\sqrt{gR^3 M}}{(2T)}$ (B) $t = \frac{\sqrt{gRM}}{(2T)}$ (C) $t = \sqrt{\frac{gR}{\pi}} \frac{M}{T}$ (D) $t = \frac{\sqrt{gRM}}{(\pi T)}$
- A satellite is launched in the equatorial plane in such a way that it can transmit signals upto 60° latitude on the earth. Then the angular velocity of the satellite is :
 (A) $\sqrt{\frac{GM}{8R^3}}$ (B) $\sqrt{\frac{GM}{2R^3}}$ (C) $\sqrt{\frac{GM}{4R^3}}$ (D) $\sqrt{\frac{3\sqrt{3}GM}{8R^3}}$
- The orbital properties of a geostationary satellite include (i) its orbit is directly over the Earth equator, and (ii) its orbital period is the same as the Earth's rotation period. Suppose a lunar stationary satellite (its orbital period is the same as the Moon's rotation period) is placed over the Moon equator, what is the value of the ratio $(R_{\text{Earth}} + H_{\text{Earth}})/(R_{\text{Moon}} + H_{\text{Moon}})$? R_{Earth} and R_{Moon} are the Earth and the Moon radii, H_{Earth} and H_{Moon} are the satellite heights from the Earth and from the Moon surfaces, respectively. You may assume (Earth mass)/(Moon mass) = 81 and rotation period of the Moon = 27 days.
 (A) $(1/9)^{1/3}$ (B) $(81 \times 27)^2$ (C) $(81/27)^3$ (D) $(1/9)^3$
- Q charge given to a uniform hemispherical charged distribution of radius ' R '. Then sum of potential at two diametrically opposite points ($V_A + V_B$) is equal to :-



- (A) $\frac{KQ}{R}$ (B) $\frac{KQ}{2R}$ (C) $\frac{2KQ}{R}$ (D) $\frac{KQ}{3R}$
- A charge q is uniformly distributed on the surface of a disc of radius R . Then a circular hole of radius $R/2$ is punched taking one of the radii as the diameter of the hole. Calculate the potential V due to remaining portion of the disc at the center of the disc. $\left[V_0 = \frac{q}{2\pi\epsilon_0 R} \left(1 - \frac{1}{\pi} \right) \right]$
 (A) $V = 2V_0$ (B) $V = V_0$ (C) $V = 3V_0$ (D) $V = \frac{V_0}{2}$
- A spherical shell of radius R carries a uniformly distributed charge q . The electrical forces arising cause the expansion of the shell. The mechanical stress in the shell is
 (A) $\frac{q^2}{32\pi^2\epsilon_0 R^4}$ (B) $\frac{q^2}{16\pi^2\epsilon_0 R^4}$ (C) $\frac{q^2}{4\pi^2\epsilon_0 R^4}$ (D) $\frac{q^2}{2\pi^2\epsilon_0 R^4}$

Multiple Correct Answer Type

3 Q. [4 M (-1)]

7. At the equator on a spherical planet a body weighs one third of that at the pole. Radius of planet is R . The time period of revolution of the planet around its axis is equal to $T = 60$ min.

(Take : $G = \frac{20}{3} \times 10^{-11} \text{ Nm}^2/\text{kg}^2$)

(A) The average density ρ of the planet is $\frac{\pi}{192} \times 10^6 \text{ kg/m}^3$

(B) The height of geostationary satellite from the surface of planet is $\left(\sqrt[3]{\frac{3}{2}} - 1\right)R$

(C) At a latitude of 60° , the weight of the body is half the weight at pole

(D) The geostationary satellite can be in a plane passing through the poles.

8. Two Earth's satellites move in a common plane along circular orbits. The orbital radius of one satellite is r while that of the other satellite is $r - \Delta r$ (Here $\Delta r \ll r$).

(A) Time interval separating the periodic approaches of the satellites to each other over the minimum

distance is $\frac{4\pi r^{5/2}}{3(GM)^{1/2} \Delta r}$.

(B) Time interval separating the periodic approaches of the satellites to each other over the minimum

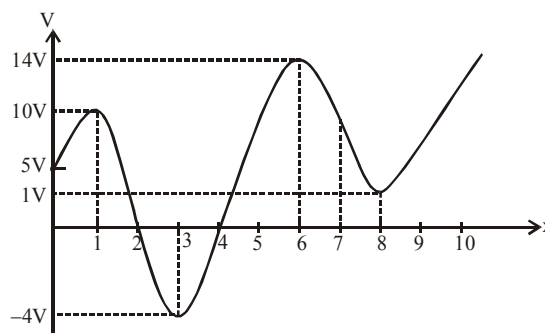
distance is $\frac{2\pi r^{5/2}}{3(GM)^{1/2} \Delta r}$.

(C) Angular velocity of approach between two satellites is $\frac{3(GM)^{1/2} \Delta r}{r^{5/2}}$.

(D) Angular velocity of approach between two satellites is $\frac{3(GM)^{1/2} \Delta r}{2r^{5/2}}$.

9. Given figure is a graph of the electric potential in a region of gravity free space where electric field is parallel to the x-axis. A charge q of mass m is projected from origin along positive x-axis with speed v_0

then $\left(\frac{q}{m} = 2\right)$



(A) minimum velocity for which charge can pass through $x = 8$ is 6 m/s

(B) minimum velocity for which charge can pass through $x = 8$ is 4 m/s

(C) if $v_0 > \sqrt{40}$ m/s particle pass through $x = 4$

(D) if $v_0 = 8$ m/s then maximum speed of particle during its forward motion is equal to 10 m/s

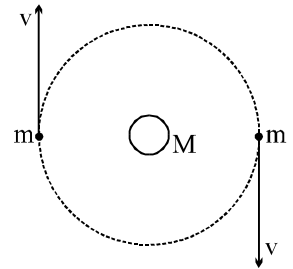
Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for question nos. 10 to 12

A triple star system consists of two stars, each of mass m , in the same circular orbit about central star with mass $M = 2 \times 10^{30}$ kg. The two outer stars always lie at opposite ends of a diameter of their common circular orbit. The radius of the circular orbit is $r = 10^{11}$ m and the orbital period of each star is

1.6×10^7 s. [Take $\pi^2 = 10$ and $G = \frac{20}{3} \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$]



10. The mass m of the outer star is

- (A) $\frac{16}{15} \times 10^{30}$ kg (B) $\frac{11}{8} \times 10^{30}$ kg (C) $\frac{15}{16} \times 10^{30}$ kg (D) $\frac{8}{11} \times 10^{30}$ kg

11. The orbital velocity of each star is

- (A) $\frac{5}{4}\sqrt{10} \times 10^3$ m/s (B) $\frac{5}{4}\sqrt{10} \times 10^5$ m/s (C) $\frac{5}{4}\sqrt{10} \times 10^2$ m/s (D) $\frac{5}{4}\sqrt{10} \times 10^4$ m/s

12. The total mechanical energy of the system is

- (A) $-\frac{1375}{64} \times 10^{35}$ J (B) $-\frac{1375}{64} \times 10^{38}$ J (C) $-\frac{1375}{64} \times 10^{34}$ J (D) $-\frac{1375}{64} \times 10^{37}$ J

Matching List Type (4 × 4)

1 Q. [3 M (-1)]

13. A satellite is to be launched and established in its orbit around the earth. For this purpose it is raised to a certain height (equal to its orbit distance) with a rocket & held stationary by some mechanism and a explosion is carried out instantly to provide the required velocity to the satellite. Let the original mass of satellite is M_s , mass of the earth M_e , radius of the required orbit R and universal gravitational constant G . As a result of explosion the satellite breaks into two parts of masses m_1 & m_2 and energy released is given by E . List-I gives certain combination of m_1 , m_2 & E while list-II gives the nature of orbit followed by m_2 . Match them suitably (If $v > v_e$; orbit is hyperbolic, $v = v_e$; orbit is parabolic and $v < v_e$; orbit is elliptical where v_e is escape velocity and v is velocity of the part) :-

List-I

List-II

(P) $m_1 = m_2$; $E = \frac{GM_e M_s}{R}$

(1) Parabolic orbit

(Q) $m_1 = 2 m_2$; $E = \frac{GM_e M_s}{4R}$

(2) Hyperbolic orbit

(R) $m_1 = 3.5 m_2$; $E = \frac{GM_e M_s}{4R}$

(3) Circular orbit

(S) $m_1 = \frac{1}{3} m_2$; $E = \frac{2GM_e M_s}{R}$

(4) Elliptical orbit

Codes :

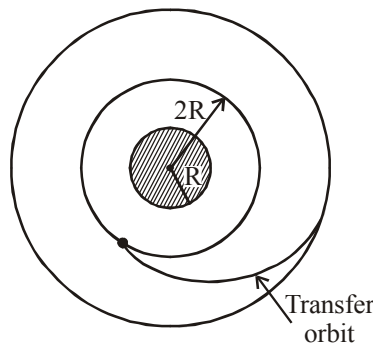
	P	Q	R	S
(A)	1	3	2	4
(B)	3	2	4	1
(C)	4	1	3	2
(D)	2	4	1	3

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

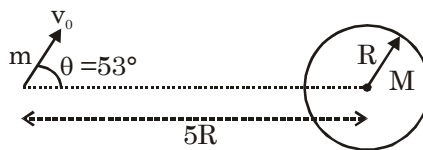
4 Q. [4 M (0)]

- On a planet which has the same density as that of earth, we construct a frictionless inclined plane of inclination 53° with the horizontal. When a block is released on the inclined plane, it is seen to accelerate at 1 m/s^2 . What is the radius of planet (in 10^5 m) ? (Take radius of Earth = 6400 km)
- An artificial satellite of mass m orbiting the earth in a circular orbit of radius twice the earth's radius R . It is to be taken in another circular orbit of radius $4R$. The transfer is accomplished through an elliptical orbit as shown in the figure. The energy needed to complete the job is calculated to be equal to $\frac{1}{k} \frac{GMm}{R}$ where M is the mass of the earth. Find the value of k .



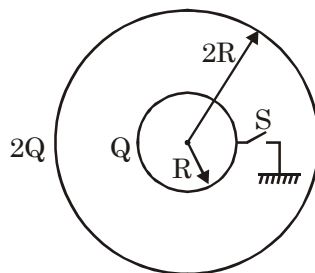
- A spaceship is sent to investigate a planet of mass M and radius R . While hanging motionless in space at a distance $5R$ from the centre of the planet, the spaceship fires an instrument package with speed v_0 as shown in the figure. The package has mass m , which is much smaller than the mass of the spaceship.

The package just grazes the surface of the planet and at that moment its speed is v . $v_0 = \sqrt{\frac{xGM}{75R}}$. Fill x in OMR sheet.



- Two conducting spherical shells of radii R & $2R$ given charges Q and $2Q$ respectively. Inner shell is provided with a switch which can ground the inner shell, as shown. Switch is initially open and energy stored in the system is U_1 . After the switch is closed, energy stored in the system is found to be U_2 . Find

$$\frac{U_1}{U_2}.$$



SECTION-IV

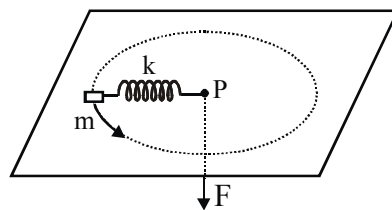
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Some situations are given in column-I while column II describes effect on different parameters.

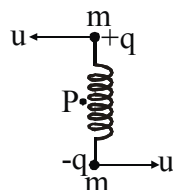
Column I

- (A) One end of an ideal spring is attached to a particle of mass 'm' and the other end is attached to a string as shown in fig. The mass m is kept on a smooth horizontal table and is provided velocity 'v' perpendicular to the spring. The string passes through a hole a point P and is being pulled vertically down. In this process

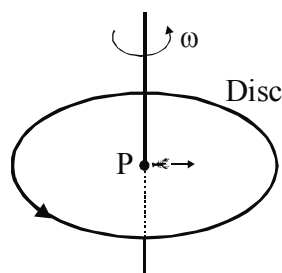


Column II

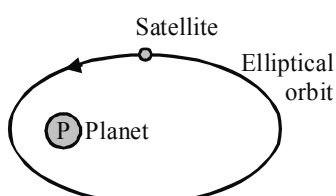
- (P) Mechanical energy is constant.
- (Q) Mechanical energy is variable.
- (R) Kinetic energy is variable.
- (S) Angular momentum about point P is conserved.
- (T) Kinetic energy is constant.
- (B) Identical and oppositely charged particles attached with a non-conducting spring are placed on a smooth horizontal surface. They are given an initial velocity as shown in figure. In this process



- (C) A disc is rotating in horizontal plane. An insect having some mass moves from centre to the circumference as shown in figure. In this process



- (D) A satellite is revolving around planet in an elliptical orbit. In this process



SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

1. A satellite of mass m is orbiting the earth in a circular orbit of radius r . It starts losing energy slowly at a constant rate k due to friction. M_e and R_e denote the mass and radius of the earth respectively, the satellite takes a limit time t to fall on the earth. The radius r is :-

(A) $\frac{GmM_e R_e}{Gm M_e - 4kR_e t}$ (B) $\frac{GmM_e R_e}{2GmM_e + 2kR_e t}$ (C) $\frac{GmM_e R_e}{GmM_e + 2kR_e t}$ (D) $\frac{GmM_e R_e}{Gm M_e - 2kR_e t}$

2. A small satellite of mass ' m ' is revolving around earth in a circular orbit of radius r_0 with speed v_0 . At certain point of its orbit, the direction of motion of satellite is suddenly changed by angle $\theta = \cos^{-1}(3/5)$ by turning its velocity vector in the same plane of motion, such that speed remains constant. The satellite, consequently goes to elliptical orbit around earth. The ratio of speed at perigee to speed at apogee is :-

(A) 3 (B) 9 (C) 1/3 (D) 1/9

3. A satellite of mass m is at a distance of ' a ' from a star of mass M . The speed of the satellite is u . Suppose the law of universal gravity is $F = -G \frac{Mm}{r^{2.1}}$ instead of $F = -G \frac{Mm}{r^2}$, find the speed of the satellite when it is at a distance b from the star.

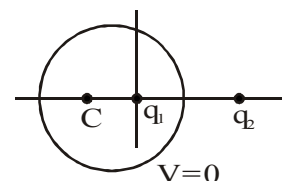
(A) $\sqrt{u^2 + 2GM \left(\frac{1}{b^{1.1}} - \frac{1}{a^{1.1}} \right)}$ (B) $\sqrt{u^2 + \frac{2}{1.1} GM \left(\frac{1}{b^{1.1}} - \frac{1}{a^{1.1}} \right)}$
(C) $\sqrt{u^2 + \frac{2}{2.1} GM \left(\frac{1}{b^{1.1}} - \frac{1}{a^{1.1}} \right)}$ (D) $\sqrt{u^2 + \frac{2}{1.1} GM \left(\frac{1}{b} - \frac{1}{a} \right)}$

4. A planet of small mass m moves around sun of mass M along an elliptical orbit such that its minimum and maximum distance from sun are r and R respectively. Its period of revolution will be

(A) $2\pi \sqrt{\frac{(r+R)^2}{6GM}}$ (B) $2\pi \sqrt{\frac{(r+R)^3}{3GM}}$ (C) $\pi \sqrt{\frac{(r+R)^3}{2GM}}$ (D) $2\pi \sqrt{\frac{(r+R)^3}{GM}}$

5. A point charge $q_1 = +6e$ is fixed at origin. Another point charge $q_2 = -10e$ is fixed at $x = 8$ nm, $y = 0$. The locus of all points in xy plane for which potential $V = 0$ is a circle centered on X axis as shown, find x coordinate of center of circle.

(A) -2 nm (B) -3 nm
(C) -4.5 nm (D) -7.5 nm



Multiple Correct Answer Type

5 Q. [4 M (-1)]

6. A body of mass M and a body of mass $m \ll M$ are in circular orbits about their center of mass under the influence of their mutual gravitational attraction. The distance between the bodies is R , which is much larger than the size of either body.

A small amount of matter $\delta m (\ll m)$ is removed from the body of mass m and transferred to the body of mass M . The transfer is done in such a way so that the orbits of the two bodies remain circular, and remain separated by a distance R . Which of the following statements is correct?

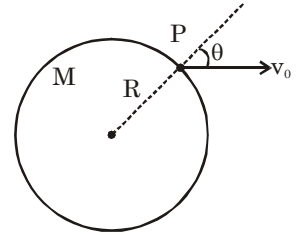
- (A) The gravitational force between the two bodies increases.
(B) The gravitational force between the two bodies remains constant.
(C) The total angular momentum of the system increases.
(D) The period of the orbit of two bodies remains constant.

7. Consider a planet of mass M and radius R as shown in the figure. A particle is projected from the surface of the planet at an angle θ from radial direction and with velocity equal to the orbital velocity of circular orbit at that point then choose the **CORRECT** statement(s) :

(A) Maximum distance of particle from centre of the planet is $(R + R \cos \theta)$
(B) Path of the particle is parabola

(C) Time taken by the particle to return to the surface of planet is $\sqrt{\frac{R^3}{GM}}$

(D) Time taken by the particle to reach the surface of planet is equal to $\pi \sqrt{\frac{R^3}{GM}}$



8. The electric field in x - y plane is given by $\vec{E} = x^2 \hat{i} + y^2 \hat{j}$. Potential at point $(3, 3)$ is taken as zero. Then :-

(A) Work done in moving a test charge q slowly from a point (a, b) to a point (b, a) is zero.
(B) Potential at any point on x - y plane can't exceed 18 unit for x & y in 1st quadrant.
(C) Potential can't be defined as field is non- conservative in nature.
(D) Equipotential curve are concentric circles with centre $(3, 3)$

9. In free space, two particles, first of mass 3×10^{-11} kg and of charge 10^{-10} C and second of mass 6×10^{-11} kg and charge 3×10^{-10} C move at speeds such that distance $d = 1.5$ cm between them remains constant. Consider only electric interaction between particles :

(A) Ratio of speeds of first particle to that of second particle is $\frac{v_1}{v_2} = \frac{1}{2}$

(B) Ratio of speeds of first particle to that of second particle is $\frac{v_1}{v_2} = 2$

(C) Speed of first particle is 20 m/s

(D) Speed of second particle is 0.2 m/s

10. Two spherical bodies of masses M and $2M$ and radii R and $2R$, respectively, start approaching each other at time $t = 0$ from rest, due to mutual gravitational attraction. They were initially very far away from each other. They collide at time $t = T$.

(A) Distance travelled by their COM till time $t = T$ is zero (COM—centre of mass)

(B) Their relative velocity at time $t = T$ is $\frac{1}{3} \sqrt{\frac{2GM}{R}}$

(C) PE of the system decreases as the bodies approach each other

(D) In COM frame, speeds of the bodies are always equal

Linked Comprehension Type

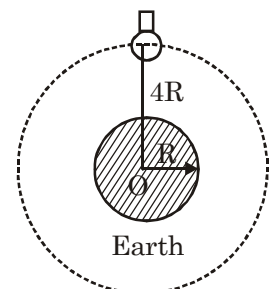
(2 Para $\times$ 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question No. 11 and 12

A canon of mass m is revolving around earth in circular orbit of radius $4R$. A shell of mass m is also equipped in canon and moving with in same orbit. At a particular instant, shell is fired from cannon in radial direction (direction directly away from the center of earth) w.r.t. canon. Kinetic energy of shell

just after firing is $\frac{3}{2}$ times of kinetic energy of shell before firing.



11. What is maximum distance of shell from center of earth :-

(A) $4R + 2\sqrt{2} R$

(B) $8R + 4\sqrt{2} R$

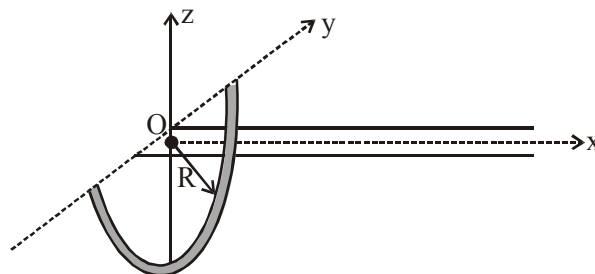
(C) $8R + 2\sqrt{2} R$

(D) $6R + 3\sqrt{2} R$

12. Minimum distances of shell and canon from center of earth are r_1 and r_2 then value of $\left(\frac{4r_1}{\sqrt{2}r_2}\right)^2$:-
 (A) 4 (B) 16 (C) 8 (D) 12

Paragraph for Question nos. 13 & 14

There is a fixed semicircular ring of radius R lying in y - z plane, with centre of arc at origin and it is uniformly charged with charge $+Q$. There is an insulated long hollow smooth pipe of very small radius fixed along x -axis from origin O as shown in figure. A small ball with charge $+q$ and mass m is projected from O in pipe with negligible velocity, ball can smoothly move in pipe. Whole arrangement lies in gravity free space.



13. The maximum acceleration of ball in pipe is :
 (A) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{mR^2}$ (B) $\frac{1}{12\sqrt{3}\pi\epsilon_0} \frac{Qq}{mR^2}$ (C) $\frac{1}{6\sqrt{3}\pi\epsilon_0} \frac{Qq}{mR^2}$ (D) None of these
14. The kinetic energy of particle when its acceleration is maximum is
 (A) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{R} \left(1 - \sqrt{\frac{2}{3}}\right)$ (B) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{R} \left(\frac{1}{2}\right)$ (C) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{R} \left(\frac{1}{\sqrt{2}}\right)$ (D) None of these

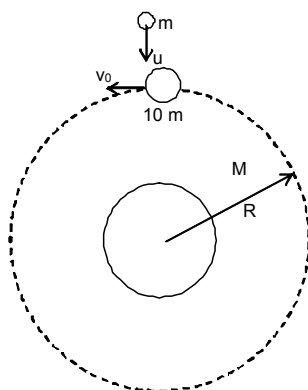
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

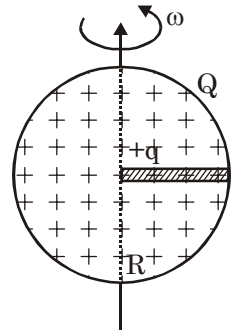
1. A meteorite of mass ' m ' strikes the satellite of mass $10m$ moving in circular path of radius ' R ' around planet of mass ' M ' ($\gg m$). Meteorite strikes perpendicular to the orbital velocity of satellite. The combined satellite and meteorite has minimum distance $R/2$ from planet's centre during subsequent motion. Velocity

u of meteorite just before the collision is $\sqrt{k \times 29 \frac{GM}{R}}$. Find the value of k .

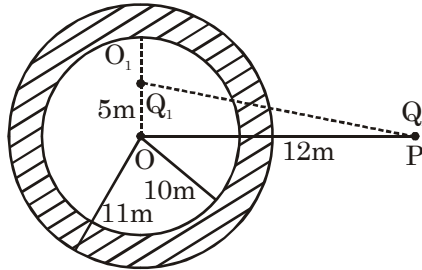


2. A trinary star system which is a system of three stars orbiting around centre of mass of system has time period of 3 years, while the distance between any two stars of the system is 2 astronomical unit. All the three stars are identical, mass of the sun is M and total mass of this multiple star system is $\frac{a}{b} \times M$. Then find $\frac{b}{3}$? (a and b are lowest possible integers) (1 astronomical unit is equal to distance between sun and earth, time period of rotation of earth around sun is 1 year)

3. A small ball of mass 1 kg and charge $\frac{2}{3} \mu\text{C}$ is placed at the centre of a uniformly charged sphere of radius 1 m and charge $\frac{1}{3} \text{mC}$. A narrow smooth groove is made in the sphere from centre to surface as shown in figure. The sphere is made to rotate about its vertical diameter at a constant rate of $\frac{1}{2\pi}$ revolutions per second. Find the speed w.r.t. ground (in m/s) with which the ball slides out from the groove. Neglect any magnetic force acting on ball.



4. A charge Q_1 is placed at O , inside a hollow conducting sphere having inner and outer radii as 10 m and 11 m as shown. The force experienced by Q_2 at P is F_1 and force experienced by Q_2 when Q_1 is placed at O_1 is F_2 . Then F_1/F_2 is equal to



5. Three concentric, thin, spherical, metallic shells have radii 1 cm, 2 cm and 4 cm and they are held at potentials 10 V, 0 V and 40 V respectively. Taking the origin at the common centre, calculate the potential (in V) at $r = 1.25$ cm.

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. A particle is projected vertically up from the surface of the earth with an speed $\sqrt{gR}$.

Column-I

- (A) Maximum height attained by the particle is αR then α is
(B) Time taken by the particle to reach the maximum height

is $\beta \sqrt{\frac{R^3}{GM}}$ then β is

- (C) Value of acceleration due to gravity at maximum height

is $\gamma \frac{GM}{4R^2}$ then γ is

- (D) Orbital velocity of a satellite revolving at maximum height

is $\delta \sqrt{\frac{GM}{2R}}$ then δ is

Column-II

(P) 2

(Q) $\frac{\pi}{4} + 1$

(R) $\frac{\pi}{2} + 1$

(S) 1

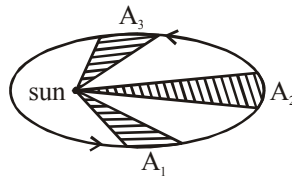
(T) None of these

SECTION-I

Single Correct Answer Type

4 Q. [3 M (-1)]

1. A planet moving around sun sweeps area A_1 in 2 days, A_2 in 3 days and A_3 in 6 days. Then the relation between A_1 , A_2 and A_3 is :-

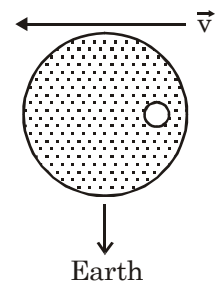


- (A) $3A_1 = 2A_2 = A_3$ (B) $2A_1 = 3A_2 = 6A_3$ (C) $3A_1 = 2A_2 = 6A_3$ (D) $6A_1 = 3A_2 = 2A_3$
2. A system consists of two stars of equal masses that revolve in a circular orbit about centre of mass midway between them. Orbital speed of each star is v & period is T . Find the mass M of each star (G is gravitational constant)

- (A) $\frac{2Gv^3}{\pi T}$ (B) $\frac{v^3 T}{\pi G}$ (C) $\frac{v^3 T}{2\pi G}$ (D) $\frac{2Tv^3}{\pi G}$

3. A small air bubble is inside a drop of water residing in a space station on an orbit around Earth. The direction to Earth is downwards and the space station is moving to the left relative to Earth, as shown. The air bubble will _____ relative to the water drop.

- (A) move to the left
(B) move to the right
(C) move up
(D) not move



4. Two satellites are launched at a distance R from a planet of negligible radius. Both satellites are launched in the tangential direction. The first satellite launches correctly at a speed v_0 and enters a circular orbit.

The second satellite, however, is launched at a speed $\frac{1}{2}v_0$. What is the minimum distance between the second satellite and the planet over the course of its orbit?

- (A) $\frac{1}{2}R$ (B) $\frac{1}{3}R$ (C) $\frac{1}{4}R$ (D) $\frac{1}{7}R$

Multiple Correct Answer Type

6 Q. [4 M (-1)]

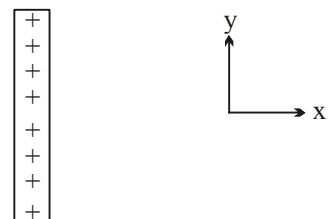
5. An insulating rod of uniform linear charge density λ and uniform linear mass density μ lies on a smooth table whose surface is xy -plane. A uniform electric field E is switched on.

- (A) If electric field is along x -axis, the speed of the rod when it has travelled

a distance d is $\sqrt{\frac{2\lambda Ed}{\mu}}$

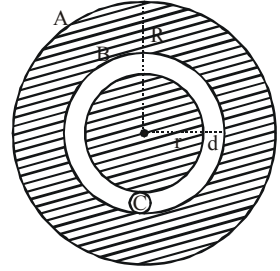
- (B) If electric field E is at an angle θ ($< 90^\circ$) with x -axis, the speed of the rod

when it has travelled a distance d is $\sqrt{\frac{2\lambda Ed \cos \theta}{\mu}}$



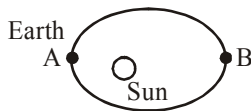
- (C) Torque on the rod due to the field about centre of mass in case B is into the plane of paper.
(D) Torque on the rod due to the field about centre of mass in case B is zero.

6. 'A' is a huge planet with uniform mass distribution, mass M , radius R . B is a circular tunnel made in it, concentric with planet. Radius of tunnel is r , ($r = \frac{R}{2}$) cross sectional diameter is d ($d \ll r$). C is a small ball of mass m ($m \ll M$) which is moving freely inside the tunnel without friction in a uniform circular motion, take acceleration due to gravity on surface of A as g :-



- (A) If contact force on C is double the gravitational force, its time period = $2\pi\sqrt{\frac{R}{3g}}$
- (B) If contact force on C is double the gravitational force, its time period = $2\pi\sqrt{\frac{R}{g}}$
- (C) If contact force is zero, time period T , radius of tunnel r , then $T \propto r$
- (D) If contact force is zero, time period T , radius of tunnel r , then T would be independent of r .
7. Because of the effects of air drag, abandoned satellites, at the end of their usual lives lose energy in the upper layers of atmosphere, before finally burning up when they reach the denser lower layers. It can be shown that satellites originally moving along circular trajectories will continue to travel in approximately circular orbits, with their orbital radii slowly decreasing. Drag force acting on a satellite can be expressed as $F_{\text{drag}} = C\rho v^2$ where ρ is density of air at the altitude of satellite and v is the speed of satellite, C is a constant. Mark the **CORRECT** statement(s) :-
- (A) During slow decrease in radius of orbit, gravitational force is not perpendicular to velocity of satellite
- (B) Speed of satellite decreases as it approaches earth due to tangential drag force.
- (C) If change in orbital radius is Δr during a single revolution then expression for density of air is $\rho = \frac{1}{4\pi C} \frac{m}{r^2} \Delta r$
- (D) If change in orbital radius is Δr during a single revolution then expression for density of air is $\rho = \frac{1}{2\pi C} \frac{m}{r^2} \Delta r$
8. An artificial satellite is in a circular orbit around the earth. The universal gravitational constant starts decreasing at time $t = 0$ at a constant rate with respect to time t . Then the satellite has its
- (A) Path gradually spiraling out away from the centre of the earth.
- (B) Path gradually spiraling in towards from the centre of the earth.
- (C) Angular momentum about the centre of the earth remains constant
- (D) Potential energy increases.
9. A planet moves round the Sun in an elliptical orbit such that its K.E. is k_1 and k_2 when it is nearest to the Sun and farthest from the Sun respectively.
- (A) If total energy of the planet is u , then ratio of largest distance (r_2) and smallest distance (r_1) between planet and the Sun is $\frac{u - k_1}{u - k_2}$.
- (B) If total energy of the planet is u , then ratio of largest distance (r_2) and smallest distance (r_1) between planet and the Sun is $\frac{u - k_2}{u - k_1}$.
- (C) if $r_2 = 2r_1$, the total energy of planet in terms of k_1 and k_2 is $2k_1 - k_2$
- (D) if $r_2 = 2r_1$, the total energy of planet in terms of k_1 and k_2 is $2k_2 - k_1$

10. The earth is moving around the sun in an elliptical orbit. Point A is the closest and point B is the farthest point in the orbit, as shown. In comparison to the situation when the earth passes through point B :



- (A) total energy of the earth-sun system is greater when the earth passes through point A.
- (B) gravitational potential energy of the earth-sun system is greater when the earth passes through point A.
- (C) kinetic energy of the earth due to the motion around the sun is greater when it passes through the point A.
- (D) magnitude of angular momentum of the earth about the sun is greater when the earth passes through point A.

Linked Comprehension Type
(Single Correct Answer Type)

(2 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 11 and 12

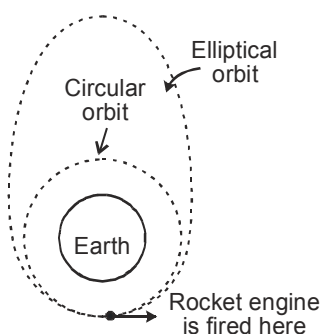
Changing from a circular to an elliptical orbit

Let us identify the system as the spacecraft and the Earth but not the portion of the fuel in the spacecraft that we use to change the orbit. In a given orbit, the mechanical energy of the spacecraft – Earth system is given by

$$E = - \frac{GMm}{2r}$$

This energy includes the kinetic energy of the spacecraft and the potential energy associated with the gravitational force between the spacecraft and the Earth. If the rocket engines are fired, the thrust force moves the spacecraft through a displacement. As a result, the mechanical energy of the spacecraft – Earth system increases.

The spacecraft has a new higher energy but is constrained to be in an orbit that includes the original starting point. It can not be in a higher energy circular orbit having a larger radius because this orbit would not contain the starting point. The only possibility is that the orbit is elliptical as shown in the figure.



$$E = - \frac{GMm}{2a}$$

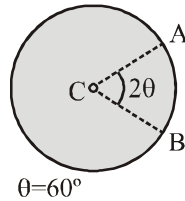
Above equation gives the energy of the spacecraft – Earth system for an elliptical orbit where a is semimajor axis. Thus if we know the new energy of the orbit, we can find the semi-major axis of the elliptical orbit. Conversely, if we know the semi-major axis of an elliptical orbit we would like to achieve, we can calculate how much additional energy is required from the rocket engines.

A spacecraft is moving in a circular orbit around the Earth (Radius 6400 km), at a height of 300 km from the surface. To place the spacecraft in an elliptical orbit, the magnitude of the mechanical energy of the spacecraft- Earth system is decreased by 10.0%.

11. If the spacecraft-earth system had initial energy $(-E_0)$, then the total mechanical energy of the system after firing the rocket will be :
 (A) $-1.1 E_0$ (B) $-0.9 E_0$ (C) $-E_0$ (D) None of these
12. Minimum distance of the spacecraft from the surface of the Earth is :
 (A) 900 km (B) 1474 km (C) 300 km (D) 1774 km

Paragraph for Question no. 13 and 14

A rocket is launched from point A on earth. It reaches point B on earth with its velocity antiparallel to its initial velocity. The angle subtended by straight line joining AB at the center is 2θ . The satellite moving close to surface of earth has time period of revolution T_0 . Radius of earth is R.



13. The time of flight of the rocket is :-
 (A) $\frac{T_0}{2}$ (B) $T_0 \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right)$ (C) $T_0 \left(\frac{1}{2} + \frac{1}{2\pi} \right)$ (D) $T_0 \left(\frac{1}{2} + \frac{\sqrt{3}}{2\pi} \right)$
14. Maximum height attained by the rocket is :-
 (A) R (B) $\frac{R}{2}$ (C) $\frac{R\sqrt{3}}{2}$ (D) $\sqrt{3}R$

SECTION-III

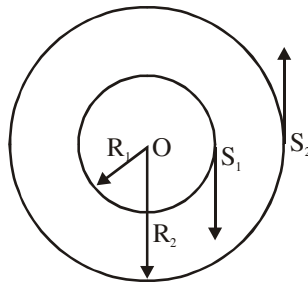
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

1. You are at a distance of $R = 1.5 \times 10^6$ m from the centre of an unknown planet. You notice that if you throw a ball horizontally it goes completely around the planet hitting you in the back 90,000 seconds later with exactly the same speed that you originally threw it. If the length of semi major axis of ball is $2R$, what is the mass of the planet. Express in scientific notation ($a \times 10^b$) kg and fill 'a' in OMR sheet.

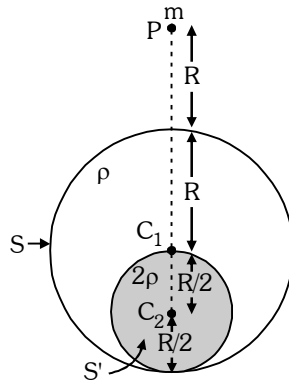
[Take $G = \frac{20}{3} \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $\pi^2 = 10$]

2. Two satellites S_1 and S_2 revolve around a planet in coplanar circular orbits in the opposite sense. The periods of revolutions are $T = \pi/3$ seconds and ηT , where η is equal to 8 respectively. Find the angular speed of S_2 (in rad/s) as observed by an astronaut in S_1 , when they are closest to each other.



3. A satellite of mass m is in a circular orbit around an airless spherical planet of radius R . An asteroid of equal mass m falls radially towards the planet, starting at zero velocity from a very large distance. The satellite and the asteroid collide inelastically and stick together, moving in a new orbit such that it just misses the planet's surface. The radius of the satellite's original circular orbit is $(\alpha + \sqrt{11})R$. Find the value of α .

4. A composite solid sphere S of radius R has density 2ρ in spherical region S' of radius $\frac{R}{2}$ while remaining region has density ρ . Gravitational force exerted by this composite body on a point mass (m) placed as shown in the figure is $\pi R \rho G m \left(\frac{K}{25} \right)$. Find K.



SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. Match the following :

For planet orbiting about sun in elliptical orbit, some incomplete statements regarding physical quantities are given in column-I, Which can be completed by using entries of column-II.

Column-I

- (A) Maximum potential energy of sun planet system
- (B) Maximum speed of planet
- (C) Minimum potential energy of sun planet system
- (D) Minimum KE of planet

Column-II

- (P) Is independent of semimajor axis of orbit
- (Q) Is at apogee
- (R) Is dependent on mass of planet
- (S) Is at perigee
- (T) depends upon mass of sun

SECTION-I

Single Correct Answer Type

6 Q. [3 M (-1)]

- Three point masses A, B, C each having mass m are kept on the x-axis at the points $(0, 0)$, $(a, 0)$ and $(2a, 0)$ respectively. Initially, they are all at rest. They start moving due to the gravitational force between themselves. Then the instantaneous acceleration of the centre of mass of the system consisting of (B + C) will be
 (A) $\frac{Gm}{a^2} \times \frac{4}{9}$ (B) $\frac{Gm}{a^2} \times \frac{8}{9}$ (C) $\frac{Gm}{a^2} \times \frac{5}{8}$ (D) zero
- A small area is removed from a uniform spherical shell of mass M and radius R . Then the gravitational field intensity near the hollow portion is :-
 (A) $\frac{GM}{R^2}$ (B) $\frac{GM}{2R^2}$ (C) $\frac{3GM}{2R^2}$ (D) 0
- Imagine a planet whose diameter and mass are one half of those of the earth. The day's temperature of this planet reaches up to 800K. (Escape velocity on the surface of the earth is 11.2 km/sec, $k = 1.38 \times 10^{-23} \text{ J/K}$). Consider that the molecules possess average kinetic energy. Among the following, choose the wrong statement
 (A) Oxygen molecules escape from the planet
 (B) Oxygen molecules cannot escape from the planet
 (C) Nitrogen molecules cannot escape from the planet
 (D) Argon molecules cannot escape from planet.
- In figure A, a stationary spacecraft of mass M is passed by asteroid A of mass m , asteroid B of the same mass m , and asteroid C of mass $2m$. The asteroids move along the indicated straight paths at the same speed; the perpendicular distances between the spacecraft and the paths are given as multiples of R . Figure B gives the gravitational potential energy $U(t)$ of the spacecraft-asteroid system during the passage of each asteroid treating time $t = 0$ as the moment when separation is minimum. Which asteroid corresponds to which plot of $U(t)$?

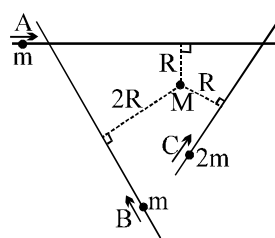


Figure A

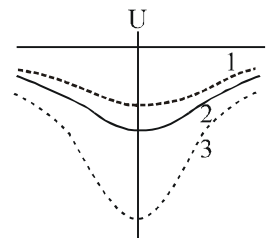


Figure B

- (A) A-1, B-2, C-3 (B) B-1, A-3, C-2 (C) B-1, A-2, C-3 (D) None
- A satellite of mass m orbits the earth in an elliptical orbit having aphelion distance r_a and perihelion distance r_p . The period of the orbit is T . The semi-major and semi-minor axes of the ellipse are $\frac{r_a + r_p}{2}$ and $\sqrt{r_p r_a}$ respectively. The angular momentum of the satellite is

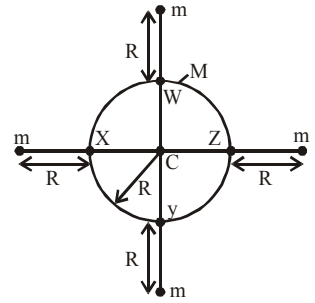
- $\frac{m\pi(r_a + r_p)\sqrt{r_a r_p}}{T}$ (B) $\frac{2m\pi(r_a + r_p)\sqrt{r_a r_p}}{T}$ (C) $\frac{m\pi(r_a + r_p)\sqrt{r_a r_p}}{2T}$ (D) $\frac{m\pi(r_a + r_p)\sqrt{r_a r_p}}{4T}$

6. If satellite is in an elliptical orbit around the earth. (The nearest position of satellite from earth is referred as perigee and farthest as apogee). Mark the correct statement
- (A) If satellite is to escape from the earth completely and if the satellite's rocket is fired at perigee it will require less kinetic energy
- (B) If satellite is to escape from the earth completely and if the satellite's rocket is fired at apogee then it will require less kinetic energy
- (C) Radius of curvature of orbit at apogee is more than at perigee
- (D) Instantaneous centripetal acceleration at apogee is more than at perigee

Multiple Correct Answer Type

4 Q. [4 M (-1)]

7. Four points W, X, Y and Z are on the surface of a hollow mass sphere. Four point masses each of mass m are located. Choose correct option(s) :
- (A) At centre C, field intensity is non zero.
- (B) At all four points W, X, Y and Z, field intensity vector are not equal
- (C) Potential is same for W, X, Y and Z.
- (D) Due to point mass, field can not exist inside the hollow mass sphere.

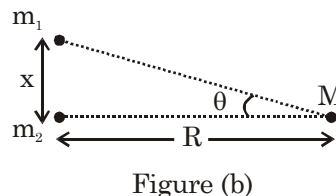
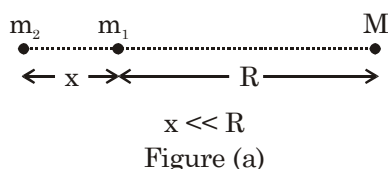


8. Mass density of a planet varies as $\rho(r) = \frac{\rho_0 r}{R}$; where R is radius of planet and r is distance from center of planet : -
- (A) A particle of mass m is to be projected from surface so as to escape the planet, its velocity of projection relative to surface is $\sqrt{2\pi G \rho_0 R^2}$.
- (B) A particle of mass m is to be projected from surface so as to escape the planet, its velocity of projection relative to surface is $2\sqrt{2\pi G \rho_0 R^2}$.
- (C) Gravitational intensity on surface of planet is $\pi G \rho_0 R$.
- (D) Time period of a satellite in circular orbit very close to planet is $2\pi \sqrt{\frac{R}{\pi G \rho_0}}$.

9. Two equal masses m are at a distance ℓ apart and interact via gravity. They are given proper tangential speed v_0 , so that they both travel in a circular of radius $\frac{\ell}{2}$ around their CM. If one of the masses is grabbed and held at rest, then the closest distance the other mass comes to it is r . Then :

(A) $v_0 = \sqrt{\frac{Gm}{2\ell}}$ (B) $v_0 = \sqrt{\frac{Gm}{\ell}}$ (C) $r = \frac{\ell}{3}$ (D) $r = 3\ell$

10. In figure (a) and (b) shown the mass M is very large and exerts a gravitational force on m_1 and m_2 . Gravitational force between m_1 and m_2 is very small as compared to that of M . Net force $\vec{F}_{\text{net}}$ on m_1 in accelerating reference frame of m_2 is called tidal force. If m_1 is to be at rest in reference frame of m_2 an additional force equal to tidal force is to be applied on m_1 . Mark the correct statement(s) using appropriate approximation :



$x \ll R$
 $\cos \theta \approx 1$
 $\sin \theta \approx \frac{x}{R}$

- (A) Tidal force on m_1 in figure (a) is given by $\frac{GMm_1x}{R^3}$
- (B) Tidal force on m_1 in figure (a) is given by $\frac{2GMm_1x}{R^3}$
- (C) Tidal force on m_1 in figure (b) is given by $\frac{GMm_1x}{R^3}$
- (D) Tidal force on m_1 in figure (b) is given by $\frac{2GMm_1x}{R^3}$

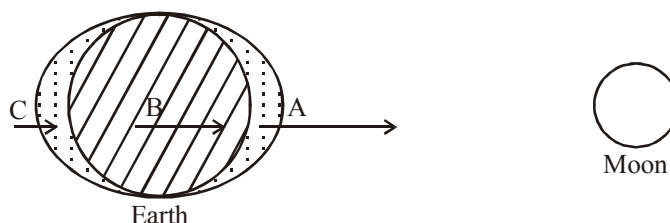
Linked Comprehension Type (1 Para × 2Q.) (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question no. 11 and 12

According to Newton's universal law of gravity, the gravitational field due to an object of mass M at a distance R is given by,

$$\vec{F} = -\frac{GM}{R^3} \vec{R}$$

Now the force hold an inverse square relation with the distance. This results in the gradient in the force field as we move away from the object. This means that your foot experiences a different force of gravity as compared to your head, however the effect is much smaller for the pair of object like earth and us but when compared for the moon and water in the ocean, effect of differential gravity becomes quite apparent and results in the occurrence of tides in the oceans. This can be illustrated in diagram as shown.



And for some of the massive objects, this differential gravity is much higher that it breaks the larger particle into smaller one. The distance at which the gravity alone can't hold the large object intact is called **Roche's limit** and any object within this radius can't be hold together only due to mutual gravitational attraction.

11. Based on the above comprehension choose the statement that doesn't result due to differential gravity:-
- (A) Ring of Saturn are composed of many small particles.
- (B) Spaghettification (elongation of object along its length) of chemically bonded object falling in black holes.
- (C) Flying off objects from the surface of very fast rotating stars.
- (D) Occurrence of low & high tides in the oceans on earth.
12. Which of the following star will have greatest tidal effect due to its own mass on its surface?
- (A) A star having mass M & radius R . (B) A star having mass $5M$ & radius $\frac{R}{2}$
- (C) A star having mass $5M$ and radius $2R$ (D) A star having mass $5M$ and radius $10R$

Paragraph for Question No. 13 to 15

A problem that has fascinated philosophers & scientists is how the universe began, if it had a beginning, and how long ago it happened. Much has been learned in the last few decades about the conditions of the early universe. Current thinking assumes that the universe began as a big-bang, at which time everything was condensed into a very small space, where there were both extremely high densities & exceptionally high temperatures (actually, it was beginning of space and time as we know them). Since that time the universe has been in continuous expansion, so that the average density & temperature have been decreasing continuously. The rate of expansion of the universe (that is, the rate at which galaxies are receding from each other) is given by Hubble's Law, discovered in 1929 by the astronomer Edwin Hubble, it states that the rate of separation of any two galaxies in the universe is directly proportional to their separation. Thus, if we have two galaxies a distance R apart, their receding relative speed is given by $v = HR$ where, H is a proportionality factor called as Hubble's parameter. The currently accepted value for H is : $22 \text{ kms}^{-1} \text{ MLyr}^{-1} = 2.32 \times 10^{-18} \text{ s}^{-1}$ where one ML_{yr} (mega light - year) is $9.46 \times 10^{21} \text{ m}$.

13. If we define a time t_H , which corresponds to the time when two galaxies have reached a separation of $2R$ from R , using Hubble's Law, then t_H is

(A) $t_H = \frac{1}{H}$ (B) $t_H = \frac{\ln(2)}{H}$ (C) $t_H = \frac{R}{H}$ (D) $t_H = \frac{H}{R} \ln(2)$

14. Some recent observations by Hubble Space Telescope has fascinated astronomers and baffled the scientists that even with the presence of gravitational attraction to retard the separation between the galaxies, it seems that they are accelerating which according to some scientists proves the existence of dark energy which is supplying the increasing kinetic energy to accelerate the expansion. To measure the amount of dark energy supply we need to calculate the rate at which separation velocity is increasing at different levels of separation, find the ratio of relative acceleration at separation of R and $2R$ respectively.

(A) 2 : 1 (B) 1 : 1 (C) 1 : 2 (D) 1 : 4

15. There is a distance called as the horizon of the universe, which is equal to the farthest distance that can be observed from earth, it corresponds to the time taken by light to reach earth almost very close to the age of universe. The oldest & farthest object observed from earth is near the constellation of ura major. It is a star like object called a quasar and is estimated to be at a distance of $1.3 \times 10^{26} \text{ m}$, therefore we see it as it was about how much years ago :

(A) 1.36×10^{10} years (B) 4.3×10^{17} years (C) 2.32×10^{18} years (D) 9.46×10^{21} years

Matching list Type ($4 \times 4 \times 4$) single option correct

(Three Columns and Four Rows)

1 Table $\times$ 3 Q. [4(-1)]

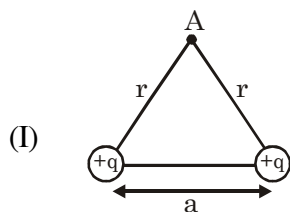
Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

Column-I represents the charges kept in gravity free space. Column-II represents the total interaction potential energy of the system and column-III represents the electric potential at point A. (Take potential at infinity to be zero).

Column-I
Point charge
of system

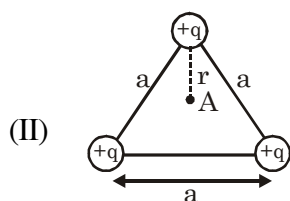
Column-II
Total interaction potential
energy of the system

Column-III
Electric potential
at point A



(i) 0

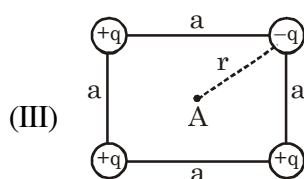
(P) 0



(ii) $\frac{kq^2}{a}$

(Q) $\frac{kq}{r}$

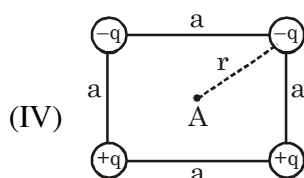
A : centroid of triangle



(iii) $\frac{-\sqrt{2}kq^2}{a}$

(R) $\frac{2kq}{r}$

A : centre of square



(iv) $\frac{-kq^2}{a}$

(S) $\frac{3kq}{r}$

A : centre of square

16. Which of the options are correctly matched :-

(A) (I) (ii) (P) (B) (I) (ii) (R) (C) (I) (iii) (Q) (D) (IV) (ii) (R)

17. Which of the options are correctly matched :-

(A) (II) (iii) (R) (B) (II) (i) (Q) (C) (II) (iv) (P) (D) (IV) (iii) (P)

18. Which of the options are correctly matched :-

(A) (III) (iii) (Q) (B) (III) (ii) (R) (C) (III) (i) (R) (D) (IV) (iv) (Q)

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. Six point masses of mass m each are at the vertices of a regular hexagon of side ℓ . The force on any one

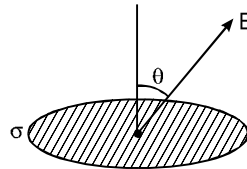
of the masses is given by $F = \frac{Gm^2}{\ell^2} \left(\frac{1}{x} + \frac{1}{\sqrt{y}} + \frac{1}{z} \right)$, then find $x + y - z$. (Take : $x < y < z$)

SECTION-I

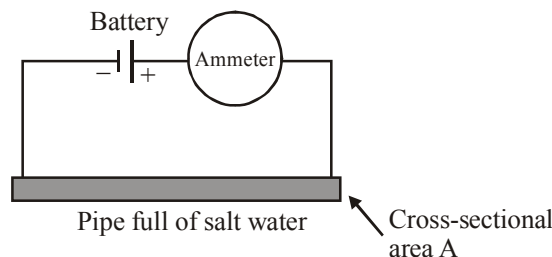
Single Correct Answer Type

6 Q. [3 M (-1)]

1. The electric field 'E' is uniform in the shaded area and makes an angle θ with the normal. The conductivity of the material in which the area 'S' lies is σ . The electric current flowing out of the shaded area is



- (A) $\sigma E S \cos\theta$ (B) $\sigma E S \cos^2\theta$ (C) $\sigma E S \sin\theta$ (D) $\sigma E S \sin^2\theta$
2. A hollow conducting sphere of inner radius R and outer radius 2R has resistivity ' ρ ' as a function of the distance 'r' from the centre of the sphere: $\rho = kr^2/R$. The inner and outer surfaces are painted with a perfectly conducting 'paint' and a potential difference ΔV is applied between the two surfaces. Then, as 'r' increases from R to 2R, the electric field inside the sphere :-
- (A) increases (B) decreases (C) remains constant (D) passes through a maxima
3. Salt water contains n sodium ions (Na^+) per cubic meter and n chloride ions (Cl^-) per cubic meter. A battery is connected to metal rods that dip into a narrow pipe full of salt water. The cross sectional area of the pipe is A. The magnitude of the drift velocity of the sodium ions is V_{Na} and the magnitude of the drift velocity of the chloride ions is V_{Cl} . Assume that $V_{\text{Na}} > V_{\text{Cl}}$ (+e is the charge of a proton). What is the magnitude of the ammeter reading?

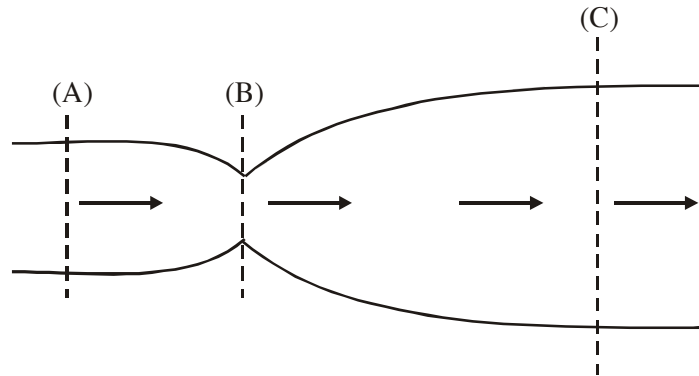


- (A) $enAV_{\text{Na}} - enAV_{\text{Cl}}$ (B) $enAV_{\text{Na}} + enAV_{\text{Cl}}$ (C) $enAV_{\text{Na}}$ (D) Zero
4. Length of a rod is ℓ and cross-section area A as shown in the figure. A steady current i flows through the rod. Number of free electrons (carrying current) per unit volume changes linearly from n_0 to $2n_0$ with distance x from left end to right end. Then average drift velocity of electrons (averaged over the distance travelled in the rod) will be
- (A) $\frac{2i}{3n_0eA}$ (B) $\frac{3i}{2n_0eA}$ (C) $\frac{i\ell n_2}{n_0eA}$ (D) $\frac{i}{n_0eA\ell n_2}$
5. Two substances are identical except that the electron mean free time for substance A is twice the electron mean free time for substance B. If the same electric field exists in both substances the electron drift speed in A is:
- (A) the same as in B (B) twice that in B (C) half that in B (D) four times that in B
(E) one-fourth that in B
6. Assuming nothing else blocks their view, approximately how far can two people stand from each other in clear atmosphere until they can not longer see each other due to the curvature of the Earth ?
[Take : radius of earth = 6400 km]
- (A) 10 m (B) 100 m (C) 1 km (D) 10 km

Multiple Correct Answer Type

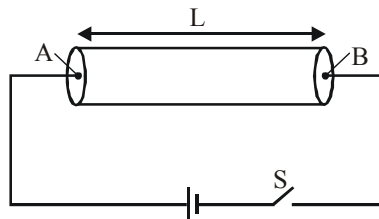
6 Q. [4 M (-1)]

7. Figure shows a copper wire of non-uniform cross section carrying current toward right. For sectors A, B and C.



- (A) $i_B > i_A > i_C$ (i = current) (B) $E_B > E_A > E_C$ (E = electric field)
(C) $V_B > V_A > V_C$ (V = drift velocity) (D) $J_B > J_A > J_C$ (J = current density)

8. Figure shows a battery, a conductor and a switch. After the switch is closed



- (A) The electric field is set up inside the conductor.

- (B) The time taken by the current to travel from A to B is $\frac{L}{v_d}$.

[v_d is drift speed of electrons]

- (C) The random motion of the electron stops and they start drifting to the left.

- (D) The random motion of the electron continues along with the drifting.

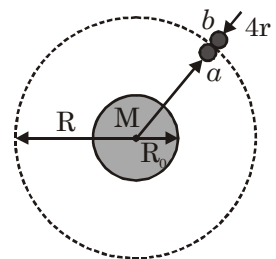
9. Two small spherical bodies 'a' & 'b' of mass m and radius ' r ' kept in contact. Both spherical body move along a circular orbit around the planet of mass M . They are kept in contact in such a way that the centers of the two bodies and the center of planet M always lies on the same line. Also, the point of contact of the two bodies is at a distance R from the center of the planet. The density of the blocks is assumed to be equal to the mean density of the planet. Assume $R_0 \gg r$:-

- (A) The angular velocity of 'a' is $\sqrt{\frac{2GM(R^2 + r^2)}{R(R^2 - r^2)^2}}$

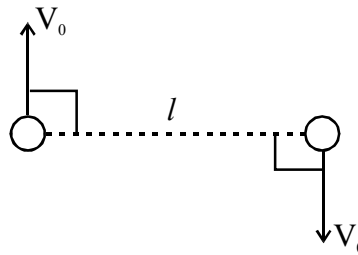
- (B) The contact force exerted by 'a' on 'b' is $\frac{3GmMr}{R^3}$

- (C) The contact force exerted by 'a' on 'b' is $\left[\frac{Gm^2}{4r^2} - \frac{GmM(3R^2r + r^3)}{R(R^2 - r^2)^2} \right]$

- (D) The value of ' R ' at which 'a' stops exerting force on 'b' is nearly $\sqrt[3]{12}R_0$



10. Two masses m separated by a distance l are given initial velocities V_0 , as shown in the diagram. The masses interact only through mutual universal gravitation. Choose **CORRECT** statement(s)



- (A) For non zero value of V_0 , particles will never collide
 (B) Minimum distance between particles can be l for a specific value of V_0
 (C) If $\frac{Gm}{V_0^2 l} > 2$, then minimum distance between particles is $d = \frac{l}{\frac{Gm}{V_0^2 l} - 1}$
 (D) If $\frac{Gm}{V_0^2 l} < 2$, then minimum distance between particles is $d = \frac{l}{\frac{Gm}{V_0^2 l} - 2}$
11. A satellite is moving in circular orbit of r with speed v_0 and time period T_1 . A particle of small mass is projected from satellite in direction of its motion with relative speed $\left(\frac{2}{\sqrt{3}} - 1\right)v_0$:-
 (A) Maximum distance of particle from center of earth is $2r$
 (B) Maximum distance of particle from center of earth is $(\sqrt{5} - 1)r$
 (C) Time period of particle around earth $> T_1$
 (D) Time period of particle around earth $< T_1$
12. A near earth orbit satellite of earth experiences very small air resistance. It's period of revolution decreases by 1.08 sec/day.
 (A) Its radius decreases by $\frac{160}{3}$ m in 1 day. (B) Its radius decreases by $\frac{160}{3}$ m in 1 revolution.
 (C) Its angular momentum decreases (D) Its mechanical energy increases.

Linked Comprehension Type

(1 Para $\times$ 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question nos. 13 to 15

A small insect crawls in the direction of electron drift along bare copper wire that carries a current of 2.56 A. It travels with the drift speed of the electron in the wire of uniform cross section area 1 mm^2 . Number of free electrons for copper = $8 \times 10^{22}/\text{cc}$ & resistivity of copper = $1.6 \times 10^{-8} \Omega \text{ m}$.

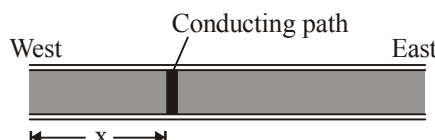
13. How much time would the insect take to crawl 1.0 cm if it crawls at the drift speed of the electrons in the wire?
 (A) 50 sec (B) 5 sec (C) 5000 sec (D) None of these
14. What is order of the average time of collision for free electrons of copper?
 (A) 10^{-14} sec (B) 10^{-17} sec (C) 10^{-11} sec (D) 10^{-8} sec
15. If the insect starts from the point of zero potential at $t = 0$, it reaches a point of ____ potential after 10 sec.
 (A) $80 \mu\text{V}$ (B) $-80 \mu\text{V}$ (C) $160 \mu\text{V}$ (D) $-160 \mu\text{V}$

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1 Q. [3(0)]

1. A 10-km-long underground cable extends east to west and consists of two parallel wires, each of which has resistance $13\Omega/\text{km}$. A short develops at distance x from the west end when a conducting path of resistance R connects the wires (figure). The resistance of the wires and the short is then 100Ω when the measurement is made from the east end, 200Ω when it is made from the west end. What is value of R (in ohm).



SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. A beam of fast moving electrons having cross-sectional area $A = 1 \text{ cm}^2$ falls normally on a flat surface. The electrons are absorbed by the surface and the average pressure exerted by the electrons on this surface is found to be $P = 9.1 \text{ Pa}$. If the electrons are moving with a speed $v = 8 \times 10^7 \text{ m/s}$, then find the effective current (in A) through any cross-section of the electron beam.
(mass of electron $= 9.1 \times 10^{-31} \text{ kg}$)

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. In column-I, certain situations are indicated whose graph can be plotted. Column-II lists the comments on the modulus of physical quantity on y-axis. If the modulus of physical quantity on x-axis is increasing then match with column-II.

Column-I

- (A) y-axis $\rightarrow$ Temperature
x-axis $\rightarrow$ Pressure
Situation : Adiabatic process on an ideal gas.

- (B) y-axis $\rightarrow$ Electric field
x-axis $\rightarrow$ Distance from centre along axis.
Situation: A uniformly charged ring.

- (C) y-axis $\rightarrow$ position of image w.r.t. lens
x-axis $\rightarrow$ position of real object w.r.t. lens
Situation : A point object kept on the optic axis of a converging lens.

- (D) y-axis $\rightarrow$ Speed
x-axis $\rightarrow$ distance from centre of earth
Situation : A particle projected with a

speed of $2\sqrt{\frac{GM_{\text{earth}}}{R_{\text{earth}}}}$ from earth vertically upwards.

Column-II

- (P) Graph passes through origin.

- (Q) Continuously increasing graph.

- (R) Continuously decreasing graph.

- (S) First increase and then decrease.

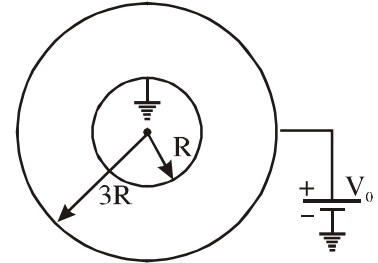
- (T) Decreases and then increasing.

SECTION-I

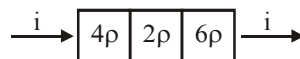
Single Correct Answer Type

7 Q. [3 M (-1)]

1. A conducting sphere with inner radius R and outer radius $3R$ has some alloys mixed in it due to which its resistivity changes with radial distance r according to the function $\rho_r = Kr$, where K is a +ve constant. Inner surface of the sphere is grounded while outer surface of sphere is maintained at potential V_0 with the help of a battery of emf V_0 . Battery is ideal. What will be the potential of the sphere at radial distance of $2R$

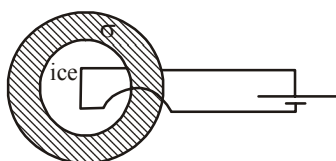


- (A) $V_0 \left(\frac{\ln 2}{\ln 3} \right)$ (B) $V_0 \left(\frac{\ln 3 - \ln 2}{\ln 3} \right)$ (C) $V_0 \left(\frac{\ln 3 - \ln 2}{\ln 2} \right)$ (D) $V_0 \left(\frac{\ln 2}{\ln 9} \right)$
2. A copper wire carries a current of 10 A. It has cross-sectional area of 0.05 cm^2 . Density of copper is 8.9 g/cm^3 . Molar mass of copper is $63.5 \times 10^{-3} \text{ kg/mol}$. The drift speed of the electrons is closest to (Assume one free electron per copper atom.)
(A) $1.5 \times 10^{-4} \text{ m/s}$ (B) $1.5 \times 10^4 \text{ m/s}$ (C) $3.5 \times 10^{-4} \text{ m/s}$ (D) 1.5 m/s
3. Two wires of same material and identical lengths are joined in parallel to a battery of constant voltage. If the radius of cross section of both the wires are in the ratio of 3:5 then the ratio of drift speed of electrons in the wire is given by :
(A) 9:25 (B) 5:3 (C) $\sqrt{3} : \sqrt{5}$ (D) None of these
4. Which of the following electric field pattern is correct inside the conductor having three materials of resistivity?



- (A) (B) (C) (D)

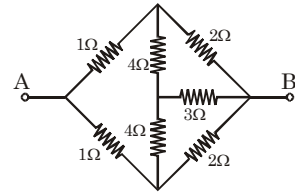
5. Figure shows a thick shell made of electrical conductivity σ and has inner & outer radii of 10 cm & 20 cm respectively and is filled with ice inside it. Its inside and outside surface are kept at different potentials by a battery of internal resistance $\frac{2}{\pi} \Omega$ & $\mathcal{E} = 5 \text{ V}$. Find value of σ for which ice melts at maximum possible rate if 25% of heat generated by shell due to joule heating is used to melt ice.



- (A) $\frac{5}{3} \text{ siemen/m}$ (B) 2 siemen/m (C) $\frac{1}{2} \text{ siemen/m}$ (D) $\frac{5}{8} \text{ siemen/m}$

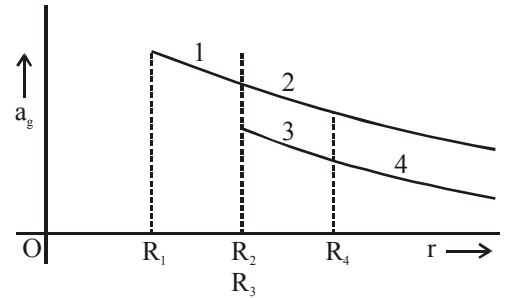
6. What is the equivalent resistance between points A & B ?

- (A) $\frac{5}{6} \Omega$ (B) $\frac{1}{2} \Omega$
(C) $\frac{4}{3} \Omega$ (D) $\frac{7}{6} \Omega$



7. Figure shows the variation of the gravitational acceleration a_g of four planets with the radial distance r from the centre of the planet for $r \geq$ radius of the planet. Plots 1 and 2 coincide for $r \geq R_2$ and plots 3 and 4 coincide for $r \geq R_4$. The sequence of the planets in the descending order of their densities is :

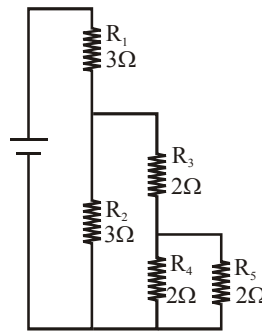
- (A) 1, 2, 3, 4
(B) 4, 3, 2, 1
(C) 2, 1, 4, 3
(D) 1, 2, 4, 3



4 Q. [4 M (-1)]

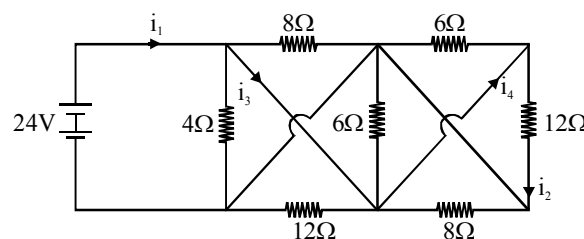
Multiple Correct Answer Type

8. In the figure shown



- (A) currents in R_4 and R_5 are equal
(B) currents in R_2 and R_3 are equal
(C) current in R_1 is equal to the sum of currents in R_3 , R_4 and R_5
(D) current in R_1 is equal to the sum of currents in R_2 , R_4 and R_5

9. For the circuit shown in figure

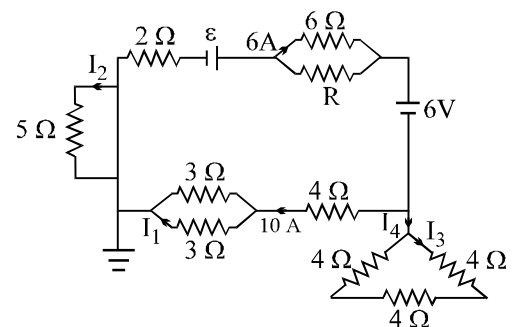


- (A) $i_1 = 24 \text{ A}$ (B) $i_2 = 2 \text{ A}$ (C) $i_3 = 15 \text{ A}$ (D) $i_4 = 6 \text{ A}$

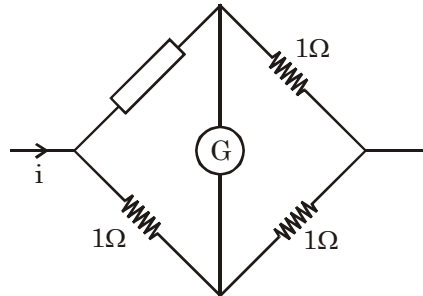
10. The figure shows the network of resistances connected across the cell of unknown emf ε .

Mark the **CORRECT** statement(s) :

- (A) The value of unknown resistance $R = 9 \Omega$
(B) $I_1 = 2 \text{ A}$, $I_3 = 10 \text{ A}$, $I_4 = 10 \text{ A}$
(C) The emf ε of the source is 117 V
(D) $I_1 = 5 \text{ A}$, $I_3 = 0$, $I_4 = 0$



11. The rectangular box is a non ohmic resistance following $I = \frac{V^4}{125}$ where I is current through it (in A) & V voltage across it (in V). At the balanced condition of wheatstone bridge.



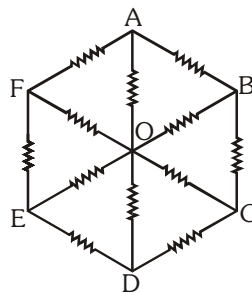
- (A) $i = 10$ A (B) $V = 5$ Volt (C) $I = 5$ A (D) $V_{\text{Source}} = 10$ Volt

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Questions no. 12 to 14

Twelve wires of equal resistance R are placed in a given arrangement as shown in figure. Then find



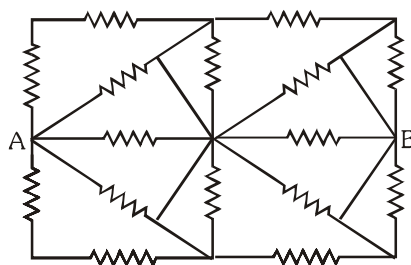
12. Equivalent resistance between A & B is
 (A) $\frac{11R}{20}$ (B) $\frac{10R}{20}$ (C) $\frac{R}{20}$ (D) $\frac{3R}{2}$
13. Equivalent resistance between A & C is
 (A) $\frac{3R}{4}$ (B) $\frac{R}{4}$ (C) $\frac{5R}{6}$ (D) $\frac{3R}{5}$
14. Equivalent resistance between A & O is
 (A) $\frac{19R}{41}$ (B) $\frac{20R}{41}$ (C) $\frac{18R}{23}$ (D) $\frac{9R}{20}$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

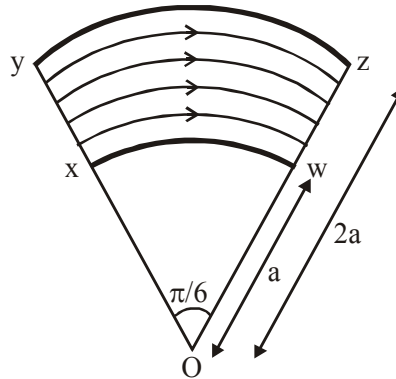
3 Q. [4 M (0)]

1. In the given circuit all resistors are identical. Each resistance is equal to 20Ω . Find the equivalent resistance between A & B in Ω .

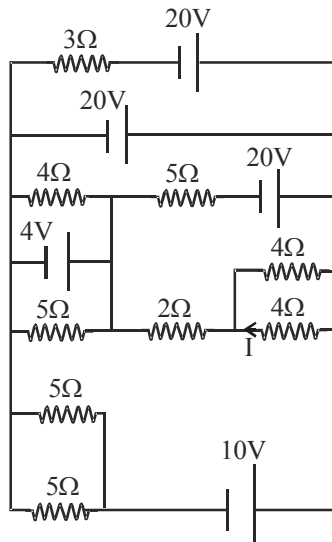


2. Calculate the resistance (in Ω) of the conducting strip xyzw having uniform conductivity

$\sigma = \frac{\pi}{6}$ mho/met, thickness $t = \frac{1}{\ln 2}$ m., current flows along the strip as shown in figure :



3. In the circuit shown find the value of current I in amp?

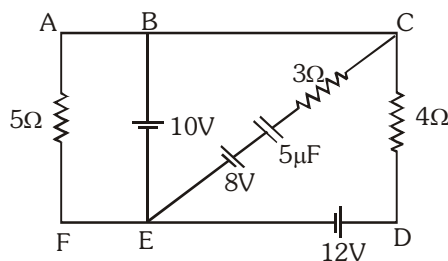


SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. A network consisting of three resistors, three batteries, and a capacitor is shown in figure.



Column-I

- (A) Current in branch EB is
- (B) Current in branch CB is
- (C) Current in branch ED is
- (D) Charge on capacitor is

Column-II

- (P) 10 μC
- (Q) 0.5 A
- (R) 1.5 A
- (S) 5 μC
- (T) None of these

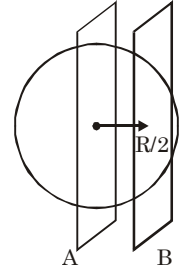
SECTION-I

Single Correct Answer Type

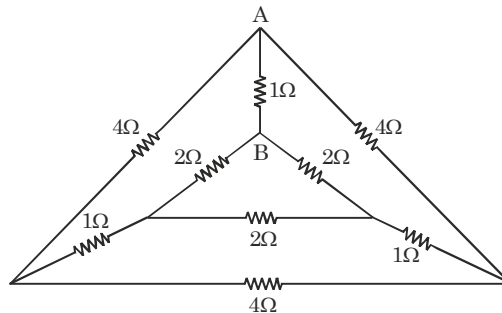
5 Q. [3 M (-1)]

1. From a sphere of electrical conductivity K two planes cut a piece such that first plane passes through the centre of sphere and second parallel to first one at distance $R/2$ from centre then resistance between A and B.

- (A) $\frac{1}{\pi KR}$ (B) $\frac{\sqrt{3}}{\pi KR}$
(C) $\frac{\ln 3}{2\pi KR}$ (D) None

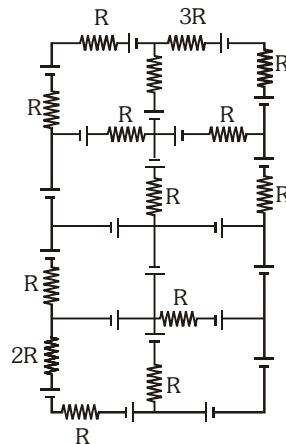


2. Consider the circuit as shown in figure. The equivalent resistance between A and B is



- (A) $\frac{9}{7}\Omega$ (B) $\frac{1}{9}\Omega$ (C) $\frac{7}{9}\Omega$ (D) None

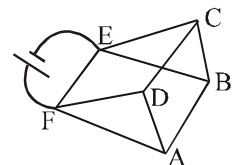
3. In the shown circuit all batteries have equal emf E . Current through the $2R$ resistance is



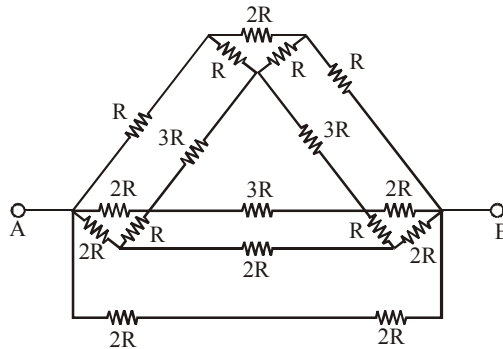
- (A) $\frac{3E}{2R}$ (B) $\frac{2E}{3R}$ (C) $\frac{E}{3R}$ (D) zero

4. A prism is made of wire mesh with each side having equal resistance R . A battery of 6 volt and zero resistance is connected across E and F as shown in the figure. The current in the branch AB, if R is equal to 5Ω , is :-

- (A) 0.6 A (B) 0.8 A
(C) 0.4 A (D) 2 A



5. Find equivalent resistance between A & B.



- (A) $\frac{R}{11}$ (B) $\frac{12R}{11}$ (C) R (D) $4R$

Multiple Correct Answer Type

4 Q. [4 M (-1)]

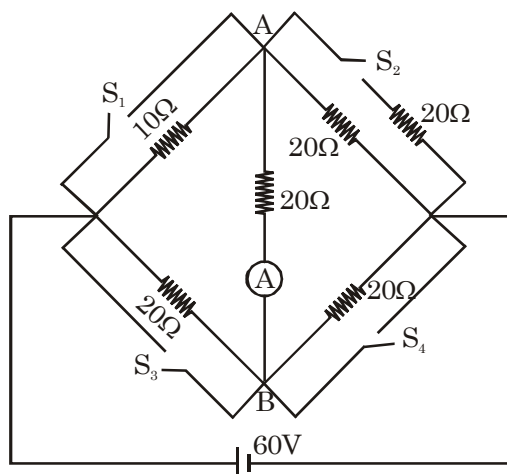
6. A metal sphere of radius a is surrounded by a concentric metal sphere of inner radius of b , where $b > a$. The space between the spheres is filled with a material whose electrical conductivity σ varies with the electric field strength E as $\sigma = kE$ where k is a constant. A potential difference V is maintained between spheres.

- (A) For $a < r < b$, current is $4\pi r^2 kE^2$
(B) For $a < r < b$, current is $2\pi r^2 kE^2$

(C) Potential difference between spheres is $V = \sqrt{\frac{I}{4\pi k}} \ln\left(\frac{b}{a}\right)$ where I is total current

(D) Potential difference between spheres is $V = \sqrt{\frac{I}{2\pi k}} \ln\left(\frac{b}{a}\right)$ where I is total current

7. Consider the circuit shown. Resistance connected between terminal 'A' and 'B' is 20Ω and ammeter is ideal. Then select the **CORRECT** statement :-

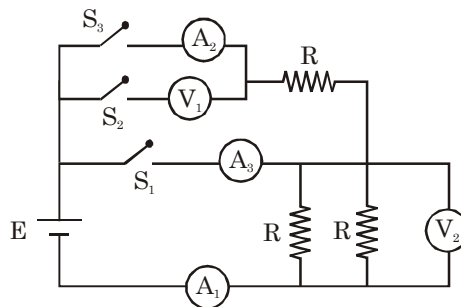
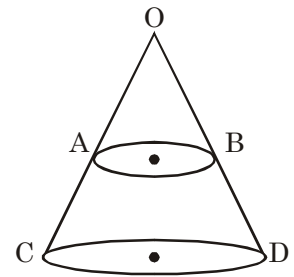


- (A) When switch ' S_1 ' and ' S_4 ' are closed, current reading of ammeter is 3 amp.
(B) When switch ' S_3 ' and ' S_2 ' are closed, current reading of ammeter is $\frac{6}{5}$ amp.
(C) When only ' S_2 ' is closed, reading of ammeter is zero.
(D) When all the switch are close, reading of ammeter is 0 amp.

8. In given figure AB and CD are diameter of rings such that $AB = 10$ cm, $CD = 20$ cm and $OA = 10$ cm. The resistance per unit length of straight segment is $1\Omega/\text{cm}$ and that of rings is $\frac{4}{\pi}\Omega/\text{cm}$. A battery of emf 60 V is

connected across AB. Select correct alternatives :

- (A) Current through AO is 4 A
(B) Current through BD is 1.5 A
(C) Potential difference across CD is 30 V
(D) Potential difference across AD is 40 V
9. All ammeter/voltmeter are ideal. Which of the statements are **CORRECT**.

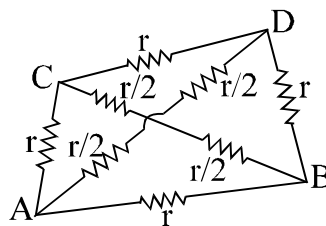


- (A) when all switch are closed, reading of A_1 and A_3 are same.
(B) When only S_2 is closed then reading of V_2 is E .
(C) When only S_2, S_3 closed then a reading of A_1 is $\frac{2E}{3R}$
(D) Reading of A_1 is independent of the function of switch S_2 .

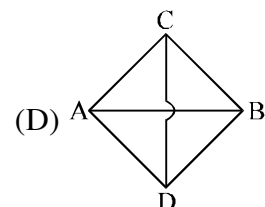
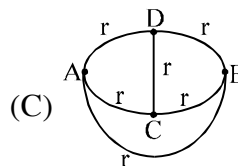
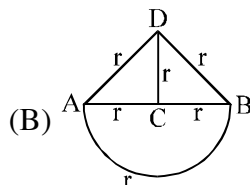
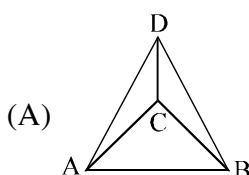
Linked Comprehension Type (1 Para $\times$ 3 Q.) [4 M (-1)]
(Multiple Correct Answer Type)

Paragraph for Questions no. 10 to 12

The figure shows a tetrahedron, each side of which has a resistance r



10. Choose the correct statement(s) related to the resistance between any two points.
(A) $R_{AB} = R_{BD} = R_{BC} = R_{CD} = R_{CA} = R_{AD}$
(B) $R_{AB} = R_{AC} = R_{AD} = R_{BD} = R_{BC} \neq R_{CD}$
(C) R_{CD} is the least
(D) $R_{AB} = R_{AC} = R_{BC}$ and $R_{CD} = R_{AD} = R_{BD}$
11. Choose the correct diagram(s), which show two-dimensional equivalent of the tetrahedron.



12. If a battery is connected between any two points of the tetrahedron, then identify the correct statement(s).
- (A) The potentials of the other two points are always equal.
 - (B) There always exists a branch through which no current flows.
 - (C) The current coming out of the battery in each case is same.
 - (D) None of these

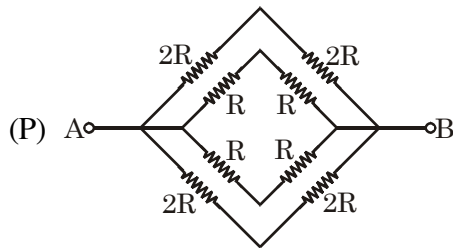
Matching List Type (4 × 4)

2 Q. [3 M (-1)]

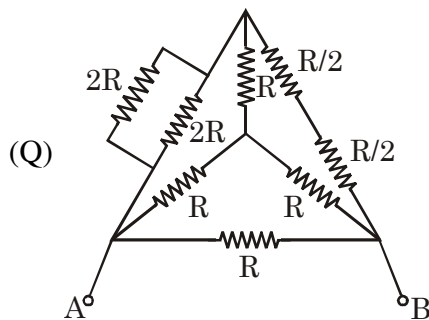
- 13.** List-I shows electric circuits and List-II shows equivalent resistance between A and B.

List-I

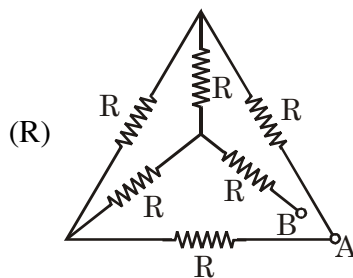
List-II



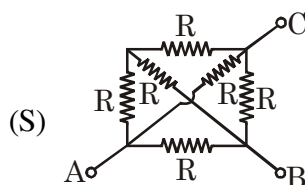
(1) $R/2$



(2) $R/3$



(3) $2R/3$



(4) 2R

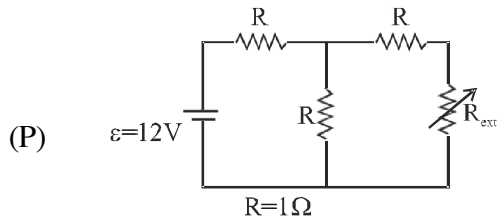
Code

	P	Q	R	S
(A)	3	1	4	2
(B)	2	1	4	2
(C)	3	1	4	1
(D)	2	1	4	3

14. Match the list-I with list-II.

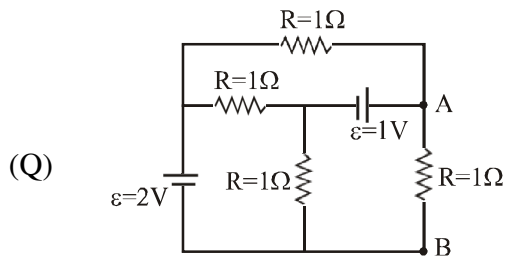
List-I

List-II



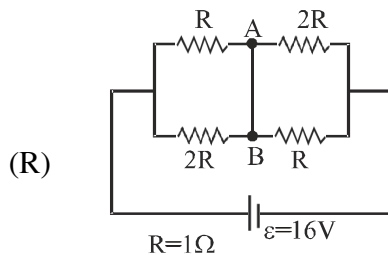
(1) $\frac{1}{2}$

Numerical value of maximum power (in watt) delivered to R_{ext} is



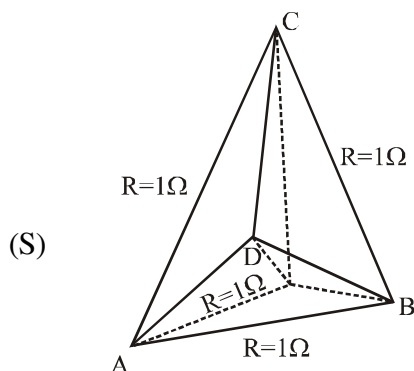
(2) $\frac{3}{2}$

Numerical value of potential difference (in volt) between points A & B is



(3) 6

Wire AB has negligible resistance. Numerical value of current (in ampere) through AB is



(4) 4

Resistances are arranged in tetrahedron configuration. All resistance are equal. Numerical value of equivalent resistance (in Ω) between A & B is

Code

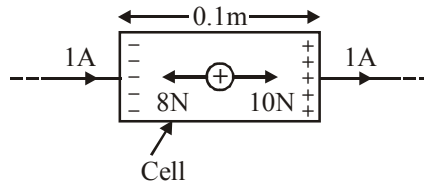
	P	Q	R	S
(A)	4	3	2	1
(B)	3	2	4	1
(C)	1	2	3	4
(D)	2	3	1	4

SECTION-III

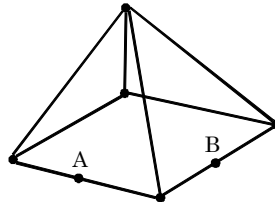
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

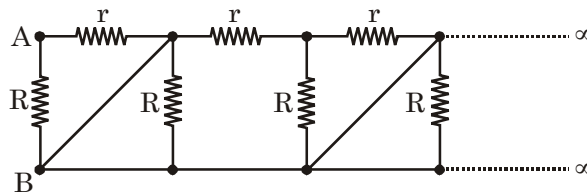
1. Figure shows a cell in which unit positive charge experiences a constant non electric force of 10N and a constant electric force of 8N in directions shown in the figure. The internal resistance of the cell is $r\Omega$. Fill $10r$ in OMR sheet.



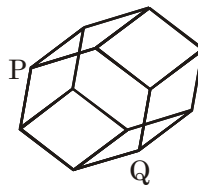
2. A regular pyramid of square is made of eight identical wires each resistance 12Ω shown in the figure. The terminals A and B are the mid-points of the corresponding sides. Find equivalent resistance between the terminals A and B in ohms.



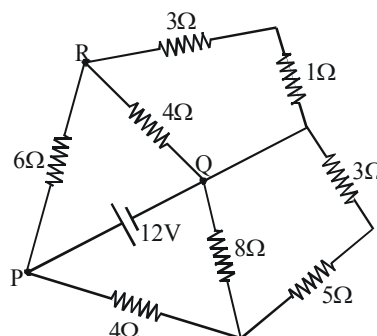
3. Find the equivalent resistance between the terminals A and B as shown on the diagram below. Put $R = 12\Omega$, $r = 6\Omega$ and neglect the resistance of leads.



4. All the wires on the front and the back hexagonal face (of the Skelton hexagonal prism) have resistance R . All the wires along the lines joining the vertices of two hexagons have resistance R . Take $R = \frac{20}{3}\Omega$. The equivalent resistance between P and Q as shown in the figure is ... Ω .



5. Find n if the total power dissipated by the circuit is $6n$ watts.

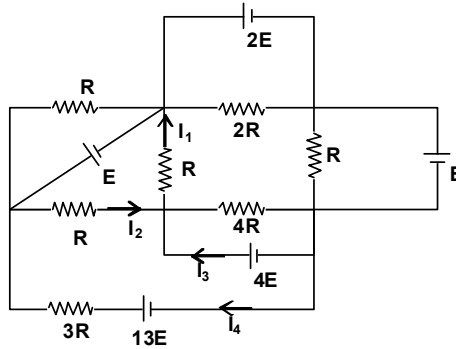


SECTION-I

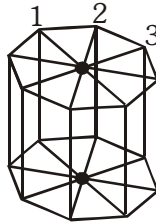
Single Correct Answer Type

6 Q. [3 M (–1)]

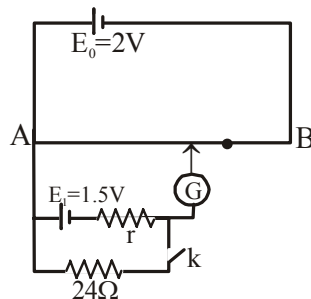
1. In the circuit shown in the figure, I_1, I_2, I_3, I_4 denote current in the branches. Choose the correct option.



- (A) $I_1 > I_2 > I_3 > I_4$ (B) $I_1 = I_2 > I_3 > I_4$ (C) $I_1 < I_3 < I_2 < I_4$ (D) $I_2 < I_1 < I_3 < I_4$
2. In the diagram shown, all the wires have resistance R . The equivalent resistance between the upper and lower dots shown in the diagram is–

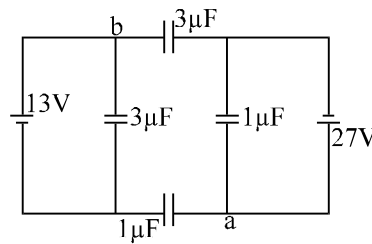


- (A) $R/8$ (B) R (C) $2R/5$ (D) $3R/8$
3. For the arrangement of the potentiometer shown in the figure, the balance point is obtained at a distance 75 cm from A when the key k is open. The second balance point is obtained at 60 cm from A when the key k is closed. Find the internal resistance (in Ω) of the battery E_1 .



- (A) 3 (B) 6 (C) 9 (D) 12
4. A galvanometer, having a resistance of 50Ω , gives a full scale deflection for a current of 0.05 A . The length in meter of a resistance wire of area of cross-section $2.97 \times 10^{-2} \text{ cm}^2$ that can be used to convert the galvanometer into an ammeter which can read a maximum of 5 A current is (Specific resistance of the wire = $5 \times 10^{-7} \Omega \text{ m}$)
- (A) 9 (B) 6 (C) 3 (D) 1.5

5. Capacity of an isolated sphere of radius R_1 is increased n times when it is enclosed by an earthed concentric sphere of radius R_2 ($R_2 > R_1$). The ratio $\frac{R_2}{R_1}$ is
- (A) $\frac{n^2}{n-1}$ (B) $\frac{n}{n-1}$ (C) $\frac{2n}{n+1}$ (D) $\frac{2n+1}{n+1}$
6. Four capacitors and two sources of e.m.f. are connected as shown in the figure. The p.d. in volts between the points a and b is :

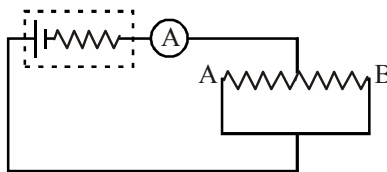


- (A) zero (B) 13 (C) 17 (D) 27

Multiple Correct Answer Type

4 Q. [4 M (-1)]

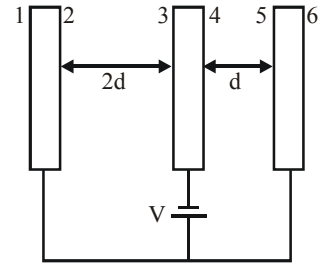
7. The circuit diagram shows a source of emf 10 V having an internal resistance of 2.5Ω connected to an ideal ammeter. The jockey can be slid to different points on the Rheostat whose total resistance is 10Ω . What can be the reading of the ammeter.



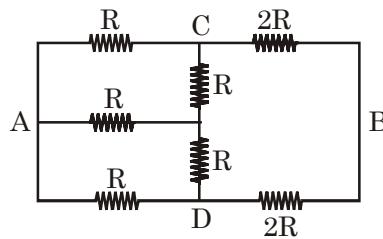
- (A) 1A (B) 2A (C) 3A (D) 4A
8. In a meter bridge experiment with the resistance R_1 in the left gap and a resistance X in the right gap, the null point obtained at 40 cm from left end. Now with the resistance R_2 in the left gap and the same resistance X in the right gap, null point is obtained at 50 cm from left end.
- (A) If R_1 and R_2 are put in series in the left gap and the right gap contains resistance X , the null point is obtained at a distance 75 cm from the left end.
- (B) If R_1 and R_2 are put in series in the left gap and the right gap contains resistance X , the null point is obtained at a distance $\frac{125}{2}$ cm from the left end.
- (C) If R_1 and R_2 are put in parallel in the right gap and the left gap contains resistance X , the null point is obtained at a distance $\frac{500}{7}$ cm from the left end.
- (D) If R_1 and R_2 are put in parallel in the left gap and the right gap contains resistance X , the null point is obtained at a distance $\frac{200}{7}$ cm from the left end.

9. In the figure shown, area of each plate is A.

- (A) Charge on plate 5 is $\left(\frac{\epsilon_0 A}{d}\right)V$
- (B) Potential difference between plates 2 and 3 is V
- (C) Potential difference between plates 2 and 5 is zero
- (D) Charge on the plate 3 is zero.



10. A part of circuit is as shown in the diagram ($R = 8\Omega$)



- (A) Equivalent resistance between A and B is 11Ω .
- (B) Equivalent resistance between A and B is 8Ω .
- (C) Equivalent resistance between C and D is 6.2Ω .
- (D) Equivalent resistance between A and B is 10Ω .

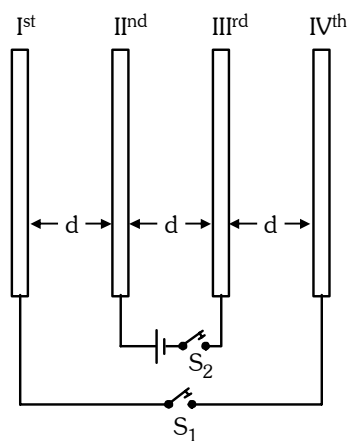
Linked Comprehension Type

(1 Para $\times$ 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question no. 11 and 12

Four large conducting plates each of area A having charge $4Q$, $3Q$, $2Q$ and Q respectively are kept parallel to each other. If switch S_1 and S_2 both are closed simultaneously. (Emf of battery $E = \frac{Qd}{A\epsilon_0}$)



11. Final charge on plate Ist is :

- (A) $2Q$ (B) $-3Q$ (C) $5Q$ (D) $4Q$

12. Total work done by battery is :

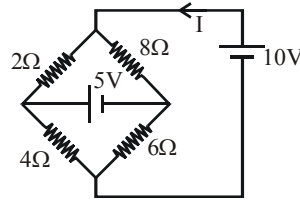
- (A) $\frac{5Q^2d}{A\epsilon_0}$ (B) $\frac{Q^2d}{A\epsilon_0}$ (C) $\frac{2Q^2d}{A\epsilon_0}$ (D) $\frac{-Q^2d}{A\epsilon_0}$

SECTION-III

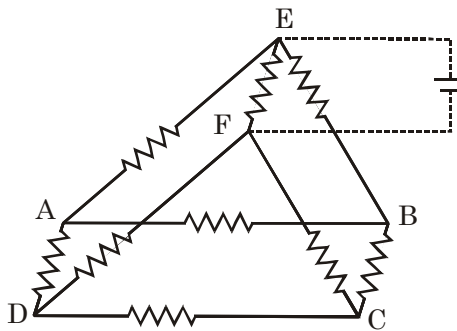
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

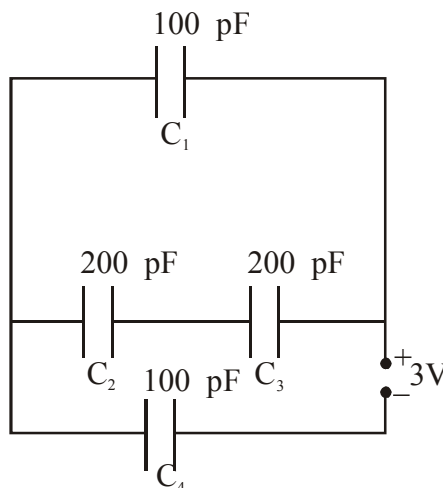
1. The current supplied to the circuit by 10 V cell is I ampere. Assume both cells as ideal find 4I.



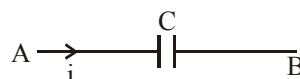
2. A prism is made of wire mesh with each side having equal resistance R. A battery of 6V and zero resistance is connected across E and F as shown in the figure. The current in the branch AD (in Ampere) if R is equal to 5Ω is α , then find 10α .



3. If a galvanometer is converted into voltmeter with help of 32Ω resistance then voltage range becomes n times of initial. If same galvanometer is converted into ammeter with help of 2Ω resistance (as a shunt) then current range become n times of initial. Find resistance (in Ω) of original galvanometer.
4. For the network having capacitances as shown in figure, the charge on 200 pF capacitor is given by $\alpha \times 10^2$ pC. Fill the value of α in OMR sheet.



5. In a circuit, a branch carrying only capacitor has variable current $i = I_0 \sin 2\pi t$ and capacitor was uncharged at $t = 0$. The maximum magnitude of potential difference across capacitor is $\frac{2I_0}{k\pi C}$. Then value of k will be :



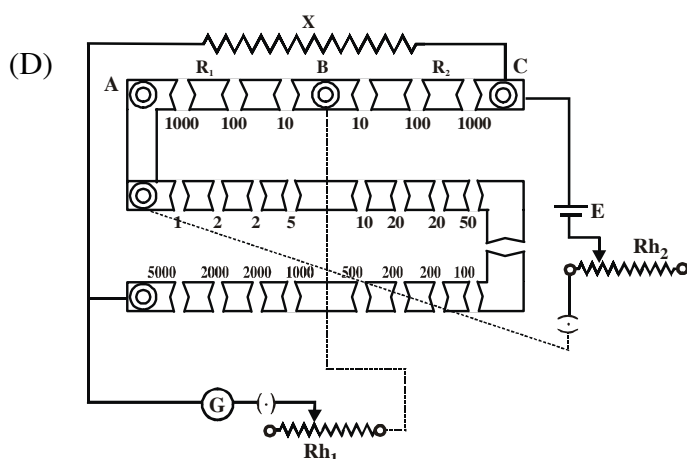
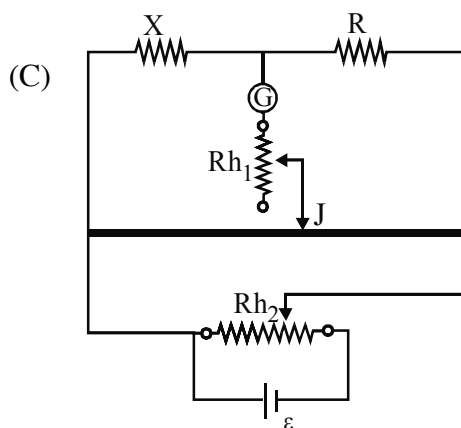
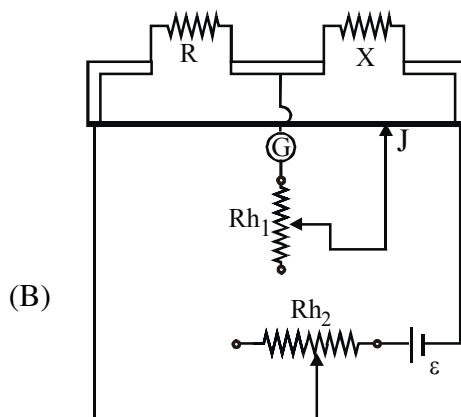
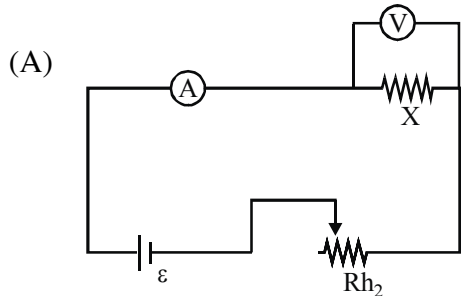
SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

1. Match the column.

Column-I



Column-II

(P) Set-up can be used to find unknown resistance X.

(Q) Set-up is used to verify Ohm's law.

(R) Wheatstone bridge

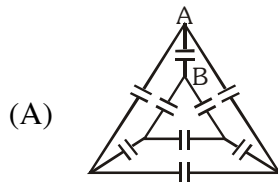
(S) Rheostat (Rh_2) is a potential divider.

(T) Rheostat (Rh_1) should be kept low for higher sensitivity to measure X.

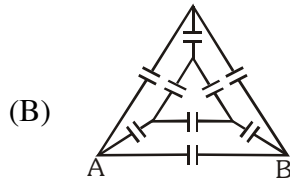
2. All capacitors have capacitance of $1\mu\text{F}$. Find equivalent capacitance between points A & B.

Column-I (Circuit)

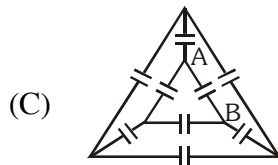
Column-II (Capacitance)



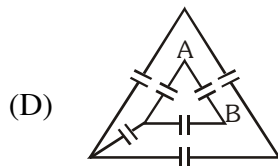
(P) $\frac{4}{3}\mu\text{F}$



(Q) $\frac{3}{2}\mu\text{F}$



(R) $\frac{15}{8}\mu\text{F}$



(S) $\frac{5}{3}\mu\text{F}$

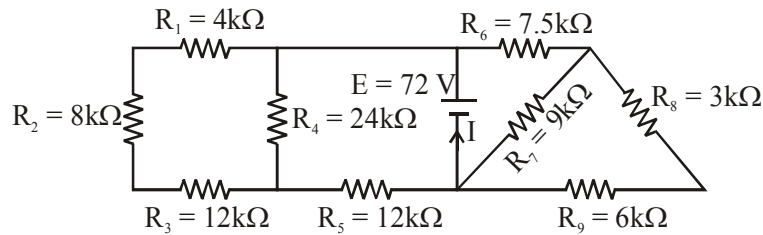
(T) None of these

SECTION-I

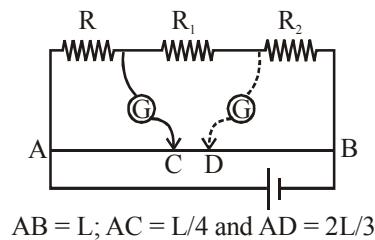
Single Correct Answer Type

6 Q. [3 M (-1)]

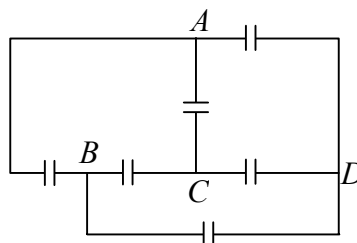
1. The current through the battery E in the network shown is



- (A) Zero (B) 9 A (C) 6 mA (D) 9 mA
2. The diagram shows a modified meter bridge, which is used for measuring two unknown resistances at the same time. When only the first galvanometer is used, for obtaining the balance point, it is found at point C. Now the first galvanometer is removed and the second galvanometer is used, which gives balance point at D. Using the details given in the diagram, find out the value of R_1 and R_2 :-

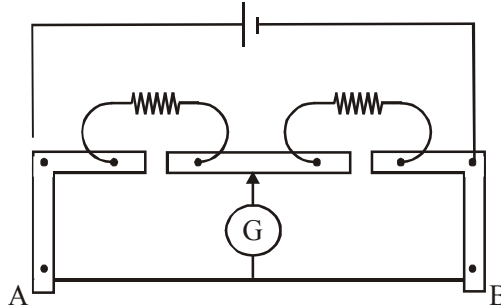


- (A) $R_1 = 5R/3$, $R_2 = 4R/3$ (B) $R_1 = 4R/3$, $R_2 = R$
(C) $R_1 = 4R/3$, $R_2 = 5R/3$ (D) $R_1 = R$, $R_2 = R/3$
3. Choose the correct statement for the given capacitor arrangement. (All the 6 capacitor are of same capacitance).

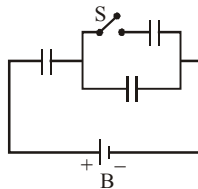


- (A) If a battery is connected across AB, all 6 capacitors get charged.
(B) If a battery is connected across AC, all 6 capacitors get charged.
(C) If a battery is connected across AD, all 6 capacitors get charged.
(D) It is not possible to charge all the 6 capacitors using single source.

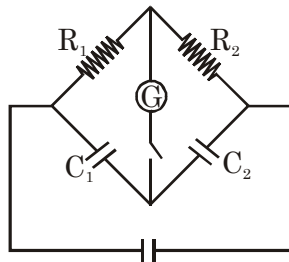
4. A meter bridge set up as shown to determine end correction at A and B. When a resistance of 15Ω is used in left gap and of 20Ω in right gap, then null point comes at a distance 42cm from A. When these resistances are interchanged null point comes at a distance 57cm from A. Values of end corrections are :-



- (A) 1 cm, 2 cm (B) 2 cm, 3 cm (C) 3 cm, 4 cm (D) 3 cm, 2 cm
5. In figure an arrangement of three identical capacitors is shown along with a switch S and a battery B. If the switch S is closed, then the ratio of the energy of the capacitors system in the final steady state to the initial state is :



- (A) 2 : 1 (B) 4 : 3 (C) 3 : 2 (D) 3 : 4
6. Circuit shown is in steady state, now when switch is closed, galvanometer shows no deflection, then correct relation is :-



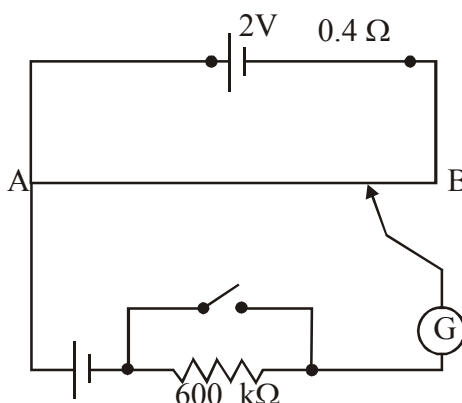
- (A) $\frac{R_1}{R_2} = \frac{C_2}{C_1}$ (B) $\frac{R_1}{R_2} = \frac{C_1}{C_2}$
- (C) $R_1 R_2 = C_1 C_2$ (D) $R_1 \sqrt{C_1} = R_2 \sqrt{C_2}$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q. & 1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 7 to 9

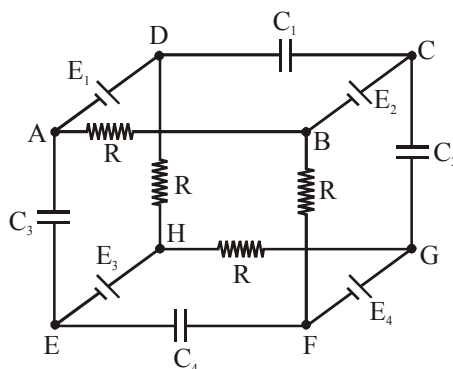
Figure shows a potentiometer with a cell of 2.0 V and internal resistance 0.40Ω maintaining a potential drop across the resistor wire AB. A standard cell which maintains a constant emf of 1.02 V (for very moderate currents upto a few mA) gives a balance point at 60 cm length of the wire. To ensure very low currents drawn from the standard cell, a very high resistance of $600\text{ k}\Omega$ is put in series with it, which is shorted close to the balance point. The standard cell is then replaced by a cell of unknown emf ϵ and the balance point found similarly, turns out to be at 80 cm length of the wire.



7. The value of emf of the unknown cell ε is
(A) 1.2 V (B) 1.36V (C) 1.5V (D) 2V
8. Would the method work in the above situation if the driver cell of the potentiometer had an emf of 1.0 V instead of 2.0 V ?
(A) Yes but the value of high resistance (600 kΩ) must be lowered.
(B) Yes it will work perfectly and no modification is needed to be done in the circuit.
(C) Yes only if the driver cell is ideal in nature.
(D) No as the driver cell has lower emf as compared to the unknown cell.
9. Would the circuit work well for determining an extremely small emf, say of the order of a few mV (such as the typical emf of a thermo-couple) ?
(A) Yes, with the same accuracy and with no further modification.
(B) Yes, but with reducing the high resistance (600 kΩ) to a lower value.
(C) Yes, but by connecting using suitable resistance in series with the driver cell.
(D) No, as the balanced wheat stone bridge will not form with the lower emf of unknown cell.

Paragraph for Question no. 10 and 11

Four batteries of emf $E_1 = 4V$, $E_2 = 8V$, $E_3 = 12V$ and $E_4 = 16V$ and four identical capacitors each of $1 \mu F$ and four identical resistance are connected in circuit as in figure. Batteries do not have any internal resistances



10. At steady state potential difference across C_2 is :
(A) 1V (B) 5V (C) 4V (D) 6V
11. If points H and B are shorted, the new charge on C_2 at steady state will be :
(A) Zero (B) $5 \mu C$ (C) $4 \mu C$ (D) $1 \mu C$

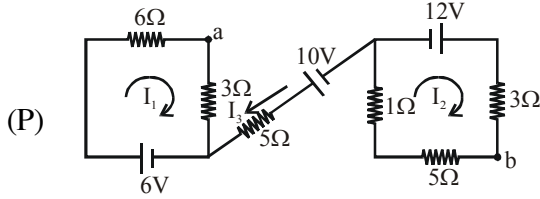
Matching List Type (4 × 4)

1 Q. [3 M (-1)]

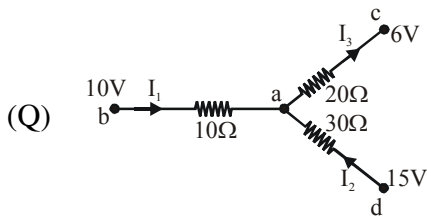
12. Consider the electric circuits given in List-I and List-II gives values of some quantities from electric circuits. Match the corresponding quantities with correct circuit.

List-I

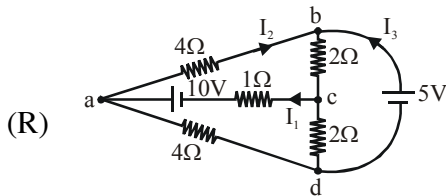
List-II



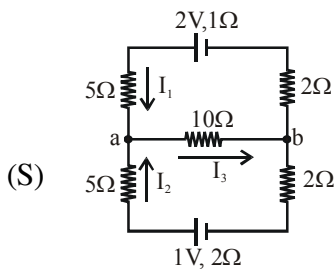
(1) $I_1 = \frac{1}{55} \text{ A}$



(2) $I_2 = \frac{-1}{121} \text{ A}$



(3) $I_3 = \frac{15}{8} \text{ A}$



(4) $V_a - V_b = 4 \text{ V}$

Code :

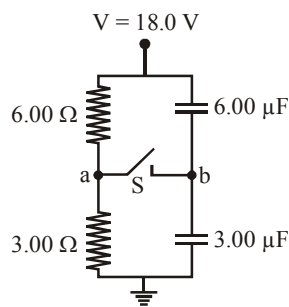
	P	Q	R	S
(A)	4	1	3	2
(B)	4	1	2	4
(C)	3	4	3	4
(D)	4	1	1	2

SECTION-III

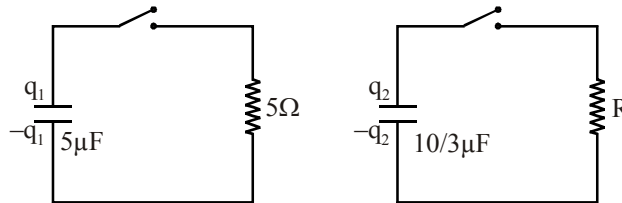
Numerical Grid Type (Ranging from 0 to 9)

6 Q. [4 M (0)]

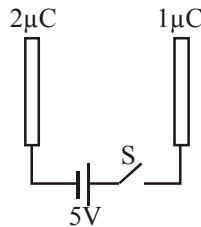
1. Initially switch S is open. Charge flows through switch S when it is closed is $6 \times Q \mu\text{C}$. Then find the value of Q ?



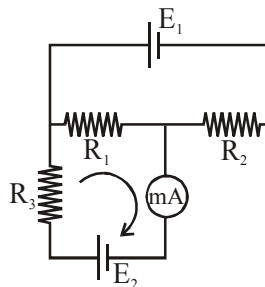
2. The figure shows two circuits with a charged capacitor that is to be discharged through a resistor as shown. The initial charge on capacitors is $\frac{q_2}{q_1} = 2$. If both switches are closed at $t = 0$, the charges become equal at $10^{-4} \ln 2$ sec. Find the resistance R (in Ω).



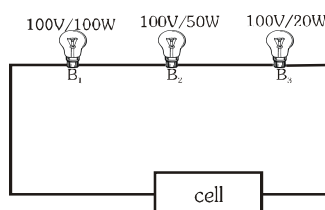
3. A parallel plate capacitor of capacitance $0.1 \mu\text{F}$ is shown in the figure. Its two plates are given charges $2 \mu\text{C}$ and $1 \mu\text{C}$. Find the value of heat dissipated (in μJ) after switch is closed.



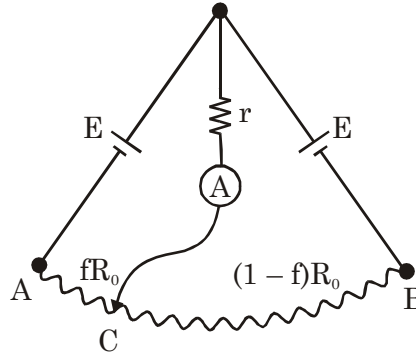
4. The circuit shown in the figure contains three resistors $R_1 = 100 \Omega$, $R_2 = 50 \Omega$ & $R_3 = 20 \Omega$ and cells of emf's $E_1 = 2\text{V}$ & E_2 . The ammeter indicates a current of 50mA . Find the emf (in V) of the second cell. (The internal resistance of the ammeter and of the cell should be neglected.)



5. Figure shows three bulbs connected in series. Their voltage rating shows that they will fuse if potential drop across them is greater than 100V . Voltage of the cell can be varied. In each step voltage of the cell increases by 20V . Initially the voltage of the cell is 100V . If voltage of the cell is gradually increased and it is found that after N^{th} step current in the circuit becomes zero. Find the value of N .



6. In Fig., the uniform-resistance wire between A and B has a total resistance R_0 . The contact at C can divide the wire into resistors fR_0 and $(1-f)R_0$. Consider the ideal ammeter for any $0 < f < 1$. Assume that the batteries are identical and have negligible internal resistance. For what value of f is the ammeter reading is minimum. Fill value of $12f$.

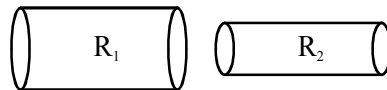


SECTION-IV

Matrix Match Type (4×5)

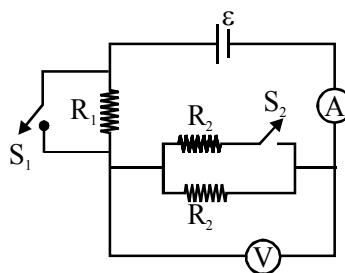
1 Q. [8 M (for each entry +2(0))]

1. Figure shows two cylindrical uniform specimen of electrical resistances with their characteristics.



Resistivity :	$\rho_1 = \rho$	$\rho_2 = 2\rho$
Length :	$\ell_1 = \ell$	$\ell_2 = 2\ell$
Area of cross-section :	$A_1 = A$	$A_2 = A/2$

R_1 and R_2 are connected in circuit as shown in figure. Voltmeter and ammeter connected in the circuit are ideal. Comparison of electric field, drift speed and power dissipated per unit volume should be done only if R_1 carries current (both the materials have same electron density).



Column-I

- (A) S_1 is open S_2 is closed
- (B) S_1 is closed S_2 is open
- (C) S_1 and S_2 both are closed
- (D) S_1 and S_2 both are open

Column-II

- (P) Reading of voltmeter is maximum
- (Q) Reading of ammeter is maximum
- (R) Magnitude of electric field in R_2 is greater than in R_1
- (S) Drift speed of electrons is greater in R_2 than in R_1
- (T) Power dissipated per unit volume is greater in R_2 than in R_1

PHYSICS

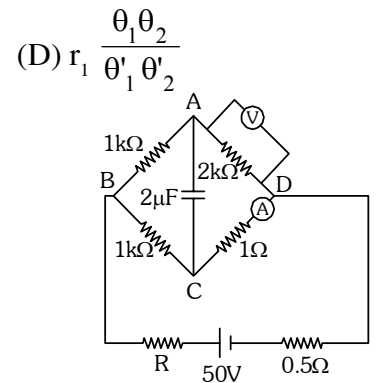
CLASS TEST # 33

SECTION-I

Single Correct Answer Type

3 Q. [3 M (-1)]

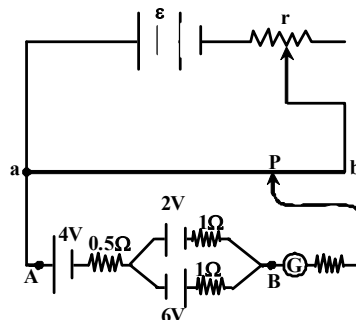
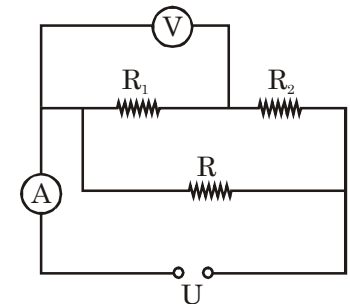
- Two ammeters, 1 and 2, have different internal resistances: r_1 (known) r_2 (unknown). Each ammeter has scale such that the angular deviation of the needle from zero is proportional to the current. Initially, the ammeters are connected in series and then to a source of constant voltage. The deviations of the needles of the ammeters are θ_1 and θ_2 , respectively. The ammeters are then connected in parallel and then to the same voltage source. This time, the deviations of the needles are θ_1' and θ_2' , respectively. r_2 in terms of r_1 , θ_1 , θ_2 , θ_1' and θ_2' is
 (A) $r_1 \frac{\theta_1 \theta_1'}{\theta_2 \theta_2'}$ (B) $r_1 \frac{\theta_2 \theta_1'}{\theta_1 \theta_2'}$ (C) $r_1 \frac{\theta_2 \theta_2'}{\theta_1 \theta_1'}$ (D) $r_1 \frac{\theta_1 \theta_2}{\theta_1' \theta_2'}$
- Calculate the energy stored in the capacitor of capacitance $2\mu\text{F}$, at the instant when the voltmeter 'V' gives a reading of 15V and the ammeter A reads 15 mA. Resistance of voltmeter is unknown and ammeter is 999Ω .
 (A) $5\mu\text{J}$
 (B) $10\mu\text{J}$
 (C) $0.5\mu\text{J}$
 (D) 0
- In a potentiometer, there are 10 identical cells (E , r -each) arranged in series in primary circuit. A cell of emf E_0 and internal resistance r_0 is balanced at ℓ_1 length of the potentiometer wire. Now, the polarity of two cells is reversed in primary circuit. The same cell of emf E_0 now gets balanced at ℓ_2 then ℓ_1/ℓ_2 is :-
 (A) $3/5$ (B) $5/3$ (C) $1/2$ (D) $2/1$



Multiple Correct Answer Type

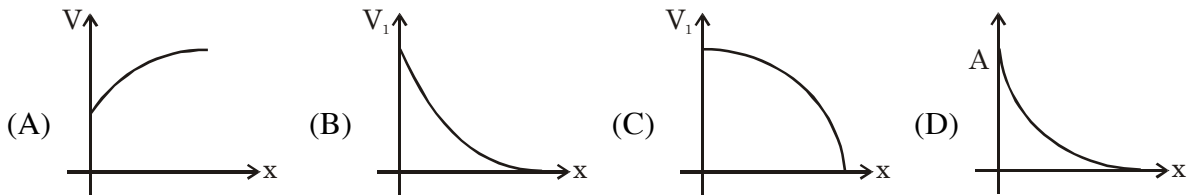
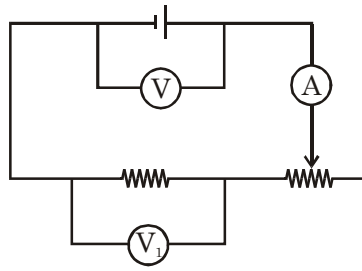
9 Q. [4 M (-1)]

- What are the resistances of the resistors R_1 and R_2 , shown in the figure, if the voltage supply is 40 V, and the power dissipated in resistor R is 80 W, the readings on the ammeter and on the voltmeter are 3 A and 30 V, respectively. (All devices are ideal)
 (A) $R_1 = 30\Omega$
 (B) $R_2 = 30\Omega$
 (C) $R_1 = 10\Omega$
 (D) $R_2 = 10\Omega$
- A potentiometer as shown in figure in which three batteries between A & B are connected when the point P is the null point, measured by the potentiometer arrangement. Then choose correct statement(s)

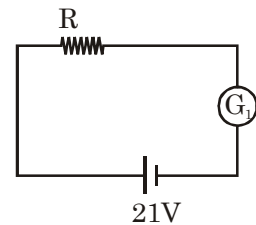


- current passing through 6 V battery is 4A
- current passing through 2 V battery is 4A
- $V_A - V_B = 2\text{V}$
- $V_A - V_B = 6\text{V}$

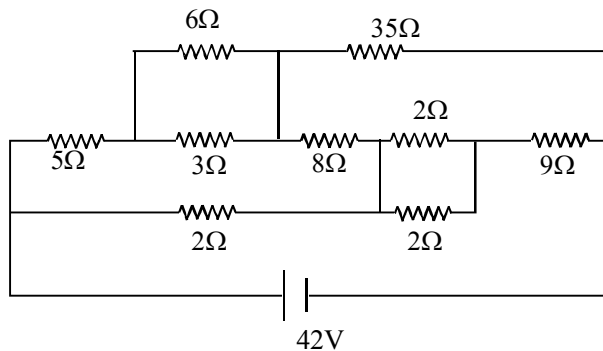
6. Two ideal voltmeters & an ideal ammeter are connected as shown. The battery is non ideal. Find the correct plot of A vs x , V vs x & V_1 vs x , where x is distance of slider on rheostat from left end.



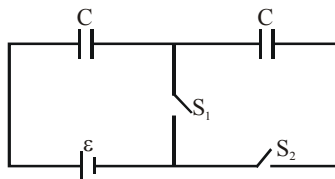
7. In the circuit shown, the battery is ideal. It has a voltage of 21 V. The galvanometer G_1 shows 0.42 A. When another identical galvanometer G_2 is connected across R , the reading of G_1 increases to 0.50 A :-



- (A) The resistance of galvanometer is 30Ω
 (B) The resistance $R = 40 \Omega$
 (C) The reading of $G_2 = 0.20$ A
 (D) If instead of this, G_2 was connected parallel to G_1 , their reading would be same = 0.21 A.
8. In given circuit :-

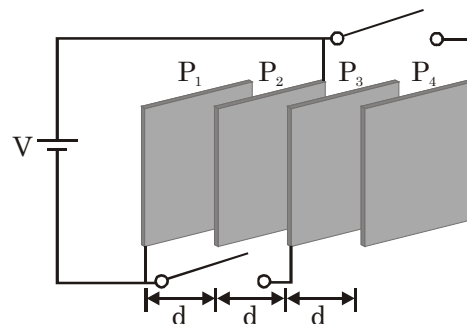


- (A) Power dissipated in 5Ω resistor is 10 W. (B) Power dissipated in 5Ω resistor is 5 W.
 (C) Power dissipated in 8Ω resistor is 0 W. (D) Power dissipated in 8Ω resistor is 2 W.
9. Figure shows two capacitors having same capacitance connected to a battery and two switches S_1 and S_2 . Initially S_1 and S_2 is closed :-



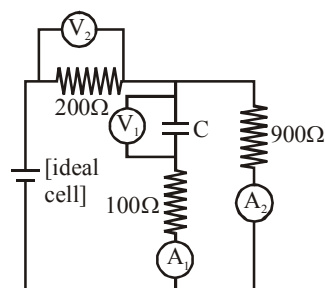
- (A) Before opening S_1 charge on left capacitor is $C\epsilon$.
 (B) After opening S_1 the final charge on both capacitor is equal to $\frac{1}{2}C\epsilon$
 (C) After opening S_1 the final charge on both capacitor is $C\epsilon$.
 (D) After opening S_1 the charge on left capacitor is $C\epsilon$.

10. Four parallel metal plates P_1, P_2, P_3 and P_4 each of area A , are separated successively by a distance d as shown in figure. P_1 is connected to the negative terminal of a battery, and P_2 is connected to the positive terminal. The battery maintains a potential difference of V . Mark the correct statement(s) :-

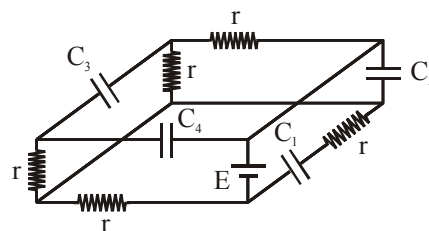


- (A) If P_3 is connected to the negative terminal, the capacitance of the three-plate system $P_1 P_2 P_3$ is $\frac{2\epsilon_0 A}{d}$.
- (B) If P_3 is connected to the negative terminal, the charge on P_2 is $2CV$, where C is $\frac{\epsilon_0 A}{d}$.
- (C) If P_3 is connected to the negative terminal and P_4 is connected to the positive terminal then the capacitance of the four-plate system $P_1 P_2 P_3 P_4$ is $3C$, where C is $\frac{\epsilon_0 A}{d}$.
- (D) If P_3 is connected to the negative terminal and P_4 is connected to the positive terminal then charge on P_4 is $\frac{\epsilon_0 A}{d} V$.

11. An electrical circuit is shown in the given figure. The resistance of each voltmeter is infinite and each ammeter is 100Ω . The charge on the capacitor of $100\mu F$ in steady state is 4 mC . Choose correct statements (s) regarding above circuit :-



- (A) Reading of voltmeter V_2 is 16 V
- (B) Reading of Ammeter A_1 is 0 A and A_2 is $1/25\text{ A}$.
- (C) Reading of voltmeter V_1 is 40 V
- (D) EMF of the ideal cell is 66 V .
12. The network has C_1, C_2, C_3, C_4 four capacitors each of capacitance $C = 1\mu F$, all resistance are identical having $r = 1\Omega$. The battery has emf $E = 30\text{ V}$. The circuit is analyzed at steady state. Mark the **CORRECT** options.
- (A) ratio of steady state charge of C_1 and C_2 is $1 : 1$
- (B) ratio of steady state charge of C_2 and C_4 is $1 : 1$
- (C) ratio of steady state charge of C_1 and C_3 is $1 : 1$
- (D) ratio of steady state charge of C_2 and C_3 is $1 : 1$

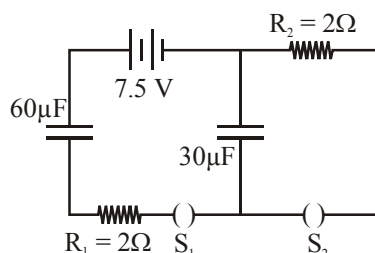


Linked Comprehension Type (1 Para × 2Q.)
(Single Correct Answer Type)

[3 M (-1)]

Paragraph for Question no. 13 and 14

In the given circuit all capacitor were uncharged initially. First switch S_1 is closed for long time when circuit reaches to steady state, switch S_1 is opened & S_2 is closed.



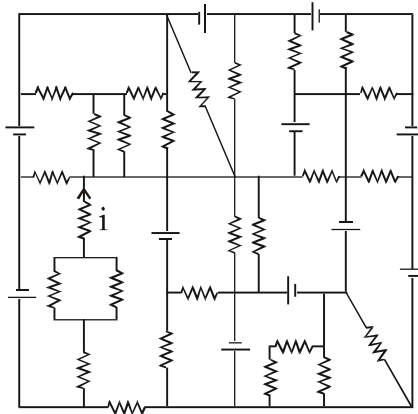
13. The current through R_2 , just after closing the switch S_2 is :-
 (A) 5A (B) 2.5 A (C) 3.75 A (D) 2A
14. Net heat produced in R_2 after closing switch S_2 :-
 (A) 375 μ J (B) 100 μ J (C) 250 μ J (D) 200 μ J

SECTION-III

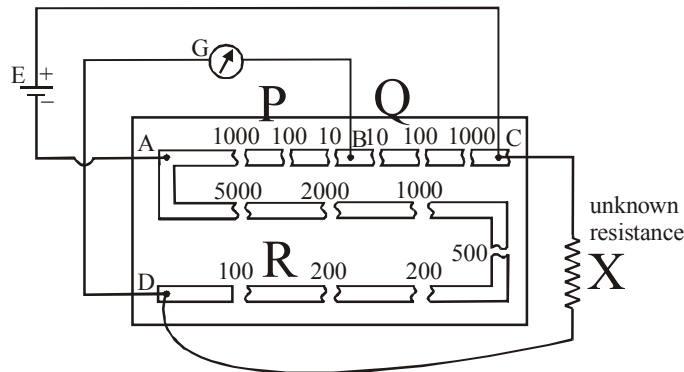
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

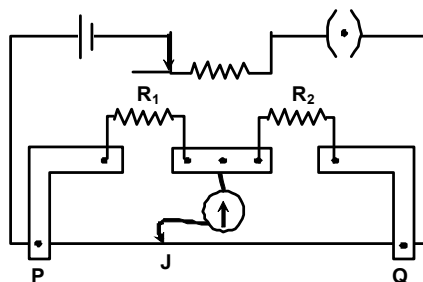
1. What is the size of current i (in A) in figure, where all resistances are 4.0Ω and all batteries have an emf of 10 V?



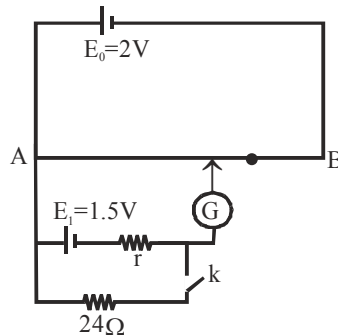
2. In the figure shown below, the maximum possible unknown resistance X (in Ω), that can be measured by the post office box are $X_{\max}$ is given by $R \times 10^5 \Omega$, then R is : (in this experiment, we take out only one plug in arm AB and only one plug in arm BC, but in arm AD we can take out many plugs).



3. The circuit diagram given in the figure shows the experimental setup for the measurement of unknown resistance by using a meter bridge. The wire connected between the points P and Q has non-uniform resistance such that the resistance per unit length varies directly as the distance from the point P. Null point is obtained with the jockey J with R_1 and R_2 in the given position. On interchanging the positions of R_1 and R_2 in the gaps, the jockey has to be displaced through a distance x from the previous position along the wire to establish the null point. If the ratio of $R_1/R_2 = 3$. The value of x is $50(\sqrt{K} - 1)$ cm. Find the value of K . (length of wire PQ = 1m)



4. For the arrangement of the potentiometer shown in the figure, the balance point is obtained at a distance 75 cm from A when the key k is open. The second balance point is obtained at 60 cm from A when the key k is closed. Find the internal resistance (in Ω) of the battery E_1 .



SECTION-IV

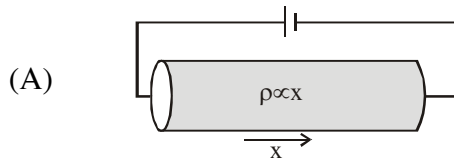
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

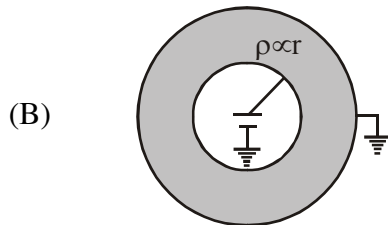
1. Column-I has some conductor across which battery is connected as shown. Variation of resistivity ρ is also indicated. Which of the quantities in column-II remain constant throughout the volume of conductor. (Assume that the free electron density is uniform in the material).

Column-I

Column-II

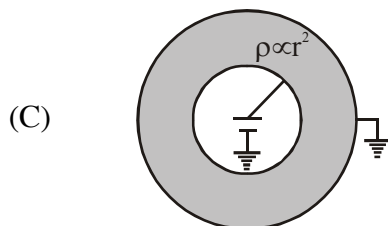


(P) Magnitude of electric field



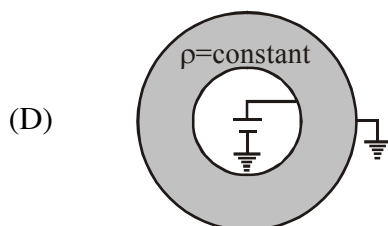
(Q) Magnitude of current density

Hollow cylinder



(R) Power dissipated per unit volume

Hollow sphere



(S) Drift speed of free electron

Hollow sphere

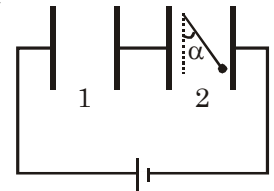
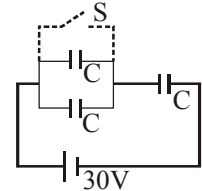
(T) Electric current

SECTION-I

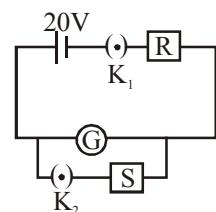
Single Correct Answer Type

7 Q. [3 M (-1)]

- The capacitor each having capacitance $C = 2 \mu\text{F}$ are connected with a battery of emf 30V as shown in figure. When the switch S is closed. Find the amount of charge flow through the switch
(A) $20\mu\text{C}$ (B) $60\mu\text{C}$
(C) $40\mu\text{C}$ (D) $10\mu\text{C}$
- Three identical metal plates of area 1 m^2 each are arranged horizontally, parallel to each other, with identical paper ($K = 4$) sheets in between of thickness 0.177 mm completely filling the space between them. Counted from the top, they are given charges equal to $+2\text{ mC}$, -1 mC and -1 mC , respectively. The electrostatic energy stored in this system is ($\epsilon_0 = 8.85 \times 10^{-12}\text{ C}^2/\text{Nm}^2$)
(A) 25 J (B) 12.5 J (C) 50 J (D) None
- Two identical parallel plate capacitors 1 and 2 are placed vertically and connected in series to a battery. In capacitor-2 there is a charged small particle attached by a thin wire to a fixed point, as shown. Ignore the effect of the charge particle on the charge distribution of the capacitor plates. At equilibrium, the angle between the wire and the vertical direction is α . Now slowly pull a plate of capacitor-1 until the distance between its two plates is doubled. After equilibrium, angle between the wire and the vertical direction is.

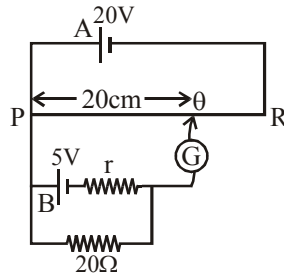


- (A) $\tan^{-1}\left(\frac{2}{3}\tan\alpha\right)$ (B) $\tan^{-1}\left(\frac{1}{3}\tan\alpha\right)$ (C) $\tan^{-1}\left(\frac{1}{2}\tan\alpha\right)$ (D) $\tan^{-1}\left(\frac{3}{2}\tan\alpha\right)$
- If galvanometer has 1000Ω resistance and $R = 10000\Omega$, then what should be the resistance connected to galvanometer in parallel to it so that its deflection reduces to half:-
(A) 455Ω (B) 500Ω
(C) 909Ω (D) 1236Ω
- When a galvanometer is shunted with a 5Ω resistance, the deflection is reduced to one-fifth. If the galvanometer is further shunted with a 3Ω wire, the further reduction (find the ratio of decrease in current to the previous current) in the deflection will be (the main current remains the same) :-

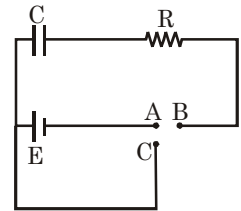


- $\left(\frac{4}{7}\right)$ of the deflection when shunted with 5Ω only
- $\left(\frac{5}{13}\right)$ of the deflection when shunted with 5Ω only
- $\left(\frac{3}{7}\right)$ of the deflection when shunted with 5Ω only
- $\left(\frac{3}{13}\right)$ of the deflection when shunted with 5Ω only

6. The figure shows a potentiometer arrangement used to measure the internal resistance of the cell B of 5V. The null point θ is located at 20 cm on a 100 cm wire PR. Find the internal resistance 'r' of the cell B :-



- (A) 1Ω (B) 2Ω (C) 3Ω (D) 5Ω
7. H_1 is the heat produced in resistance R when A and B are connected for a long time with no charge on the capacitor initially. Heat produced in the resistance is H_2 when B and C are connected for a long time thereafter. The energy supplied by the battery during charging is :-
- (A) H_1 (B) H_2
(C) $2H_1$ (D) $2(H_1 + H_2)$



Multiple Correct Answer Type

5 Q. [4 M (-1)]

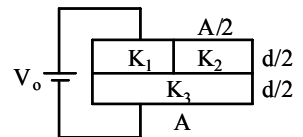
8. An ideal parallel plate capacitor of area A is filled with three dielectric slabs having dielectric constants $K_1 = 3.0$, $K_2 = 5.0$ and $K_3 = 2.0$ as shown in the figure. Choose the **CORRECT** option(s) :-

(A) the equivalent capacitance of the system is $\frac{8\epsilon_0 A}{3d}$

(B) the electric field in K_1 is more than the electric field in K_2 .

(C) the electric field on the left half of K_3 is less than the electric field on the right half of K_3 .

(D) the electric field on the left half of K_3 is more than the electric field on the right half of K_3 .



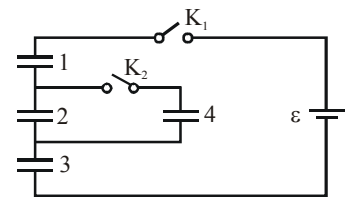
9. Four identical capacitor each of capacitance $5\mu\text{F}$ are connected as shown in Figure. They are attached to the battery of emf $E = 9\text{ V}$. Initially the key K_2 is open, and the switch K_1 is closed. Then key K_1 is opened and K_2 is closed.

(A) The initial potential difference across capacitor 2 is 3 V.

(B) The final potential difference across capacitor 2 is 1.8 V.

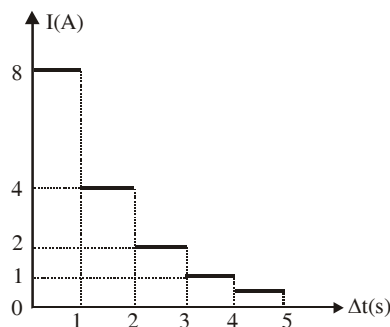
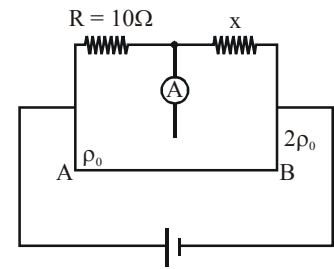
(C) The charge flown through the battery after the reversal of switches is $3\mu\text{C}$.

(D) The charge flown through the battery after the reversal of switches is zero.



10. An electric kettle has two heating coils. When one coil is switched on, the water in the kettle begins to boil after 15 minutes and when the other is switched on, water boils after 30 minutes. Water in the kettle will boil
- (A) In 45 minutes if coils are connected in series
(B) In 90 minutes if coils are connected in series
(C) In 10 minutes if coils are connected in parallel
(D) In 5 minutes if coils are connected in parallel

11. In a meter bridge experiment, the metre wire has been chosen such that its cross section is uniform but its resistivity increase, linear from ρ_0 to $2\rho_0$ as you go from left end to right end neglect any end correction, choose the correct statement(s) :-
- (A) If $x = 10\Omega$, Null point is obtained closer to A on wire
 (B) If null point is obtained at mid point of AB wire x will be more than 10Ω
 (C) For given value of x null point will not change, even if we change the magnitude of ρ_0
 (D) We may get two null points for a particular value of x
12. The current in a wire of resistance 10Ω varies as shown. Select correct alternative(s).



- (A) The heat produced in the wire in 4 s is 852.5 J
 (B) The heat produced in the wire in 4 s is 850 J
 (C) The charge flows across any section of the wire in 4 s is 15 C
 (D) The charge flows across any section of the wire in 4 s is 15.5 C

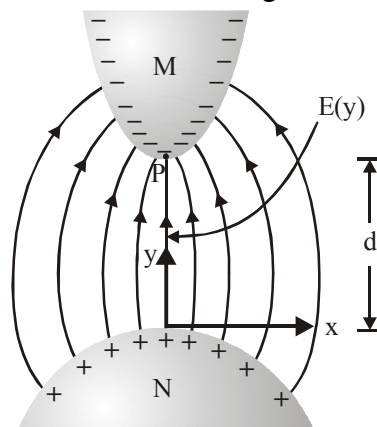
Linked Comprehension Type (1 Para × 2Q.)
(Single Correct Answer Type)

[3 M (-1)]

Paragraph for Questions 13 and 14

Two conductors carry charges $+Q$ and $-Q$. They create a nonuniform electric field as shown in figure. A coordinate system is placed on the positive conductor, and the electric field strength along the y -axis

is found to vary according to the function $E(y) = QA\epsilon_0 \left(1 + \frac{y^2}{B}\right)$ where A and B are constant. The separation between the conductors is d as shown in figure.



13. The equivalent capacitance of the system is

- (A) $\frac{1}{A\epsilon_0 \left(d - \frac{d^3}{3B}\right)}$ (B) $\frac{1}{A\epsilon_0 \left(d + \frac{d^3}{3B}\right)}$ (C) $A\epsilon_0 \left(d + \frac{d^3}{3B}\right)$ (D) $\frac{A\epsilon_0}{\left(d + \frac{d^3}{3B}\right)}$

14. Surface charge density at point P on conductor M is

(A) $QA \epsilon_0^2 \left(1 + \frac{d^2}{B}\right)$ (B) $2QA \epsilon_0 \left(1 + \frac{d^2}{B}\right)$ (C) $2QA \epsilon_0^2 \left(1 + \frac{d^2}{B}\right)$ (D) $QA \epsilon_0 \left(1 + \frac{d^2}{B}\right)$

Matching list based comprehension Type ($4 \times 4 \times 4$)

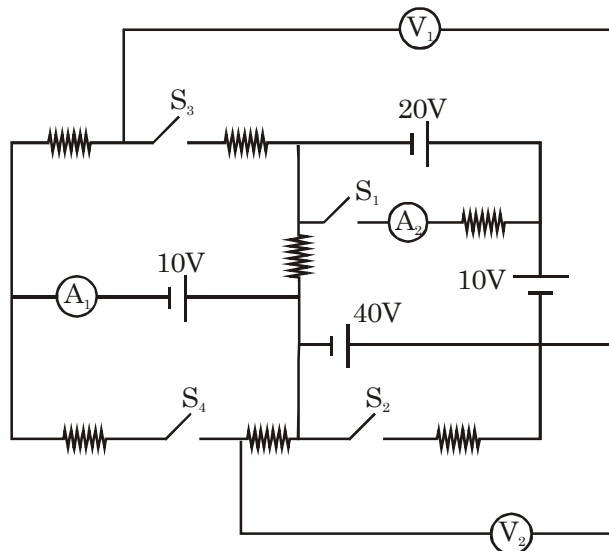
1 Table $\times$ 3 Q. [3(-1)]

Single option correct

(Three Columns and Four Rows)

Answer Q.15, Q.16 and Q.17 by appropriately matching the information given in the three columns of the following table.

In given circuit, all batteries, voltmeters and ammeters are ideal. Voltmeters and ammeters have only +ive scale for reading. Resistance of each resistor is 5Ω . Reading of voltmeters are V_1 and V_2 and of ammeters are A_1 and A_2 . Initially all switches are closed.



Column-I

$|A_1 - A_2|$

- (I) 3 A
(II) 0 A
(III) 5 A
(IV) 1 A

Column-II

V_1

- (i) 20 V
(ii) 30 V
(iii) 50 V
(iv) 40 V

Column-III

V_2

- (P) 45 V
(Q) 35 V
(R) 40 V
(S) 25 V

15. Only switch S_2 is open

- (A) (I) (iii) (S) (B) (IV) (ii) (P) (C) (I) (iii) (P) (D) (IV) (ii) (S)

16. Only switch S_3 is open :

- (A) (III) (iii) (R) (B) (IV) (ii) (S) (C) (I) (iii) (P) (D) (II) (iv) (Q)

17. Only switch S_1 is open :

- (A) (III) (iii) (P) (B) (I) (iii) (R) (C) (IV) (ii) (P) (D) (III) (ii) (P)

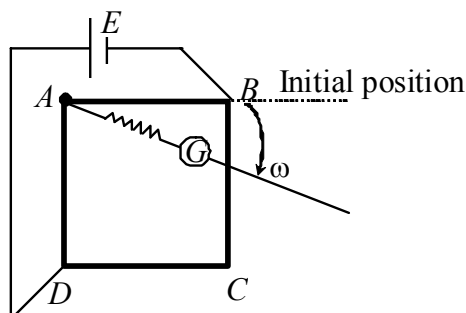
SECTION-II

Numerical Answer Type Question

1Q.[3M(0)]

(upto second decimal place)

1. $ABCD$ is a square frame made from different wires of same length and each having different uniform resistance per unit length. Resistances of wires forming sides AB , BC , CD and DA are 100Ω , 400Ω , 500Ω and 200Ω respectively. An ideal cell is connected across B and D . A straight conducting wire containing a resistance and a galvanometer in series starts rotating about pivoted point A from initial position as shown with uniform angular velocity $\omega = \frac{\pi}{360}$ rad/sec. One end of the straight wire (rotating) is pivoted at A and other end always in sliding contact with a side of the square. The time (in second) after start when galvanometer shows zero deflection is.



SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

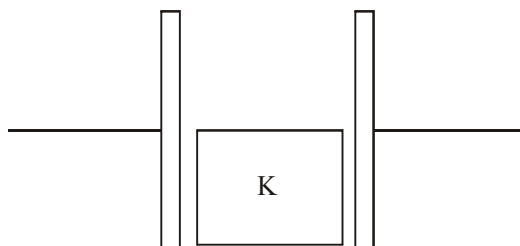
1. Two identical parallel plate capacitors of capacitance $C = 0.01$ F each are connected in parallel and charged to voltage 300 V and disconnected from the power source. The plates of one of the capacitors were spaced at a distance of twice the original. What is the charge passed (in C) during this time through the connecting wires?

SECTION-I

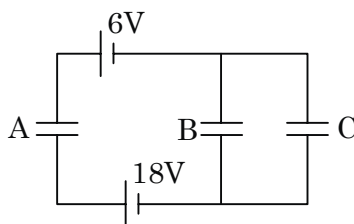
Single Correct Answer Type

1 Q. [3 M (-1)]

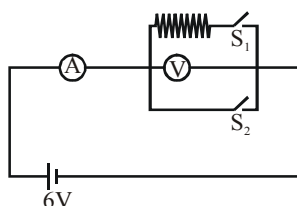
1. A parallel plate capacitor is charged from a battery and then isolated from it. Now a dielectric slab of dielectric constant K is introduced in the region between plates, filling half of it as shown. The electric field intensity in dielectric region is E_1 & that in air between plates is E_2 .



- (A) $E_1 = \frac{E_2}{K}$ (B) $E_1 = \frac{E_2}{K-1}$ (C) $E_1 = E_2 \left(1 - \frac{1}{K}\right)$ (D) $E_1 = E_2$
2. Three capacitors A, B and C each of capacitance $1 \mu\text{F}$ are connected in a circuit as shown in figure. The energy stored in the capacitor C is

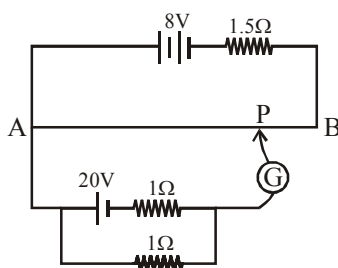


- (A) $4 \mu\text{J}$ (B) $6 \mu\text{J}$ (C) $8 \mu\text{J}$ (D) $2 \mu\text{J}$
3. In a planar vacuum diode, the dependence of the current on voltage has the form $I = cV^{3/2}$ where c is a constant. How many times will the pressure on the anode arising from the impact of electrons on its surface increase, if the voltage on the diode were to double? Initial velocity of electrons emitted from the cathode is negligible.
- (A) four times (B) two times (C) $\sqrt{2}$ times (D) $2\sqrt{2}$ times
4. An ammeter and a voltmeter are initially connected in series to a battery of zero internal resistance. When switch S_1 is closed the reading of the voltmeter becomes half of the initial, whereas the reading of the ammeter becomes double. If now switch S_2 is closed then reading of ammeter becomes :-



- (A) $\frac{3}{2}$ times the initial value (B) $\frac{3}{2}$ times the value after closing S_1
- (C) $\frac{3}{4}$ times the value after closing S_1 (D) $\frac{3}{4}$ times the initial value

5. Between the plates of a parallel-plate capacitor, there is a metallic plate whose thickness takes up 60% of the capacitor gap. When that plate is absent the capacitor has a capacity $C = 20\text{nF}$. The capacitor is connected to a dc voltage source of voltage $=100\text{ V}$. The metallic plate is slowly extracted from gap. Find the mechanical work (in μJ) performed in the process of plate extraction.
(A) 150 (B) 100 (C) 200 (D) 250
6. A potentiometer wire has length of 10 m and resistance 10Ω as shown in figure. A circuit is setup as shown. For null deflection in galvanometer length AP is :

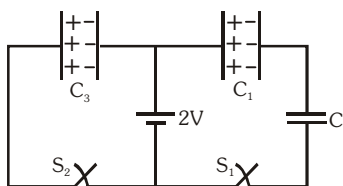


- (A) 2m (B) 4.5 m (C) 6.5 m (D) None of these

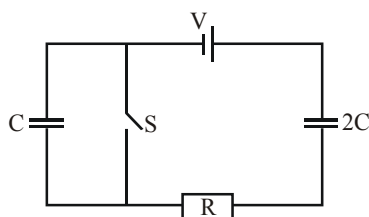
Multiple Correct Answer Type

5 Q. [4 M (-1)]

7. Three capacitors $C_1 = 2\mu\text{F}$, $C_2 = 2\mu\text{F}$ and $C_3 = 3\mu\text{F}$ having initial charges $4\mu\text{C}$, zero and $1\mu\text{C}$ connected through a battery of emf 2V as shown, on closing the switches S_1 and S_2 .



- (A) charge on capacitor C_1 is $4\mu\text{C}$ (B) charge on capacitor C_2 is 0
(C) charge on capacitor C_3 is $6\mu\text{C}$ (D) charge flown through S_2 is $5\mu\text{C}$
8. A series circuit consists of two capacitors, a resistor, and an ideal voltage source.



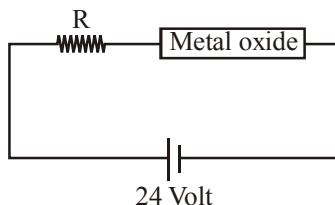
(A) The charge flown through the battery after closing the switch is $\frac{4CV}{3}$.

(B) Work done by the battery after closing the switch is $\frac{4CV^2}{3}$

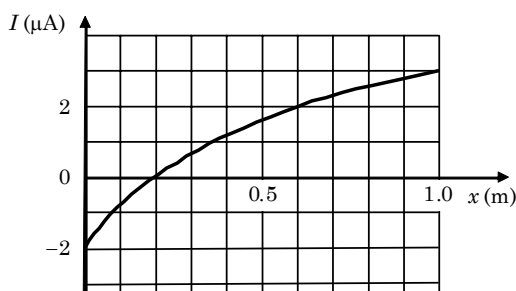
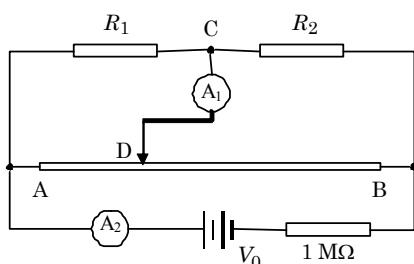
(C) The change in potential energy of the capacitors after closing the switch is $\frac{2CV^2}{3}$

(D) Heat generated in the circuit after closing the switch is $\frac{2CV^2}{3}$

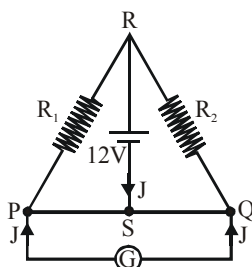
9. In the circuit shown potential difference across the rod of metal oxide varies as $V = Ki^2$. Where i is current in the rod and $K = 1 \text{ volt/amp}^2$. It is given that power dissipated in rod is double the power dissipated in R . Then :



- (A) Value of $R = 2\Omega$ (B) Current in the circuit is 2A
(C) Power dissipated in R is 32 W (D) If R is increased current in the circuit decreases.
10. In the network shown, R_1 and R_2 are unknown resistances, uniform rod AB is of length 1.0 m and resistance $1.0 \text{ k}\Omega$ and A_1 and A_2 are ideal ammeters. The adjacent graph shows relation between current in the ammeter A_1 and distance x of the jockey D from the end A. Based on the above information, mark **CORRECT** statements.



- (A) Resistance R_1 is closest to $0.2 \text{ k}\Omega$ (B) Ratio of resistance R_1 to R_2 is equal to 1 : 4.
(C) Emf V_0 of the battery is closest to 3.6 V (D) None of the above is correct.
11. In the circuit shown $R_1 = R_2 = 10\Omega$ and resistance per unit length of wire PQ = $1\Omega/\text{cm}$ and length PQ = 10 cm. If R_2 is made 20Ω then to get zero deflection in galvanometer. S is midpoint of wire PQ :-



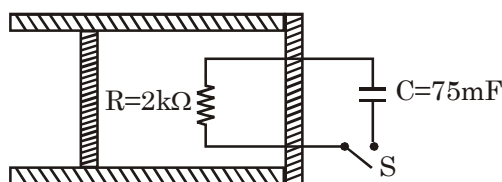
- (A) The jockey at P can be moved towards S by 2 cm
(B) The jockey at Q can be moved towards S by 2cm
(C) The jockey at S can be moved towards P by a distance of $5/3 \text{ m}$
(D) The jockey at all position fixed and R_1 should be made 20Ω

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]

Paragraph for Question Nos. 12 to 14

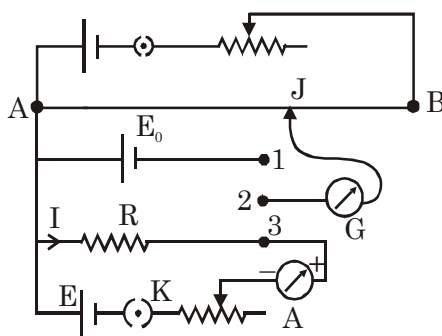
A cylinder fitted with a piston which can slide without friction contains one mole of an ideal gas. The walls of the cylinder and piston are adiabatic. The cylinder contains a resistor of resistance $R = 2 \text{ k}\Omega$ which is connected to a capacitor of capacity $C = 75 \text{ mF}$. Initially potential difference across capacitor is $\left(\frac{640}{3}\right) \text{ V}$ and switch is open. When switch is closed for $(2.5 \ln 4) \text{ min}$, the gas expands isobarically and its temperature increases by 72 K . Heat loss through the wires is negligible. ($R = 8.3 \text{ J mole}^{-1} \text{ K}^{-1}$)



12. Work done by the gas is approximately
(A) 0.2 kJ (B) 0.4 kJ (C) 0.6 kJ (D) 0.8 kJ
13. Increment in internal energy of gas is
(A) 1 kJ (B) 2 kJ (C) 3 kJ (D) 4 kJ
14. The value of adiabatic exponent γ for gas is
(A) 1.1 (B) 1.3 (C) 1.6 (D) 2.5

Paragraph from Question no. 15 and 16

A potentiometer can also be used to calibrate an ammeter or a voltmeter. Here we require a standard cell of which emf is known say E_0 . This cell is connected to secondary branch of potentiometer and we balance this standard cell against a length ℓ_0 on potentiometer wire. (by connecting 1 & 2). Now standard cell is removed and the galvanometer end 2 is connected to point 3 on circuit and balanced against length ℓ_1 . Current I can be obtained using principle of potentiometer as well as from ammeter. I as given by potentiometer is true value and ammeter reading may have error.



15. If resistance R is 15Ω , $E_0 = 5 \text{ V}$, $\ell_0 = 2.5 \text{ m}$, $\ell_1 = 3 \text{ m}$, current I as read by ammeter is 0.45 A
(A) The ammeter reads more than true value by 0.05 A
(B) The ammeter reads less than true value by 0.01 A
(C) The ammeter reads less than true value by 0.05 A
(D) The ammeter reads more than true value by 0.01 A
16. If the same error is obtained for another circuit where $E_0 = 10 \text{ V}$, $\ell_0 = 3 \text{ m}$, $\ell_1 = 1.5 \text{ m}$, current I in ammeter is 0.55 A , the resistance R used is :

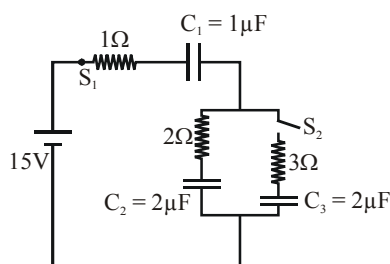
- (A) 10Ω (B) 8Ω (C) $\frac{100}{9} \Omega$ (D) 8.1Ω

SECTION-III

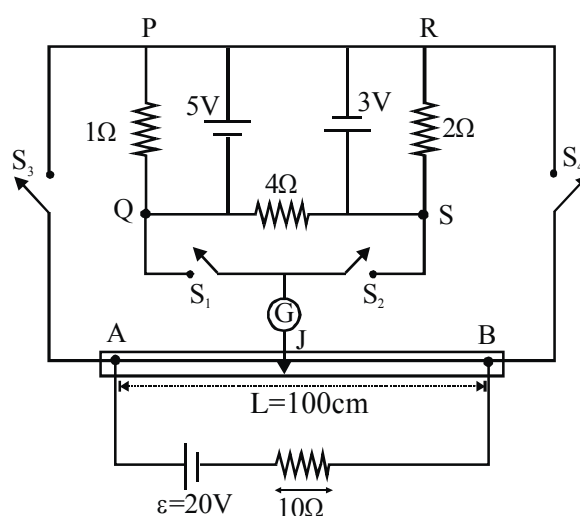
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1. In the following circuit, switch S_1 remains closed for a long time and switch S_2 remains open. The system reaches a steady state. Now we close switch S_2 also, keeping S_1 close. In the new steady state the magnitude of change in charge on capacitor C_2 of $2\ \mu\text{F}$ is $q\ \mu\text{C}$. Find q .



2. Two wires of the same length but of different square cross sections are made from the same material. The sides of the cross sections of the first and second wires are $d_1 = 1\ \text{mm}$ and $d_2 = 4\ \text{mm}$. The current required to fuse the first wire is $I_1 = 1.0\ \text{A}$. Determine the least current I_2 (in A) required to fuse the second wire, assuming that the amount of heat dissipated to the ambient per second obeys the law $Q = kS(T - T_{\text{am}})$, where S is the surface area of the wire, T is its temperature, T_{am} is the temperature of the ambient away from the wire, and k is the proportionality factor which is the same for the two wires.
3. Figure shows a potentiometer connected to an external circuit. At an instant either switch S_1 and S_3 is closed or S_2 and S_4 is closed. When switch S_1 and S_3 is closed null point is attained at $J_1(AJ_1 = \ell_1)$ and when S_2 and S_4 is closed it is attained at $J_2(BJ_2 = \ell_2)$. Find the value of $3\left(\frac{\ell_1}{\ell_2}\right)$.

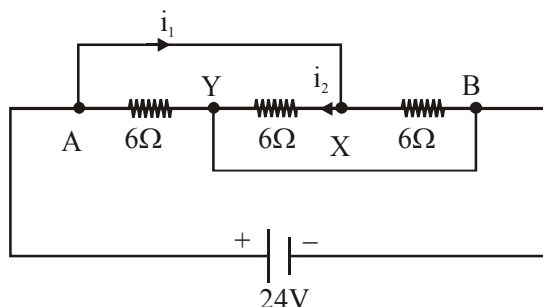


SECTION-I

Single Correct Answer Type

7 Q. [3 M (-1)]

1. In the figure shown, the current in the wires which join the points A and X is i_1 and the points X and Y is i_2 . Then i_1 & i_2 are respectively ?



- (A) $i_1 = 12 \text{ A}, i_2 = 4 \text{ A}$ (B) $i_1 = 8 \text{ A}, i_2 = 4 \text{ A}$ (C) $i_1 = 4 \text{ A}, i_2 = 4 \text{ A}$ (D) $i_1 = 3 \text{ A}, i_2 = 12 \text{ A}$

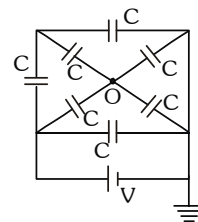
2. In the given circuit find potential at point 'O' :-

(A) $\frac{4V}{11}$

(B) $\frac{3V}{11}$

(C) $\frac{9V}{11}$

(D) None of these



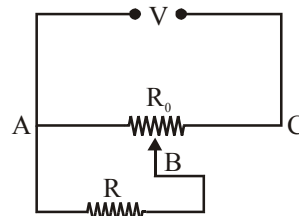
3. A resistance of $R \Omega$ draws current from a potentiometer. The potentiometer has a total resistance $R_0 \Omega$ (Figure). A voltage V is supplied to the potentiometer. An expression for the voltage across R when the sliding contact is in the middle of the potentiometer is

(A) $V_1 = \frac{2VR}{R_0 + 2R}$

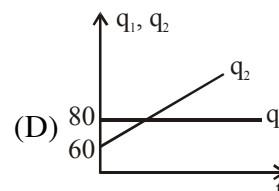
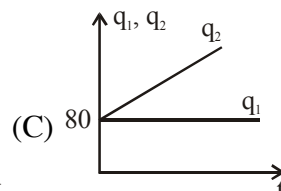
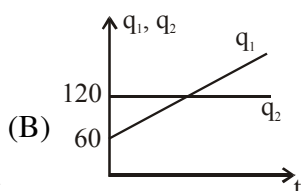
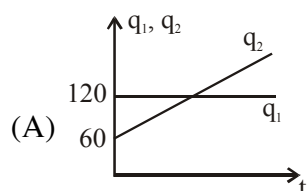
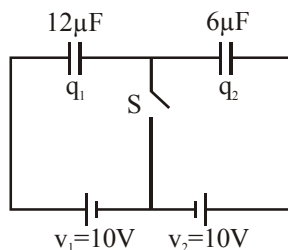
(B) $V_1 = \frac{2VR}{R_0 + 4R}$

(C) $V_1 = \frac{VR}{R_0 - 2R}$

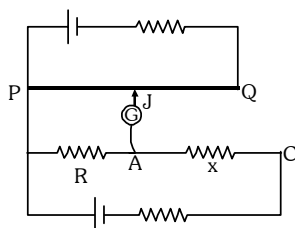
(D) $V_1 = \frac{VR}{R_0 + R}$



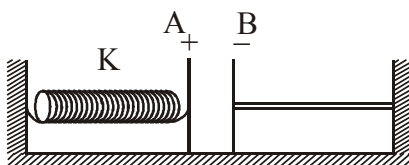
4. In the figure switch S is closed and steady state is reached. Now at $t = 0$ voltage V_2 is increased according to equation $V = t + 10$; upto the breakdown stage. The correct plot of charges q_1 and q_2 with respect to t will be :-



5. Circuit for the measurement of resistance by potentiometer is shown. The galvanometer is first connected at point A and zero deflection is observed at length $PJ = 10$ cm. In second case it is connect at point C and zere deflection is observed at a length 30cm from P. Then the unknown resistance X is



- (A) $2R$ (B) $\frac{R}{2}$ (C) $\frac{R}{3}$ (D) $3R$
6. A parallel plate capacitor is charged to potential V_0 . Separation between plates is d_0 . Now the distance between plates is changed periodically $d = d_0 + d_1 \sin \omega t$. Find value of d_1 if voltage of capacitor is given by $V = V_0 (1 + \frac{1}{2} \sin \omega t)$.
- (A) $d_1 = d_0$ (B) $d_1 = \frac{d_0}{2}$ (C) $d_1 = 2d_0$ (D) $d_1 = 0$
7. The figure present an arrangement in horizontal plane. It consists of a parallel plate capacitor with one of its square plate fixed by means of an insulating support. While the other plate is attached to the free end of a spring made of insulating material of force constant K . If the capacitor carries charge Q and battery is disconnected, mass and side length of plate A be M and L respectively, time period of oscillation of A (assuming that it does not strike B) would be(Ignore any friction)

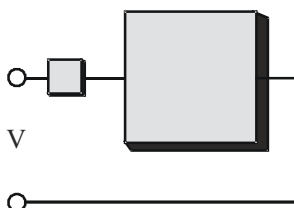


- (A) $2\pi \sqrt{\frac{M}{k}}$ (B) $2\pi \sqrt{\frac{k}{M}}$ (C) $2\pi \sqrt{\frac{2M\epsilon_0 L^3}{Q^2}}$ (D) $2\pi \sqrt{\frac{M}{k - \left(\frac{Q^2}{2\epsilon_0 L^3}\right)}}$

Multiple Correct Answer Type

4 Q. [4 M (-1)]

8. Square copper plates of equal thickness and having, area 1 cm^2 and 1 m^2 , are connected in series as shown in the Figure.



- (A) The resistance of the both the plates are equal
(B) The potential drop across both the plates are equal
(C) The current in both the plates are equal
(D) The total heat dissipated in both the plates per unit time are equal.

9. We have a parallel plate capacitor made of two conducting, circular sheets of radius $2R$ each. They are kept at a small separation d between them. Suppose, we insert a circular shape dielectric of radius R and thickness d , between the gap with its centre coinciding with the plate centres the dielectric constant is k . The correct statement is/are :

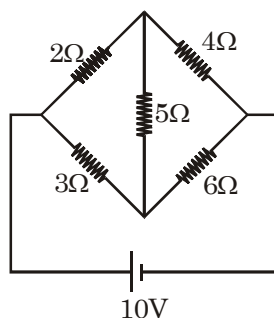
(A) Capacitance of the capacitor is $\frac{(k+3)\epsilon_0 \pi R^2}{d}$

(B) Capacitance of the capacitor is $\frac{k\epsilon_0 \pi (4R^2)}{d}$

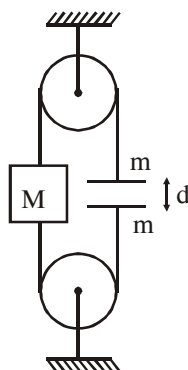
(C) Ratio of electric field in the region with dielectric & without dielectric is 1

(D) Ratio of electric field in the region with dielectric & without dielectric is 1 : k

10. Consider a circuit as shown in figure. Then choose **CORRECT** statement(s) :-



- (A) If 4Ω resistance is shorted (resistance less wire connected in parallel to it), the current through battery will increase.
- (B) If 4Ω resistor is shorted, the current through 5Ω resistance will increase.
- (C) If instead of 4Ω , 5Ω , resistor is shorted, current through battery will decrease.
- (D) If instead of 4Ω , 3Ω resistor is shorted, current through battery will decrease.
11. Two plates of mass m each are connected to mass M by two light inextensible strings. The pulleys are ideal. To hold system in equilibrium, we need to charge the plates by a battery of at least 8 volts. If distance between plates is $1\mu\text{m}$, capacitance formed is $1\mu\text{F}$, ($g = 10 \text{ m/s}^2$)



- (A) $M = 12.8 \text{ kg}$ (B) $m = 3.2 \text{ kg}$ (C) $M = 6.4 \text{ kg}$ (D) $m = 1.6 \text{ kg}$

Linked Comprehension Type

(1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question no. 12 and 13

When a battery of emf E and internal resistance r is connected to a load resistance R , current $I = \frac{E}{r + R}$

start in the circuit and battery start working. Chemical energy of battery start converting into heat, some part through load resistance and other part through internal resistance. Power supplied by battery to load resistance is $= I \Delta V$; ΔV is potential difference across the load resistance.

For two different value of resistance R_1 and R_2 , power supplied by battery to the load is same. Internal resistance of battery is 8Ω . We can choose only integer value of load resistance.

12. Maximum value of $R_1 + R_2$ is :

- (A) 16Ω (B) 65Ω (C) 8Ω (D) 13Ω

13. $R_1 + R_2$ can't be equal to :

- (A) 34Ω (B) 65Ω (C) 20Ω (D) 32Ω

Matching list based comprehension Type (4 × 4 × 4)

1 Table × 3 Q. [3(-1)]

Single option correct (Three Columns and Four Rows)

Answer Q.14, Q.15 and Q.16 by appropriately matching the information given the three columns of the following table.

A parallel plate capacitor has a plate area 'A' and plate separation 'd'. A dielectric slab of thickness ' $t \leq d$ ' is placed between the plates having same area of its face. The dielectric constant of the slab is 'k'. The dimensions of plate are given in **column-I**. In **column-II** is the values of k & t and in **column-III** is the value of charge Q on the plates.

Column-I

(I) $A = 10 \text{ cm}^2$, $d = 1 \text{ cm}$

(II) $A = 10 \text{ cm}^2$, $d = 0.1 \text{ cm}$

(III) $A = 20 \text{ cm}^2$, $d = 0.2 \text{ cm}$

(IV) $A = 100 \text{ cm}^2$, $d = 1 \text{ cm}$

Column-II

(i) $k = 1$, $t = d$

(ii) $k = 2$, $t = \frac{d}{2}$

(iii) $k = 10$, $t = \frac{d}{2}$

(iv) $k = 2$, $t = d$

Column-III

(P) $Q = 100 \mu\text{C}$

(Q) $Q = 10 \mu\text{C}$

(R) $Q = 1000 \mu\text{C}$

(S) $Q = 50 \mu\text{C}$

14. In which of the case the electric field within the dielectric slab will be the maximum ?

- (A) (IV), (iii), (Q) (B) (II), (ii), (Q) (C) (IV), (iii), (R) (D) (III), (iv), (S)

15. In which of the case the potential drop across the dielectric slab is the minimum

- (A) (II), (i), (P) (B) (I), (i), (Q) (C) (III), (iii), (R) (D) (IV), (iv), (S)

16. In which of the case the energy stored in the capacitor is the maximum ?

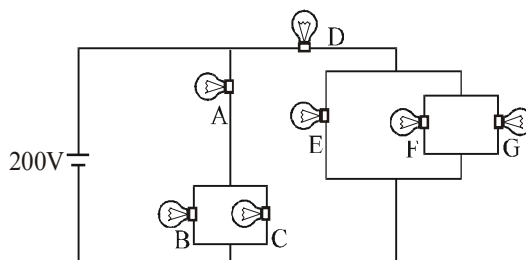
- (A) (I), (iii), (Q) (B) (II), (iv), (S) (C) (III), (ii), (S) (D) (IV), (i), (Q)

SECTION-III

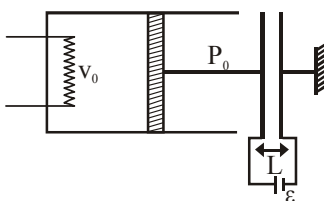
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

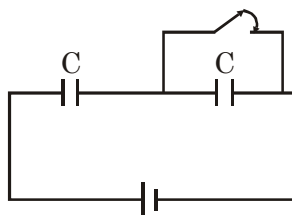
1. Circuit shows seven identical bulbs (A to G) connected through a battery of emf 200 V. The bulbs are rated as 200 V, 100 Watt. The power dissipated in the bulb which glow brightest is given by P watt. Fill the value of $\frac{2}{3}\sqrt{P}$.



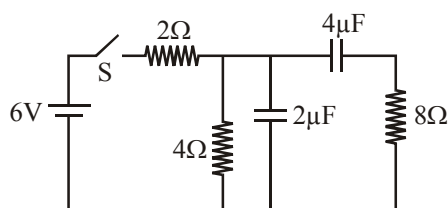
2. An ideal gas is enclosed in a cylinder fitted with a frictionless piston. The piston is connected with a light rod to one plate of capacitor whose other plate is fixed as shown. Initially the volume of the gas inside the cylinder is V_0 , pressure is P_0 , atmospheric pressure is P_0 , separation between the plates of capacitor is L , area of the piston as well as of the capacitor plates is A and emf of battery is ϵ . A heater supplies necessary heat to the gas such that equilibrium is achieved when pressure of the gas is given as $P = P_0 - \frac{n\epsilon_0\epsilon^2}{L^2}$ and piston is displaced towards right by a distance $\frac{L}{2}$. Find the value on n .



3. In the circuit shown, initially switch is closed and energy stored in the system (consisting of both capacitors) is U_1 . After opening the switch, energy stored in the system is U_2 . Then $\frac{U_1}{U_2}$?



4. In the circuit shown in figure the switch S is initially open and both the capacitors are initially uncharged. Find the ratio of current through 2Ω resistor, just after the switch S is closed and a long time after the switch S is closed.



PHYSICS

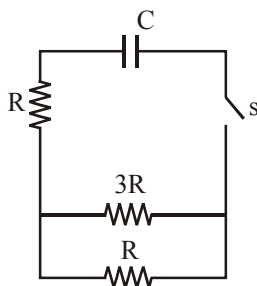
CLASS TEST # 37

SECTION-I

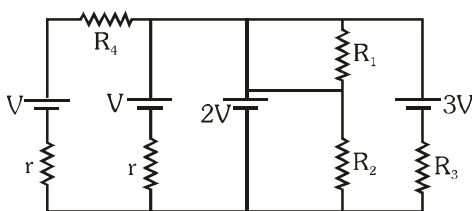
Single Correct Answer Type

6 Q. [3 M (-1)]

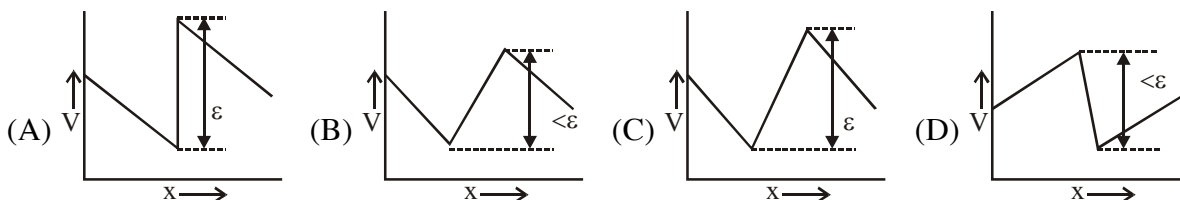
- An isolated parallel-plates capacitor has circular plates of radius 4.0 cm. Its gap is filled with a partially conducting material of dielectric constant K and conductivity $5.0 \times 10^{-4} \Omega^{-1} \text{ m}^{-1}$. When the capacitor is charged to a surface charge density of $15 \mu\text{C}/\text{m}^2$, the initial current between the plates is 1.0 A. Determine the value of dielectric constant K (approximately).
(A) $K = 1.32$ (B) $K = 4.26$ (C) $K = 8.74$ (D) $K = 2.1$
- Two parallel plate capacitors of capacitance C_0 and $2C_0$ are connected in parallel and charged to a potential difference V_0 . The battery is disconnected and the region between the capacitor plates of capacitance C_0 is completely filled with a dielectric of dielectric constant K . The potential difference across the capacitors, now becomes :
(A) $\frac{V_0}{(3K+1)}$ (B) $\frac{3V_0}{(K+2)}$ (C) $\frac{2V_0}{K}$ (D) $\frac{V_0}{(2K+1)}$
- In the figure, capacitor is completely charged and switch is closed at $t = 0$. The time after which the current from the capacitor becomes $1/4^{\text{th}}$ of its maximum value will be :-



- (A) $2 RC \ln 2$ (B) $\frac{7RC}{4} \ln 2$ (C) $RC \ln 2$ (D) $\frac{7RC}{2} \ln 2$
- In the circuit shown, if $R_1 : R_2 : R_3 : R_4 = 1 : 2 : 3 : 4$ then

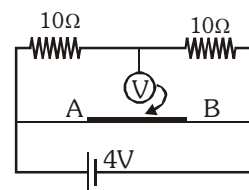


- (A) Ratio of current in R_2 and R_3 is 3 : 1 (B) Ratio of current in R_1 and R_3 is 1 : 2
(C) Ratio of current in R_1 and R_2 is 4 : 5 (D) current in R_1 is more than one in R_2
- The two ends of a uniform conductor are joined to a cell of emf ϵ and some internal resistance. Starting from the midpoint P of the conductor, we move in the direction of the current and return to P . The potential V at every point on the path is plotted against the distance covered (x). Which of the following best represents the resulting curve?



6. In the adjacent circuit, AB is a potentiometer wire of length 40 cm and resistance per unit length $50 \Omega/\text{m}$. As shown in the figure, the free end of an ideal voltmeter is touching the potentiometer wire. What should be the velocity of the jockey as a function of time so that reading in the voltmeter varies with time as $(2 \sin \pi t)$?

- (A) $(10 \pi \sin \pi t) \text{ cm/s}$ (B) $(10 \pi \cos \pi t) \text{ cm/s}$
(C) $(20 \pi \sin \pi t) \text{ cm/s}$ (D) $(20 \pi \cos \pi t) \text{ cm/s}$



Multiple Correct Answer Type

4 Q. [4 M (-1)]

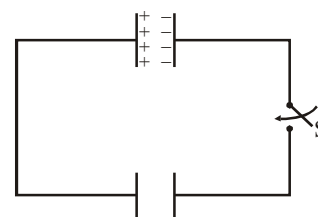
7. In the given figure, two identical capacitors are shown, with one of them having charge Q and the other being uncharged.

Process (A) : A dielectric of dielectric constant k is introduced between the plates of the charged capacitor and then the switch S is closed.

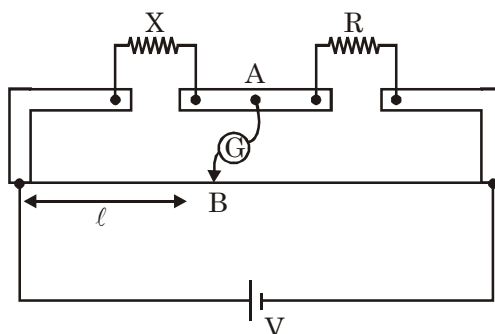
Process (B) : Switch S is closed without inserting the dielectric.

Find the **CORRECT** statement(s) :-

- (A) Energy lost during process (A) is less as compared to process (B)
(B) Charge transferred to the uncharged capacitor in process (A) is more as compared to process (B)
(C) Common potential difference after redistribution of charge is more in process (B) as compared to process (A)
(D) In process (B), electric field between the plates, in both the capacitors, becomes same, after redistribution of charge.



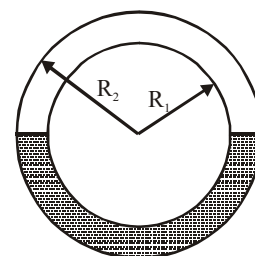
8. A meter bridge is used to predict the value of unknown resistance X , it is observed that current is flowing from A to B initially (when jockey is fixed at a point B). If the temperature of X is increased, then select possible option/s :



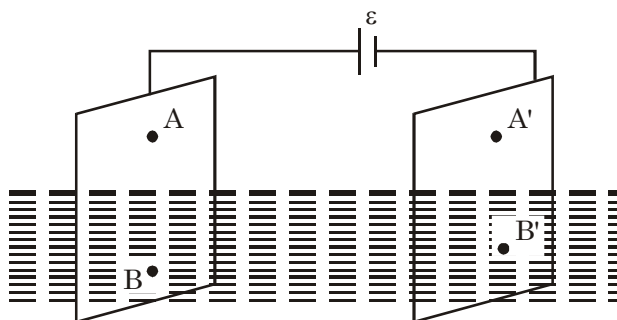
- (A) Current reduces in magnitude
(B) Current remain constant (for any raise in temperature)
(C) Current become zero
(D) Current flows in opposite direction.

9. A spherical capacitor is half filled with an oil of dielectric constant $k = 7$ and the rest is in vacuum. The inner and outer radii of the surfaces are $R_1 = 5 \text{ cm}$ and $R_2 = 6 \text{ cm}$ respectively. The inner and outer surface of capacitor is connected to a battery of emf ξ .

- (A) The potential difference across the dielectric side is same as that across vacuum side.
(B) The potential difference across the dielectric side is less than that across vacuum side.
(C) The electric field in the dielectric is same as that in the vacuum.
(D) The electric field in the dielectric is less than that in the vacuum.



10. A parallel plate capacitor is dipped in a dielectric liquid vertically, so that half of capacitor is in liquid. Plates of capacitor are joined to an ideal battery of emf ε . A, B, A' and B' are four points on plate at facing surface as shown. Which of the following is/are correct?



- (A) $V_A - V_{B'} = V_A - V_{A'}$
 (B) $V_A - V_{B'} = V_B - V_{A'}$
 (C) Electric field around A in air = Electric field around B' in liquid.
 (D) Electric field around B in liquid = Electric field around B' in liquid.

Linked Comprehension Type

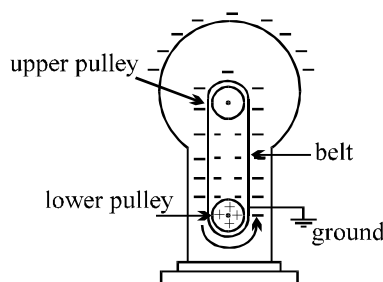
(1 Para × 5Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Questions No. 11 to 15

Van de Graaff generators like the one shown in figure are used to produce very high voltages. In the figure, the + signs represent positive charge and the – signs represent negative charge. In this common Van de Graaff generator, charge is separated by the frictional contact of the belt and the lower pulley shown. Positive charge collects on the lower pulley and an equal amount of negative charge spreads out along the inside of the belt. Electrons from the ground are attracted to the outside of the belt by the net positive charge on the lower portion of the belt pulley system. These electrons travel up the belt and are transferred to the dome, which is a hollow metal sphere. A high negative charge density can be built up on the dome, because the electrons from the outside of the belt experience no net force from the charge built up on the outside of the sphere.

The electric potential of the dome is $V = Er$ where E is electric field just outside the dome and r is the radius. The charges on the surface of the dome do not affect the electric field inside the cavity. The potential that can build up on the dome is limited by the dielectric strength of the air, which is about 30,000 V/cm for dry air at room temperature. When the electric field around the dome reaches the dielectric strength of the air, air molecules are ionized. This enables the air to conduct electricity.



Van de Graaff generators are routinely used in college physics laboratories. When a student gets within a few inches of a Van de Graaff generator, she may draw a spark with an instantaneous current of 10 amps and remain uninjured. An instantaneous current is the transfer of charge within 1 μ s

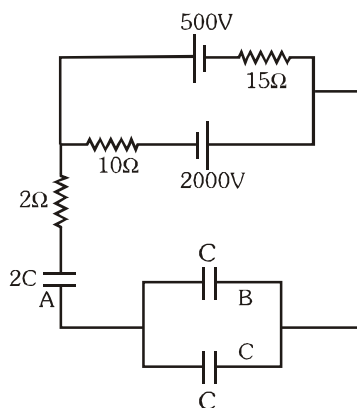
11. The 660 V rails on a subway can kill a person upon contact. A 10,000V Van de Graaff generator, however, will only give a mild shock. Which of the following best explains this seeming paradox?
 - (A) The generator provides more energy per charge, but since it has few charges it transfers a lesser amount of energy.
 - (B) The generator provides more energy, but since there is little energy per charge the current is small.
 - (C) Most of the energy provided by the generator is dissipated in the air because air presents a smaller resistance than the human body
 - (D) Most of the energy flows directly to the ground without going through the human body since the generator is grounded.
12. What is the maximum potential the dome, with a radius of 10 cm, can sustain in dry air?
 - (A) 3 kV
 - (B) 5 kV
 - (C) 300 kV
 - (D) 500 kV
13. Why is the potential of the dome limited by the dielectric strength of the air?
 - (A) Once the potential of the dome reaches the dielectric strength of the air, charge from the belt is repelled by the charge on the dome.
 - (B) Once the potential of the dome reaches the dielectric strength of the air, the air heats the metal of the dome, and it is no longer a good conductor.
 - (C) Once the air molecules become ionized, charge on the dome can leak into the air.
 - (D) Once the air molecules become ionized, they no longer conduct electricity.
14. Why does negative charge from the outside of the belt continue to build up on the outside of the dome instead of being repelled by the charge that is already there?
 - (A) The potential is zero inside the dome.
 - (B) The conducting dome shields the effects of the charges on the surface.
 - (C) There is only positive charge on the outside of the dome.
 - (D) Charge does not build up on the outside of the dome.
15. What is the work required to move a charge q from the top of the belt to the surface of the dome, if the amount of charge on the dome is Q and q was the only charge on the belt?
 - (A) 0
 - (B) $kQq/2r$
 - (C) kQq/r
 - (D) kq/r

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. If C is equal to $2000 \mu\text{F}$ then find the summation of charges on all capacitors after steady state.



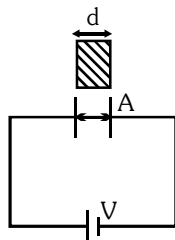
2. A cell of emf ϵ and internal resistance r is connected to an external resistance R . Efficiency of the power supplied by cell increases to double when R resistance changes from 1Ω to 3Ω , find r (in Ω)

SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

1. Some events related to a capacitor are listed in column-I. Match these events with their effect (s) in column-II.



Column-I (Events)

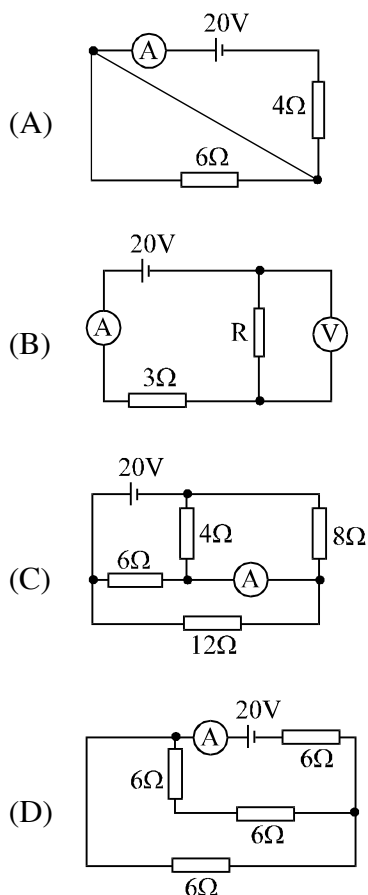
- (A) Insertion of dielectric while battery remain attached.
- (B) Removal of dielectric while battery is not present
- (C) Slow decrease in separation between plates while battery is attached.
- (D) Slow increase of separation between plates while battery is not present.

Column-II (Effects)

- (P) Electric field between plates changes
- (Q) Charge present on plates changes
- (R) Energy stored in capacitor increases
- (S) Work done by external agent is positive
- (T) None of these

2. Column I has four circuits each having an ammeter. Column II has four values of current in the ammeter. The ammeter has zero resistance. The voltmeter, in (B) has infinite resistance and a reading 8V. The resistance R has not been specified. Match the circuit with its correct ammeter reading.

Column I



Column II

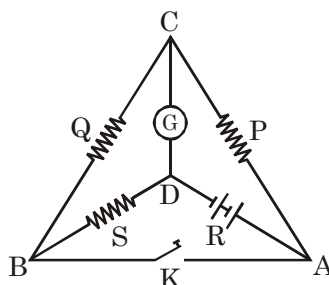
- (P) 0
- (Q) 2 Ampere
- (R) 4 Ampere
- (S) 5 Ampere

SECTION-I

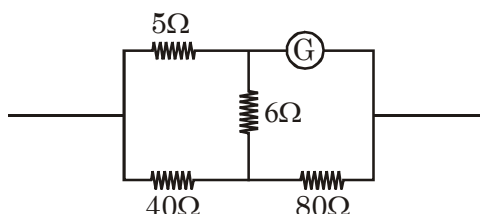
Single Correct Answer Type

4 Q. [3 M (-1)]

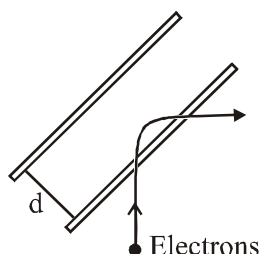
1. A network is arranged as in Mance's experiment. This is used to find the internal resistance of the battery R. The resistance of galvanometer, and of each of the other three arms, is 4 ohms. The current in the galvanometer with the key open, as well as when it is closed is 0.2A. The internal resistance of battery is :-



- (A) 4Ω (B) 2Ω (C) 8Ω (D) 1Ω
2. A galvanometer of $i_g = 1\text{mA}$ & resistance = 10Ω is connected in a circuit as shown. This combination (circuit) can be used as an ammeter or a voltmeter also. The range of ammeter (R_A) and voltmeter (R_V) are :-



- (A) $\frac{9}{8}\text{mA}$, 15 mV (B) 1 mA, 12 mV (C) $\frac{8}{3}\text{mA}$, 20 mV (D) $\frac{4}{3}\text{mA}$, 10 mV
3. A problem of practical interest is to make a beam of electrons turn at 90° corner. This can be done with the parallel plates shown in figure. An electron with kinetic energy $8.0 \times 10^{-17}\text{J}$ enters through a small hole in the bottom plate. The strength of electric field that is needed if the electron is to emerge from an exit hole 1.0cm away from the entrance hole, traveling at right angles to its original direction is..... $\times 10^5\text{N/C}$.



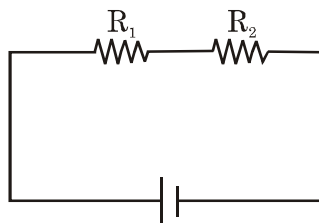
- (A) 3 (B) 1 (C) 6 (D) 2.5
4. A random shaped charged conducting body is placed in poor conducting dielectric medium of resistivity ρ and dielectric constant k . Discharging of body through medium is best explained by equation

(A) $q = Q_0 e^{-\frac{Kt}{\epsilon_0 \rho}}$ (B) $q = Q_0 e^{-\frac{t}{k\epsilon_0 \rho}}$ (C) $q = Q_0 e^{-\frac{\rho t}{\epsilon_0 k}}$ (D) $q = Q_0 e^{-\frac{t}{k^2 \epsilon_0 \rho}}$

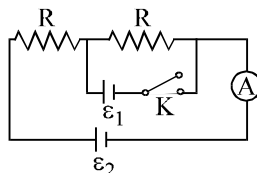
Multiple Correct Answer Type

7 Q. [4 M (-1)]

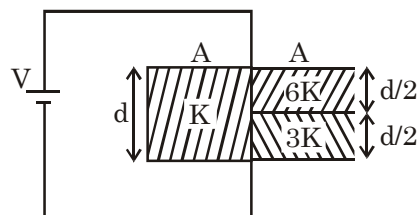
5. The circuit shown in the figure contains an ideal battery and two resistors, R_1 and R_2 . A voltmeter is used to measure the voltage across R_1 then across R_2 , then across the battery. Its reading are, respectively 2.0V, 3.0V and 6.0V. The actual voltage (before connecting voltmeter) across R_1 and R_2 , are V_1 and V_2 , respectively. Choose the **CORRECT** option(s)



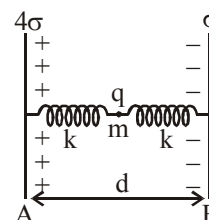
- (A) $V_1 = 2.4 \text{ V}$ (B) $V_1 = 3.6 \text{ V}$ (C) $V_2 = 2.4 \text{ V}$ (D) $V_2 = 3.6 \text{ V}$
6. In the given circuit, when key K is open, reading of ammeter is I. Now key K is closed then the **INCORRECT** statement(s) is/are:



- (A) If $\epsilon_1 > IR$, reading of the ammeter is less than I
(B) If $IR < \epsilon_1$, reading of the ammeter is greater than I
(C) If $\epsilon_1 < 2IR$, reading of the ammeter will be zero
(D) Reading of ammeter will not change
7. 3 dielectric slabs (each of area A) are filled between capacitor of plate area 2A. Capacitor is connected with battery of emf V. Electric field and electric field energy stored in dielectric K, 6K, 3K are E_1, E_2, E_3 and U_1, U_2, U_3 respectively :

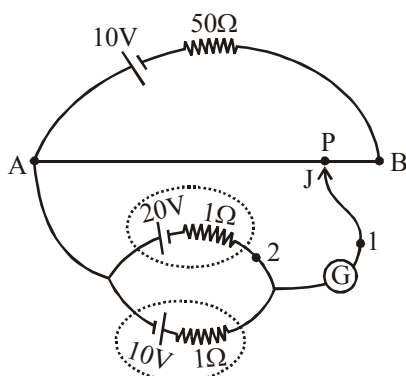


- (A) Total energy stored in capacitors is $\frac{5AK\epsilon_0 V^2}{2d}$
(B) $E_1 : E_2 : E_3 :: 3 : 2 : 4$
(C) $U_1 : U_2 : U_3 :: 3 : 4 : 8$
(D) Capacitance of capacitor is $\frac{5AK\epsilon_0}{d}$.
8. Two large conducting plates having surface charge densities 4σ and $-\sigma$ respectively, are fixed 'd' distance apart. A small test charge q of mass m is attached to two identical non conducting springs as shown in figure with spring in natural length. Then q will [neglect gravity] :-

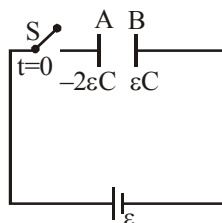


- (A) Perform S.H.M. with angular frequency $\sqrt{\frac{2k}{m}}$
(B) Perform S.H.M. with amplitude $\frac{5\sigma q}{2k\epsilon_0}$
(C) Not perform S.H.M., but will have a periodic motion
(D) Perform S.H.M. with amplitude $\frac{5\sigma q}{4k\epsilon_0}$

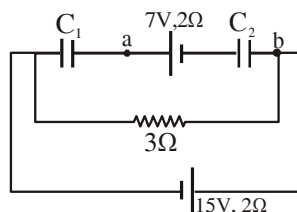
9. Figure shows the circuit diagram of a potentiometer apparatus. P is the point, where jockey would touch potentiometer. (Length of potentiometer wire = 10 m, resistance of potentiometer wire = $450\ \Omega$ and 10V battery is ideal.)



- (A) Potential gradient of potentiometer wire at null deflection would be 0.9 V/m .
 (B) For any location of P, we can not have null deflection point.
 (C) For a particular point P, current at point 1 can be made zero, but never at point 2.
 (D) For P coinciding with B current at point 1 would be in upwards direction.
10. A parallel plate capacitor of capacitance 'C' has charges on its plates initially as shown in the figure. Now at $t = 0$, the switch 'S' is closed. Select the **CORRECT** alternative(s) for this circuit diagram.



- (A) In steady state the charges on the outer surfaces of plates 'A' and 'B' will be same in magnitude and sign.
 (B) In steady state the charges on the outer surfaces of plates 'A' and 'B' will be same in magnitude and opposite in sign.
 (C) In steady state the charges on the inner surfaces of plates 'A' and 'B' will be same in magnitude and opposite in sign.
 (D) The work done by the battery by the time steady state is reached is $\frac{5\varepsilon^2 C}{2}$
11. In the figure shown, $C_1 = 11\ \mu\text{F}$ and $C_2 = 5\ \mu\text{F}$, then at steady state:



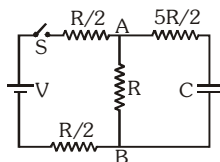
- (A) the potential difference across C_1 is 5V
 (B) the potential difference across C_2 is 2V
 (C) the potential difference between points a and b is -4V
 (D) the potential difference between the terminals of 15 V battery is 9V

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Questions no. 12 to 14

In the circuit shown in figure, the battery is an ideal one with emf V . The capacitor is initially uncharged. The switch S is closed at time $t = 0$.



12. The charge Q on the capacitor at time t is—

- (A) $\frac{CV}{2} \left(1 - e^{-\frac{t}{RC}} \right)$ (B) $\frac{CV}{2} \left(1 - e^{-\frac{t}{3RC}} \right)$ (C) $\frac{CV}{2} \left(1 - e^{-\frac{2t}{5RC}} \right)$ (D) $\frac{CV}{2} \left(1 - e^{-\frac{2t}{9RC}} \right)$

13. The current in AB at time t is—

- (A) $\frac{V}{2R} \left(1 - e^{-\frac{t}{3RC}} \right)$ (B) $\frac{2V}{R} \left(1 - e^{-\frac{t}{3RC}} \right)$ (C) $\frac{2V}{R} \left(1 - e^{-\frac{t}{6RC}} \right)$ (D) $\frac{V}{2R} \left(1 - e^{-\frac{t}{6RC}} \right)$

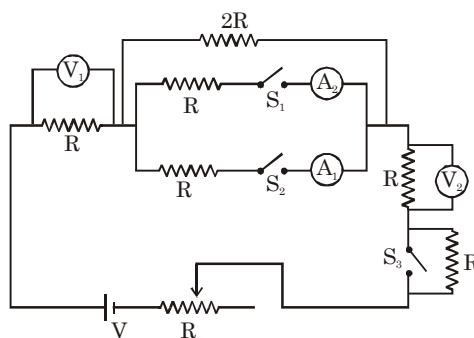
14. What is its limiting value i_{AB} at $t \rightarrow \infty$?

- (A) $\frac{V}{2R}$ (B) $\frac{V}{R}$ (C) $\frac{2V}{R}$ (D) $\frac{V}{3R}$

Matching List Type (4 × 4)

1 Q. [3 M (-1)]

15. All ammeters and voltmeters are ideal. The Rheostat can have maximum resistance R . The switch mentioned in list-I is kept closed and the reading of voltmeter/ammeter is observed while changing the Rheostat.



List-I

- (P) When only S_1 is closed and Rheostat is decreased
(Q) When S_1 and S_2 are closed and Rheostat is increased.
(R) When only S_3 is closed and Rheostat is decreased
(S) When all switches are closed and Rheostat is increased

List-II

- (1) Reading of voltmeter V_2 will decrease.
(2) Reading of ammeter A_2 will remain unchanged.
(3) Reading of voltmeter V_1 lies between $\frac{V}{5} \leq V_1 \leq \frac{V}{3}$.
(4) Reading of A_1 and A_2 are same.
(5) Reading of V_1 and V_2 are same.

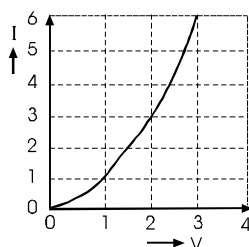
- (A) $P \rightarrow 2,4$; $Q \rightarrow 1,2,3,5$; $R \rightarrow 1,2,3$; $S \rightarrow 2,4$
(B) $P \rightarrow 1,5$; $Q \rightarrow 1,3,5$; $R \rightarrow 3,4,5$; $S \rightarrow 1,4,5$
(C) $P \rightarrow 3,5$; $Q \rightarrow 1,3,4,5$; $R \rightarrow 2,3,4,5$; $S \rightarrow 1,4,5$
(D) $P \rightarrow 1,3$; $Q \rightarrow 1,2,5$; $R \rightarrow 1,2,3$; $S \rightarrow 1,3,4,5$

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1 Q. [3(0)]

1. A cell of emf 3V and internal resistance 0.75Ω is connected to a non-linear conductor whose V-I graph is shown in figure. Obtain graphically the terminal voltage of the cell (in V).

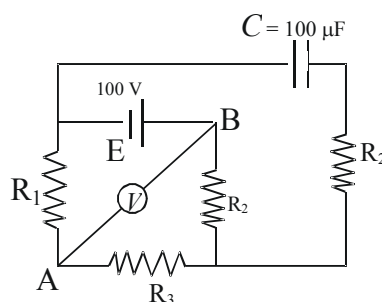


SECTION-III

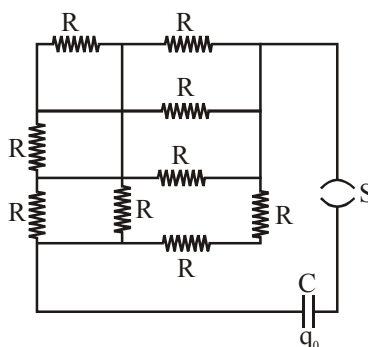
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

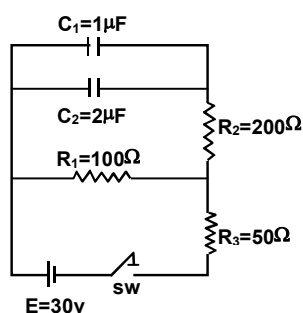
1. In the circuit shown in figure E is a battery of emf 100 V. Resistance $R_1 = 100\Omega$, $R_2 = 200\Omega$ and $R_3 = 200\Omega$. The voltmeter resistance is 200Ω . The voltmeter shows $400/x$ volt in steady state. Find the value of x.



2. At $t = 0$ switch is closed. The time at which charge on capacitor will become half of its initial value is t, find t ($RC \ln 2 = 3.4$ sec).



3. The circuit is closed for long time so that the capacitors are fully charged. Switch is opened at a certain instant. The heat developed in resistor R_2 after that instant is $K \times 10^{-4}$ J. Then find the value of K.

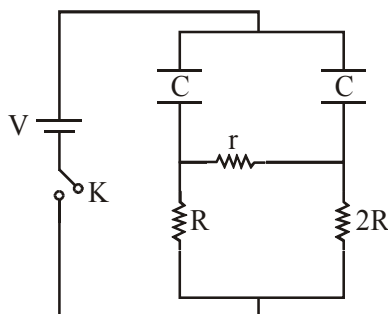


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Consider the situation shown in diagram. Both capacitors are having same capacitance C . Capacitors are initially uncharged. Switch K is closed at time $t = 0$ & after a long time capacitors are fully charged. Column-I represents value of r and Column-II represents charge passing through r during complete charging process. Now match the followings :



Column-I

- (A) $r = 0$
(B) $r = R$
(C) $r = 2R$
(D) $r = 3R$

Column-II

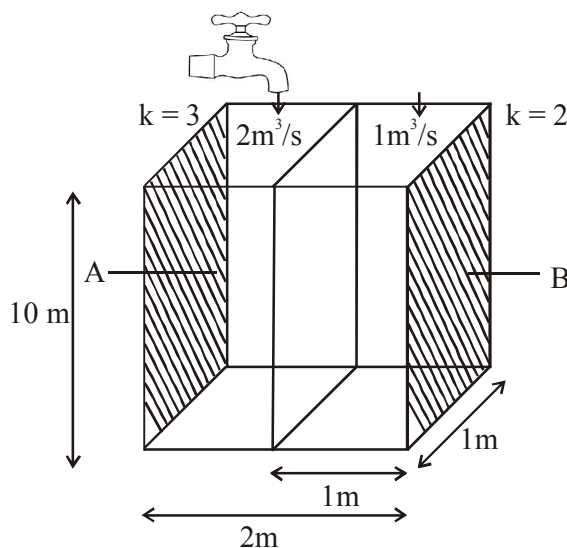
- (P) $\frac{CV}{6}$
(Q) $\frac{CV}{5}$
(R) $\frac{CV}{4}$
(S) $\frac{CV}{3}$
(T) $\frac{CV}{2}$

SECTION-I

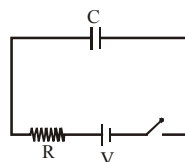
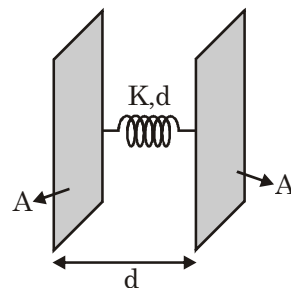
Single Correct Answer Type

5 Q. [3 M (-1)]

1. A thin metallic partition of negligible thickness is inserted between two shaded metallic plates as shown. The remaining ends are then packed with insulating plates to form a container like structure. 2 taps shown are opened at $t = 0$ and finally closed at $t = 5$ s. Find capacitance of system between A and B after closing taps. (Assume liquid to be non conducting) Volumetric flow rates and dielectric constants of liquid are given.

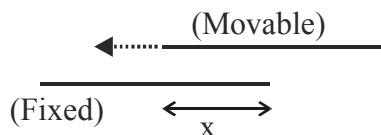


- (A) $8.62 \times 10^{-11} \text{F}$ (B) $8.85 \times 10^{-10} \text{F}$ (C) $4.42 \times 10^{-10} \text{F}$ (D) $4.42 \times 10^{-11} \text{F}$
2. Consider the figure below, initially distance between plates is 'd' and 'A' is area of plates. In this state spring is at its natural length. Now, charges $+Q$ & $-Q$ are given to plates and system is allowed to reach mechanical equilibrium. Find potential difference between plates in this state of equilibrium.
- (A) $V = \frac{Qd}{A\epsilon_0} \left(1 - \frac{Q^2}{2\epsilon_0 AKd} \right)$ (B) $V = \frac{Qd}{2A\epsilon_0} \left(1 - \frac{Q^2}{2\epsilon_0 AKd} \right)$
- (C) $V = \frac{Qd}{A\epsilon_0} \left(1 - \frac{Q^2}{4\epsilon_0 AKd} \right)$ (D) $V = \frac{Qd}{A\epsilon_0} \left(1 - \frac{Q^2}{\epsilon_0 AKd} \right)$
3. In an RC circuit, the time required for the charge on a capacitor to build up to a given fraction of its equilibrium value, is independent of :
- (A) the value of the applied emf to the circuit
(B) the value of C
(C) the value of R
(D) none of the above
4. In the circuit shown capacitor is initially uncharged and switch is closed at $t = 0$. Find the time at which power dissipated in resistor is equal to rate of energy change in capacitor :-



- (A) RC (B) $RC \ln 2$ (C) $RC \ln 3$ (D) $2RC \ln 2$

5. A parallel-plate capacitor consists of a fixed plate and a movable plate that is allowed to slide in the direction parallel to the plates. Let x be the distance of overlap, as shown in the figure. The separation between the plates is fixed. Assume that the plates are electrically isolated, so that their charges $\pm Q$ are constant. Force on the movable plate is proportional to :-

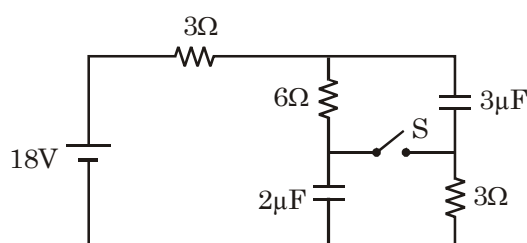


- (A) x^{-2} (B) x^{-1} (C) x^0 (D) x

Multiple Correct Answer Type

5 Q. [4 M (-1)]

6. The given circuit is in steady state with the switch closed. At $t = 0$ switch is opened, which of the following is/are correct ?



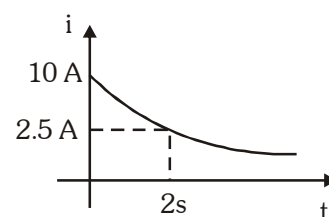
- (A) Charge on $2\mu\text{F}$ before the switch is open is $9\mu\text{C}$
 (B) Charge on $3\mu\text{F}$ before the switch is open is $27\mu\text{C}$
 (C) voltage across $3\mu\text{F}$, a long time after the switch is opened is 18V .
 (D) voltage across $2\mu\text{F}$, a long time after the switch is opened is 18V
7. The figure, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance of resistance 10Ω .

- (A) The initial potential difference across the capacitor is 100 volt .

- (B) The capacitance of the capacitor is $\frac{1}{10\ell n 2}\text{F}$.

- (C) The total heat produced in the circuit will be $\frac{500}{10\ell n 2}\text{ joules}$.

- (D) The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ell n 2}\text{ second}$.



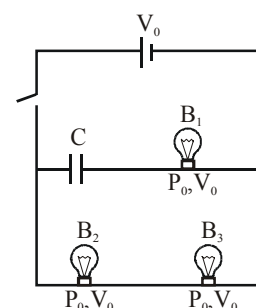
8. An uncharged capacitor is connected in circuit as shown in figure. Power ratings of bulbs are given. At $t = 0$ switch is closed :-

- (A) At $t = 0$, power consumption in circuit is $\frac{3P_0}{2}$

- (B) After a long time power consumption in circuit is $\frac{P_0}{2}$

- (C) Brightness of B_1 decreases with time

- (D) Initially brightness of B_2 is less than B_1 but later B_2 will be brighter



9. Figure shows the polarity and charge on capacitor with K open.

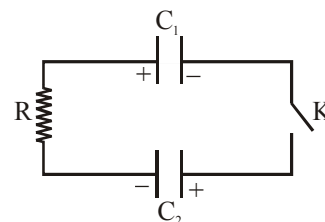
(Given : $C_1 = 2F$, $Q_1 = 4C$, $C_2 = 4F$, $Q_2 = 16C$ and $R = 10\Omega$)

(A) After closing the key, electrostatic electric field between plates of one of capacitor momentarily becomes zero at some instant.

(B) Overall, the circuit would suffer heat loss.

(C) Final charge on $2F$ capacitor would be $4C$ (one of the plates) with polarity reversed with respect to initial condition.

(D) Final charge on $4F$ capacitor would be $8C$ with polarity reversed with respect to initial condition.



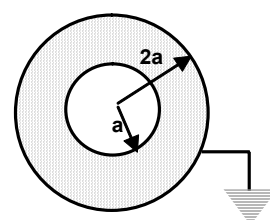
10. The space between two concentric conducting spherical shells of radius a and $2a$ is filled with a dielectric medium having dielectric constant K and resistivity ρ . The inner sphere is given a charge $+Q$ at $t = 0$ and outer is earthed then

(A) charge on inner sphere as a function of time is $Qe^{-\frac{t}{\epsilon_0 K \rho}}$

(B) charge on inner sphere as a function of time is $Qe^{-\frac{2t}{\epsilon_0 K \rho}}$

(C) resistance is $\frac{\rho}{8\pi a}$

(D) resistance is $\frac{\rho}{4a}$



Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Questions no. 11 to 13

It is possible to take a high quality photograph of a very fast moving object by illuminating the object for quite a small fraction of a second. You may have come across photographs of a bullet penetrating a banana or an apple in many text books or magazines. This is called 'Stop action' photography because the fast moving object travels a very short distance during the time of illumination. Harold Edgerton, the inventor of stroboscope, was a pioneer of this kind of photography.

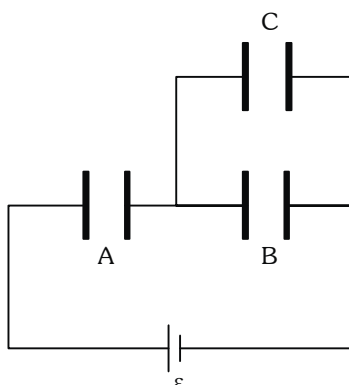
A normal photographic plate works properly if it receives an energy of $4J$ during the exposure. To release this energy in a very small fraction of time, huge amount of power is required. Such huge power can not be generated directly from a battery because of its high internal resistance. To produce such power a capacitor is used. The time in which a capacitor discharges can be very short. Although, theoretically it would take a long time for a capacitor to discharge completely, it discharges almost completely in about 10 time constants. Consider the following situation. A capacitor of $200\mu F$, storing $4J$ energy is made to discharge through a flash light in $2ms$. This setup is used to take the picture of a bullet moving at a speed of $100 m/s$. Assume that the flash light acts as a resistor and there is no other resistance in the circuit.

11. If we use a lens of power 10 diopters, the lens to photographic plate distance is $15 cm$ and the bullet moves perpendicular to the principal axis, what is the distance covered by bullet as seen on photographic plate in $2ms$?
 (A) $1 cm$ (B) $5 cm$ (C) $10 cm$ (D) $20 cm$
12. What is the order of energy delivered to the flash light in $2 ms$ (approx.) ?
 (A) $0.4 J$ (B) $1.83 J$ (C) $2.74 J$ (D) $3.45 J$
13. What is the initial current in the circuit ?
 (A) $200 A$ (B) $120 A$ (C) $700 A$ (D) $3700 A$

Matching List Type (4 × 4)

1 Q. [3 M (–1)]

14. Three capacitor A, B and C are connected with battery of emf ε . All capacitors are identical initially.



List-I

- (P) If dielectric slab is inserted between plates of capacitor B slowly with the help of external force, then
- (Q) If dielectric slab is inserted between plates of capacitor A slowly with help of external force.
- (R) If capacitor C is removed from the circuit, (assuming without any work)
- (S) If separation between plates of capacitor A is slowly increased by external force, then

Code :

	P	Q	R	S
(A)	3	3	1,3	3,4
(B)	3,4	4	1,3	2,3,4
(C)	3	4	1,4	2,3,4
(D)	3,4	3	1,4	3,4

List-II

- (1) Chemical energy of battery remains unchanged
- (2) Positive work is done on the given system by external force.
- (3) Force between plates of capacitor C decreases.
- (4) Magnitude of change of charge on B and C is same.

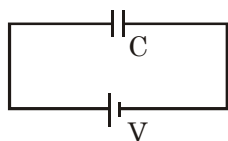
Matching List Type (4 × 5)

1 Q. [3 M (–1)]

15. Match the condition given in list-I to the corresponding changes in list-II.

List-I

- (P) A capacitor C is connected to a battery of potential difference V. Now a dielectric of dielectric constant K is placed keeping the battery connected



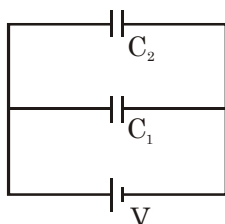
- (Q) A capacitor C is charged to a potential difference V and isolated from the battery. Now a dielectric of dielectric constant K is placed between the plates of the capacitor.



List-II

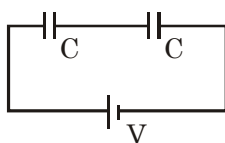
- (1) Electric field between the plates of capacitor in which dielectric is placed decreases.
- (2) Potential difference between the plates of capacitor in which dielectric is placed remains unchanged.

- (R) Two identical capacitors each of capacitance C are connected in parallel to a battery of potential difference V and isolated from the battery. Now a dielectric of dielectric constant K is placed between one of the capacitors keeping the system isolated.



- (3) Capacitance of the capacitor in which dielectric is placed increases.

- (S) Two identical capacitors are connected in series with a battery of potential difference V . Now a dielectric constant K is placed between the plates of one of the capacitors



- (4) Energy of capacitor in which dielectric is placed increases.

- (5) Energy stored in the capacitor in which dielectric is placed decreases.

- (A) $P \rightarrow 2,3,4$; $Q \rightarrow 1,3,4$; $R \rightarrow 1,3,5$; $S \rightarrow 1,3,4$
 (B) $P \rightarrow 2,3,4$; $Q \rightarrow 1,3,5$; $R \rightarrow 1,3,5$; $S \rightarrow 1,3,5$
 (C) $P \rightarrow 2,3$; $Q \rightarrow 1,5$; $R \rightarrow 3,5$; $S \rightarrow 1,2,5$
 (D) $P \rightarrow 1,2,3,4$; $Q \rightarrow 1,2,3,5$; $R \rightarrow 2,3,5$; $S \rightarrow 1,3,4$

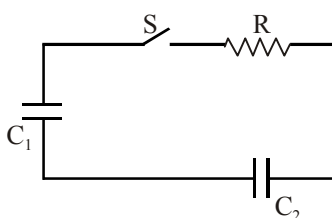
SECTION-II

Numerical Answer Type Question

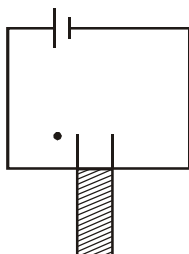
3 Q. [3(0)]

(upto second decimal place)

- An uncharged capacitor and a resistor of big resistance are connected in series to a battery of electromotive force 4.5 V . The voltage across the capacitor is 3 V one minute after the closing the circuit. What will the voltage across the capacitor be 2 min. after closing circuit (in V) ?
- Consider the shown network, the capacitor $C_1 (= 6 \mu\text{F})$ has an initial charge $q_0 = \frac{30e}{e-1} \mu\text{C}$, $C_2 = 4 \mu\text{F}$ and $R = 80 \Omega$. Initially C_2 is uncharged. At $t = 0$, the switch S is closed. Obtain the charge on C_2 (in μC) at $t = 192 \mu\text{s}$.



3. A dielectric slab of dielectric constant k , mass m , thickness d and area $L \times L$ is hanging vertically in equilibrium under the influence of gravity and electrostatic pull of a capacitor connected to a battery of voltage V . The capacitor has plates of area $L \times L$ and distance between plates is d . The capacitor is half filled by the dielectric. Suddenly a mass of m is attached to dielectric without any impulse on the system. The slab falls off in time t . Evaluate t (in sec). Take $k = 2$, $V = 4$ volts, $L = 80$ cm, $d = 0.1$ mm.



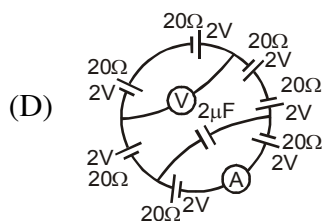
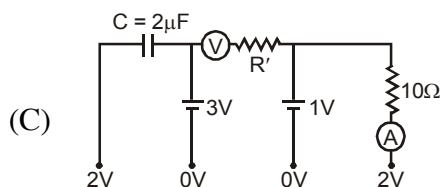
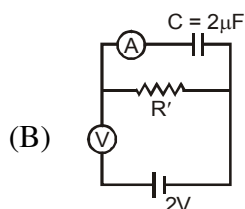
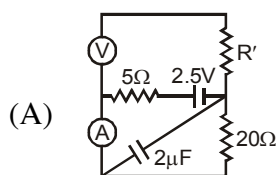
SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Column-I gives some electrical circuits in steady state, column-II gives some statements regarding the circuits match appropriately.

Column-I



Column-II

(P) Reading of voltmeter is 2 V

(Q) Reading of ammeter is 0.1 A

(R) Current through R' is zero

(S) Charge on capacitor is $2\mu\text{C}$

(T) Current through R' is non-zero

SECTION-I

Single Correct Answer Type

4 Q. [3 M (-1)]

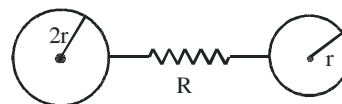
1. Two conducting spheres of radius r and $2r$ at very large separation. Each sphere is having charge Q . These sphere are connected with a conducting wire of resistance R . Then, the current flowing in the wire at $t = 2$ ms. (where $r = 30$ cm, $R = 90$ M Ω and $Q = 10$ mC)

(A) $\frac{100}{3e}$ amp

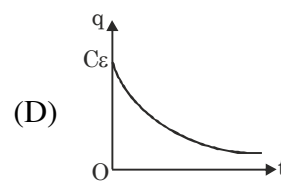
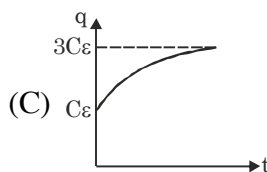
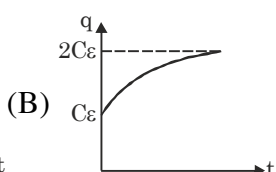
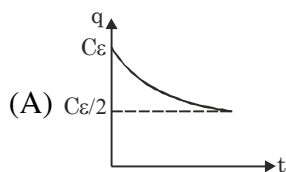
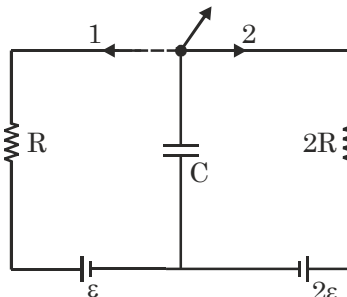
(B) $\frac{5}{3e}$ amp

(C) $\frac{50}{3e}$ amp

(D) $\frac{25}{3e}$ amp



2. In the circuit shown, the switch is shifted from position 1 $\rightarrow$ 2 at $t = 0$. The switch was initially in position 1 since a long time. The graph between charge on capacitor C and time ' t ' is :



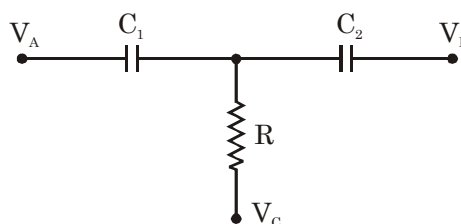
3. For given circuit charge on capacitor C_1 and C_2 in steady state will be equal to :-

(A) $C_1(V_A - V_C)$, $C_2(V_C - V_B)$ respectively

(B) $C_1(V_A - V_B)$, $C_2(V_A - V_B)$ respectively

(C) $(C_1 + C_2)(V_A - V_B)$ on each capacitor

(D) $\left(\frac{C_1 C_2}{C_1 + C_2} \right) (V_A - V_B)$ on each capacitor



4. In the following RC circuit, the capacitor was charged in two different ways.

(i) The capacitor was first charged to 5V by moving the toggle switch to position P and then it was charged to 10V by moving the toggle switch to position Q.

(ii) The capacitor was directly charged to 10V, by keeping the toggle switch at position Q.

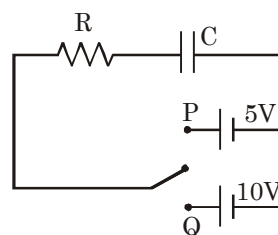
Assuming the capacitor to be ideal, which one of the following statement is correct?

(A) The energy dissipation in cases (i) and (ii) will be equal and non-zero.

(B) The energy dissipation for case (i) will be more than that for case (ii).

(C) The energy dissipation for case (i) will be less than that for case (ii).

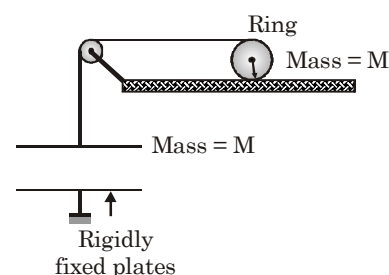
(D) The energy will not be dissipated in either case.



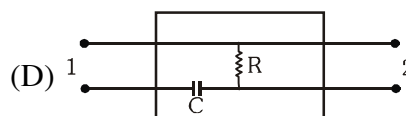
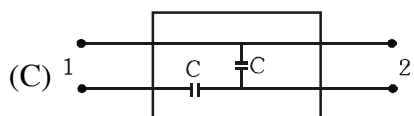
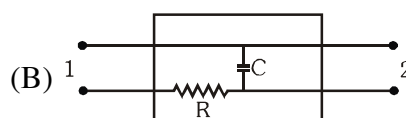
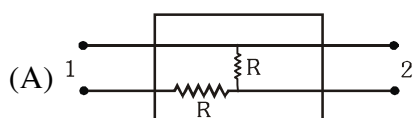
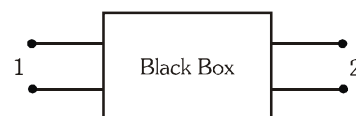
Multiple Correct Answer Type

5. If ring of radius 'R' is placed on smooth floor connected by a string at the top. Other end of string is connected to upper plate of capacitor of same mass "M" and charge density σ and area A. Lower plate of capacitor is rigidly fixed. Then which of the following statements are correct if upper plate of capacitor is falling freely with acceleration 'g'.

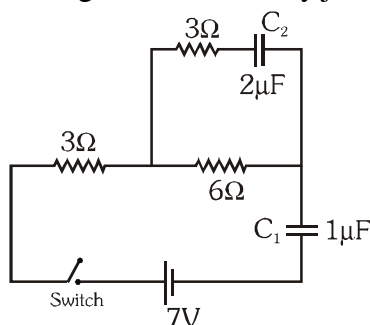
6 Q. [4 M (-1)]



- (A) Charge density of plates of capacitor must be $\sigma = \sqrt{2Mg} \epsilon_0$
- (B) The gain in kinetic energy of the ring and the plate of capacitor is equal to the loss in gravitational potential energy
- (C) Charge of plates of capacitor must be $q = \sqrt{Mg} \epsilon_0 A$
- (D) The gain in kinetic energy of the ring and the plate of capacitor is 1.5 times the loss in gravitational potential energy
6. If the battery of voltage V is connected across terminals 1, the voltmeter across terminal 2 reads V/2. If same battery is connected across terminal 2, the voltmeter connected across terminal 1 gives reading V. Which of the following black box is correct



7. For the arrangement shown in figure, the switch is closed at $t = 0$. C_2 is initially uncharged while C_1 has charge of $2\mu C$. Then the current coming out of the battery just after switch is closed, will be



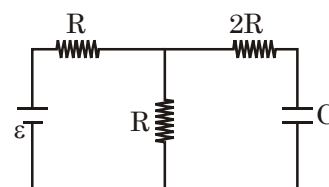
- (A) 1 (B) 0 (C) 1.8 (D) 1.4
8. Circuit shown in the figure is in steady state. Now the capacitor is suddenly filled with medium of dielectric constant $K = 2$.

(A) Current through battery just after this moment is $\frac{11\epsilon}{20R}$.

(B) Current through '2R' just after this moment is $\frac{\epsilon}{10R}$

(C) Current through '2R' just after this moment is $\frac{\epsilon}{15R}$

(D) Potential difference across capacitor just after this moment is $\frac{\epsilon}{4}$



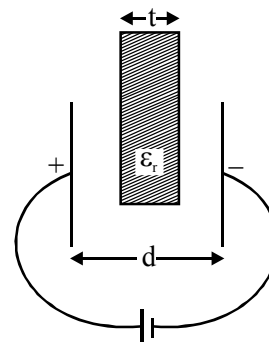
9. A dielectric slab of relative permittivity ϵ_r and thickness t is inserted into the capacitor. Then

(A) the capacitance of the system increases by $\frac{\epsilon_0 A}{t \left(1 - \frac{1}{\epsilon_r}\right)}$

(B) $\frac{q_{\text{free}}}{q_{\text{bound}}} = \frac{\epsilon_r}{\epsilon_r - 1}$

(C) the fraction change in the energy stored is $\epsilon_r - 1$

(D) the plates are moved apart by a relative distance $t \left(1 - \frac{1}{\epsilon_r}\right)$ to recover the original energy stored (or original capacitance)



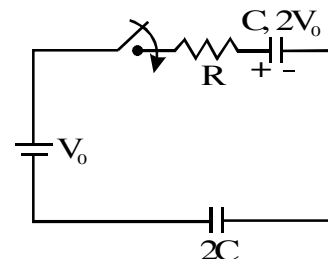
10. Two capacitors are connected to a resistance and battery as given. Capacitor with capacitance C is charge upto $2V_0$, where V_0 is potential of battery and capacitor of $2C$ capacitance is initially uncharged. If switch is closed at $t = 0$ then choose **CORRECT** statement(s)

(A) Ratio of final charge on C to $2C$ is 2

(B) total work done by battery after switch is closed is $\frac{-2CV_0^2}{3}$

(C) total work done by battery after switch is closed is $\frac{+2CV_0^2}{3}$

(D) total heat loss across resistance after closing switch is $\frac{1}{3}CV_0^2$



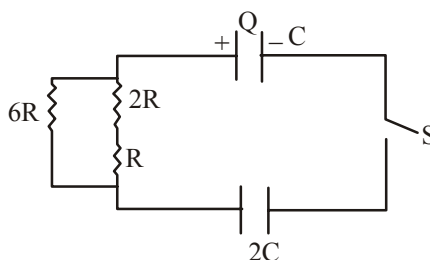
Linked Comprehension Type (1 Para $\times$ 2 Q.)

[4 M (-1)]

(Multiple Correct Answer Type)

Paragraph for Questions 11 and 12

In the given figure switch 'S' is closed at $t = 0$



11. Mark the **CORRECT** option(s) :

(A) Charge on capacitor $2C$ as function of time is $\frac{2Q}{3} \left(1 - e^{-\frac{3t}{4RC}}\right)$.

(B) Charge on capacitor C as function of time is $\frac{Q}{3} \left(1 - e^{-\frac{3t}{4RC}}\right)$.

(C) Current through the resistance $6R$ as a function of time is $\frac{Q}{2RC} e^{-\frac{3t}{4RC}}$

(D) Current through the resistance $6R$ as a function of time is $\frac{Q}{6RC} e^{-\frac{3t}{4RC}}$

12. Mark the **CORRECT** option(s) :

- (A) Amount of heat produced in resistance 'R' in long time is $\frac{2Q^2}{27C}$.
- (B) Amount of heat produced in resistance '2R' in long time is $\frac{4Q^2}{27C}$.
- (C) Amount of heat produced in resistance '6R' in long time is $\frac{Q^2}{9C}$.
- (D) Amount of heat produced in resistance '6R' in long time is $\frac{Q^2}{18C}$.

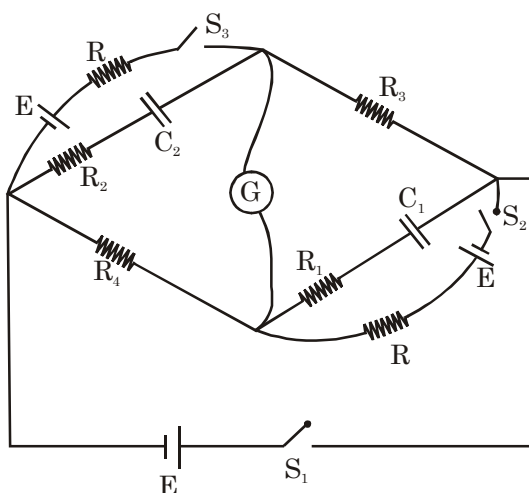
Matching list based comprehension Type (4 × 4 × 4) 1 Table × 3 Q.[3M(-1)]

Single option correct

(Three Columns and Four Rows)

Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table.

In the shown DC-RC circuit galvanometer of resistance 'R' is connected as shown. (Initially at $t = 0$ all switches are open & capacitors are uncharged)



Condition of switch, resistance & capacitance is given in column (I), (II), (III) respectively.

Column-I

Column-II

Column-III

- | | | |
|---------------------------------------|--|-------------------------|
| (I) S_1 closed, S_2 & S_3 open | (i) $R_1 = 2R, R_2 = 2R$
$R_3 = R, R_4 = R$ | (P) $C_1 = C_2 = C$ |
| (II) S_1 open, S_2 & S_3 closed | (ii) $R_2 = R, R_4 = 3R$
$R_3 = 2R, R_1 = 6R$ | (Q) $C_1 = C_2 = 2C$ |
| (III) S_1, S_2 closed S_3 open | (iii) $R_1 = R_2 = R_3 = R_4 = R$ | (R) $C_1 = C, C_2 = 2C$ |
| (IV) S_1, S_3 open, S_2 closed | (iv) $R_1 = R_2 = 2R, R_4 = R_3 = R$ | (S) $C_1 = 2C, C_2 = C$ |

13. Initially both capacitors are uncharged. In which of the following case galvanometer shown no deflection at $t = 0$ (just after switches is closed) ?

- (A) (IV) (ii) (S) (B) (I) (i) (R) (C) (I) (ii) (S) (D) (IV) (iii) (R)

14. In which of the following case galvanometer shown no deflection at ($t = \infty$) (steady state)

- (A) (II) (iv) (P) (B) (IV) (i) (S) (C) (II) (ii) (S) (D) (IV) (ii) (P)

15. In which of the following case final charge on capacitor C_2 is zero :-

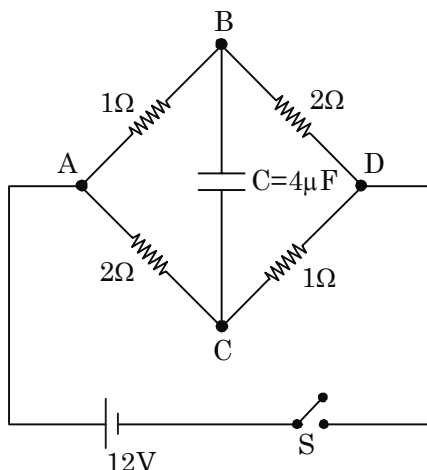
- (A) (III) (iv) (R) (B) (III) (iii) (Q) (C) (II) (ii) (Q) (D) (IV) (ii) (R)

SECTION-II

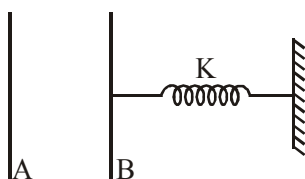
Numerical Answer Type Question (upto second decimal place)

3Q.[3(0)]

1. An uncharged capacitor of capacitance C is connected in given circuit diagram, switch S is closed at $t = 0$. If current in branch BC as function of time is given by $I = I_0(\text{amp}) e^{-\frac{t}{\tau(\text{in } \mu\text{S})}}$ then find the numerical value of $I_0\tau$.



2. Initially the plates of the parallel plate capacitor shown in the figure are A and B containing charge $+2\mu\text{C}$ and $-2\mu\text{C}$ respectively. Negatively charged plate B is connected to a spring of spring constant 5000N/m at one end and the other end of the spring is connected to a rigid support. Now the plate A is fixed and the system is released from rest. The maximum elongation produced in the spring is $\frac{\alpha \times 10^{-12}}{\epsilon_0}$ m. The area of the plates is 2 cm^2 , the medium between the plates is air and Electric permittivity of air is ϵ_0 (in SI). Then the value of α .



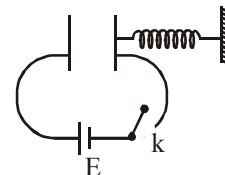
3. Between the plates of a parallel-plate capacitor there is a metallic plate whose thickness takes up $\eta = 0.60$ of the capacitor gap. When that plate is absent the capacitor has a capacity $C = 20\text{ nF}$. The capacitor is connected to a dc voltage source $V = 100\text{ V}$. The metallic plate is slowly extracted from the gap. Find the mechanical work performed in the process of plate extraction (in μJ). If your answer is N fill value $N/30$.

SECTION-I

Single Correct Answer Type

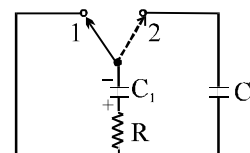
6 Q. [3 M (-1)]

1. A plate A of a parallel plate capacitor is fixed, while the plate B is attached to the wall by a spring and can move, remaining parallel to the plate A (see figure). The capacitor is charged, plate B starts moving & comes to rest in an equilibrium position. The separation between the plates d , decreases by 10%. What will be the decrease in the plate separation if the charging is done in a very short time that the plate B couldn't shift noticeably.



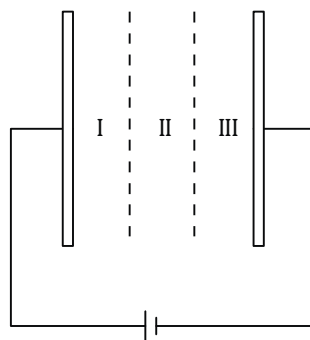
- (A) 10 % (B) 8.1 % (C) 9 % (D) 6.3 %

2. A charged capacitor C_1 is discharged through a resistor R . When the discharge current attains the value I_0 , the key is shifted from 1 to 2. Then the amount of heat Q liberated in the resistor starting from this moment, is



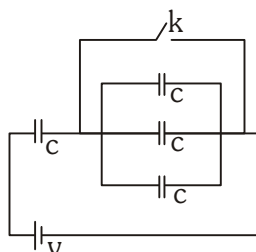
- (A) $\frac{I_0^2 R^2 C_1 C_2}{2(c_1 + c_2)}$ (B) $\frac{I_0^2 R^2 C_1 C_2}{4(c_1 + c_2)}$ (C) $\frac{I_0^2 R^2}{2} \left(\frac{C_1 + C_2}{C_1 C_2} \right)$ (D) None

3. The figure shows a capacitor having three layers of equal thickness and same area as that of plate. Layer I is vacuum, layer II is conductor and layer III is dielectric of dielectric constant K . The ratio of energy stored in region III to total energy stored in capacitor is :-



- (A) $\frac{1}{K+1}$ (B) $\frac{3}{K+1}$ (C) $\frac{4}{K+3}$ (D) $\frac{4}{3K+1}$

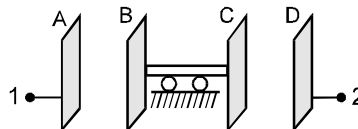
4. The charge flown through the cell on closing the key k is equal to



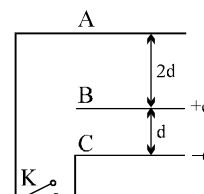
- (A) $\frac{CV}{4}$ (B) CV (C) $\frac{4}{3}CV$ (D) $\frac{3}{4}CV$

5. A, B, C, D are large conducting plates kept parallel to each other. A and D are fixed. Plates B and C, connected to each other by a rigid conducting rod can slide over frictionless rails as shown. Initially the distance between plates A and B is same as that between plates C and D. If now the rod (along with plates B and C) is slightly moved towards right, the capacitance between the terminals 1 and 2.

- (A) remains unchanged
(B) increases
(C) decreases
(D) nothing can be said



6. Three identical large metal plates of area A are at distances d and 2d from each other as shown. Metal plate A is uncharged, while metal plates B and C have respective charges +q and -q. Metal plates A and C are connected by switch K through a wire. How much energy is lost when switch is closed?

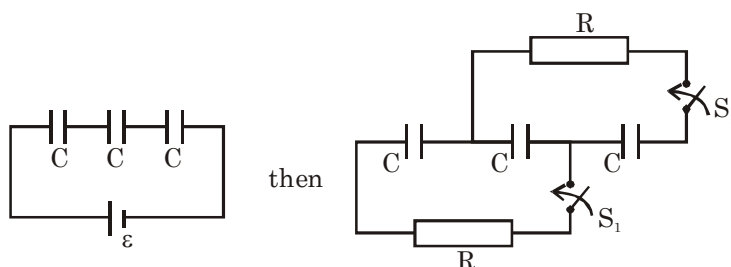


- (A) $\frac{2q^2 d}{3\epsilon_0 A}$ (B) $\frac{q^2 d}{6\epsilon_0 A}$ (C) $\frac{q^2 d}{3\epsilon_0 A}$ (D) none of these

Multiple Correct Answer Type

5 Q. [4 M (-1)]

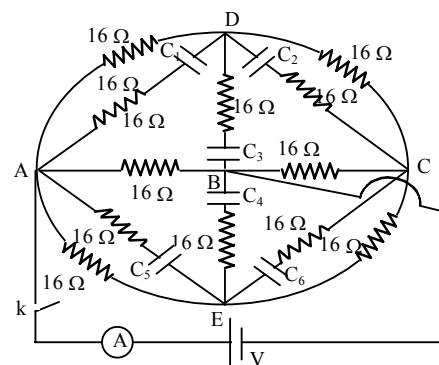
7. Three identical charge-less capacitors of capacitance C each are connected in series. The capacitors are charged by connecting a battery of electromotive force ϵ to the terminal as shown in figure. Next, the battery is disconnected, and two resistors of resistance R are connected simultaneously as shown in figure below. If switches S_1 & S_2 are closed simultaneously at time $t = 0$, then choose the **CORRECT** option(s):



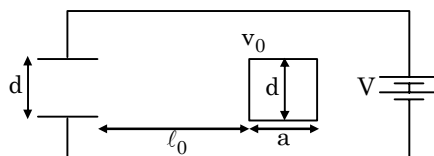
- (A) Current in any resistor R just after $t = 0$ is $\frac{2\epsilon}{3R}$.
(B) Final magnitude of charge on all the capacitors will be same.
(C) Total heat loss through any one resistor is $\frac{4}{54} C\epsilon^2$.
(D) Initial total energy stored in the system is $\frac{3}{2} C\epsilon^2$.

8. The resistance each of 16Ω and capacitance of each $100 \mu\text{F}$ are arranged as shown in the figure. A battery of emf 12V is joined across A and B. Then

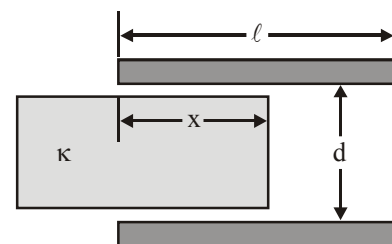
- (A) reading of the ammeter just after key closed is 2A.
(B) reading of the ammeter long time after key closed is $\frac{9}{8} \text{ A}$
(C) reading of the ammeter just after key closed is 1A.
(D) reading of the ammeter long time after key closed is $\frac{3}{8} \text{ A}$



9. A parallel plates capacitor is made of square conducting plates of side a and the separation between plates is d . The capacitor is connected with battery of emf V volt as shown in the figure. There is a dielectric slab of dimension $a \times a \times d$ with dielectric constant k . At $t = 0$, dielectric slab is given velocity v_0 towards capacitor as shown in the figure. (Neglect the effect of gravity and electrostatic force acting on the dielectric when dielectric is out side of capacitor. Also ignore any type frictional force acting on the dielectric during its motion) let the x be the length dielectric inside the capacitor at $t = t$ sec. [$\lambda_0 \gg a$]



- (A) Motion of dielectric slab is periodic but not simple harmonic motion
(B) Motion of dielectric slab is simple harmonic motion
(C) At any time, the slope of graph of total energy verses x is twice the slope of graph of potential energy verses x .
(D) The value of maximum energy stored in the system is $\frac{1}{2}mv_0^2 + \frac{\epsilon_0 a^2 V^2}{2d}(2k - 1)$
10. To measure the capacitance of a conductor, it is first charged to a potential $V_0 = 1350$ V. It is then connected by a conducting wire to a distant metal sphere of radius $r = 3$ cm. As a result the conductor potential drops to $V_1 = 900$ V.
Choose the correct option (s)
(A) Initial charge on the conductor is 9 nC
(B) Capacitance of the conductor is $\frac{1}{150}$ nF
(C) Charge on the metal sphere after connection is 6 nC
(D) Electrostatic potential energy decreases after redistribution of charges.
11. Two square plates of sides ℓ are placed parallel to each other with separation d as suggested in figure. You may assume d is much less than ℓ . The plates carry uniformly distributed static charges $+Q_0$ and $-Q_0$. A block of metal has width ℓ , length ℓ , and thickness slightly less than d . It is inserted at distance x into the space between the plates. The charges on the plates remain uniformly distributed as the block slides in. In a static situation, a metal prevents an electric field from penetrating inside it. The metal can be thought of as a perfect dielectric, with $\kappa \rightarrow \infty$.



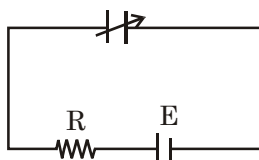
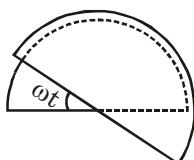
- (A) The stored energy as a function of x is given by $\frac{Q_0^2 d (\ell - x)}{(2\ell^3 \epsilon_0)}$
(B) The magnitude of the force that acts on the metallic block is given by $\frac{Q_0^2 d}{(\ell^3 \epsilon_0)}$ to the right .
(C) The area of the advancing front face of the block is essentially equal to ℓd , the stress (force per unit area) on it is $\frac{Q_0^2}{(2\ell^4 \epsilon_0)}$.
(D) The energy density in the electric field between the charged plates in terms of Q_0 , ℓ , d , and ϵ_0 is $\frac{Q_0^2}{(2\ell^4 \epsilon_0)}$

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Question no. 12 and 13

A variable condensator or also known as varco is a capacitor that its capacitance can be varied. One type is varied by turning one of its plates clockwise. Suppose that a varco consists of a pair of half circle plate with diameter D separated by a distance d . the varco plate is rotating with angular velocity ω . The other plate is fixed. The plates may have some initial charge.



This varco is connected in a circuit as shown. We will consider $t = 0$ to $t = \frac{\pi}{\omega}$.

12. If the current is constant.
- (A) The angular velocity is increasing with time
 - (B) The angular velocity is decreasing with time
 - (C) The angular velocity is constant with time
 - (D) It is not possible.
13. If $R = 0$ what torque needs to be applied to rotate the plate with constant small angular velocity?
- (A) $\frac{\epsilon_0 E^2 D^2}{16d}$ clockwise
 - (B) $\frac{\epsilon_0 E^2 D^2}{16d}$ anticlockwise
 - (C) $\frac{\epsilon_0 E^2 D^2}{8d}$ clockwise
 - (D) $\frac{\epsilon_0 E^2 D^2}{8d}$ anticlockwise

Linked Comprehension Type
(Multiple Correct Answer Type)

(1 Para × 2 Q.) [4 M (-1)]

Paragraph for Question no. 14 and 15

A parallel plate capacitor placed in a cylindrical tank is filled with a liquid of dielectric constant k . The area of cross section of tank is A and height of liquid is equal to the length of the square plate of plate area ℓ^2 . The separation of plates is d . A very small hole of area 'a' is opened at the bottom of the tank at $t = 0$. If the capacitor in the process remains connected with a battery of emf E . Assume that the level of liquid in the capacitor remains same as outside and Bernoulli's equation is valid for the situation.

14. Choose the **CORRECT** statement(s) :

- (A) The current in circuit as a function of time is $i = \frac{\epsilon_0 \ell E}{d} (k-1) \frac{a}{A} \left(\sqrt{2g\ell} - \frac{a}{A} gt \right)$
- (B) The current in circuit as a function of time is $i = \frac{\epsilon_0 \ell E}{2d} (k-1) \frac{a}{A} \sqrt{2g\ell}$.
- (C) The capacitance of capacitor decreases with time.
- (D) The potential difference between the plates of capacitor remains constant.

15. Choose the **CORRECT** statement(s) :

- (A) The rate at which the level of liquid decreases is $\frac{dh}{dt} = \frac{a}{2A} \sqrt{2g\ell}$
- (B) The rate at which the level of liquid decreases is $\frac{dh}{dt} = \frac{a}{A} \left(\sqrt{2g\ell} - \frac{a}{A} gt \right)$
- (C) The power supplied by the battery increases with time.
- (D) When the container gets empty, the power delivered by the battery becomes zero.

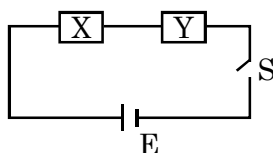
Matching list based comprehension Type (4 × 4 × 4)

1 Table × 3 Q. [3(-1)]

Single option correct (Three Columns and Four Rows)

Answer Q.16, Q.17 and Q.18 by appropriately matching the information given the three columns of the following table.

Following is a circuit containing an ideal battery of emf E in which the elements X & Y are unknown & may be a capacitor C , a resistor R , a battery of emf E , or may be their combination as mentioned in each case. The connecting wires are perfectly conducting.



The switch S is initially open & it is closed at $t = 0$.

Column-1

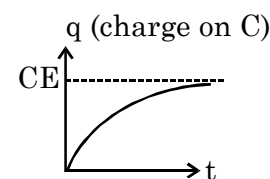
Column-2

Column-3

- (I) Element X is R
Element Y is C

- (i) Entire energy (non-zero) supplied by batteries is dissipated as heat

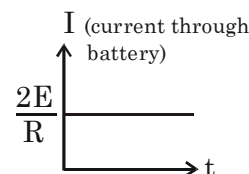
(P)



- (II) Both elements X & Y are R each

- (ii) Total charge drawn from the battery(s) after long time is $\frac{CE}{2}$

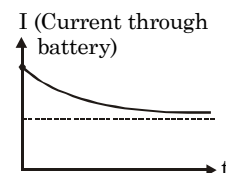
(Q)



- (III) Element X is R ,
element Y is a parallel combination of R and C

- (iii) Total heat produced in circuit is less than the energy supplied by the battery(s)

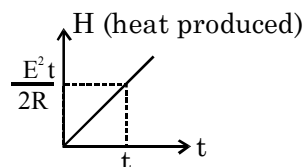
(R)



- (IV) Element X is R
element Y is battery E joined with same polarity

- (iv) $I = 0$ just after S is closed as well as after long time

(S)



16. Of the given 4 situations, which one has maximum value of current just after closing the switch ?

(A) I, iii, P

(B) IV, i, Q

(C) II, iii, P

(D) III, i, S

17. In which case the charge drawn from the battery(s) during $t = 0$ to $t = RC$ is the least ?
 (A) II - i - S (B) I, iii, P (C) III, ii, Q (D) IV, i, Q
18. In which case the current through battery(s) has different non zero values just after closing the switch & after a long time ?
 (A) II - i - S (B) I, iv, Q (C) IV, i, Q (D) III - iii- R

SECTION-III

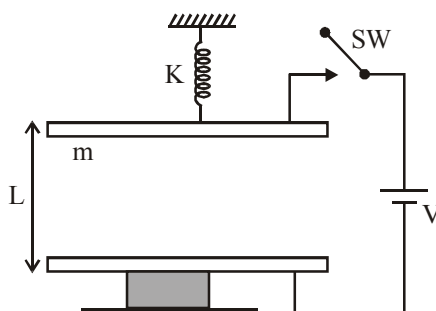
Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

1. A parallel plate capacitor having its lower end fixed and upper end is attached with spring having spring constant K . Upper plate is in equilibrium before switch is closed. After switch is closed, the condition on

the potential of battery so that the system can acquire new equilibrium position is $V \leq \sqrt{\left(\frac{p}{q}\right)^r \frac{L^3 K}{\epsilon_0 A}}$.

Then $p + q + r$ is ? [p & q are smallest possible integers and V is potential difference across the battery]

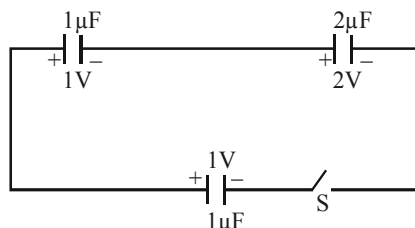


SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

1. Figure shows 3 different capacitors, separately charged by batteries, and then connected as shown, with initially, switch open. When switch is closed, the charge :-



- (A) On all capacitors get equalized (B) becomes zero on all capacitors
(C) On $2\mu\text{F}$ capacitor increases by 20% (D) On $2\mu\text{F}$ capacitor decreases by 20%

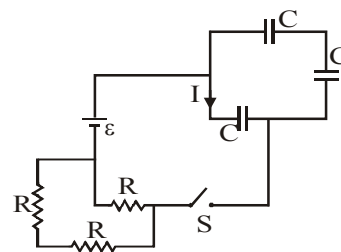
2. In the given circuit switch S is closed at $t = 0$. The current I in the figure at time t is

(A) $i = \frac{\epsilon e^{-t/RC}}{2R}$

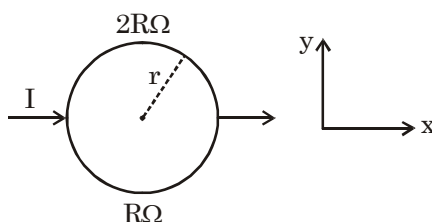
(B) $i = \frac{2\epsilon e^{-t/RC}}{3R}$

(C) $i = \frac{\epsilon e^{-\frac{2t}{3RC}}}{2R}$

(D) $i = \frac{\epsilon e^{-t/RC}}{R}$



3. Two wires are bent (shown as joint-circle) with radius r (in xy plane). The upper half has resistance of $2R\Omega$ and the lower half of $R\Omega$. A current I is passed into circle as shown. The magnetic field at centre is :



(A) $\frac{\mu_0 I}{r}$

(B) $\frac{\mu_0 I}{2r}(\hat{z})$

(C) zero

(D) $\frac{\mu_0 I}{12r}\hat{z}$

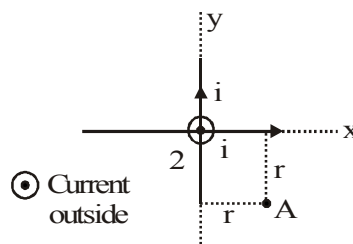
4. Three infinite wires are arranged in space in three dimensions (along x , y and z axis) as shown. Each wire carries current i . Find magnetic field at A.

(A) $-\frac{\mu_0 i}{2\pi r}\hat{i} - \frac{\mu_0 i}{2\pi r}\hat{j} - \frac{\mu_0 i}{\pi r}\hat{k}$

(B) $-\frac{\mu_0 i}{4\pi r}\hat{i} + \frac{\mu_0 i}{4\pi r}\hat{j} - \frac{\mu_0 i}{\pi r}\hat{k}$

(C) $\frac{\mu_0 i}{4\pi r}\hat{i} + \frac{\mu_0 i}{4\pi r}\hat{j} - \frac{\mu_0 i}{\pi r}\hat{k}$

(D) $\frac{\mu_0 i}{2\pi r}\hat{i} + \frac{\mu_0 i}{2\pi r}\hat{j} - \frac{\mu_0 i}{\pi r}\hat{k}$



5. A charged particle enters into a uniform magnetic field with velocity v_0 perpendicular to it, the length of magnetic field is $x = \frac{\sqrt{3}}{2} R$, where R is the radius of the circular path of the particle in the field. The magnitude of change in velocity of the particle when it comes out of the field is :-

- (A) $2v_0$ (B) $\frac{v_0}{2}$ (C) $\frac{\sqrt{3}v_0}{2}$ (D) v_0

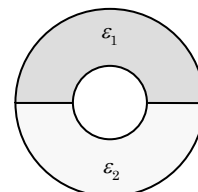
Multiple Correct Answer Type

4 Q. [4 M (-1)]

6. Radii of the inner and outer concentric conducting spheres of a spherical capacitor are a and b . One half of the space between the spheres is filled with a linear isotropic dielectric of permittivity ϵ_1 and the other half with another linear isotropic dielectric of permittivity ϵ_2 as shown in the figure.

The outer sphere is given a total charge $+Q$ and the inner sphere a total charge $-Q$. Which of the following statements are correct?

- (A) Electric fields in both the dielectrics at the same radial distance are not equal.
(B) Electric field intensities in both the dielectrics at radial distance r from center



are given by $E_1 = E_2 = \frac{Q}{2\pi(\epsilon_1 + \epsilon_2)r^2}$.

- (C) Potential difference between the spheres is given by $V = \frac{Q(b-a)}{2\pi(\epsilon_1 + \epsilon_2)ab}$.

- (D) Charge densities on both the halves of a sphere are equal.

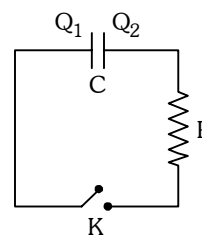
7. The two plates of a capacitor of capacitance C are given charges Q_1 and Q_2 . This capacitor is connected across a resistance R as shown key is closed at $t = 0$. Find the charges on the plates after time t .

- (A) Total charge of the right plate $q_2 = \frac{Q_1 + Q_2}{2} - \left(\frac{Q_1 - Q_2}{2}\right)e^{-\frac{t}{RC}}$

- (B) Total charge on the left plate $q = \frac{Q_1 + Q_2}{2} + \left(\frac{Q_1 - Q_2}{2}\right)e^{-\frac{t}{RC}}$

- (C) Initial potential difference across the plates is given by $\frac{Q_1 - Q_2}{2C}$

- (D) Initial potential difference across the plates is given by $\frac{Q_1 + Q_2}{2C}$



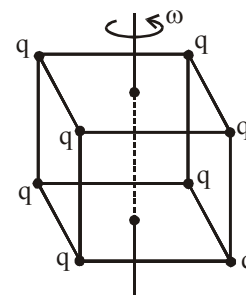
8. Consider a cube of side 'a' as shown. Eight point charges are placed at the corners. The cube is rotated about the axis with constant angular velocity ' ω ':

- (A) Net magnetic field at the centre of cube is zero

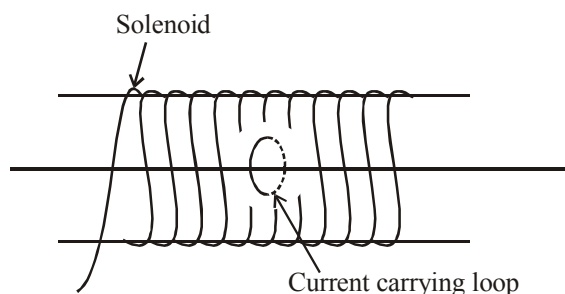
- (B) Net magnetic field at the centre of cube is $\frac{\sqrt{2}\mu_0 q \omega}{\pi a}$

- (C) Net magnetic field at the centre of cube is $\frac{8}{3\sqrt{3}} \frac{\mu_0 q \omega}{\pi a}$

- (D) If polarity of any four charges are reversed, then magnetic field at the centre of cube will be zero.



9. A single circular loop of wire with radius 0.02 m carries a current of 8.0 A. It is placed at the centre of a long solenoid that has length 0.65m, radius 0.080 m and 1300 turns. (Axis of circular loop coincide with axis of solenoid). Let 'i' be the current in solenoid.



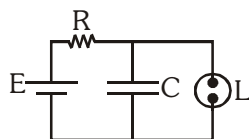
- (A) If $i = 100$ mA, the magnetic field at the center of coil can be zero.
 (B) If $i = 44$ mA, the magnetic field at the center of coil can be zero.
 (C) If $i = 100$ mA, the magnetic field at the center of coil can be $8\pi \times 10^{-5}$ T.
 (D) If $i = 100$ mA, the magnetic field at the center of coil can be $16\pi \times 10^{-5}$ T.

Linked Comprehension Type
(Single Correct Answer Type)

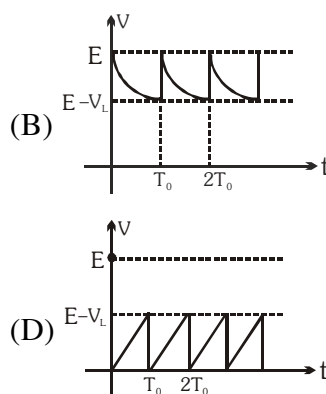
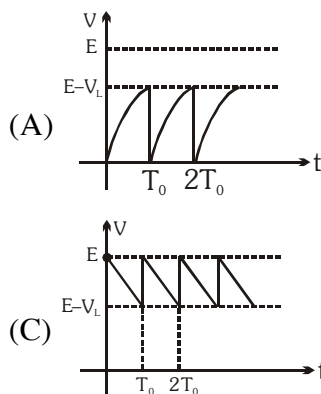
(1 Para $\times$ 3Q. & 1 Para $\times$ 2Q.) [3 M (-1)]

Paragraph for Questions no. 10 to 12

Figure shows the circuit of a flashing lamp, used at construction sites. The fluorescent lamp L, having negligible capacitance, is connected in parallel across the capacitor C of an RC circuit. There is a current through the lamp only when the potential difference across it reaches the breakdown voltage V_L . In this event, the capacitor discharges completely through the lamp and lamp flashes briefly.



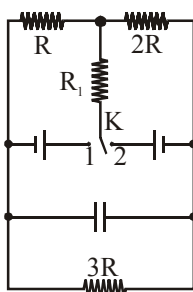
10. Consider an instant, when the capacitor has just discharged through the flash light. Taking this instant as $t = 0$, the time after which the lamp flashes briefly is given by :-
 (A) $RC \ln \left(\frac{E}{E - V_L} \right)$ (B) $RC \left[1 - \ln \left(\frac{E}{V_L} \right) \right]$ (C) $RC \ln \left(\frac{E}{V_L} \right)$ (D) $RC \ln \left(\frac{E - V_L}{E} \right)$
11. The number of flashes per second produced by the arrangement is (neglecting the time of flashing or discharging of capacitor) :-
 (A) $\frac{1}{RC \ln \left(\frac{E}{E - V_L} \right)}$ (B) $\frac{1}{RC \ln \left(\frac{E}{V_L} \right)}$ (C) $\frac{1}{RC \ln \left(\frac{E - V_L}{E} \right)}$ (D) $\frac{1}{RC \left[1 - \ln \left(\frac{E}{V_L} \right) \right]}$
12. Which of the following graph represents the variation of potential drop across the resistor ?



Paragraph for Question no. 13 and 14

When the switch is shifted from position 1 to position 2 (as shown in figure), the steady state current through the resistance R_1 does not change its direction, but decreases to $\frac{2}{3}$ times of its previous value.

The batteries are ideal and identical :-



13. Value of R_1 is :-

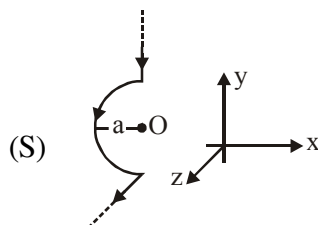
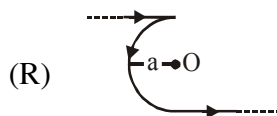
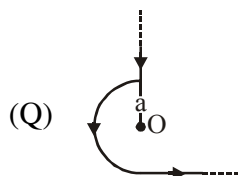
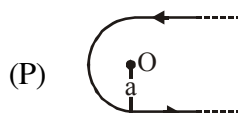
- (A) $\frac{R}{2}$ (B) $\frac{R}{6}$ (C) $\frac{R}{3}$ (D) $\frac{R}{4}$

14. What is the ratio of the steady state charge on the capacitor before switching and after switching ?

- (A) $\frac{3}{5}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$

Matching List Type (4 × 4)

15. List-I
Current Carrying Wires



1 Q. [3 M (-1)]

List-II
 B_0

(1) $\frac{\mu_0 i}{4a} \left(1 + \frac{1}{\pi} \right)$

(2) $\frac{\mu_0 i}{4a} \sqrt{1 + \frac{1}{\pi^2}}$

(3) $\frac{\mu_0 i}{2a} \left(\frac{1}{2} + \frac{1}{\pi} \right)$

(4) $\frac{\mu_0 i}{4a}$

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	4	2	3	1
(C)	2	3	1	4
(D)	1	4	2	3

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1 Q. [3(0)]

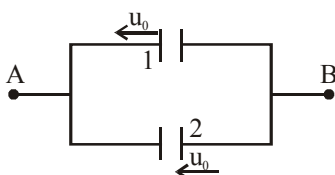
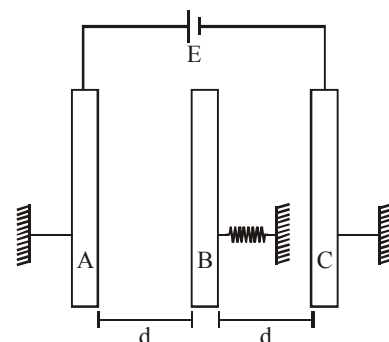
1. A 20 ohms resistor is connected in series with a capacitor of 0.01 farad and *e.m.f.* E volts given by $40e^{-3t} + 20e^{-6t}$. If $q = 0$ at $t = 0$, the maximum charge on the capacitor is (in C).

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

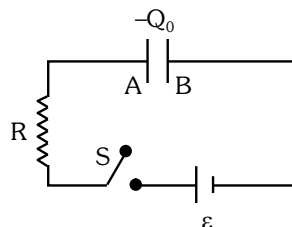
4 Q. [4 M (0)]

1. A, B, C are three identical neutral conducting plates, A and C are rigidly attached to fixed supports whereas plate B is attached to a spring having constant K . The supports and spring are very small in size and their effect on induced charges on plates A, B, C is negligible. The separations between plates are d as shown. If plate B is displaced by x ($x \ll d$) in the plane and released it is found that plate B undergoes SHM. If $K = 2$ N/m, mass of plate B is 8 kg, emf of battery = 2V and system is in gravity free space, the time period of B is found to be $T = \alpha\pi$. Find value of α .
2. Two identical capacitors having plate separation 1mm are connected parallel to each other across points A and B as shown in figure. Total charge of $4 \mu\text{C}$ is imparted to the system by connecting a battery across A and B and battery is removed. Now first plate of first capacitor and second plate of second capacitor starts moving with constant velocity 3 m/s towards left. Find the magnitude of current (in mA) flowing in the loop initially.

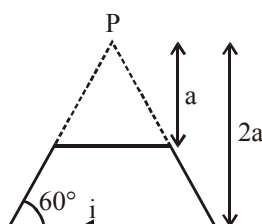


3. The figure shows a RC circuit with a parallel plate capacitor. Before switching on the circuit, plate A of the capacitor has a charge $-Q_0$ while plate B has no net charge. Now, at $t = 0$, the circuit is switched on. How much time (in second) will elapse before the net charge on plate A becomes zero.

(Given $C = 1 \mu\text{F}$, $Q_0 = 1 \text{mC}$, $\epsilon = 1000$ V and $R = \frac{2 \times 10^6}{\ln 3} \Omega$)



4. Figure shows a conducting loop in shape of a trapezium carrying a current $i = 10\text{A}$. Find the magnetic field B (in μT) at a point P. It is given that $a = 10$ cm.



SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

1. A particle of mass m and charge q , accelerated by a potential difference V enters a region of a uniform transverse magnetic field B in a direction perpendicular to boundary. If d is the thickness of the region of B , the angle θ through which the particle deviates from the initial direction on leaving the region is given by :-

(A) $\sin \theta = Bd \left(\frac{q}{2mV} \right)^{1/2}$

(B) $\cos \theta = Bd \left(\frac{q}{2mV} \right)^{1/2}$

(C) $\tan \theta = Bd \left(\frac{q}{2mV} \right)^{1/2}$

(D) $\cot \theta = Bd \left(\frac{q}{2mV} \right)^{1/2}$

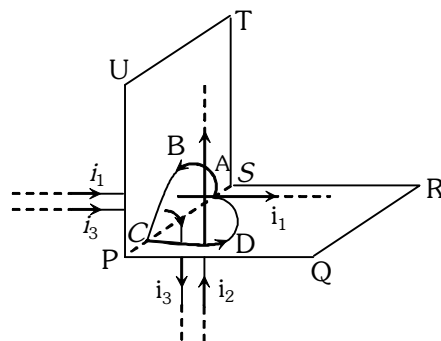
2. Figure shows an Amperian path ABCDA. Part ABC is in vertical plane PSTU while part CDA is in horizontal plane PQRS. Direction of circulation along the path is shown by an arrow near point B and D. $\oint \vec{B} \cdot d\vec{\ell}$ for this path according to Ampere's law will be

(A) $(-i_1 - i_2 + i_3)\mu_0$

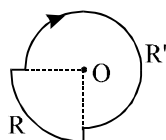
(B) $(-i_1 + i_2)\mu_0$

(C) $i_3\mu_0$

(D) $(i_1 + i_2)\mu_0$



3. A current of i ampere is flowing through each of the bent wires as shown the magnitude and direction of magnetic field at O is



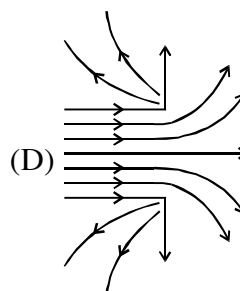
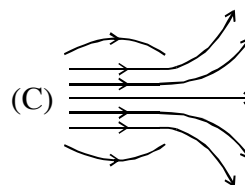
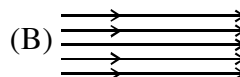
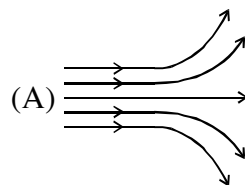
(A) $\frac{\mu_0 i}{4} \left(\frac{1}{R} + \frac{2}{R'} \right)$

(B) $\frac{\mu_0 i}{4} \left(\frac{1}{R} + \frac{3}{R'} \right)$

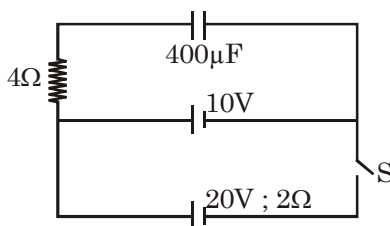
(C) $\frac{\mu_0 i}{8} \left(\frac{1}{R} + \frac{3}{2R'} \right)$

(D) $\frac{\mu_0 i}{8} \left(\frac{1}{R} + \frac{3}{R'} \right)$

4. Which of the following most appropriately represent magnetic field lines near the end of a long solenoid



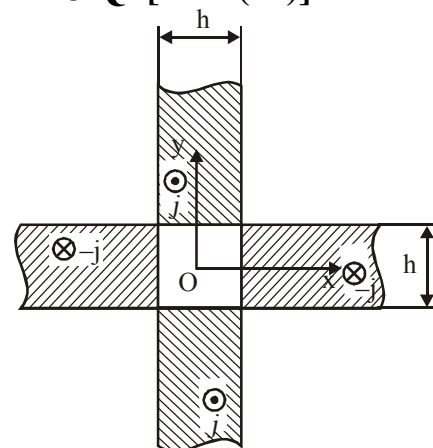
5. Total heat generated in 4Ω resistance after switch S is closed :- (Initially circuit was in steady-state)



- (A) 80 mJ (B) zero (C) 40 mJ (D) 120 mJ

Multiple Correct Answer Type

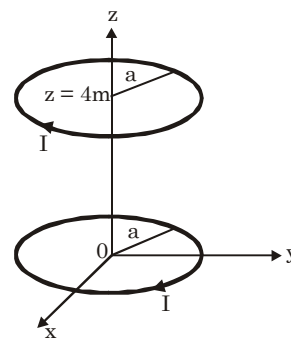
6. Figure shows two infinite sheets of thickness h each, carrying current of density (current/area) ' J '. The direction of current is indicated in the diagram. Point 'O' is at the centre of cuboid formed due to intersection of sheets.



- (A) The magnitude of magnetic field at point O is $\sqrt{2}\mu_0 J$
(B) The magnitude of magnetic field at point O is zero.
(C) The magnitude of magnetic field at any finite distance outside the sheet is zero.
(D) The magnitude of magnetic field at any finite distance

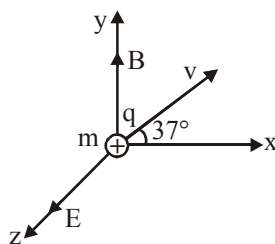
outside the sheet is $\frac{\mu_0 Jh}{\sqrt{2}}$

7. Two parallel, circular loops carrying a current of 40 A each are arranged as shown in figure. The first loop is situated in the x - y plane with its center at the origin, and the second loop's center is at $z = 4$ m. If the two loops have the same radius $a = 3$ m, determine the magnetic field at:



- (A) $z = 0, \vec{B} = \frac{-2432\pi \times 10^{-7}}{75} \hat{k} \text{ T}$
(B) $z = 4, \vec{B} = \frac{-2432\pi \times 10^{-7}}{75} \hat{k} \text{ T}$
(C) $z = 0, \vec{B} = \frac{-1685\pi \times 10^{-7}}{24} \hat{k} \text{ T}$
(D) $z = 4, \vec{B} = \frac{-1685\pi \times 10^{-7}}{24} \hat{k} \text{ T}$

8. A charge particle of charge q , mass m is projected with velocity $\vec{v}$. The electric field $\vec{E} = E\hat{k}$ and magnetic field $\vec{B} = B\hat{j}$ is applied. The acceleration of the particle is



- (A) $\frac{4qvB}{5m} \hat{k}$ (B) $\frac{3qE}{5m} \hat{k}$ (C) $\frac{q(5E + 4vB)}{5m} \hat{k}$ (D) $\frac{q(5E - 4vB)}{5m} \hat{k}$

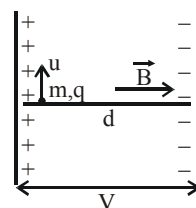
9. A particle of specific charge $\frac{q}{m} = \pi \times 10^{10} \text{ Ckg}^{-1}$ is projected from the origin along the positive x-axis with a velocity of 10^5 ms^{-1} in a uniform magnetic field $\vec{B} = -2 \times 10^{-3} \hat{k}$ tesla. Choose correct alternative(s)
- (A) The centre of the circle lies on the y-axis
(B) The time period of revolution is 10^{-7} s .
(C) The radius of the circular path is $\frac{5}{\pi} \text{ mm}$
(D) The velocity of the particle at $t = \frac{1}{4} \times 10^{-7} \text{ s}$ is $10^5 \hat{j} \text{ ms}^{-1}$
10. There is uniform magnetic field between large parallel plates of a capacitor having separation d and potential difference V . A charge particle is projected very near to +ve plate parallel to it with velocity u . Due to magnetic and electric field resultant motion of particle would be helical.

(A) Time taken by particle to reach at -ve plate is $\sqrt{\frac{2md^2}{qV}}$.

(B) Time taken by particle to reach at -ve plate is $\sqrt{\frac{md^2}{qV}}$

(C) Number of revolutions completed by particle just before reaching at -ve plate $\sqrt{\frac{qB^2d^2}{\pi^2mV}}$

(D) Number of revolutions completed by particle just before reaching at -ve plate $\sqrt{\frac{qB^2d^2}{2\pi^2mV}}$

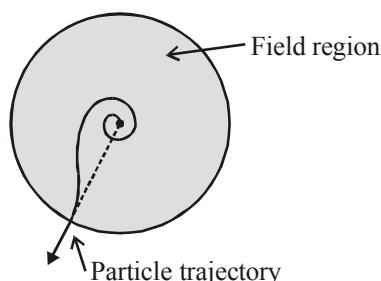


Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 2Q.) [3 M (-1)]

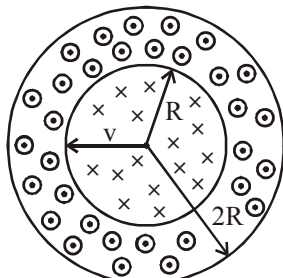
Paragraph for Question no. 11 and 12

A circularly symmetrical magnetic field B (depends only on the distance from their axis), pointing perpendicular to the page, occupies the shaded region in figure. It can be shown that if the total flux ($\int \vec{B} \cdot d\vec{a}$) is zero, then a charged particle that starts out at the centre will emerge from the field region on a radial path (provided it escapes at all). On the reverse trajectory a particle fired towards the centre from outside will hit its target at the centre though it may follow a weird route getting there.



11. If particle in the described situation is projected with speed v_0 from the centre then it will emerge out from the field with speed
- (A) greater than v_0
(B) less than v_0
(C) equal to v_0
(D) depends on the variation (with distance from centre) & value of magnetic field in the region

12. Two concentric circularly symmetrical magnetic field are shown in the diagram. Both are uniform and have same magnitude and pointing perpendicular to the page, one is extended in the region from zero to R & pointing into the page & other is extended from R to $2R$ pointing out to the page. A charge particle is projected from centre with speed v , its distance from the centre when it is moving in radial outward direction is :

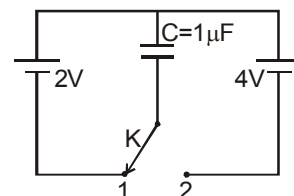


- (A) $\frac{R}{2}$ (B) R (C) $\sqrt{2} R$ (D) $2 R$

Matching List Type (4×4)

1 Q. [3 M (-1)]

13. The circuit involves two ideal cells connected to a $1 \mu\text{F}$ capacitor via a key K. Initially the key K is in position 1 and the capacitor is charged fully by 2V cell. The key is pushed to position 2. Column-I gives physical quantities involving the circuit after the key is pushed from position 1. Column-II gives corresponding results. Match the statements in Column-I with the corresponding values in Column-II and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the OMR.



Column-I

- (P) The net charge crossing the 4 volt cell in μC is
(Q) The magnitude of work done by 4 Volt cell in μJ is
(R) The gain in potential energy of capacitor in μJ is
(S) The net heat produced in circuit in μJ is

Column-II

- (1) 2
(2) 6
(3) 8
(4) 16

Codes :

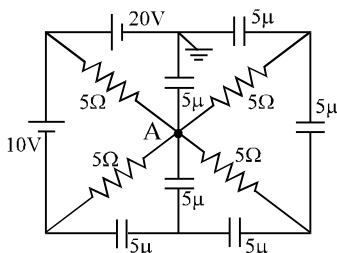
	P	Q	R	S
(A)	4	3	2	1
(B)	4	1	2	3
(C)	1	2	3	4
(D)	1	3	2	1

SECTION-II

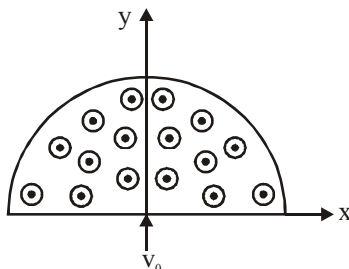
Numerical Answer Type Question (upto second decimal place)

2Q.[3(0)]

1. Find the potential of point A (in Volt) at steady state in the circuit shown in the figure.



2. Magnetic field given by $(100\hat{k})$ T in the region defined by $x^2 + y^2 = 16$ and $y > 0$. A charge particle of mass $m = 10^{-19}$ kg, $q = 1.6 \times 10^{-19}$ C, enters at origin with a velocity $(640\hat{j})$ m/s. What is the angle of deviation (in degree) suffered by charge particle in the magnetic field.

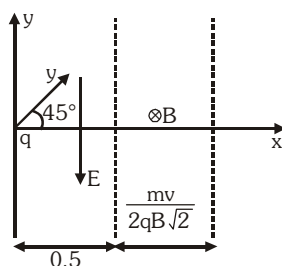


SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

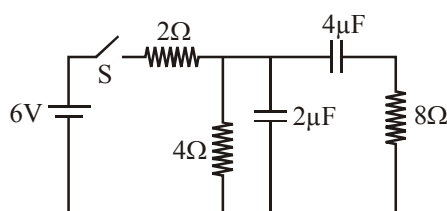
1. A charge particle of charge q and mass m is projected in a region which contains electric and magnetic field as shown in the figure with velocity v at an angle 45° with x direction. If $v = \sqrt{\frac{qE}{m}}$, then net deviation in particle motion will be θ (in degree) then find the value of $\theta/15$. (neglect the effect of gravity)



2. An electron is shot into one end of a solenoid. As it enters the uniform magnetic field within the solenoid, its speed is 800 m/s and its velocity vector makes an angle of 30° with the central axis of the solenoid. The solenoid carries 4.0 A current and has 8000 turns along its length. Find number of revolutions made by the electron within the solenoid by the time it emerges from the solenoid's opposite end. (Use charge to mass ratio $\frac{e}{m}$ for electron $= \sqrt{3} \times 10^{11}$ C/kg). Fill your answer of y if total number of revolution is

$$\frac{y}{5} \times 10^6.$$

3. In the circuit shown in figure the switch S is initially open and both the capacitors are initially uncharged. Find the ratio of current through 2Ω resistor, just after the switch S is closed and a long time after the switch S is closed.



SECTION-IV

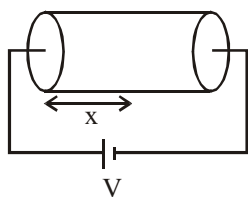
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

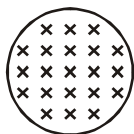
1. Column-I indicated certain situations and column-II indicates the values of electric field (induced or electrostatic). Match the correct options :

Column-I

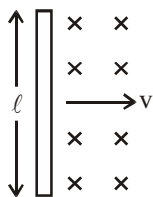
- (A) A uniformly charged infinite cylinder of radius R . x is distance from axis of cylinder
- (B) Current flowing through uniform cylindrical resistor. x is distance from left end. x is less than the length of resistor



- (C) An ideal solenoid of radius R . Current through solenoid is increasing with time. x is distance from axis of solenoid



- (D) A conducting rod of length ℓ is moving in a direction perpendicular to the magnetic field with uniform velocity. x is distance from upper end. ($x < \ell$)



Column-II

- (P) $E = 0$
- (Q) $|\vec{E}|$ increases with increase in x , reaches a maximum and then decreases.
- (R) $|\vec{E}|$ decreases with increase in x , reaches a minimum and then increases.
- (S) $|\vec{E}|$ is constant
- (T) $|\vec{E}|$ continuously decreases with increase in x

SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

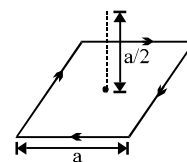
1. The magnetic field due to a current carrying square loop of side a at a point located symmetrically at a distance of $a/2$ from its centre (as shown is)

(A) $\frac{\sqrt{2} \mu_0 i}{\sqrt{3} \pi a}$

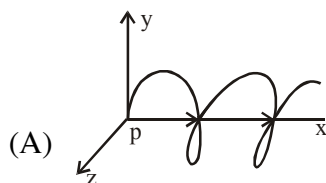
(B) $\frac{\mu_0 i}{\sqrt{6} \pi a}$

(C) $\frac{2 \mu_0 i}{\sqrt{3} \pi a}$

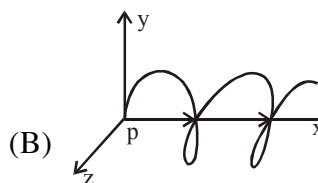
(D) zero



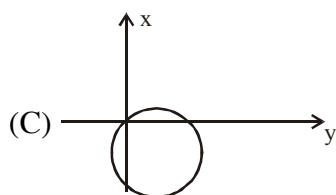
2. An electron is projected from origin with velocity $3.2\hat{i} + \hat{j}$ (m/s). A uniform magnetic field of $2\hat{k}$ tesla is present. The path followed by electron is correctly shown in ($m_e = 9 \times 10^{-31}$ kg)



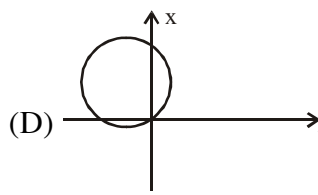
Helix on +ve side of z-axis



Helix on -ve side of z-axis



circle in xy plane



circle in xy plane

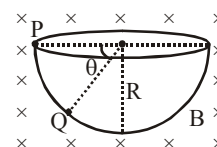
3. A charged sphere of mass m and charge $-q$ starts sliding along the surface of a smooth hemispherical bowl, at position P. The region has a transverse uniform magnetic field B . Normal force by the surface of bowl on the sphere at position Q is :-

(A) $mg \sin \theta + qB \sqrt{2gR \sin \theta}$

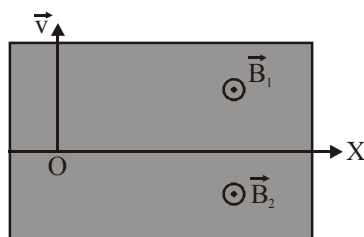
(B) $3 mg \sin \theta + qB \sqrt{2gR \sin \theta}$

(C) $mg \sin \theta - qB \sqrt{2gR \sin \theta}$

(D) $3 mg \sin \theta - qB \sqrt{2gR \sin \theta}$



4. A charged particle is projected at a speed $V = 10^6$ m/s perpendicular to the boundary OX of two homogeneous magnetic fields. Magnetic induction vector fields are parallel to each other and perpendicular to the particle velocity. The average velocity of the particle till it completes a semicircle in both magnetic fields is 10^5 m/s. Then :-



(A) $B_1 : B_2 = 40 - \pi : 40 + \pi$

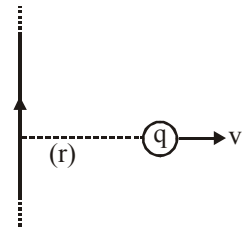
(B) $B_1 : B_2 = 20 - \pi : 20 + \pi$

(C) $B_1 : B_2 = 1 : 1$

(D) $B_1 : B_2 = 2\pi : 2 + \pi$

5. A charge (q, m) is thrown perpendicularly with speed v from a point at a distance r from an infinite long current (I) carrying wire. If its maximum distance from wire is R then :

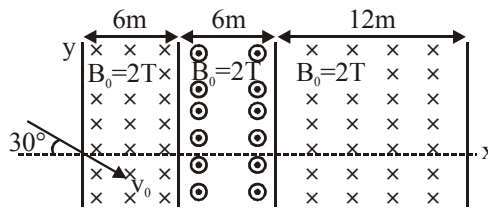
- (A) R will be infinite
(B) R will be $\left(re^{\frac{2\pi mv}{\mu_0 q I}} \right)$
(C) R will be $\left(re^{\frac{4\pi mv}{\mu_0 q I}} \right)$
(D) R will be $\left(re^{\frac{\pi mv}{\mu_0 q I}} \right)$



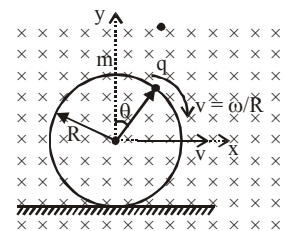
Multiple Correct Answer Type

2 Q. [4 M (-1)]

6. A particle having charge 1C and mass 1 kg enters a region having uniform magnetic field of strength ' 2T ' with a speed of 12 m/s , as shown in figure, then the **CORRECT** statement(s) is/are :-



- (A) The time for which the charge particle remains in magnetic field is $\frac{4\pi}{3}\text{ s}$.
(B) The velocity of charged particle becomes parallel to x -axis 6 time during its motion
(C) The distance between the point where the charge particle enters the uniform magnetic field and the point where it emerges out is 6 m
(D) The deviation of the charge particle when it emerges out of the magnetic field is $\frac{2\pi}{3}\text{ rad}$.
7. A ring of mass m and radius R is set into pure rolling on horizontal rough surface, in a uniform magnetic field of strength B as shown in the figure. A point charge q of negligible mass is attached to rolling ring. Friction is sufficient so that it does not slip at any point of its motion. (θ is measured in clockwise from positive y -axis) :-
- (A) Ring will continue to move with constant velocity
(B) The value of friction acting on ring is $Bqv \cos \theta$
(C) The value of friction acting on ring is $Bqv \sin \theta$



- (D) Ring will lose contact with ground if v is greater than $\left(\frac{mg}{2qB} \right)$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q. & 1 Para $\times$ 2Q.) [3 M (-1)]

Paragraph for Questions no. 8 to 10

A charge particle of mass m and charge q is projected on a rough horizontal XY plane. Both electric and magnetic fields are given by $\vec{E} = -10\hat{k}\text{ N/C}$ and magnetic field $\vec{B} = -5\hat{k}\text{ tesla}$ are present in the region. The particle enters into the magnetic field at $(4, 0, 0)\text{ m}$ with a velocity $50\hat{j}\text{ m/sec}$. The particle

starts into a curved path on the plane. If coefficient of friction $\mu = \frac{1}{3}$ between particle and plane, then

($qE = 2mg, g = 10\text{m/s}^2$)

8. Radius of curvature of the path followed by particle, initially, is

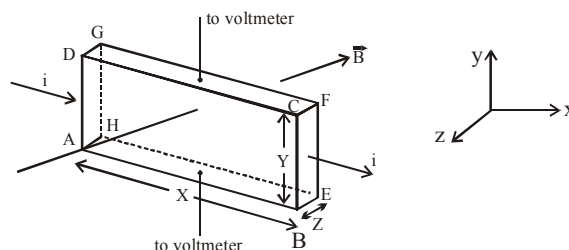
- (A) 5m (B) 2.5m (C) 1.25m (D) 10m

9. The time after which particle comes to rest, is :-
(A) 5s (B) 4s (C) 3s (D) 1s
10. Total work done by electric force on the particle is
(A) 250J (B) zero (C) 125J (D) none

Paragraph for Question no. 11 and 12

When a conductor carries a current i the charge carriers move with a drift velocity v_d directed along the length of conductor. A uniform magnetic field B is applied perpendicular to current as shown in figure.

Magnetic force deflects charge carriers. As a resultant of magnetic deflection a potential difference called Hall voltage is established in transverse direction to current. In usual notations n = number of electron per unit volume, σ = conductivity of conductor, q = magnitude of charge carrier.



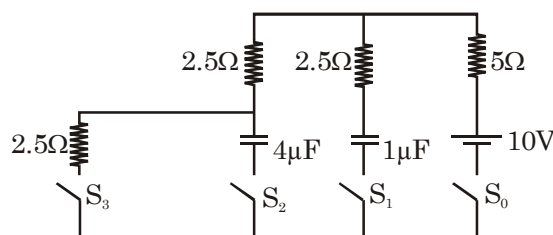
11. Mark the **CORRECT** option :
(A) In steady state face CFGD becomes positively charged
(B) Hall voltage developed across conductor has magnitude $\frac{1}{nq} \frac{iB}{Y}$
(C) An electric field is established from face ABCD towards EFGH
(D) By decreasing Z , voltage developed can be increased.
12. Direction of resultant electric field inside conductor from x-axis is given by :

- (A) $\tan \alpha = \frac{\sigma B}{nq}$ (B) $\tan \alpha = \frac{iB}{\sigma n}$ (C) $\tan \alpha = \frac{\sigma n}{iB}$ (D) $\tan \alpha = \frac{ni}{\sigma B}$

Matching List Type (4×4 & 4×5)

13. The circuit consists of four switches S_0, S_1, S_2, S_3 as shown below. All switches are initially open. Consider these four events in following order :-

- Switches S_0, S_1, S_2 are closed. Switch S_3 remains open
- Now switch S_3 is also closed, so that all four switches are closed
- Now switches S_0 and S_3 are opened simultaneously. Switches S_1 and S_2 are left closed.
- Now S_2 is also opened and S_0 is closed. After a long time, a dielectric slab ($k = 3$) is inserted slowly between the plates of $1\mu\text{F}$ capacitor completely filling the gap.



1 Q. [3 M (-1)]

- | List-I | List-II |
|---|---------|
| (P) Charge (in μC) on $1\mu\text{F}$ capacitor long time after event (i) is | (1) 10 |
| (Q) Charge (in μC) on $4\mu\text{F}$ capacitor long time after event (ii) is | (2) 15 |
| (R) Charge (in μC) on $1\mu\text{F}$ capacitor long time after event (iii) is | (3) 3 |
| (S) Charged supplied (in μC) by the battery during event (iv) is | (4) 20 |
| | (5) 0 |

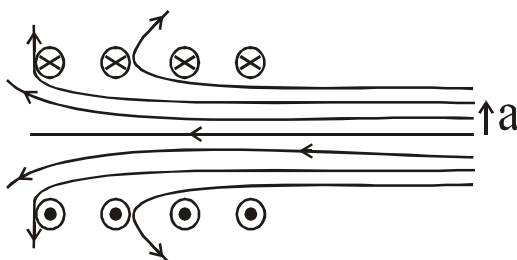
- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$ (B) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$
(C) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$ (D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

SECTION-III

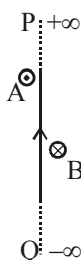
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

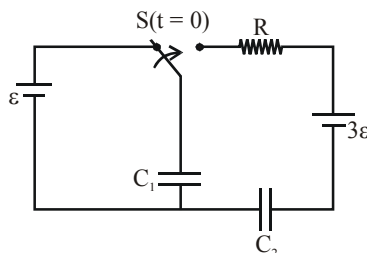
1. A direct current flows in a solenoid of length L and radius R , ($L \gg R$), producing a magnetic field of magnitude B_0 inside the solenoid. Magnetic field line which leaves end perpendicularly is at distance 'a' from axis of solenoid. If $a = \frac{R}{\sqrt{n}}$, find n



2. Two infinitely long wires are placed perpendicular to the plane of paper. Current in wire A is $4i_0$ outward the plane of paper and current in 'B' is i_0 inward the plane of paper. The $\int_{-\infty}^{+\infty} \vec{B} \cdot d\vec{\ell}$ along the QP is $\frac{K}{2} \mu_0 i_0$. Find the value of K .



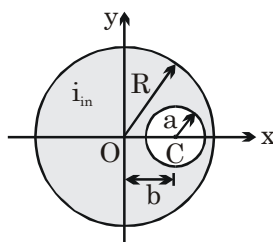
3. A capacitor $C_1 (= C)$ is charged to a potential difference ε is connected to a charging circuit by changing the switch S as shown. Assume the instant of switching as $t = 0$. Capacitor C_2 is, initially uncharged, then, the charge on $C_2 (= C)$ is changing according to equation ; $q_2 = Q_2(1 - e^{-t/\delta})$. where, δ is called time constant and Q_2 is the steady state charge on C_2 . Let, $Q_2 = n_2(C\varepsilon)$ and $\delta = n_1(RC)$. Find $\left(\frac{n_2}{n_1}\right)$.



4. A galvanometer of resistance 40Ω , shunted by a resistance of 50Ω gives a deflection of 50 divisions when joined in series with a resistance of $\frac{1000}{9}\Omega$ and a 2 volt battery, what is the current sensitivity of galvanometer (in div/mA)?

5. A very long straight conductor has a circular cross-section of radius R and carries a current density J . Inside the conductor there is a cylindrical hole of radius a whose axis is parallel to the axis of the conductor and at a distance b from it. Let the z -axis be the axis of the conductor, and let the axis of the hole be at $x = b$. Find the x component of magnetic field on the y -axis at $y = 2R$. If your answer is

$$B_x = \mu_0 J R \left(\frac{1}{A} - \frac{a^2}{BR^2 + b^2} \right) \text{ fill value of } |A| + |B|.$$



SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. In the column-I, there are certain situations depicted. Match them with their description in column-II.

Column-I

- (A) A point charge is projected at an acute angle to uniform magnetic field in gravity free space
- (B) A satellite is moving around earth in an elliptical orbit
- (C) A charge is released from rest in uniform electric and magnetic field parallel to each other (gravity free space)
- (D) A charge is released from rest in uniform electric and magnetic field perpendicular to each other (gravity free space)

Column-II

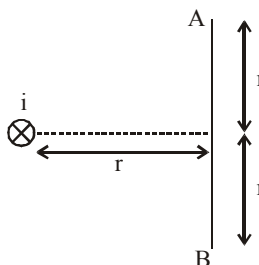
- (P) Speed changes with time
- (Q) Momentum changes with time
- (R) Radius of curvature of path changes with time
- (S) Acceleration is constant in magnitude
- (T) Acceleration changes its direction with time

SECTION-I

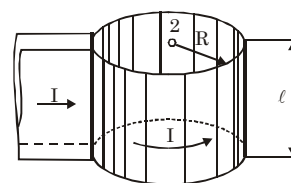
Single Correct Answer Type

7 Q. [3 M (-1)]

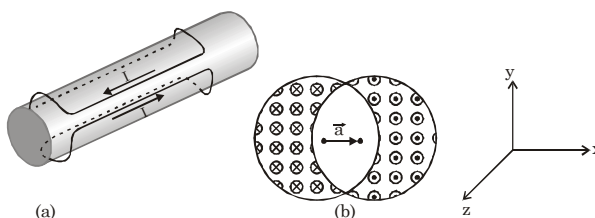
1. What will be the value of $\int_A^B \vec{B} \cdot d\vec{\ell}$, where AB is line perpendicular to an infinite current carrying wire as shown in diagram ? (where $\vec{B}$ is magnetic field due to wire carrying current i)



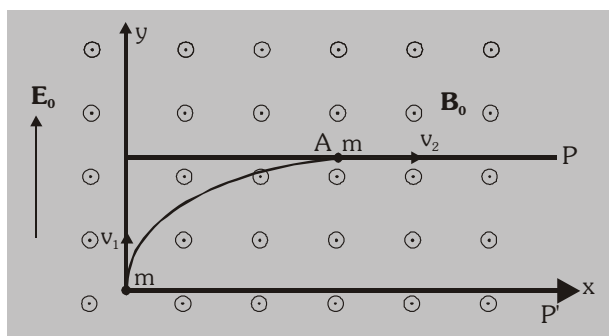
- (A) $\frac{\mu_0 i}{\pi}$ (B) $\frac{\mu_0 i}{\sqrt{2}}$ (C) $\frac{\mu_0 i}{4}$ (D) $\frac{\mu_0 i}{2}$
2. Thin tape of width $\ell = 40$ cm is rolled into a tube of radius $R = 30$ cm. On the tape flows evenly distributed current of $I = 200$ A as shown in figure. The magnetic induction B on the axis of the tube at point 2 is :-
- (A) $100 \pi \mu T$ (B) $80 \pi \mu T$
(C) $60 \pi \mu T$ (D) $120 \pi \mu T$



3. To produce a uniform magnetic field directed parallel to a diameter of a cylindrical region, one can use the saddle coils illustrated in figure. The loops are wrapped over a somewhat flattened tube. Assume the straight sections of wire are very long. The end view of the tube shows how the windings are applied. The overall current distribution is the superposition of two overlapping, circular cylinders of uniformly distributed current, one toward you and one away from you. The current density J is the same for each cylinder. The position of the axis of one cylinder is described by a position vector $\vec{a}$ relative to the other cylinder. The magnetic field inside the hollow tube is.



- (A) $\frac{\mu_0 J a}{2}$ along x-axis (B) $\frac{\mu_0 J a}{2}$ along y-axis (C) $\mu_0 J a$ along z-axis (D) $\mu_0 J a$ along y-axis
4. There are constant electric field $E_0 \hat{j}$ & magnetic field $B_0 \hat{k}$ present between plates P and P'. A particle of mass m is projected from plate P' along y axis with velocity v_1 . After moving on the curved path, it passes through point A just grazing the plate P with velocity v_2 . The magnitude of impulse (i.e. $\vec{F} \Delta t = \Delta \vec{p}$) provided by magnetic force during the motion of particle from origin to point A is :-



- (A) $m|v_2 - v_1|$ (B) $m\sqrt{v_1^2 + v_2^2}$ (C) mv_1 (D) mv_2

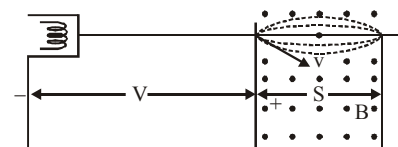
5. Figure shows an electron gun that emits electrons at very small angle with axis. The paraxial electrons emitted from the cathode (filament) are refocused at a distance s after accelerating through a potential difference V , s is given by (Take $\pi^2 = 10$) :-

(A) $s = 2\sqrt{\frac{20mV}{eB^2}}$

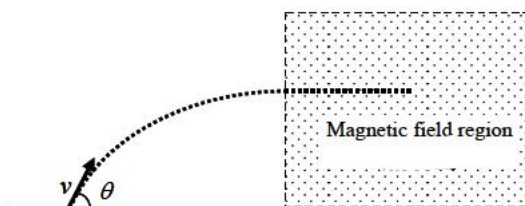
(B) $s = \sqrt{\frac{20mV}{eB^2}}$

(C) $s = \sqrt{\frac{10mV}{eB^2}}$

(D) None of these

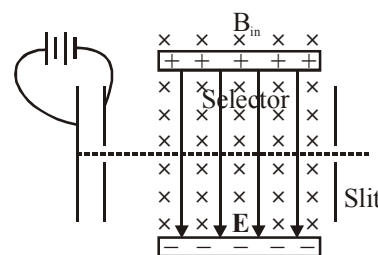


6. A particle of mass M and carrying charge Q is launched with initial speed v and at an angle of θ relative to the horizontal direction. When it reaches the maximum height it enters a region of uniform magnetic field. In the region it moves at constant velocity in the horizontal direction. Determine the direction and strength of the magnetic field.



- (A) outward $\frac{Mg}{Qv \cos \theta}$ (B) inward $\frac{Mg}{Qv \cos \theta}$ (C) outward $\frac{Mg}{Qv \sin \theta}$ (D) inward $\frac{Mg}{Qv \sin \theta}$

7. The given figure shows a set-up for accelerating protons from rest, through a potential (fixed) by the battery, and then through a velocity selector. The E and B fields are carefully adjusted so that the accelerated protons go straight through the slit after exiting the selector. You want to modify this so that electrons instead of protons are used. You reverse the leads on the accelerating battery. What do you need to do to the potential difference between the plates that create the electric field in the selector?

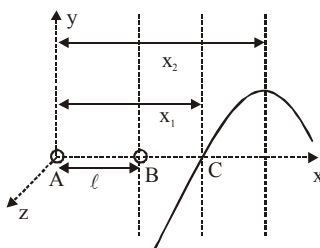


- (A) Nothing (B) Reduce its magnitude
(C) Increase its magnitude (D) Reverse the polarity, and reduce its magnitude.

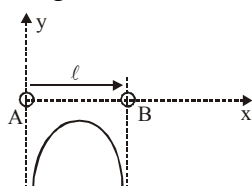
Multiple Correct Answer Type

4 Q. [4 M (-1)]

8. Two infinite long wires A and B carrying currents $2I$ and I are placed parallel to each other at $x = 0$ and $x = l$ respectively. The magnetic field intensity to the right of the wire B on the line that passes through the two wires varies according to a law that is represented systematically as shown in figure.

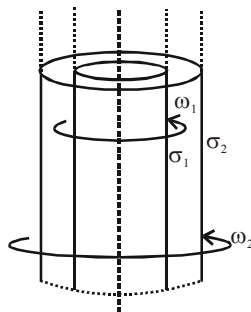


- (A) Current in A is along positive z -axis and in B is along negative z -axis
(B) Leaving point C and infinity, the magnetic field intensity is not zero in the complete region. (including region between A and B and region left to A)
(C) Magnetic field intensity between A and B on the line joining A and B varies as



- (D) Value of x_1 and x_2 are $2l$ and $l(\sqrt{2} + 2)$ respectively

9. Two **VERY LONG** hollow nonconducting cylindrical shells (of different radius) are placed co-axially, one inside other. They carry non-zero surface charge densities σ_1 & σ_2 , as shown. The shells begin to rotate about their common axis with non-zero angular speeds either clockwise or anticlockwise. If magnetic field on the axis is zero then choose right possibility(ies) :-



- (A) $\sigma_1 \neq \sigma_2, \omega_1 \neq \omega_2$ (B) $\sigma_1 \neq \sigma_2, \omega_1 = \omega_2$ (C) $\sigma_1 = \sigma_2, \omega_1 \neq \omega_2$ (D) $\sigma_1 = \sigma_2, \omega_1 = \omega_2$
10. An infinite current carrying wire carries current i in negative x direction. Force on a charged particle at a point having coordinate $(6, 2, 3)$ is zero if velocity of the particle is
- (A) $-4\hat{i} + 8\hat{j}$ (B) $4\hat{j} + 3\hat{k}$ (C) $-6\hat{j} + 4\hat{k}$ (D) $6\hat{i} + 3\hat{j} - 2\hat{k}$
11. A particle ($q/m = 1$) is released from rest at origin. Choose correct statement
[Given $\vec{B} = 4e^{-6x}\hat{j} + \sin 2y\hat{i}$ and $\vec{E} = 4x\hat{i}$]
- (A) Speed of the particle at $x = 1$ is $v = 2$ m/s (B) Particle will move in x - y plane only
(C) Speed of the particle at $x = 2$ is $v = 2$ m/s (D) None of these

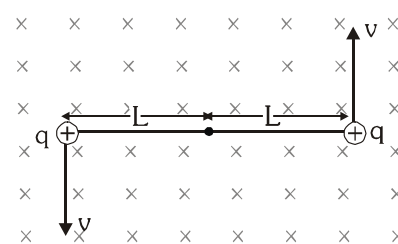
Linked Comprehension Type

(1 Para $\times$ 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Questions 12 to 14

Two charge particles each of mass ' m ', carrying charge $+q$ and connected with each other by a massless inextensible string of length $2L$ are describing circular path in the plane of paper, each with speed $v = \frac{qB_0L}{m}$ (where B_0 is constant) about their centre of mass in the region in which a uniform magnetic field $\vec{B}$ exists into the plane of paper as shown in figure. Neglect any effect of electrical & gravitational forces.

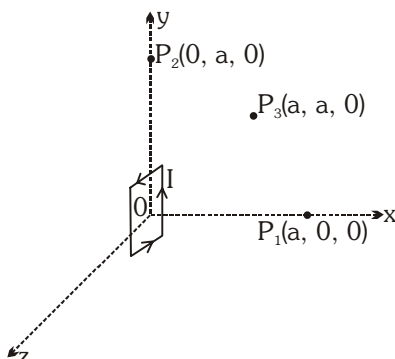


12. The magnitude of the magnetic field such that no tension is developed in the string will be
- (A) $\frac{B_0}{2}$ (B) B_0 (C) $2B_0$ (D) 0
13. If the actual magnitude of magnetic field is half to that of calculated in part (i) then tension in the string will be
- (A) $\frac{3}{4} \frac{q^2 B_0^2 L}{m}$ (B) zero (C) $\frac{q^2 B_0^2 L}{2m}$ (D) $\frac{2q^2 B_0^2 L}{m}$
14. Given that the string breaks when the tension is $T = \frac{3}{4} \frac{q^2 B_0^2 L}{m}$. Now if the magnetic field is reduced to such a value that the string just breaks then find the maximum separation between the two particles during their motion
- (A) $16L$ (B) $4L$ (C) $14L$ (D) $2L$

Matching List Type (4 × 4)

1 Q. [3 M (-1)]

15. A very small current carrying square loop (current I) of side ' L ' is placed in y - z plane with centre at origin of the coordinate system (shown in figure). In column-I the coordinate of the points are given & in column-II magnitude of strength of magnetic field is given. Then :-



List-I

- (P) At point O (0, 0, 0)
- (Q) At point P_1 (a, 0, 0) (here $a \gg L$)
- (R) At point P_2 (0, a, 0) (here $a \gg L$)
- (S) At point P_3 (a, a, 0) (here $a \gg L$)

List II

- (1) $\frac{\mu_0 I L^2}{2\pi a^3}$
- (2) $\frac{2\sqrt{2}\mu_0 I}{\pi L}$
- (3) $\frac{\mu_0 \sqrt{5} I L^2}{16\pi a^3}$
- (4) $\frac{\mu_0 I L^2}{4\pi a^3}$

Code :

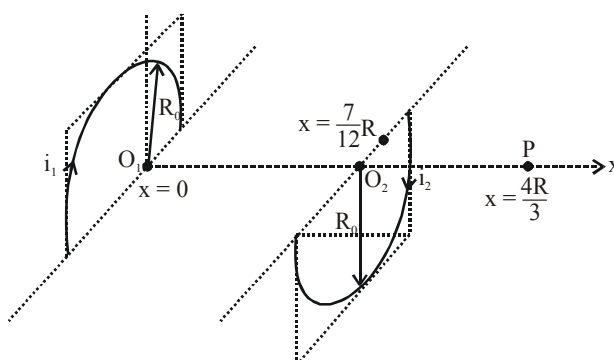
	P	Q	R	S
(A)	4	2	3	1
(B)	1	1	2	3
(C)	4	1	2	3
(D)	2	1	4	3

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

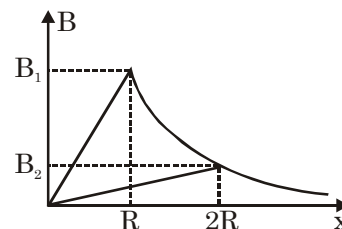
3 Q. [4 M (0)]

1. Two semicircular ring of same radius (R) having current i_1 & i_2 are kept parallel to each other in y - z plane as shown in the figure. Centre of first ring is kept at $x = 0$ and that of second ring is kept at $x = \frac{7}{12}R$. If magnetic field at point P ($x = \frac{4R}{3}$) is along the x -axis then its value is $\frac{n\mu_0 i_1}{20R}$. Find the value of n .

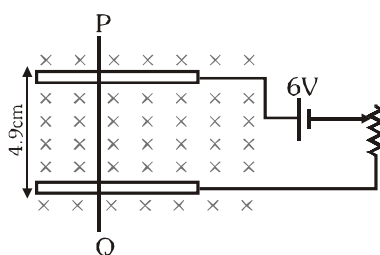


2. We have two separate long cylindrical wires both having different but uniform current densities. The radius of one of the wires is twice of the other. The figure shows the graph of magnetic field (B) with radial distance (r) from their axis. The curved parts of the two graphs

are overlapping. If the ratio $\left(\frac{B_1}{B_2}\right) = x$ then give $2x$.



3. A wire PQ of mass 10 g is at rest on two parallel metal rails. The separation between the rails is 4.9 cm. A magnetic field of 0.80 tesla is applied perpendicular to the plane of the rails, directed downwards. The resistance of the circuit is slowly decreased. When the resistance decreases to below 20 ohm, the wire PQ begins to slide on the rails. The coefficient of friction between the wire and the rails is found to be $\frac{\alpha\beta}{100}$, where α and β are single digit integers. Then find the value of $\alpha + \beta$.

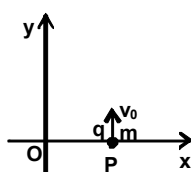


SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. A negative charged particle of mass 'm' having charge q enters in magnetic field $\vec{B} = B_0 \hat{k}$ at point P (3m, 0, 0) with velocity $\vec{v}_0 = 3\hat{j} + 4\hat{k}$, at t = 0 as shown in the figure $\left[\text{Given } \frac{m}{B_0 q} = 1 \right]$ [No other field is present]



Column I

- (A) Pitch of the motion of particle
(B) $\frac{24\pi}{25}$ × radius of curvature of particle during motion at time t = t sec
(C) Component of velocity of particle in x-y plane.
(D) Acceleration of particle

Column II

- (P) $(-3 \sin t \hat{i} + 3 \cos t \hat{j})$ unit
(Q) $(-3 \cos t \hat{i} - 3 \sin t \hat{j})$ unit
(R) 8π unit
(S) Constant
(T) Variable

SECTION-I

Single Correct Answer Type

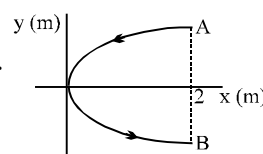
7 Q. [3 M (-1)]

1. Two long concentric cylindrical conductors of radii a & b ($b < a$) are maintained at a potential difference V & carry equal & opposite currents I . An electron with a particular velocity " U " parallel to the axis will travel undeviated in the evacuated region between the conductors. Then $U =$

(A) $\frac{4\pi V}{\mu_0 I \ln\left(\frac{b}{a}\right)}$ (B) $\frac{2\pi V}{\mu_0 I \ln\left(\frac{a}{b}\right)}$ (C) $\frac{2\pi V}{\mu_0 I \ln\left(\frac{b}{a}\right)}$ (D) $\frac{8\pi V}{\mu_0 I \ln\left(\frac{a}{b}\right)}$

2. A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current $i = 2$ A as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ Tesla. The magnetic force on the wire is (in newton)

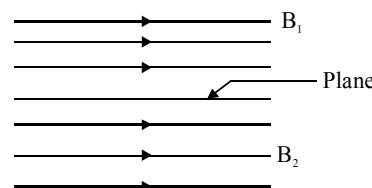
(A) $-16\hat{i}$ (B) $32\hat{i}$ (C) $-32\hat{i}$ (D) $16\hat{i}$



3. Two uniform magnetic fields B_1 and B_2 exist above and below the plane. The magnetic force acting on unit area of the plane is :-

(A) $\frac{B_2^2 - B_1^2}{2\mu_0} \uparrow$

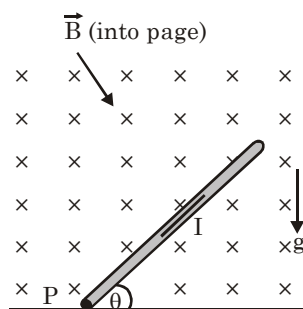
(B) $\frac{|B_1 - B_2|^2}{2\mu_0} \downarrow$



(C) $\frac{|B_1 + B_2|^2}{2\mu_0} \uparrow$

(D) None of the above

4. The drawing shows a thin, uniform rod that has a length of 0.45 m and a mass of 0.09 kg. This rod lies in the plane of the paper and is attached to the floor by a hinge at point P. A uniform magnetic field of 0.36 T is directed perpendicularly into the plane of the paper. There is a current $I = 4.0$ A in the rod, and rod does not rotate clockwise or counterclockwise. Find the angle θ .



(A) $\cos^{-1}(0.36)$ (B) $\cos^{-1}(0.72)$ (C) $\cos^{-1}(0.46)$ (D) $\cos^{-1}(2/3)$

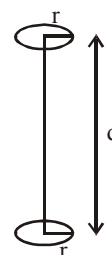
5. Two current carrying coil having radius ' r ' are separated by distance ' d ' as shown in diagram. If $r \ll d$. Then find the force between two ring. Current in both the ring is clockwise and equal to i .

(A) $\frac{6\mu_0 i^2 \pi^2 r^4}{4\pi d^4}$

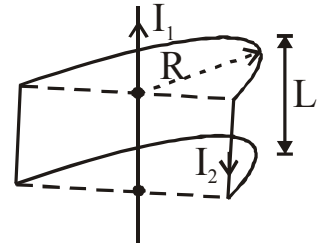
(B) $\frac{\mu_0 i^2 \pi^2 r^4}{4\pi d^4}$

(C) $\frac{3\mu_0 i^2 \pi^2 r^4}{4\pi d^4}$

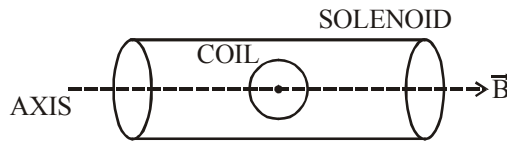
(D) $\frac{\mu_0 i^2 \pi^2 r^4}{2\pi d^4}$



6. An infinitely long straight wire carrying a current I_1 is partially surrounded by a loop, as shown in figure. The loop has a length L and radius R , current I_2 . The axis of loop coincides with the wire. Find force between two system



- (A) zero
(B) $\frac{\mu_0 I_1 I_2 L}{2\pi R}$
(C) $\frac{\mu_0 I_1 I_2 L}{\pi R}$
(D) $\frac{\mu_0 I_1 I_2 L}{R}$
7. The torque required to hold a small circular coil of 10 turns, area 1 mm^2 and carrying a current of $\left(\frac{21}{44}\right) \text{ A}$ in the middle of a long solenoid of 10^3 turns/m carrying a current of 2.5 A , with its axis perpendicular to the axis of the solenoid is

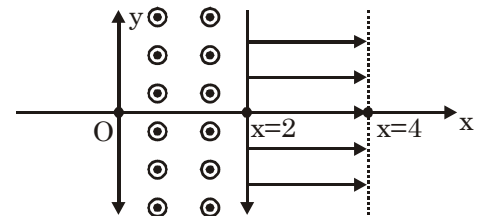


- (A) $1.5 \times 10^{-6} \text{ N-m}$ (B) $1.5 \times 10^{-8} \text{ N-m}$ (C) $1.5 \times 10^{+6} \text{ N-m}$ (D) $1.5 \times 10^{+8} \text{ N-m}$

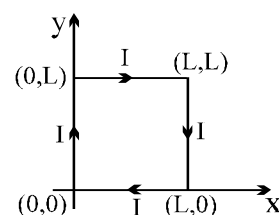
Multiple Correct Answer Type

5 Q. [4 M (-1)]

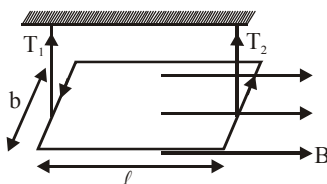
8. There exist uniform magnetic field $\vec{B} = 5\text{T}(\hat{k})$ from $x = 0$ to $x = 2\text{m}$. From $x = 2\text{m}$ to $x = 4\text{m}$ a uniform electric field $\vec{E} = 50\text{N/C}(\hat{i})$. A point charge ($q = 2\text{C}$, $m = 1\text{kg}$) is projected from origin with velocity $\vec{V} = 20\text{m/s}(\hat{j})$. Which of the following is/are correct :



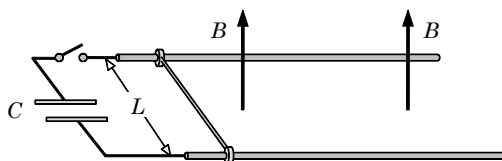
- (A) Time spend by the charge particle in magnetic field is $\frac{\pi}{20} \text{ s}$
(B) Time spend by the charge particle in electric field is $\frac{1}{5} \text{ s}$
(C) Charge particle will leave the electric field with speed $20\sqrt{2} \text{ m/s}$
(D) Charge particle will leave the electric field with velocity $20\hat{i} + 20\hat{j}$
9. Figure shows a square loop carrying current I is present in the magnetic field which is given by $\vec{B} = \frac{B_0 z}{L} \hat{j} + \frac{B_0 y}{L} \hat{k}$ where B_0 is positive constant. Which of the following statement(s) is/are correct?
- (A) force on side $(0,0)$ to $(0,L)$ is $\left(\frac{B_0 I L}{2}\right) \hat{i}$.
(B) force on side $(0,L)$ to (L,L) is $-B_0 I L \hat{j}$.
(C) Net magnetic force on loop is zero.
(D) force on side $(L,0)$ to $(0,0)$ is zero.



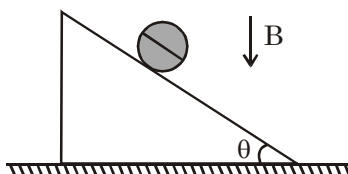
10. A uniform conducting rectangular loop of sides ℓ , b and mass m carrying current i is hanging horizontally with the help of two vertical strings. There exists a uniform horizontal magnetic field B which is parallel to the longer side of loop. Choose the **CORRECT** option(s) :-



- (A) The value of $T_1 = T_2 = \frac{mg}{2}$ (B) The value of $T_1 = \frac{mg - 2ibB}{2}$
- (C) The value of $T_2 = \frac{mg + 2ibB}{2}$ (D) The value of $T_1 < \text{value of } T_2$
11. In a region of space, where a uniform vertical magnetic field of induction B exists, a light rod of length L and mass m rigidly connected between two sleeves can slide on two horizontally fixed very long parallel conducting rails. A capacitor of capacitance C charged to voltage V_0 is connected through a switch to the rails as shown. Initially the switch is open and the rod is at rest. Now the switch is closed. Assume the rails to be long enough and ignoring electrical resistance and frictional forces. Which of the following conclusion can you make?



- (A) The steady-state speed acquired by the rod is $\frac{BlCV_0}{m + CB^2L^2}$.
- (B) Efficiency of conversion of electrical energy of the capacitor into mechanical energy is $\frac{CB^2L^2}{m + 2CB^2L^2}$.
- (C) Heat released in the resistance of the jumper is $\frac{1}{2}CV_0^2 \left(\frac{m}{m + CB^2L^2} \right)$.
- (D) None of the above is correct.
12. In the figure shown a coil of single turn is wound on a sphere of radius R and mass M . The plane of the coil is parallel to the inclined plane and lies in the equatorial plane of the sphere. Current in the coil is i . The sphere is in equilibrium.



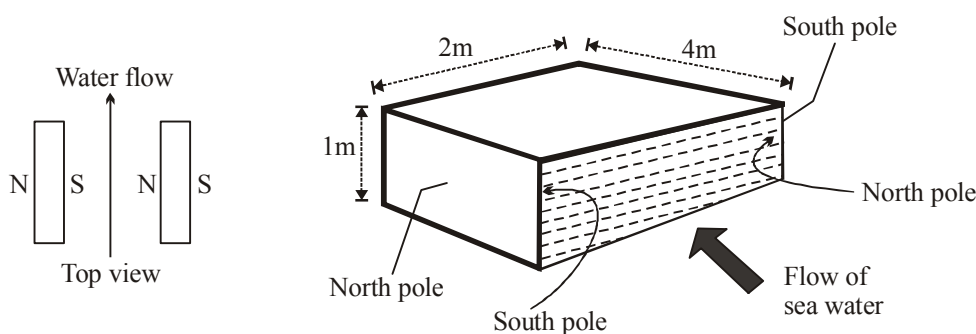
- (A) The value of B is $\frac{mg \tan \theta}{\pi i R}$
- (B) The current in coil is clockwise as seen from top.
- (C) Minimum friction coefficient required is $\tan \theta$.
- (D) If the surface is smooth, sphere will slide down without rotation.

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for question nos. 13 to 15

The dead-quiet “caterpillar drive” for submarines in the movie "The Hunt for Red October" is based on a magnetohydrodynamic (MHD) drive ; as the submarines moves forward, seawater flows through multiple channels in a structure built around the rear of the hull. Figure shows the essentials of a channel. Magnets, positioned along opposite sides of the channel with opposite poles facing each other, create a magnetic field within the channel. Electrodes (not shown) create an electric field across the channel. The electric field derives a current across the channel and through the water ; the magnetic force on the current propels the water toward the rear of the channel, thus propelling the submarines forward.



13. In figure what should be the direction of electric field ?
(A) upward (B) downward (C) left ward (D) rightward
14. If the value of magnetic field is 100T and current flowing is 50 A what is the force with which the water is pushed out through this channel?
(A) 500 N (B) 5000N (C) 10000N (D) 20000N
15. What should be the value of electric field to achieve the 50 A current given that resistivity of sea water is $8\Omega\text{-m}$.
(A) 25 V/m (B) 50 V/m (C) 100 V/m (D) 200 V/m

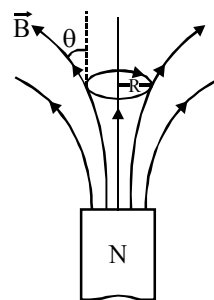
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

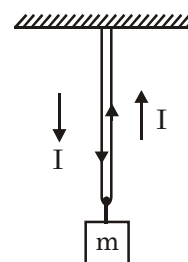
1. A superconducting current carrying ring of mass $m = 40\text{ gm}$ and $R = 2\text{cm}$ floats in the magnetic field of a magnet. If the field lines make an angle $\theta = 30^\circ$ with vertical and the ring is placed co-axially, assuming $i = 10\text{A}$,

find the magnetic field $\vec{B}$. if answer is $\frac{n}{3\pi}$. Write the value of n.

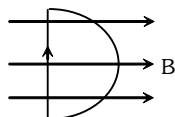


2. Consider a loop of freely deformable conducting wire with insulation of length 2ℓ , the two ends of which are fixed (permanently) to the ceiling. A load of mass 'm' is fixed to the middle of the wire (the mass of the wire is negligible). There is also a outward magnetic field of induction B; free fall acceleration is g. A current I is flowing through the wire which starts increasing slowly to value I_0 . How large current I_0 is needed to lift the

load by $\Delta h_0 = \ell \left(1 - \frac{3}{\pi}\right)$. $m = \frac{3\sqrt{3}}{\pi}\text{kg}$, $\ell = 2\text{m}$, $B = 1\text{ tesla}$.



3. A semicircular loop of mass per unit length $\frac{\pi}{10}$ kg/m is placed on a surface with its plane parallel to the surface in a uniform magnetic field of magnitude 1 Tesla as shown in figure. Find the minimum amount of current (in A) that should be passed in the loop so that it just start to rotate?



SECTION-IV

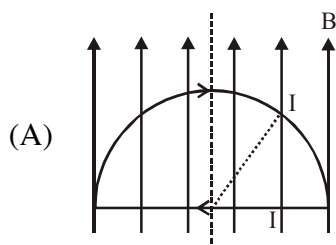
Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

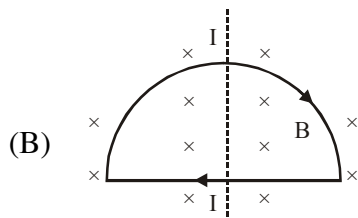
1. Find correct match for the figure in column I as shown with the items given in column II :
Match the following column :

Column-I

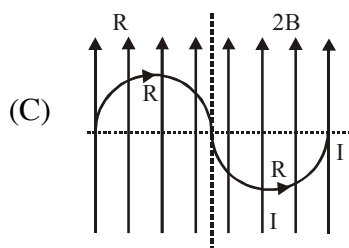
Column-II



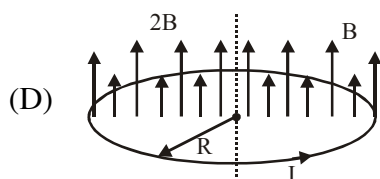
(P) $F \neq 0$



(Q) $F = 0, \tau \neq 0$



(R) $F = 0, \tau = 0$



(S) $F \neq 0, \tau = 0$

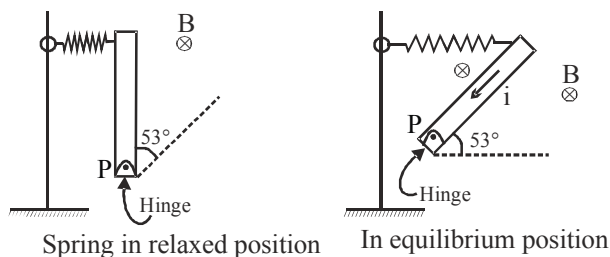
(T) $F \neq 0, \tau \neq 0$

SECTION-I

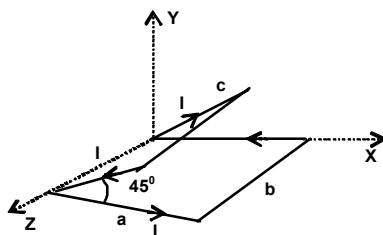
Single Correct Answer Type

6 Q. [3 M (-1)]

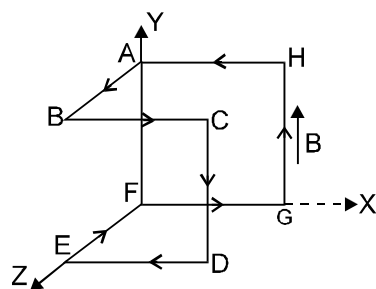
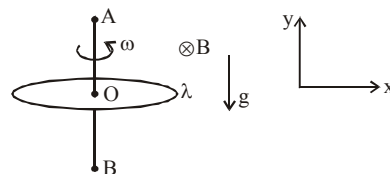
1. A thin uniform rod with negligible mass and length ℓ is attached to the floor by a frictionless hinge at point P. A horizontal spring with force constant k connects the other end to wall. The rod is in a uniform magnetic field B directed into the plane of paper. What is extension in spring in equilibrium when a current i is passed through the rod in direction shown. Assuming spring to be in natural length initially.



- (A) $\frac{5i\ell B}{8k}$ (B) $\frac{3i\ell B}{8k}$ (C) $\frac{5i\ell B}{4k}$ (D) $\frac{5i\ell B}{6k}$
2. Find the magnetic dipole moment of the rectangular loop shown in the figure. Sides, $a = 3\text{m}$, $b = 4\text{m}$ and $c = 2\text{m}$, current in the loop $I = 1\text{ Amp}$

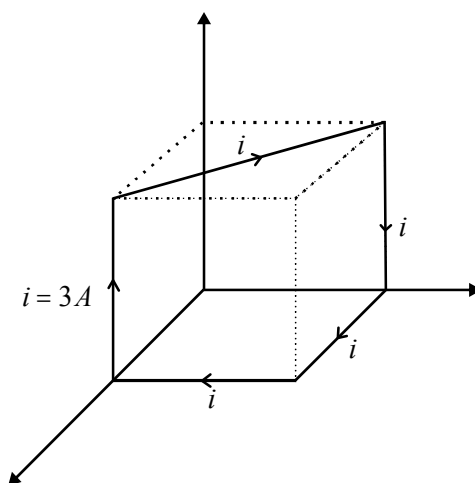


- (A) 11.8 Am^2 (B) 15.8 Am^2 (C) 13.8 Am^2 (D) 12.8 Am^2
3. Diagram shows a charged disc rigidly fixed to a rotation axis AB (That passes through the centre of mass of disc and is perpendicular to the plane containing disc). Disc is set into rotation with constant angular speed ' ω ' in a magnetic field as shown in the figure. Choose the INCORRECT statement :-
- (A) Torque on disc due to reaction force of axis is along positive x-axis.
(B) Torque of magnetic field on the disc is same about any axis.
(C) Torque of gravity about the axis of rotation is non-zero.
(D) Net torque on disc is zero.
4. The given fig. shows a coil bent with $AB = BC = CD = DE = EF = FG = GH = HA = 1\text{ m}$ and carrying current 1 A . There exists in space a vertical uniform magnetic field of 2 T in the y-direction. Then find out the torque (in vector form) on the loop.

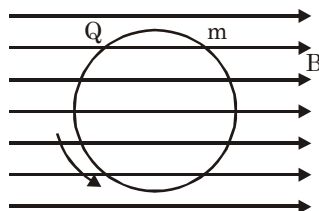


- (A) $2\hat{k}$
(B) $-2\hat{k}$
(C) $2\hat{j}$
(D) $-2\hat{j}$

5. A wire carrying 3A current is wrapped around a non - conducting cube of side 1m as shown below. The magnitude of net magnetic moment due to the loop will be :-



- (A) $\frac{9}{2}$ unit (B) $\frac{1}{\sqrt{3}}$ unit (C) 9 unit (D) None of these
6. A thin uniform ring of mass m and having electric charge Q uniformly distributed rotates around an axis perpendicular to its plane and going through its center. The angular momentum of the ring is $7.5 \times 10^{-4} \text{ kg-m}^2/\text{s}$. The ring is in a homogeneous magnetic field of a field strength of 0.1 T and the lines of the magnetic induction are parallel with the plane of the ring. Torque exerted on the ring is ? [The specific charge (charge-mass ratio) of the ring is $Q/m = 10^{-5} \text{ C/kg}$].



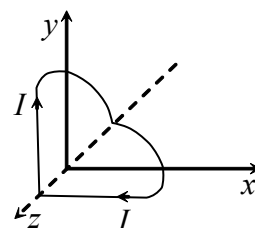
- (A) $3.75 \times 10^{-10} \text{ (N-m)}$ upward along the plane
(B) $3.75 \times 10^{-10} \text{ (N-m)}$ downward along the plane
(C) $7.5 \times 10^{-10} \text{ (N-m)}$ upward along the plane
(D) none of these

Multiple Correct Answer Type

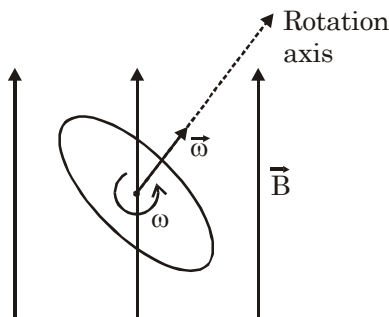
2 Q. [4 M (-1)]

7. A circular current carrying loop of radius R is bent about its diameter by 90° and placed in a magnetic field $\vec{B} = B_0(\hat{i} + \hat{j})$ as shown in figure.

- (A) The torque acting on the loop is zero
(B) The magnetic moment of the loop is $\frac{I\pi R^2}{2}(-\hat{i} - \hat{j})$
(C) The angular acceleration of the loop is non zero.
(D) The magnetic moment of the loop is $\frac{I\pi R^2}{2}(-\hat{i} + \hat{j})$



8. A uniform, thin, uniformly charged disk of mass m radius R and uniform surface charge density σ rotates with angular speed ω about an axis through its centre and perpendicular to disc. The disk is in region with a uniform magnetic field B that makes angle θ with rotation axis. Mark the **CORRECT** statement :-



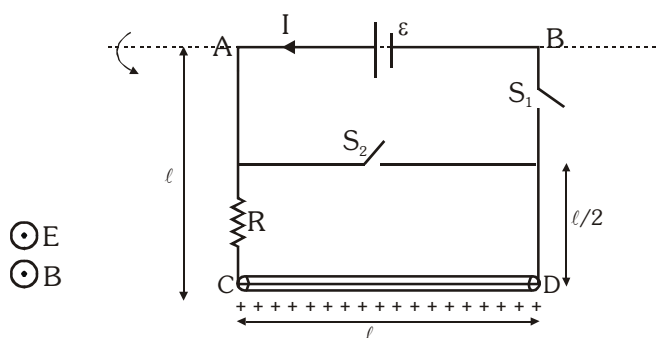
- (A) Torque exerted on the disk by magnetic field is $\frac{1}{4} \pi \sigma R^4 \omega B \sin \theta$
- (B) Frequency with which angular velocity vector rotates is given by $\frac{\pi \sigma R^2 B}{4m} \sin \theta$.
- (C) For an observer looking from above the angular velocity vector begins to rotate anticlockwise sense.
- (D) For an observer looking from above the angular velocity vector begins to rotate clockwise sense.

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 3Q. & 1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions no. 9 to 11

Consider the circuit shown with the two switches S_1 and S_2 . The battery has an emf of e and the internal resistor has a resistance. The bottom wire is encased in an insulated wrapping of mass M which carries a positive static charge of Q . Assume that the masses of all other wires and components are very small as compared to this charge bearing wrapping. The dimensions of the circuit are ℓ . The bridge (or middle connecting) wire, with S_2 in it, is $\ell/2$ above the bottom wire. The device is suspended in such a way that it can rotate in vertical circle about axis AB. The circuit maintains its shape regardless of the switch configuration. A magnetic field B , and an electric field (E), both point out of the page, exist in the space. Magnetic interaction between wires of the circuit is negligible.



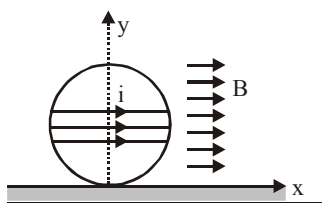
9. With both switches open, find the equilibrium angle θ that the plane of the circuit makes with the vertical as it swings upwards due to the presence of the electric field.
- (A) $\tan^{-1}(QE/3Mg)$ (B) $\tan^{-1}(4QE/3Mg)$ (C) $\tan^{-1}(QE/6Mg)$ (D) $\tan^{-1}(QE/Mg)$
10. With S_1 closed and S_2 open, find the equilibrium angle θ (assuming current through battery as I)
- (A) $\tan^{-1} QE/(3Mg + ILB)$ (B) $\tan^{-1} 4QE/(3Mg + ILB)$
- (C) $\tan^{-1} QE/(6Mg + ILB)$ (D) $\tan^{-1} QE/(Mg + ILB)$

11. With S_1 and S_2 both closed, magnetic force on the wire CD is (assuming current through battery as I)

(A) $IB\ell$ (B) $\frac{1}{2}B\ell$ (C) zero (D) $\frac{2}{3}IB\ell$

Paragraph for Question no. 12 and 13

A person wants to roll a solid non-conducting spherical ball of mass m and radius r on a surface whose coefficient of static friction is μ . He placed the ball on the surface wrapped with n turns of closely packed conducting coils of negligible mass at the diameter. By some arrangement he is able to pass a current i through the coils either in the clockwise direction or in the anticlockwise direction. A constant horizontal magnetic field $\vec{B}$ is present throughout the space as shown in the figure. (Assume μ is large enough to help rolling motion)



12. If current i is passed through the coils the maximum torque in the coil is :-

(A) $-\pi n i r^2 B \hat{k}$ (B) $\pi n i r^2 B \hat{j}$ (C) $-\pi n i r^2 B \hat{j}$ (D) $\pi n i r^2 B \hat{k}$

13. The angular velocity of the ball when it has rotated through an angle θ is ($\theta < 180^\circ$) is :-

(A) $\sqrt{\frac{10}{7} \frac{\pi n i B}{m}} \sin \theta$ (B) $\sqrt{\frac{5}{14} \frac{\pi n i B}{m}} \sin \theta$ (C) $\sqrt{\frac{5}{14} \frac{\pi n i B}{m}} \cos \theta$ (D) $\sqrt{\frac{5}{7} \frac{\pi n i B}{m}} \sin \theta$

SECTION-II

Numerical Answer Type Question

1 Q. [3(0)]

(upto second decimal place)

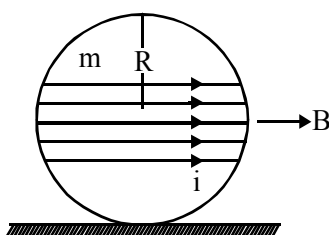
1. Two coaxial long solenoids having number of turns 3000 & 1500 per metre, radii 2.0 m & 1.0 m and carry current of 2A & 4A in opposite sense respectively. Find the magnetic energy stored per unit length (in J/m). [Take $\pi^2 = 10$]

SECTION-III

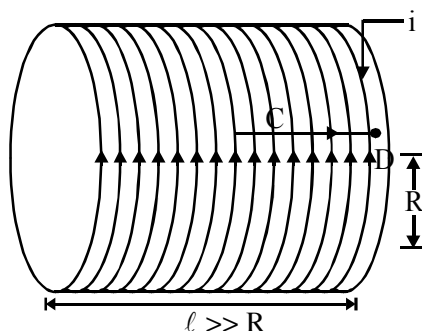
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

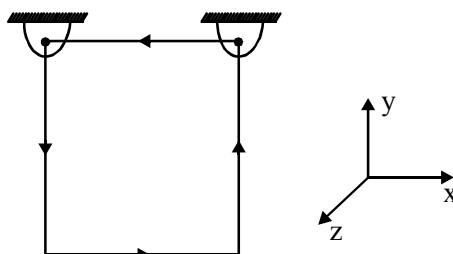
1. A wire is wrapped $N = 10$ times over a solid sphere of mass $m = 5\text{kg}$, current $I = 2\text{A}$, which is placed on a smooth horizontal surface. A horizontal magnetic field of induction $|\vec{B}| = 10\text{T}$ is present. Find the angular acceleration experienced by the sphere. Assume that the mass of the wire is negligible compared to the mass of the sphere. If answer is $20n\pi$. Write value of n .



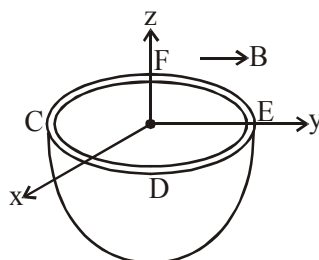
2. A small circular loop of radius $a = 2\text{cm}$ carrying a current $i_0 = 1\text{A}$ is placed co-axially with an ideal solenoid of radius $R = 5\text{cm}$, and current $i = 1\text{A}$ flowing through its windings. Number of turns per unit length is $n = 100$. If the loop is displaced from the centre C of the solenoid to the end of the solenoid slowly by an external agent, find the magnitude of work done by the external agent. If answer is $x \left(\frac{\pi \mu_0 \times 10^{-2}}{3} \right)$ (in J). Then the value of x is.



3. A current carrying uniform square frame is suspended from hinged supports as shown in the figure such that it can freely rotate about its upper side. The length and mass of each side of the frame is 2m and 4kg respectively. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{j})$ is applied. When the wire frame is rotated to 45° from vertical and released it remains in equilibrium. If the magnitude of current (in A) in the wire frame is I then find $\left(\frac{3}{5} \right) I$.



4. A rigid circular massless frame carrying current of 6A is fixed to a uniform nonconducting hemispherical shell of mass 3kg and the system lies in a gravity free space. The wire frame lies in x - y plane and a uniform magnetic field (2 Tesla) is switched on along y -axis. If the angular acceleration (in rad/s^2) of the shell at this instant is $x \times \pi$ then find the value of x .



5. A long and thin metallic tube of radius $\frac{2}{\sqrt{\pi}}\text{cm}$ carries a current of 4000A along its length. Calculate the magnetic pressure (in kPa) on the tube that tries to compress it.

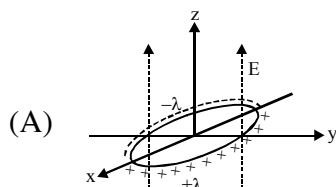
SECTION-IV

Matrix Match Type (4 × 5)

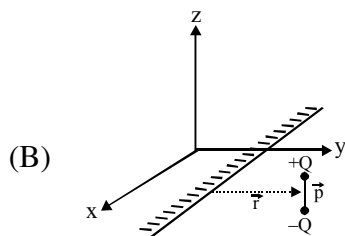
1 Q. [8 M (for each entry +2(0))]

1. Column-I shows some charge distributions and current distributions accompanied by their descriptions. Column-II shows the instantaneous characteristics. Here (S) symbolizes the system on which results are to be obtained.

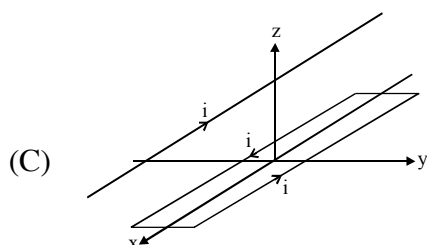
Column-I



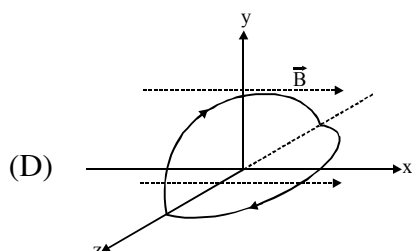
Circular ring (S) half positive and other half negative placed in a uniform electric field, with centre at origin.



Dipole (S) is placed in front of a long uniformly negatively charged wire parallel to x-axis, such that $\vec{p}$ is perpendicular to $\vec{r}$ and dipole is kept parallel to z-axis.



A square current carrying coil (S) is placed in xy-plane with centre at origin and sides parallel to x-axis and y-axis, and a long wire placed parallel above square on z-axis and parallel to x-axis.



A circular current coil (S) with one half in yz-plane other half in xz-plane, placed in a uniform magnetic field in x-direction.

Column-II

- (P) Net force on S is zero
- (Q) Net force on S have no x-component
- (R) Net torque on S is along x-axis
- (S) Net torque on S is zero
- (T) Direction of magnetic dipole moment or electric dipole moment is in x-y plane.

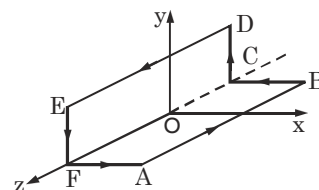
SECTION-I

Single Correct Answer Type

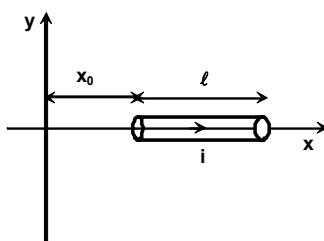
6 Q. [3 M (-1)]

1. In the figure shown ABCDEFA was a square loop of side ℓ , but is folded in two equal parts so that half of it lies in x-z plane and the other half lies in the y-z plane. The origin 'O' is centre of the frame. The loop carries current 'i'. The magnetic field at the centre 'O' is :-

- (A) $\frac{\mu_0 i}{2\sqrt{2}\pi\ell}(\hat{i} - \hat{j})$ (B) $\frac{\mu_0 i}{2\pi\ell}(-\hat{i} + \hat{j})$
(C) $\frac{\sqrt{2}\mu_0 i}{\pi\ell}(\hat{i} + \hat{j})$ (D) $\frac{\mu_0 i}{\sqrt{2}\pi\ell}(\hat{i} + \hat{j})$



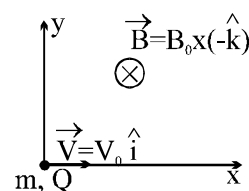
2. A wire of length ℓ , carrying a current i is kept along x-axis as shown in the figure.



- (A) Unit vector along magnetic field due to the wire at (a, b, c) is independent on x_0 .
(B) Unit vector along magnetic field due to the wire at (a, b, 0) is dependent on x_0 .
(C) Unit vector along magnetic field due to the wire at (0, b, c) is independent of b and c
(D) Unit vector along magnetic field due to the wire at (a, 0, c) is independent of a and c
3. In a non uniform magnetic field having strength $\vec{B} = B_0 x(-\hat{k})$ (here B_0 is constant). A particle of mass m and charge Q is projected horizontally with velocity $\vec{V} = V_0 \hat{i}$ from the origin at time $t = 0$. If

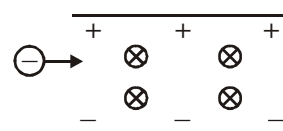
$\frac{QB_0}{m} = 1$ & at any instant 't' the velocity vector of the particle is found to be $\frac{V_0}{2} \hat{i} + a\hat{j} + b\hat{k}$ then value of a & b will be (neglect gravity)

- (A) $a = \frac{V_0}{2}, b = 0$ (B) $a = 0, b = 0$
(C) $a = \frac{\sqrt{3}V_0}{2}, b = 0$ (D) $a = 0, b = \frac{\sqrt{3}V_0}{2}$



4. An electron enters the space between the plates of a charged capacitor as shown. The charge density on the plate is σ . Electric field intensity in the space between the plates is E. A uniform magnetic field B also in that space perpendicular to the direction of E. The electron moves perpendicular to both $\vec{E}$ and $\vec{B}$ without any change in direction. The time taken by the electron to travel a distance ℓ in that space is.....

- (A) $\frac{\epsilon_0 \ell B}{\sigma}$ (B) $\frac{\epsilon_0 \ell}{\sigma B}$
(C) $\frac{\sigma \ell}{\epsilon_0 B}$ (D) $\frac{\sigma B}{\epsilon_0 \ell}$



5. A particle of charge per unit mass α is released from origin with velocity $\vec{V} = v_0 \hat{i}$ in a magnetic field

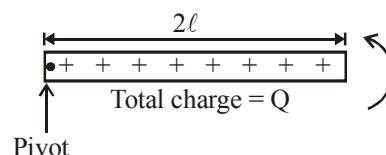
$\vec{B} = -B_0 \hat{k}$ for $x \leq \frac{\sqrt{3}}{2} \frac{v_0}{B_0 \alpha}$ and $\vec{B} = 0$ for $x > \frac{\sqrt{3}}{2} \frac{v_0}{B_0 \alpha}$. The x-coordinate of the particle at time

$t > \left(\frac{\pi}{3B_0 \alpha} \right)$ would be

- (A) $\frac{\sqrt{3}}{2} \frac{v_0}{B_0 \alpha} + \frac{\sqrt{3}}{2} v_0 \left(t - \frac{\pi}{3B_0 \alpha} \right)$ (B) $\frac{\sqrt{3}}{2} \frac{v_0}{B_0 \alpha} + v_0 \left(t - \frac{\pi}{3B_0 \alpha} \right)$
(C) $\frac{\sqrt{3}}{2} \frac{v_0}{B_0 \alpha} + \frac{v_0}{2} \left(t - \frac{\pi}{3B_0 \alpha} \right)$ (D) $\frac{\sqrt{3}}{2} \frac{v_0}{B_0 \alpha} + \frac{v_0 t}{2}$

6. A rod has a total charge Q uniformly distributed along its length 2ℓ . If the rod rotates with angular velocity ω about its one of the end, the magnetic moment of the rotating rod is :-

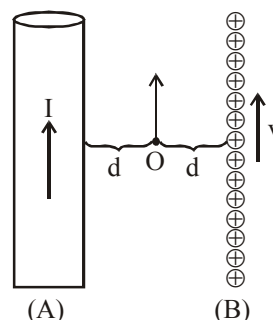
- (A) $\frac{Q\omega\ell^2}{6}$ (B) $\frac{2Q\omega\ell^2}{3}$
(C) $\frac{4Q\omega\ell^2}{3}$ (D) 0



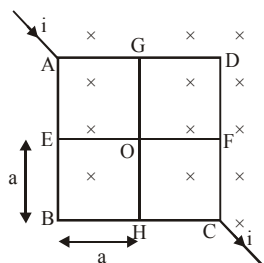
Multiple Correct Answer Type

6 Q. [4 M (-1)]

7. Figure shows a long straight conductor A, carrying a current I , where in electrons are drifting at velocity v . B represents a narrow beam of protons (in vacuum) moving with speed v (same as that of electrons) and result in an equivalent current, same as that of current in the conductor (for any stationary observer). O is an observer situated midway between the two conductors and is moving with same velocity as that of electrons or protons.



- (A) O would observe a net magnetic field
(B) O would not observe a net magnetic field
(C) According to O, magnetic field of both A and B is zero.
(D) According to O, magnetic field of both A and B is cancel each other.
8. In the figure, there is a uniform conducting structure in which each small square has side a . The structure is kept in uniform magnetic field B :-



- (A) The magnetic force on the structure is $2\sqrt{2} iBa$
(B) The potential of point B = potential of point D.
(C) Potential of point O = potential of point B
(D) The magnetic force on the structure is $\sqrt{2} iBa$

9. A nonconducting disc can have a radius up to $R = \frac{\pi}{2}$. It carries a charge of density σ given by

$$\sigma = A \cos r - B \sin r \quad 0 \leq r \leq R$$

where r is radial distance from centre. The disc is rotated about an axis perpendicular to its plane and passing through its centre with an angular velocity $\omega = 2 \text{ rad/s}$:-

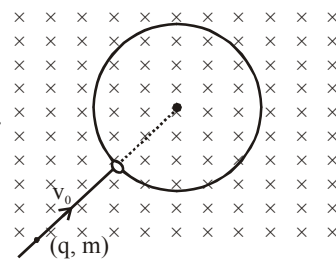
- (A) Radius of disc r for which magnetic field at centre is maximum is $r = \tan^{-1} \frac{A}{B}$
- (B) Radius of disc r for which magnetic field at centre is maximum is $r = \tan^{-1} \frac{B}{A}$
- (C) Maximum magnetic field at centre is $\mu_0 (\sqrt{A^2 + B^2} - B)$
- (D) Maximum magnetic field at centre is $\mu_0 (A^2 + B^2)$
10. A circular beam of charge of radius a consists of electrons moving with a constant speed u along the $+z$ direction. The beam's axis is coincident with the z axis and the electron charge density is given by $\rho = cr^2 \text{ (C/m}^3\text{)}$ where c is a constant and r is the radial distance from the axis of the beam :-
- (A) The current crossing the xy -plane is $\pi c u a^4$
- (B) The electric field at distance r ($r < a$) from z axis is proportional to r^3
- (C) The magnetic field at distance r ($r < a$) from z axis is proportional to r^3
- (D) The electric field at distance r ($r < a$) from z axis is inversely proportional to r .
11. A non-conducting cylindrical shell of radius R is placed in uniform magnetic field. The axis of the cylinder is parallel to the direction of magnetic field. A hole is drilled in cylinder and a particle having charge q and mass m is projected with velocity v_0 perpendicular to magnetic field and directed towards the axis of cylinder as shown in diagram. The particle collides elastically with inner wall of the cylinder and rebounds.

- (A) The minimum no. of collisions after which it will emerge is two

- (B) The maximum speed v_0 for which it can come out is $v_0 = \left(\frac{qBR}{m} \right) \sqrt{3}$.

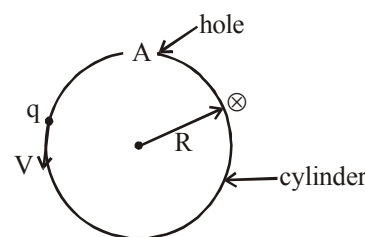
- (C) The minimum time in which it can come out will be $\left(\frac{\pi m}{qB} \right)$

- (D) If the particle is projected with a velocity $\frac{v_0}{3}$ where $v_0 = \left(\frac{qBR}{m} \right) \sqrt{3}$ it will come out after five collisions.



12. A smooth cylinder is fixed on a smooth surface with its axis vertical and radius 1 cm. If a downward magnetic field of magnetic field intensity $1 \times 10^5 \text{ T}$ exists in a region and charge $(q) = 100 \mu\text{C}$ of mass 100 gm is projected with velocity 4 m/s in anticlockwise direction then :

- (A) Normal exerted by the walls of the cylinder is 120 N
- (B) Normal exerted by walls of cylinder is zero
- (C) Particle will fly radially outward when it reaches position A on the cylinder.
- (D) Radius of curvature at position A will be 4 cm



SECTION-III

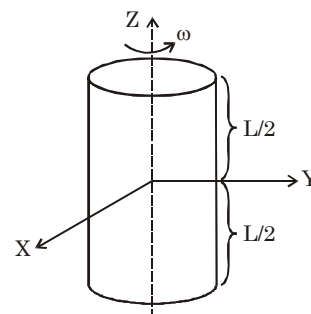
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

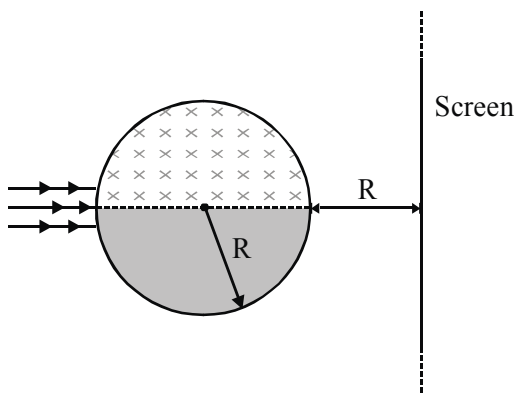
1. A thin glass rod of radius R and length L carries a uniform surface charge σ . It is set spinning about its axis, at an angular velocity ω . The magnetic field at a distance $S \gg R$ from the axis, in the x - y

plane is given by $\frac{\mu_0 \omega \sigma L R^\alpha}{\beta \left[S^2 + \left(\frac{L}{2} \right)^2 \right]^{\frac{\gamma}{2}}}$, where α , β & γ are integers. Fill

the value of $\alpha + \frac{\beta}{2} + \gamma$ in OMR sheet.



2. Figure shows circular region of radius $R = \sqrt{3}$ m in which upper half has uniform magnetic field $\vec{B} = 0.2(-\hat{k})T$ and lower half has uniform magnetic field $\vec{B} = 0.2\hat{k}T$. A very thin parallel beam of point charges each having mass $m = 2$ gm, speed $v = 0.3$ m/sec and charge $q = +1$ mC are projected along the diameter as shown in figure. A screen is placed perpendicular to initial velocity of charges as shown. If the distance between the points on screen where charges will strike after being deflected by the magnetic field is $4X$ meters, then calculate X .

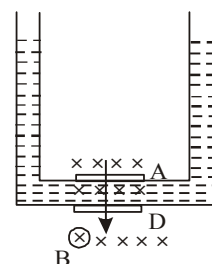


3. Consider 3 straight, infinitely long, equally spaced wires (with almost zero radius), each carrying a current I in the same direction and separated by distance d . Mass per unit length of each wire is m . If the middle wire is displaced a small distance x in the same plane, then find out minimum time interval in

which wire will return to its equilibrium position. If your answer is $N \sqrt{\frac{\pi m d}{\mu_0 I}}$. Fill the value of N .

4. Mercury is contained in U-tube of uniform square cross-section of side $a = 4$ mm. Electrodes A and D are sealed inside as shown in the fig. An electric current $I = 10$ A passes between the electrodes from A to D and magnetic field $B = 5.44$ wb/m² is applied across the horizontal arm perpendicular into the plane of the figure. Here density of mercury $= 13.6 \times 10^3$ kg/m³, $g = 10$ m/s², Neglect surface tension. The difference (in mm) in mercury level in limbs of U tube is

$\frac{400}{x}$. Find x .



SECTION-IV

Matrix Match Type (4×5)

1. Match the column.

Column-I

- (A) The pressure on walls due to quantity 'X' is tending to contract them

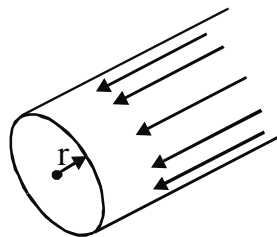
- (B) The pressure on walls due to quantity 'X' is tending to expand them

- (C) The energy/volume inside the walls is more than outside.

- (D) The energy /volume outside the walls is more than inside.

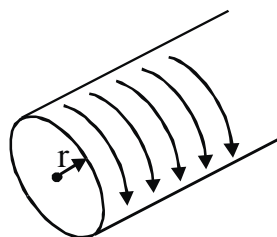
Column-II

- (P) Current (X) is flowing axially on the surface of a long thin hollow pipe of radius r .



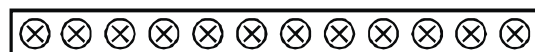
Energy : Magnetic field energy

- (Q) Current (X) is flowing tangentially on the surface of a long thin hollow pipe of radius r .



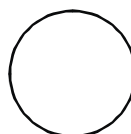
Energy : Magnetic field energy

- (R) Two current (X) carrying infinite thin sheets as shown in figure.



Energy : Magnetic field energy.

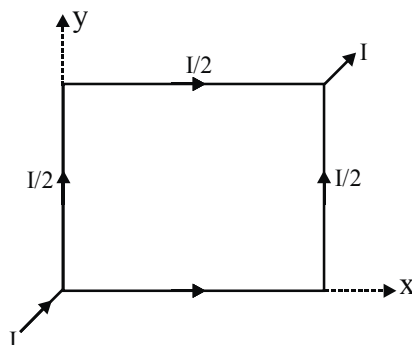
- (S) A soap bubble present in air containing air. The temperature inside and outside air is the same. Surface tension is (X)



Energy : internal energy of air

- (T) Point source of light (X) is present at the centre of a spherical shell with both surfaces perfectly black. Energy : light energy.

2. A square loop of uniform conducting wire is as shown in figure. A current- I (in amperes) enters the loop from one end and exists the loop from opposite end as shown in figure. The length of one side of square loop is ℓ metre. The wire has uniform cross section area and uniform linear mass density. In four situations of column-I, the loop is subjected to four different uniform and constant magnetic field. Under the conditions of column-I, match the column-I with corresponding results of column-II (B_0 in column I is a positive non-zero constant).



Column I

- (A) $\vec{B} = B_0 \hat{i}$ in tesla
(B) $\vec{B} = B_0 \hat{j}$ in tesla
(C) $\vec{B} = B_0 (\hat{i} + \hat{j})$ in tesla
(D) $\vec{B} = B_0 \hat{k}$ in tesla

Column II

- (P) Magnitude of net force on loop is $\sqrt{2}B_0 I \ell$ newton
(Q) Magnitude of net force on loop is zero
(R) Magnitude of net torque on loop about its centre is zero
(S) Magnitude of net force on loop is $B_0 I \ell$ newton
(T) Magnitude of net torque on a loop about its centre is non-zero.

SECTION-I

Single Correct Answer Type

1. Constant current is flowing in toroid & top-view of toroid section in which loop ABCD is present as shown in figure. Then choose the **CORRECT** statement.

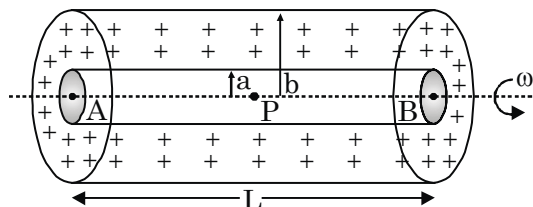
(A) $\int \vec{B} \cdot d\vec{\ell}$ is more for AB than DC

(B) $\int \vec{B} \cdot d\vec{\ell}$ is less for AB than DC

(C) $\int \vec{B} \cdot d\vec{\ell}$ is equal for both AB and DC

(D) Can't say anything without knowing about cross section shape of toroid

2. A thick uniformly charged hollow cylinder of inner radius a and outer radius b rotates with constant angular speed ω about its axis APB, charge density = ρ



Given that

$$L \gg a$$

$$L \gg b$$

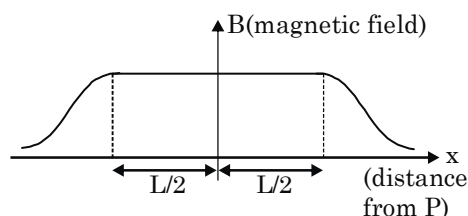
and P is mid point of AB. Choose the **incorrect** option.

(A) Magnetic field at point P is $\frac{\mu_0 \omega \rho (b^2 - a^2)}{2}$

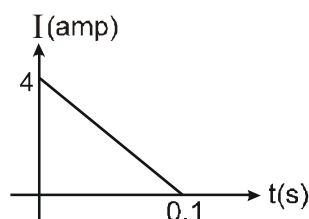
(B) Magnetic field at point B is $\frac{\mu_0 \omega \rho (b^2 - a^2)}{4}$

(C) Direction of magnetic field at A, B, P is same.

(D) Graph of magnetic field on the axis of cylinder with distance from point P is



3. When magnetic flux through a coil is changed, the variation of induced current in the coil with time is as shown in graph. If resistance of coil is 10Ω , then the total change in flux of coil will be

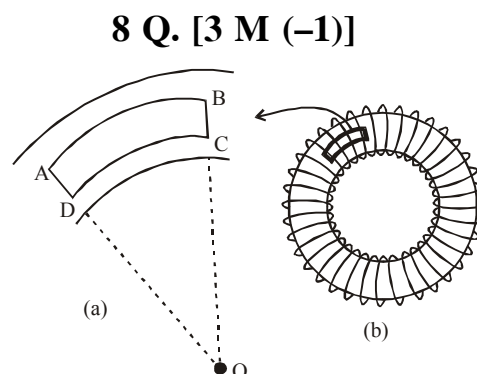


(A) 4

(B) 8

(C) 2

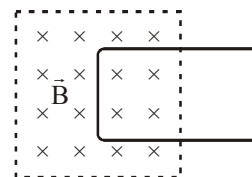
(D) 6



8 Q. [3 M (-1)]

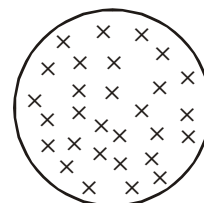
4. A conducting loop is halfway into a magnetic field. Suppose the magnetic field begins to increase rapidly in strength. What happens to the loop ?

(A) The loop is pushed upward, toward the top of the page.
(B) The loop is pushed downward, toward the bottom of the page.
(C) The loop is pulled to the left, into the magnetic field.
(D) The loop is pushed to the right, out of the magnetic field.

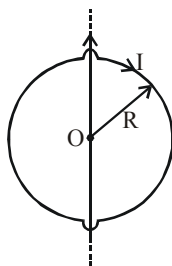


5. A uniform magnetic field $B = kt$ is passing perpendicular to plane of a ring of resistance R . k is a constant and t is time. If resistance of the ring is constant, the current flowing through the ring is :-

(A) Clockwise and constant
(B) Anticlockwise and constant
(C) Zero
(D) Anticlockwise and increasing with time

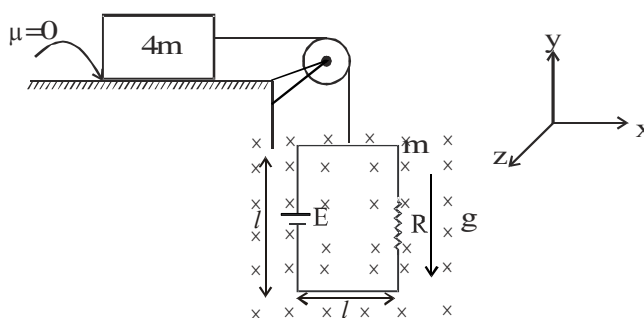


6. A circular conducting loop of radius R carries a current I . Another straight infinite conductor carrying current I passes through the diameter of this loop as shown in the figure. The magnitude of force exerted by the straight conductor on the loop is :-



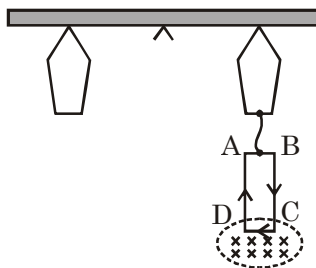
(A) $\pi\mu_0 I^2$ (B) $\mu_0 I^2$ (C) $\frac{\mu_0 I^2}{2\pi}$ (D) $\frac{\mu_0 I^2}{\pi}$

7. A block of mass $4m$ and a conducting square loop of mass m (with a cell of emf E and resistance R) are connected by a light insulating string passing over a frictionless pulley as shown in the figure. The plane of the loop (initially at rest) is perpendicular to magnetic field. As the system is released :-



(A) The block does not move.
(B) The loop starts rotating about y-axis.
(C) The loop moves down with constant velocity
(D) The acceleration of the block is constant but not zero.

8. A 100 turn rectangular coil ABCD (in XY plane) is hung from one arm of a balance (Figure). A mass 500g is added to the other arm to balance the weight of the coil. A current 4.9 A passes through the coil and a constant magnetic field of 0.2 T acting inward (in xz plane) is switched on such that only arm CD of length 1 cm lies in the field. How much additional mass 'm' must be added to regain the balance? ($g = 9.8 \text{ m/s}^2$)



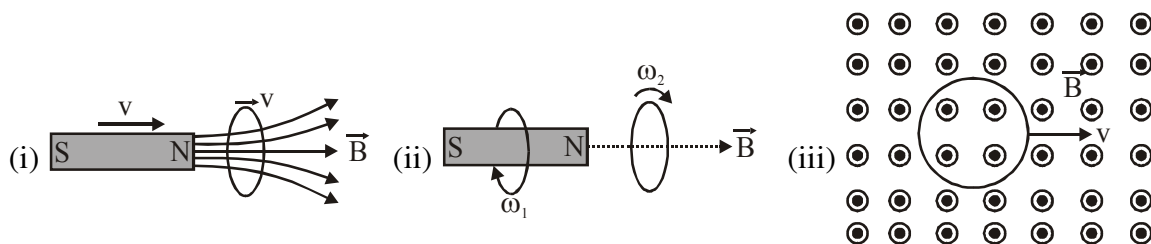
- (A) 500 gm (B) 1 gm (C) 100 gm (D) 10 gm

Multiple Correct Answer Type

5 Q. [4 M (-1)]

9. In the following figures, the rate of change of magnetic flux linked with area is measured as

$\left(\frac{d\phi}{dt}\right)_A$, $\left(\frac{d\phi}{dt}\right)_B$ and $\left(\frac{d\phi}{dt}\right)_C$ respectively at the instant shown in diagram then :-



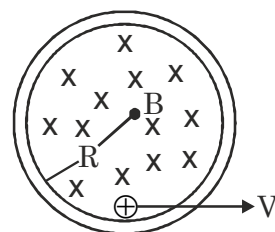
- (A) $\left(\frac{d\phi}{dt}\right)_A = \left(\frac{d\phi}{dt}\right)_B = 0$ (B) $\left(\frac{d\phi}{dt}\right)_A = \left(\frac{d\phi}{dt}\right)_C = 0, \left(\frac{d\phi}{dt}\right)_B \neq 0$
(C) $\left(\frac{d\phi}{dt}\right)_C = 0$ (D) $\left(\frac{d\phi}{dt}\right)_C = \left(\frac{d\phi}{dt}\right)_B = 0 \& \left(\frac{d\phi}{dt}\right)_A \neq 0$

10. A ring of radius 20 cm has a total resistance of 0.04Ω . A uniform magnetic field varying with time $B = (0.4t) \text{ T}$ is perpendicular to the plane of the ring.

- (A) Induced current in the ring is $\frac{2\pi}{5} \text{ A}$.
(B) The ring will be in tension.
(C) The ring will be in compression.
(D) The magnetic field due to ring at the center of the ring is in direction opposite to applied magnetic field.

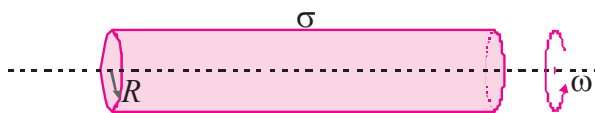
11. A charge particle of charge q is projected with a speed v in a downward magnetic field in the smooth tube. Neglecting the effect of gravity:

- (A) the particle moves in a circle of radius R.
(B) if $v \leq \frac{qBR}{m}$, $r < R$ and $N = 0$, where r = radius of the circular path
(C) if $v > \frac{qBR}{m}$; $r = R$ and $N = \frac{mv^2}{R} - Bqv$, where N = reaction force
(D) all circles must pass through the point of projection



12. A thin long cylindrical shell of radius R carries a uniform surface charge density σ . It rotates with a constant angular velocity ω in a gravity free space. Its mass per unit surface area is λ . The sheet from which the cylinder is made can withstand a maximum stress of T N/m. under longitudinal loading. It is

known that the relative permeability of cylinder is 1. [Magnetostatic energy density is given by $\frac{B^2}{2\mu_0}$]

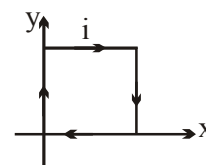


- (A) Magnetostatic energy density inside the cylinder will be $\frac{\mu_0 \sigma^2 \omega^2 R^2}{2}$
- (B) Magnetostatic energy density outside the cylinder will be $\frac{\mu_0 \sigma^2 \omega^2 R^2}{2}$
- (C) Maximum angular velocity with which the cylinder can be rotated without developing a crack on it is $\omega_0 = \sqrt{\frac{2T - \sigma^2 / \epsilon_0}{2\lambda R + \mu_0 \sigma^2 R^2}}$
- (D) Maximum angular velocity with which the cylinder can be rotated without developing a crack on it is $\omega_0 = \sqrt{\frac{2T}{2\lambda R + \mu_0 \sigma^2 R^2}}$

13. Figure shows a square current carrying coil of edge length L . The magnetic field on the coil is given by

$$\vec{B} = \frac{B_0 y}{L} \hat{i} + \frac{B_0 x}{L} \hat{j} \text{ where } B_0 \text{ is a positive constant. (A is area of coil)}$$

- (A) If coil is free to rotate about x axis torque on the coil is given by $\frac{1}{2} i A B_0 \hat{i}$.
- (B) If coil is free to rotate about y - axis torque on coil is given by $-\frac{1}{2} i A B_0 \hat{j}$.
- (C) Resultant force on coil is zero.
- (D) Equation for the torque $\vec{\mu} \times \vec{B}$ where μ is magnetic moment of coil is not valid on the coil if any of the side is fixed as axis.

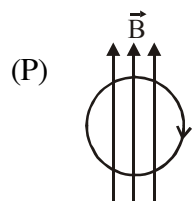


Matching List Type (4 × 4)

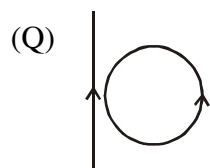
1Q.[3 M (-1)]

14. Match the options given in List-I with those given in List-II.

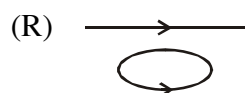
List-I



A current carrying loop placed in uniform magnetic field in plane of the loop.



Current carrying loop placed near an infinite current carrying wire (both in same plane).



A small current carrying loop placed vertically below the infinite current carrying wire.

List-II

- (1) Net force as well as net torque on the loop is zero.
- (2) Net force as well as net torque on the loop is non zero.
- (3) Net force on the loop is zero but net torque on the loop is non zero

(S)



A small current carrying loop placed in plane perpendicular to infinite current carrying wire as shown.

(4) Net force on the loop is non zero but net torque on the loop is zero.

Code :

	P	Q	R	S
(A)	3	4	1	2
(C)	3	2	4	1

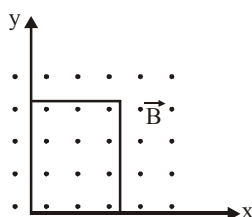
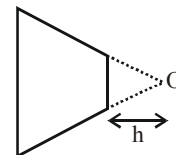
	P	Q	R	S
(B)	3	4	2	1
(D)	3	1	4	2

SECTION-III

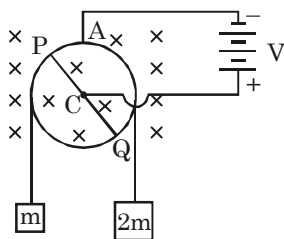
Numerical Grid Type (Ranging from 0 to 9)

5 Q. [4 M (0)]

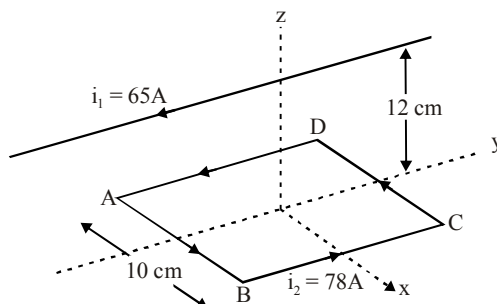
- A current of $I = \sqrt{2}$ A flows in a circuit having the shape of an isosceles trapezium. If magnetic field at point O in the plane of trapezium is 2×10^{-6} T and if length of smaller base of trapezium is $\ell = 100$ mm and distance $h = 50$ mm as shown then find ratio of length of parallel sides of trapezium.
- The square loop has sides of length 2cm. A magnetic field points out of the page and its magnitude is given by $B = (4t^2y)$ T. Emf induced in loop is given by $\frac{x}{10} \mu\text{V}$ at $t = 2.5$ s. Find the value of x.



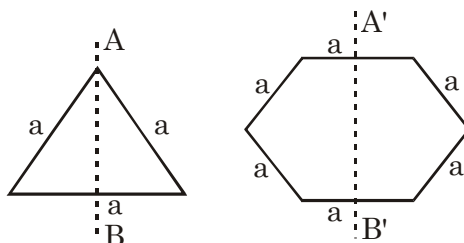
- A conducting ring of mass m and radius r has a weightless conducting rod PQ of length $2r$ and resistance $2R$ attached to it along its diameter. It is pivoted at its centre C with its plane vertical, and two blocks of mass m and $2m$ are suspended by means of a light in-extensible string passing over it as shown in the figure. The ring is free to rotate about C and the system is placed in a magnetic field B (into the plane of the ring). A circuit is now completed by connecting the ring at A and C to a battery of emf V . Find the value of V (in volt) so that the system remains static. Take $B = 0.1$ T, $m = 10$ gm, $r = 0.5$ m, $R = 1\Omega$.



- Figure shows a square loop 10 cm on each side in the x-y plane with its centre at the origin. An infinite wire is at $z = 12$ cm above y-axis. What is torque on loop due to magnetic force? If torque is expressed as $12x \times 10^{-6}$ N-m, fill value of x.



5. The magnetic dipole moment of a uniformly charged lamina in shape of equilateral triangle of side a is M , when rotated about axis AB with uniform ω (see figure). If a hexagonal lamina of side a and having same charge density is rotated about axis $A'B'$ (with same angular speed ω). The diopole moment generated is $5\alpha M$. Find α .

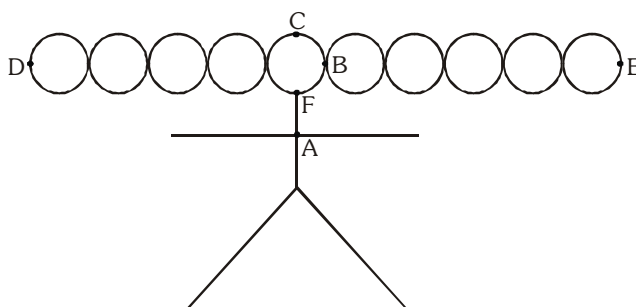


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. **Ravana-modified** : Consider a two dimensional network of uniform wire (non-zero resistivity). Point B, D, E and centers of all the 'heads' are collinear and so are A, C, F and the centre 'head'. 'Heads touch each other symmetrically. Assume dc source unless stated otherwise,



Column – I

- (A) Magnetic field at the centre of the central 'head' when battery is connected between A and B
- (B) Magnetic field at the centre of the central 'head' when battery is connected between D and E
- (C) Magnetic field at the centre of the central 'head' when battery is connected between A and C
- (D) Magnetic field at the centre of the central 'head' when battery is connected between C and E

Column – II

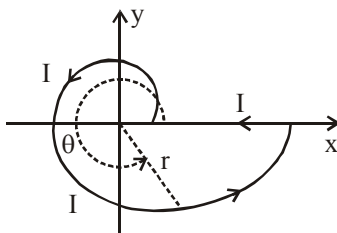
- (P) Zero
- (Q) Non-zero
- (R) Zero or non-zero behaviour of average field over a long period is same when a.c. source is used
- (S) Can't be defined if the points are short circuited using ideal wire of shortest length.
- (T) Non-zero if the points are short circuited using ideal wire of shortest length.

SECTION-I

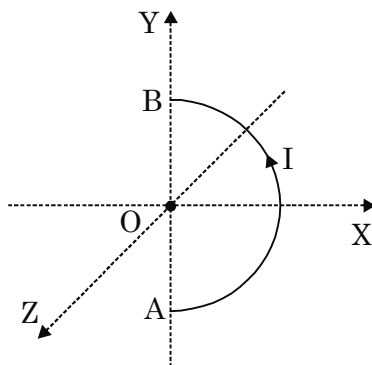
Single Correct Answer Type

7 Q. [3 M (-1)]

1. A wire carrying a current I is bent into the shape of an exponential spiral, $r = e^\theta$, from $\theta = 0$ to $\theta = 2\pi$ as shown in figure. To complete a loop, the ends of the spiral are connected by a straight wire along the x-axis. Find the magnitude of B at the origin.

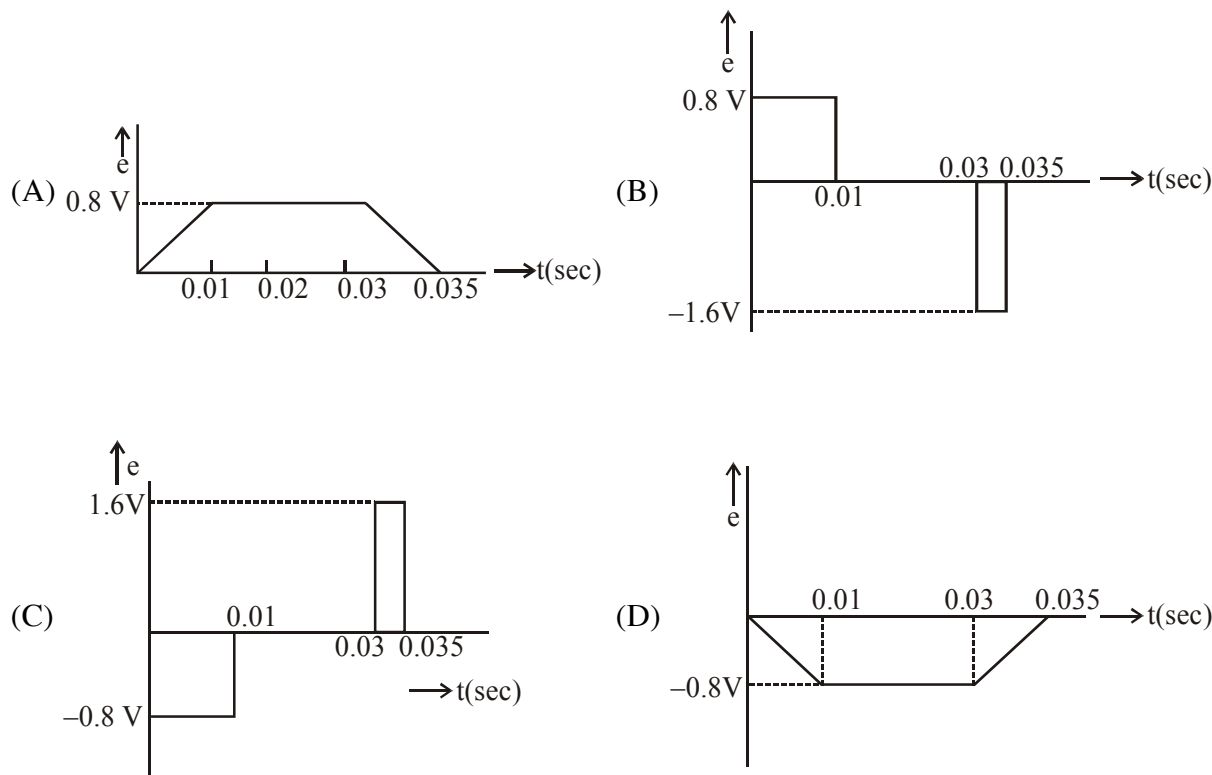


- (A) $\frac{\mu_0 I}{2\pi\sqrt{2}} [1 - e^{-2\pi}]$ (B) $\frac{\mu_0 I}{4\pi} \sqrt{2} [1 - e^{-2\pi}]$
- (C) $\frac{\mu_0 I}{2\pi} [1 - e^{-2\pi}]$ (D) $\frac{\mu_0 I}{4\pi} [1 - e^{-2\pi}]$
2. An infinitely long cylindrical wire of radius ' R ' is carrying a current with current density $j = \alpha r^3$ (where α is constant and ' r ' is the radial distance from the axis of the wire). If the magnetic field at $r = \frac{R}{2}$ is ' B_1 ' and at $r = 2R$ is ' B_2 ' then the ratio $\frac{B_2}{B_1}$ is:-
- (A) 2 (B) 4 (C) 6 (D) 8
3. A conductor carrying current 60A is in the form of a semicircle AB of radius R and lying in xy-plane with its centre ' O ' at origin as shown in the figure. The magnitude of $\oint \vec{B} \cdot \vec{d\ell}$ for the circle $x^2 + z^2 = 3R^2$ in xz-plane due to current in curve AB is $10n\mu_0$. Find the value of n (n is an integer)

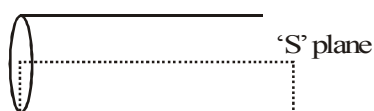


- (A) 7 (B) 3 (C) 5 (D) 8

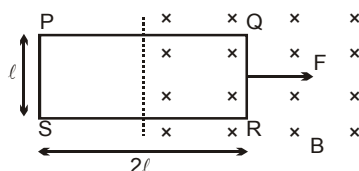
4. The flux linked with a coil of 200 turns, rises at a uniform rate from 0 to $40 \mu \text{ Wb}$ in 0.01 s, then remains constant for next 0.02 s and then falls to zero at a uniform rate in 0.005 s. Which one of the following graph represents the induced emf in coil with time ?



5. A long copper wire of circular cross-section carries a current of 10A. "S" is a plane surface as shown in figure wholly inside the wire passing through axis of cylinder. The magnetic flux per meter of wire, for the plane surface is, in wb/m .

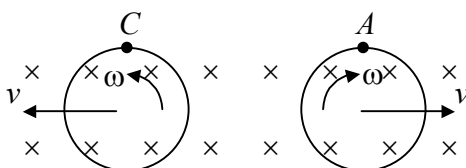


- (A) 1×10^{-5} (B) 1.5×10^{-6} (C) 1×10^{-6} (D) 1.5×10^{-5}
6. A rectangular loop PQRS, is being pulled with constant speed into a uniform transverse magnetic field by a force F (as shown). E.m.f. induced in side PS and potential difference between points P and S respectively are (Resistance of the loop = r)



- (A) Zero, $\frac{Fr}{B\ell}$ (B) zero, Zero
- (C) Zero, $\frac{Fr}{6B\ell}$ (D) $\frac{Fr}{6B\ell}$, $\frac{Fr}{6B\ell}$

7. Two metallic rings of radius R are rolling on a metallic rod. A magnetic field of magnitude B is applied in the region. The magnitude of potential difference between point A and point C on the two rings (as shown), will be :-



- (A) 0 (B) $4B\omega R^2$ (C) $8B\omega R^2$ (D) $2B\omega R^2$

Multiple Correct Answer Type

5 Q. [4 M (-1)]

8. A wire of 3 mm radius is made up of an inner material ($0 < r < 2$ mm) for which conductivity $\sigma = 10^7$ SI units, and an outer material ($2 \text{ mm} < r < 3$ mm) for which conductivity $\sigma = 4 \times 10^7$ SI units. If the wire carries a total current of 100 mA dc,

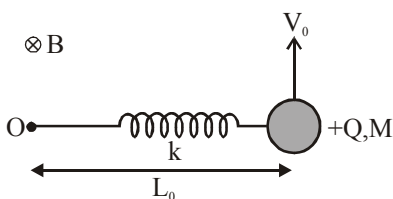
- (A) The current in both the material is the same
(B) The current in the inner material is less than that in the outer material.

- (C) The magnetic field in the inner material as function of r is $\frac{r}{1200}$ T

- (D) The magnetic field in the outer wire as a function of r is $3 \times 10^{-6} \left(\frac{10^5 r^2}{6} - \frac{1}{20} \right)$ T

9. In zero gravity region, a point charge $+Q$ of mass M is connected to a spring of natural length L_0 and spring constant k . A uniform magnetic field B is directed inside the plane of paper. The point charge is given an initial velocity v_0 perpendicular to the length of spring as shown. It is given that $B = \frac{Mv_0}{QL_0}$.

Then which of the following quantities remain(s) conserved during the motion of particle?

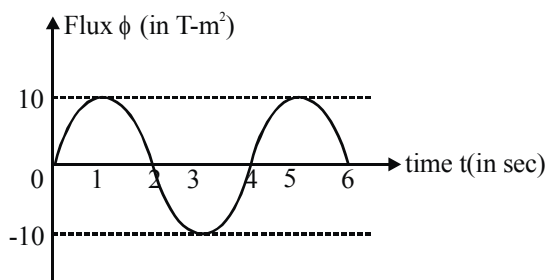


- (A) Kinetic energy (B) Potential energy
(C) Total energy of system (D) Angular momentum of particle about O
10. In a region having an electric field and a magnetic field, a positive charge is released with the below given parameters

$$\begin{cases} \vec{E} = a\hat{i} \\ \vec{B} = b\hat{i} + c\hat{j} + d\hat{k} \\ \vec{v} = x\hat{i} + y\hat{j} \end{cases}$$

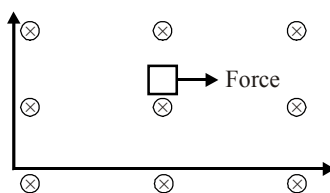
- (A) If $(yb + cx)\hat{k} + xd\hat{j} = 0$ charged particle moves along straight line
(B) For $c = 0, d = 0, y = 0$ charged particle moves along straight line with increasing speed
(C) For $c = 0, d = 0, y \neq 0$ charged particle moves along helix of varying pitch
(D) For $a = 0, \vec{v} \cdot \vec{B} = 0$ charged particle moves along circle

11. A closed conducting loop, having resistance R , is being rotated about an axis perpendicular to the magnetic field. Magnetic flux through the closed conducting loop is continuously changing according to the graph shown in the adjacent figure. Then, which of the following statement(s) is/are correct?



- (A) The electric current through the loop is minimum (zero) at $t = 1$ s, 3s and 5s.
 (B) The electric current through the loop is minimum (zero) at $t = 0$ s, 2s and 6s.
 (C) Total charge flown through any cross-section of a closed conducting loop between 0 and 6 s is zero
 (D) Total work done in rotating the loop in the magnetic field is zero
12. A square loop of finite resistance at time $t = 0$ starts moving parallel to x-axis with a constant acceleration 1 m/s^2 as shown in a space where magnetic field is changing with position and time according to relations

$$\frac{dB}{dt} = 10^{-3} \text{ T/s}, \quad \frac{dB}{dx} = -10^{-3} \text{ T/m}.$$



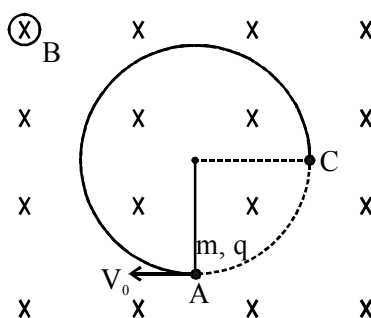
- (A) Initially current in loop is anticlockwise
 (B) Initially current in loop is clockwise
 (C) At $t = 1$ s, current in loop is zero
 (D) After $t > 1$ s current is anticlockwise

Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q.) [3 M (-1)]

Paragraph For Questions 13 to 15

A small ball of mass m and charge q is attached to the bottom end of a piece of negligible-mass thread of length l , whose top end is fixed. The system formed by the thread and ball is in vertical plane and is in uniform horizontal magnetic field B , which is perpendicular to the plane of figure and points into the paper



The ball is started with a velocity v_0 from lower most point of circle in a direction perpendicular both to the magnetic induction and to direction of thread. The ball moves along a circular path such that thread remains tight during the whole motion. Neglect any loss of energy.

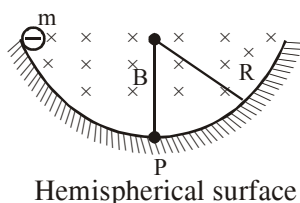
13. What is the magnitude of magnetic induction B, if the minimum initial speed at which the described motion of ball (complete vertical circular motion) occurs is $v_0 = \frac{1}{2}\sqrt{17gL}$
- (A) $\frac{3mg}{2q\sqrt{gL}}$ (B) $\frac{mg}{2\sqrt{gL}}$
- (C) $\frac{2mg}{3q\sqrt{gL}}$ (D) Situation is not possible
14. By what factor is the force acting on the thread (tension in thread) at point A is greater than at point C, when speed of the ball is the above stated one?
- (A) $\frac{21+\sqrt{17}}{18}$ (B) $\frac{21+2\sqrt{17}}{18}$ (C) $\frac{21+3\sqrt{17}}{18}$ (D) None of these
15. Choose **CORRECT** statement
- (A) If direction of magnetic field is reversed then value of B will be same as calculated in above questions of this paragraph
- (B) If B is reversed, then conservation of energy will still be applicable
- (C) If magnetic field were not present then particle will complete vertical circle for any value of v_0
- (D) All of above are correct

SECTION-III

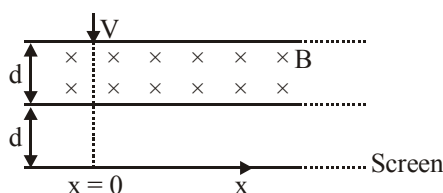
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

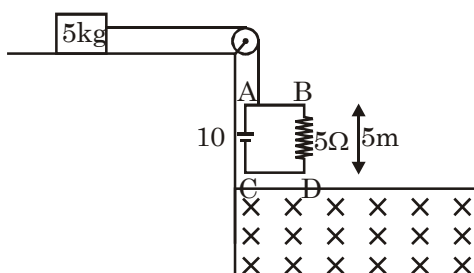
1. A charge particle of mass 'm' and charge $-q$ is released from rest from the given position. In the presence and absence of uniform horizontal magnetic field normal reaction acting on the charge at point P are N_1 and N_2 respectively. Neglect friction. The value of $N_1 - N_2$ is given by $\frac{5Bq\sqrt{2gR}}{k}$. Find k



2. A moving-coil galvanometer has 100 turns and each turn has an area 2.0 cm^2 . The magnetic field produced by the magnet is 0.01 T . The deflection in the coil is 2 radian when a current of 10 mA is passed through it. If the torsional constant of the suspension wire is $N \times 10^{-6} \text{ N-m/rad}$, then find N.
3. A non relativistic positive charge particle of charge q and mass m is projected perpendicular to uniform magnetic field B as shown. Neglecting gravity calculate X-coordinate of point on screen at which the charge particle will hit : $d = \frac{r\sqrt{3}}{2}$, where $r = \frac{mV}{qB}$. If the x-coordinate is nr then find n.



4. Square wire frame ABCD of mass 5kg is connected by a block of mass "5kg" placed on a rough surface having coefficient of friction equals to 0.6. String is massless and passes over a frictionless pulley. Frame consist of Battery of E.M.F. 10 volt and a resistance 5Ω is connected in a loop as shown. Length of side of loop is 5m and magnetic field is 0.5 T. Find the time taken by loop to enter completely into magnetic field (Neglect self induction of the coil ABCD & consider wire CD just inside the magnetic field at the time of release)

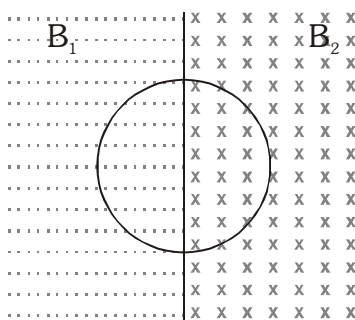


SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. A circular conducting coil of radius 1m and resistance 1Ω is placed such that half of its area is in a region of uniform magnetic field $B_1(\hat{k})$ and the remaining half is in another region of uniform magnetic field $B_2(-\hat{k})$ as shown in figure. Column I shows values of B_1 and B_2 as a function of time t , (for $t \geq 0$), while column II predicts possible directions of current and magnitude of force.



Column-I

- (A) $B_1 = 0$; $B_2 = 10$
(B) $B_1 = 10$; $B_2 = 10t + 14$
(C) $B_1 = t^2$; $B_2 = t^2 + 4$
(D) $B_1 = t^2$; $B_2 = t + 4$

Column-II

- (P) Current either always clockwise or always zero or always anticlockwise
(Q) Current sometimes clockwise, sometimes anticlockwise, and at some time zero
(R) Force may be non zero
(S) force must be non-zero

PHYSICS

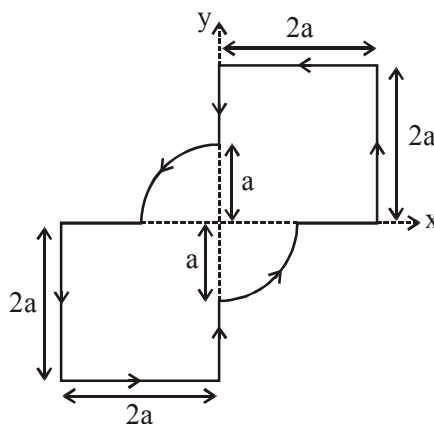
CLASS TEST # 51

SECTION-I

Single Correct Answer Type

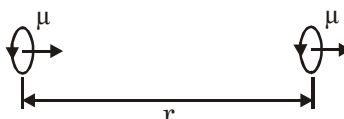
7 Q. [3 M (-1)]

1. A current 'I' flows through a thin wire as shown in the figure. If there exists an external magnetic field B in the same plane of the wire. The torque acting on the coil is :-



- (A) $I \left(\frac{\pi a^2}{2} + 8a^2 \right) B$ (B) $I \left(\frac{\pi a^2}{2} + 4a^2 \right) B$ (C) $I (\pi a^2 + 8a^2) B$ (D) 0

2. Two small magnetic dipoles with the same magnetic moments μ are located at a distance r from each other. Determine force of their interaction.

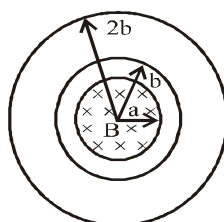


- (A) $F = \frac{3\mu_0\mu^2}{2\pi r^4}$ (B) $F = \frac{\mu_0\mu^2}{\pi r^4}$ (C) $F = \frac{1\mu_0\mu^2}{2\pi r^4}$ (D) $F = \frac{\mu_0\mu^2}{2\pi r^4}$

3. A wire of radius R, carries current uniformly distributed over its cross section. The total current carried by the wire is I. The magnetic energy per unit length inside the wire is

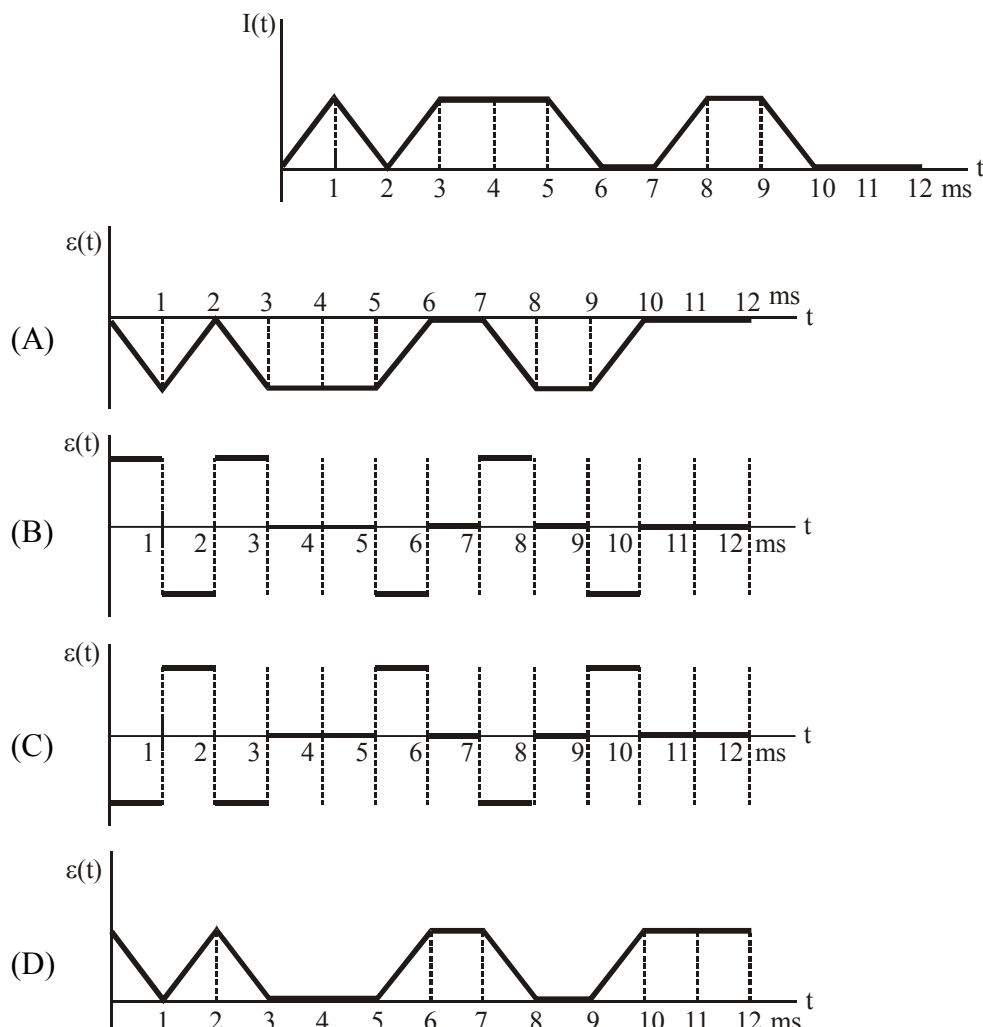
- (A) $\frac{\mu_0 I^2}{4\pi}$ (B) $\frac{\mu_0 I^2}{8\pi}$ (C) $\frac{\mu_0 I^2}{2\pi}$ (D) $\frac{\mu_0 I^2}{16\pi}$

4. A uniform magnetic field B that is perpendicular to the plane of the page now passes through the loops, as shown. The field is confined to a region of radius a, where $a < b$, and is changing at a constant rate. The induced emf in the wire loop of radius b is ε . What is the induced emf in the wire loop of radius 2b :-



- (A) 0 (B) $\varepsilon/2$ (C) ε (D) 2ε

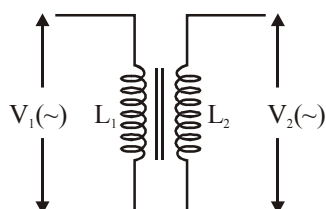
5. A spy taps the serial link between two computers by wrapping a small coil around the current-carrying wire connecting the computers. The current versus time for the transmission between the two computers is a signal as shown in figure. Then the sketch of the induced emf in the coil versus time as detected by the spy will be :-



6. Consider a toroid of circular cross-section of radius b , major radius R much greater than minor radius b . Find the total energy stored in toroid. (I is current, N is total number of turns) :-

(A) $\frac{\mu_0 N^2 I^2 b^2}{2R}$ (B) $\frac{\mu_0 N^2 I^2 b^2}{3R}$ (C) $\frac{\mu_0 N^2 I^2 b^2}{6R}$ (D) $\frac{\mu_0 N^2 I^2 b^2}{4R}$

7. For an ideal transformer, ratio of V_1 & V_2 is equal to



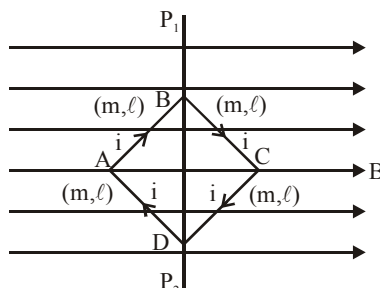
Where L_1 & L_2 are self inductances of primary and secondary windings. (For an ideal transformer coefficient of mutual induction for pair of coils is $M = \sqrt{L_1 L_2}$) :-

(A) $\frac{L_1}{L_2}$ (B) $\sqrt{\frac{L_1}{L_2}}$ (C) $\frac{L_2}{L_1}$ (D) $\sqrt{\frac{L_2}{L_1}}$

Multiple Correct Answer Type

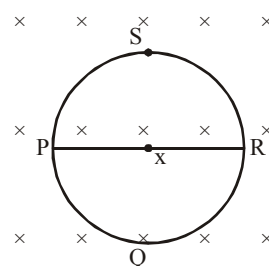
3 Q. [4 M (-1)]

8. Rod AB, BC, CD & DA form a square loop having current i , mass and length of each rod is m and ℓ respectively, is situated in a uniform magnetic field B as shown in the figure and it can rotate about axis P_1P_2 , then

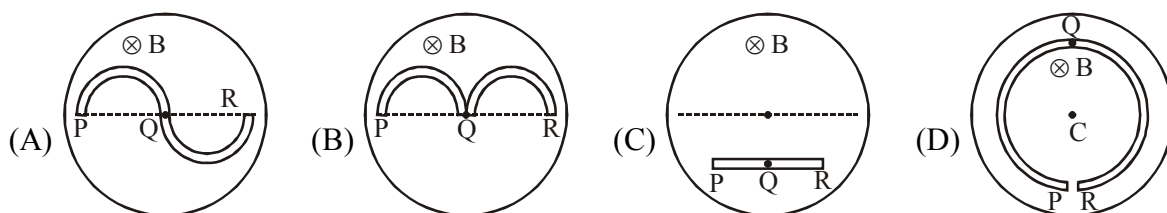


- (A) Angular acceleration of the square loop at $t = 0$, is $\frac{3 Bi}{2 m}$
- (B) Angular velocity when square loop rotated by 30° , is $\sqrt{\frac{3iB}{2m}}$
- (C) Torque on the loop when it rotated by 60° , is $\frac{iB\ell^2}{2}$
- (D) Angular acceleration of the square loop when it is rotated from starting to 90° , decreases
9. The radius of the circular loop is 'a'. Magnetic field is increasing at the constant rate α . Magnetic field is confined to a cylindrical region and axis of magnetic field coincides with the axis of the loop. Resistance per unit length of the wire of loop is ρ . Choose the **CORRECT** option(s) :-

- (A) Current in the loop PQRS is $\frac{a\alpha}{2\rho}$ anticlockwise
- (B) Current in the loop PQRS is $\frac{a\alpha}{\rho}$ clockwise
- (C) Current in the wire PR is zero
- (D) Current in the wire PR is $\frac{\pi a\alpha}{2\rho}$



10. In each of the following diagrams, the magnetic field in the circular region is inside the plane of the paper and is increasing with time. Four different shaped metallic conductors 'PQR' are placed as shown. The end P will be more positively charged compared to R in : (Q is the centre in options AB).

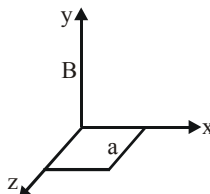


Linked Comprehension Type
(Single Correct Answer Type)

(2 Para × 2Q.) [3 M (-1)]

Paragraph for Question 11 and 12

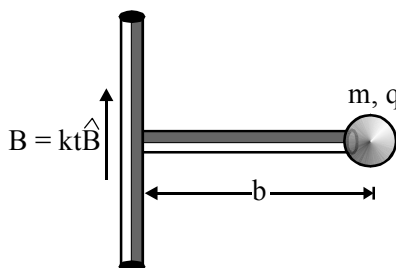
A massless square loop of side a is kept in xz plane as shown. Magnetic field in space is non uniform given by $B = \frac{B_0 y}{a} \hat{k}$. The loop is rotated about x -axis with constant angular velocity ω .



11. e.m.f. induced in the loop as function of time is equal to :-
- (A) $\frac{B_0 a^2}{2} \omega \sin 2\omega t$ (B) $\frac{B_0 a^2 \omega}{2} (1 + \sin 2\omega t)$
(C) $2B_0 a^2 \omega \cos \omega t$ (D) $B_0 a^2 \omega \cos^2 \omega t$
12. Torque required to rotate the loop with constant angular velocity (as a function of time). Take resistance of loop = R .
- (A) $\frac{B_0^2 a^4 \omega}{4R} (1 + \sin 2\omega t)^2$ (B) $\frac{B_0^2 a^4 \omega}{4R} \sin^2 2\omega t$
(C) $\frac{4B_0^2 a^4 \omega}{R} \cos^2 \omega t$ (D) $\frac{B_0^2 a^4 \omega}{R} \cos^4 \omega t$

Paragraph for Question Nos. 13 & 14

In the figure a small dielectric body of mass m carries charge q . The body lies at the end of a rod of length b , having negligible mass. The other end of the rod is connected to a vertical shaft mounted on frictionless bearings so that the rod is perpendicular to the shaft and thus free to swing in a horizontal plane. At time $t = 0$, the rod is at rest, and a uniform magnetic field is turned on whose magnitude increases linearly with time : $B = kt$, where k is a constant. The direction of the field is parallel to the shaft.



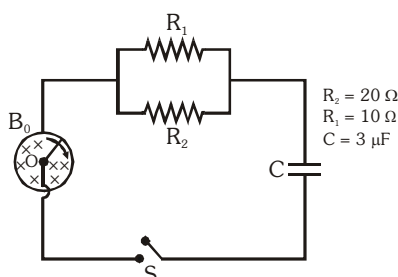
13. What is the angular acceleration of the body about the shaft.
- (A) $\alpha = \frac{qk}{2m}$ (B) $\alpha = \frac{2qk}{m}$ (C) $\alpha = \frac{qk}{m}$ (D) $\alpha = \frac{2qk}{3m}$
14. What is the speed v of the body as a function of time ?
- (A) $v = \frac{2qkbt}{2m}$ (B) $v = \frac{qkbt}{2m}$ (C) $v = \frac{2qkbt}{m}$ (D) $v = \frac{2qkbt}{3m}$

SECTION-III

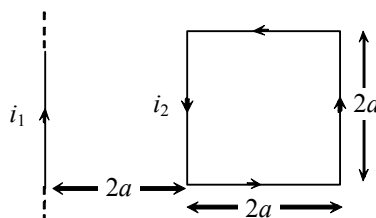
Numerical Grid Type (Ranging from 0 to 9)

6 Q. [4 M (0)]

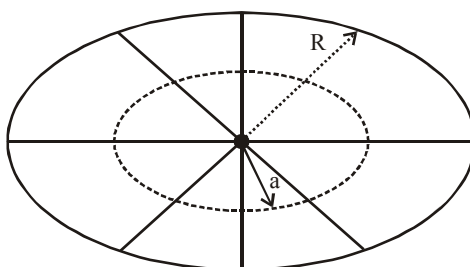
1. There is a metallic ring of radius 1m and having negligible resistance placed perpendicular to a constant magnetic field of magnitude 1T as shown in figure. One end of a resistanceless rod is hinged at the centre of ring O and other end is placed on the ring. Now rod is rotated with constant angular velocity 4 rad/s by some external agent and circuit is connected as shown in the figure, initially switch is open and capacitor is uncharged. If switch S is closed at $t = 0$, then calculate heat loss (in μJ) from the resistor R_1 from $t = 0$ to the instant when voltage across the capacitor becomes half of steady state voltage. (Assume plane of ring to be horizontal and friction to be absent at all the contacts).



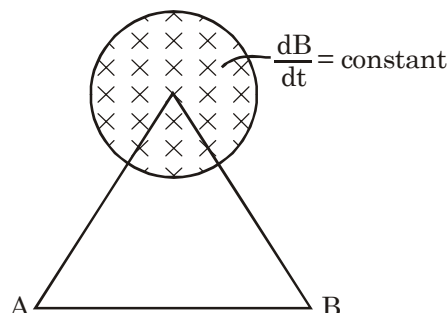
2. An infinitely long straight wire carries a current $i_1 = 1$ amp. A rectangular current loop placed in the same plane of shown dimensions carries a current $i_2 = 1$ amp anticlockwise. Magnitude of net force on the loop is found to be $\beta \frac{\mu_0}{4\pi}$. Find value of β .



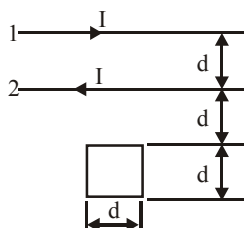
3. A line charge $\lambda = 10^{-6}$ C/m is fixed on the rim of a wheel of radius 'R' = 1m which is then suspended horizontally, so that it is free to rotate (the spokes are made of wood). In the central region up to radius $a = \frac{1}{2}$ m there is uniform magnetic field, $|\vec{B}_0| = 1\text{T}$ pointing up. Now suddenly the field is turned off. If the moment of inertia (I) is $= 0.25 \text{ kg/m}^2$, the final angular velocity ' ω ' of the wheel is $n\pi \times 10^{-6}$ rad/s. Find n.



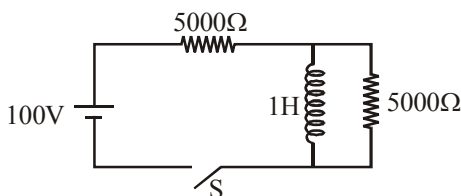
4. An uniform magnetic field is present in a cylindrical region as shown. This field is increasing uniformly with time. An equilateral loop is placed in such a way that its vertex coincide with centre of cylindrical region. Resistances of sides BC and CA are negligible whereas that of AB is 2Ω . If a current of magnitude 2A ampere flows into the triangular loop, due to induced emf, find potential difference between points A and B (in volt).



5. Two infinite parallel wires separated by a distance d carry equal currents I in opposite directions, with I increasing at the rate $\frac{dI}{dt} = 2 \text{ A/s}$. A square loop of wire of length d on a side lies in the plane of the wires at a distance $d = 2\pi \text{ (m)}$ from one of the parallel wires, as illustrated in figure. Find the emf induced (in V) in the square loop. If your answer is $n\mu_0 \ell n \frac{2}{3}$ write value of n .



6. Find the value of instantaneous power (in W) supplied by battery at the moment after the switch is closed.



PHYSICS

CLASS TEST # 52

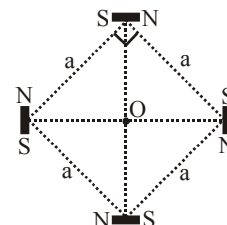
SECTION-I

Single Correct Answer Type

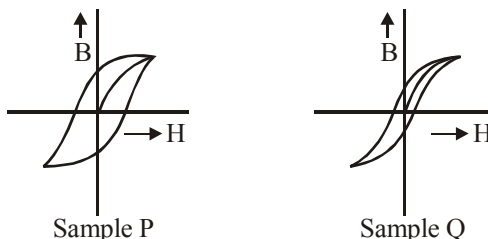
6 Q. [3 M (-1)]

1. Four small identical bar magnets each of magnetic dipole moment M are placed on vertices of a square of side a such that diagonals of the square coincide with perpendicular bisectors of respectively magnets. The net magnetic field at the centre of the square is :-

- (A) 0
(B) $\frac{\mu_0}{\sqrt{2}\pi} \frac{M}{a^3}$
(C) $\frac{2\sqrt{2}\mu_0}{\pi} \cdot \frac{M}{a^3}$
(D) $\frac{\mu_0}{\pi} \cdot \frac{M}{a^3}$

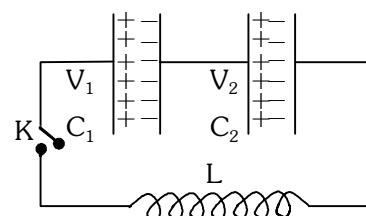


2. If the B-H curves of two samples of P and Q of iron are as shown below, then which one of the following statements is **CORRECT** ?



- (A) Both P and Q are suitable for making permanent magnet
(B) P is suitable for making permanent magnet and Q for making electromagnet
(C) P is suitable for making electromagnet and Q is suitable for permanent magnet
(D) Both P and Q are suitable for making electromagnets

3. Two capacitors of capacitance C_1 and C_2 are charged to a potential difference of V_1 and V_2 respectively and are connected to an inductor of inductance L as shown in the figure. Initially key k is open. Now key k is closed and current in the circuit starts increasing. When current in the circuit is maximum

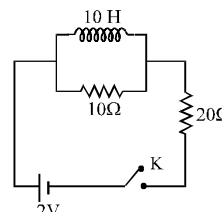


- (A) charge on both the capacitors is same
(B) induced emf in the inductor is zero
(C) potential difference across both the capacitors is half of the induced emf
(D) electrostatic potential energy stored in both the capacitors is same
4. A wire of fixed length is wound in such a way that it forms a solenoid of length ' ℓ ' and radius ' r '. Its self inductance is found to be L . Now if same wire is wound in such a way that it forms a solenoid of length

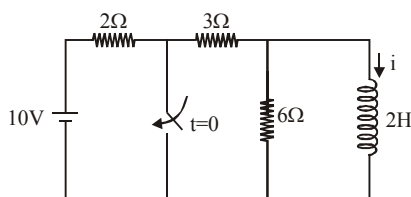
$\frac{\ell}{2}$ and radius $\frac{r}{2}$, then the self inductance will be:

- (A) $2L$ (B) L (C) $4L$ (D) $8L$
5. Two resistors of 10Ω and 20Ω and an ideal inductor of $10H$ are connected to a $2V$ battery as shown. The key K is inserted at time $t = 0$. The initial ($t = 0$) and final ($t \rightarrow \infty$) currents through battery are

- (A) $\frac{1}{15}A, \frac{1}{10}A$ (B) $\frac{1}{10}A, \frac{1}{15}A$
(C) $\frac{2}{15}A, \frac{1}{10}A$ (D) $\frac{1}{15}A, \frac{2}{25}A$



6. Find the initial current through the inductor and the current through it after a long time. Switch is closed at $t = 0$.



(A) 2A, 2A

(B) 10/11 A, 0 A

(C) 2A, 0A

(D) 0A, 2A.

Multiple Correct Answer Type

5 Q. [4 M (-1)]

7. A cylindrical space is having uniform magnetic field. The field varies with time as $B = B_0 + \alpha t$. From point A to B there are three different paths as shown in figure.

Path 1 is a semi circle of radius a , with AB as diameter.

Path 2 contain two straight lines AO and OB.

Path 3 is a single straight line AB.

If $\int \vec{E} \cdot d\vec{\ell}$ from A to B along paths 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then mark the **INCORRECT** options. :-

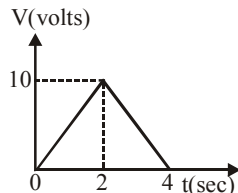
(A) $E_1 = E_2 = E_3$

(B) $E_1 = E_3 > E_2$

(C) $E_1 > E_3 > E_2$

(D) $E_3 > E_2 > E_1$

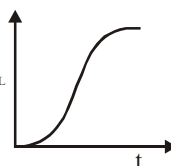
8. The potential difference across a 2H inductor as a function of time is shown in the figure. At time $t = 0$, current is zero. Choose the correct statement(s) :-



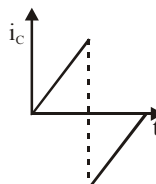
(A) Current at $t = 2$ sec is 5A

(B) Current at $t = 2$ sec is 10A

(C) Current vs time graph across inductor will be i_L



(D) Current vs time graph across inductor will be i_c



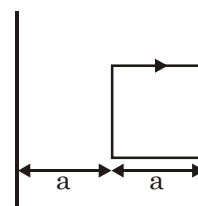
9. A very long straight conductor and a square conducting frame lie in a plane and are separated from each other as shown in the figure $a = 10$ cm. If current in the square frame is increasing a rate of 2 A/s in clockwise direction :-

(A) Mutual inductance of the two loops is $2\ln 2 \times 10^{-8}$ H

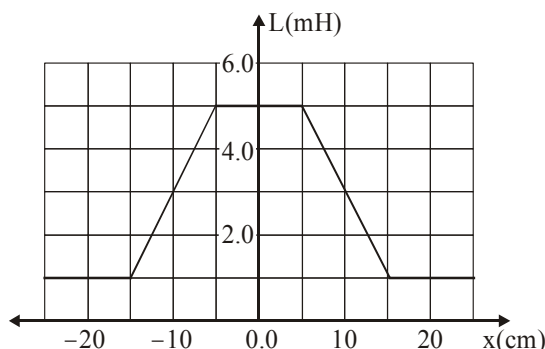
(B) The current induced in the wire is upwards.

(C) The current induced in the wire is downwards.

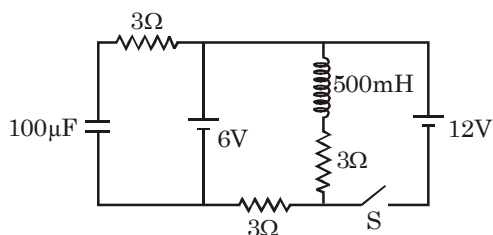
(D) The current induced in the wire is zero.



10. A coil is wound on a hollow insulating cylinder, which contains in it a laminated iron core. The dependence of inductance L with the displacement x of the iron core is shown in the figure. In the initial state $x = 0$ (the core is fully inserted into the coil), the current in the coil is 1.0 A.



- (A) The current in the coil immediately after the core is quickly taken out of the coil is 5 amp.
(B) The current in the coil immediately after the core is quickly taken out of the coil is 2.5 amp.
(C) Magnetic flux associated with coil initially is 5 milli Tesla m^2 .
(D) Magnetic flux associated with coil just after quick removal is 5 mili Tesla m^2 .
11. In the circuit shown below, switch S is kept open and circuit is in steady state. At $t = 0$, the switch S is closed and new steady state is reached after some time. Mark the **CORRECT** option(s) :



- (A) Current in the inductor when the circuit reaches the new steady state is 4A.
(B) The net change in flux in the inductor is 1.5 Wb.
(C) The potential difference across the inductor is 9 V when the circuit reaches a new steady state.
(D) The charge stored in the capacitor in the new steady state is 1.2 mC.

Linked Comprehension Type

(1 Para × 3Q.) [3 M (-1)]

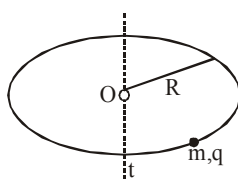
(Single Correct Answer Type)

Paragraph for Question Nos. 12 to 14

A small bead of mass m and charge q is threaded on a thin horizontal ring of radius R made of insulating material. The bead can move on the circular track without friction and is initially at rest. A magnetic field that is cylindrically symmetric (about axis t) is created, in which the component of magnetic induction that is perpendicular to the plane of the track depends only on the distance r measured from the centre and time t :

$$B(r, t) = \frac{E_0}{r} \cdot t,$$

where E_0 is a given constant. (In a negligibly small neighbourhood of $r = 0$ the induction has some finite value)



12. Mark the **CORRECT** statement :-
 (A) Bead experiences an electric force that has decreasing magnitude and radially outward
 (B) Bead experiences an electric force that has increasing magnitude and radially outward
 (C) Bead experiences electric force of constant magnitude
 (D) Bead experiences a conservative electric force
13. The velocity-time function of the bead is :-
 (A) $\frac{E_0 q}{2mR} t$ (B) $\frac{E_0 q}{2m} t$ (C) $\frac{E_0 q}{m} t$ (D) $\frac{E_0 q R}{m} t$
14. How does the radial component of the normal force between the bead and the track change as function of time ?
 (A) $\frac{E_0^2 q^2 t^2}{mR}$ (B) $\frac{E_0^2 q^2 t^2 R}{m}$
 (C) $\frac{E_0^2 q^2 t^2}{mR}$ (D) Wire does not exert any radial force on bead

Linked Comprehension Type

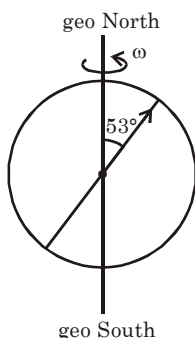
(1 Para × 2 Q.) [4 M (-1)]

(Multiple Correct Answer Type)

Paragraph for Question no. 15 and 16

The cause of earth's magnetic field has been investigated by scientists for last 4 centuries. One speculation is that this magnetic field is caused by charges circulating in the molten core of earth. In this paragraph, we will try to explore this possibility. Assume that the circulating charges are moving in a circular path about the centre of earth. Their circle is assumed to be in the same plane as magnetic equator. The axis of this circle (and imaginary magnet of earth) are assume to be parallel the earth's rotational axis. We assume that the circulation charges are rotating in the same sense/opposite sense to the earth's rotation. At the magnetic equator, the magnetic field of earth is observed to be 4×10^{-3} T. The radius of earth is assumed to be 6×10^6 m.

15. Choose **CORRECT** statement(s) :
 (A) If field was produced by moving positive charges they would circulate in direction of earth's rotation.
 (B) If field was produced by moving negative charges, they would circulate in direction of earth's rotation.
 (C) The magnetic moment of earth is 864×10^{22} J/T.
 (D) The magnetic field of earth at poles would be 8×10^{-3} T.
16. If sun produces magnetic field of 10^{-2} T at an angle of 53° to axis of rotation of earth :



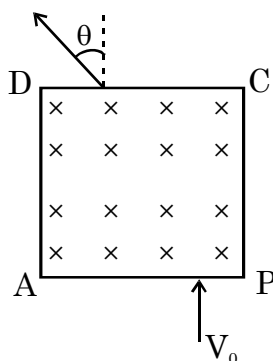
- (A) The torque on earth is approx 5.2×10^{20} Nm
 (B) The potential energy of earth-sun magnetic field is +ve.
 (C) The potential energy of earth-sun magnetic field is -ve.
 (D) The torque on earth is approx 6.9×10^{22} Nm.

SECTION-III

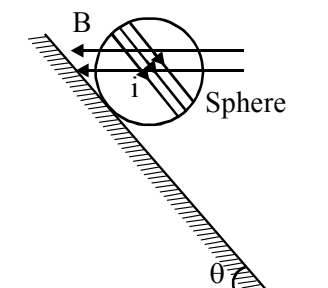
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

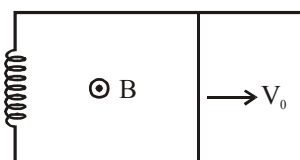
1. A uniform magnetic field B exists in a square region of side 'a' in inward direction. A particle having charge q and mass m enters the region with velocity v_0 perpendicular to side AP and exits from side CD and an angle θ with normal to side CD as shown in diagram. If $\sin\theta = \frac{qBa}{xmv_0}$ then find x .



2. A solid sphere of mass $m = 1\text{ kg}$ and radius $R = 10\text{ cm}$ is being wrapped by N turns of a massless wire, is kept on a rough inclined plane. If the wire carries a current i , the sphere remains in equilibrium in a horizontal magnetic field $\vec{B}$. If $\vec{B}$ is suddenly reversed, find the angular acceleration of the sphere. If answer is $n \times 100\text{ rad/s}^2$. Write the value of n . (Given $\theta = 37^\circ$)



3. A loop is formed by two parallel conductors connected by a solenoid with inductance $L = 10^3\text{ H}$ and a conducting rod of mass $m = 1\text{ gm}$ which can freely slide (without friction) over a pair of conductors. The conductors are located in a horizontal plane in a uniform magnetic field $B = 4\text{ T}$ in the direction shown. The distance between conductors is equal to $\ell = 1\text{ m}$. At the moment $t = 0$, the rod is imparted an initial velocity V_0 to the right. Determine angular frequency (in rad/s) of oscillation of rod.



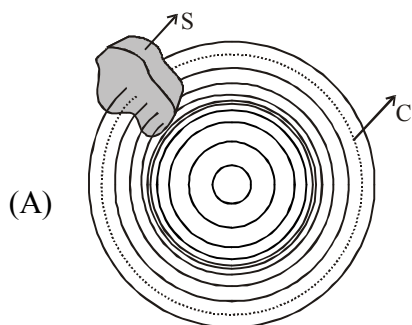
SECTION-IV

Matrix Match Type (4 × 5)

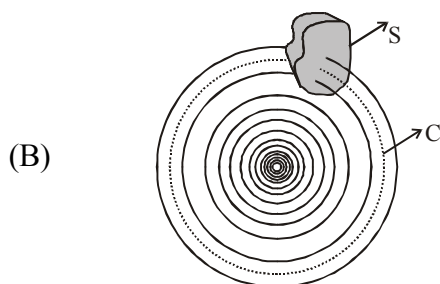
1 Q. [8 M (for each entry +2(0))]

1. Column-I shows some curves, and their properties in part of space are given in column-II.
Here C – represents a closed loop, S – represents a closed surface
Match the Column-I with Column-II.

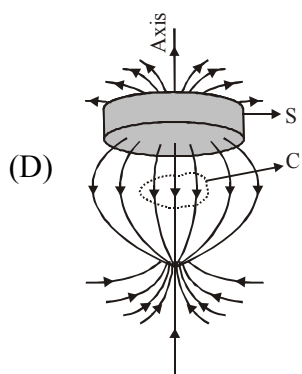
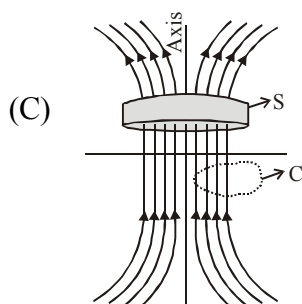
Column-I



Density of field lines first increases then decreases as we move radially outward



Density of field lines continuously decreases as we move radially outward



Column-II

- (P) Such curves can represent both electric and magnetic field lines

(Q) $\oint_C \vec{H} \cdot d\vec{\ell} \neq 0$

where $\vec{H}$ represents electric or magnetic field

(R) $\oint_C \vec{H} \cdot d\vec{\ell} = 0$

where $\vec{H}$ represents electric or magnetic field

(S) $\oint_S \vec{H} \cdot d\vec{A} = 0$

where $\vec{H}$ represents electric or magnetic field

(T) $\oint_S \vec{H} \cdot d\vec{A} \neq 0$

where $\vec{H}$ represents electric or magnetic field

PHYSICS

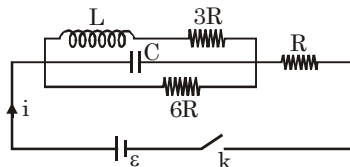
CLASS TEST # 53

SECTION-I

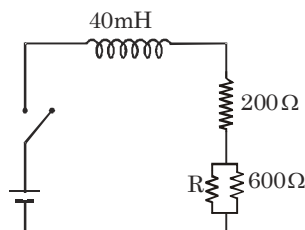
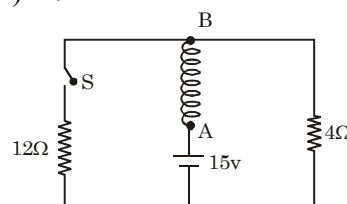
Single Correct Answer Type

8 Q. [3 M (-1)]

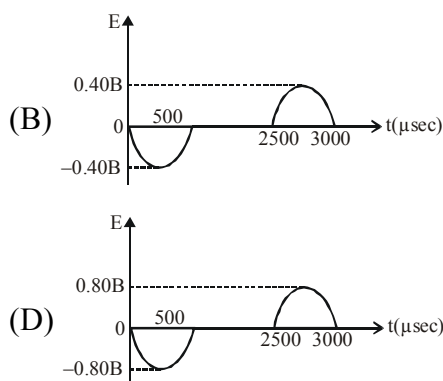
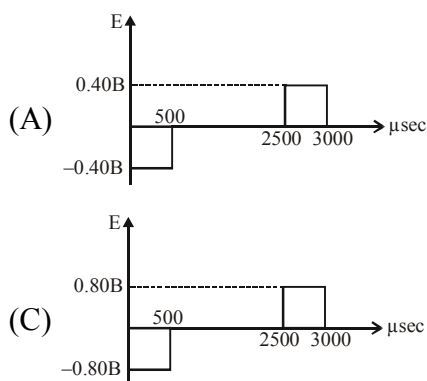
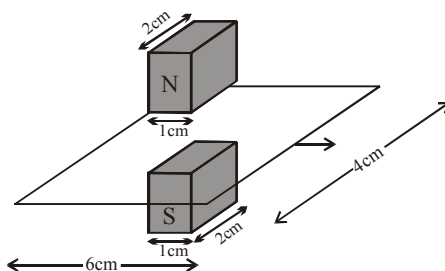
1. In the given circuit diagram, the key K is switched on at $t = 0$. The ratio of the current i through the cell at $t = 0$ and at $t = \infty$ will be :-



- (A) 3 : 1 (B) 1 : 3 (C) 1 : 2 (D) 2 : 1
2. The circuit is in steady state with switch S closed. Now if switch is opened, the voltage developed across the inductor will be :-
 (A) 5V with B at higher potential
 (B) 5V with A at higher potential
 (C) 3V with B at higher potential
 (D) 10V with B at higher potential
3. When switch S is closed, then for what value of R will the time constant will be $100 \mu\text{s}$.



- (A) 400Ω (B) 300Ω (C) 200Ω (D) 100Ω
4. A magnetic field (B), uniform between two magnets can be determined by measuring the induced voltage in the loop as it is pulled through the gap at uniform speed 20 m/sec . Size of magnet is $2 \text{ cm} \times 1 \text{ cm} \times 2 \text{ cm}$ and size of coil $4 \text{ cm} \times 6 \text{ cm}$ as shown in figure. The correct variation of induced emf with time is : (Assume at $t = 0$, the coil enters the magnetic field) :-



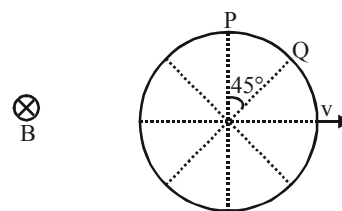
5. A conducting ring of radius R is placed in uniform inward magnetic field $\vec{B}$ as shown. If ring is moving with velocity $\vec{v}$ in its plane, the induced emf across arc PQ will be :-

(A) $\frac{vBR}{2} \left(1 + \frac{1}{\sqrt{2}} \right)$

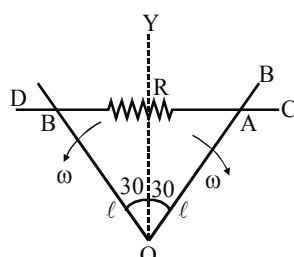
(B) $vBR \left(1 - \frac{1}{\sqrt{2}} \right)$

(C) $vBR \left(1 + \frac{1}{\sqrt{2}} \right)$

(D) $\frac{vBR}{2} \left(1 - \frac{1}{\sqrt{2}} \right)$



6. In the shown figure uniform magnetic field B_0 is pointing out of the plane in the region. Wire CD is fixed and has resistance R , while OA & OB are conducting wires rotating with angular velocity ω about O as shown. If at some instant $OA = OB = \ell$ and each wire makes angle $\theta = 30^\circ$ with Y-axis of the current through resistance R is :



(A) $\frac{B\omega\ell^2}{R}$

(B) $\frac{\sqrt{3}B\omega\ell^2}{2R}$

(C) $\frac{\sqrt{3}B\omega\ell^2}{4R}$

(D) $\frac{B\omega\ell^2}{2R}$

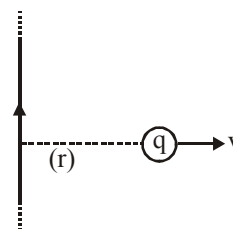
7. A charge (q , m) is thrown perpendicularly with speed v from a point at a distance r from an infinite long current (I) carrying wire. If its maximum distance from wire is R then :

(A) R will be infinite

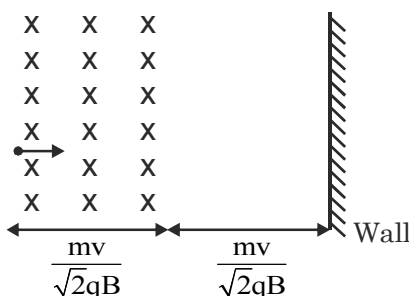
(B) R will be $\left(re^{\frac{2\pi mv}{\mu_0 q I}} \right)$

(C) R will be $\left(re^{\frac{4\pi mv}{\mu_0 q I}} \right)$

(D) R will be $\left(re^{\frac{\pi mv}{\mu_0 q I}} \right)$



8. A particle of mass m and charge q enters a region of magnetic field (as shown) with speed v . There is a region in which the magnetic field is absent, as shown. The particle after entering the region collides elastically with a rigid wall. Time after which the velocity of particle becomes anti parallel to its initial velocity is :



(A) $\frac{m}{2qB}(\pi + 4)$

(B) $\frac{m}{qB}(\pi + 2)$

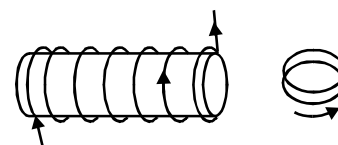
(C) $\frac{m}{4qB}(\pi + 2)$

(D) $\frac{m}{4qB}(2\pi + 3)$

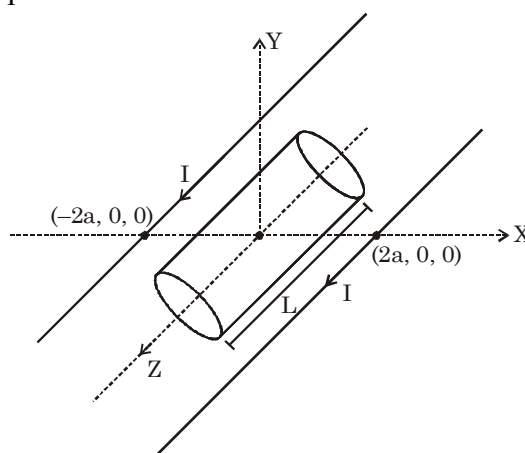
Multiple Correct Answer Type

5 Q. [4 M (-1)]

9. At a small distance from a solenoid carrying a current there is placed a circular coil with a current in such a manner that the solenoid's axis lies in the plane of the circular coil. The directions of the currents in solenoid and circular coil are shown by arrows. Mark the **CORRECT** statement(s) :



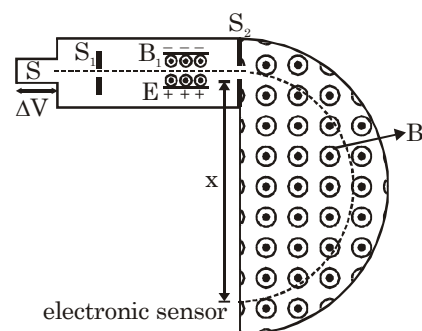
- (A) Circular coil rotates counter clockwise and begins to moves towards the solenoid.
(B) Circular coil rotates clockwise and begins to moves away from the solenoid.
(C) If the direction of current in the circular coil is opposite to that in figure then circular coil rotates counter clockwise and begins to moves away the solenoid.
(D) If the direction of current in the circular coil is opposite to that in figure then circular coil rotates clockwise and begins to moves towards the solenoid.
10. Two long wires parallel to Z-axis, having current I each in same direction are at distance $4a$ from each other. A cylinder of radius a and length L has its axis as Z-axis and centre the origin of coordinates. If the net upward magnetic flux through half of the curved cylindrical surface above X-Z plane be ϕ . Then mark the **INCORRECT** options.



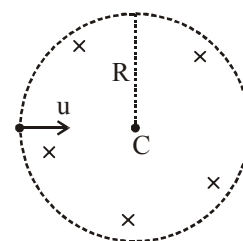
- (A) $\phi = \frac{\mu_0 I L \ln 2}{\pi}$ (B) $\phi = \frac{\mu_0 I L \ln 3}{\pi}$ (C) $\phi = \frac{-\mu_0 I L \ln 2}{\pi}$ (D) $\phi = \frac{-\mu_0 I L \ln 3}{\pi}$

11. Shown below are the essentials of a commercial mass spectrometer. This device is used to measure the composition of gas samples, by measuring the abundance of species of different masses. An ion of mass m and charge $q = +e$ is produced in source S , a chamber in which a gas discharge is taking place. The initially stationary ion leaves S , accelerated by a potential difference $\Delta V > 0$, and then enters a selector chamber, S_1 , in which there is an adjustable magnetic field $\vec{B}_1$, pointing out of the page and a deflecting electric field $\vec{E}$, pointing from positive to negative plate. Only particles of a uniform velocity $\vec{v}_1$ leave the selector. The emerging particles at S_2 , enter a second magnetic field $\vec{B}_2$, also pointing out of the page. The particle then moves in a semicircle, striking an electronic sensor at a distance x from the entry slit :-

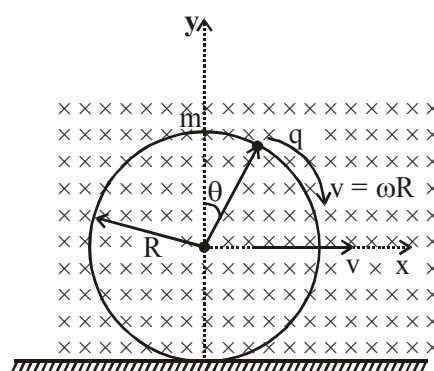
- (A) the mass of the particle is $\frac{eB_2^2 x^2}{8\Delta V}$
(B) the mass of the particle is $\frac{eB_2^2 x^2}{4\Delta V}$
(C) the magnetic field in the selector chamber that is needed to ensure that the particle travels straight through is $\frac{ExB_2}{2\Delta V}$
(D) the magnetic field in the selector chamber that is needed to ensure that the particle travels straight through is $\frac{ExB_2}{4\Delta V}$.



12. Figure shows a uniform magnetic field B (directed inwards), cylindrical in shape, with radius R . Dotted line shows the boundary of magnetic field. A point charge q , mass m , moving with velocity u , directed towards the centre of cylindrical region, enters the field :



- (A) If u increases, time spent by charge inside field increases then decreases.
(B) If u increases, time spent by charge inside field decreases.
(C) Time spent inside is directly proportional to m .
(D) Time spent inside is not directly proportional to m .
13. A ring of mass m and radius R is set into pure rolling on horizontal rough surface in a uniform magnetic field of strength B as shown in the figure. A point charge q of negligible mass is attached to rolling ring. Friction is sufficient so that it does not slip at any point of its motion. (θ is measured in clockwise from positive y -axis)



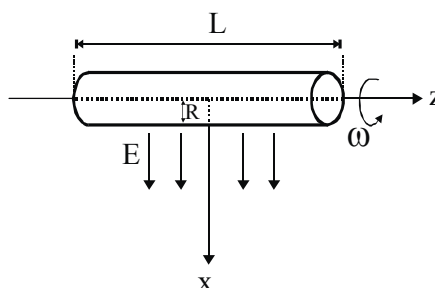
- (A) Ring will continue to move with constant velocity for the case when it does not lose contact
(B) The value of friction acting on ring is $Bqv \cos \theta$
(C) The value of friction acting on ring is $Bqv \sin \theta$
(D) Ring will lose contact with ground if v is greater than $\left(\frac{mg}{2qB} \right)$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 2Q.) [3 M (-1)]

Paragraph for Question no. 14 and 15

A very long insulating cylinder (dielectric constant ϵ) of length L and radius R ($L \gg R$) has a charge Q uniformly distributed over its outside surface. Cylinder is rotated about its symmetry axis with angular velocity ω (see figure). [Ignore edge effects]

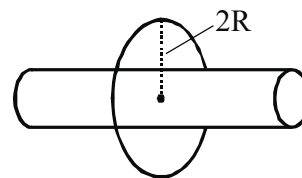


14. The magnetic field inside the cylinder is

- (A) $\frac{\mu_0 Q \omega}{2\pi R}$ (B) $\frac{\mu_0 Q \omega}{2\pi L}$ (C) $\frac{\mu_0 Q \omega}{R}$ (D) $\frac{\mu_0 Q \omega}{L}$

15. A single-turn coil of radius $2R$ and resistance ρ is wrapped around the cylinder as shown in figure, and the rotation of the cylinder is slowed down linearly ($\omega(t) = \omega_0(1-t/t_0)$) as a function of time. What is the magnitude of the induced current in the coil? In what direction does the current flow?

- (A) $\frac{\mu_0 Q R^2 \omega_0}{2\pi \rho L t_0}$ in the direction of rotation
- (B) $\frac{\mu_0 Q R^2 \omega_0}{2\pi \rho L t_0}$ in the direction opposite to the direction of rotation
- (C) $\frac{\mu_0 Q R^2 \omega_0}{2\rho L t_0}$ in the direction of rotation
- (D) $\frac{\mu_0 Q R^2 \omega_0}{2\rho L t_0}$ in the direction opposite to the direction of rotation



Linked Comprehension Type

(1 Para $\times$ 2 Q.) [4 M (-1)]

(Multiple Correct Answer Type)

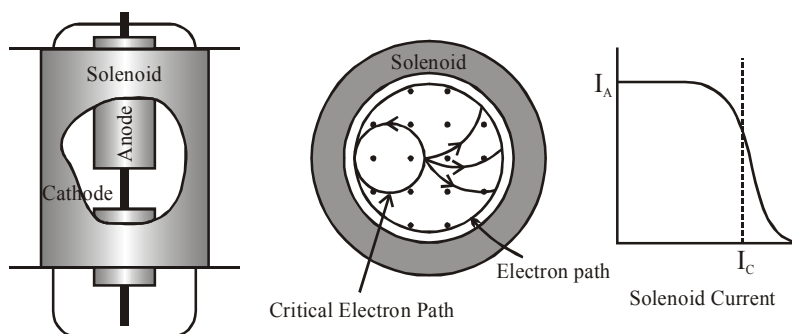
Paragraph for Question no. 16 and 17

The magnetron

This is a valve with a cylindrical anode surrounding a linear cathode (Figure-1). This type of valve provides a simple way to determine e/m . A solenoid is placed over the valve with its axis coincident with that of the cathode. In the magnetron we must assume that the cathode is very narrow, so that the electric field at its surface is high. If we do this then we can also assume that the electron experience most be their acceleration close to the cathode and so their velocity across the rest of the valve to the anode is uniform.

With no current in the coil of the solenoid the electron emitted from the cathode (due to Electric field) travel in straight line along a radius and are collected by the anode – a current (I_A) flows in the valve. Now if a current is passed through the solenoid a magnetic field will be produced and the emitted electron will begin to curve in the field - the radii of their paths becoming smaller and smaller as the current, and therefore the magnetic field is increased. Up to this point the anode current remains constant. Eventually a point will be reached for a solenoid current I_C where the diameter of their path is just equal to the radius of the anode, any further increase in the solenoid current and the electrons will not hit the anode. There will then be a sudden drop in the anode current.

Assume that the voltage difference between the cathode and the anode is V , the magnetic field due to the solenoid is B and radius of the valve is R .



16. Which of the following changes to the setup will affect the result of critical value of I_C ?
- (A) Taking a solenoid of larger radius
- (B) Taking a cathode of larger radius
- (C) Taking an anode of larger radius (but still inside the solenoid)
- (D) Taking a solenoid of larger number of turns per unit length.

17. Mark the correct statement(s) :

(A) If instead of electrons, the cathode would emit protons. The protons would travel in a larger radius and I_A would become zero at a larger value of I_C .

(B) If instead of electrons, the cathode would emit protons. The protons would travel in a smaller radius and I_A would become zero at a larger value of I_C .

(C) Value of B at which the current I_A become zero is $\sqrt{\frac{2mV}{qR^2}}$

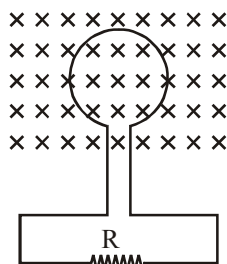
(D) Value of B at which the current I_A become zero is $\sqrt{\frac{8mV}{qR^2}}$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1 In the figure, the flux due to magnetic field through the loop perpendicular to the plane of the coil and directed into the paper is varying according to the relation, $\phi = t^2 + 7t + 1$ where ϕ is in weber and t is in seconds. The magnitude of the emf induced in the loop at $t = 1$ s is (in volt).



2. A small ball of mass m carrying positive charge +Q is dropped in uniform horizontal magnetic field B.

The vertical depth of the deepest point of its path from initial position is $\frac{(3k-1)m^2g}{Q^2B^2}$ Find k.

3. A disk, having homogenous mass and charge distribution has mass m, charge Q, radius r is rotating on its own axis with $(\omega)\hat{j}$ in zx plane with centre at origin in a free space (no hinge). P is a point on the circumference of disk. At an instant when P is at (0, 0, r), a magnetic field

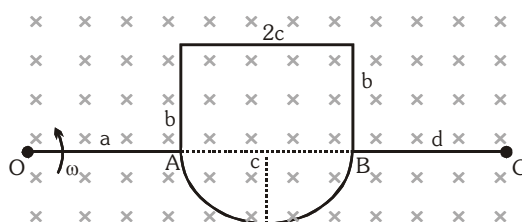
$\left[2\sqrt{n} \frac{\omega m}{Q}\right](-\hat{k})$ is switched on in the region. As a result, the acceleration of P becomes double momentarily. Write n.

SECTION-I

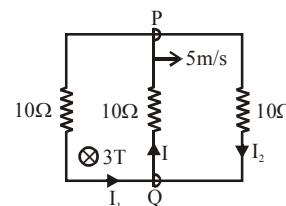
Single Correct Answer Type

8 Q. [3 M (-1)]

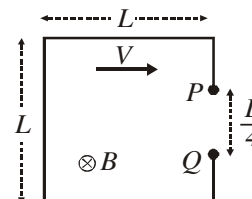
1. A frame is rotating about hinge at O in a uniform transverse magnetic field as shown in the figure. Dimensions of various sections of the rod are shown. Some points are marked on the rod C choose the *incorrect* alternative



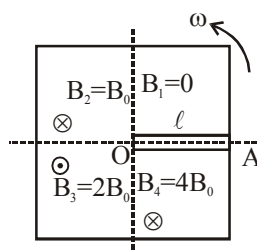
- (A) Potential difference between O and A can never be zero
(B) Potential difference between A and C may be zero
(C) Potential difference between O and C does not depend on b
(D) Potential difference between A and B does not depend on a
2. A rectangular loop has a sliding connector PQ of length 2 m and resistance $10\ \Omega$ and it is moving with a speed 5 m/s as shown. The set-up is placed in a uniform magnetic field 3T going into the plane of the paper. The three currents I_1 , I_2 and I are :-
- (A) $I_1 = I_2 = 3\text{A}$, $I = 1\text{A}$ (B) $I_1 = I_2 = 5\text{A}$, $I = 2\text{A}$
(C) $I_1 = I_2 = 1\text{A}$, $I = 2\text{A}$ (D) $I_1 = I_2 = I = 2\text{A}$
3. The loop shown moves with a constant velocity 'V' in a uniform magnetic field of magnitude 'B' directed into the paper. The potential difference between P and Q is :-



- (A) $e = \frac{3}{4}BLV$, Q is positive with respect to P
(B) $e = \frac{1}{4}BLV$, P is positive with respect to Q
(C) $e = 0$
(D) $e = \frac{1}{4}BLV$, Q is positive with respect to P



4. A thin conducting rod OA of length ℓ is rotated about end O with constant angular frequency ω . The magnetic field in the quadrants are $B_1 = 0$; $B_2 = B_0$; $B_3 = 2B_0$; $B_4 = 4B_0$. B_2 and B_4 are into the plane of the paper whereas B_3 is out of the plane. The average emf induced in one cycle is :



- (A) $\frac{13}{16}\omega B_0 \ell^2$ (B) $\frac{3}{8}\omega B_0 \ell^2$ (C) $\frac{1}{6}\omega B_0 \ell^2$ (D) $\frac{3}{4}\omega B_0 \ell^2$

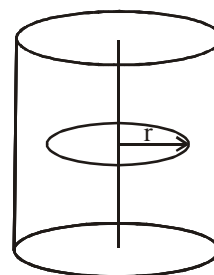
5. A point charge is moving in clockwise direction in a circle with constant speed. Consider the magnetic field produced by the charge at a point P (not centre of the circle) on the axis of the circle.
- (A) it is constant in magnitude only
(B) it is constant in direction only
(C) it is constant in direction and magnitude both
(D) it is not constant in magnitude and direction both
6. The magnetic field inside a solid conducting long wire at distance r from its axis is given as $B = B_0 r^3$ where B_0 is constant. Which of the following relations correctly represents current enclosed in the loop of radius r shown in the figure:

(A) $\frac{2\pi B_0 r^3}{5\mu_0}$

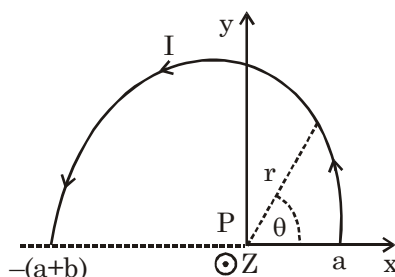
(B) $\frac{\pi B_0 r^4}{2\mu_0}$

(C) $\frac{2\pi B_0 r^2}{\mu_0}$

(D) $\frac{2\pi B_0 r^4}{\mu_0}$



7. A wire segment is bent into the shape of an Archimedes spiral (see figure).



The equation that describes the curve in the range $0 \leq \theta \leq \pi$ is

$$r(\theta) = a + \frac{b}{\pi} \theta, \text{ for } 0 \leq \theta \leq \pi$$

where θ is the angle from x-axis in radians. Point P is at the origin. I is the current. Magnetic field at point P is

(A) $\frac{\mu_0 I b}{2a}$

(B) $\frac{\mu_0 I}{2\pi} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

(C) $\frac{\mu_0 I}{4} \left(\frac{1}{a} - \frac{1}{b} \right)$

(D) $\frac{\mu_0 I}{4b} \ln \left(1 + \frac{b}{a} \right)$

8. Consider two identical circular loops of same radius, with the second (right side) coil rotated slightly clockwise relative to the first when looked from above as shown in figure (a). A large current is suddenly injected into the left side loop. What happens to the right side loop?

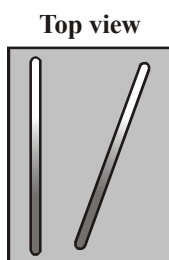


figure (a)

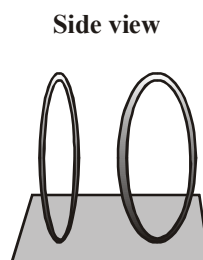


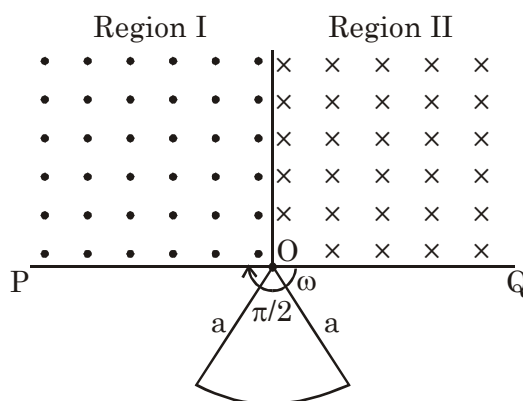
figure (b)

- (A) Force to the left, torque rotates clockwise (in top view)
(B) Force to the right, torque rotates clockwise (in top view)
(C) Force to the right, torque rotates counterclockwise (in top view)
(D) Can't tell without knowing which direction current injected into left loop

Multiple Correct Answer Type

1 Q. [4 M (-1)]

9. A wire frame is in the shape of a quadrant of a circle of radius a , having resistance R and is free to rotate about O about axis perpendicular to plane of paper. Above line PQ a uniform magnetic field exist having magnitude B and direction out of plane for region I and inside plane for region II. If frame rotates with constant ω . Mark the **CORRECT** options :-



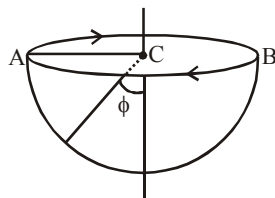
- (A) As frame goes from region I to region II, the thermal energy dissipated is $\frac{B^2 \omega \pi a^4}{2R}$
- (B) As frame goes from region I to region II, the thermal energy dissipated is $\frac{B^2 \omega \pi a^4}{4R}$
- (C) Total thermal energy dissipated in one cycle is $\frac{3B^2 \omega \pi a^4}{8R}$
- (D) Average power is $\frac{3B^2 \omega^2 a^4}{8R}$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 10 and 11

A non conducting hollow hemisphere of radius R is rotated with constant angular velocity ω about a fixed vertical axis as shown in the figure. The surface charge density on the sphere is varying as $\sigma = \sigma_0 \cos \phi$, ϕ being measured with the vertical axis.



10. The magnetic induction at the centre C is

- (A) $\left(\frac{\mu_0 \sigma_0 \omega R}{8} \right)$ (B) $\left(\frac{\mu_0 \sigma_0 \omega R}{4} \right)$
- (C) $\left(\frac{3\mu_0 \sigma_0 \omega R}{8} \right)$ (D) $\left(\frac{3\mu_0 \sigma_0 \omega R}{4} \right)$

11. If a small conducting ring of radius a is held co-axially at the centre C , and the hemisphere starts rotating with the constant angular acceleration α , then find the current flowing in the ring. Given that the resistance of the ring is r .

(A) $\frac{\mu_0 \sigma_0 \pi a^2 R \alpha}{4r}$

(B) $\frac{3\mu_0 \sigma_0 \pi a^2 R \alpha}{4r}$

(C) $\frac{3\mu_0 \sigma_0 \pi a^2 R \alpha}{8r}$

(D) $\frac{\mu_0 \sigma_0 \pi a^2 R \alpha}{8r}$

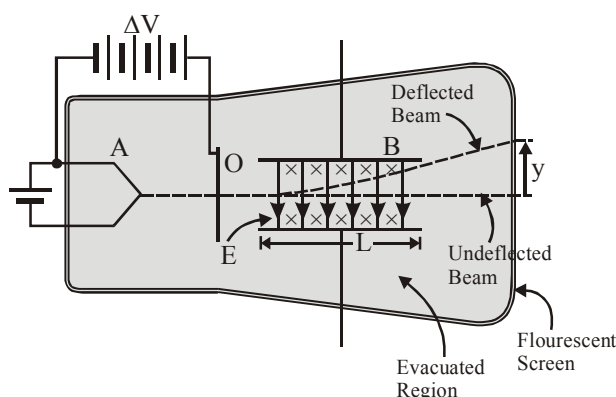
SECTION-II

Numerical Answer Type Question

1 Q. [3(0)]

(upto second decimal place)

1. In 1897, J.J. Thomson measured the charge-to-mass ratio (e/m) of an electron by using a device similar to that illustrated in figure. Electrons from the heated filament A are accelerated by a potential difference ΔV through a small opening O . The electrons pass through a region where perpendicular electric and magnetic fields can be applied; these electrons eventually collide with a fluorescent screen kept just after the deflecting plates, where they are observed. Thomson measured the vertical deflection 5 cm at the screen that a certain known electric field 1.6×10^4 N/C produced, and then he applied a magnetic field of strength 10^{-4} T that returned the beam to its undeflected position. If 1 m is the length of the deflecting plates, he calculated the charge to mass ratio as $\alpha \times 10^9$ (in S.I. units). Fill the value of α in OMR sheet.



SECTION-III

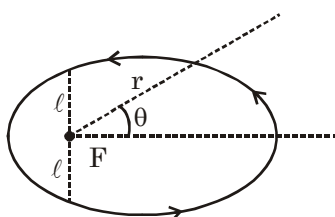
Numerical Grid Type (Ranging from 0 to 9)

6 Q. [4 M (0)]

1. Find the magnetic field due to current i flowing in an elliptical loop at it's focus. The equation of ellipse

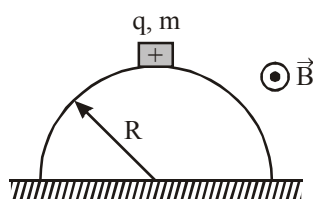
(in polar coordinates as shown) is $\frac{\ell}{r} = (1 + e \cos \theta)$. Here e is eccentricity which is a constant.

Take $\ell = 50$ cm, $e = 0.8$, $i = 2$ A, if your answer is $n\pi \times 10^{-7}$ T, fill n in OMR sheet.



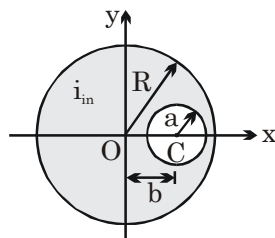
2. A small body of mass m , having a positive charge q begins to slide from the top of a smooth fixed half-cylinder of radius $R = 10$ m. At what height (in m), measured from the base of the half-cylinder, the body detaches itself from cylinder? Movement occurs in a uniform magnetic field B directed perpendicular

to the plane of the drawing and the observer. Take $B = \frac{m}{2q} \sqrt{\frac{g}{R}}$.



3. A very long straight conductor has a circular cross-section of radius R and carries a current density J . Inside the conductor there is a cylindrical hole of radius a whose axis is parallel to the axis of the conductor and a distance b from it. Let the z -axis be the axis of the conductor, and let the axis of the hole be at $x = b$. Find the x component of magnetic field on the y -axis at $y = 2R$. If your answer is

$$B_x = \mu_0 J R \left(\frac{1}{A} - \frac{a^2}{BR^2 + b^2} \right) \text{ fill value of } |A| + |B|.$$

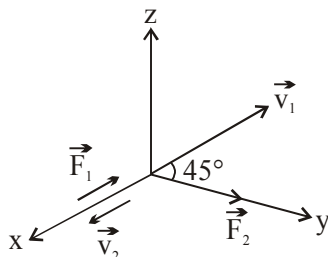


4. A non conducting rod of length $2m$ is hinged at one end and a charge of $\frac{1}{10} C$ is distributed uniformly over it. At $t = 0$, it is released from the position shown. There exist a uniform magnetic field of $\sqrt{15} T$ inside the plane of motion of rod. If mass of rod is 100 g, then find the value of hinge force (in N) when rod is rotated by $\frac{\pi}{2}$ due to gravity.



5. A neutral atom of atomic mass number 100 which is stationary at the origin in gravity free space emits an α -particle (A) in z -direction. The product ion is P . A uniform magnetic field exists in the x -direction. Disregard the electro magnetic interaction between A and P . If the angle of rotation of A after which A and P will meet for the first time is $\frac{12n\pi}{25}$ radians, what is the value of n ?

6. A particle has a charge of 4nC . When it moves with a velocity $v_1 = 3 \times 10^4 \text{ m/s}$ at 45° above y axis in yz plane, a uniform magnetic field exerts a force of F_1 along $-ve$ x axis on it. When it moves with a velocity of $v_2 = 2 \times 10^4 \text{ m/s}$ along x axis, the same magnetic field exerts a force F_2 of $16 \times 10^{-5} \text{ N}$ along y axis. What is the magnitude of the magnetic field (in T)?



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

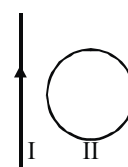
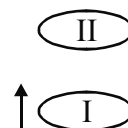
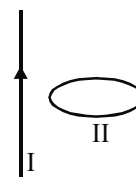
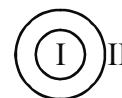
1. In all the situations current in loop-I is i_1 and current in loop-II is i_2 . Consider the infinite wire as the side of infinite large loop. Column-II describes the variation in current i_1 in different arrangements and column-I describes the various effects.

Column I

- (A) Current is clockwise in loop-II.
- (B) Current is anticlockwise in loop-II.
- (C) Flux of current i_1 through loop-II is less than flux of current i_1 through loop-I
- (D) Loop-II tends to reduce its area due to magnetic force applied by magnetic field of i_1

Column-II

- (P) Current i_1 is clockwise and decreasing at constant rate.
- (Q) Current i_1 is clockwise and decreasing at constant rate.
- (R) Current in infinite wire is i_1 and decreasing at constant rate. The wire is perpendicular to plane of loop.
- (S) Loop-I having constant current in clockwise direction moving upward with retardation. Both loops are co-axial.
- (T) Current in infinite wire is i_1 upward and decreasing at constant rate. The wire is parallel to the plane of loop.



PHYSICS

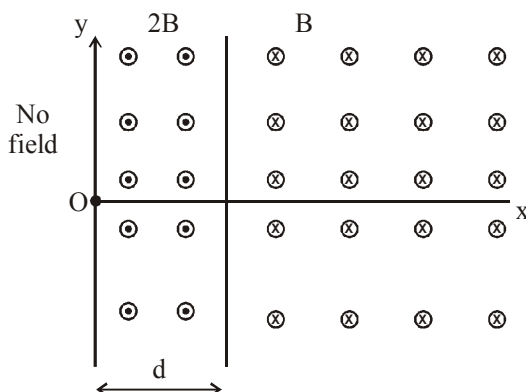
CLASS TEST # 55

SECTION-I

Single Correct Answer Type

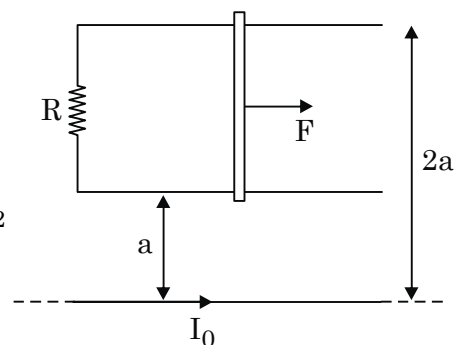
5 Q. [3 M (-1)]

1. A particle of charge q & mass m is projected in x - y plane in region containing magnetic fields as shown with velocity $(v_i + v_j)$ m/s from origin given that $d = \frac{mV}{qB}$. Find the coordinate where particle crosses the y -axis again.



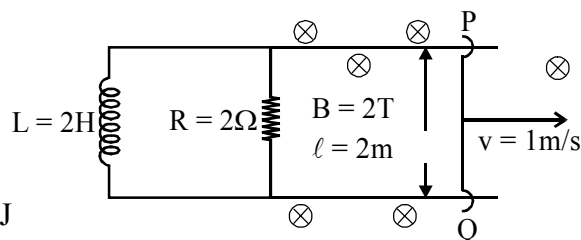
- (A) $(0, d)$ (B) $(0, \sqrt{2}d)$ (C) $(0, 2d)$ (D) $(0, d/\sqrt{2})$

2. A long straight wire carries a current I_0 , is placed near a loop with resistance R , as shown in the figure. Two wires of loop are at distance, a , $2a$ respectively from a wire. A conducting rod with zero resistance and mass m is placed across the loop. What is the force needed to move the rod with constant speed v .



- (A) $\frac{\mu_0^2 I_0^2 v}{\pi^2 R} (\ln 2)^2$ (B) $\frac{\mu_0^2 I_0^2 v}{4\pi^2 R} (\ln 2)^2$
(C) $\frac{\mu_0^2 I_0^2 v}{2\pi^2 R} (\ln 2)^2$ (D) $\frac{\mu_0^2 I_0^2 v}{2\pi^2 R}$

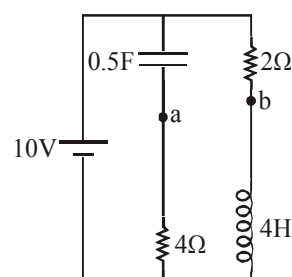
3. The given figure shows an inductor and resistor fixed on a conducting wire. A movable conducting wire PQ starts moving on the fixed rails from $t = 0$ with constant velocity 1 m/s. The work done by the external force on the wire PQ in 2 seconds is :-



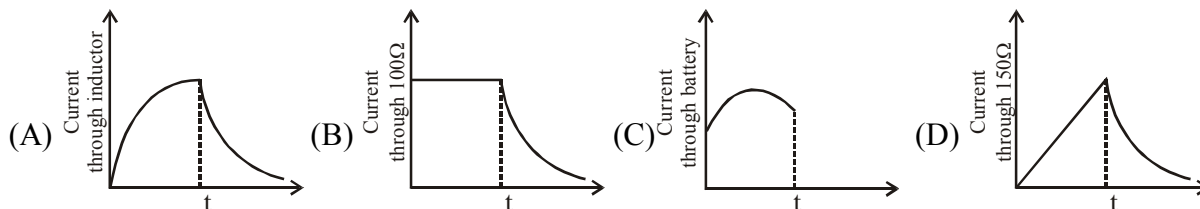
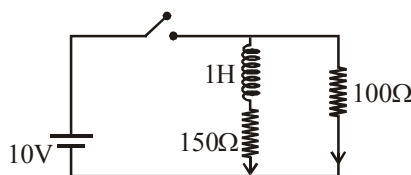
- (A) 16 J (B) 32 J
(C) 48 J (D) 64 J

4. Regarding the given circuit, the correct statement is

- (A) $(V_a - V_b)$ is increasing with time
(B) $(V_a - V_b)$ is decreasing with time
(C) $(V_a - V_b) = 10$ V
(D) $(V_a - V_b) = \text{zero}$



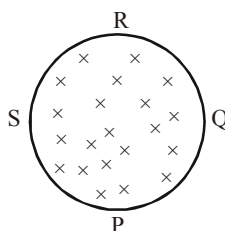
5. In the circuit, the switch is closed and then opened after a long time t . Find the correct graph(s) :-



Multiple Correct Answer Type

1 Q. [4 M (-1)]

6. A conducting ring is placed in a uniform external magnetic field present in the space within the ring and perpendicular to plane of ring. The magnetic field changes at a constant rate due to which a current of magnitude 4 amp flows in the ring. The resistance of parts PQR and PSR are 4Ω and 8Ω respectively. Then :-



- (A) The emf induced in the ring 48 V
(B) Potential difference between P & R is 24 V
(C) Potential difference between P & R is 8V
(D) Potential difference between P & R is zero as conservative electric field in the ring wire is zero

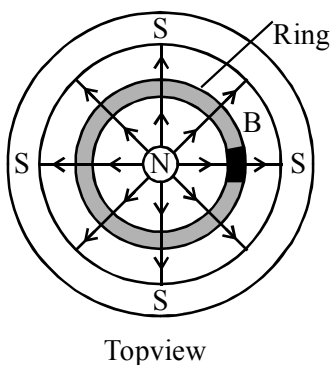
Linked Comprehension Type

(3 Para $\times$ 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question Nos. 7 & 8

Figure shows a thin circular horizontal laminated aluminium ring, with square cross-section of very small area A , containing horizontal laminae. It is released from rest and falls vertically under gravity through horizontal radial magnetic field B produced by north, N and south, S magnetic poles. It has mass m , radius r and resistivity ρ . The centre of the ring falls down the vertical line of symmetry of the system.



7. What is magnetic force on the ring :-

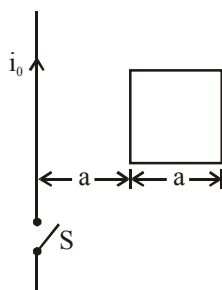
- (A) $\frac{2\pi AB^2 vr}{\rho}$, downward (B) $\frac{2\pi AB^2 vr}{\rho}$, upward
(C) $\frac{\pi AB^2 vr}{\rho}$, downward (D) $\frac{\pi AB^2 vr}{\rho}$, upward

8. Obtain an expression for terminal velocity of the ring.

- (A) $v_T = \frac{mg\rho}{\pi AB^2 r}$ (B) $v_T = \frac{2mg\rho}{\pi AB^2 r}$ (C) $v_T = \frac{2mg\rho}{3\pi AB^2 r}$ (D) $v_T = \frac{mg\rho}{2\pi AB^2 r}$

Paragraph for Question No. 09 and 10

Consider a square loop of wire with dimension a . One side of which is parallel to another long straight wire carrying current i_0 , at a distance a . The resistance of the loop is R . Neglect the self inductance of the loop. Initially switch S was closed and at $t = 0$ it is opened.



9. What is the net amount of charge that will flow after $t = 0$ through a fixed cross-section of the wire of the loop?

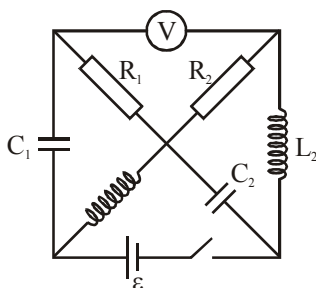
- (A) $\frac{\mu_0 i_0 \ell \ln(2)a}{\pi R}$ (B) $\frac{2\mu_0 i_0 \ell \ln(2)a}{\pi R}$ (C) $\frac{\mu_0 i_0 \ell \ln(2)a}{2\pi R}$ (D) $\frac{4\mu_0 i_0 \ell \ln(2)a}{\pi R}$

10. What is the net momentum P given to the loop during the switching-off of the current :-

- (A) $\frac{\mu_0^2 i_0^2 \times \ell \ln(2)a}{2\pi^2 R}$ (B) $\frac{\mu_0^2 i_0^2 \times \ell \ln(2)a}{4\pi^2 R}$ (C) $\frac{\mu_0^2 i_0^2 \times \ell \ln(2)a}{\pi^2 R}$ (D) $\frac{\mu_0^2 i_0^2 \times \ell \ln(2)a}{16\pi^2 R}$

Paragraph for Question no 11 and 12

For the circuit shown in figure, $R_1 = 3R$, $R_2 = R$, $C_1 = C_2 = C$, and $L_1 = L_2 = L$. The electromotive force of the battery is ε . Initially the switch is closed and the system is operating in a stationary regime.



11. Reading of the voltmeter immediately after the switch is open is equal to :-

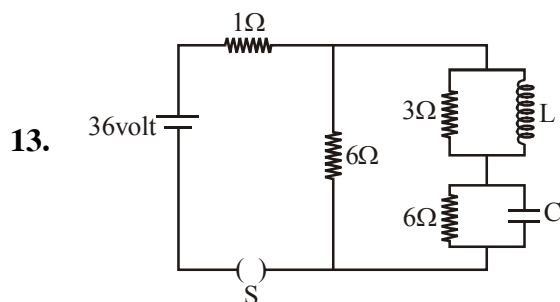
- (A) ε (B) 2ε (C) $\frac{2\varepsilon}{3}$ (D) None of these

12. Total amount of heat dissipated after opening the switch, until a new equilibrium state is achieved, is equal to :-

(A) $\frac{1}{2}C\varepsilon^2 + \frac{1}{2}\frac{L\varepsilon^2}{R^2}$ (B) $\frac{1}{2}C\varepsilon^2 + \frac{L\varepsilon^2}{R^2}$ (C) $C\varepsilon^2 + \frac{L\varepsilon^2}{R^2}$ (D) $2C\varepsilon^2 + \frac{L\varepsilon^2}{R^2}$

Matching List Type (4 × 4)

1 Q. [3 M (-1)]



List-I

- (P) Current through battery just after closing of switch (in ampere)
(Q) Current through capacitor just after closing of switch (in ampere)
(R) Switch is closed and when steady state of current is achieved the current through battery (in ampere)
(S) After steady state switch is re-opened just after reopening of switch the ratio of current through inductor and capacitor

List-II

- (1) 12
(2) 8
(3) 9
(4) 1

Codes :

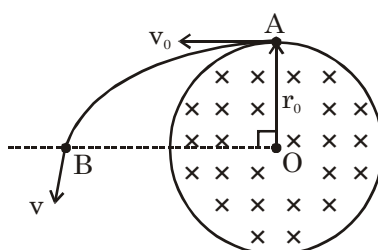
	P	Q	R	S
(A)	1	2	3	4
(B)	3	4	2	1
(C)	4	3	1	2
(D)	2	1	4	3

SECTION-III

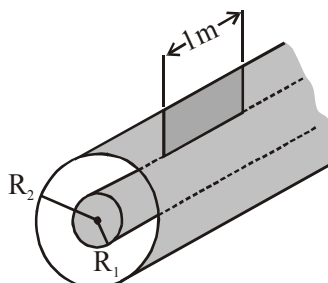
Numerical Grid Type (Ranging from 0 to 9)

6 Q. [4 M (0)]

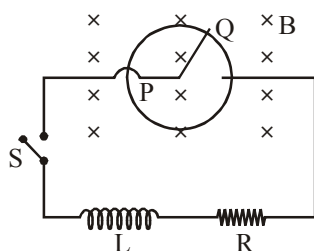
1. Consider a cylindrical region of uniform but time varying magnetic field. Now consider a plane perpendicular to this field. Top view of this cylindrical magnetic field on the plane is shown in the diagram which is a circular region of radius $r_0 = 1\text{ m}$. A particle of mass $\pi\text{ kg}$ & charge $2C$ is projected with velocity of $v_0 = \sqrt{5}\text{ m/s}$, tangential to this circle from point A. This particle intersects another radial line OB at point B with velocity v . If the magnetic field is increasing at the rate of 4 T/s then find the value of v .



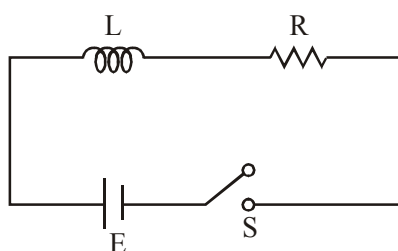
2. A transmission line consists of two concentric tubes of thin copper sheet metal with radii $R_1 = R$ and $R_2 = 2R$ (see figure). The current flows to the left along the central tubes and back along the other, completing a circuit; the current is uniformly distributed over the surface of each tube. The self-inductance per meter length of this transmission line is $\left(\frac{1}{10}\right) \ln(X)$ micro henry (μH). Find the value of X .



3. Two small current carrying loops each of radius r are kept in same plane such that separation between their centres is a ($a \gg r$), then their mutual inductance will be $\frac{\mu_0 \pi r^4}{ka^3}$. Find out the value of k .
4. A ring made of a superconductor is placed into a homogeneous magnetic field whose intensity grows from zero to $B_0 = 2\text{T}$. The plane of the ring is perpendicular to the force lines of the field. The induced current appearing in the ring is $1/x$ (amp). The radius of the ring $r = \frac{1}{\sqrt{\pi}} \text{cm}$ and its inductance $L = 1\text{mH}$. Find x .
5. The diagram shows a circuit having a coil of resistance R and inductance L connected to a conducting rod PQ which can slide on a perfectly conducting circular ring of radius 10 cm with its centre at ' P '. Assume that friction & gravity are absent and a constant uniform magnetic field of 5 T exists as shown in figure. At $t = 0$, the circuit is switched on and simultaneously a time varying external torque is applied on the rod so that it rotates about P with a constant angular velocity 40 rad/s . Find magnitude of this torque when current reaches half of its maximum value. Neglect the self inductance of the loop formed by the circuit. Resistance $R = 1\Omega$. If your answer is $x \times 10^{-4} \text{ Nm}$ fill value of $x/25$.



6. Consider the circuit given in the diagram. Switch S is closed at time $t = 0$. The current in the circuit increases with time. At some instant rate of energy storage in the inductor will become maximum. At that time find ratio of power supplied by battery and thermal power generated in resistor.



SECTION-I

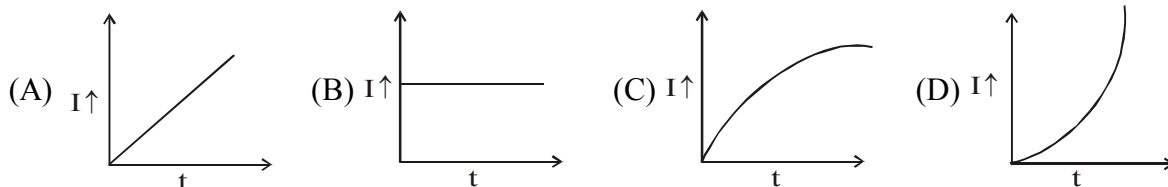
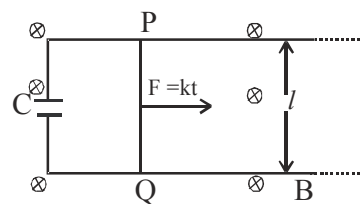
Single Correct Answer Type

4 Q. [3 M (-1)]

1. A current I is flowing in a circular ring 1 of radius R . A second ring 2 whose radius r is much smaller than that of the first, is moving with constant velocity V along the axis in such a manner that the plane of the ring remains parallel to the plane of the ring 1 during the course of the motion. Find the maximum emf induced in the ring 2.

(A) $\frac{24}{25\sqrt{5}} \frac{\mu_0 \pi I r^2 V}{R^2}$ (B) $\frac{24}{25} \frac{\mu_0 \pi I r^2 V}{R^2}$ (C) $\frac{2}{3\sqrt{3}} \frac{\mu_0 \pi I r^2 V}{R^2}$ (D) $\frac{48}{25\sqrt{5}} \frac{\mu_0 \pi I r^2 V}{R^2}$

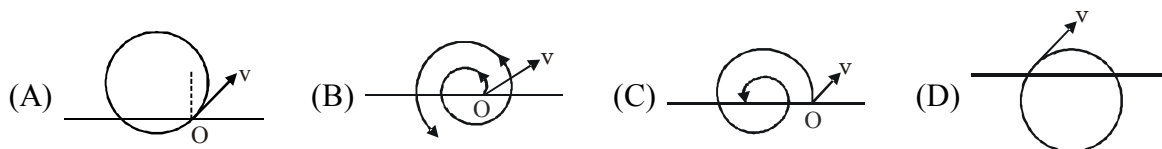
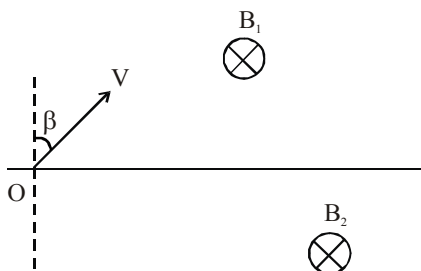
2. A wire of mass m can slide without friction on two parallel rails as shown with a capacitor. A time dependent force $F = kt$ is applied parallel to the rod PQ. A magnetic field $B \otimes$ exist perpendicular to the plane of rails. Which of the following represents variation of current with time?



3. A particle of positive charge q and mass m enters with velocity $\hat{v}_j$ at the origin in a magnetic field B ($-\hat{k}$) which is present in the whole space. The charge makes a perfectly inelastic collision with identical particle at rest but free to move at its maximum Y-coordinate. After collision the combined charge will move on trajectory (where $r = \frac{mv}{qB}$) :-

(A) $y = \frac{mv}{qB} (-\hat{i})$ (B) $(X + r)^2 + (Y - r/2)^2 = r^2/4$
(C) $(X - r)^2 + (Y - r)^2 = r^2$ (D) $(X - r)^2 + (Y + r/2)^2 = r^2/4$

4. In the diagram shown, a particle of charge $+Q$ and mass M is projected making an angle β with the vertical line. Draw the possible path on which the charge will move. Above the dark line magnetic field is B_1 and below it is B_2 . (Consider all possible cases for values of B_1 and B_2)

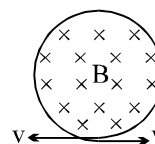


Multiple Correct Answer Type

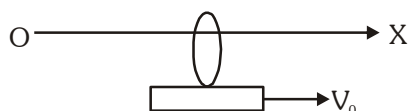
7 Q. [4 M (-1)]

5. A circular conducting loop of radius r_0 and having resistance per unit length λ as shown in the figure is placed in a magnetic field B which is constant in space and time. The ends of the loop are crossed and pulled in opposite directions with a velocity v such that the loop always remains circular and the radius of the loop goes on decreasing, then

- (A) Radius of the loop changes with r as $r = r_0 - vt/\pi$
 (B) EMF induced in the loop as a function of time is $e = 2Bv[r_0 - vt/\pi]$
 (C) Current induced in the loop is $I = \frac{Bv}{2\pi\lambda}$
 (D) Current induced in the loop is $I = \frac{Bv}{\pi\lambda}$



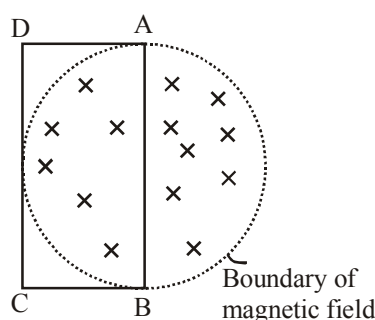
6. Magnetic field along x-axis varies according to the relation $\vec{B} = B_0 x \hat{i}$. Given a coil of area A with its axis along x-axis is connected over the top of a plastic trolley which moves along x-axis with velocity v . If the resistance of coil is R , then (at $t = 0$, coil is at $x = 0$ and $v = v_0$)



- (A) The flux linked with the coil at any position x is $B_0 x A$
 (B) An observer at origin 'O' finds the induced current as $\frac{B_0 A v_0}{R}$ anticlockwise if trolley moves with constant velocity $v = v_0$.
 (C) If the trolley has acceleration α then the induced current as a function of time t is given as $\frac{B_0 A (v_0 + \alpha t)}{R}$ anticlockwise
 (D) If the trolley has acceleration α then the induced current as function of position x is given as

$$\frac{B_0 A (\sqrt{v_0^2 + 2\alpha x})}{R} \text{ anticlockwise}$$

7. Diagram shows a cylindrical magnetic field increasing at B_0 per second. ABCD is a rectangular loop. AB being diameter ($2R$) of field. EMF (magnitude) across :-



- (A) AB is zero (B) DC is zero (C) DC is $\frac{\pi R^2 B_0}{4}$ (D) AD is $\frac{\pi R^2 B_0}{8}$

8. Consider a metal rod of length L that is given a uniform acceleration as shown in figure (a) and an identical rod rotating with constant angular velocity in figure (b).

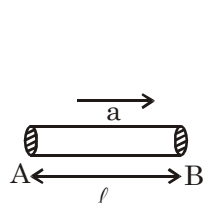


Figure (a)

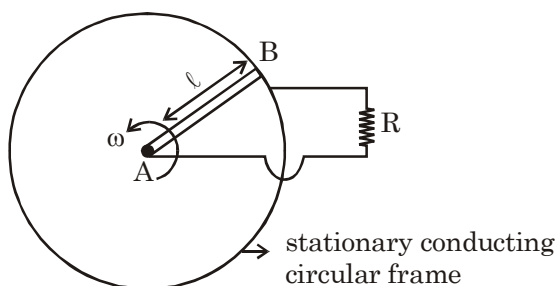
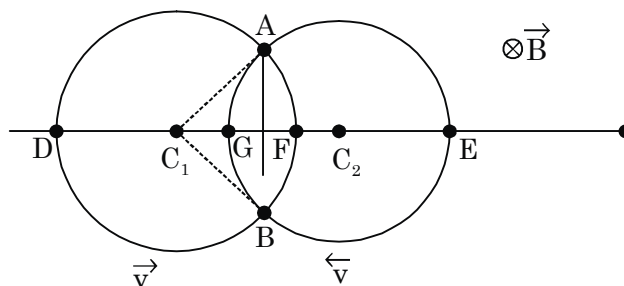


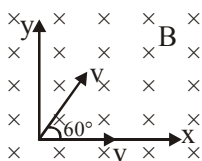
Figure (b)

- (A) If V_A and V_B are potential of end A and B respectively, then $V_A < V_B$ in figure (a)
 (B) If V_A and V_B are potential of end A and B respectively, then $V_A > V_B$ in figure (b)
 (C) Electric field inside rod has magnitude $\frac{ma}{e}$ in figure (a)
 (D) Electric field inside rod has magnitude $\frac{m\omega^2 r}{e}$ in figure (b)
9. Two circular rings of radius a , having resistance per unit length λ are moving with uniform speed v along x-axis one over other as shown in the figure in a uniform magnetic field $\vec{B} = B_0(-\hat{k})$. At $t = 0$, separation between centre is just greater than $2a$.

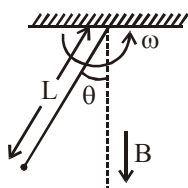


- (A) the magnetic force on any circular ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{72B^2av}{5\pi\lambda}$.
 (B) the magnetic force on any circular ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{18B^2av}{5\pi\lambda}$.
 (C) the current in the branch BDA of the ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{3Bv}{5\pi\lambda}$.
 (D) the current in the branch BDA of the ring at the time t , when $\angle AC_1B = 60^\circ$ is $\frac{2Bv}{5\pi\lambda}$.

10. Two identical charged particles (of mass m and having charge q) are simultaneously projected from origin (as shown) with equal speeds v , in a region of transverse magnetic field.



- (A) If $\vec{r}_1$ and $\vec{r}_2$ are the position vectors of the particles at time $t = \frac{\pi m}{qB}$ then $\vec{r}_1 \cdot \vec{r}_2$ equals $2 \left(\frac{mv}{qB} \right)^2$
- (B) If $\vec{r}_1$ and $\vec{r}_2$ are the position vectors of the particles at time $t = \frac{\pi m}{qB}$ then $\vec{r}_1 \cdot \vec{r}_2$ equals $\frac{1}{2} \left(\frac{mv}{qB} \right)^2$
- (C) Magnitude of relative speeds of the particles after they are projected remains constant
- (D) Magnitude of relative angular momentum of the particles after they are projected first increases, then decreases
11. A charged particle having its charge to mass ratio as β goes in a conical pendulum of length L making an angle θ with vertical and angular velocity ω . If a magnetic field B is directed vertically downwards (see figure) :



- (A) $B = \frac{1}{\beta} \left[\omega - \frac{g}{\omega L \cos \theta} \right]$
- (B) Angular momentum of the particle about the point of suspension remains constant.
- (C) If the direction of B were reversed maintaining same ω and L , then θ will remain unchanged.
- (D) Rate of change of angular momentum of the particle about the point of suspension is not a constant vector.

Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question Nos. 12 to 14

A metal cylinder is rotating at an angular velocity ω around its axis of symmetry. The cylinder is in a uniform magnetic field with the induction vector $\vec{B}$ parallel to its axis.

12. If magnetic induction vector $\vec{B}$ and $\vec{\omega}$ vector are oppositely directed, what is magnitude of electric field in the metal cylinder
- (A) Zero (B) $\frac{(e\omega B + m\omega^2)r}{e}$ (C) $\frac{(e\omega B - m\omega^2)r}{e}$ (D) $\frac{(m\omega^2 - e\omega B)r}{e}$
13. In previous question, determine the charge density in the interior of the cylinder.
- (A) Zero (B) $\rho = \frac{\epsilon_0 \omega (eB - m\omega)}{2e}$
- (C) $\rho = \frac{2\epsilon_0 \omega (eB + m\omega)}{e}$ (D) $\rho = \frac{\epsilon_0 \omega (eB + m\omega)}{2e}$

14. If $\vec{B}$ and $\vec{\omega}$ point in the same direction, mark the correct statement

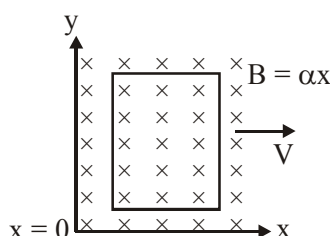
- (A) If $\omega > \frac{eB}{m}$; electric field vector is radially inward, the outer surface of the cylinder carries positive charge
- (B) If $\omega > \frac{eB}{m}$; electric field vector is radially outward, the outer surface of the cylinder carries negative charge
- (C) If $\omega = \frac{eB}{m}$; electric field vector is radially inward, the outer surface of the cylinder carries positive charge
- (D) If $\omega < \frac{eB}{m}$; electric field vector is radially outward, the outer surface of the cylinder carries negative charge

SECTION-II

Numerical Answer Type Question
(upto second decimal place)

1 Q. [3(0)]

1. A square loop of length $L = 0.2$ m and resistance $R = 80$ m Ω lies with one of its length along the x-axis as shown in the figure. It moves with speed $V = 10$ m/s in the positive x-direction in a magnetic field which is into the page and has a magnitude that varies with x according to $B = \alpha x$ where $\alpha = 0.2$ Tm⁻¹. The magnitude of induced current is I ampere. Then the value of $10I$ is. (Neglect self inductance of the loop)



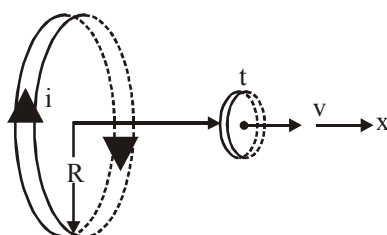
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

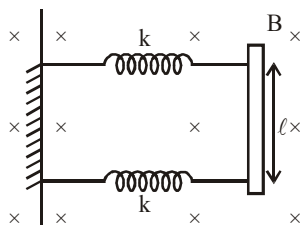
4 Q. [4 M (0)]

1. A very small loop of radius r moves co-axially with a constant velocity v , to another bigger loop of radius R which is carrying a steady current i . It is assumed that the smaller loop begins to move from the centre of the bigger loop. Find the expression for emf induced in the smaller loop in the function of time.

If your answer is $N \left(\frac{\mu_0 v^2 \pi i t R^2 r^2}{(R^2 + v^2 t^2)^P} \right)$. Give the value of $N + P$



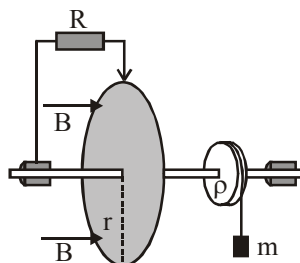
2. A loop is formed with two springs and a conducting rod of length $\ell = 1\text{ m}$ and mass $m = 1\text{ kg}$. A uniform magnetic field $B = 0.01\text{ T}$ is directed perpendicular to the plane of the loop. At $t = 0$ the rod is released when the spring is extended by a distance 4 cm . What is the maximum value of the emf (in mV) (spring constant $k = 200\text{ N/m}$) ?



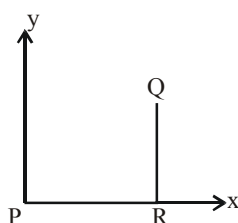
3. A disc of radius r is made of a material of negligible resistance and can rotate about a horizontal shaft. A smaller disc of radius ρ is fixed onto the same shaft and has a massless cord wrapped around it, which is attached to a small object of mass m as shown. Two ends of a resistor of resistance R are connected to the perimeter of the disc and to the shaft by wiping contacts. The system is then placed into a uniform horizontal magnetic field B and mass m is released. Find the constant angular velocity (in rad/s) with

which the disc will rotate after a certain time. If your answer is ω give value of $\frac{\omega}{20}$. Data : $r = 10\text{ cm}$,

$\rho = 2\text{ cm}$, $R = 0.01\ \Omega$, $B = 0.2\text{ T}$, $m = 50\text{ g}$.



4. An inverted L shaped conductor PRQ is made by joining two perpendicular conducting rods, each of length $1.5L$, at end R. This structure is moving in x-y plane containing variable magnetic field, $\vec{B} = -3x\hat{k}$ with a velocity $\hat{v}_i + \hat{v}_j$. If potential of P is V_P and that of Q is V_Q , then, value of $|V_P - V_Q|$ at the instant when P is at origin as shown in figure is $\alpha \left(\frac{9vL^2}{8} \right)$. Fill the value of ' α ' in OMR sheet.



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Column I shows certain system at $t = 0$. Column II shows its behavior there after. Neglect dissipative forces in all cases.

Column I

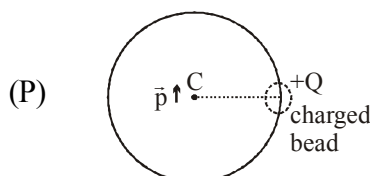
- (A) System experiences net torque initially about C.

- (B) Angular velocity about C first increases then decreases

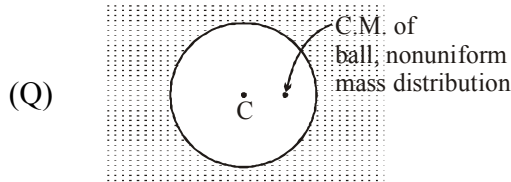
- (C) System executes periodic motion

- (D) Acceleration of centre of mass of system is zero initially.

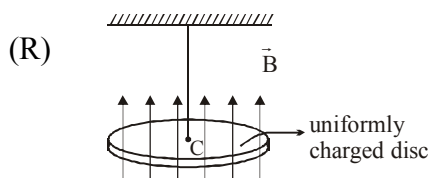
Column II



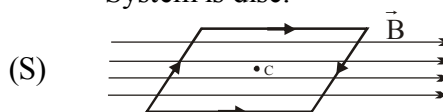
A charged bead is constrained to move along a non conducting ring in gravity free space. A dipole is fixed at the centre of ring. Charge is released from perpendicular bisector of dipole at $t = 0$. System is bead.



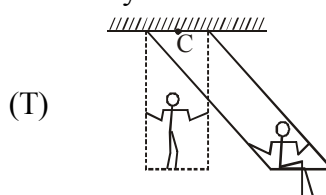
A spherical ball made of wood is released inside water from rest in the configuration shown at $t = 0$. Point C is geometric centre of ball. System is ball.



Disc hangs with an elastic wire of some torsional stiffness. Uniform magnetic field is suddenly switched on at $t = 0$. System is disc.



A current carrying coil is placed in a uniform magnetic field parallel to the plane of coil at $t = 0$, coil lies in a gravity free space. System is coil.



A boy on a swing small angular amplitude. When swing gets to mean position ($t = 0$) boy instantaneously stands up. System is swing & boy.

PHYSICS

CLASS TEST # 57

SECTION-I

Single Correct Answer Type

7 Q. [3 M (-1)]

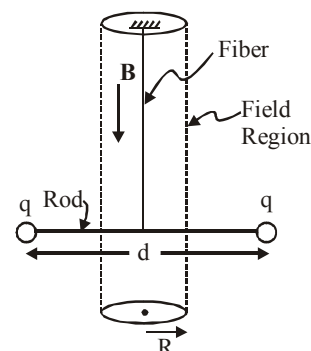
1. A current I flows along the length of a thin walled, long metallic hollow cylinder of radius R , distributed uniformly on its surface then pressure on the wall will be :

(A) $\left(\frac{\mu_0 I^2}{8\pi^2 R^2}\right)$ (B) $\left(\frac{\mu_0 I^2}{4\pi^2 R^2}\right)$ (C) $\left(\frac{\mu_0 I^2}{\pi^2 R^2}\right)$ (D) None of these

2. A smooth insulating ring of radius R , with a bead having charge q is placed horizontally and in a uniform magnetic field of strength B_0 and perpendicular to the ring plane. Starting from $t = 0$, the magnetic field is changed to $B(t) = B_0 + \alpha t$, where α is a positive constant. The contact force between the ring and bead as a function of time is (Neglect gravity)

(A) $\frac{\alpha q^2 R t}{m} (2B_0 + \alpha t)$ (B) $\frac{\alpha q^2 R t}{4m} (2B_0 + \alpha t)$
(C) $\frac{\alpha q^2 R t}{4m} (B_0 + \alpha t)$ (D) $\frac{4\alpha q^2 R t}{m} (2B_0 + \alpha t)$

3. Two small pith balls, each carrying a charge q , are attached to the ends of a light rod of length d , which is suspended from the ceiling by a thin torsion-free fiber, as shown in the figure below. There is a uniform magnetic field B , pointing straight down, in the cylindrical region of radius R around the fiber. The system is initially at rest. If the magnetic field is turned off, which of the following describes what happens to the system ?



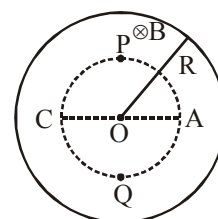
(A) It rotates with angular momentum qBR^2 .

(B) It rotates with angular momentum $\frac{1}{4}qBd^2$.

(C) It rotates with angular momentum $\frac{1}{2}qBRd$.

(D) It does not rotate because to do so would violate conservation of angular momentum.

4. A uniform magnetic field B increasing with time exists in a cylindrical region of centre O and radius R . The direction of magnetic field is inwards the paper as shown. The work done by external agent in taking a unit positive charge slowly from A to C via paths APC , AOC and AQC be W_{APC} , W_{AOC} and W_{AQC} respectively. Then :



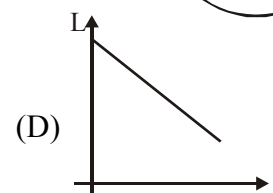
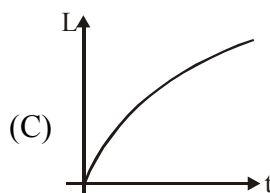
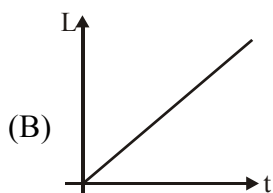
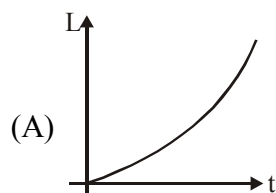
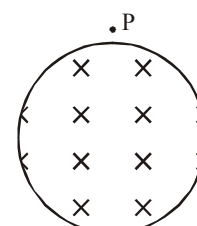
(A) $W_{APC} = W_{AOC} = W_{AQC}$

(B) $W_{APC} > W_{AOC} > W_{AQC}$

(C) $W_{APC} < W_{AOC} < W_{AQC}$

(D) $W_{APC} = W_{AQC} < W_{AOC}$

5. A uniform time varying magnetic field is present in a cylindrical region as shown. Magnetic field strength increases linearly with time and a charged particle is released from a point P , just outside the cylindrical region. If its angular momentum about center is L , then which of the graph correctly demonstrate its behaviour as function of time ? (Neglect gravity)



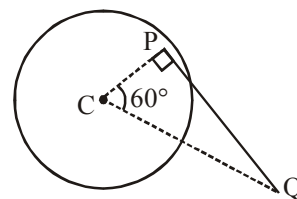
6. A magnetic field, confined in a cylindrical region of radius R , is changing at the rate of 4 T/s . A conducting rod PQ of length $\sqrt{\frac{3}{2}}R$ is placed in the region as shown. The induced emf across the rod will be :

(A) $\frac{R^2}{24}(\pi + 6)V$

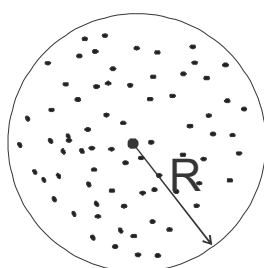
(B) $\frac{R^2\pi}{24}V$

(C) $\frac{R^2\pi}{6}V$

(D) $\frac{R^2}{6}(\pi + 6)V$



7. In the diagram a time varying magnetic field passes through a circular region of radius " R " and the magnetic field directed outwards and it is a function of radial distance ' r ' and time ' t ' as $B = B_0 r t$. The electric field strength at a radial distance $R/2$ from the center ?



(A) $\frac{B_0 R^2}{12}$

(B) $\frac{B_0 R^2}{6}$

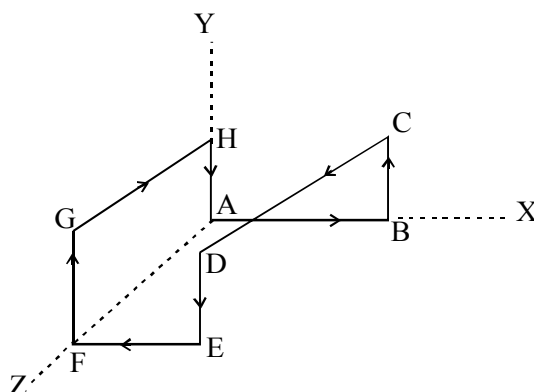
(C) $\frac{2B_0 R^2}{3}$

(D) $\frac{B_0 R^2}{16}$

Multiple Correct Answer Type

4 Q. [4 M (-1)]

8. A current I flows through a loop ABCDEFGHA along the edge of the cube of width l meters as shown in the figure. One corner of the loop lies at origin.



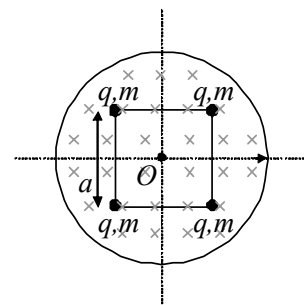
Then mark the **CORRECT** statement(s) : -

- (A) This current path ABCDEFGHA can be treated as a superposition of three square loops carrying current I . The three loops are FGHA, FABEF, FGDEF
- (B) The unit vector in the direction of magnetic field at the centre of cube is $-\hat{j}$.
- (C) Now if a uniform external magnetic field $\vec{B} = B_0 \hat{j}$ is switched on, then the unit vector in the direction of torque due to external magnetic field ($\vec{B}$) acting on the current carrying loop (ABCDEFGHA) is

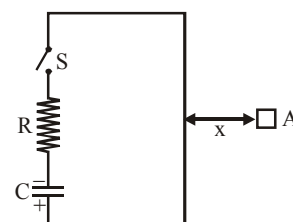
$\frac{2\hat{i} - \hat{j}}{\sqrt{5}}$

- (D) None of these

9. Four identical charge particles each of mass 0.1kg and charge $2C$ connected to each other via massless non-conducting rods of equal length. The whole arrangement is placed in a cylindrical region carrying a uniform magnetic field as shown in the figure ($B_0 = 4\text{T}$, $a = 1\text{m}$). Suddenly the magnetic field is switched off. Then choose the correct statement(s) :-
- (A) the angular momentum of the system is 8 Nm/s
 (B) the angular momentum of the system is $8\sqrt{2}\text{ Nm/s}$
 (C) Angular velocity of the system is 40 rad/s
 (D) the magnetic field at point O after the magnetic field is switched off is non zero.



10. In figure the switch is closed at $t = 0$, with the capacitor of capacity $1\text{ }\mu\text{F}$ and having initial charge of $20\text{ }\mu\text{C}$ (the polarity shown). The left square loop has very large dimensions as compared to the distance between loops and a resistance of $10\text{ }\Omega$ is connected in series with capacitor as shown. A wire of length 8 mm and resistance per unit length of $0.5\text{ }\Omega/\text{mm}$ is bent in the form of square loop A and placed at a distance $x = 1\text{ m}$ from left loop.



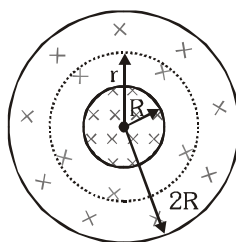
- (A) The mutual inductance of the system will be $8 \times 10^{-13}\text{ H}$
 (B) The induced current in the loop A will be clockwise
 (C) The induced current in the loop A will be anticlockwise
 (D) The induced current in the loop A at time $t = 2RC \ln 2$ will be 10^{-8} A
11. The primary winding of step-down transformer with a turn ratio 10 has a source voltage of 120 V connected to it. The resistance of the secondary and primary winding is negligible, the current in secondary is $I = 5\text{ A}$.
- (A) The current in primary is 0.5 A
 (B) The power delivered to the load is 60 W
 (C) The load resistance is 24 ohms
 (D) The voltage across the load is 1200 V .

Linked Comprehension Type (Single Correct Answer Type)

(2 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question 12 to 14

A variable magnetic field exist in a cylindrical region of space which is perpendicular & into the plane of paper. Magnetic field in region (I) $0 < r \leq R$ is uniform and increasing with constant rate. Magnetic field in the region (II) $R < r \leq 2R$ is also uniform and increasing with constant rate which is $(1/3)$ times of that in region (I). At $t=0$, field in both region is found zero and at $t=t_0$, field in the region (II) is found B_0 .



12. Induced electric field at $r=R$ & $t=t_0$ is :-
- (A) $\frac{3RB_0}{2t_0}$ (B) $\frac{2RB_0}{t_0}$ (C) $\frac{3RB_0}{t_0}$ (D) $\frac{RB_0}{3t_0}$
13. Induced electric field at $r=2R$ & $t=t_0$ is :-
- (A) $\frac{3RB_0}{2t_0}$ (B) $\frac{2RB_0}{t_0}$ (C) $\frac{3RB_0}{t_0}$ (D) $\frac{RB_0}{3t_0}$
14. As a result of increasing magnetic field a charge particle initially kept at rest in region (II) starts moving in circular trajectory of radius r . Value of r is :-
- (A) $\sqrt{3}\text{ R}$ (B) $\sqrt{2}\text{ R}$ (C) 2 R (D) $\frac{\text{R}}{\sqrt{3}}$

Paragraph for Question 15 to 17

A non conducting thin ring of radius R and mass m is placed on a rough horizontal insulating plane.

Charge Q is uniformly distributed over the ring. At $t=0$ an uniform vertical magnetic field $B = \frac{B_0 t^2 \sqrt{3}}{2}$

is switched on. The coefficient of friction is $\mu = \frac{QB_0 R}{2mg}$

15. What is $\int_{\text{over the ring}} |d\vec{F}|$ where $d\vec{F}$ is force applied by electric field on an infinitesimal element of ring.

(A) $QB_0 Rt$ (B) $\frac{QB_0 Rt}{2}$ (C) $QB_0 Rt\sqrt{3}$ (D) $\frac{QB_0 Rt\sqrt{3}}{2}$

16. Time (in seconds) at which ring will just start rotating is :-

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$

17. Magnetic field is switched off at $t = \sqrt{3}$ sec. Find the angular velocity just after this:

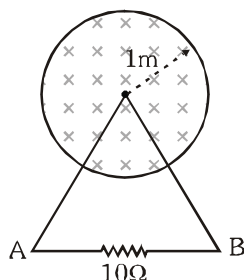
(A) $\frac{QB_0}{m\sqrt{3}}$ (B) $\frac{2QB_0}{m}$ (C) $\frac{QB_0}{2m}$ (D) $\frac{2QB_0}{m\sqrt{3}}$

SECTION-III

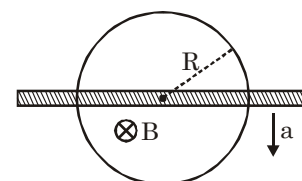
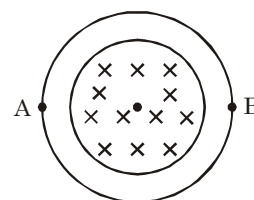
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1. A uniform magnetic field of intensity $B = \frac{6}{\pi} \sin 2t$ directed into the plane of the paper exists in the cylindrical region of radius 1 m. A loop of resistance 10Ω folded in the form of equilateral triangle of side length 2m is placed as shown in figure. Find the maximum potential drop in wire AB.



2. A resistance less ring has 2 bulbs A & B rated at 2 V, 10 W and 2 V, 20W respectively. The ring encloses an ideal solenoid whose magnetic field is as shown. The radius of solenoid is 1m and the number of turns/length = 1000/m. The current changes at rate of 9 A/sec. Find the power dissipated P in bulb B. Fill $P \times 10^4$ in nearest integer.
3. A uniform ring of radius (R) is fixed in a horizontal plane, ring has resistance ($\lambda\Omega/m$). There is a perpendicular magnetic field (B) in the region. A perfectly conducting rod of length $\ell(> 2R)$ is kept along the diameter of ring and is made to accelerate uniformly in the horizontal plane with acceleration (a) as shown.



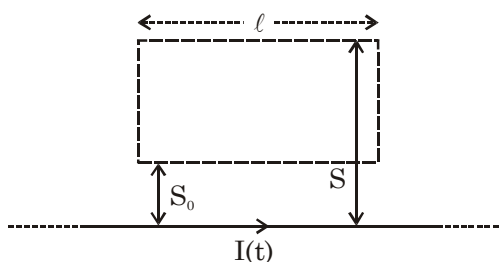
When the rod has travelled a distance equal to $\left(\frac{R}{2}\right)$ at that instant, current (in amp) through the rod is (Assume $\lambda = 3$, $B = 4\pi T$, $R = 1m$, $a = 3 m/s^2$)

SECTION-I

Single Correct Answer Type

8 Q. [3 M (-1)]

1. An infinitely long straight wire carries a slowly varying current $I(t)$. The line integral of electric field along the loop shown in figure is given by.



(A) $-\frac{\mu_0 \ell}{2\pi} \frac{dI}{dt} (\ln S - \ln S_0)$

(B) $-\frac{\mu_0 \ell}{2\pi} \frac{dI}{dt} \frac{(S - S_0)}{(S_0 + S)}$

(C) $-\frac{\mu_0 \ell}{2\pi} \frac{(S + S_0)}{(S - S_0)} \frac{dI}{dt}$

(D) $\frac{\mu_0 \ell}{4\pi} \frac{dI}{dt} (\ln S - \ln S_0)$

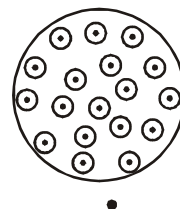
2. As shown, a uniform magnetic field B pointing out of the paper plane is confined in the shaded area of radius r . At a distance R ($R > r$) from the center of the shaded area there is a point particle of mass m and carrying charge q . Speed of the particle if the magnetic field in the region is quickly changed to zero will be.

(A) $V = \frac{qBr^2}{2mR}$

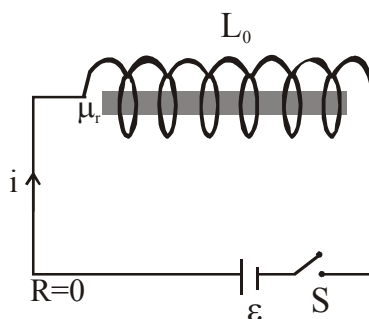
(B) 0

(C) $V = \frac{qBr^2}{mR}$

(D) $v = \frac{2qBr^2}{mR}$



3. At $t = 0$, switch is closed. At $t = \tau$ sec core is dragged out suddenly by an external agent. Find the value of current just before and just after pulling the core. Resistance of the circuit is very small and can be neglected.



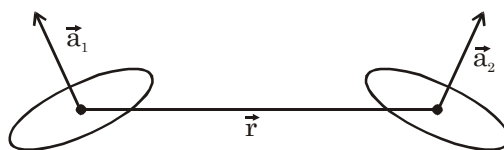
(A) $i_{\text{before}} = \frac{2\varepsilon\tau}{L_0}, i_{\text{after}} = \mu_r \frac{\varepsilon\tau}{2L_0}$

(B) $i_{\text{before}} = \frac{\varepsilon\tau}{L_0}, i_{\text{after}} = \mu_r \frac{\varepsilon\tau}{L_0}$

(C) $i_{\text{before}} = \frac{\varepsilon\tau}{L_0}, i_{\text{after}} = \mu_r \frac{\varepsilon\tau}{2L_0}$

(D) $i_{\text{before}} = \frac{\varepsilon\tau}{4L_0}, i_{\text{after}} = \mu_r \frac{\varepsilon\tau}{L_0}$

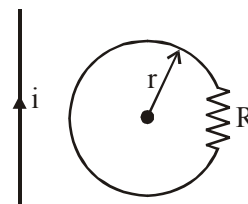
4. Two very tiny wire loops, with areas $\vec{a}_1$ and $\vec{a}_2$ are situated a displacement $\vec{r}$ apart (see figure). Their mutual inductance is given by



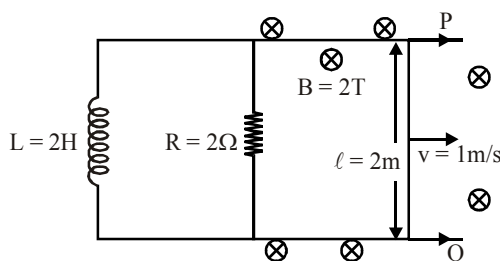
- (A) $\frac{\mu_0}{4\pi r^3} [3(\vec{a}_1 \cdot \hat{r})(\vec{a}_2 \cdot \hat{r}) + \vec{a}_1 \cdot \vec{a}_2]$ (B) $\frac{3\mu_0}{4\pi r^3} [3(\vec{a}_1 \cdot \hat{r})(\vec{a}_2 \cdot \hat{r}) + \vec{a}_1 \cdot \vec{a}_2]$
(C) $\frac{3\mu_0}{4\pi r^3} [3(\vec{a}_1 \cdot \hat{r})(\vec{a}_2 \cdot \hat{r}) - \vec{a}_1 \cdot \vec{a}_2]$ (D) $\frac{\mu_0}{4\pi r^3} [3(\vec{a}_1 \cdot \hat{r})(\vec{a}_2 \cdot \hat{r}) - \vec{a}_1 \cdot \vec{a}_2]$

5. In the given figure, the mutual inductance of coil and the very long straight wire is M , the coil has resistance R and the self inductance L . The wire lies in the same plane as that of the coil. The current in the wire varies according to the law $i = at$, where a is a constant and t is the time. The time dependence of current in the coil is (consider clockwise current in loop as positive) :-

- (A) $-\frac{Ma}{R} e^{-tR/L}$ (B) $-\frac{Ma}{R} (1 - e^{-tR/L})$
(C) $\frac{Ma}{R} e^{-tR/L}$ (D) $\frac{Ma}{R} (1 - e^{-tR/L})$

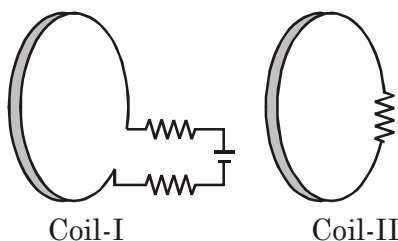


6. The given figure shows an inductor and resistance fixed on a conducting wire. A movable conducting wire PQ starts moving on the fixed rails from $t = 0$ with constant velocity 1 m/s . The work done by the external force on the wire PQ in 2 seconds is :-



- (A) 16 J (B) 32 J (C) 48 J (D) 64 J

7. In the shown figure the mutual inductance of two coils is M , the coil II has resistance R and the self inductance L . The current in the coil I varies according to the law $i = at$, where a is a constant and t is time. Find the current in coil II as function of time (except M for the pair and L for the coil II ignore any other effect of induction)



- (A) $\frac{Ma}{R} e^{-\frac{Rt}{L}}$ (B) $\frac{M}{R} \left(1 - e^{-\frac{Rt}{L}} \right)$ (C) $\frac{Ma}{R} \left(1 - e^{-\frac{Rt}{L}} \right)$ (D) $\frac{Ma}{R}$

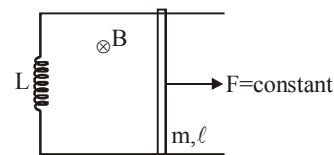
8. A loop is formed by two parallel conductors connected by a solenoid with inductance L and a conducting rod of mass m , which can freely (without friction) slide over the conductors. The conductors are located in a horizontal plane in a uniform vertical magnetic field with induction B . The distance between the conductors is equal to ℓ . At the moment $t = 0$ a constant force F starts acting on the rod. Angular frequency of motion of rod is ω . If F and m are doubled then :

(A) ω become $\frac{1}{\sqrt{2}}$ times

(B) ω become $\sqrt{2}$ times

(C) ω become 2 times

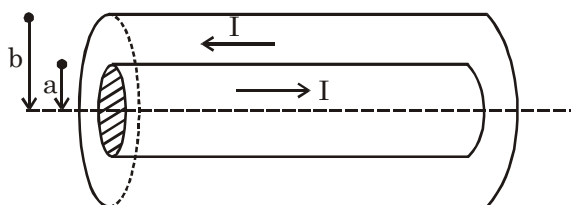
(D) ω become $\frac{1}{2}$ times



Multiple Correct Answer Type

5 Q. [4 M (-1)]

9. A long co-axial cable carries current I as shown in figure. (Current flows down the surface of inner cylinder of radius a and back along the outer cylinder of radius b .) Choose the correct statement(s) :



(A) The magnetic energy stored in a section of length ℓ is $\frac{\mu_0 I^2 \ell}{4\pi} \ln\left(\frac{b}{a}\right)$

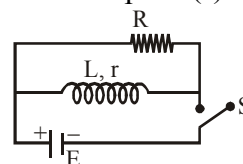
(B) The magnetic field between the cylinders have magnitude of $\frac{\mu_0 I}{2\pi(b-a)}$

(C) Self inductance of the cable of length ℓ is $\frac{\mu_0 \ell}{4\pi} \ln\left(\frac{b}{a}\right)$

(D) Magnetic field outside the cylinder is zero.

10. A solenoid of inductance L and resistance r is connected in parallel to a resistance R . A battery of emf E and of negligible internal resistance is connected across this parallel combination as shown in figure. The circuit is in steady state. At time $t = 0$, switch S is opened. Choose the **CORRECT** option(s).

(A) Current in the inductor just after opening of the switch is $= \frac{E(r+R)}{rR}$



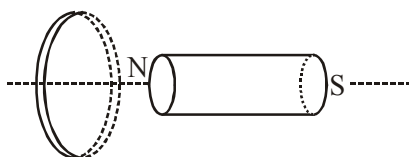
(B) Total energy dissipated in the solenoid and the resistor long time after opening of the switch is

$$\frac{1}{2} L \frac{E^2 (R+r)^2}{r^2 R^2}$$

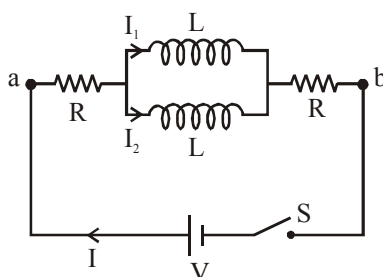
(C) The amount of heat generated in the solenoid after opening of switch is $\frac{E^2 L}{2r(r+R)}$

(D) The amount of heat generated in the solenoid after opening of the switch is $\frac{E^2 L}{2R(r+R)}$

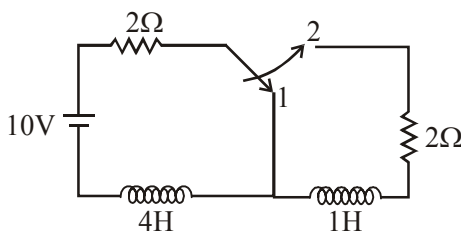
11. A magnet is placed co-axially with the conducting loop :-



- (A) when the magnet is pushed towards right, a clockwise current is observed
 (B) if the flux passing through the loop is ϕ and its resistance is R , the total charge flown after the magnet is thrown to infinity is $\frac{\phi}{R}$
 (C) if the loop's resistance is neglected and inductance is assumed to be L , the flux passing through the loop is conserved.
 (D) if we bring the resistanceless loop from infinity, it will carry a current $i = \frac{\phi}{L}$, where L = inductance of the loop and ϕ = flux passing through the coil by the magnet.
12. In the circuit shown switch S is closed at $t = 0$. At time (t) which of the following is/are **CORRECT**:



- (A) $I = \frac{V}{2R} \left(1 - e^{-\frac{4Rt}{L}} \right)$
 (B) $I_1 = I_2 = \frac{V}{2R} \left(1 - e^{-\frac{4Rt}{L}} \right)$
 (C) At $t = 0$, $\frac{dV_{ab}}{dt} = 0$
 (D) $I = \sqrt{I_1^2 + I_2^2}$
13. A network is given as shown. Switch was at position 1 for long time. at $t = 0$ it is shifted to position 2, then



- (A) Current just after switching is $\sqrt{20}$ Amp
 (B) Current just after switching is 4 Amp
 (C) Current in the circuit varies $i = \frac{5}{2} + \frac{3}{2} e^{-4t/5}$
 (D) Current just after switching is 0

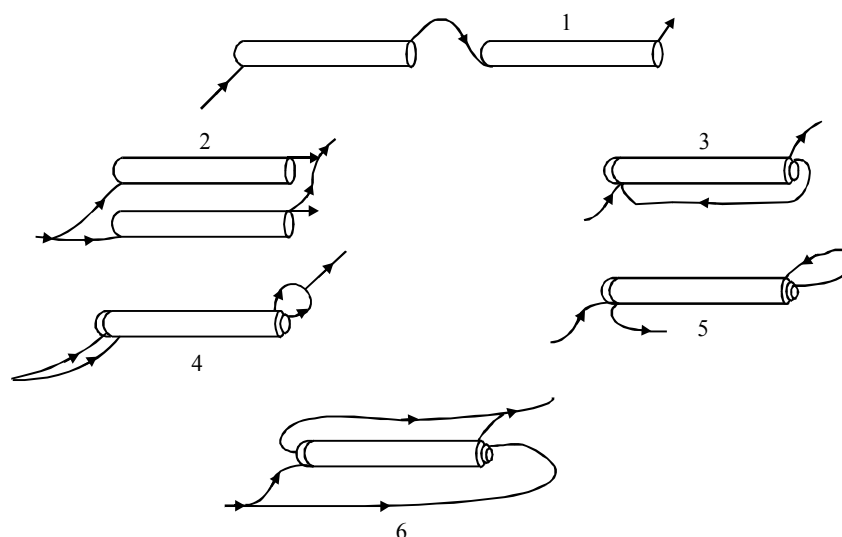
Linked Comprehension Type
(Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Questions 14 and 15

Suppose that we have two solenoids of the same length. Their diameters differ only to the extent to which one can be fitted onto the other. The inductances of the two solenoids can be considered the same and equal to L_0 . Here are the ways in which the solenoids can be connected :

- (1) the solenoids are connected in series and are separated by a large distance ;
- (2) the solenoids are connected in parallel and are separated by a large distance ;
- (3) the solenoids are connected in series, one is fitted onto the other, and the senses of the turns coincide ;
- (4) the solenoids are connected in parallel, one is fitted onto the other, and the senses of the turns coincide ;
- (5) the solenoids are connected in series, one is fitted onto the other, and the senses of the turns are opposite;
- (6) the solenoids are connected in parallel, one is fitted onto the other, and the senses of the turns are opposite.



14. Mark the **CORRECT** statement :-

- (A) in case (i) net emf induced $\varepsilon = -2L_0 \frac{dI}{dt}$
- (B) in case (ii) net emf induced $\varepsilon = -L_0 \frac{dI}{dt}$
- (C) in case (i) net emf induced $\varepsilon = -L_0 \frac{dI}{dt}$
- (D) in case (ii) net emf induced $\varepsilon = -2L_0 \frac{dI}{dt}$

15. Mark the **INCORRECT** statement :-

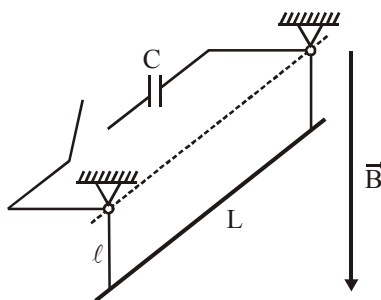
- (A) In case (iii) inductance of system is $4L_0$
- (B) In case (iv) inductance of system is L_0
- (C) In case (v & vi) inductance of system is zero
- (D) In case (v & vi) inductance of system is L_0

SECTION-II

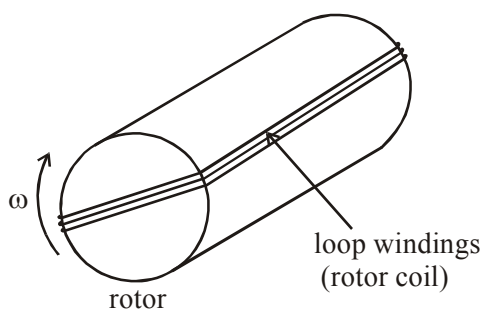
Numerical Answer Type Question (upto second decimal place)

2Q.[3(0)]

1. Metal rod with a mass $m = 10$ g, and length $L = 0.2$ m is suspended by two light wires length $\ell = 1$ cm in a magnetic field induction $B = 1$ T which is directed vertically downwards (Figure). A capacitor of capacitance $C = 100$ μ F charged to a voltage of 100 V is connected as shown. Determine the maximum deflection (in degree) of the rod from the initial position after the switch is closed. Resistance of wire and rod is not taken into account.



2. In a motor, a rotor is fitted with the armature that has current of 10 A. The rotor rotates with angular speed of 3 rad/s. Magnetic field of magnitude 2 T varies in direction in such a way that it is always perpendicular to the loop area. If the rotor coil has N number of turns and area of each loop is 0.45 m^2 then find the value N . Given that motor consumes 2106 W power and there are no losses.

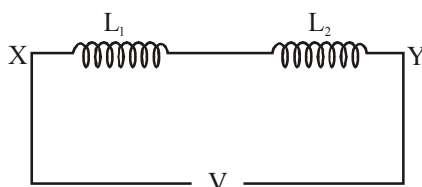


SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

1 Q. [4 M (0)]

2. In the circuit shown current is changing at the rate of $\frac{di}{dt} = 2$ A/s. If $L_1 = L_2 = 1$ H and $M_{12} = \frac{1}{2}$ H, find emf V (in volt) between terminals X & Y.

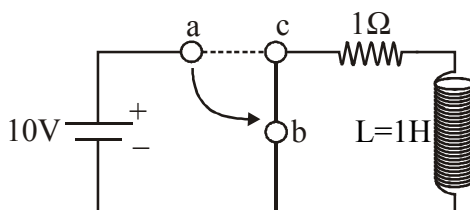


SECTION-I

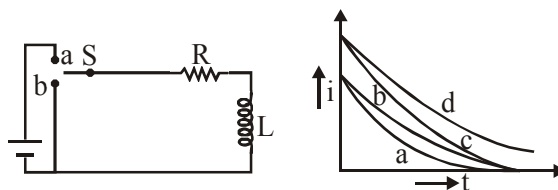
Single Correct Answer Type

6 Q. [3 M (-1)]

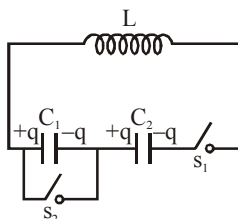
1. In the given circuit, the switch is closed to the position bc from the earlier position of ac at $t = 0$. The current in the inductor after 2s of closing the switch between b and c is :



- (A) $\frac{1}{10e^2}$ A (B) $10e^{-2}$ A (C) $10e$ A (D) $10e^2$ A
2. In the figure shown below, the switch S has been closed on 'a' for a long time. It is then thrown to 'b'. The resulting current through the inductor is sketched in figure on right for four sets of values of R and L. Which curve goes with which set of values?
- (1) R_0, L_0 (2) $2R_0, L_0$ (3) $R_0, 2L_0$ (4) $2R_0, 2L_0$

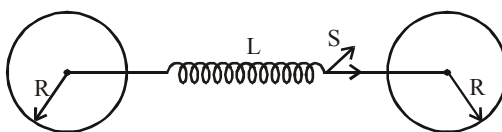


- (A) a, 2; b, 4; c, 1; d, 3 (B) a, 2; b, 1; c, 4; d, 3 (C) a, 4; b, 2; c, 1; d, 3 (D) a, 4; b, 2; c, 3; d, 4
3. A long solenoid is made of a superconducting wire carrying a current I_0 . The solenoid is slowly stretched so that its cross section does not change but its length changes from L_0 to L_1 . What is the new current in the solenoid assuming the solenoid is still tightly wound ?
- (A) $\frac{L_0 I_0}{L_1}$ (B) $\frac{L_1 I_0}{L_0}$ (C) $\frac{L_0^2 I_0}{L_1^2}$ (D) $\frac{L_1^2 I_0}{L_0^2}$
4. Two capacitors C_1 & C_2 of capacitance $\frac{1}{\pi^2} \times 10^{-2}$ F each and an inductor L of inductance 2×10^{-2} H are connected in series as shown in the figure. Initially charge on each capacitors are $4\sqrt{3}$ μ C. At $t = 0$ switch S_1 is closed and at $t = \frac{1}{400}$ sec, switch S_2 is also closed. The maximum charge on capacitor C_2 during LC oscillation is :-

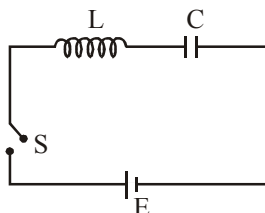


- (A) 3 μ C (B) $6\sqrt{2}$ μ C (C) $3\sqrt{3}$ μ C (D) $6\sqrt{3}$ μ C

5. Two conducting spheres of equal radii R are placed far apart and are connected by an ideal inductor as shown. Initially one of the spheres has a charge Q while other is uncharged. Switch is closed at time, $t = 0$. Then :-



- (A) Minimum time after which all charge can be transferred to other sphere is $\sqrt{2\pi^3 \epsilon_0 LR}$
 (B) Minimum time after which all charge can be transferred to other sphere is $\sqrt{8\pi^3 \epsilon_0 LR}$
 (C) First sphere will acquire a negative charge at some instant of time
 (D) At some instant of time both spheres will have zero charge
6. An ideal DC source of emf E is connected with an uncharged capacitor and inductor. If switch is closed at $t = 0$, what is the maximum charge on the capacitor in subsequent flow of charge ?

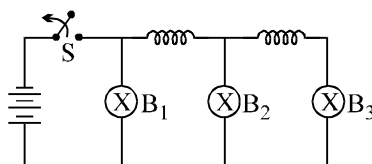


- (A) $2CE$ (B) CE (C) $1.5 CE$ (D) $4CE$

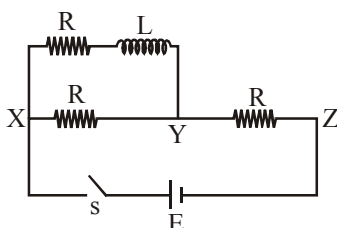
Multiple Correct Answer Type

5 Q. [4 M (-1)]

7. The circuit shown in figure consisting of three identical lamps and two coils is connected to a direct current source. The ohmic resistance of the coils is negligible. After some time switch S is opened. Which of the following statement(s) is/are **correct** for the instant immediately after opening the switch?

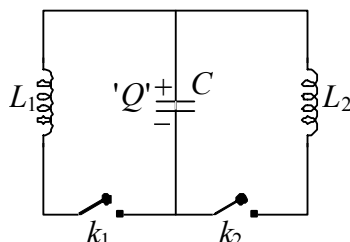


- (A) All the lamps are turned off (B) Brightness of B_2 & B_3 remains unchanged
 (C) Brightness of B_1 suddenly increases (D) Insufficient data to draw any conclusion.
8. The switch is closed at $t = 0$ in the adjoining circuit. Which of the following is/are **CORRECT** ?

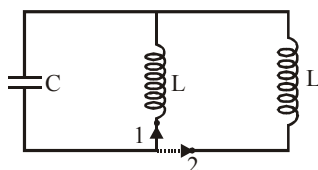


- (A) The potential difference across YZ at $t = 0$ is $2E/3$
 (B) The potential difference across XY at $t = \infty$ is $E/2$
 (C) The potential difference across YZ at $t = 0$ is $E/2$
 (D) The potential difference across XY at $t = \infty$ is $2E/3$

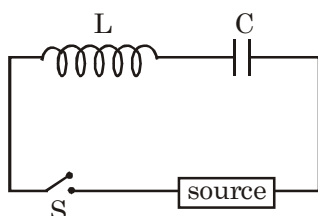
9. The given arrangement carries a capacitor with capacitance 40mF and two inductors $L_1 = 25\text{ H}$ and $L_2 = 100\text{ H}$. If the capacitor initially carries a charge of 10 mC , then :-



- (A) the maximum current through the inductor L_1 when key k_1 is closed is 20 mA
 (B) the maximum current through the inductor L_2 when key k_2 is closed is 5mA
 (C) the maximum current through inductor L_2 when both the keys are closed is $\sqrt{5}\text{ mA}$
 (D) time period of oscillation of charge is minimum when both the keys are closed
10. A capacitor with charge Q on it is connected to an inductor L as shown in diagram at $t = 0$. When the switch is flipped from position 1 to 2, the current in the circuit is observed to be at half of its maximum value. Then :-



- (A) Charge on capacitor at the time was $\frac{Q}{2}$ (B) Charge on capacitor at that time was $\frac{Q\sqrt{3}}{2}$
 (C) Total energy lost is $\frac{Q^2}{4C}$ (D) Total energy lost is $\frac{Q^2}{8C}$
11. In the circuit shown source may be AC or DC. At $t = 0$ switch S is closed. Choose the correct statement(s) :-



- (A) After a long time charge on the capacitor will be zero if source is DC
 (B) After a long time charge on the capacitor will be zero if source is AC
 (C) After a long time charge on the capacitor will not be constant if source is DC
 (D) After a long time charge on the capacitor will not be constant if source is AC

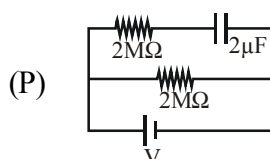
Matching List Type (4 × 4)

1Q.[3 M (-1)]

12. Match the physical quantity of List-I with value given in List-II.

List-I

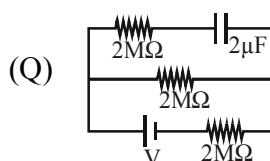
List-II



time constant

$$\tau = \dots \text{ sec}$$

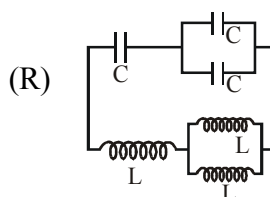
(1) 100



time constant,

$$\tau = \dots \text{ sec}$$

(2) 1.33

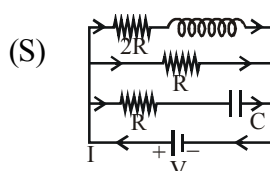


$$C = 100 \mu\text{F}, L = 1\text{H}$$

angular frequency

$$\omega = \dots \text{ rad/sec}$$

(3) 4



$$\frac{I(0)}{I(\infty)} = \dots$$

$I(0)$ and $I(\infty)$ are current

drawn from the battery

at $t = 0$ and $t \rightarrow \infty$

(4) 6

Codes :

	P	Q	R	S
(A)	2	1	4	3
(C)	3	4	1	2

	P	Q	R	S
(B)	3	1	4	2
(D)	4	2	3	1

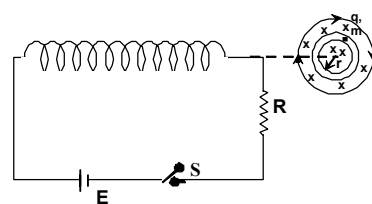
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

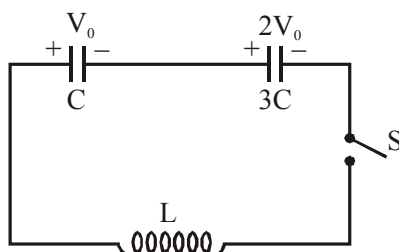
5 Q. [4 M (0)]

1. There are two concentric coplanar rings where outer ring has radius 4m. The inner ring has radius 1mm, resistance 1Ω & self inductance $2 \mu\text{H}$. Initially the outer and inner rings have current of $5 \times 10^6 \text{ A}$ & 2 A respectively in clockwise sense. Later on current in inner ring is found to be 3 A clockwise whereas for outer ring it is $5 \times 10^6 \text{ A}$ anticlockwise. If total charge flown through the inner ring till this moment is $Q_0 \mu\text{C}$ then find value of Q_0 . (Take $\pi^2 = 10$)

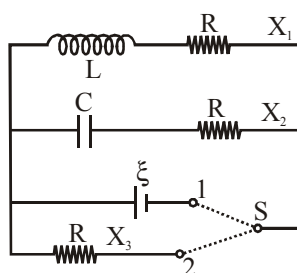
2. A small charged particle (charge q and mass m) is released at $t = 0$ inside a smooth hollow circular tube (it's like tube of wheel) placed inside a long solenoid (n turns per unit length). Cross sectional view of the arrangement is given. Axis of tube is coinciding with axis of solenoid. Radius of tube is very small i.e. r such that we can take magnetic field which is passing through the tube is same as at axis of solenoid. Cross-sectional area of tube is also very small such that charged particle can just move inside it. [Neglect the resistance of solenoid.] If the switch S is closed at $t = 0$. The speed of the particle after a long time is $\frac{\mu_0 n E q r}{k m R}$. Find the value of k .



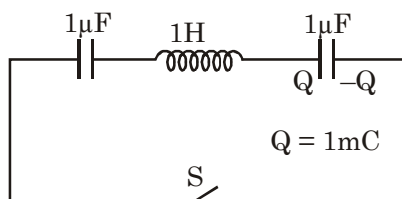
3. Two capacitors of capacitance C and $3C$ are charged to potential difference V_0 and $2V_0$ respectively and connected to an inductor of inductance L as shown in the figure. Initially the current in the inductor is zero. Now the switch S is closed. Find the maximum current (in mA) in the inductor. (Take $C = 12 \mu\text{F}$, $L = 1\text{H}$ & $V_0 = 1\text{V}$)



4. Figure shows a circuit containing three resistor X_1 , X_2 & X_3 having resistance R each, an inductor, capacitor and an emf source having inductance ' L ', capacitance C & emf ξ respectively (Given $R = \sqrt{\frac{L}{C}}$). The switch is first connected to position-1. When charge on capacitor becomes half of its maximum possible value then switch S is connected to position-2. Current in resistance X_3 just after shifting the switch from position-1 to position-2 is (given $L = 20\text{mH}$, $C = 2\mu\text{F}$, $R = 100\Omega$, $\xi = 5\text{V}$)



5. Two capacitors of equal capacity $1 \mu\text{F}$ each are joined in series with an inductor of inductance 0.5H . One of the capacitors has an initial charge 1mC and other is uncharged. Switch is closed at $t = 0$. Find maximum current (in A) through the inductor.

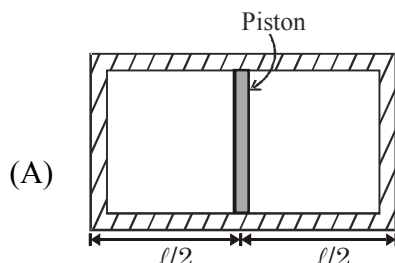


SECTION-IV

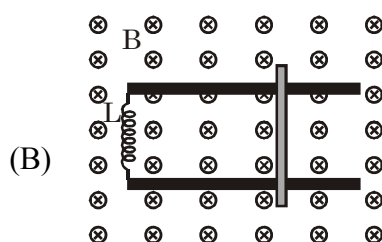
Matrix Match Type (4×5)

1. Match the following Questions.

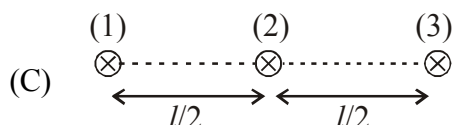
Column-I



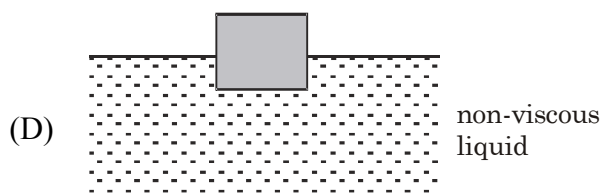
Thermally insulated container. Friction is absent. Helium gas is present on both sides of the thermally insulated piston. Piston is slightly displaced from equilibrium & released. Piston is A.



Rod & parallel rails are conducting. Uniform magnetic field B is pointing into the plane of paper. B is time independent. Friction is absent between rod and parallel rails. Inductor L is ideal. At equilibrium rod is given an impulse perpendicular to its length. Rod is A. (Neglect self inductance of rectangular loop)



Infinite long parallel current carrying wires are placed as shown. Wire (1) and (3) are fixed and wire (2) can move freely. Wire is slightly displaced from equilibrium & released. Wire 2 is A.



A cubical block floats in a non-viscous liquid as shown. Block is slightly displaced in vertical direction from equilibrium and released. Block is A.

Column-II

(P) Equilibrium of A can be unstable

(Q) Acceleration of A increases on moving away from equilibrium position

(R) Graph of kinetic energy of A as a function of position measured from equilibrium can be parabola.

(S) Motion of A can be oscillatory

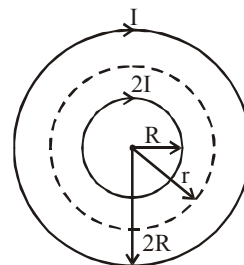
(T) Graph of velocity of A as a function of position measured from equilibrium can be ellipse.

SECTION-I

Single Correct Answer Type

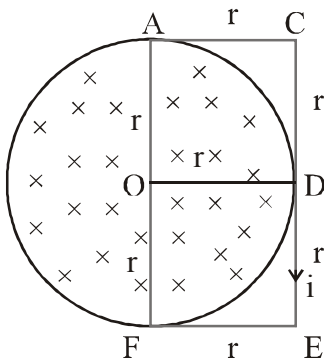
4 Q. [3 M (-1)]

1. A long solenoid contains another coaxial solenoid (whose radius R is half of its own). Their coils have same numbers of turns per unit length and initially both carry no current. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their directions are the same. As a result of the increasing current a charged particle initially at rest between the solenoids, starts moving along a circular trajectory. The radius of the circular trajectory is :

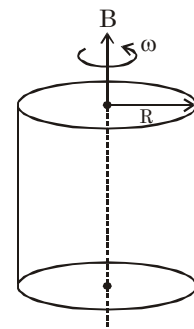


- (A) $\frac{5}{4}R$ (B) $\sqrt{2}R$ (C) $\frac{7}{6}R$ (D) $\sqrt{3}R$

2. There is a cylindrical region of variable magnetic field $\frac{dB}{dt} = \alpha$ having a wire frame inside as shown each part of the wire frame has resistance r . If $i_{DE} = \left\{ \frac{-me}{nr} \right\}$ then find the value of $(m+n)$. (Where e is the e.m.f. developed in the rod AC)



- (A) 3 (B) 4 (C) 5 (D) 7
3. A very long solid conducting cylinder of radius R is placed in a constant and uniform magnetic field which is parallel to the axis of the cylinder. Now the cylinder is rotated with a constant angular velocity ω as shown in the figure. Charge and mass of electron can be taken as e and m respectively.



- (A) The magnitude of electric field inside the cylinder as a function of radial

distance r is $\left| \omega B - \frac{m\omega^2}{e} \right| r$.

- (B) The potential difference between the axis of cylinder and circumference is $\frac{B\omega R^3}{2}$

- (C) The magnitude of volume charge density of the cylinder is $\omega \left| B - \frac{m\omega}{e} \right| \epsilon_0$.

- (D) The magnitude of volume charge density of the cylinder is $\omega \left| \frac{B}{2} - \frac{m\omega}{e} \right| \epsilon_0$.

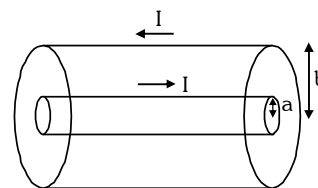
4. A long coaxial cable carries current I (along its surface). The current flows down the surface of inner cylinder of radius a and back along the outer cylinder of radius b .

(A) Energy stored in magnetic field per unit length is $\frac{\mu_0 I^2}{8\pi} \ln \frac{b}{a}$

(B) Energy stored in magnetic field per unit length is $\frac{\mu_0 I^2}{2\pi} \ln \frac{b}{a}$

(C) Self inductance of given arrangement per unit length is $\frac{\mu_0}{2\pi} \ln \left(\frac{b}{a} \right)$

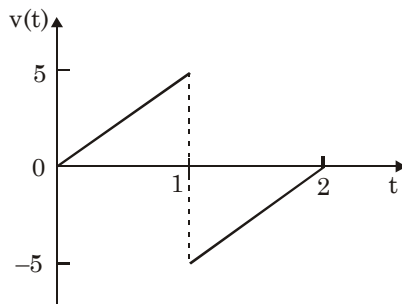
(D) Self inductance of given arrangement per unit length is $\frac{\mu_0}{4\pi} \ln \left(\frac{b}{a} \right)$



Multiple Correct Answer Type

1 Q. [4 M (-1)]

5. If the voltage waveform in figure is applied to a 10-mH inductor, find the inductor current. Assume $i(0) = 0$.



- (A) The current through the inductor at $t = 1$ seconds is 5×10^{-2} A
(B) The current through the inductor at $t = 2$ seconds is 0 A
(C) The current through the inductor at $t = 1$ seconds is 250 A
(D) The current through the inductor at $t = 2$ seconds is 5×10^{-3} A

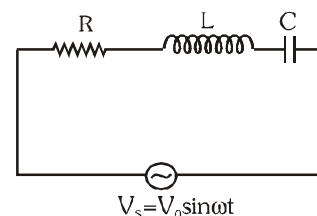
Linked Comprehension Type

(1 Para $\times$ 3Q.) (2 Para $\times$ 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 6 to 8

Many ac circuits used in practical electronic systems involve resistance, inductive reactance, and capacitive reactance. A simple example is a series circuit containing a resistor, an inductor, a capacitor, and an ac source, as shown in figure. To analyze this and similar circuits, we will use a phasor diagram that includes the voltage and the current phasors for each of the components. In this circuit, because of Kirchhoff's loop rule, the instantaneous total voltage V_{ad} across all the three components is equal to the source voltage at that instant. We will show that the phasor representing this total voltage is the vector sum of the phasors for the individual voltages.



Impedance of circuit $Z = \sqrt{R^2 + X^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$

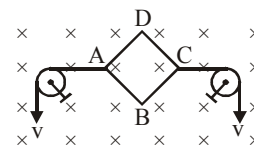
6. If ω is small then impedance will be mainly decided by

- (A) R and ωL (B) R and $\frac{1}{\omega C}$ (C) ωL and $\frac{1}{\omega C}$ (D) None of these

7. If ω is large then impedance will be mainly decided by
 (A) R and ωL (B) R and $\frac{1}{\omega C}$ (C) ωL and $\frac{1}{\omega C}$ (D) None of these
8. If $V_s = 200 \sin 100t$, $R = 500 \Omega$, $L = 10 \text{ H}$ and $C = 20 \mu\text{F}$, find the value of impedance of circuit.
 (A) 500Ω (B) $500\sqrt{2} \Omega$ (C) 1000Ω (D) 825Ω

Paragraph for Question no. 9 and 10

Figure shows a metallic square frame of edge a and total resistance R . A uniform magnetic field B exists in the space in the direction perpendicular to the plane of paper. Two strings are attached to the opposite corners of the square. The strings start moving with constant speed v at $t = 0$ and square starts to deform into a rhombus.



9. Find the induced emf in the frame at the instant when the angles at corners A and C reduce to " 2θ " :-
 (A) $2Bav \left(\frac{\cos 2\theta}{\sin \theta} \right)$ (B) $Bav \left(\frac{\cos 2\theta}{\sin \theta} \right)$ (C) $\frac{1}{2} Bav \left(\frac{\sin \theta}{\cos 2\theta} \right)$ (D) $\frac{1}{4} Bav \left(\frac{\sin \theta}{\cos 2\theta} \right)$
10. Find the value of magnitude of magnetic force on side DC when angles at corners A and C reduce to 60° :-
 (A) $\frac{B^2 a^2 v}{R}$ (B) $\frac{2B^2 a^2 v}{R}$ (C) $\frac{\sqrt{3}}{2} \frac{B^2 a^2 v}{R}$ (D) $\frac{B^2 a^2 v}{2R}$

Paragraph for Question No. 11 and 12

A smooth insulating ring of radius R , with a bead of mass m and charge q is placed horizontally and in a uniform magnetic field of strength B_0 and perpendicular to the ring plane. Starting from $t = 0$, the magnetic field is changed to $B(t) = B_0 + \alpha t$, where α is a positive constant & t is time. Neglect gravity.

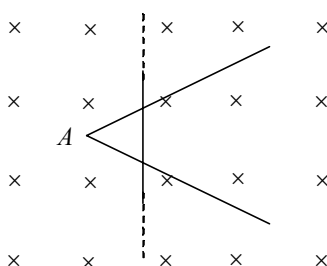
11. The contact force between the ring and bead as a function of time t is :
 (A) $\frac{\alpha q^2 R t}{4m} (2B_0 + \alpha t)$ (B) $\frac{\alpha q^2 R t}{2m} (B_0 + \alpha t)$ (C) $\frac{\alpha q^2 R t}{m} (B_0 + \alpha t)$ (D) $\frac{qR}{2m} (2B_0 + \alpha t)$
12. The velocity of bead as function of time t
 (A) $\frac{qR}{m} \alpha t$ (B) $\frac{qR}{2m} \alpha t$ (C) $\frac{2qR}{m} \alpha t$ (D) $\frac{qR}{2m} (B_0 + \alpha t)$

SECTION-III

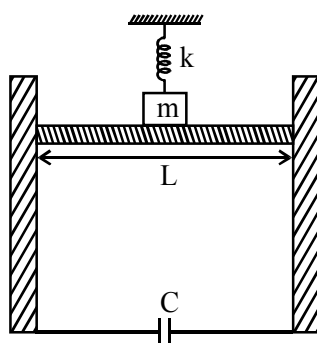
Numerical Grid Type (Ranging from 0 to 9)

6 Q. [4 M (0)]

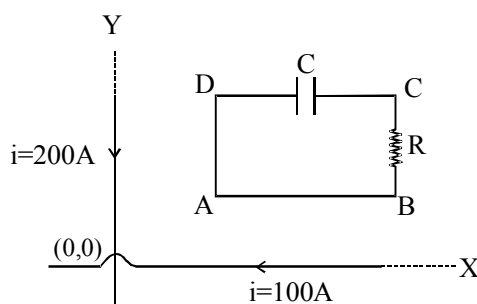
1. A long wire bend into the shape of a right angle is held stationary on a horizontal frictionless plane. A very long rod of mass 1 kg initially starts with velocity $v_0 = 4 \text{ m/s}$ from the apex A of the bend wire. The resistance per unit length of the wire and the rod is $(\sqrt{2} - 1) \times 10^{-2} \Omega/\text{m}$. The whole arrangement is put in a region of uniform magnetic field of 0.05 T directed normally into the horizontal plane. Find the distance travelled by the rod before it comes to rest.



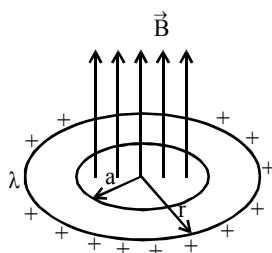
2. A block of mass 300 g is attached to the ceiling by a spring that has a force constant $k = 200 \text{ N/m}$. A conducting massless rod is rigidly attached to the block and can slide without friction along two vertical parallel rails which are at a distance $L = 1 \text{ m}$ apart. A capacitor of known capacitance $C = 500 \mu\text{F}$ is attached to the rails by the wire and the entire system is kept in magnetic field $B = 20 \text{ T}$ as shown in figure. Neglect the self inductance and electrical resistance of all wire and rod. If ' ω ' is angular frequency (in rad/sec) of vertical oscillation of block then $\frac{\omega}{10}$ is equal to :-



3. Two straight infinitely long current wires lying along the x-axis and the y-axis carry currents 100 A and 200 A as indicated in the adjoining figure. A rigid square metallic wire frame ABCD of side 10 cm and resistance ($R = 10 \Omega$) is connected to a capacitor $C_1 = 1 \text{ mF}$ as shown in the figure. The wire frame, which lies in the x-y plane, moves with a constant velocity $30 (\hat{i} + \hat{j}) \text{ m/s}$. At the instant, when the point A is located at (20 cm, 20 cm) the charge q_0 on the capacitor (as shown in the figure) is $0.2 \mu\text{C}$. If, at the above mentioned instant, the current flowing in the circuit is given by 10^{-n} amp . Fill the value of n in nearest integer.



4. A non conducting ring of mass 4 kg is uniformly charged with $\lambda = 4 \text{ C/m}$ and kept on rough horizontal surface with friction coefficient $\mu = \pi/4$. A time varying magnetic field $B = B_0 t^2$ is applied in a circular region of radius a ($a < r$) perpendicular to the plane of ring as shown in figure. Find out the time (in sec) when ring just starts to rotate on surface. (take $a = 5 \text{ cm}$ and $g = 10 \text{ m/s}^2$, $B_0 = 125 \text{ SI Unit}$).



5. A superconducting round ring of radius $a = 20 \text{ cm}$ and inductance $L = 0.1 \text{ H}$ was located in a uniform magnetic field of induction $B = 10 \text{ T}$. The plane of ring was parallel to the vector B , and the current in the ring was equal to zero. Then the ring was turned through 90° so that its plane became perpendicular to the field. The work (in J) performed during the turn is (Assume $\pi^2 = 10$) :-

6. A long single layer tightly wound solenoid with a wire of resistivity $\frac{\pi}{2} \times 10^{-8}$ ohm-m has 1000 turns per meter. The thickness of wire insulation is negligible. The radius of solenoid is 5 cm. If the phase difference between current and alternating voltage applied to the solenoid with frequency 100 Hz is $\frac{\pi}{x}$, find x (Take $\pi^2 = 10$).

SECTION-IV

Matrix Match Type (4 × 5)

1. Match the column

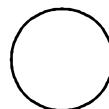
Column-I

- (A) The pressure on walls is tending to contract them.
- (B) The pressure on walls is tending to expand them
- (C) The energy/volume inside the walls is more than outside.
- (D) The energy/volume outside the walls is more than inside.

1 Q. [8 M (for each entry +2(0))]

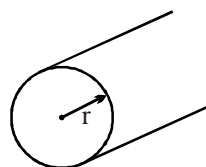
Column-II

- (P) A soap bubble present in air containing air. The temperature inside and outside air is the same.



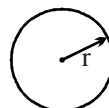
Energy : internal energy of air.

- (Q) Current flowing in a thin hollow pipe of radius r .



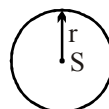
Energy : Magnetic field energy.

- (R) A hollow conducting shell with a charge on it.



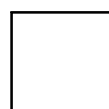
Energy : electrostatic field energy.

- (S) Point source of light is present at the centre of a spherical shell with both surfaces perfectly black.



Energy : light energy.

- (T) A cubical container has adiabatic walls. The temperature of air inside is less than the temperature of air outside. The density of air is same on both sides.



Energy : internal energy of air

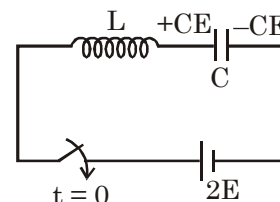
SECTION-I

Single Correct Answer Type

10 Q. [3 M (-1)]

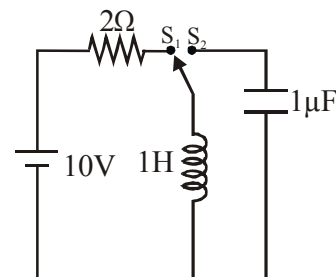
1. Maximum charge on capacitor after switch is closed :-

(A) $2CE$
(B) $4CE$
(C) $3CE$
(D) None of these



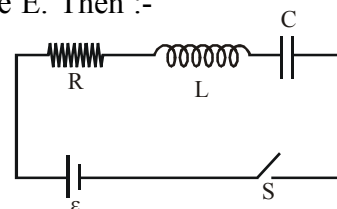
2. For the circuit shown, the switch was closed at switch S_1 for a long time till steady state condition reached. At time $t = 0$, the switch S_1 is opened and S_2 is closed. The current through inductor as a function of time will be

(A) $5 \cos 100 t$
(B) $10 \cos 100 t$
(C) $5 \cos 1000 t$
(D) $10 \cos 1000 t$



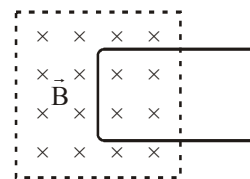
3. A resistor of resistance R , an inductor of inductance L and a capacitor of capacitance C are connected in series and the combination is connected to a battery of emf ε as shown. At $t = 0$, the switch S is closed. Let, after a very long time, the thermal energy developed in circuit be Q , energy stored in circuit be U and energy supplied by the battery be E . Then :-

(A) $Q : U : E = 1 : 1 : 2$
(B) $Q : U : E = 1 : 3 : 4$
(C) $Q : U : E = 3 : 1 : 4$
(D) $Q : U : E = 1 : 2 : 3$

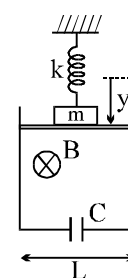


4. A conducting loop is halfway into a magnetic field. Suppose the magnetic field begins to increase rapidly in strength. What happens to the loop ?

(A) The loop is pushed upward, toward the top of the page.
(B) The loop is pushed downward, toward the bottom of the page.
(C) The loop is pulled to the left, into the magnetic field.
(D) The loop is pushed to the right, out of the magnetic field.



5. A block is attached to the ceiling by a spring that has a force constant $k = 200 \text{ N/m}$. A conducting rod is rigidly attached to the block. The combined mass of the block and the rod is $m = 0.3 \text{ kg}$. The rod can slide without friction along two vertical parallel rails, which are a distance $L = 1 \text{ m}$ apart. A capacitor of known capacitance $C = 500 \mu\text{F}$ is attached to the rails by the wires. The entire system is placed in a uniform magnetic field $B = 20 \text{ T}$ directed as shown. Find the angular frequency (in rad/sec) of the vertical oscillations of the block. Neglect the self-inductance and electrical resistance of the rod and all wires.

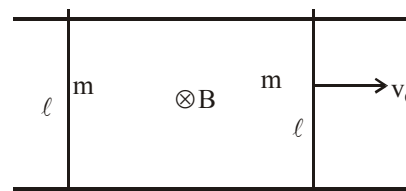


(A) $20\sqrt{5}$ (B) 20 (C) $\frac{20\sqrt{5}}{3}$ (D) none

6. The magnetic field in a region is given by $\vec{B} = \vec{k} \frac{B_0}{L} y^2$ where L is a fixed length. A conducting rod of length L lies along the Y -axis between the origin and the point $(0, L, 0)$. If the rod moves with a velocity $\vec{V} = V_0 \hat{i}$, what is the emf induced between the ends of the rod?

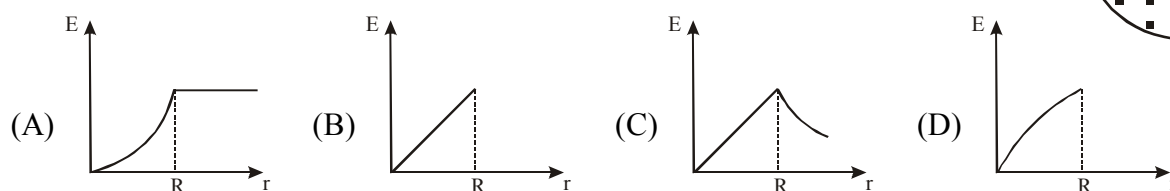
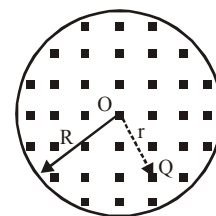
(A) $\frac{B_0 V_0 \ell^2}{3}$ (B) $\frac{B_0 V_0 \ell^2}{2}$ (C) $\frac{2B_0 V_0 \ell^2}{3}$ (D) $\frac{B_0 V_0 \ell^2}{6}$

7. Two parallel conducting rails, separated by a distance ℓ , are placed horizontally in a region of uniform magnetic field $\vec{B}$, perpendicular to the plane of the rails as shown in the figure. Two conducting wires, each of length ℓ are placed so as to slide on parallel conducting rails without friction. One of the wires is given a velocity v_0 parallel to the rails. Loss in kinetic energy of the system till the steady state is achieved will be :



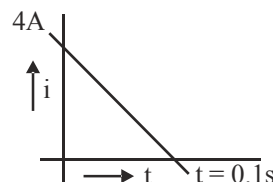
(A) 0 (B) $\frac{3}{4}mv_0^2$ (C) $\frac{3}{8}mv_0^2$ (D) $\frac{1}{4}mv_0^2$

8. A uniform magnetic field exists in cylindrical region of radius R with axis passing through O . The field is made to vary at a constant rate. The variation of the induced electric field E with distance r from the axis is given by :



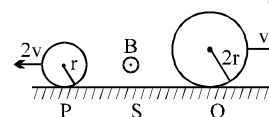
9. Some magnetic flux is changed from a coil of resistance 10Ω . As a result an induced current is developed in it which varies with time as shown in figure. The magnitude of change in flux through the coil in weber is :-

(A) 2 (B) 4
(C) 6 (D) None of these



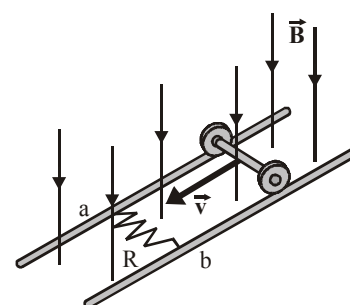
10. Two conducting rings P and Q of radii r and $2r$ rotate uniformly in opposite directions with centre of mass velocities $2v$ and v respectively on a conducting surface S . There is a uniform magnetic field of magnitude B perpendicular to the plane of the rings. The potential difference between the highest points of the two rings is

(A) 0 (B) $4Bvr$
(C) $8Bvr$ (D) $16Bvr$



Multiple Correct Answer Type

11. In figure the rolling axle, of length ℓ is pushed along horizontal rails at a constant speed v . A resistor R is connected to the rails at points a and b , directly opposite each other. The wheels make good electrical contact with the rails, so the axle, rails, and R form a closed-loop circuit. The only significant resistance in the circuit is R . A uniform magnetic field B is vertically downward. Mark the **CORRECT** statement(s)



4 Q. [4 M (-1)]

(A) The induced current I in the resistor is $\frac{B\ell v}{R}$.

(B) Horizontal force F is required to keep the axle rolling at constant speed is $\frac{B^2 \ell^2 v^2}{R}$.

(C) End of the resistor, a is at the higher electric potential than b .

(D) After the axle rolls past the resistor, the current in R reverse direction.

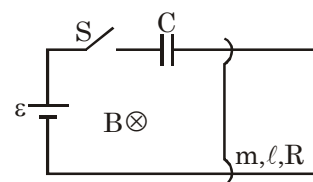
12. In the figure shown a conducting rod of mass m , length ' ℓ ' and resistance ' R ' can smoothly move along parallel rails in horizontal plane. Initially the rod is at rest. A uniform magnetic field $\vec{B}$ perpendicular to the plane of motion exists in the region. Now switch S is closed at $t = 0$, then :

(A) The charge on the capacitor in steady state is $m c \epsilon / (m + c B^2 \ell^2)$

(B) The charge on the capacitor in steady state is $2 m c \epsilon / (m + c B^2 \ell^2)$

(C) Velocity of conducting rod in steady state is $2 B \ell c \epsilon / (m + c B^2 \ell^2)$

(D) Velocity of conducting rod in steady state is $B \ell c \epsilon / (m + c B^2 \ell^2)$



13. A massless frame is present in uniform magnetic field and a block of mass m hangs on the frame as shown in figure. When

a constant current $I > \frac{mg}{\ell B}$ is maintained in the frame, it gets

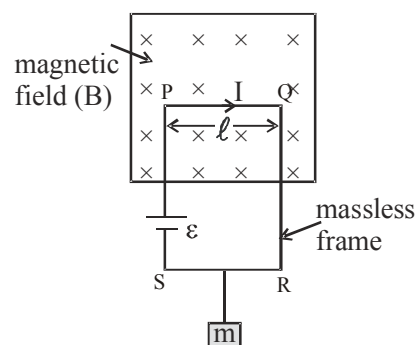
displaced by ' h ' in some time interval. If ϵ is the emf of battery then which of the following is/are **CORRECT** ?

(A) Work done by magnetic force is $I \ell B h$

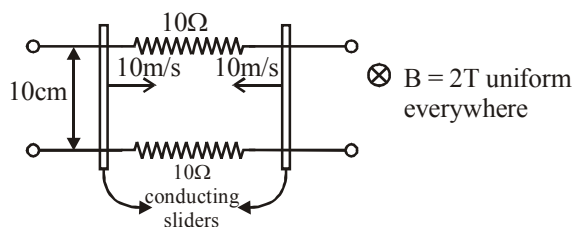
(B) Work done by battery is $I \ell B h$

(C) Velocity of block when it gets displacement by h is $\sqrt{\frac{2(I \ell B - mg)h}{m}}$

(D) Force on charges in the segment PQ while the frame moves up is in the vertical direction



14. The circuit below shows two parallel rails separated by distance of 10 cm. The rails has 10Ω resistor each at its middle. The region of space contains magnetic field which is uniform throughout the space. There are two conducting wires on the parallel rails moving towards each other with speed of 10 m/s.



(A) Current in the circuit is 0.2 A.

(B) Power lost in the circuit is 0.8 W.

(C) The electric field inside the wires is non-conservative in nature.

(D) The electric field inside the wires is conservative in nature.

Matching list based comprehension Type ($4 \times 4 \times 4$)

1 Table $\times$ 3 Q. [3(-1)]

Single option correct

(Three Columns and Four Rows)

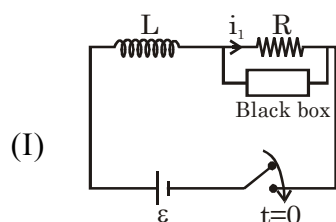
Answer Q.15, Q.16 and Q.17 by appropriately matching the information given in the three columns of the following table.

Column-1 gives various circuit diagram with a black box which can contain either inductor or resistor. Column-2 lists time constant for the various circuit while column-3 lists the value of current (i_1) at either $t = 0$ or $t \rightarrow \infty$. Initially before switching on inductor does not carry any energy.

Column-1

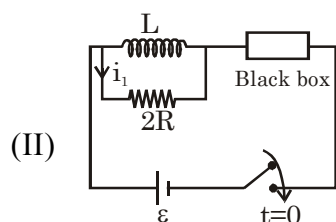
Column-2

Column-3



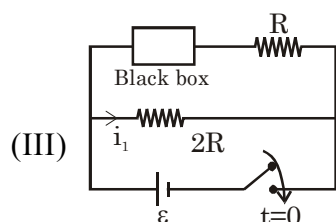
(i) $\frac{L}{2R}$

(P) $i_1 = 0$ at $t = 0$



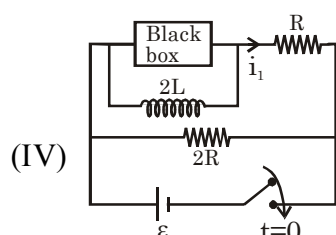
(ii) $\frac{2L}{3R}$

(Q) $i_1 = \frac{\varepsilon}{R}$ at $t \rightarrow \infty$



(iii) $\frac{3L}{R}$

(R) $i_1 = \frac{\varepsilon}{2R}$ at $t = 0$



(iv) $\frac{3L}{2R}$

(S) $i_1 = \frac{\varepsilon}{3R}$ at $t = 0$

15. If the black box consists of $\frac{2R}{\text{resistor}}$, then correct matching will be :

- (A) (I) (iv) (Q) (B) (II) (i) (P) (C) (I) (iv) (R) (D) (IV) (iii) (P)

16. If the black box consists of $\frac{R}{\text{resistor}}$, then correct matching will be :

- (A) (I) (ii) (P) (B) (IV) (i) (Q) (C) (II) (iv) (S) (D) (II) (iii) (R)

17. If the black box consists of _____ (connecting wire), then correct matching will be :

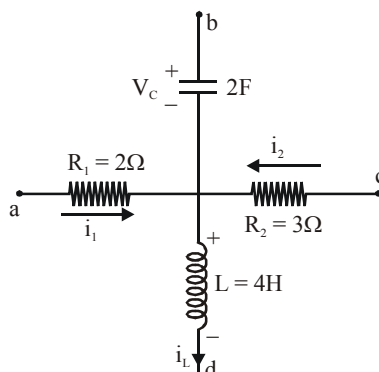
- (A) (II) (iii) (Q) (B) (II) (i) (R) (C) (II) (i) (P) (D) (II) (ii) (Q)

SECTION-III

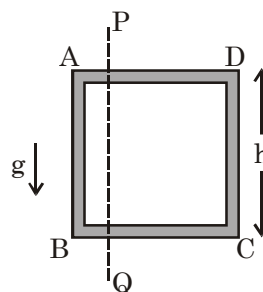
Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. In the figure shown $i_1 = 10e^{-2t}$ A, $i_2 = 4$ A and $V_c = 3e^{-2t}$ V. If $V_L = ae^{-bt}$ then find $\left| \frac{a}{b} \right|$:-



2. ABCD is a square frame of conductor of electrical resistivity ρ . The frame lies in a vertical plane. PQ is an imaginary boundary separating space into two parts. Left of PQ, a uniform gravitational field $\vec{g}$ exists (figure) whereas no gravitational field is present right of PQ. The electrical potential difference between A and B will be $\frac{k}{4} \frac{mgh}{e}$. e is charge on an electron and m is mass of electron. Find k .



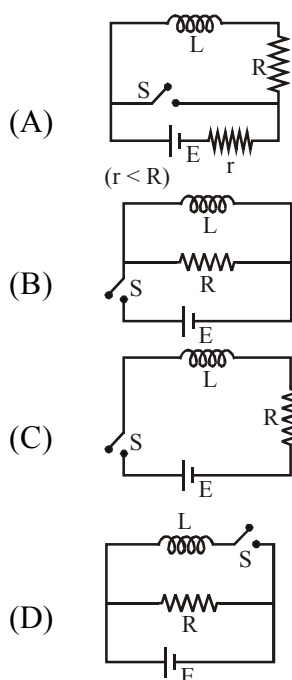
SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. In column-I some circuit are given. In all the circuits except in (A) switch S remains closed for long time and then it is opened at $t = 0$ while for (A), the situation is reversed. Column-II tells something about the circuit quantities. Match the entries of column-I with the entries of column-II.

Column-I



Column-II

- (P) Voltage across inductor can be greater than E at $t = 0$.
- (Q) Voltage across inductor would be less than E at $t = 0$.
- (R) After long time, energy stored in inductor is zero.
- (S) After long time, energy stored in inductor is non-zero.
- (T) Voltage across inductor increases as time progress.

SECTION-I

Single Correct Answer Type

4 Q. [3 M (–1)]

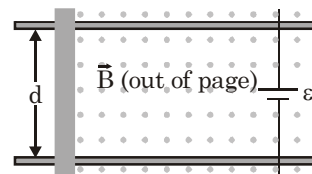
1. A bar of mass m , length d and resistance R slides without friction in a horizontal plane, moving on parallel rails as shown in figure. A battery that maintains a constant emf ε is connected between the rails, and a constant magnetic field $\vec{B}$ is directed perpendicularly to the plane of the page. Assuming the bar starts from rest, Find speed at time t :-

(A) $v = \frac{\varepsilon}{Bd} \left(1 - e^{-\frac{B^2 d^2 t}{mR}} \right)$

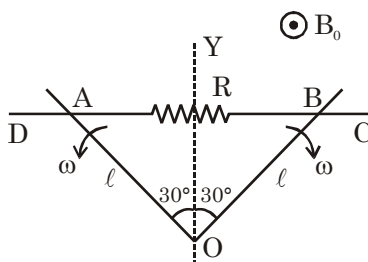
(B) $v = \frac{\varepsilon}{Bd} \left(1 - e^{-\frac{B^2 d^2 t}{2mR}} \right)$

(C) $v = \frac{\varepsilon}{Bd} \left(1 - e^{-B^2 d^2 t / mR} \right)$

(D) $v = \frac{\varepsilon}{Bd} \left(1 - e^{-\frac{Bdt}{mR}} \right)$



2. In the shown figure, uniform magnetic field B_0 is pointing out of the plane in the region. Wire CD is fixed and has resistance R , while OA and OB are conducting wires rotating with angular velocity ω about O as shown. If at some instant $OA = OB = \ell$ and each wire makes angle $\theta = 30^\circ$ with y-axis, then current through resistance R is



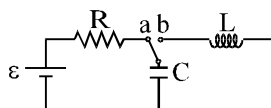
(A) $\frac{B_0 \omega \ell^2}{R}$

(B) $\frac{\sqrt{3} B_0 \omega \ell^2}{2R}$

(C) $\frac{\sqrt{3} B_0 \omega \ell^2}{4R}$

(D) Zero

3. The switch in the circuit pictured is in position a for a long time. At $t = 0$ the switch is moved from a to b. The energy in inductor will becomes half of its maximum value in a minimum time t equals



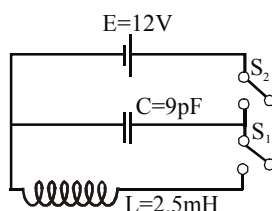
(A) $2\pi\sqrt{LC}$

(B) $\frac{\pi}{4}\sqrt{LC}$

(C) $\frac{\pi}{2}\sqrt{LC}$

(D) $\pi\sqrt{LC}$

4. In the circuit shown, the capacitor is initially charged with a 12 V battery, when switch S_1 is open and switch S_2 is closed. S_1 is then closed and, at the same time, S_2 is opened. The maximum value of current in the circuit is



(A) 0.38 mA

(B) 0.84 mA

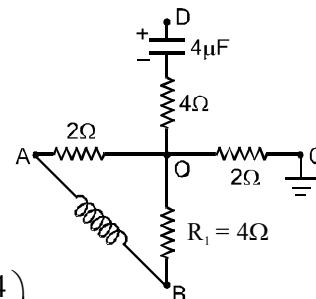
(C) 0.72 mA

(D) 0.1 mA

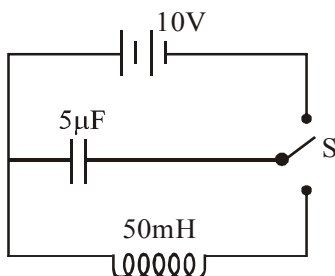
Multiple Correct Answer Type

3 Q. [4 M (-1)]

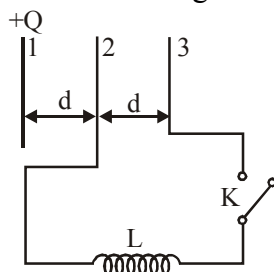
5. Figure shows a part of a circuit. At a particular instant of time the potentials of points 'A', 'B' and 'D' are 8V, 16V and 4V respectively and voltage across the capacitor at that instant is 4V with polarity as shown. Choose the **CORRECT** statement(s) :-



- (A) The potential of point O at the given instant is $\left(\frac{16}{3}\right)$ V
- (B) The charge on the capacitor at the given instant is decreasing at a rate $\left(\frac{4}{3}\right)$ C/s
- (C) The charge on the capacitor at the given instant is increasing at a rate $\left(\frac{4}{3}\right)$ C/s
- (D) The electric current at the given instant of time in resistance R_1 is flowing from O to B having magnitude $\left(\frac{8}{3}\right)$ A
6. In the circuit shown, switch S is first pushed up to charge the capacitor. Switch S is then pushed down.



- (A) Maximum value of energy stored in the inductor is 250×10^{-6} J
- (B) Maximum value of energy stored in the capacitor is 250×10^{-6} J
- (C) Maximum current in the circuit is 100 mA
- (D) Maximum charge on the capacitor is 5 μC
7. Three identical fixed metal plates are arranged in the air at equal distances d from each other. The area of each of the plates equals A . The plate 1 has charge $+Q$, while plates 2 and 3 are not charged and are connected through a key K to the inductor L . The distance between the plates is small compared with their size. The ohmic resistance of the coil can be neglected :-



- (A) Just before the switch K is closed, the charge on right surface of plate 2 is $Q/2$
- (B) Just before the switch K is closed, the charge on right surface of plate 3 is $Q/2$
- (C) After the switch K is closed, the maximum current in the circuit is $\frac{Q}{2} \sqrt{\frac{d}{L\epsilon_0 A}}$
- (D) After the switch K is closed, the maximum current in the circuit is $Q \sqrt{\frac{d}{L\epsilon_0 A}}$

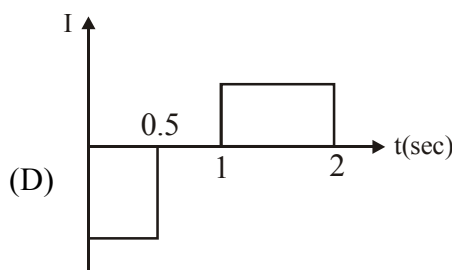
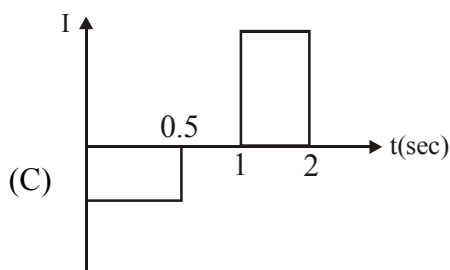
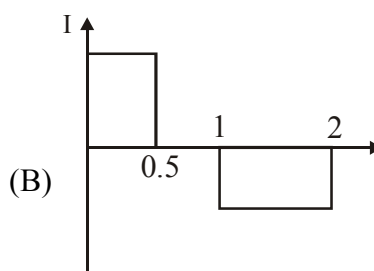
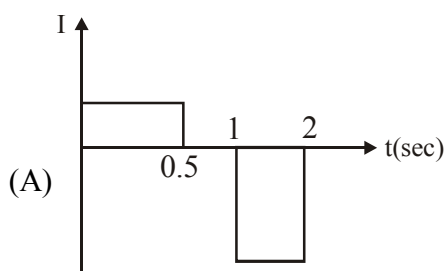
Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Question no. 08 and 9

A circular coil of wire has a radius of 0.5m and a total resistance of 1.57Ω . The coil lies in the x-y plane. The coil is in uniform magnetic field $\vec{B}$ that is in $-z$ direction, which is directed away from you as you view the coil. The magnitude B of the field depends on time as follows : It increases at a constant rate from 0 at $t = 0$ to 0.8 T at $t = 0.5$ sec, is constant at 0.8T from $t = 0.5$ sec to $t = 1$ sec and decreases at a constant rate from 0.8T at $t = 1$ sec to 0 at $t = 2$ sec.

8. Draw a graph of the current I induced in the coil versus t for 0 to 2 sec. Let counter clockwise currents be positive and clockwise currents be negative.



9. What is the maximum induced electric field magnitude in the coil during 0 to 2 sec interval :-

(A) 0.4 V/m (B) 0.2 V/m (C) 0.6 V/m (D) 8 V/m

Matching list based comprehension Type (4 × 4 × 4)

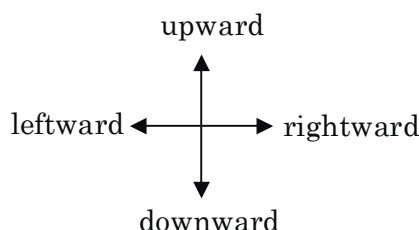
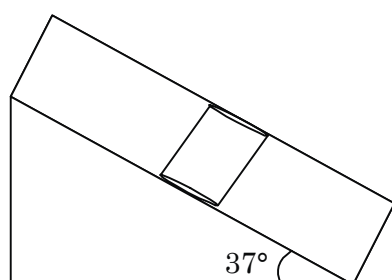
1 Table × 3 Q. [3(-1)]

Single option correct

(Three Columns and Four Rows)

Answer Q.10, 11 & 12 by appropriately matching the information given in the three columns of the following table.

Figure shows a solid wooden cylinder of mass $m = 0.250$ kg and length $L = 0.100$ m with $N = 10$ turns of wire wrapped around it longitudinally, so that the plane of wire coil contains the long central axis of the cylinder. The cylinder is released on a plane inclined at an angle θ to the horizontal, with the plane of the coil parallel to the inclined plane. A magnetic field B of different magnitudes and directions as per column-1 is applied in the region. The value of current through the coil is I and such that the direction of field in all cases produced by coil makes an acute angle with vertically upward direction. There is enough friction to ensure that the cylinder rolls without slipping. (take $g = 10 \text{ m/s}^2$)



Answer the questions given below on the basis of given matrix.

Column-1

(B)

(I) 0 T

(II) 0.5 T upward

(III) 0.5 T downward

(IV) 0.5 T leftward

Column-2

(I)

(i) 0 Amp

(ii) 1 Amp

(iii) 15/8 Amp

(iv) 5/2 Amp

Column-3

(Acceleration of C.M of cylinder)

(P) 0 m/s²

(Q) $\frac{2g}{15}$ m/s²

(R) $\frac{4g}{5}$ m/s²

(S) $\frac{5.6g}{3}$ m/s²

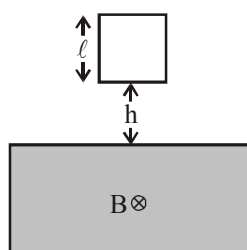
10. In which case among the following cases will the cylinder remain stationary
(A) (II) (iii) (P) (B) (III) (ii) (P) (C) (II) (iv) (P) (D) (III) (iv) (P)
11. In which case will the induced emf rise at the maximum rate initially ?
(A) (III) (iv) (R) (B) (III) (iii) (S) (C) (III) (iv) (S) (D) (II) (iv) (Q)
12. Among the given situations, when will the friction be a maximum ?
(A) (IV) (iii) (P) (B) (III) (iv) (S) (C) (II) (iii) (S) (D) (IV) (iv) (Q)

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1Q.[3(0)]

1. A square loop of side length ℓ ($= 25$ cm) falls into a finite and uniform magnetic field ($B = 0.5$ T) confined to a region as shown. Find the height 'h' (in m) through which the loop should be dropped, so that its velocity does not change during the period it enters the magnetic field. Assume mass of loop $m = 25$ gm, the resistance $R = 2.5 \Omega$ and $g = 10$ m/s².



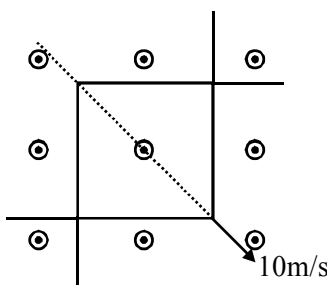
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

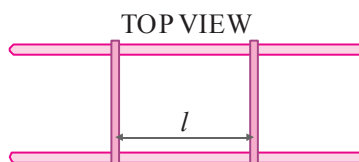
5 Q. [4 M (0)]

1. The L-shaped conductor as shown in figure moves with 10 m/s across a stationary L-shaped conductor in a 0.10 T magnetic field. The two vertices overlap so that the enclosed area is zero at time $t = 0$. The conductor has resistance of 0.010 ohms per meter. If current at time $t = 0.10$ sec. is x amp, find the value

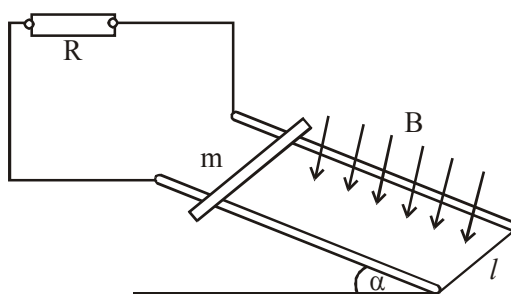
of $\left(\frac{x}{25}\right)^2$.



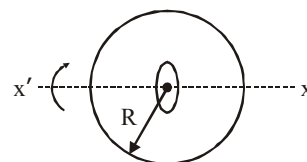
2. Two identical conducting bars rest on two horizontal parallel conducting rails. The bars are perpendicular to the rails and parallel to each other as shown. The distance between the bars is l . At a certain moment, a uniform vertical upward magnetic field is turned on. The field quickly reaches its maximum magnitude and then remains constant. Assume that the resistance of each bar is much greater than the resistance of the rails. Neglecting friction, find the new distance 'd' between the bars. If the new distance comes out to be $\frac{l}{\alpha}$; fill α in OMR sheet.



3. A homogeneous field of magnetic induction $\mathbf{B}$ is perpendicular to a track of gauge ℓ which is inclined at an angle $\alpha = 30^\circ$ to the horizontal. A frictionless conducting rod of mass m straddles the two rails of the track as shown in the figure. If maximum speed of the conductor is $= \frac{mgR}{nB^2\ell^2}$. Then find n

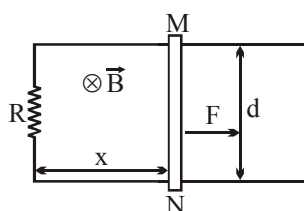


4. A non-conducting ring of radius R having uniformly distributed charge Q starts rotating about $x-x'$ axis passing through diameter with an angular acceleration α as shown in the figure. Another small conducting ring having radius a ($a \ll R$) is kept fixed at the centre of bigger ring in such a way that axis xx' is passing through its centre and perpendicular to its plane. If the resistance of small ring is $r = 1\Omega$, find the induced current in it in amperes.



(Given $q = \frac{16 \times 10^2}{\mu_0} \text{ C}$, $R = 1 \text{ m}$, $a = 0.1 \text{ m}$, $\alpha = 8 \text{ rad/s}^2$)

5. Two long parallel rails, a distance $d = 1 \text{ m}$ apart and each having a resistance per unit length $\lambda = 2\Omega/\text{m}$ are joined at one end by a resistance $R = 1\Omega$. A perfectly conducting rod MN of mass $m = 1 \text{ kg}$ is free to slide along the rails without friction. There is a uniform magnetic field of induction $B = 1 \text{ T}$ normal to the plane of the paper and directed into the paper. A variable force F is applied to the rod MN such that, as the rod moves, a constant current $i = 1 \text{ Amp}$. flows through R . Find the applied force F (in N) at a distance $x = 1 \text{ m}$ of the rod from R .

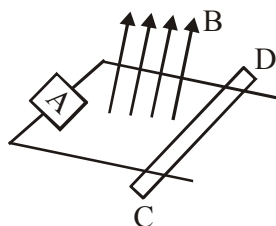


SECTION-IV

Matrix Match Type (4×5)

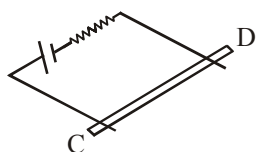
1 Q. [8 M (for each entry +2(0))]

1. A homogeneous magnetic field B is perpendicular to a sufficiently long track of width ℓ which is horizontal. A frictionless rod of mass m straddles the two rail of the track as shown in the figure. Entire arrangement lies in horizontal plane. For the situation suggested in column-II match the appropriate entries in column-I. The rails are also resistanceless.

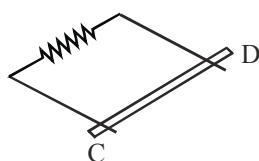


Column-I

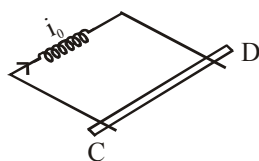
- (A) A is a battery of emf V and internal resistance R . The conducting rod is initially at rest.



- (B) A is a resistance. The non-conducting rod is projected to the right with a velocity V_0



- (C) A is an inductor having with initial current i_0 . It is having no resistance and CD is a conducting rod.



- (D) A is a resistance. The conducting rod is projected to the right with a velocity V_0 .

Column-II

- (P) Energy is dissipated during the motion.

- (Q) The rod moves with a constant velocity after a long time.

- (R) After a certain time interval rod will change its direction of motion.

- (S) If a constant force is applied on the rod to the right, it can move with a constant velocity.
(T) The rod stops after some time in absence of an external force.

SECTION-I

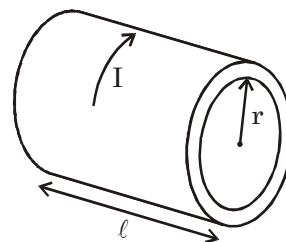
Single Correct Answer Type

4 Q. [3 M (-1)]

1. A capacitor and a coil in series are connected to a 6 volt ac source. By varying the frequency of the source, maximum current of 600 mA observed. If the same coil is now connected to a cell of emf 6 volt and internal resistance of 2Ω , the steady-state current through it will be :-

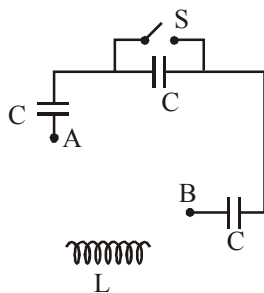
(A) 0.5 A (B) 0.6 A (C) 1.0 A (D) 2.0 A

2. A hollow cylinder has length ℓ , radius r , and thickness d , where $\ell \gg r \gg d$, and is made of a material with resistivity ρ . A time-varying current I flows through the cylinder in the tangential direction. Assume the current is always uniformly distributed along the length of the cylinder. The cylinder is fixed so that it cannot move; assume that there are no externally generated magnetic fields during the time considered for the problems below. Assume current at $t = 0$ to be I_0 . What is current $I(t)$ for $t > 0$



(A) $I = I_0 e^{-\frac{\rho}{2\mu_0 r d} t}$ (B) $I = I_0 e^{-\frac{2\rho}{\mu_0 r d} t}$ (C) $I = I_0 e^{-\frac{2\mu_0 r d}{\rho} t}$ (D) $I = I_0 e^{-\frac{\mu_0 r d}{2\rho} t}$

3. Three capacitors of capacitance C each are connected in series as shown in the figure. Initially switch S is open. Now capacitors are charged by a battery of emf V by connecting between terminal A and B. After long time battery is disconnected and inductor of inductance L is connected between A and B at time $t = 0$ so that an oscillatory circuit is formed. Now at an instant t_0 switch S is closed, then find the amplitude of charge oscillations of the remaining capacitors.



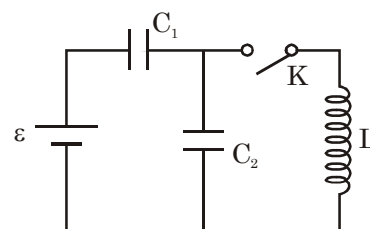
(A) $CV \sqrt{\frac{1}{6} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$

(B) $\frac{CV}{3} \sqrt{\frac{1}{5} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$

(C) $CV \sqrt{\frac{1}{4} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$

(D) $\frac{CV}{3}$

4. Consider the electrical circuit given below; initially chargeless capacitors C_1 and C_2 were connected to a battery and at certain moment, the key K will be closed. After that moment, current and voltage will start oscillating. For these oscillations :

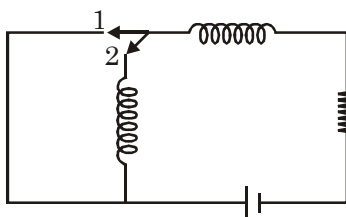


- (A) When charge on C_2 is maximum, current through C_1 is same as current in L.
(B) When charge on C_1 is maximum, power delivered by battery is positive.
(C) Charge on C_2 is maximum when current through L is maximum.
(D) None of these

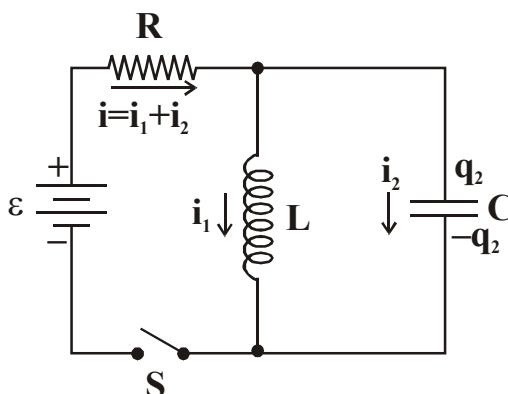
Multiple Correct Answer Type

7 Q. [4 M (-1)]

5. In the circuit shown, switch can be placed in position 1 or 2.



- (A) If switch is in position 1 for a very -2 long time and then shifted to 2 at $t = 0$ then current will grow with time for $t > 0$
(B) If switch is in position 1 for a very -2 long time and then shifted to 2 at $t = 0$ then current will decay with time for $t > 0$
(C) If switch is in position 2 for a very -2 long time and then shifted to 1 at $t = 0$ then current will decay with time for $t > 0$
(D) If switch is in position 2 for a very -2 long time and then shifted to 1 at $t = 0$ then current will not change for $t > 0$
6. Consider the circuit shown in figure. At time $t = 0$, switch S is closed. The current through the inductor is i_1 , the current through the capacitor branch is i_2 , and the charge on the capacitor is q_2 . Then choose the **CORRECT** statements :-



- (A) At any instant according to Kirchoff rule, $R(i_1 + i_2) + L \left(\frac{di_1}{dt} \right) = \varepsilon$
(B) At any instant according to Kirchoff rule, $R(i_1 + i_2) - \frac{q_2}{C} = \varepsilon$
(C) Initial values of i_1 & i_2 will be zero and $\frac{\varepsilon}{R}$ respectively
(D) Initial values of i_1 & i_2 will be $\frac{\varepsilon}{R}$ and zero respectively

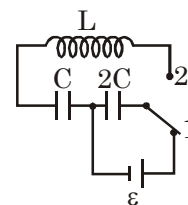
7. In the circuit shown in the figure, the switch is initially at position 1 for a long time. Now, at $t = 0$, the switch is turned to position 2. Initially charge on capacitor 'C' is zero. Choose the **CORRECT** statements:-
(A) After the switch is thrown to position 2, the circuit performs LC oscillation

with angular frequency $\sqrt{\frac{3}{2LC}}$

(B) Just after the switch is turned, the current through the circuit is $E\sqrt{\frac{2C}{3L}}$

(C) The maximum charge in capacitor 'C' is equal to $\frac{4CE}{3}$

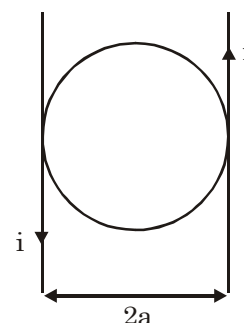
(D) The maximum charge in capacitor 'C' is equal to $\frac{CE}{3}$



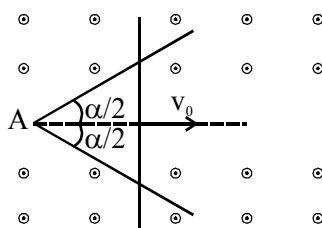
8. Two conductors of infinite length carry a current i each, which varies with time according to relation $i = i_0 e^{\left(\frac{2}{\mu_0}\right)t}$. They are parallel and separated by a distance

2a. A circular conducting ring of radius a and having a resistance $1 \Omega/\text{length}$, in the plane of wires lies between the two straight conductors and insulated from them as shown :-

- (A) Mutual inductance of system is $2 \mu_0 a$.
(B) Mutual inductance of system is $\mu_0 a$.
(C) Net magnetic field at the centre of circle is always zero.
(D) Net magnetic field at the centre of circle increases with time.



9. A long straight wire of negligible resistance is bent into V shape, its two arms making an angle α with each other and placed horizontally in a vertical, homogeneous field B . A rod of total mass m , and resistance r per unit length, is placed on V shaped conductor, at a distance x_0 from its vertex A, and perpendicular to the bisector of angle α (see figure)



The rod is started off with an initial velocity v_0 in the direction of bisector and away from vertex A. The rod is long enough not to fall off the wire during the subsequent motion, and the electrical contact between the two is good although friction between them is negligible. Choose **CORRECT** statement(s)

(A) At any position x , let velocity of wire is v then $\frac{B^2}{r} x_0^2 \tan \frac{\alpha}{2} + mV_0 = \frac{B^2}{r} x^2 \tan \frac{\alpha}{2} + mv$

(B) maximum value of x coordinate of wire is $x_{\max} = \sqrt{x_0^2 + \frac{mv_0 r}{B^2 \tan \frac{\alpha}{2}}}$

(C) As x increases, v decreases

(D) Whatever is the direction of vertical magnetic field, the rod will ultimately stop.

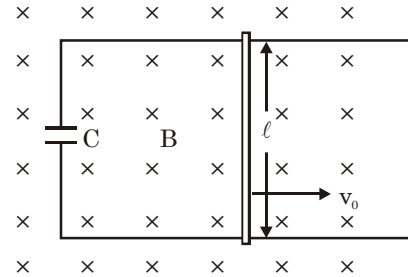
10. Two fixed parallel conducting rails of negligible resistance are connected at one end by a capacitor C . Distance between the rails is ℓ . Arrangement is kept on a horizontal plane with vertical uniform magnetic field as shown in the figure. Initially capacitor is uncharged & a rod of resistance R & mass m is laid perpendicularly on to the rails & given a velocity v_0 . Choose the **CORRECT** option(s), provided that the rail is long enough & homogeneous field extends far enough (Friction & effect of self induction is negligible)

(A) Final velocity of the wire is $\frac{B^2 \ell^2 C}{m + B^2 \ell^2 C} v_0$.

(B) Final charge on the capacitor is $\left[\frac{m B \ell C}{m + B^2 \ell^2 C} \right] v_0$.

(C) Final current in the circuit is $\frac{B v_0 \ell}{2R}$.

(D) Ratio of final kinetic energy & initial kinetic energy is $\left(\frac{m}{m + B^2 \ell^2 C} \right)^2$.



11. A circular conducting loop of radius R and resistance per unit length λ is pulled out from the region of uniform magnetic field with constant velocity v . The situation shown in the figure corresponds to time $t = 0$. Mark out the **CORRECT** statement(s) :

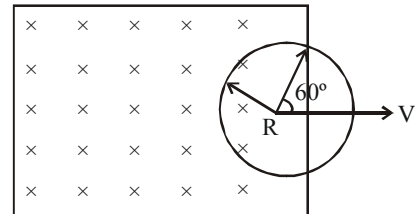
- (A) Just after $t = 0$ i.e., the motion starts, the induced

current in the loop is $\frac{\sqrt{3} B v}{2 \pi \lambda}$

- (B) Current will be induced in the loop for $\frac{3R}{2v}$ sec

- (C) At any time t , the current induced in the loop is given by $\frac{B v \sqrt{3R^2 - 4v^2 t^2 + 4Rvt}}{\lambda \times 2 \pi R}$ and is in clockwise direction.

- (D) Induced current is in clockwise direction for $t = 0$ to $\frac{R}{2v}$ and thereafter it becomes in anticlockwise direction.



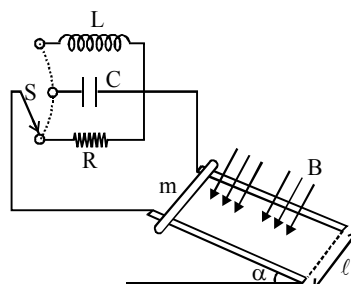
Linked Comprehension Type (Single Correct Answer Type)

(1 Para $\times$ 2 Q.) [3 M (-1)]

Paragraph for Question Nos. 12 & 13

A homogeneous field of magnetic induction $\vec{B}$ is perpendicular to a track of width ℓ which is inclined at an angle α to the horizontal. A frictionless conducting rod of mass m straddles the two rails of the track as shown in figure. Now the rod is being released from rest, if the circuit formed by the rod and the track is closed by :-

- (i) a capacitor of capacitance C , or (ii) a coil of inductance L or (iii) a resistor ?



12. Mark the **INCORRECT** statement :-
- (A) If switch is connected to resistor then rod eventually attains terminal velocity.
 (B) If switch connected to capacitor velocity continuously increases.
 (C) If switch connected to inductor then rod executes simple harmonic motion with initial position as equilibrium position.
 (D) If resistor is connected then long time after power dissipated in resistor equals power delivered by gravity
13. Mark the correct statement :-

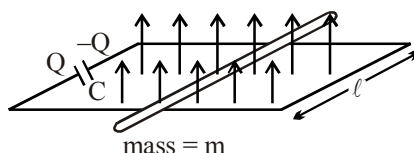
- (A) with inductor angular frequency of oscillation is given by $\omega^2 = \frac{B^2 \ell^2}{2mL}$
- (B) with capacitor acceleration of rod is $a = \frac{mg \sin \alpha}{m + B^2 \ell^2 C \cos^2 \alpha}$
- (C) with inductor equilibrium position is given by $x_0 = \frac{mgL \sin \alpha}{B^2 \ell^2}$
- (D) displacement of rod as function of time is given by $x(t) = A \cos \omega t$ (where origin is at initial position)

SECTION-III

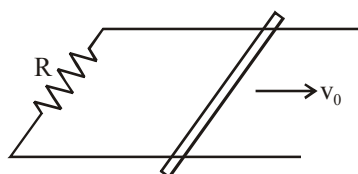
Numerical Grid Type (Ranging from 0 to 9)

7 Q. [4 M (0)]

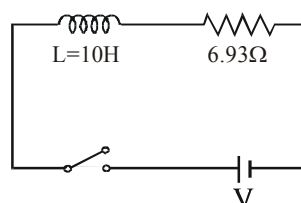
1. An arrangement of smooth pair of long conducting rails which are resistanceless are joined to a charged capacitor. The rails are connected by a rod of mass m and length l and resistance R . Vertical magnetic field is uniform. The rod is initially at rest. It starts moving due to the influence of magnetic field. What should be the mass of the rod (in mg) so that its kinetic energy in steady state is maximum ?
 [Take : $l = 1\text{m}$, $B = 2\text{T}$, $C = 1\mu\text{F}$, $R = 1\Omega$]



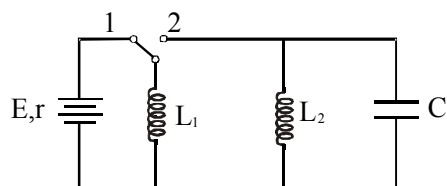
2. A rod of mass $m = 2\text{kg}$ slides without friction along two parallel rails at distance $d = 1.3\text{m}$ from each other (see figure). The rails are joined by a resistor to a resistance $R = 0.32\Omega$ and placed in a vertical magnetic field of induction $B = 0.4\text{ T}$. The rod is pushed with velocity $v_0 = 3.38\text{ m/s}$. Find the distance (in m) covered by the rod until it stops.



3. A resistor and an inductor in series are connected to a battery through a switch. After the switch has been closed, at what time (in sec) will the Joule heat dissipated in the resistor change at the fastest rate?



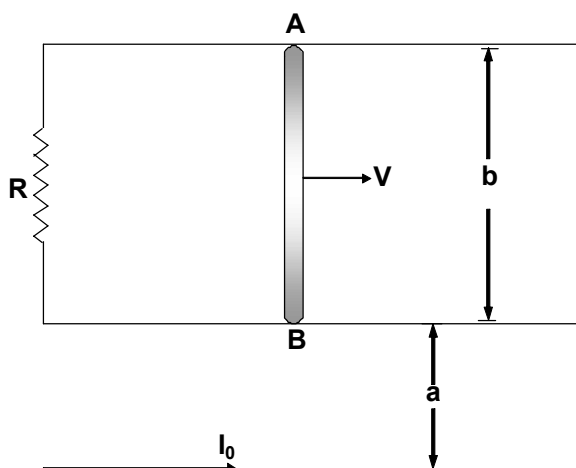
4. A circuit shown consists of two inductors of inductances L_1 and L_2 , a capacitor of capacitance C , a battery of electromotive force E and internal resistance r and a switch. Initially the switch was in position 1 for a long time. Find the maximum current (in A) in the inductor L_2 after the switch is thrown to position 2 ?



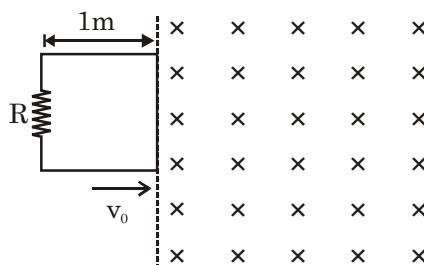
(Given : $E = 5V$, $L_1 = 4mH$, $L_2 = 1mH$, $r = 2\Omega$)

5. A long straight conductor carries a current I_0 . At distances a and $(a + b)$ from it, there are two identical wires, each having resistance λ per unit length, which are inter-connected by a resistance R as shown in Figure. A conducting rod AB of length b can slide along the wires without friction. At $t = 0$, the rod is in extreme left position and starts to move to the right without friction and with constant velocity v . The force function (in terms of time t) required to maintain velocity of rod constant to V m/s is

$$\frac{\mu_0^2 I_0^2 V}{K\pi^2(R + 2Vt\lambda)} \left[\log_e \left(\frac{a+b}{a} \right) \right]^2. \text{ The value of } K \text{ is}$$



6. A rigid square loop having resistance R and mass 100 gm has side length 1m. It is projected with velocity $v_0 = 5$ m/s in a region having uniform magnetic field of strength 10 T as shown in figure. Find the value of $\frac{R}{100}$ (in ohms) so that the loop comes to rest when it is completely inside the magnetic field region. (ignore gravity)



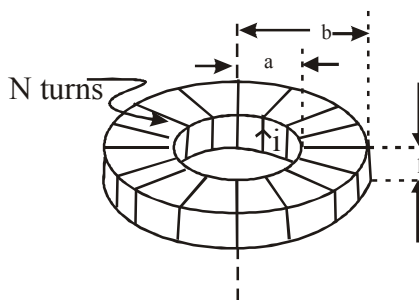
7. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field $B = Kxt^2$, where $K = \text{constant}$, x is the distance (in metre) from the centre of the disc and t is the time (in second) is switched on perpendicular to the plane of the disc. The torque (in N-m) acting on the disc after 15 sec. (Take $2KQ = 1$ S.I. unit and $R = 1$ metre).

SECTION-I

Single Correct Answer Type

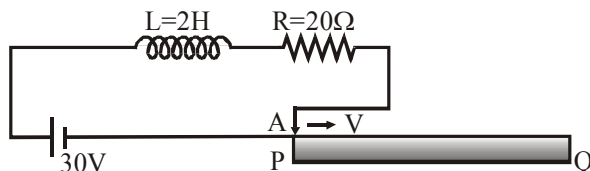
8 Q. [3 M (-1)]

1. Complete toroidal coil of rectangular cross section is shown in figure. Current i is flowing in each turn and no. of turns are N . Find the value of coefficient of self inductance.

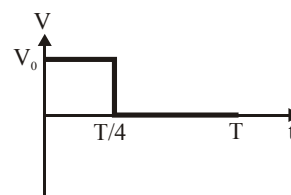


(A) $L = \frac{\mu_0 N^2 h}{4\pi} \ln\left(\frac{b}{a}\right)$ (B) $L = \frac{3\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$ (C) $L = \frac{3\mu_0 N^2 h}{4\pi} \ln\left(\frac{b}{a}\right)$ (D) $L = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$

2. In the circuit shown in figure, A is a sliding contact which can move over a smooth rod PQ. Resistance per unit length of the rod PQ is 1 ohm/m. Initially slider is just left to the point P and circuit is in the steady state. At $t = 0$ slider starts moving with constant velocity $v = 5$ m/s towards right. Current in the circuit at $t = 2$ sec is :-



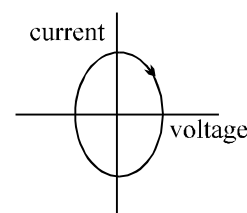
- (A) 1 amp (B) less than 0.5 amp
(C) more than 1 amp (D) in between 0.5 to 1.0 amp
3. The power factor in a circuit connected to an AC power supply has a value which is
(i) zero when the circuit contains an ideal inductance only
(ii) unity when the circuit contains an ideal resistance only
(iii) zero when the circuit contains an ideal resistance only
(iv) zero when the circuit contains an ideal capacitance only
(A) (ii), (iii), (iv) (B) (i), (ii), (iv) (C) (i), (iii), (iv) (D) (ii), (iii)
4. A periodic voltage V varies with time t as shown in the figure. T is the time period. The r.m.s. value of the voltage is :-



(A) $\frac{V_0}{8}$ (B) $\frac{V_0}{2}$
(C) V_0 (D) $\frac{V_0}{4}$

5. The graph shows the current versus the voltage in a driven RLC circuit at a fixed frequency. The arrow indicates the direction that this curve is drawn as time progresses. In this plot, the

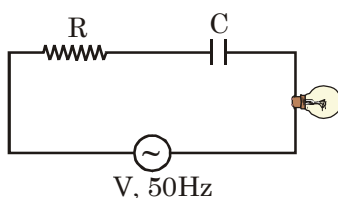
- (A) Current lags the voltage by about 90 degrees
(B) Current leads the voltage by about 90 degrees
(C) Current and voltage are in phase
(D) Current and voltage are 180 degrees out of phase



6. A charged capacitor discharges through a resistance R with time constant τ . The two are now placed in series across an AC source of angular frequency $\omega = \frac{1}{\tau}$. The impedance of the circuit will be-

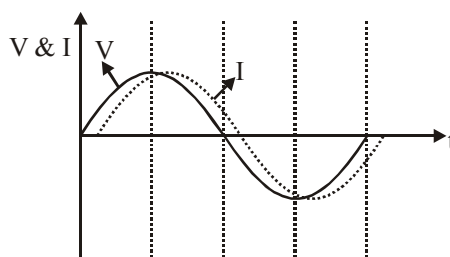
- (A) $\frac{R}{\sqrt{2}}$ (B) R (C) $\sqrt{2}R$ (D) $2R$

7. In the R-C circuit, if the capacitor is filled with a dielectric, then



- (A) Impedance of the circuit increases
(B) Brightness of the bulb increases
(C) Resonance will be occur at lower frequency.
(D) The value of rms current will decrease.

8. Graph shows voltage (V) and current (I) against time for an AC circuit where all elements will be in series. Circuit may be contain :-

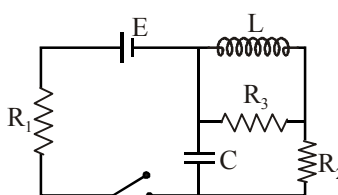


- (A) L, C only (B) L only (C) L, R only (D) R, C only

Multiple Correct Answer Type

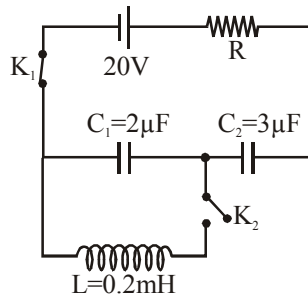
5 Q. [4 M (-1)]

9. Current in R_3 in the shown circuit

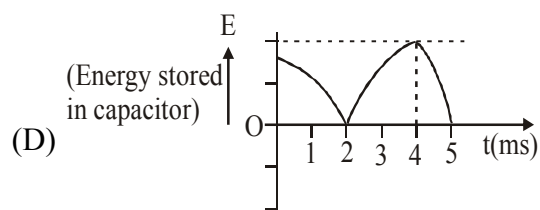
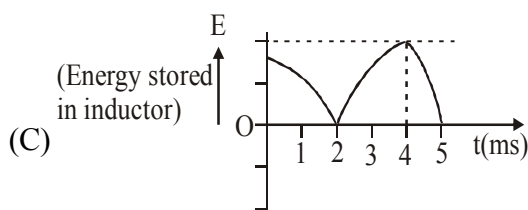
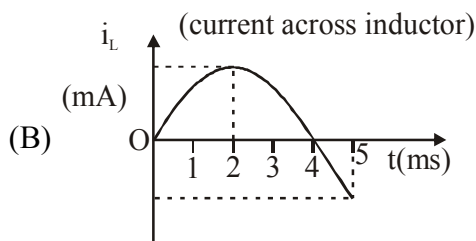
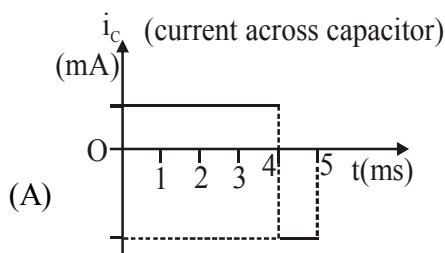
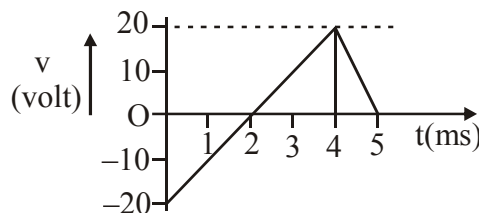


- (A) just after closing the switch is zero (B) long after closing the switch is zero
(C) just after closing the switch is $\frac{E}{R_3}$ (D) long after closing the switch is $\frac{E}{R_3}$

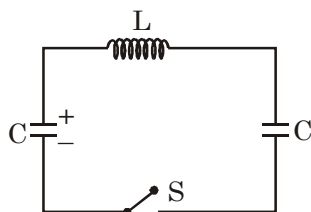
10. A circuit containing capacitors C_1 and C_2 , shown in the figure is in the steady state with key K_1 closed. At the instant $t = 0$, K_1 is opened and K_2 is closed. The ratio of energy in the inductor and that of capacitor C_1 becomes one third at time 't' for the first time. If charge on capacitor C_1 at that instant is q_1 and that on capacitor C_2 is q_2 then choose the **CORRECT** statement(s) :-



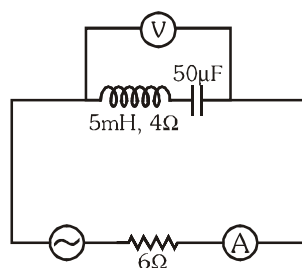
- (A) Angular frequency of oscillations of the L.C. circuit will be 5×10^4 rad/s.
 (B) Time 't' will be 21 μ s.
 (C) Charge on capacitor C_1 at that instant will be, $q_1 = 12\sqrt{3}\mu\text{C}$.
 (D) Charge on capacitor C_2 at that instant will be, $q_2 = 24\mu\text{C}$.
11. The voltage shown in the figure is applied to a $2.5 \mu\text{F}$ capacitor and a 0.5 H inductor separately. choose correct graph



12. Figure shows an electric circuit with negligibly small active resistance. Initially left capacitor is charged to a potential V_0 and then the switch was closed.



- (A) Charge on right capacitor is given by $\frac{CV_0}{2} (1 - \cos \omega t)$; $\omega = \sqrt{\frac{2}{LC}}$
- (B) At the instant when charge on capacitor plates have same magnitude, total energy in capacitor and inductor is equal.
- (C) Maximum current in the circuit has magnitude $V_0 \sqrt{\frac{C}{2L}}$
- (D) Maximum magnitude of induced emf in circuit is V_0 .
13. In the circuit shown in the figure, the A.C. source gives a voltage $V = 20 \cos(2000t)$ volt. Neglecting source resistance, the voltmeter and ammeter readings will be :-



- (A) 0V, 2A (B) 0 V, 1.4A (C) 5.6 V, 0.47 A (D) 1.68 V, 0.47 A

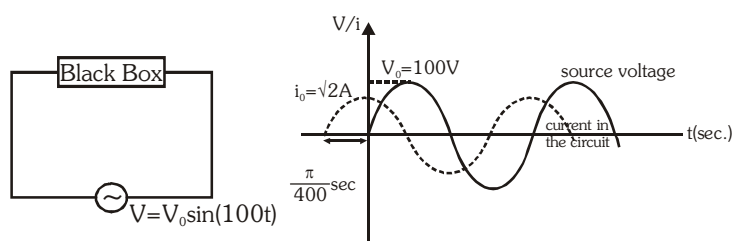
Linked Comprehension Type

(1 Para × 3Q. & 1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Questions no. 14 to 16

A voltage source $V = V_0 \sin(100t)$ is connected to a black box in which there can be either one element out of L, C, R or any two of them connected in series.



At steady state the variation of current in the circuit and the source voltage are plotted together with time, using an oscilloscope, as shown.

14. The element(s) present in black box is/are :
(A) only C (B) LC (C) L and R (D) R and C
15. Values of the parameters of the elements, present in the black box are :
(A) $R = 50\Omega$, $C = 200 \mu F$ (B) $R = 50\Omega$, $L = 2mH$
(C) $R = 400 \Omega$, $C = 50\mu F$ (D) None of these

16. If AC source is removed, the circuit is shorted and then at $t = 0$, a battery of constant EMF is connected across the black box. The current in the circuit will-
- (A) Increases exponentially with constant = 0.02 sec.
(B) Decrease exponentially with time constant = 0.01 sec.
(C) Oscillate with angular frequency 20 sec^{-1}
(D) First increase and then decrease.

Paragraph for Question no. 17 & 18

A student constructs a series RLC circuit. While operating the circuit at a frequency f , he uses an AC voltmeter and measures the potential difference across each device as $(\Delta V_R)_{\max} = 4.8 \text{ V}$, $(\Delta V_C)_{\max} = 29 \text{ V}$, and $(\Delta V_L)_{\max} = 20 \text{ V}$.

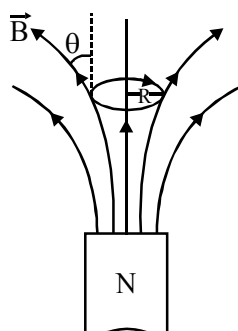
17. How should the frequency of this circuit be changed to increase the current i_m through the circuit?
- (A) Increase f
(B) Decrease f
(C) The current is already at a maximum
(D) There is not enough information to answer the question
18. What will happen to the value of $(\Delta V_L)_{\max}$ if the frequency is adjusted to increase the current through the circuit?
- (A) $(\Delta V_L)_{\max}$ will increase
(B) $(\Delta V_L)_{\max}$ will decrease
(C) $(\Delta V_L)_{\max}$ will remain the same regardless of any changes to f .
(D) The current is already at a maximum
(E) There is not enough information to answer the question

SECTION-III

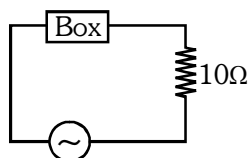
Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. A superconducting current carrying ring of mass $m = 40 \text{ gm}$ and $R = 2 \text{ cm}$ floats in the magnetic field of a magnet. If the field lines make an angle $\theta = 30^\circ$ with vertical and the ring is placed co-axially, assuming $i = 10 \text{ A}$, find the magnetic field $\vec{B}$. if answer is $\frac{n}{3\pi}$. Write the value of n .



2. In the circuit shown power factor of box is given by 0.5 and power factor of circuit is given by $\frac{\sqrt{3}}{2}$. Current leading the voltage. The effective resistance of the box is made by using 5 resistance of equal value. Find the value of each resistance (Assume all the combination are in series)



PHYSICS

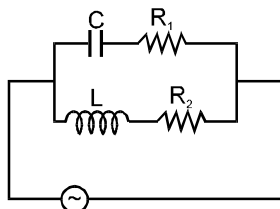
CLASS TEST # 65

SECTION-I

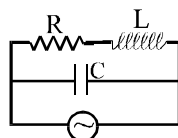
Single Correct Answer Type

3 Q. [3 M (-1)]

1. In the circuit the rms value of voltage across the capacitor C, inductor L and resistor R_1 are respectively $5\sqrt{5}$, 10V and 5V. Then the peak voltage across R_2 is



- (A) 7 V (B) 10 V (C) 15 V (D) $5\sqrt{2}$ V
2. The A.C. circuit shown in figure. Find the frequency (ω_0) of the AC voltage source so that current through the source will be in same phase as of voltage of source.



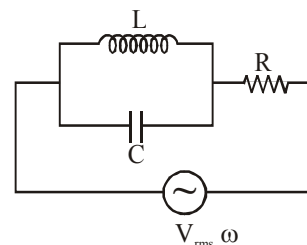
- (A) $\omega_0 = \sqrt{\frac{1}{LC} + \frac{R^2}{L^2}}$ (B) $\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$ (C) $\omega_0 = \frac{1}{\sqrt{LC}}$ (D) $\omega_0 = \sqrt{\frac{1}{LC} + \frac{R^2}{4L} + \frac{R}{2L}}$
3. In the given L-C-R circuit the value of rms current through the a.c. source is

(A) $\frac{V_{rms}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

(B) $\frac{V_{rms}}{\sqrt{R^2 + \left(\frac{\omega L}{1 - \omega^2 LC}\right)^2}}$

(C) $\frac{V_{rms}}{\sqrt{R^2 + (\omega^2 LC - \omega L)^2}}$

(D) $\frac{V_{rms}}{\sqrt{(R + \omega L)^2 + \frac{1}{\omega^2 C^2}}}$



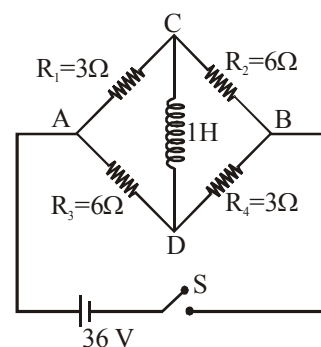
Multiple Correct Answer Type

5 Q. [4 M (-1)]

4. For an AC series circuit which of the following options are correct :
- (A) When i is maximum then V is maximum then circuit must be purely resistive.
 (B) When i is maximum then V is maximum then circuit may consist series R-L-C.
 (C) When i is maximum then V is zero then circuit may be purely inductive.
 (D) When i is maximum then V is zero then circuit may be purely capacitive.

5. The switch S is closed at $t = 0$ as shown in the figure.

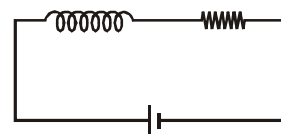
- (A) The current as function of time in the inductor is $3(1 - e^{-4t})$ A.
 (B) The current through R_1 just after closing of switch is 8A
 (C) The power supply by the battery as function of time is $108(1 - e^{-4t})$ W
 (D) The current through R_2 after a long time closing of switch is 3A



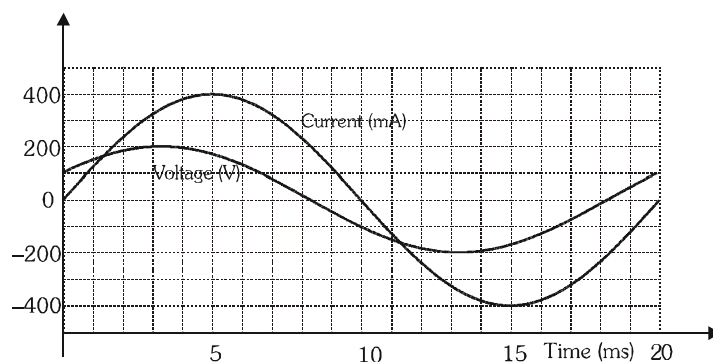
6. An ideal battery is connected to a resistor and an ideal inductor as shown.

Considering steady state :-

- (A) EMF across inductor is equal to emf across battery.
(B) Potential difference across inductor is equal to emf across inductor in magnitude.
(C) Potential difference across resistor is equal to emf across inductor in magnitude.
(D) EMF across resistor is zero.



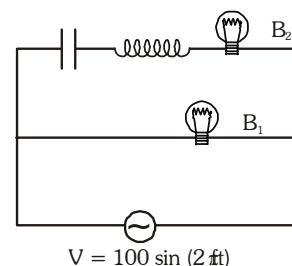
7. The given graph shows variation with time in the source voltage and steady state current drawn by a series RLC circuit. Which of the following statements is/are correct?



- (A) Current lags the voltage. (B) Resistance in the circuit is $250\sqrt{3} \Omega$.
(C) Reactance in the circuit is 250Ω . (D) Average power dissipation in the circuit is $20\sqrt{3} \text{ W}$.

8. Two similar bulbs of rating $(100\text{W}, 50\sqrt{2} \text{ V})$ are connected in circuit as shown when frequency of source is varied, brightness of B_2 changes. It is observed that at a 50 Hz frequency both the bulbs have same brightness. Select correct alternative/s.

- (A) Current in both the bulbs will be in same phase when both have same brightness.
(B) Current through source will be 2A when both the bulbs have same brightness.
(C) Current through source will be $2\sqrt{2} \text{ A}$ when both the bulbs have same brightness.
(D) B_1 will be more brighter for all frequencies except 50 Hz



Linked Comprehension Type (2 Para $\times$ 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Questions no. 9 to 11

An electric current that reverses its direction with a constant frequency is called alternating current. The

average or mean value of ac is defined for half cycle as $I_{\text{mean}} = \frac{\int_0^{T/2} Idt}{\int_0^{T/2} dt}$. The effective, virtual or rms value

of ac is defined as $I_{\text{rms}} = \sqrt{\frac{\int_0^T I^2 dt}{\int_0^T dt}}$

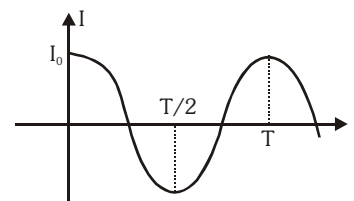
9. The mean and rms value of an alternatic current as shown in figure are

(A) $\frac{2I_0}{\pi}, I_0\sqrt{2}$

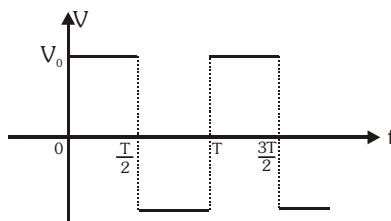
(B) $\frac{4I_0}{\pi}, I_0$

(C) $\frac{I_0}{\pi}, \frac{I_0}{\sqrt{2}}$

(D) $\frac{2I_0}{\pi}, \frac{I_0}{\sqrt{2}}$



10. The mean and rms value of an alternating voltage as shown in figure then



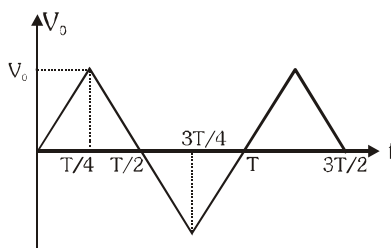
(A) V_0, V_0

(B) $\frac{V_0}{2}, V_0$

(C) $\frac{3V_0}{2}, \frac{V_0}{2}$

(D) $\frac{V_0}{4}, \frac{V_0}{2}$

11. The mean and rms value of an alternating voltage as shown in figure then



(A) $\frac{V_0}{3}, \frac{V_0}{2}$

(B) $\frac{V_0}{\sqrt{2}}, \frac{V_0}{\sqrt{3}}$

(C) $\frac{V_0}{2}, \frac{V_0}{3}$

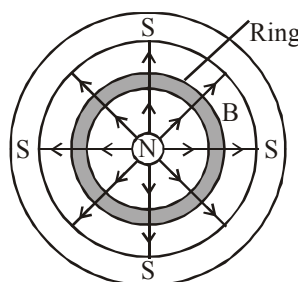
(D) $\frac{V_0}{2}, \frac{V_0}{\sqrt{3}}$

Paragraph for Question Nos. 12 to 14

There is a thin aluminium doughnut of square cross-section A and radius r. It is released from rest and falls vertically under gravity through horizontal radial magnetic field B produced by north, N and south, S magnetic poles as shown in figure. The ring has mass m, radius r and resistivity ρ . The centre of the ring falls down the vertical line of symmetry of the system.



Crosssectional view of ring



Top view of ring falling through radial magnetic field

12. What is magnetic force on the ring at the instant when speed of ring is v :-

(A) $\frac{2\pi AB^2 vr}{\rho}$, downward

(B) $\frac{2\pi AB^2 vr}{\rho}$, upward

(C) $\frac{\pi AB^2 vr}{\rho}$, downward

(D) $\frac{\pi AB^2 vr}{\rho}$, upward

13. Obtain an expression for terminal velocity of the ring.

(A) $v_T = \frac{mg\rho}{\pi AB^2 r}$ (B) $v_T = \frac{2mg\rho}{\pi AB^2 r}$ (C) $v_T = \frac{2mg\rho}{3\pi AB^2 r}$ (D) $v_T = \frac{mg\rho}{2\pi AB^2 r}$

14. Heat dissipated per unit time after achieving terminal velocity of ring :

(A) $\frac{m^2 g^2 \rho}{\pi AB^2 r}$ (B) $\frac{2m^2 g^2 \rho}{\pi AB^2 r}$ (C) $\frac{2m^2 g^2 \rho}{3\pi AB^2 r}$ (D) $\frac{m^2 g^2 \rho}{2\pi AB^2 r}$

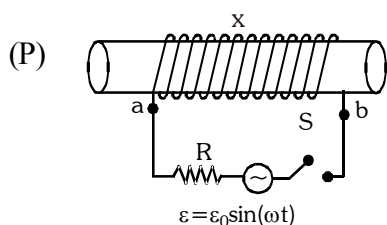
Matching List Type (4 × 4)

1 Q. [3 M (-1)]

15. Column-I contains some circuits and column-II contains some properties. In column-II current signifies instantaneous current and potential difference refers instantaneous potential difference.

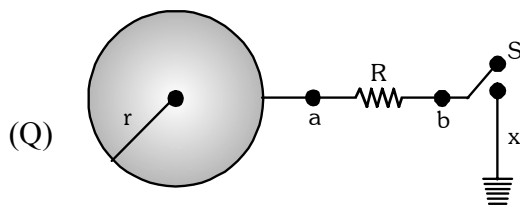
Column – I Charge System

Column – II Properties



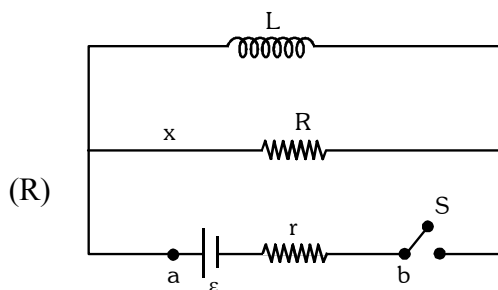
(1) In branch **x**, current varies

A closely separated conducting wire is wrapped over an insulating cylinder. Wire is connected to a resistance R and an alternating source via a switch S , which is closed at $t = 0$



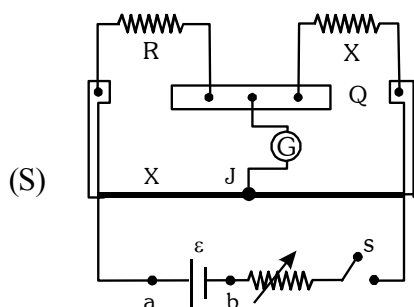
(2) Between points **a** and **b** magnitude of potential difference increases always.

A charged metallic sphere of radius r is earthed with a resistance R and a switch S , which is closed at $t = 0$.



(3) In branch **x**, current is constant

An inductor L and resistor R is connected to a cell of emf ε and internal resistance r with a switch(s), which is closed at $t = 0$



A circuit (used to measure an unknown resistance) is in balanced conditions (Galvanometer reads zero)

(4) Between points **a** and **b**, of potential difference zero always.

Code :

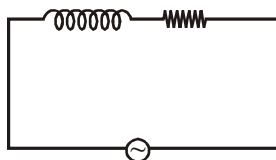
	P	Q	R	S
(A)	1	1	1	3
(B)	1	2	3	4
(C)	2	1	3	4
(D)	2	3	1	4

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1 Q. [3(0)]

- In a series LR circuit connected to an alternating voltage source, it is observed that at the instant voltage across the source is maximum, voltage across the inductor is 3V and voltage across the resistance is 4V. If resistance is 2Ω , what is reactance (in Ω) of the inductor ?

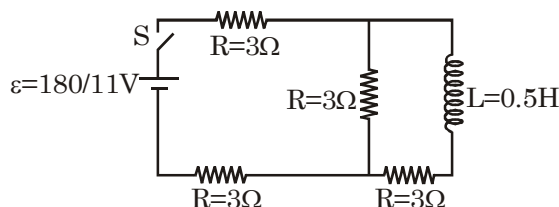


SECTION-III

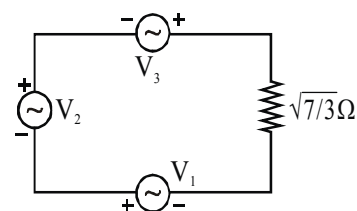
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

- In the circuit shown, find the current in Amp through the battery at $t = 0.1 \ln 2$ sec, if switch is closed at $t = 0$.



- Three alternating voltage sources $V_1 = 3\sin\omega t$ volt, $V_2 = 5\sin(\omega t + \phi_1)$ volt and $V_3 = 5\sin(\omega t - \phi_2)$ volt connected across a resistance $R = \sqrt{\frac{7}{3}}\Omega$ as shown in the figure (where ϕ_1 and ϕ_2 corresponds to 30° and 127° respectively). Find the peak current (in Amp) through the resistor.



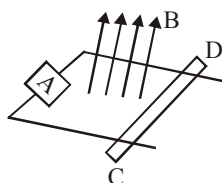
3. A coil, a capacitor and an AC source of voltage 24 V (rms) are connected in series. By varying the frequency of the source, a maximum rms current of 6 A is observed. If this coil is connected to a DC battery of emf 12V and internal resistance 2Ω , then find current (in amp), through it.

SECTION-IV

Matrix Match Type (4×5)

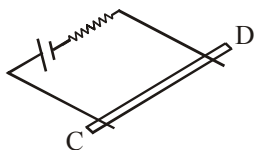
1 Q. [8 M (for each entry +2(0))]

1. A homogeneous magnetic field B is perpendicular to a sufficiently long fixed track of width ℓ which is horizontal. A frictionless rod of mass m straddles the two rail of the track as shown in the figure. Entire arrangement lies in horizontal plane. For the situation suggested in column-II match the appropriate entries in column-I. The rails are also resistanceless.

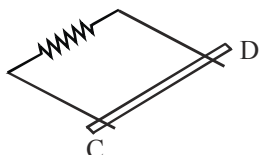


Column-I

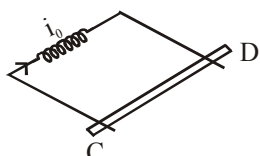
- (A) A is a battery of emf V and internal resistance R . The conducting rod is initially at rest.



- (B) A is a resistance. The non-conducting rod is projected to the right with a velocity V_0 .



- (C) A is an inductor having with initial current i_0 . It is having no resistance and CD is a conducting rod. (consider no loss of energy)



- (D) A is a resistance. The conducting rod is projected to the right with a velocity V_0 .

Column-II

- (P) Energy is dissipated during the motion.

- (Q) The rod moves with a constant velocity after a long time.

- (R) After a certain time interval rod will change its direction of motion.

- (S) If a constant force is applied on the rod to the right, it can move with a constant velocity from starting itself

- (T) The rod stops after some time permanently in absence of an external force.

SECTION-I

Single Correct Answer Type

6 Q. [3 M (–1)]

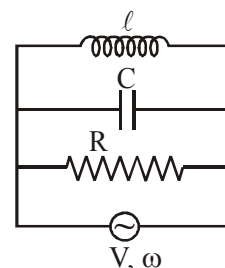
- A coil (which can be modelled as a series circuit) has been designed for high Q performance at a rated voltage and a specified frequency. If the frequency of operation is doubled and the coil is operated at the same rated voltage, then the Q factor and the active power P consumed by the coil will be affected as below
 (A) P is doubled, Q is halved (B) P is halved, Q is doubled
 (C) P remains constant but Q increases 4 times (D) P decreases 4 times but Q is doubled
- Consider L, C, R circuit as shown in figure, with a.c. source of peak value V and angular frequency ω . Then the peak value of current through the ac source

(A) $\frac{V}{\sqrt{\frac{1}{R^2} + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

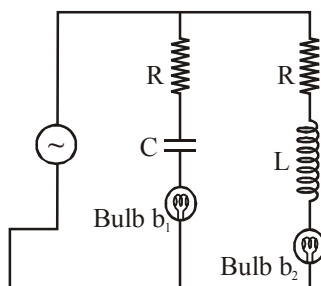
(B) $V \sqrt{\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2}$

(C) $\frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

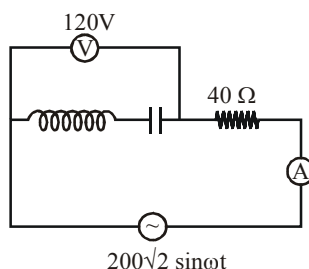
(D) $\frac{VR\omega C}{\sqrt{\omega^2 C^2 + R(\omega^2 C^2 - 1)^2}}$



- Two identical incandescent light bulbs are connected as shown in the figure. When an AC voltage source of frequency f is applied in circuit, which of the following observations will be **CORRECT**?



- (A) Maximum brightness of both bulbs does not depend on frequency f, if $f = \frac{1}{2\pi}(\sqrt{1/LC})$
 - (B) Both bulbs will glow with same brightness provided frequency $f = \frac{1}{2\pi}(\sqrt{1/LC})$
 - (C) Bulb b_1 will light up initially and goes off, bulb b_2 will be ON constantly
 - (D) Bulb b_1 will blink and bulb b_2 will be ON constantly
- In the given LCR series circuit find the reading of the hot wire ammeter.



(A) 2 A

(B) 4 A

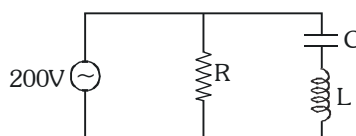
(C) 5 A

(D) 9 A

5. A coil with inductance L and resistance R is connected to an alternating source. What should be the capacity of a capacitor connected in series with the coil such that active power of circuit does not change.

(A) $C = \frac{1}{\omega^2}$ (B) $C = \frac{1}{\omega^2 L}$ (C) $C = \frac{1}{2\omega^2 L}$ (D) $C = \frac{\omega^2}{L}$

6. In the circuit diagram shown, $X_C = 100\Omega$, $X_L = 200\Omega$ & $R = 100\Omega$. Effective current through the source is :-

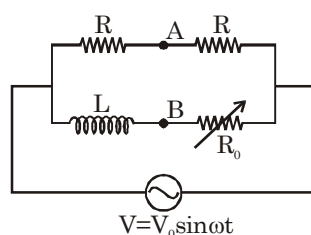


(A) $\sqrt{2}$ A (B) $2\sqrt{2}$ A (C) 2 A (D) $\frac{1}{2}$ A

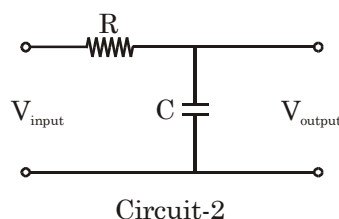
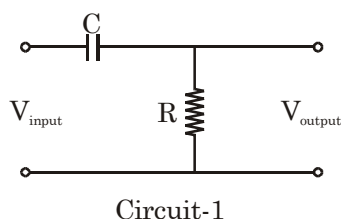
Multiple Correct Answer Type

5 Q. [4 M (-1)]

7. Consider the A.C. circuit as shown in the figure and choose **CORRECT** option(s). R.M.S. value of potential difference between A and B is V_{AB} :



- (A) V_{AB} will increase on increasing R_0
 (B) V_{AB} will decrease on decreasing R_0
 (C) V_{AB} does not depend on non-zero value of R_0
 (D) V_{AB} does not depend on non-zero value of ω .
8. Figure shows two circuits in each case V_0 denotes peak value of input voltage. Input voltage is given by $V = V_0 \sin \omega t$ and output voltage by $V = V_M \sin(\omega t + \delta)$:



- (A) In circuit-1 maximum output voltage is $V_M = \frac{V_0}{\sqrt{1 + (\omega RC)^{-2}}}$
 (B) In circuit-2 maximum output voltage is $V_M = \frac{V_0}{\sqrt{1 + (\omega RC)^2}}$
 (C) In circuit-1 average power dissipated is given by $P_{av} = \frac{V_0^2}{2R\sqrt{1 + (\omega RC)^{-2}}}$
 (D) Phase angle δ for circuit-2 is given by $\tan \delta = \omega RC$

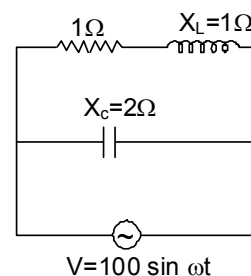
9. In the given a.c. circuit, choose the **CORRECT** statement(s).

(A) Impedance of circuit is 2Ω

(B) Power factor of circuit is $\frac{1}{\sqrt{2}}$

(C) Peak value of current through resistance is $50\sqrt{2}$ A

(D) Average power supplied by source is 2500 W.



10. In an LCR series AC circuit, voltage across resistance is 2 volt. The inductive and capacitive reactances are 10Ω and 5Ω respectively. The phase difference between voltage across the circuit and current is 30° , then pick up the **CORRECT** statement(s) :

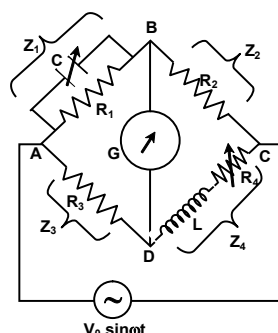
(A) the resistance of resistor is $5\sqrt{3}\Omega$.

(B) the current in the circuit is $\frac{2}{5\sqrt{3}}$ A

(C) the voltage of AC source is $\frac{4}{\sqrt{3}}$ volts

(D) the voltage of AC source is $4\sqrt{3}$ volts

11. If given circuit is balance wheat stone bridge of the AC bridge then (Balance condition of the AC bridge is $\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4}$)



(A) it will be independent of source frequency

(B) it will be dependent of source frequency

(C) $L = CR_2R_3$

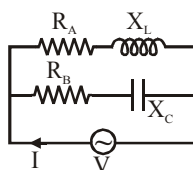
(D) $L = CR_3R_4$

Linked Comprehension Type
(Single Correct Answer Type)

(1 Para $\times$ 2Q. & 1 Para $\times$ 3Q.) [3 M (-1)]

Paragraph for Question no. 12 and 13

Figure shows a two branched parallel circuit with $R_A = 20\Omega$, $X_L = 15\Omega$, $R_B = 6\Omega$ and $X_C = 8\Omega$. The total current supplied by alternating source V is $I = 14$ A.



12. Current passing through the inductor (X_L) is :-

(A) Zero

(B) 4A

(C) 10 A

(D) 14 A

13. Power dissipated in the circuit is :-

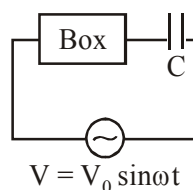
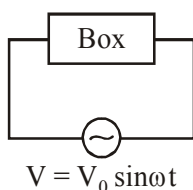
(A) Zero

(B) 500 W

(C) 920 W

(D) 1400 W

Paragraph for Questions no. 14 to 16



When a box is connected with a source $V = V_0 \sin \omega t$, then power factor of circuit is $\frac{1}{\sqrt{2}}$. If a capacitor is connected in series with the box, then power factor of circuit becomes unity. (Assume that box contains only series combination of elements)

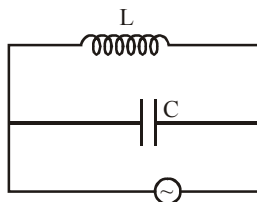
14. Box may contains resistance in series with an
(A) capacitor (B) inductor (C) another resistance (D) none of the above
15. Impedance of box with same source is :-
(A) $\frac{\sqrt{2}}{\omega C}$ (B) $\frac{1}{\omega C}$ (C) ωC (D) $\sqrt{2} \omega C$
16. In box ratio of resistance and reactance is
(A) equal to 1 (B) greater than 1 (C) less than 1 (D) none of above

SECTION-III

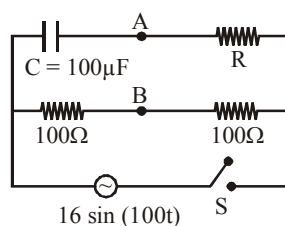
Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

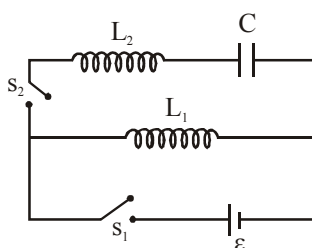
1. An alternating voltage $V_0 = 100$ V with angular frequency ω is connected across the capacitor and inductor having $X_L = 5\Omega$ and $X_C = 10\Omega$. Find the ratio of current through inductor to AC source.



2. An uncharged capacitor $C = 100 \mu\text{F}$ is connected in series with 50Ω and with AC source as shown in the figure. If switch S is closed at $t = 0$, the maximum value of $(V_A - V_B)$ is K volt. Calculate K.



3. Diagram shows a circuit. Switch S_1 was closed for a long time with S_2 open. At $t = 0$ switch S_2 was closed & S_1 was opened. The time period of L-C oscillation after $t = 0$ is given by $2\pi\alpha \times 10^{-4}$ s. Fill the value of α . (Given $C = 30 \mu\text{F}$, $L_1 = L_2 = 6 \text{ mH}$).

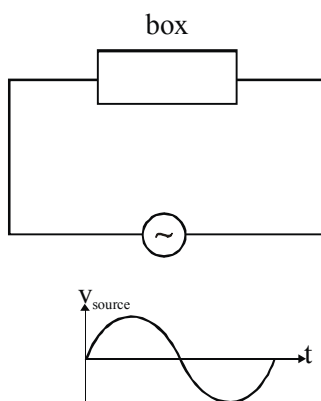


SECTION-IV

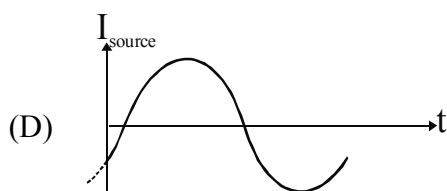
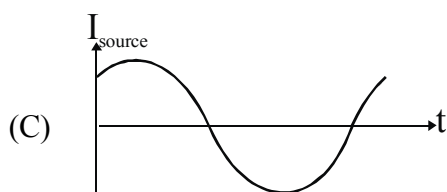
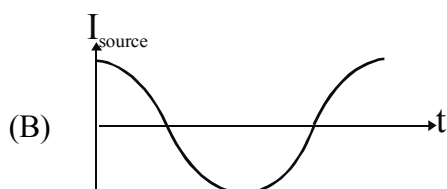
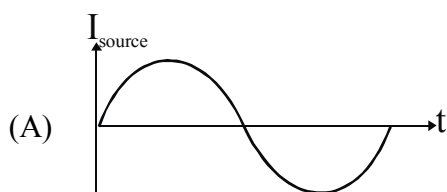
Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Box may have any series combination of L, C and R. Column-I represents source current and column-II represents possible statements.



Column-I



Column-II

(P) Box may contain LCR.

(Q) Box may contain only LR.

(R) Power factor of box is zero.

(S) Box may contain only CR.

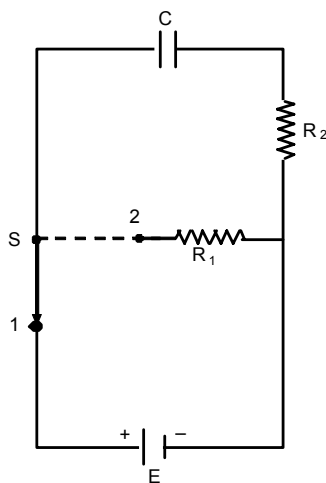
(T) State of resonance.

SECTION-I

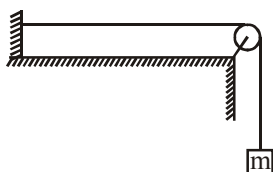
Single Correct Answer Type

5 Q. [3 M (–1)]

- Consider a wave represented by $y = a \cos^2(\omega t - kx)$ where symbols have their usual meaning. This wave has (here $k = \frac{2\pi}{\lambda}$) :-
 (A) An amplitude a , frequency ω and wave length λ
 (B) An amplitude a , frequency 2ω and wave length 2λ
 (C) An amplitude $\frac{a}{2}$, frequency 2ω and wave length $\frac{\lambda}{2}$
 (D) An amplitude $\frac{a}{2}$, frequency 2ω and wave length λ
- An inductor of reactance 1Ω and a resistor of 2Ω are connected in series to the terminals of a 6 V (rms) a.c. source. The power dissipated in the circuit is :
 (A) 8 W (B) 12 W (C) 14.4 W (D) 18 W
- Switch S of circuit shown in figure is in position 1 for a long time. At instant $t = 0$, it is thrown from position 1 to 2. The thermal power $P_1(t)$ generated in resistance R_1 .



- $\frac{E^2 R_1}{(R_1 + R_2)^2} \cdot e^{-2t/(R_2 + R_2)C}$
 - $\frac{E^2 R_1}{(R_1 + R_2)^2} \cdot e^{-t/(R_2 + R_2)C}$
 - $\frac{E^2 R_1}{(R_1 + R_2)^2} \cdot e^{-t/2(R_2 + R_2)C}$
 - $\frac{E^2 R_1}{2(R_1 + R_2)^2} \cdot e^{-t/(R_2 + R_2)C}$
- A transverse wave is passing through a string shown in figure. Mass density of the string is 1 kg/m^3 and cross section area of string is 0.01 m^2 . Equation of wave in string is $y = 2\sin(20t - 10x)$. The hanging mass is (in kg) [t is in second and x is in meter]:-



- 40
- 0.2
- 0.004
- 0.4

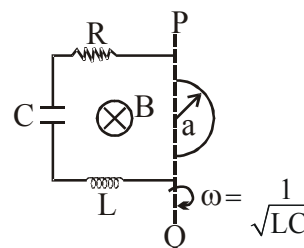
5. A semi circular wire is given angular velocity $\omega = \frac{1}{\sqrt{LC}}$, with the help of an external agent. A uniform magnetic field B exists in space and is directed into the plane of the figure. (circuit part remains at rest) (Left part is at rest). then the **incorrect** statement for given arrangement is :

(A) The rms value of current in the circuit is $\frac{\pi Ba^2}{R\sqrt{2LC}}$

(B) The rms value of current in the circuit is $\frac{\pi Ba^2}{2R\sqrt{2LC}}$

(C) The maximum energy stored in the capacitor is $\frac{\pi^2 B^2 a^2}{8R^2 C}$

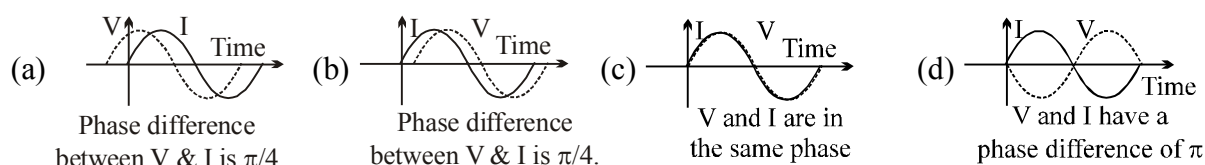
(D) The maximum power delivered by the external agent is $\frac{\pi^2 B^2 a^4}{8LCR}$



Multiple Correct Answer Type

4 Q. [4 M (-1)]

6. The diagram shows the variation of V and I in an AC circuit. The circuit only be a series RC or series RL or series LC or series RLC. Consider the four different combinations of V and I graphs. Pick the correct combination/combinations for each graph. Solid curves represent I and broken curves represent V .



Take the angular frequency of the AC voltage source to be 100 rad/s (This is ω)

(I) : $R = 1 \text{ k}\Omega$, $L = 1 \text{ H}$ and $C = 100 \mu\text{F}$

(II) : $R = 1 \Omega$ and $L = 10^{-2} \text{ H}$

(III) : $R = 100 \Omega$ and $C = 10^{-2} \mu\text{F}$

(IV) : $L = 1 \text{ H}$, $C = 100 \mu\text{F}$ and $R = 1 \Omega$

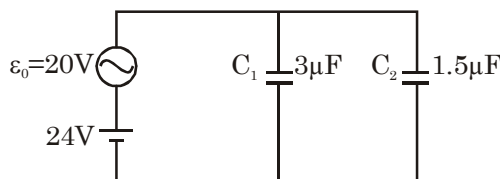
(A) $a \rightarrow \text{II}$; $d \rightarrow \text{I, IV}$

(B) $c \rightarrow \text{I}$; $d \rightarrow \text{none}$

(C) $b \rightarrow \text{III}$; $c \rightarrow \text{IV}$

(D) $a \rightarrow \text{II, III}$; $b \rightarrow \text{II, III}$

7. A circuit consists of two capacitors, a 24 V battery and an AC source connected as shown in figure. The AC voltage is given by $\varepsilon = (20 \cos 120\pi t) \text{ V}$, where t is in second.



- (A) Charge on capacitor C_2 as function of time $Q_2 = (30 \mu\text{C}) \cos (120\pi t) + 36 \mu\text{C}$
- (B) Steady state current is $33.9 \text{ mA} \sin (120\pi t + \pi)$
- (C) Maximum energy stored in capacitor C_1 & C_2 is 36 mJ
- (D) Minimum energy stored in capacitors C_1 and C_2 is $36 \mu\text{J}$

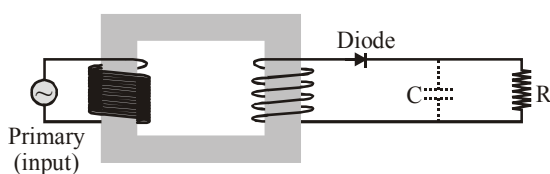
8. A circuit, containing an inductance and a resistance connected in series, has an AC source of 200 V, 50Hz connected across it. An AC current of 10A rms flows through the circuit and the power loss is measured to be 1 kW.
- (A) The inductance of the circuit is $\frac{\sqrt{3}}{10\pi}$ H .
- (B) The frequency of the AC when the phase difference between the current and emf becomes $\pi/4$, with the above components is $\frac{50}{\sqrt{3}}$ Hz .
- (C) The frequency of the AC when the phase difference between the current and emf becomes $\pi/3$, with the above components is $\frac{25}{\sqrt{3}}$ Hz .
- (D) The frequency of the AC when the phase difference between the current and emf becomes $\pi/4$, with the above components is $\frac{25}{\sqrt{3}}$ Hz .
9. Equation of a travelling wave is $y = 5e^{(-ax^2 - bt^2 - 2\sqrt{ab}xt)}$ where x & y are in meter and t is in sec.
 $a = 25 \text{ m}^{-2}$ and $b = 9 \text{ sec}^{-2}$
- (A) Travelling wave propagates along (+) x direction.
- (B) Travelling wave propagates along (–) x direction.
- (C) Speed of wave is $\frac{3}{5}$ m/sec.
- (D) Maximum displacement of particle is 5m.

Linked Comprehension Type (Single Correct Answer Type)

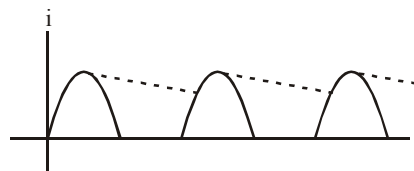
(1 Para × 3Q. & 1 Para × 2Q.) [3 M (-1)]

Paragraph for Question Nos. 10 to 12

Portable electronic devices such as radios and compact disc players are often powered by direct current supplied by batteries. Many devices come with AC-DC converters such as that shown in figure (i). Such a converter contains a transformer that steps the voltage down from 120 V to, typically, 9V and a circuit that converts alternating current to direct current. The AC-DC converting process is called rectification, and the converting device is called a rectifier.

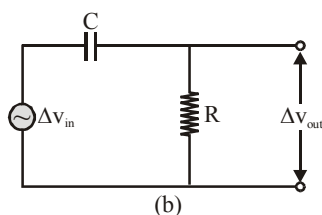


(i) A half-wave rectifier with an optional filter capacitor

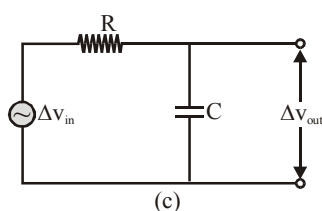


(ii) Current versus time in the resistor. The solid curve represents the current with no filter capacitor, and the dashed curve is the current when the circuit includes the capacitor.

We can also design filters that respond differently to different frequencies. Consider the simple series RC circuit shown in figure. The input voltage is across the series combination of the two elements. The output is the voltage across the resistor.



Now consider the circuit shown in figure, where we have interchanged the resistor and capacitor and where the output voltage is taken across the capacitor. At low frequencies, the reactance of the capacitor and the voltage across the capacitor is high. As the frequency increases, the voltage across the capacitor drops. Therefore, this filter is an RC low-pass filter.



10. Consider the filter circuit shown in Figure (b), the ratio of the input, output voltage is :-

(A) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(B) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{R}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(C) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(D) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{R}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

11. What value of $\frac{\Delta v_{out}}{\Delta v_{in}}$ does approach as the frequency decreases toward zero ?

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) Zero

12. Consider the filter circuit shown in figure (c). The ratio of the input, output voltage is :-

(A) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{1/\omega C}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

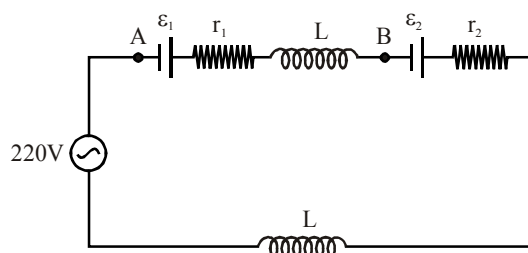
(B) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{1/\omega C}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(C) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{1/\omega C}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(D) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{1/\omega C}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

Paragraph for Question 13 and 14

In AC circuits, deflection based galvanometers always fail to record the current, as coil of galvanometer experiences an alternating torque of fast frequency and hence gives zero reading. AC ammeters are based on heating effect and can measure alternating current. Although an AC ammeter measures i^2 but is calibrated to give i_{rms} . In a circuit two coils of equal inductances and two cells of emfs $\varepsilon_1 = 50 \text{ V}$ and $\varepsilon_2 = 100 \text{ V}$ are connected to a 220 V AC source. The internal resistances of the cells are r_1 and r_2 . In the circuit current lags the voltage by 45° . A DC voltmeter connected between points A and B reads zero.



13. Value of $\frac{r_2}{r_1}$ is :-

(A) 2 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

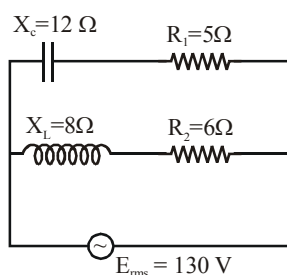
14. When an AC voltmeter is connected across A and B, its reading in steady state is :-
(A) 220 V (B) 153.5 V (C) 107 V (D) 93.5 V

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

1. The amount of power delivered by the ac source in the capacitor & resistance R_1 of the circuit given below is $\frac{X}{8} \times 10^3$ watt, then x is :



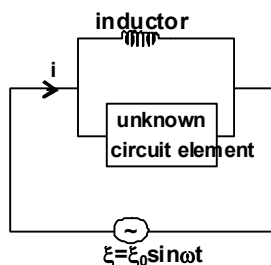
2. A DC ammeter and an AC thermal ammeter are connected to a circuit in series. When a DC is passed through the circuit, the DC ammeter shows $I_1 = 6 \text{ A}$. When an AC flows through the circuit, the AC ammeter shows $I_2 = 8 \text{ A}$. What will the difference in final readings (in amp) of ammeters, if the DC and the AC flow simultaneously through the circuit?
3. The earth has a radius of 6400 km . The inner core of 1000 km radius is solid. Outside it, there is a region from 1000 km to a radius of 3500 km which is in molten state. Then again from 3500 km to 6400 km the earth is solid. Only longitudinal (P) waves can travel inside a liquid. Assume that the P wave has a speed of 10 km/s in the densest part and a speed of 8 km/sec in the other solid layer. An earthquake occurs at some place close to the surface of the earth and was recorded in a seismometer at a diametrically opposite point on the earth (waves travel along the diameter) after 32 minute 5 second. Find the speed (in km/s) of the P-wave in the liquid layer. Give your answer to the nearest integer.

SECTION-IV

Matrix Match Type (4 × 5)

2 Q. [8 M (for each entry +2(0))]

1. Match the following column :



Column-I

(If current i in the circuit is)

(A) $i = i_0 \sin\left(\omega t - \frac{\pi}{4}\right)$

(B) $i = i_0 \sin\left(\omega t + \frac{\pi}{4}\right)$

(C) $i = i_0 \sin \omega t$

(D) $i = i_0 \cos \omega t$

Column-II

(Unknown circuit element may be)

(P) L – C series combination

(Q) L – R series combination

(R) R – C series combination

(S) L – C – R series combination

2. A sine wave $y = A \sin (2\pi x - 2\pi t + \pi/3)$ is propagating in the medium. Match the description of the motion of particles of the medium with entries in column I.

Column-I

(A) $x = \frac{1}{3} \text{ m}, t = \frac{1}{3} \text{ sec}$

(B) $x = \frac{1}{3} \text{ m}, t = 1 \text{ sec}$

(C) $x = 1 \text{ m}, t = \frac{1}{3} \text{ sec}$

(D) $x = 1 \text{ m}, t = 1 \text{ sec}$

Column-II

(P) Velocity is in positive y direction

(Q) Velocity is in negative y direction

(R) Particle is stationary

(S) Particle has positive displacement

(T) Particle has negative displacement

SECTION-I

Single Correct Answer Type

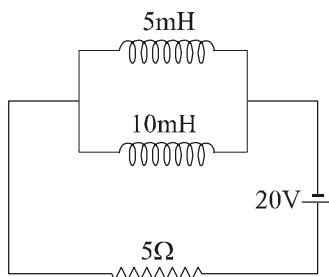
1. In the figure shown hot wire voltmeter and hot wire ammeter are ideal. The reading of voltmeter is :-

(A) $5\sqrt{2}$ V
(B) 5 V
(C) 10 V
(D) None of these

2. In the circuit diagram shown, $X_C = 100$ ohms, $X_L = 200$ ohms & $R = 100$ ohms. The effective current through the source is :

(A) 2 A
(B) $2\sqrt{2}$ A
(C) 0.5 A
(D) None of these

3. In the given circuit, the current through 5mH inductor in steady state is



(A) $4/3$ Amp (B) $8/3$ AMP (C) 4 Amp (D) $2/3$ Amp

4. When a transverse plane wave transverse a medium, individual particles execute periodic motion

given by equation $y = 4 \sin \frac{\pi}{2} \left(2t + \frac{x}{8} \right)$. The phase difference for two position of same particle which

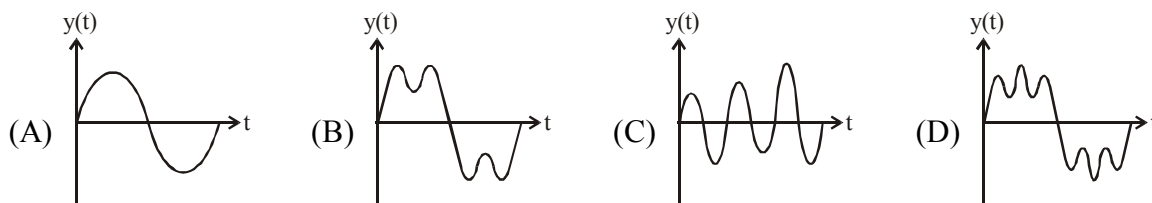
are occupied by time intervals 0.4 s apart is

(A) 72° (B) 135° (C) 45° (D) 108°

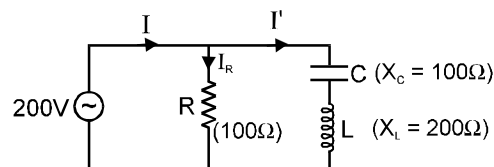
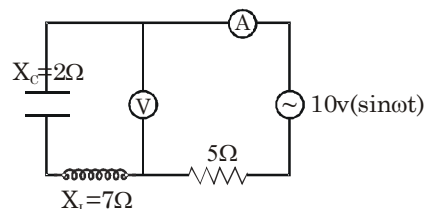
5. A composite string is made up by joining two strings of different masses per unit length μ and 4μ respectively. The composite string is under the same tension. A transverse wave pulse $Y = (6 \text{ mm}) \sin(5t + 40x)$, where t is in seconds and x in metres, is sent along the lighter string towards the joint. The joint is at $x = 0$. The equation of the wave pulse reflected from the joint is

(A) $(2\text{mm}) \sin(5t - 40x)$ (B) $(4\text{mm}) \sin(40x - 5t)$
(C) $-(2\text{mm}) \sin(5t - 40x)$ (D) $(2\text{mm}) \sin(5t - 10x)$

6. A complex wave is represented by an expression of the form $y(t) = 1 \sin \omega t + \frac{1}{2} \sin 3\omega t$, where $\omega = \frac{2\pi}{T}$ is the angular frequency and T is the period of the wave. Which of the following sketches best represent the wave ?



8 Q. [3 M (-1)]



7. String 1 is connected with string 2. The mass per unit length in string 1 is μ_1 and the mass per unit length in string 2 is $4\mu_1$. The tension in the strings is T . A travelling wave is coming from the left. What fraction of the energy in the incident wave goes into string 2?



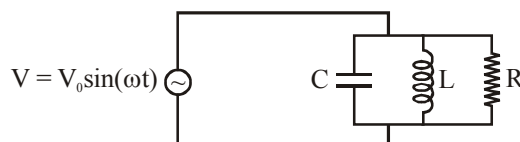
- (A) $1/8$ (B) $4/9$ (C) $2/3$ (D) $8/9$
8. An ant with mass m is sitting peacefully on top of a horizontal, stretched rope. The rope has mass per unit length μ and is under tension T . Some one starts a sinusoidal transverse wave of wavelength λ propagating along the rope. The motion of the rope is in a vertical plane. What minimum wave amplitude will make the ant become momentarily weightless? Assume that m is so small that the presence of the ant has no effect on the propagation of the wave.

- (A) $\frac{\mu g \lambda^2}{2\pi^2 T}$ (B) $\frac{\mu g \lambda^2}{\pi^2 T}$ (C) $\frac{2\mu g \lambda^2}{\pi^2 T}$ (D) $\frac{\mu g \lambda^2}{4\pi^2 T}$

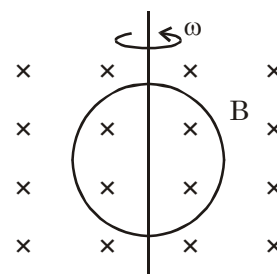
Multiple Correct Answer Type

10 Q. [4 M (-1)]

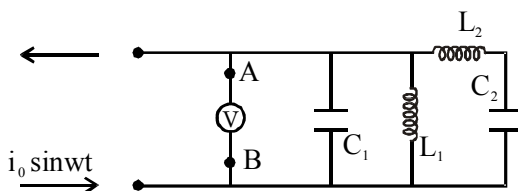
9. In the following circuit, the AC source is an ideal voltage source. Resonance is defined as a situation where the current through the source and the voltage across the source are in the same phase. At resonance, which of the following will be in the same phase as that of the applied voltage? (V_R voltage across resistor, I_R current across resistor, I_C current across capacitor, I_L current across inductor)



- (A) V_R (B) I_C (C) I_L (D) I_R
10. A ring of thin wire with active resistance 30Ω and inductance $1H$ rotates with constant angular velocity 40 rad/s in an external uniform magnetic field perpendicular to the rotation axis. In the process, the magnetic flux due to external magnetic field alone across the ring varies with time as $\phi = 5 \sin(40t)$, where all physical quantities are in S.I. units. Then :
- (A) Maximum value of current in the ring is 4 amp.
(B) Current will be zero when flux will be maximum.
(C) Average power loss is 400 W.
(D) Power loss will be maximum when flux will be maximum.



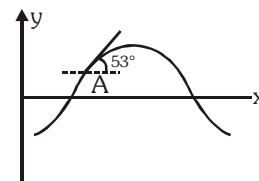
11. A source of current $i_0 \sin \omega t$, with i_0 as constant, is connected to the circuit shown in the diagram. The frequency ω is controllable. The inductors L_1 , L_2 and capacitors C_1 , C_2 are all ideal. An ideal voltmeter is connected between points A and B. Choose **CORRECT** option(s).



- (A) For very small ω (but non zero), reading of voltmeter is $\frac{i_0 \omega L_1}{\sqrt{2}}$
- (B) For very small ω (but not zero), reading of voltmeter is $\frac{i_0}{\sqrt{2} \omega C_1}$
- (C) For very large ω , reading of voltmeter is $\frac{i_0 \omega L_1}{\sqrt{2}}$
- (D) For very large ω , reading the voltmeter is $\frac{i_0}{\sqrt{2} \omega C_1}$

12. One end of a string of length L is tied to the ceiling of a lift accelerating upwards with an acceleration $2g$. The other end of the string is free. The linear mass density of the string varies linearly from 0 to λ from bottom to top.
- (A) The velocity of the wave in the string will be 0.
 (B) The acceleration of the wave on the string will be $3g/4$ every where.
 (C) The time taken by a pulse to reach from bottom to top will be $\sqrt{8L/3g}$.
 (D) The time taken by a pulse to reach from bottom to top will be $\sqrt{4L/3g}$.

13. Curve 'y-x' at an instant for a wave travelling along x-axis on a string is shown. Slope at the point A on the curve, as shown, is 53° .
- (A) Transverse velocity of the particle at point A is positive if the wave is travelling along positive x-axis.
 (B) Transverse velocity of the particle at point A is positive if the wave is travelling along negative x-axis of the particle at point A
 (C) Magnitude of transverse velocity of the particle at point A is greater than wave speed.
 (D) Magnitude of transverse velocity of the particle at point A is lesser than wave speed.



14. The figures represent two snaps of a travelling wave on a string of mass per unit length, $\mu = 0.25 \text{ kg/m}$. The two snaps are taken at time $t = 0$ and at $t = \frac{1}{24} \text{ s}$. Then the possible solution for wave are :

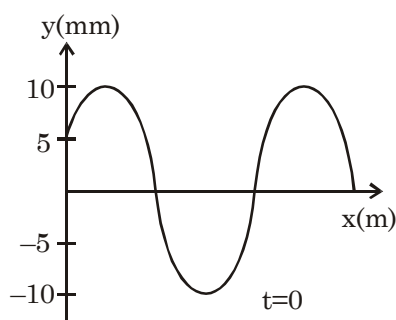


Figure-1

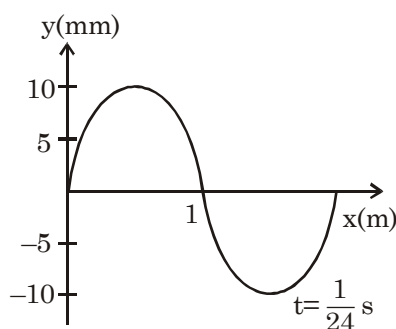


Figure-2

- (A) speed of wave is 4 m/s.
 (B) the tension in the string is 4 N
 (C) the equation of the wave is $y = 10 \sin \left(\pi x - 4\pi t + \frac{\pi}{6} \right)$
 (D) the maximum velocity of the particle = $\frac{\pi}{25} \text{ m/s}$
15. A long string of mass per unit length 0.2 kg m^{-1} is stretched to a tension of 500 N. Choose the **CORRECT** option(s):
- (A) speed of transverse waves on string is 50 ms^{-1}
 (B) speed of transverse waves on string is 25 ms^{-1}
 (C) mean power required to maintain a travelling wave of amplitude 10 mm and wavelength 0.5 m is 197 W
 (D) mean power required to maintain a travelling wave of amplitude 10 mm and wavelength 0.5 m is 150 W

16. Consider a travelling simple harmonic wave on a string of mass per unit length μ and tension T . Kinetic energy per unit length is given by $u_k = \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2$ and potential energy per unit length is given by $u_p = \frac{1}{2}T\left(\frac{\partial y}{\partial x}\right)^2$. Mark the **CORRECT** option(s).
- (A) Power transmitted by wave equals $(u_k + u_p)\sqrt{\frac{T}{\mu}}$.
 (B) u_k and u_p simultaneously attain their maximum and minimum values.
 (C) Total energy per unit length of a string is constant when a harmonic wave travels on it.
 (D) A small part of string has maximum potential energy when it is at its equilibrium position.
17. Two waves given by equation $y_1 = A \sin(\omega t - kx)$ and $y_2 = 4A \sin(2\omega t - 4kx)$ are traveling in two different strings having same mass per unit length and same cross-sectional area. Tension in first string is four times tension in the second string.
- (A) Ratio of average energy density U_1/U_2 is $1/64$ (B) Ratio of average intensity I_1/I_2 is $1/32$
 (C) Ratio of average power P_1/P_2 is $1/16$ (D) Ratio of velocity of wave v_1/v_2 is $1/2$
18. A sinusoidal wave of amplitude A and wavelength λ is incident from heavier string on a joint between two strings of which one is heavy and another is light. Choose correct option.
- (A) the amplitude of transmitted wave is more than that of the incident wave
 (B) the wavelength of the transmitted wave is more than that of the incident wave.
 (C) the amplitude of the reflected wave is less than the amplitude of incident wave.
 (D) the wavelength of the reflected wave is same as that of the incident wave.

SECTION-II

Numerical Answer Type Question (upto second decimal place)

1 Q. [3(0)]

1. A rope AB of length L placed on a smooth horizontal surface and pulled by a horizontal force F at end B. The mass of rope is m . If a transverse wave travels from A to B in time T , the value of $\frac{9T^2F}{mL}$ is.



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

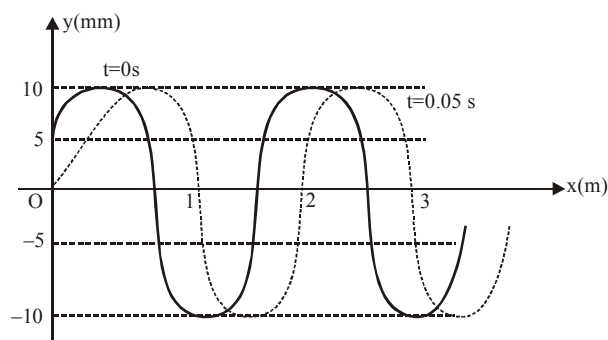
1. The figure represents two snaps of a travelling wave on a string of mass per unit length $\mu = 0.25 \text{ kg/m}$. The first snap is taken at $t = 0$ and the second is taken at $t = 0.05 \text{ s}$. Match the entries in column I with column II. [All the entries are in SI unit]

Column-I

- (A) Speed of the wave.
 (B) Frequency of the wave.
 (C) Maximum speed of the particle.
 (D) Tension in the string

Column-II

- (P) $\frac{25}{9}$
 (Q) $\frac{5}{3}$
 (R) $\frac{10\pi}{3}$
 (S) $\frac{10}{3}$
 (T) $\frac{\pi}{30}$



SECTION-I

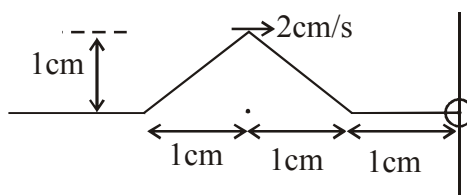
Single Correct Answer Type

4 Q. [3 M (-1)]

- Statement 1 :** $y = 2A \sin kx \cos \omega t$ refers to a travelling wave along -ve x-direction.
and
Statement 2 : When a continuous travelling wave interacts with its reflection from a rigid support, forms a standing wave.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.
- A standing wave given by the equation $y = 2A \sin \left(\frac{\pi x}{L} \right) \sin \omega t$ is formed between the points $x = -L$ and $x = L$. What is the minimum separation between two particles whose maximum velocity is half the maximum velocity of the antinodes :-

(A) $\frac{L}{12}$ (B) $\frac{L}{6}$ (C) $\frac{L}{3}$ (D) $\frac{2L}{3}$
- A triangular pulse moving at 2cm/s on a rope approaches an end at which it is free to slide on vertical pole. What is the particle speed at the free end at $\frac{3}{4}$ sec from the instant shown.



- (A) 2cm/s (B) 1cm/s (C) 3cm/s (D) 4cm/s
- The equation of a plane progressive wave is $y = 0.02 \sin 8\pi \left[t - \frac{x}{20} \right]$. When it is reflected at a rarer medium (medium with higher velocity) at $x = 0$, its amplitude becomes 75% of its previous value. The equation of the reflected wave is :

(A) $y = 0.02 \sin 8\pi \left[t - \frac{x}{20} \right]$ (B) $y = 0.02 \sin 8\pi \left[t + \frac{x}{20} \right]$
(C) $y = +0.015 \sin 8\pi \left[t + \frac{x}{20} \right]$ (D) $y = -0.015 \sin 8\pi \left[t + \frac{x}{20} \right]$

Multiple Correct Answer Type

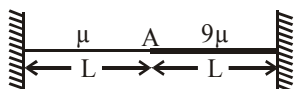
6 Q. [4 M (–1)]

5. A triangular wave pulse on a taut string travels in positive x-direction with speed v . The tension in the string is F , and linear mass density of string is μ . At $t = 0$, the shape of pulse is given by

$$y(x, 0) = \begin{cases} 0 & \text{if } x < -L \\ \frac{h(L+x)}{L} & \text{for } -L < x < 0 \\ \frac{h(L-x)}{L} & \text{for } 0 < x < L \\ 0 & \text{for } x > L \end{cases}$$

Choose the **INCORRECT** statement(s)

- (A) Magnitude of instantaneous power is zero for $-L < (x - vt) < 0$
- (B) Magnitude of instantaneous power is $Fv \left(\frac{h}{L} \right)^2$ for $(x - vt) < -L$
- (C) Magnitude of instantaneous power is $Fv \left(\frac{h}{L} \right)^2$ for $(x - vt) > L$
- (D) Magnitude of instantaneous power is $Fv \left(\frac{h}{L} \right)^2$ for $0 < (x - vt) < L$
6. Which of the following statement(s) are **TRUE** about a stationary wave ?
- (A) No particle will move with same amplitude.
- (B) All particles between two successive nodes reach their extreme position together in phase.
- (C) Displacement and velocity nodes co-exist.
- (D) Velocity nodes and pressure nodes co-exist.
7. String of length L whose one end is at $x = 0$, vibrates according to the relations given by different equations. Choose the **CORRECT** statement(s).
- (A) $y = A \sin \frac{\pi x}{L} \sin \omega t$ has 1 antinodes, 2 nodes
- (B) $y = A \cos \frac{\pi x}{L} \sin \omega t$ has 2 antinodes, 1 nodes
- (C) $y = A \sin \frac{2\pi x}{L} \sin \omega t$ has 3 nodes, 2 antinodes
- (D) $y = A \cos \frac{2\pi x}{L} \sin \omega t$ has 3 antinodes, 2 nodes
8. A string of length L fixed at both ends is vibrating in lowest mode of its vibration for which particle at $x = L/6$ from one end is having maximum displacement. The frequency of vibration of this mode is 120Hz. For this situation, mark out **CORRECT** statement(s) :
- (A) The particle at $x = L/2$ will have maximum displacement
- (B) The given mode is 2nd overtone
- (C) The next higher overtone for which same particle at $x = L/6$ has maximum displacement is 6th one
- (D) The next higher overtone for which same particle at $x = L/6$ has maximum displacement is 8th one
9. A light string is tied at one end to fixed support and to a heavy string of equal length L at the other end as shown in figure. Mass per unit length of the strings are μ and 9μ and the tension is T . Find the possible values of frequencies such that point A is an antinode.



(A) $\frac{1}{4L} \sqrt{\frac{T}{\mu}}$

(B) $\frac{3}{4L} \sqrt{\frac{T}{\mu}}$

(C) $\frac{3}{2L} \sqrt{\frac{T}{\mu}}$

(D) $\frac{7}{6L} \sqrt{\frac{T}{\mu}}$

10. For a wave travelling along a straight line, the distance between two points in the same phase is 5 m, while the distance between two points that are in the opposite phase is 1.5m.
- (A) Distance between points in same phase may be 0.5 m
(B) Distance between points in opposite phase may be 0.5 m
(C) Distance between points in same phase may be 1m
(D) Distance between points in opposite phase may be 1m

Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Questions no. 11 to 13

Two pulses traveling on the same string are described by the functions

$$y_1 = \frac{5}{(3x - 4t)^2 + 2} ; y_2 = \frac{-5}{(3x + 4t - 6)^2 + 2} \text{ where } y_1, y_2 \text{ and } x \text{ are in metre and } t \text{ in second}$$

11. In which direction does each pulse travel ?
(A) y_1 travels along +x direction while y_2 travels along -x direction
(B) y_1 travels along -x direction while y_2 travels along +x direction
(C) Both y_1 and y_2 travel along +x direction
(D) Both y_1 and y_2 travel along -x direction
12. At what time resultant displacement of particle will be zero due to superposition of both pulse?
(A) $t = 0.50$ sec (B) $t = 0.75$ sec (C) $t = 1.50$ sec (D) $t = 1.75$ sec
13. At what point the resultant displacement of particles is zero due to superposition of both pulse
(A) $x = 1.0$ m (B) $x = 2.0$ m (C) $x = 3.0$ m (D) $x = 4.0$ m

SECTION-II

Numerical Answer Type Question

3 Q. [3(0)]

(upto second decimal place)

1. Sinusoidal progressive wave of amplitude 10 cm is to be sent along a string that has linear mass density of 4×10^{-2} kg/m and a tension of 100 N. The source can deliver a maximum average power of 40 W. What is the maximum frequency (in Hz) at which source can operate? Take $\pi^2 = 10$
2. A string is stretched along the x-axis. There is a transverse disturbance along the string :

$$\psi(x, t) = 4 \sin\left(\frac{\pi x}{3}\right) \cos(10\pi t) [\text{cm}]$$

where x is measured in centimeter and t in seconds.

Assume the velocities and amplitudes are equal for the two waves.

The velocity (in cm/s) of the two waves moving towards each other which create this disturbance is

3. A wave generator at one end of a very long string creates a wave given by

$$y_1 = (6.0 \text{ cm}) \cos \frac{\pi}{2} [(2.00 \text{ m}^{-1})x + (8.00 \text{ s}^{-1})t]$$

and a wave generator at the other end creates the wave

$$y_2 = (6.0 \text{ cm}) \cos \frac{\pi}{2} [(2.00 \text{ m}^{-1})x - (8.00 \text{ s}^{-1})t].$$

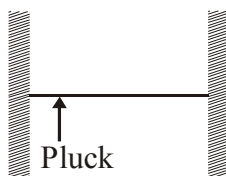
for $x > 0$, what is the location of the node (in cm) having the smallest value of x.

SECTION-III

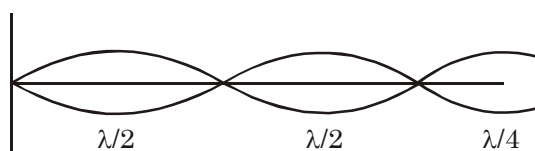
Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. A steel wire is rigidly fixed at both ends. Its length, mass and cross-sectional area are 1m, 0.1kg and 10^{-6} m^2 respectively. Then the temperature of the wire is lowered by 20°C . If transverse waves are set up by plucking the wire at 0.25 m from one end and assuming that the wire vibrates with minimum number of loops possible for such a case. Find the frequency of vibrations (in Hz). If your answer is N fill value of N/11. Given the coefficient of linear expansion of steel $= 1.21 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, and Young's modulus $= 2 \times 10^{11} \text{ N/m}^2$.



2. A string of length ℓ fixed at one end and free at another end is vibrating in 5th harmonic. Maximum displacement of particle is A. Amplitude of oscillation of a particle at distance $\frac{\ell}{10}$ from free end is $\frac{A}{\sqrt{\alpha}}$ find value of α .

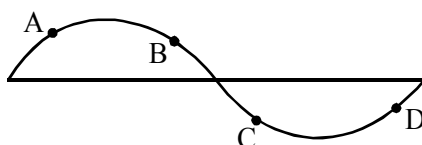


SECTION-IV

Matrix Match Type (4 × 5)

2 Q. [8 M (for each entry +2(0))]

1. Figure shows a snapshot of a wave travelling towards right. Four points A, B, C, D have been shown on string.



Column-I

- (A) Point-A
(B) Point-B
(C) Point-C
(D) Point-D

Column-II

- (P) Energy density is increasing with time
(Q) Speed is increasing with time
(R) Speed is decreasing with time
(S) Power is being transmitted towards right
(T) Strain is increasing with time

2. In case of mechanical wave a particle oscillates and during oscillation its kinetic energy and potential energy changes.

Column-I

- (A) When particle of travelling wave is passing through mean position.
(B) When particle of travelling wave is at extreme position.
(C) When particle between node and antinode in standing wave is passing through mean position
(D) When particle between node and antinode in standing wave is at extreme position

Column-II

- (P) Kinetic energy is maximum
(Q) Potential energy is maximum
(R) Kinetic energy is minimum
(S) Potential energy is minimum

SECTION-I

Single Correct Answer Type

7 Q. [3 M (-1)]

1. A string of length L consists of two distinct sections. The left half has linear mass density $\mu = \mu_0/2$, while the right half has linear mass density $\mu' = 3\mu_0$. Tension in the string is F_0 . The time required for a transverse wave pulse to travel from one end of the string to the other is

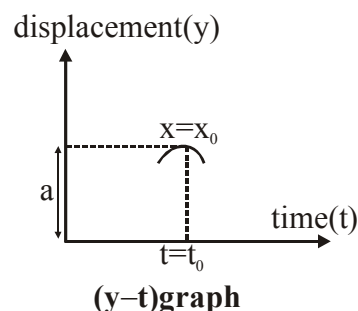
(A) $\frac{L}{4} \sqrt{\frac{\mu_0}{F_0}} (\sqrt{2} + \sqrt{6})$ (B) $\frac{L}{2} \sqrt{\frac{2\mu_0}{F_0}} (1 + \sqrt{3})$ (C) $\frac{L}{2} \sqrt{\frac{\mu_0}{2F_0}} (\sqrt{2} + \sqrt{6})$ (D) $\frac{L}{2} \sqrt{\frac{\mu_0}{2F_0}} (1 + \sqrt{6})$

2. A string on a musical instrument is held under tension T and extends from the point $x = 0$ to point $x = L$. The string is overwound with a wire in such a way that its mass per unit length $\mu(x)$ increases uniformly from μ_0 at $x = 0$ to μ_L at $x = L$. Find the time interval required for a transverse pulse to travel the length of the string is

(A) $\Delta t = \frac{3L(\mu_L + \mu_0 + \sqrt{\mu_L\mu_0})}{2\sqrt{T}(\sqrt{\mu_L} - \sqrt{\mu_0})}$ (B) $\Delta t = \frac{3L(\mu_L + \mu_0 - \sqrt{\mu_L\mu_0})}{2\sqrt{T}(\sqrt{\mu_L} + \sqrt{\mu_0})}$

(C) $\Delta t = \frac{3L(\mu_L + \mu_0 + \sqrt{\mu_L\mu_0})}{2\sqrt{T}(\sqrt{\mu_L} + \sqrt{\mu_0})}$ (D) $\Delta t = \frac{2L(\mu_L + \mu_0 + \sqrt{\mu_L\mu_0})}{3\sqrt{T}(\sqrt{\mu_L} + \sqrt{\mu_0})}$

3. A progressive simple harmonic wave is moving in air along the x -axis. The part of this wave at a given point $x = x_0$ from the source and at a certain instant $t = t_0$ has the waveform shown below in the displacement time (y - t) graph. Velocity of the wave has value v_0 and its angular velocity is ω . Which of the following equations will correctly represent the complete wave at x_0 agreeing with above wave forms?



(A) $y = -a \left[\cos \left\{ \frac{2\pi}{T} (t - t_0) \right\} - \frac{\pi}{2} \right]$ (B) $y = -a \left[\sin \left\{ \frac{2\pi}{T} (t - t_0) \right\} + \frac{\pi}{2} \right]$

(C) $y = a \left[\sin \left\{ \frac{2\pi}{T} (t - t_0) \right\} + \frac{\pi}{2} \right]$ (D) $y = a \left[\sin \left\{ \frac{2\pi}{T} (t - t_0) \right\} - \frac{\pi}{2} \right]$

4. A heavy but uniform rope of length L is suspended from a ceiling. A particle is dropped from the ceiling at the instant when the bottom end is given a transverse wave pulse. Where will the particle meet the pulse

(A) At a distance $\frac{2L}{3}$ from the bottom (B) At a distance $\frac{L}{3}$ from the bottom

(C) At a distance $\frac{3L}{4}$ from the bottom (D) None of the above

5. A string with a mass density of $4 \times 10^{-3} \text{ kg/m}$ is under tension of 360 N and is fixed at both ends. One of its resonance frequencies is 375 Hz. The next higher resonance frequency is 450 Hz. The mass of the string is :-

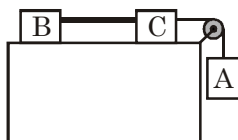
(A) $2 \times 10^{-3} \text{ kg}$ (B) $3 \times 10^{-3} \text{ kg}$ (C) $4 \times 10^{-3} \text{ kg}$ (D) $8 \times 10^{-3} \text{ kg}$

6. A string is under tension and stationary vibrations are produced in it. A, B, C, D are positions of consecutive nodes. Choose the **INCORRECT** option.
- (A) The distance $AB = \frac{\lambda}{2}$
- (B) The phase difference between a vibrating point on AB and another on BC is zero.
- (C) The amplitudes of vibration of different points between A and the midpoint of A and B are different.
- (D) When every point of AB is having maximum displacement then every point of BC is having maximum displacement in opposite direction.
7. A wave equation is represented as $\psi = A \sin\left(\alpha\left(\frac{x-y}{2}\right)\right) \cos\left(\omega t - \alpha\left(\frac{x+y}{2}\right)\right)$, where x & y are in meter and t is second then :-
- (A) The wave is stationary wave
- (B) The wave is a progressive wave propagating along +x axis
- (C) The wave is a progressive wave propagating at right angle to the +x axis
- (D) All point lying on line $y = x + \frac{4\pi}{\alpha}$ are always at rest.

Multiple Correct Answer Type

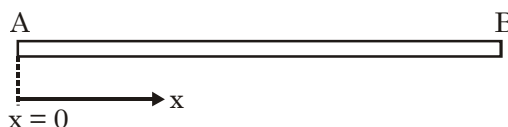
7 Q. [4 M (-1)]

8. The wave function for the wave pulse is $y(x, t) = \frac{0.1C^3}{C^2 + (x - vt)^2}$, with $C = 4$ cm at $x = 0$ the displacement $y(x, t)$ is observed to decrease from its maximum value to half of its maximum value in time $t = 2 \times 10^{-3}$ s.
- (A) $y(x, t)$ represents the motion of travelling pulse moving along positive X direction
- (B) $y(x, t)$ does not represent the motion of a travelling pulse
- (C) Speed of wave pulse is 20 m/s
- (D) Speed of wave pulse is 10 m/s
9. A block string system is shown in the figure. Mass of block A, B and C is 5 Kg, 2 Kg, and 2 Kg respectively. Mass of rod is given as 1 kg and its length is 1 m. A wave transverse is transmitted in the rod in between the block B and C once in forward and once backward direction. Choose the **CORRECT** statements.

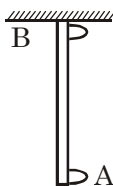


- (A) Time for string wave to reach one end to other end in forward direction is $\frac{2}{\sqrt{5}}(\sqrt{3} - \sqrt{2})$ sec
- (B) Time for waves to reach one end to other end is same for both in forward and backward direction
- (C) Time for wave to reach one end to other end is different for forward and backward direction
- (D) Time for string wave to reach one to other end in backward direction is $\frac{1}{10}(15)^{1/2}$ sec.

10. Consider a rope AB of non-uniform density as shown in figure. If we consider direction A to B as the direction of increasing x , with $x = 0$, at A, linear mass density of rope is given by $\mu = \left(\frac{x}{\ell}\right)^n \mu_0$ where n is a constant ($n \geq 0$) and ℓ is the length of the rope.



Now the rope is hanged vertically from a roof keeping end B attached to the roof and one pulse is produced at the lower end A and other pulse is attached at upper end B as shown in figure.



Choose the **CORRECT** statement(s) :-

- (A) Time taken by both pulse to reach other end is same.
 (B) Time taken by pulse to reach other end increases as value of n increases.
 (C) Acceleration of both pulse is constant during their journey from A to B or B to A for all values of n .
 (D) Acceleration of pulse does not depend on value of n .
11. In which case is the power being delivered by a given progressive sinusoidal wave on a given string is doubled?
 (A) The wave amplitude is doubled (keeping the frequency the same).
 (B) The wave frequency is cut in half (keeping the amplitude the same).
 (C) The string is made four time as taut (keeping its linear density the same).
 (D) The diameter of the string is doubled (keeping the tension the same).
12. Two very long strings are tied together at the point $x = 0$. In the region $x < 0$, the wave speed is v_1 , while in the region $x > 0$, the speed is v_2 . A sinusoidal wave is incident on the knot from the left ($x < 0$); part of the wave is reflected and part is transmitted. For $x < 0$, the displacement of the wave is described by $y(x, t) = A \sin(k_1 x - \omega t) + B \sin(k_1 x + \omega t)$, while for $x > 0$, $y(x, t) = C \sin(k_2 x - \omega t)$, where $\omega/k_1 = v_1$ and $\omega/k_2 = v_2$. Which of the following is/are **CORRECT** :

(A) $\frac{C}{A} = \frac{2v_2}{(v_1 + v_2)}$ (B) $\frac{B}{A} = \frac{(v_2 - v_1)}{(v_1 + v_2)}$ (C) $B^2 + \frac{v_1}{v_2} C^2 = A^2$ (D) $A^2 + \frac{v_1}{v_2} C^2 = B^2$

13. A waveform : $y_1 = A \sin\left(2x - 4t + \frac{\pi}{3}\right)$ is superposed with a second waveform, to produce a standing wave with a node at $x = 0$. The equation of the second waveform can be :-

(A) $y_2 = A \sin\left(2x + 4t + \frac{5\pi}{3}\right)$ (B) $y_2 = A \sin\left(2x - 4t + \frac{\pi}{3}\right)$
 (C) $y_2 = A \sin\left(2x + 4t - \frac{\pi}{3}\right)$ (D) $y_2 = A \sin\left(2x + 4t + \frac{\pi}{3}\right)$

14. Consider standing wave formed due to superposition of two sinusoidal plane waves having the same amplitude, frequency and moving in opposite direction. Mark the correct statements :
- (A) the energy of a standing wave transforms completely into potential energy and at some other instant into kinetic energy during a part of complete cycle.
- (B) kinetic energy and potential energy attain their maximum and minimum values simultaneously
- (C) there is transition of energy from each node to its adjacent antinodes and back.
- (D) the time averaged energy flux in any cross section of the wave is zero.

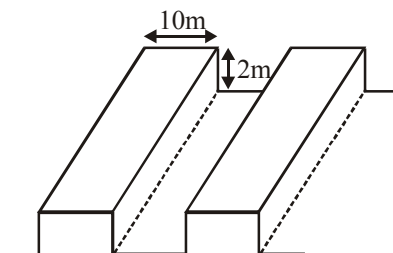
Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Questions No. 15 and 16

The world is focusing its attention on renewable sources of energy like solar energy, wind energy, tidal energy and wave energy. These sources are non-polluting, do not cause the emission of greenhouse gases, or cause any large scale damage to the ecology or environment.

Waves on the surface of the ocean are a good source of power. To illustrate this, we calculate the mechanical energy carried by an average wave of crest 1m, wavelength 20 m and a period of 5s. The wave profile is taken as approximately step-like, instead of a sinusoidal function.

A simple minded calculation gives us a contribution of 200 kW from the release of potential energy by such a wave over a 1 m wavefront.



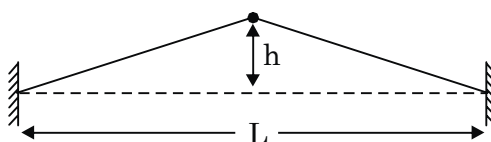
15. The speed of the wave is
(A) 100 ms^{-1} (B) 4 ms^{-1} (C) 0.25 ms^{-1} (D) none of these
16. Wave energy provides an inexpensive source of power. In the paragraph above, only the potential energy carried by the wave was calculated. The contribution to power due to kinetic energy assuming that all the water in the crest is moving forward at the speed of the wave gives us, over a 1 m wavefront approximately, of the order of
(A) 100 W (B) 10^3 W (C) 10^4 W (D) 10^5 W

SECTION-II

Numerical Answer Type Question
(upto second decimal place)

1Q.[3(0)]

1. Consider an elastic string stretched between two fixed ends A and B as shown in figure. The speed of transverse waves in the string is v and its linear mass density is μ . The string is plucked and held in a triangular form with maximum height $h \ll L$ at its middle point. At $t = 0$, the plucked string is released. All effects due to gravity may be neglected. Let T be the period of string's vibrations. If total mechanical energy of the vibrating string can be expressed either $\frac{P\mu v^2 h^2}{L}$ or $\frac{Q\mu L h^2}{T^2}$ where P and Q are integer. Find $P \times Q$.

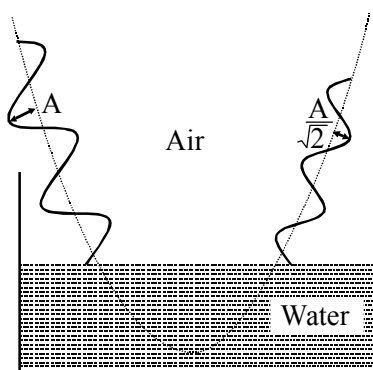


SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. A thin string of density 0.1 kg/m is held at one end and the other end is oscillated according to the equation $y(x=0, t) = 20 \text{ (cm)} \sin(40t)$ where t is in seconds. The string is under a tension of 10 N . The string passes through a bath filled with $\frac{40}{21} \text{ kg}$ of water. Due to friction, heat is transferred through the bath and a snapshot at sometime is given below. After 100α second the temperature of the bath rises by one Kelvin. Then find the value of α . [Specific heat capacity of water = 4.2 kJ/kgK]



2. In a wave, at any instant there is an incident wave and a partially reflected wave present. The ratio of maximum to minimum displacement of particles is $\frac{3}{2}$. What percent of incident energy is reflected ?

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. A triangular wave pulse on a taut string travels in the positive x -direction with speed v . The tension in the string is F , and the linear mass density of the string is μ . At $t = 0$, the shape of the pulse is given by

$$y(x, 0) = \begin{cases} 0 & \text{if } x < -L \\ h(L+x)/L & \text{for } -L < x < 0 \\ h(L-x)/L & \text{for } 0 < x < L \\ 0 & \text{for } x > L \end{cases}$$

Match information of column-I with column-II (Assume h to be positive).

Column-I

- (A) For $(x - vt) < -L$
(B) For $-L < (x - vt) < 0$
(C) For $0 < (x - vt) < L$
(D) For $(x - vt) > L$

Column-II

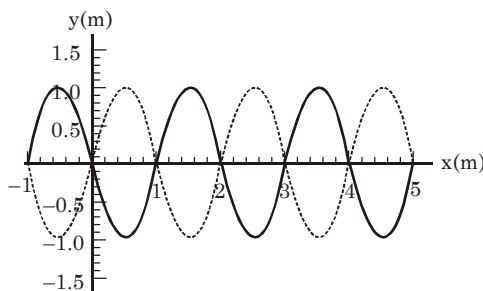
- (P) Slope of string is negative.
(Q) Instantaneous power transmitted is non-zero constant.
(R) Particle velocity is positive.
(S) Particle velocity is negative.
(T) Slope of string is zero.

SECTION-I

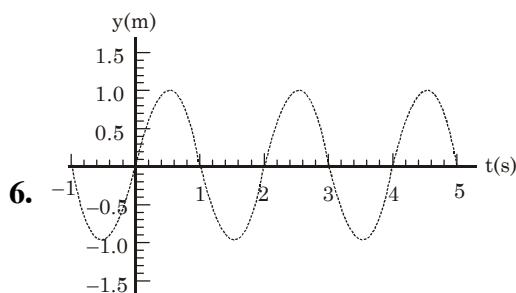
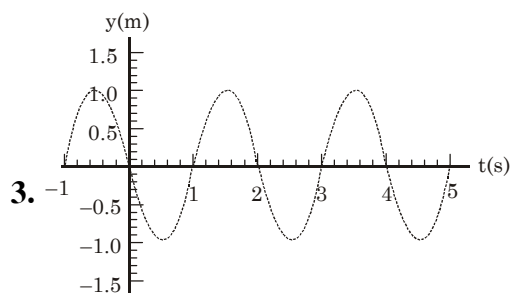
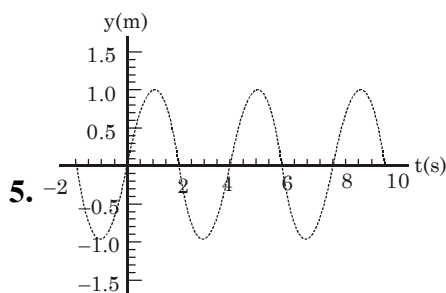
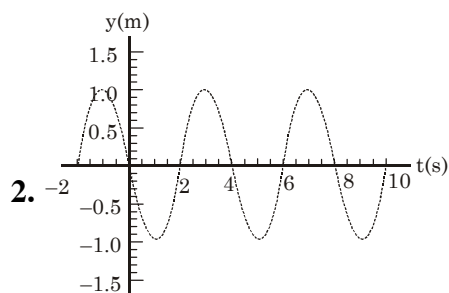
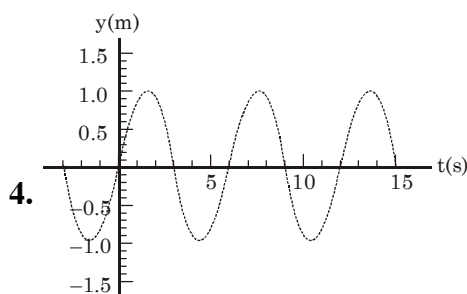
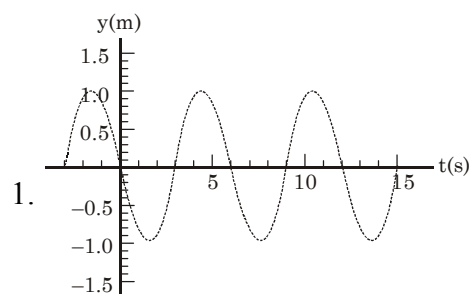
Single Correct Answer Type

4 Q. [3 M (–1)]

1. The figure to the right shows a travelling wave moving in the positive x-direction. At time $t = 0$ s, the wave plotted as a function of position has the shape shown by the solid curve, and at $t = 3$ s, the shape shown by the dashed curve.



The following six panels show plots as a function of time, at the position $x = 0$ m.



Which of the panels represent possible shapes of the travelling wave ?

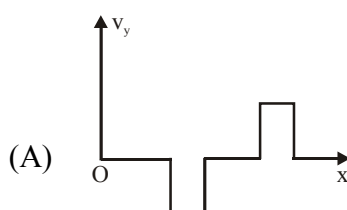
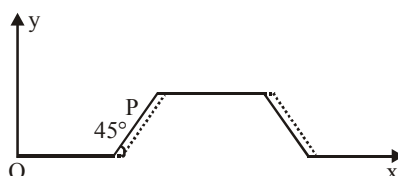
- (A) 4 & 6 (B) 1 & 3 (C) 1, 2 & 3 (D) 4, 5 & 6

2. How long will it take for the sound wave to travel a distance 1m from point P to point Q, If the temperature between these two points changes linearly from T_0 to $4T_0$ and the velocity of sound propagation in air is $V = k\sqrt{T}$, where K is a constant :
- (A) $\frac{1}{2k\sqrt{T_0}}$ (B) $\frac{\sqrt{T_0}}{2k}$ (C) $\frac{2}{3k\sqrt{T_0}}$ (D) $\frac{1}{3kT_0}$
3. 200 students are taking an examination in a room, and the sounds of pens scratching on paper, sighs, groans, and muttered imprecations has created a more or less continuous sound level of this noise of 60dB. Assuming each student contributes equally to this noise and nothing else changes or adds to it, what will the sound level in the room be when only 50 students are left ? ($\log_{10} 2 = 0.3$)
- (A) 50 dB (B) 15 dB (C) 66 dB (D) 54 dB
4. Standing waves are set up in a string of length 240cm clamped horizontally at both ends. The separation between any two consecutive points where displacement amplitude is $3\sqrt{2}$ cm is 20cm. The standing waves were set by two travelling waves of equal amplitude of 3 cm. The overtone in which the string is vibrating will be :-
- (A) 2nd (B) 3rd (C) 4th (D) 5th

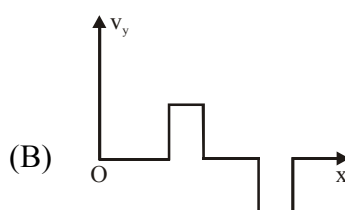
Multiple Correct Answer Type

7 Q. [4 M (-1)]

5. Sound waves from different source are interfering and pressure at same point in space is given as –
 $P = P_0 \sin (200 \pi t + \pi/2) \cos (3 \pi t + \pi/3)$
- (A) The frequency of one source is above 100Hz and other source is below 100Hz.
 (B) Beat frequency is 3 Hz
 (C) Beat frequency is 200 Hz
 (D) Beat frequency can be 3 Hz as well as 200 Hz
6. The figure shows a taut string, initially aligned with the x-axis, which carries a waveform travelling towards the right without any change in its shape. The bold line and the dotted line indicate the displacement of the string (i.e., shape of the waveform) at time = t second and at time = (t + dt) second, respectively. The speed of particle at point P at t = 2 sec is 1 cm/s. Choose the **CORRECT** statement(s) from the following (assume that dt is positive) :-



Particle velocity as a function of position



Particle velocity as a function of position

- (C) Speed of waveform is 1 cm/s and it is moving along positive x-axis
 (D) Speed of waveform is 1 cm/s and it is moving along negative x-axis.

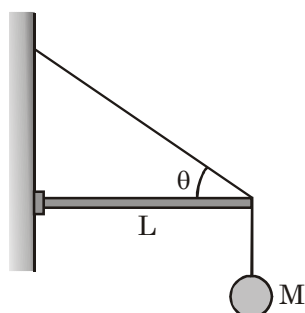
7. A standing wave is set up in a string of variable length and tension by a vibrator of variable frequency. Both ends of the string are fixed. When the vibrator has a frequency f , in a string of length L and under tension T , n antinodes are set up in the string. Mark the correct statement(s) :

- (A) If the length of the string is doubled, the frequency should be changed by a factor of $\frac{1}{2}$ so that same number of antinodes are produced.
 (B) If the length of the string is doubled, the frequency should be changed by a factor of 2 so that same number of antinodes are produced.
 (C) If the frequency and length are held constant, tension needed to produce $n + 1$ antinodes is given by

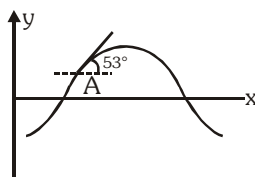
$$\left[\frac{n}{(n+1)} \right]^2 T$$

- (D) If the frequency is tripled and the length of the string is halved, then tension should be changed by factor of $\frac{9}{16}$ so that twice as many antinodes are produced.

8. A sphere of mass M is supported by a string that passes over a light horizontal rod of length L . Given that the angle is θ and that f represents the fundamental frequency of standing waves in the portion of the string above the rod. This frequency is excited using a tuning fork of frequency f .



- (A) the mass of this vibrating portion of the string is $\frac{Mg \cos \theta}{4f^2 L}$
 (B) if we increase θ without changing other parameters of the problem, the wire can oscillate in its first overtone.
 (C) if we decrease the mass M without changing other parameters of the problem, the wire can oscillate in its first overtone.
 (D) if we decrease the length L without changing other parameters of the problem, the wire can oscillate in its first overtone.
9. Curve 'y-x' at an instant for a wave travelling along x-axis on a string is shown. Slope at the point A on the curve, as shown, is 53° .



- (A) Transverse velocity of the particle at point A is positive if the wave is travelling along positive x-axis.
 (B) Transverse velocity of the particle at point A is positive if the wave is travelling along negative x-axis of the particle at point A
 (C) Magnitude of transverse velocity of the particle at point A is greater than wave speed.
 (D) Magnitude of transverse velocity of the particle at point A is lesser than wave speed.

10. A harmonic pressure wave produced by a distant source is travelling through your vicinity and wavefronts that travel through your vicinity are vertical planes. Let +x direction is to be east and +y direction be toward north. The wave function of wave is :

$$p(x, y, t) = A \cos(k_x x + k_y y - \omega t)$$

(A) Direction of wave propagation is at an angle $\theta = \cot^{-1} \frac{k_y}{k_x}$ with +x axis

(B) Direction of wave propagation is at an angle $\theta = \tan^{-1} \frac{k_y}{k_x}$ with +x axis

(C) Wave speed is $\frac{\omega}{k_x + k_y}$

(D) Wave speed is $\frac{\omega}{\sqrt{k_x^2 + k_y^2}}$

11. Two coherent waves represented by $y_1 = A \sin\left(\frac{2\pi x_1}{\lambda} - \omega t + \frac{\pi}{4}\right)$ and $y_2 = A \sin\left(2\pi \frac{x_2}{\lambda} - \omega t + \frac{\pi}{6}\right)$ are superposed. The two waves will produce :

(A) constructive interference at $(x_1 - x_2) = 2\lambda$ (B) constructive interference at $(x_1 - x_2) = \frac{23}{24} \lambda$

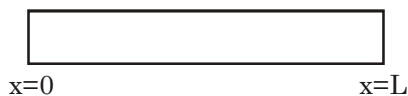
(C) destructive interference at $(x_1 - x_2) = 1.5\lambda$ (D) destructive interference at $(x_1 - x_2) = \frac{11}{24} \lambda$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Question no. 12 and 13

A uniform rod of length L is free at one end ($x = L$) and is subjected to a sinusoidal displacement $S = S_0 \sin \omega t$ at the other end ($x = 0$). As a result, each point on the rod oscillates with same frequency ω but with different amplitude. Here Y represents Young's modulus and ρ is density of the rod. Normally, the motion will be damped and the amplitude will be less than the values predicted. But here we will be ignoring this effect. Using these conditions,



12. The displacement s at (x, t) is given by

(A) $s = s_0 \cos\left(\frac{\omega x}{v}\right) \left[\cos\left(\frac{\omega L}{v}\right) + \tan\left(\frac{\omega x}{v}\right) \times \sin\left(\frac{\omega L}{v}\right) \right] \sin \omega t$

(B) $s = s_0 \cos\left(\frac{\omega x}{v}\right) \left[1 + \tan\left(\frac{\omega L}{v}\right) \tan\left(\frac{\omega x}{v}\right) \right] \sin \omega t$

(C) $s = s_0 \cos\left(\frac{\omega x}{v}\right) \left[1 + \sin\left(\frac{\omega L}{v}\right) \tan\left(\frac{\omega x}{v}\right) \right] \sin \omega t$

(D) $s = s_0 \cos\left(\frac{\omega x}{v}\right) \left[1 + \tan\left(\frac{\omega L}{v}\right) \sin\left(\frac{\omega x}{v}\right) \right] \sin \omega t$

13. If $Y = 10^{12} \text{ N/m}^2$, $\rho = 10^4 \text{ kg/m}^3$, $L = 10 \text{ m}$, resonant frequencies (at which s becomes very large) are
 (A) 500 Hz, 1000 Hz, (B) 250 Hz, 750 Hz,
 (C) 125 Hz, 375 Hz, (D) 250 Hz, 500 Hz,

Matching list based comprehension Type (4 × 4 × 4)

1 Table × 3 Q. [3(-1)]

Answer Q.14, Q.15 and Q.16 by appropriately matching the information given in the three columns of the following table.

Coloumn-1 Equation of wave	Column-2 Energy in one wavelength	Column-3 Average power transmitted at point X
(I) $y = A \sin\left(2\pi vt - \frac{2\pi}{v}vx + \frac{5\pi}{6}\right)$	(i) $2\pi^2\mu A^2v^2\lambda$	(P) 0
(II) $y = A \sin\left(\frac{5\pi}{2\ell}x\right)\cos(2\pi vt)$	(ii) $4\pi^2\mu A^2v^2\lambda$	(Q) $2\pi^2\mu v A^2v^2$
(III) $y = A \sin\left(\frac{4\pi}{\ell}x\right)\cos(2\pi vt)$	(iii) $\pi^2\mu A^2v^2\lambda$	(R) $\frac{\pi^2\mu v A^2v^2}{2}$
(IV) $y = A \sin\left(\frac{2\pi}{v}vx - 2\pi vt + \frac{\pi}{6}\right)$	(iv) $\frac{\pi^2\mu A^2v^2\lambda}{2}$	(S) $\frac{\pi\mu v A^2v^2}{4}$

14. A progressive wave is moving in the positive x direction with speed v. The amplitude of the wave is A and frequency of the source is v. At t = 0, the particle at $x = \frac{\lambda}{3}$ is located at $y = +\frac{A}{2}$ and moving away

from the mean position. Point X is located at a distance $\frac{3\lambda}{4}$ from the source.

(A) (II) (iv) (P) (B) (I) (i) (Q) (C) (III) (ii) (R) (D) (IV) (iii) (S)

15. A string of length ℓ fixed at both ends and vibrating in its 3rd overtone. The maximum amplitude of the particle at anti-node is A. At t = 0, the particle at $x = \frac{\ell}{6}$ is located at its positive extreme. Point X is

located at a distance $\frac{\ell}{2}$ from the source.

(A) (II) (iii) (Q) (B) (III) (iv) (P) (C) (III) (iii) (P) (D) (II) (iv) (R)

16. A string is vibrating in its 5th harmonic with fixed end at x = 0 and free end at x = ℓ . The maximum amplitude of the particle at antinode is A. At t = 0, the particle at $x = \frac{\ell}{4}$ is at the positive extreme. Point

X is located at a distance $\frac{\ell}{5}$ from the source.

(A) (III) (ii) (P) (B) (II) (ii) (R) (C) (II) (iv) (P) (D) (II) (iii) (P)

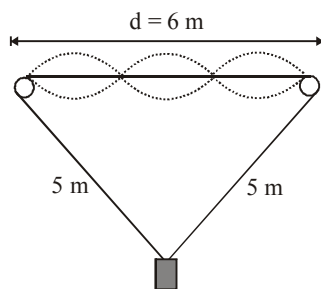
SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

1. A steel wire of length 1m, mass 0.1 kg and uniform cross-section 1 mm² is fixed rigidly at both ends that there is no tension in the wire initially. The temperature of the wire is lowered by 20°C and vibrations are set up in the wire such that three antinodes are formed between the ends. An observer moving at 70 ms⁻¹ towards the wire observes the frequency of emitted sound to be 40 Hz. Find Young's modulus of steel (in 10¹¹ N/m²). The coefficient of linear expansion of steel = 1.21 × 10⁻⁵ K⁻¹ and the velocity of sound in air is 330 ms⁻¹.

2. A string fixed at both the ends of length 2m, vibrating in its 7th overtone. Equation of the standing wave is given by $y = A \sin kx \cos (\omega t + \pi/3)$. All the symbols have their usual meaning. Mass per unit length of the string is 0.5 gm/cm. Given that $A = 1$ cm and $\omega = 100 \pi$ rad/sec. If after time t_0 starting from $t = 0$ sec, energy of vibration is completely potential for the second time. Find $120 t_0$. (Use $\pi^2 = 10$)
3. A 16 kg object hangs in equilibrium from a string with a total length of $L = 16$ m and a linear density of $\mu = 0.01$ kg/m. The string is wrapped around two light small frictionless pulleys that are separated by distance $d = 6$ m. At what frequency (in Hz) must the string between the pulleys vibrate to form the standing wave pattern with three loops on the string. If your answer is n then fill value of $\sqrt{n}$.



4. A ball of radius $R = 100$ cm is located in the way of propagation of a plane sound wave. The sonic wavelength is $\lambda = 2$ cm, the frequency is $\nu = 100\pi$ Hz, the pressure oscillation amplitude in air is $(\Delta p)_m = 4$ Pa. Find the mean energy flow rate (in watt), averaged over an oscillation period, reaching the surface of the ball. Take density of air 1 kg/m^3 .

PHYSICS

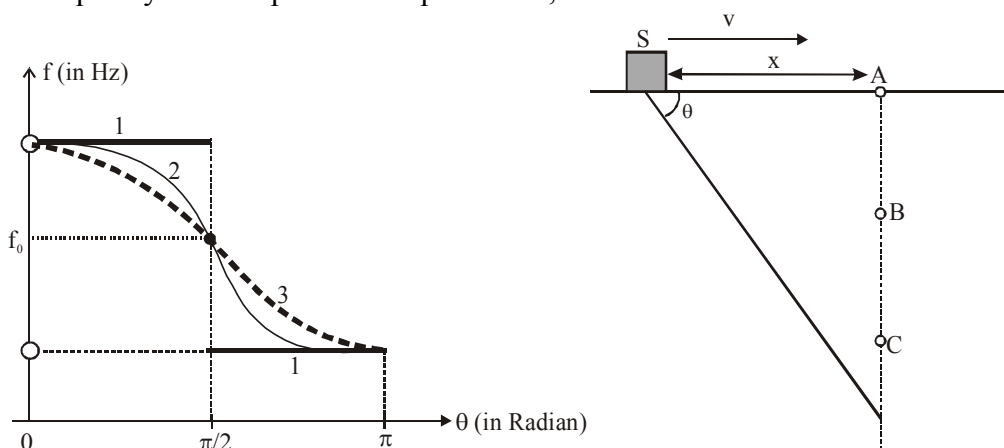
CLASS TEST # 72

SECTION-I

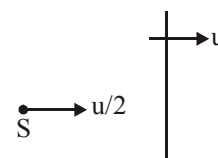
Single Correct Answer Type

6 Q. [3 M (-1)]

- A boy is running in a stadium shaped in form of circle of radius 100 m. At the centre of stadium on ground a loudspeaker emits waves of frequency 165 Hz. The ceiling of stadium is 5 m high. Waves coming directly from speaker and after one reflection from ceiling interfere. What are possible radii of boy's running circle so that he hears maximum sound. $v_{\text{sound}} = 330$ m/s.
 (A) $\left[x = \frac{25 - 2n^2}{2n} \right] n = 1, 2, 3, 4$ (B) $\left[x = \frac{50 - 2n^2}{4n} \right] n = 1, 2, 3, 4$
 (C) $\left[x = \frac{50 - 4n^2}{4n} \right] n = 1, 2, 3, 4$ (D) $\left[x = \frac{100 - 4n^2}{4n} \right] n = 1, 2, 3, 4$
- A student is performing the experiment of Resonance Column. The diameter of the column tube is $\frac{20}{3}$ cm. The frequency of the tuning fork is 1050 Hz. The air temperature is 38°C in which the speed of sound is 336 m/s. The zero of the meter scale coincides with the top end of the Resonance column tube. When the first resonance occurs, the reading of the water level in the column is
 (A) 8 cm (B) 10 cm (C) 6 cm (D) 12 cm
- A pipe open at the top end is held vertically with some of its lower portion dipped in water. At a certain depth of immersion, the air column of length $\frac{3}{8}$ m in the pipe resonates with a tuning fork of frequency 680 Hz. The speed of sound in air is 340 m/s. The pipe is now raised up by a distance 'x' until it resonates in the 'next overtone' with the same tuning fork. The value of 'x' is
 (A) 20 cm (B) 40 cm (C) 50 cm (D) 25 cm
- A source is moving with a constant speed u on a straight line, emitting a sound of frequency f_0 . There are three observers A, B and C. A on track, B at a perpendicular distance of d from the track and C at a perpendicular distance of $2d$ from the track as shown in the figure. The variation of the observed frequency with respect to the position x ,



- (A) A - 3, B - 2, C - 1 (B) A - 2, B - 3, C - 1 (C) A - 1, B - 2, C - 3 (D) A - 1, B - 3, C - 2
- A wall is moving with velocity u and a source of sound moves with velocity $u/2$ in the same direction as shown in figure. Assuming that the sound travels with velocity $10u$. The ratio of incident sound wavelength on the wall to the reflected sound wavelength by the wall, is equal to (assume observer for reflected sound is at rest) :
 (A) 9 : 11 (B) 11 : 19 (C) 4 : 5 (D) 5 : 4



6. Three musicians experiment with the Doppler effect. Musician A rides in a car at a speed u directly away from musician B who is stationary. Musician C rides in a car directly toward B and travels at the same speed as A. Musician A plays a note at frequency f_A on his trumpet. B hears the note, adjusts his trumpet, and plays the same note he heard. Choose the incorrect statement :

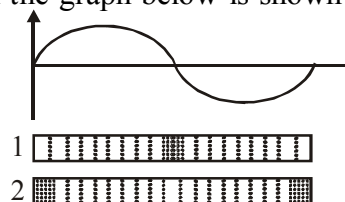


- (A) The note heard by C coming from B will be the same pitch as f_A .
(B) The note heard by A coming from B will be higher in pitch than f_A .
(C) The note heard by A coming from B will be lower in pitch than f_A .
(D) The note heard by B coming from A will be lower in pitch than f_A .

Multiple Correct Answer Type

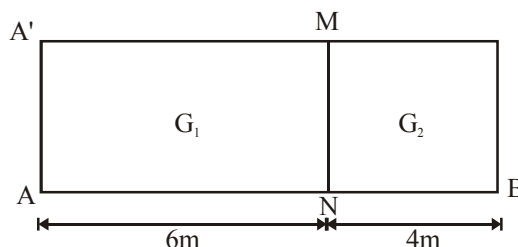
4 Q. [4 M (-1)]

7. Displacement vs position graph is shown for a longitudinal wave. In the graph below is shown pressure profile of medium. Choose the **CORRECT** option(s) :-



- (A) If wave is moving towards right then pressure profile is 2
(B) If wave is moving towards left then pressure profile is 2
(C) If wave is moving towards right then pressure profile is 1
(D) If wave is moving towards left then pressure profile is 1
8. Which of the following statements is/are valid?
- (A) The sound waves from two tuning forks of the same frequency will interfere constructively at some points.
(B) Two waves, A and B, have the same wave velocity. If A has a higher frequency than B, then A will have a shorter wavelength
(C) In a transverse wave, the motion of the particle of the medium is in the same direction as the wave velocity.
(D) The propagation speed (or wave velocity) of a wave can be found by dividing its wavelength by its period.

9. Figure shows a closed pipe (closed at both ends) of length 10m. MN is a membrane (thin sheet) capable of doing simple harmonic motion of amplitude 1mm along left and right from shown situation. On two sides of membrane are two gases G_1 and G_2 in which sound can travel at 300 m/s and 100 m/s respectively.

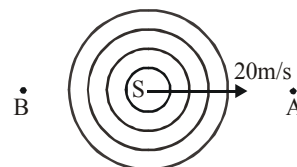


- (A) If resonance is obtained in right side (G_2), side (G_1) might not resonate.
(B) Resonance in both gases will be possible for some specific frequencies only.
(C) For angular frequency 75π rad/sec., a particle, $\frac{2}{3}$ m right of AA' would be oscillating with maximum displacement 0.5 mm.

- (D) For angular frequency 75π rad/sec., a particle, $\frac{2}{3}$ m right of AA' would be oscillating with maximum displacement $\frac{\sqrt{3}}{2}$ mm.

10. A sound source S moves with speed of 20 m/s with respect to the medium. There are two detectors A & B located as shown in figure. If wavelength emitted by source are in the range of 20 to 50 cm then choose correct statement(s) :-

- (A) A wavelength detected by A may be 20 cm, 30 cm, 50 cm
(B) A wavelength detected by A may be 19 cm, 30 cm, 41 cm
(C) A wavelength detected by B may be 20 cm, 30 cm, 50 cm
(D) A wavelength detected by B may be 22 cm, 51 cm, 53 cm

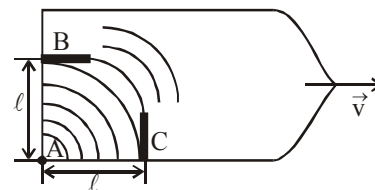


Linked Comprehension Type (1 Para × 2Q.)
(Single Correct Answer Type)

[3 M (-1)]

Paragraph for Questions 11 and 12

A ship swims at constant velocity $\vec{v}$ on a windless ocean. A short sound pulse is emitted from a sound source located at point A of the open deck of the ship, the speed of sound is v_0 . The sound is reflected from wall B that is at distance ℓ from point A and is parallel to the direction of travel. The sound is also reflected from wall C that is also at distance ℓ but is perpendicular to the direction of travel. The sound reflected from wall B is received at A after time t_B and similarly from wall C after time t_C .



11. If $t_B = nt_C$, then the ratio of $\frac{v_0}{v}$ would be
- (A) $\frac{1}{n}$ (B) $\frac{1}{\sqrt{n}}$ (C) $\frac{1}{\sqrt{1-n^2}}$ (D) $\frac{1}{1-n^2}$
12. If rather than pulse, a continuous sound wave of frequency f is produced by source, then the beat frequency as observed by an observer at A would be
- (A) $\frac{f}{n^2}$ (B) $\frac{f}{1-n^2}$ (C) $\frac{f}{n}$ (D) None of these

Linked Comprehension Type (1 Para × 3 Q.)
(Multiple Correct Answer Type)

[4 M (-1)]

Paragraph for Q. No. 13 to 15

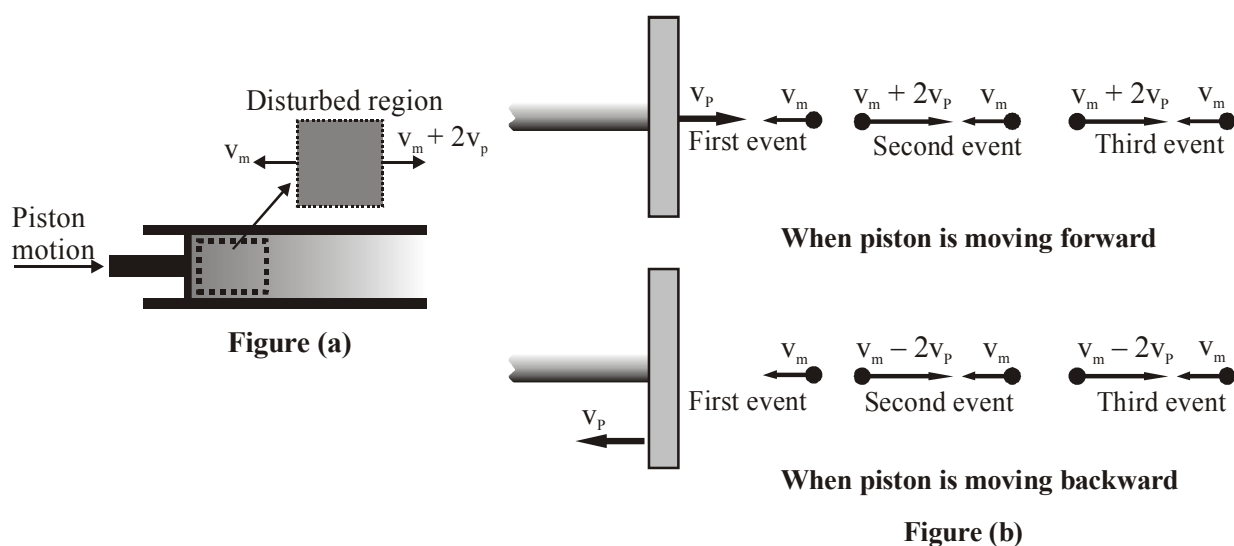
Figure shows a cylinder piston arrangement in which a piston is moved.

For molecules striking the piston, the speed of approach is $= v_p + v_m$.

Speed of separation is $= v_m + 2v_p$ after striking the piston.

As shown in figure each molecule that strikes the moving piston has the quantity $2v_p$ added to its speed. Thus, kinetic energy is added to the gas.

An identical molecule moving at v_m towards the piston that strikes head-on another molecule moving away from the piston with speed $v_m + 2v_p$. The one originally moving toward the piston with velocity v_m now moves away with velocity $v_m + 2v_p$, and the other molecule heads back toward the piston with speed v_m . The net result of all this collision is that the excess velocity $2v_p$ is passed on from one molecule to the next. The rightmost molecule with the extra velocity is at the leading edge of a propagating disturbance as shown in figure. If we look within the disturbed region, at any given moment we can expect that half the molecules have velocity $-v_m$ toward the piston and half have velocity $+v_m + 2v_p$ away from the piston as shown in figure (b). When the piston stops moving, the molecules returning to it with speed v_m simply have their velocities reversed, and we are back where we started, except for the propagating disturbance, which is now traveling down the tube with a speed of $v_m + 2v_p \approx v_m$. In the disturbed region the gas molecules have excess average speed v_p , but that region of excess speed propagates with much higher speed v_m . A compressed region is formed whenever the piston is pushed into the tube. A rarefactions region is formed whenever the piston is pushed back into the tube. As the piston oscillates sinusoidally, regions of compression and rarefaction are continuously set up. The pressure variation is a maximum when the displacement from equilibrium is zero, and the displacement from equilibrium is a maximum when the pressure variation is zero.



13. Consider for the case piston is pushed inward. Mark the **CORRECT** statement(s) :-
- The gas in the region between the piston and the leading edge of the disturbance has acquired an average velocity v_p to the right
 - Each gas molecule hit by the piston gains velocity $2v_p$ in the x-direction, thus increasing the kinetic energy of the gas.
 - Extra kinetic energy is passed on from molecule to molecule at roughly the speed v_m which is much faster than the piston speed v_p
 - The molecules that were pushed out of the way by the piston also create a region of increased density that itself also propagates down the tube along with the excess kinetic energy.
14. Mark the **CORRECT** statement(s) :-
- A compressed region is formed whenever the piston is pushed into the tube. This compressed region, moves through the tube with speed of sound in the medium.
 - A rarefaction region is formed whenever the piston is pushed back into the tube. This rarefaction region also moves through the tube with speed of sound in the medium.
 - When a compressed region is formed density of the medium increases.
 - When a rarefaction region is formed density of the medium decreases.
15. Consider that piston oscillates to and fro in simple harmonic motion. Mark the **CORRECT** statement(s) about longitudinal pressure wave.
- The density variation of the particles have their maxima and minima at the position of equilibrium of the particles.
 - At the position of maximum displacement of particles the density is at the ambient (environment) value.
 - Greatest compression occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation.
 - Greatest rarefaction occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation.

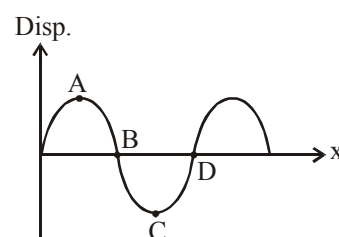
Matching List Type (4×4)

16. Figure shows displacement vs x-graph for a longitudinal wave at time t travelling along +x-direction.

- List-I**
- Point A
 - Point B
 - Point C
 - Point D

- List-II**
- Maximum pressure
 - Minimum pressure
 - Acceleration positive
 - Acceleration negative

1Q.[3 M (-1)]



Code :-

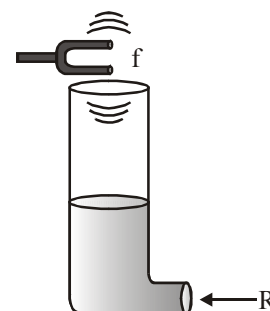
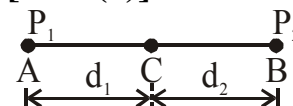
	P	Q	R	S
(A)	4	1	3	2
(B)	1	3	2	4
(C)	2	3	1	4
(D)	4	1	2	3

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

- As shown in the figure, two loudspeakers are located at point A and B. Both are vibrating in phase at a frequency ν and P_1 and P_2 are their respective power outputs. Point C lies on a line joining the two loudspeakers at a distance of d_1 from A and d_2 from B. With both speakers switched on what is the power (in W/m^2) at point C. Take velocity of sound = 300 ms^{-1} , frequency $\nu = 100 \text{ Hz}$, $d_1 = 1 \text{ m}$ and $d_2 = 1.5 \text{ m}$, $P_1 = 8\pi$ watts and $P_2 = 18\pi$ watts. Also assume that loudspeakers behave like isotropic sources. (emit sound uniformly in all directions).
- A source plays a tone with frequency $f_0 = 440 \text{ Hz}$. What is the speed of a car moving straight towards the source, if the driver hears a frequency of 462 Hz . If your answer is v_0 (in m/s) then fill value of $\frac{v_0 + 1}{2}$. (Speed of sound in the air is $v = 340 \text{ ms}^{-1}$, the effects of the material of the car can be neglected)
- As shown in figure water is pumped into a tall vertical cylinder at a volume flow rate R . The radius of the cylinder is r , and at the open top of the cylinder a tuning fork is vibrating with a frequency f . As the water rises, time interval elapse between successive resonances is given by $t = \frac{\pi r^2}{R} \frac{v}{\alpha f}$ where v is speed of sound in air. Find α .



SECTION-IV

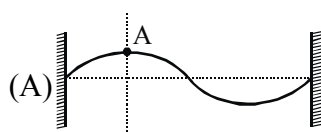
Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

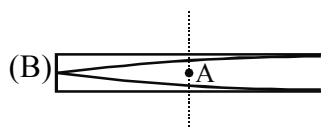
- Column I** shows four systems, for producing standing waves. Match each system with statements given in **Column II** describing the nature and different parameters of the standing waves.

Column I

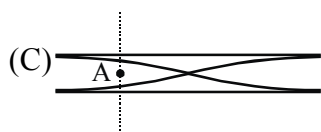
Column II



(P) Longitudinal standing wave



(Q) Transverse standing wave



(R) Power transfer can take place across point A



(S) All overtones are possible

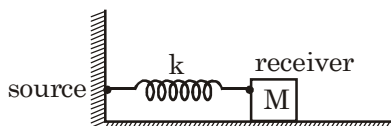
(T) KE of particle A is always zero

SECTION-I

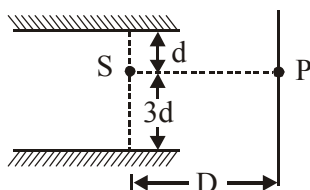
Single Correct Answer Type

7 Q. [3 M (-1)]

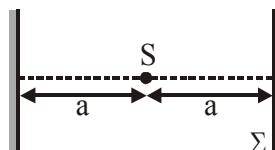
1. In the given arrangement a source is placed at the fixed end of the spring with spring constant 100 N/m and the receiver is attached to the moving end of the spring, which in turn is connected to a box of mass 4kg, which is oscillating horizontally with an amplitude of oscillation 2m. If the source emits a frequency of 990 Hz then (speed of sound = 330 m/s) :-



- (A) maximum frequency received by the receiver is 990 Hz
(B) minimum frequency received by the receiver is 990 Hz
(C) frequency band width received by the receiver is 960 Hz - 1020 Hz
(D) maximum and minimum frequencies are received at the extreme position.
2. Consider the optical system shown in the figure that follows. The point source of light S is having wavelength equal to λ . The light is reaching screen only after reflection. For point P to be 2nd maxima, the value of λ would be ($D \gg d$ and $d \gg \lambda$) :-



- (A) $\frac{12d^2}{D}$ (B) $\frac{6d^2}{D}$ (C) $\frac{8d^2}{D}$ (D) $\frac{24d^2}{D}$
3. A point source of light S is placed in front of a perfectly reflecting mirror as shown in the figure. Σ is a screen. The intensity at the centre of screen is found to be I. If the mirror is removed, then the intensity at the centre of screen would be :-



- (A) I (B) $\frac{10I}{9}$ (C) $\frac{9I}{10}$ (D) 2 I
4. A lens of diameter 5.0 cm and focal length $f = 20.0$ cm was cut along the diameter into two identical halves. In the process, a thin layer of lens ($a = 1.0$ mm) is lost. Then the halves were put together to form a composite lens. A source slit is placed 10 cm from the lens, emitting light of wavelength $\lambda = 600$ nm. Behind the lens, a screen wall located at a distance $b = 80$ cm from lens. Find the fringe width :-
(A) 0.15 mm (B) 0.3 mm (C) 0.6 mm (D) 1.2 μ m
5. In a regular YDSE, when thin film of refractive index μ is placed in front of the lower slit then it is observed that the intensity at the central point becomes half of the original intensity. It is also observed that the initial 3rd maxima is now below the central point and the initial 4th minima is above the central point. Now instead, a film of refractive index μ_1 and thickness same as the above film, is put in the front of the lower slit. It is observed that whole fringe pattern shifts by one fringe width. What is the value of μ_1 ?
(A) $(4\mu + 9)/12$ (B) $(4\mu + 9)/13$ (C) $(9\mu + 4)/13$ (D) None

6. In the young's double slit experiment a point P from the central bright fringe is such that the intensity at the point P is $\frac{3}{4}$ times the maximum intensity. Distance between the slits is d and wavelength is λ . The angular position of P is -

(A) $\sin^{-1}\left(\frac{\lambda}{d}\right)$ (B) $\sin^{-1}\left(\frac{\lambda}{12d}\right)$ (C) $\sin^{-1}\left(\frac{\lambda}{3d}\right)$ (D) $\sin^{-1}\left(\frac{\lambda}{6d}\right)$

7. A thin oil film of refractive index 1.2 floats on the surface of water ($\mu = \frac{4}{3}$). When a light of wavelength $\lambda = 9.6 \times 10^{-7}$ m falls normally on the film from air, then it appears dark when seen normally. The minimum change in its thickness for which it will appear bright in normally reflected light by the same light is :-

(A) 10^{-7} m (B) 2×10^{-7} m (C) 3×10^{-7} m (D) 5×10^{-7} m

Multiple Correct Answer Type

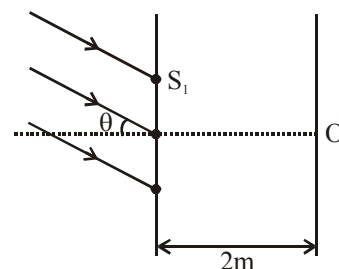
3 Q. [4 M (-1)]

8. A woman stands at rest in front of a large, smooth wall. She holds a vibrating tuning fork of frequency f_0 directly in front of her (between her and the wall). The woman now runs towards the wall with speed v_w . She detects f_1 beats per second due to the interference between the sound waves reaching her directly from the fork and those reaching her after being reflected from the wall. If the woman instead runs away from the wall, holding the tuning fork at her back so it is between her and the wall, she detects f_2 beats per seconds. (Velocity of sound is v)

(A) $f_1 = f_0 \left(\frac{2v_w}{v - v_w} \right)$ (B) $f_2 = f_0 \left(\frac{2v_w}{v - v_w} \right)$ (C) $f_1 = f_0 \left(\frac{v - v_w}{2v_w} \right)$ (D) $f_2 = f_0 \left(\frac{2v_w}{v + v_w} \right)$

9. A parallel beam of light ($\lambda = 5000 \text{ \AA}$) is incident at an angle $\theta = 30^\circ$ with the normal to the slit plane in a Young's double slit experiment. The intensity due to each slit is I_0 . Point O is equidistant from S_1 and S_2 . The distance between slits is 1 mm, then :-

- (A) The intensity at O is $4 I_0$.
(B) The intensity at O is zero
(C) The intensity at a point on the screen 4mm above O is $4 I_0$.
(D) The intensity at a point on the screen 4 mm above O is zero.



10. Four trials of Young's double slit experiment are conducted. (Given : $\lambda_{\text{blue}} = 450 \text{ nm}$ and $\lambda_{\text{red}} = 700 \text{ nm}$)
(i) In the first trial, blue light passes through two fine slits $400 \mu\text{m}$ apart and forms an interference pattern on a screen 4m away.
(ii) In a second trial, red light passes through the same slits and falls on the same screen.
(iii) A third trial is performed with red light and the same screen, but with slits $800 \mu\text{m}$ apart
(iv) A final trial is performed with red light, slits $800 \mu\text{m}$ apart, and a screen 8m away.

If angle between the first maxima and central maxima in each trial is θ_i , θ_{ii} , θ_{iii} & θ_{iv} respectively & distance between the central maxima and first maxima is y_i , y_{ii} , y_{iii} & y_{iv} respectively then:

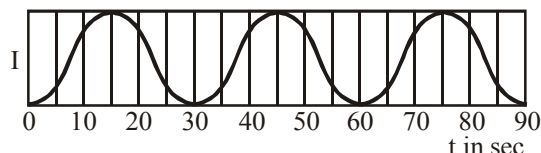
- (A) $\theta_i > \theta_{ii} > \theta_{iii} > \theta_{iv}$ (B) $y_i > y_{ii} > y_{iii} > y_{iv}$
(C) $\theta_{ii} > \theta_i > \theta_{iii} = \theta_{iv}$ (D) $y_{ii} = y_{iv} > y_i > y_{iii}$

Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 3Q.) [3 M (-1)]

Paragraph for Question no. 11 to 13

In the front of the upper slit of YDSE apparatus, a thin film of a liquid of refractive index 1.40 is placed. It is a hot day the liquid starts evaporating from the surface. A beam of light at wavelength 560 nm is incident onto the YDSE apparatus and the intensity I at the centre of the screen is monitored. Figure gives intensity I as a function of time t . The intensity changes because of evaporation from the two sides of the film. Assume that the film is flat and has parallel sides. Also assume that the film's thickness decreases at a constant rate.



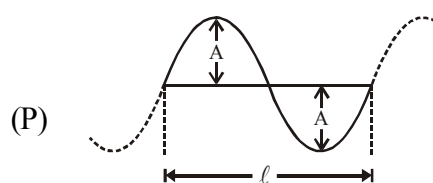
11. What can not be the initial thickness of the film?
(A) $7 \mu\text{m}$ (B) $4.9 \mu\text{m}$ (C) $7.7 \mu\text{m}$ (D) $9.1 \mu\text{m}$
12. If the maximum intensity is I_0 , then :-
(A) Intensity at $t = 10$ sec is $\frac{I_0}{\sqrt{2}}$ (B) Intensity at $t = 10$ sec is $\frac{3I_0}{4}$
(C) Intensity at $t = 10$ sec is $\frac{I_0}{4}$ (D) Intensity at $t = 10$ sec, $\frac{I_0}{2\sqrt{2}}$
13. The rate of change of thickness can be :
(A) $\frac{140}{3}$ nm/sec (B) 70 nm/sec (C) $\frac{28}{3}$ nm/sec (D) $\frac{56}{3}$ nm/sec

Matching List Type (4 × 4)

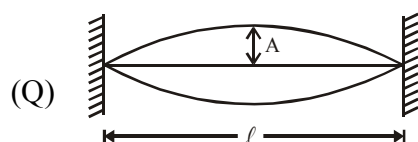
1Q.[3 M (-1)]

14. List-I shows four system of string each of same length, for producing travelling wave or standing wave. In each case velocity of travelling wave on string is same i.e. v . Match each system of list-I with expressions given in list-II describing the energy of the wave.

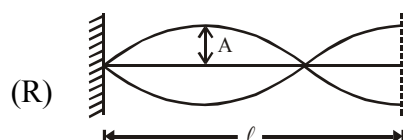
List-I



Travelling wave



Standing wave of one loop with amplitude of antinode equal to A



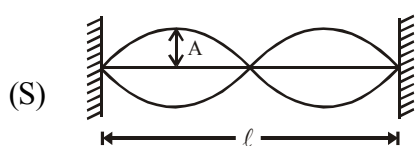
Standing wave of one plus half loop with amplitude of antinode equal to A

List-II

(1) $\frac{\mu A^2 \pi^2 v^2}{4\ell}$

(2) $\frac{9\mu A^2 \pi^2 v^2}{16\ell}$

(3) $\frac{\mu A^2 \pi^2 v^2}{\ell}$



Standing wave of 2 loop with amplitude of antinode equal to A.

(4) $\frac{2\mu A^2 \pi^2 v^2}{\ell}$

Codes :

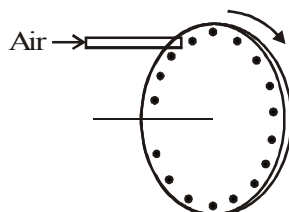
	P	Q	R	S
(A)	4	1	2	3
(B)	4	2	1	4
(C)	2	2	3	1
(D)	1	1	2	3

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

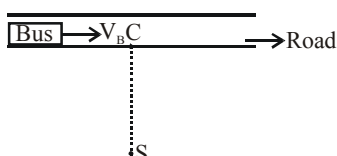
5 Q. [4 M (0)]

1. The diagram above shows the basic idea behind a disk siren. It consists of a disk in which there are 16 equally spaced holes, all at the same distance from its axle. When a jet of air is directed at the holes and the disc is rotated at a particular constant rate, the frequency of the note produced is 320 Hz. When a disk containing 24 holes is rotated at $\frac{4}{3}$ times the rate then frequency of note produced is 320 n Hz. Find the value of n

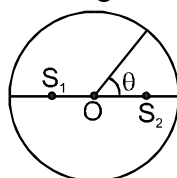


2. A bus is moving with a velocity v_B along positive x-axis along a road as shown in the figure. A shooter S is at a distance ℓ from the road. He has a detector which can detect signals only of frequency 1500 Hz. The bus blows horn of frequency 1000 Hz. When detector detects a signal, shooter immediately shoots towards the road along SC and bullet hits the bus. If velocity of sound in medium is 360 m/s and

$\frac{v_{bus}}{v_{sound}} = \frac{2}{3\sqrt{3}}$, then velocity of bullet (in m/s) is $(12)^n$ m/s. Find n.



3. Two coherent sources S_1 and S_2 separated by distance 2λ emit light of wavelength λ in phase as shown in the figure. A circular wire of radius 100λ is placed in such a way that S_1S_2 lies in its plane and the mid-point of S_1S_2 is at the centre of wire. The angular positions on the wire for which intensity reduces to half of its maximum value for the first time is given as θ . Find the value of $32 \cos\theta$.



- One slit of a Young's experiment is covered by a glass plate ($\mu_1 = 1.4$) and the other by another glass plate ($\mu_2 = 1.7$) of the same thickness. The point of central maxima on the screen, before the plates were introduced is now occupied by the third bright fringe. Find the thickness of the plates (in μm), when the wavelength of light used is $4000 \text{ \AA}$.
- Solar cell devices that generate electricity when exposed to sunlight are often coated with a transparent thin film of silicon monoxide (SiO_2 , refractive index = 2) to minimize the reflective losses from the surface. A silicon solar cell ($n = 3.5$) is coated with a thin film of silicon monoxide of thickness 60 nm . As an engineer, you have to coat an additional layer of silicon monoxide film so that reflection is least for a wave of wavelength 880 nm , then the minimum thickness of additional coat is $(x \times 10^2) \text{ \AA}$. Find the value of x .

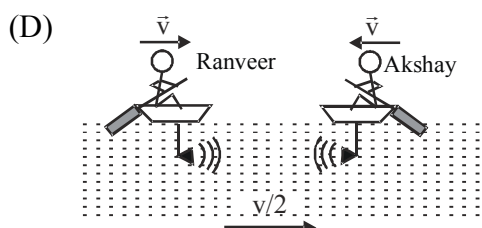
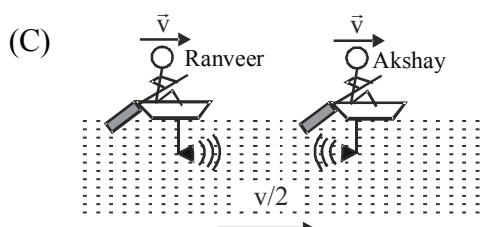
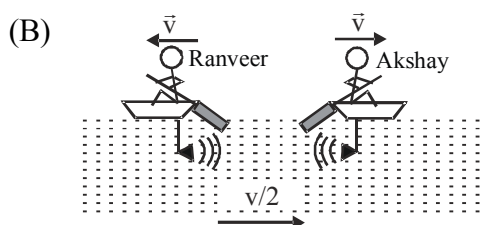
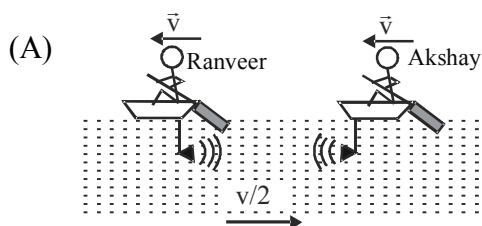
SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

- Ranvir and Akshay are rowing two boats in a river with the same speed v relative to the river. Ranvir is upstream relative to Akshay. The river is flowing with speed $v/2$. Each boat sends, through the water, a signal to the other. The frequencies f_0 of the generated continuous sonic signals are the same. Each observer observes the wave sent by the other person. In each of the situation in column-I, Match the appropriate entries.

Column I



Column II

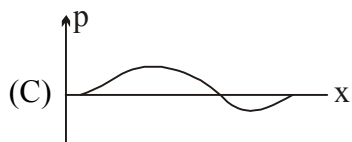
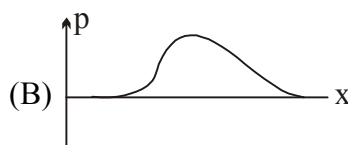
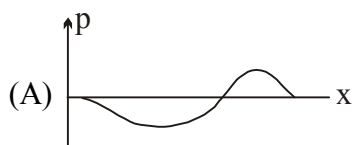
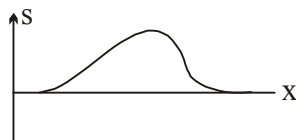
- (P) The times it takes for the signals to travel from one boat to the other will be the same.
- (Q) The wavelength observed by Akshay is greater than the wavelength observed by Ranvir.
- (R) The frequency observed by Akshay is greater than the frequency observed by Ranvir.
- (S) The wavelength observed by Akshay is lesser than the wavelength observed by Ranvir.
- (T) The frequency observed by Akshay is equal to the frequency observed by Ranvir.

SECTION-I

Single Correct Answer Type

8 Q. [3 M (-1)]

1. For displacement(s)-x graph shown for a sound wave, select appropriate excess pressure(p)-x graph.



2. A vertical tube is completely filled with water. A small sound source of constant frequency is held a little above the open upper end and water is run out from the lower end. A number of resonance positions are detected. The first of these occurs when the water surface is 7cm below the top of the tube and another occurs at 39cm. At which of the following distances should resonance also be detected?

(A) 14cm (B) 18cm (C) 23cm (D) 31cm

3. A rod of specific gravity 0.7 is found to be stretched by one millimetre when suspended vertically and loaded with a weight of lead of specific gravity 11, the volume of the weight being equal to that of the rod. What fundamental (approximately) frequency will the rod alone emit when clamped at the middle and rubbed longitudinally ?

(A) $\frac{1}{2} \sqrt{\frac{1100g}{7}}$ (B) $\frac{1}{2} \sqrt{\frac{11 \times 10^4 g}{7}}$ (C) $\frac{1}{2} \sqrt{\frac{770g}{3}}$ (D) $\frac{1}{2} \sqrt{\frac{7 \times 10^3 g}{11}}$

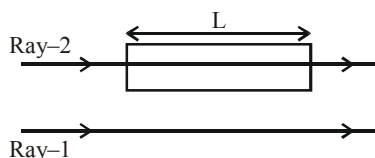
4. A source of frequency f is rotating in a circular path of radius R . An observer in the same plane is also rotating in a circular path of radius $2R$. The centers of both circular paths are separated by $6R$. Both these are rotating with same angular velocity ω in same direction. Find the difference between maximum and minimum frequency received by observer. 'C' is speed of sound.

(A) $\frac{2C\omega Rf}{C^2 - (\omega R)^2}$ (B) $\frac{6C\omega Rf}{C^2 - (\omega R)^2}$ (C) $\frac{3C\omega Rf}{C^2 - (\omega R)^2}$ (D) $\frac{4C\omega Rf}{C^2 - (\omega R)^2}$

5. In Young's double slit experiment with light of wavelength $\lambda = 600$ nm, intensity of central fringe is I_0 . Now one of the slit is covered by glass plate of refraction index 1.4 and thickness $t = 5 \mu\text{m}$, the new intensity at the same point on screen will be :-

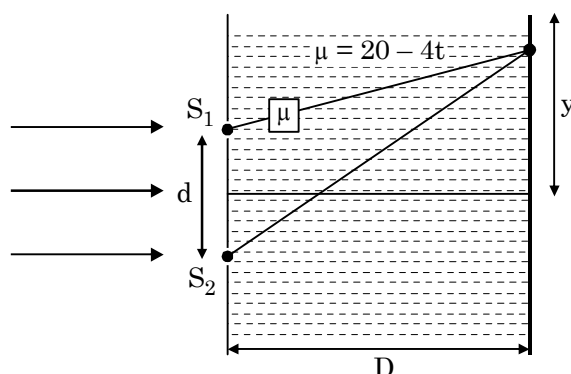
(A) $\frac{I_0}{4}$ (B) $\frac{3I_0}{4}$ (C) I_0 (D) $\frac{I_0}{2}$

6. As shown in arrangement waves with identical wavelength and amplitudes and that are initially in phase travel through different media. Ray-1 travels through air and Ray-2 through a transparent medium for equal length L in four different situations. In each situation the two rays reach a common point on the screen. The number of wavelength in length L is N_1 for Ray-1 and N_2 for Ray-2. The order of situation according to the intensity of the light at the common position in descending order :-



Situation	1	2	3	4
N_1	2.25	1.80	3.00	3.25
N_2	2.75	2.80	3.25	4.00

- (A) $I_3 = I_4 > I_2 > I_1$ (B) $I_1 > I_3 = I_4 > I_2$ (C) $I_1 > I_2 > I_3 > I_4$ (D) $I_2 > I_3 = I_4 > I_1$
7. In Young's double slit experiment the slits are 0.5 mm apart and interference is observed on a screen placed at a distance of 100 cm from the slits. It is found that the 9th bright fringe is at a distance of 9.0 mm from the second dark fringe from the centre of the fringe pattern. What is the wavelength of light used :-
- (A) 2000 Å (B) 4000 Å (C) 6000 Å (D) 8000 Å
8. A parallel beam of monochromatic light of wavelength $\lambda = 100$ (Å) is incident on the slits separated by distance $d = 2$ mm. There is a screen at a distance $D = 1$ m from slit. If R.I. of the medium between slits and screen is varying with time as $\mu = 20 - 4t$ until it becomes 1. A glass slab of R.I. $\mu = 5$ and thickness 0.2 mm is placed in front of one of the slit S_1 as shown in figure. In figure y represent position of central maxima on the screen from its geometrical centre. Then calculate value of y (in cm) at $t = 5$ sec.

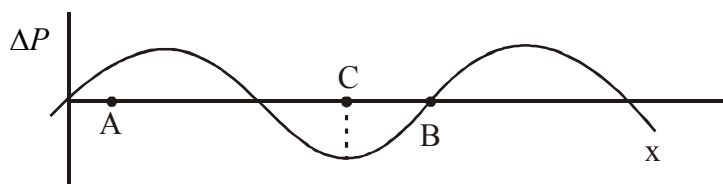


- (A) 40 (B) 7.5 (C) 20 (D) 15

Multiple Correct Answer Type

5 Q. [4 M (-1)]

9. Sound wave in form of sine wave is travelling along +x axis, $\Delta P - x$ graph is shown. Choose correct options.



- (A) Particle on just left side of A and just right side of A are moving in opposite directions.
(B) Speed of particle on just left of A is less than that on right of A.
(C) Particles on just left and just right of B have opposite direction of motion.
(D) Particles on just left and just right of C are moving in same direction

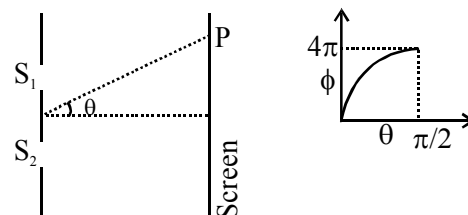
10. In a YDSE setup, we plot the phase difference (ϕ) between both waves at point P on the screen against the angular position (θ) of point P on the screen. The graph is as shown below.

(A) The distance $S_1S_2 = 2\lambda$

(B) There are a total of 4 minima on the screen

(C) The first maxima above the centre is at $\theta = \frac{\pi}{4}$

(D) At $\theta = \frac{\pi}{6}$, intensity is maximum



11. In a YDSE arrangement, white light is used to illuminate the slits. At point on the screen directly in front of the slits, which of these wavelengths is/are missing ?

(λ = wavelength of light used, d = distance between the slits, D = separation between the slits and the screen, $d \ll D$) :-

(A) $\frac{d^2}{D}$

(B) $\frac{d^2}{3D}$

(C) $\frac{d^2}{4D}$

(D) $\frac{d^2}{8D}$

12. In a Young's double slit experiment two thin transparent sheets are used in front of the slits S_1 and S_2 .

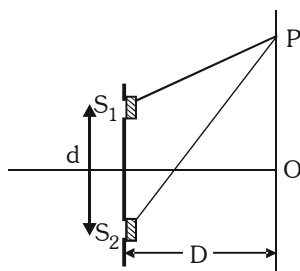
One of thin sheet has refractive index $\mu_1 = 1.6$ and other has $\mu_2 = 1.4$. Both sheets have thickness $\frac{t_1 + t_2}{2}$

the central maxima is observed at a distance of 5mm from centre O. Now the sheets are replaced by two

sheets of same material of refractive index $\frac{\mu_1 + \mu_2}{2}$ but having thickness t_1 and t_2 . Now central maxima

is observed at a distance 8 mm from centre O. Find the thickness of two sheets. Given $d = 1$ mm, $D = 1$ m.

Then :-



(A) $t_1 = 1.2 \times 10^{-5}$ m (B) $t_1 = 3.3 \times 10^{-5}$ m (C) $t_2 = 1.7 \times 10^{-5}$ m (D) $t_2 = 2 \times 10^{-5}$ m

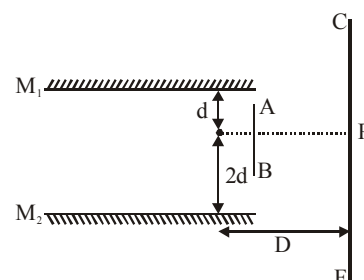
13. M_1 and M_2 are two plane mirror & S is monochromatic source. AB is a 'stop' which stops direct light to reach CF and allows reflected light from M_1 and M_2 to reach CF. P is a point on CF such that SP is parallel to M_1 & M_2 & perpendicular to CF :-

(A) Circular fringes will be formed on CF

(B) P will be central maxima

(C) P will be a point of maxima if $\lambda = \frac{6d^2}{D}$

(D) P will be a point of maxima if $\lambda = \frac{d^2}{D}$

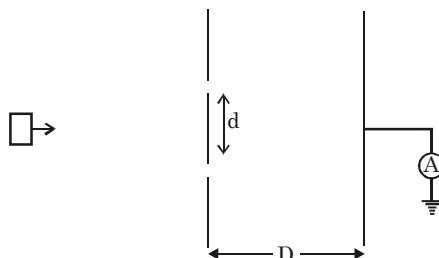


Linked Comprehension Type (Single Correct Answer Type)

(1 Para × 2Q.) [3 M (-1)]

Paragraph for Question No. 14 and 15

Let us consider an interference experiment with electrons. They are emitted from a box after being accelerated to a voltage V . The slits are both of same width. To detect the electron waves, we use a small metal patch connected to an ammeter as shown. At the centre of the screen, it shows current i_0 .



14. If current shown by ammeter is $i_0/2$ if patch is at :-

(A) $\frac{Dh}{2d\sqrt{2mqV}}$ (B) $\frac{3Dh}{2d\sqrt{2mqV}}$ (C) $\frac{3Dh}{4d\sqrt{2mqV}}$ (D) $\frac{Dh}{8d\sqrt{2mqV}}$

15. In front of the upper slit, we have produced a thin slab of thickness w in which potential is ΔV higher than surroundings. Which of the following effects will be observed :-

- (A) fringe pattern shifts up. (B) fringe pattern shifts down.
(C) fringe width increases. (D) fringe width decreases.

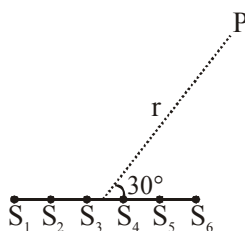
SECTION-II

Numerical Answer Type Question (upto second decimal place)

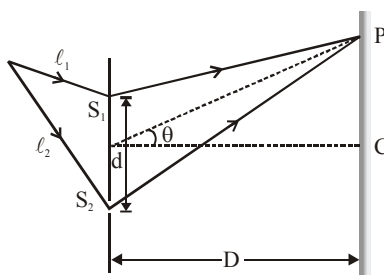
2Q.[3(0)]

1. Each light source ($S_1, S_2, S_3, S_4, S_5, S_6$) in a line has same power, same wavelength (λ) and all are in phase. Distance between consecutive sources is a if smallest value of a is a' such that point $P(r \gg a)$ has

zero light intensity, find the value of $\left(\frac{\lambda}{a'}\right)^2$.



2. In a Young experiment the light source is at distance $\ell_1 = 20\mu\text{m}$ and $\ell_2 = 40\mu\text{m}$ from the slits. The light of wavelength $\lambda = 500\text{ nm}$ is incident on slits separated at a distance $10\mu\text{m}$. A screen is placed at a distance $D = 2\text{m}$ away from the slits as shown in figure. How many maxima will appear on the screen.



SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

- The outer surface of a transparent glass slab of refractive index $\mu_g = 1.5$ is coated by a thin layer of transparent medium of refractive index $\mu_c = 1.6$. Orange light of wave length $6400 \text{ \AA}$ fall normally on the coat. The reflected light at the upper surface and at the lower surface of the coat interfere destructively. If thickness of the coat is $5K \times 10^{-8} \text{ m}$, calculate the minimum value of K.
- The fundamental frequency of an open pipe would be independent of small variation in temperature at a temperature $T_0 = \frac{\eta}{14\alpha}$, where α is the coefficient of linear expansion of the material of the tube. Fill the value of η in OMR sheet.

SECTION-IV

Matrix Match Type (4 × 5)

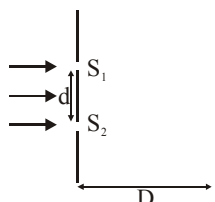
1 Q. [8 M (for each entry +2(0))]

- Column I represents how a quantity can change in each situation mentioned in column II. In each situation three different physical quantity denoted by option 1, 2 & 3 have been chosen.

COLUMN I

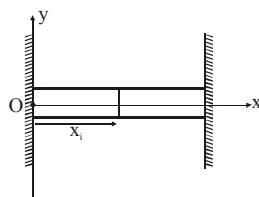
(A) Property in option 1 increases.

(P)



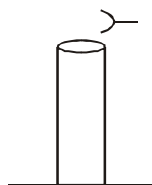
(B) Property in option 2 decreases

(Q)



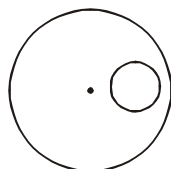
(C) Property in option 3 constant

(R)

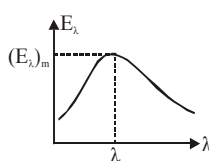


(D) Property in option 3 increases

(S)



(T)



COLUMN II

In a standard YDSE the slits are created in a metallic sheet.

Temperature increases.

Option 1 : Fringe width

Option 2 : Intensity at maxima

Option 3 : Position of zero order maxima

Two rods of same cross-section area are rigidly fixed to a wall.

Y, α are usual notations.

For $Y_1\alpha_1 > Y_2\alpha_2$

Temperature is increased

Option 1 : Co-ordinate of interface x_1 .

Option 2 : x-Co-ordinate of centre of mass of composite rod.

Option 3 : Stress at any cross-section

A metallic cylinder resonates with a tuning fork (constant frequency).

Temperature is increased.

Option 1 : Fundamental frequency of cylinder

Option 2 : Velocity of sound.

Option 3 : End correction.

A charged conducting sphere with spherical cavity as shown in figure.

Temperature is increased

Option 1 : Potential outside the sphere.

Option 2 : Potential at centre of cavity.

Option 3 : Electric field outside the sphere.

Spectral emissive power v/s wavelength graph of a black body.

Temperature is increased.

Option 1 : $(E_\lambda)_m$

Option 2 : λ_m

Option 3 : Area under graph.

PHYSICS

CLASS TEST # 75

SECTION-I

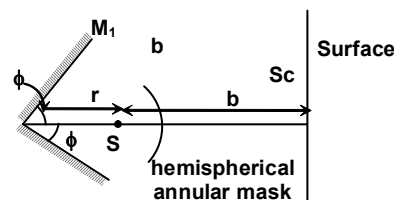
Single Correct Answer Type

7 Q. [3 M (-1)]

1. Consider a ray of light incident from air onto a slab of glass (refractive index n) of width d , at an angle θ . The phase difference between the ray reflected by the top surface of the glass and the bottom surface is (λ is wavelength of light in vacuum)

(A) $\frac{n4\pi d}{\lambda} \left(1 - \frac{1}{n^2} \sin^2 \theta\right)^{1/2} + \frac{\pi}{2}$ (B) $\frac{n4\pi d}{\lambda} \left(1 - \frac{1}{n^2} \sin^2 \theta\right)^{1/2} + \pi$
(C) $\frac{4\pi d}{\lambda} \left(1 - \frac{1}{n^2} \sin^2 \theta\right)^{1/2} + \pi$ (D) $\frac{4\pi d}{\lambda} \left(1 - \frac{1}{n^2} \sin^2 \theta\right)^{1/2} + \frac{\pi}{2}$

2. Two mirrors M_1 and M_2 make an angle ϕ with line AB. A point source S is kept at a distance r from the point of intersection of mirrors. A small hemispherical annular mask is kept close to S so that no ray emanating from S directly reaches the screen kept at a distance b from the source. Interference pattern of rays reflected from M_1 and M_2 is observed on the screen. Find the fringe width of this interference pattern. λ = wavelength of light being emitted by source.



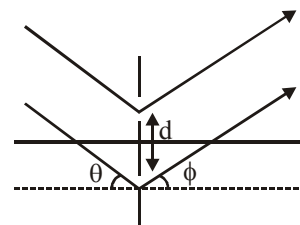
- (A) $\frac{\lambda(b+2r \sin^2 \phi)}{2r \sin 2\phi}$ (B) $\frac{\lambda(b+2r \cos 2\phi)}{2r \sin 2\phi}$ (C) $\frac{\lambda(b+r \cos 2\phi)}{2r \sin 2\phi}$ (D) $\frac{\lambda(b+2r \cos 2\phi)}{r \sin 2\phi}$
3. A monochromatic light is used in Young's double slit experiment when one of the slits is covered by a transparent sheet of thickness 1.8 mm, made of material of refractive index μ_1 number of fringes which shift is 18. when another sheet of thickness 3.6 mm, made of material of refractive index μ_2 is used, number of fringes which shift is 9. Relation between μ_1 and μ_2 is given by.
- (A) $4\mu_2 - \mu_1 = 3$ (B) $4\mu_1 - \mu_2 = 3$ (C) $3\mu_2 - \mu_1 = 4$ (D) $2\mu_1 - \mu_2 = 4$
4. A light with wavelength ($\lambda = 550$ nm) from a distant source falls normally on the surface of a glass wedge with refractive index 1.5. A fringe pattern whose neighbouring maxima on the wedge are separated by a distance ($\Delta x = 11$ μ m) is observed in reflected light. The angle between the wedge faces is :-

(A) $\frac{1}{30}$ rad (B) $\frac{1}{60}$ rad (C) $\frac{1}{90}$ rad (D) None of these

5. In the given figure light is incident at an angle θ with the normal to a plane containing two slits of separation d . Select the expression that correctly describes the positions of the interference maxima in terms of the incoming angle θ and outgoing angle ϕ .

(A) $\sin \phi + \sin \theta = \left(m + \frac{1}{2}\right) \frac{\lambda}{d}$ (B) $d \sin \theta = m\lambda$

(C) $\sin \phi - \sin \theta = (m+1) \frac{\lambda}{d}$ (D) $\sin \phi + \sin \theta = m \frac{\lambda}{d}$



6. When monochromatic light falls normally on the surface of the soap film in air, the intensity of the reflected light depends on the wavelength: it has a maximum at $\lambda_1 = 630$ nm and the minimum at $\lambda_2 = 525$ nm. What can be the film thickness? Refractive index of the film $n = 4/3$.
- (A) 590.625 nm (B) 640 nm (C) 562.625 nm (D) 466.66 nm

7. Two polaroid sheets are placed one over the other with their axes inclined to each other at 30° . The ratio of the intensity of the unpolarised incident light and the polarised emergent light is :-

- (A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{8}{3}$ (D) $\frac{3}{8}$

Multiple Correct Answer Type

3 Q. [4 M (-1)]

8. In young's experiment, the upper slit is covered by a thin glass plate

of refractive index $\frac{4}{3}$ and of thickness 9λ , where λ is the wavelength

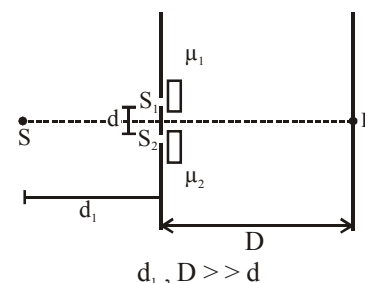
of light used in the experiment. The lower slit is also covered by another

glass plate of thickness 2λ & refractive index $\frac{3}{2}$, as shown in figure. If

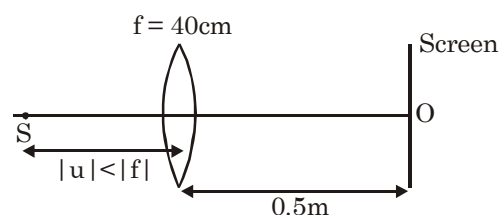
I_0 is the intensity at point P due to slits S_1 & S_2 each, then :

- (A) Intensity at point P is $4I_0$
(B) Two fringes have been shifted in upward direction after insertion of both the glass plates together.
(C) Optical path difference between the waves from S_1 & S_2 at point P is 2λ .
(D) If the source S is shifted upwards by a small distance d_2 then the fringe originally at P after inserting

the plates, shifts downward by $D\left(\frac{d_2}{d_1}\right)$.



9. A lens of focal length $f = 40$ cm is cut along the diameter into two equal halves. In this process, a layer of thickness $t = 1$ mm is lost, then halves are put together to form a composite lens. In between focal plane and the composite lens a narrow slit is placed very close to the focal plane $|u| < |f|$. The slit is emitting monochromatic light of wavelength $0.6 \mu\text{m}$. Behind the lens a screen is located at a distance $L = 0.5$ m from it as shown :-



- (A) Fringe width is 0.12 mm (B) Fringe width is 0.24 mm.
(C) Length of interference pattern is $1/8$ cm (D) Length of interference pattern is $1/16$ cm

10. Mark the **CORRECT** statement(s) :

- (A) Direction of wave propagation is along the normal to wavefront.
(B) For a point source of light, the shape of wavefront can be considered to be plane at very large distance from the source.
(C) A point source of light is placed at the focus of a thin spherical lens, then the shape of the wavefront for emergent light may be plane.
(D) The shape of the wavefront for the light incident on a thin spherical lens (kept in vacuum) is plane, the shape of the wavefront corresponding to emergent light would be always spherical.

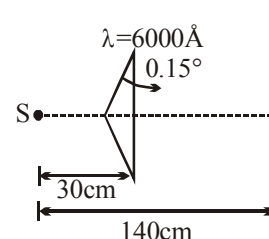
Linked Comprehension Type

(1 Para $\times$ 5Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Questions no. 11 to 15

Refracting angles of biprism are both 0.15° . Distance of biprism from source is 30 cm. Distance between source and screen is 140 cm. The Fresnel's biprism arrangement shown in the figure uses a source of light of wavelength $6000\text{\AA}$. When a converging lens is moved between the biprism and the screen two images of S are focused on the screen for two positions of the lens. The separations between the two images for the two positions are found to be 0.9mm and 1.6mm



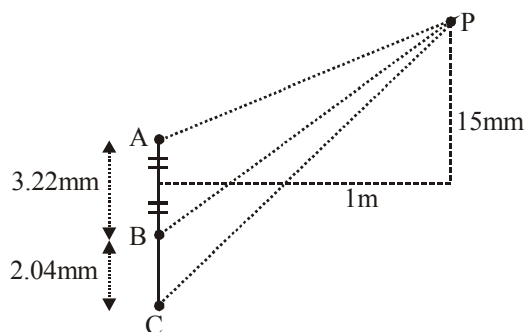
11. The focal length of the lens is :-
(A) $\frac{120}{7}$ cm (B) $\frac{240}{7}$ cm (C) Data not sufficient (D) None
12. Gap between two position of the lens is :-
(A) 80 cm (B) 60 cm (C) 40 cm (D) 20 cm
13. Refractive index of the prism material is approximately :-
(A) 1.46 (B) 1.56 (C) 1.66 (D) 1.76
14. The fringe width of the pattern on screen (after the lens has been removed) is :-
(A) 0.7 m (B) 0.35 mm (C) 1.4 mm (D) None
15. The total number of fringes obtained on the screen :-
(A) $\frac{4400}{7}$ (B) $\frac{400}{7}$ (C) $\frac{44}{7}$ (D) None

SECTION-III

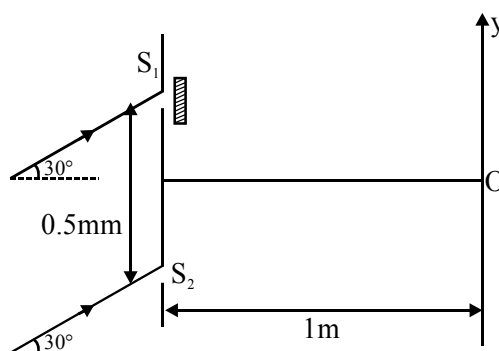
Numerical Grid Type (Ranging from 0 to 9)

4 Q. [4 M (0)]

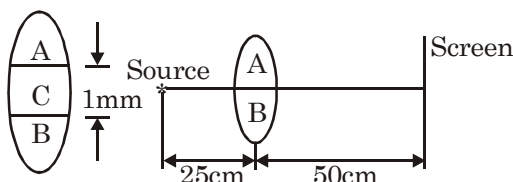
1. The intensity received at the focus of the lens is I when no glass slab has been placed in front of the slit. Both the slits are of the same dimension and the plane wavefront incident perpendicularly on them, has wavelength λ . On placing the glass slab, the intensity reduces to $3I/4$ at the focus. Find out the minimum thickness of the glass slab (in $1000\text{\AA}$) if its refractive index is $3/2$. Given $\lambda = 6000\text{\AA}$, $\mu = 1.5$.
2. For the arrangement given in the following figure, the coherent light sources A, B and C radiating in phase have individual intensities of 2 mW/m^2 , 2 mW/m^2 and 5 mW/m^2 , respectively at point P. The wavelength of each of the sources is 600 nm . The resultant intensity at point P (in mW/m^2) is.



3. In Young's double slits experiment parallel monochromatic electromagnetic waves of wavelength $3 \times 10^{-5} \text{ m}$ fall on the slits at an angle of 30° with the normal to the plane of slits as shown in the figure. A transparent sheet of thickness $5 \times 10^{-5} \text{ m}$ and refractive index 1.5 is introduced near the slit S_1 . The distance between the two slits is 0.5 mm and the distance between the plane of slits and the screen is 1 m . Find the y-coordinate (in cm) of the closest maxima to the origin on the positive y-axis.



4. A convex lens of focal length 50 cm is cut along the diameter into two identical halves A and B and in the process a layer C of the lens thickness 1 mm is lost. Then the two halves A and B are put together to form a composite lens. Now in front of this composite lens a source of light emitting wavelength $\lambda = 6000 \text{ \AA}$ is placed at a distance of 25 cm as shown in the figure. Behind the lens there is a screen at a distance 50 cm from it. The fringe width (in mm) of the interference pattern obtained on the screen is x . $10x$ is equal to



SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

1. Light from source S ($|u| < |f|$) falls on lens and screen is placed on the other side. The lens is formed by cutting it along principal axis into two equal parts and are joined as indicated in column II.

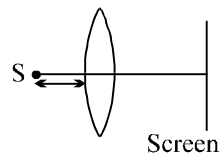
Column I

Column II

- (A) Plane of image move towards screen if $|f|$

is increased

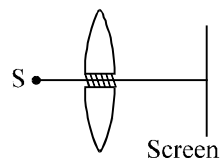
(P)



Small portion of each part near pole is removed. The remaining parts are joined.

- (B) Images formed will be virtual

(Q)

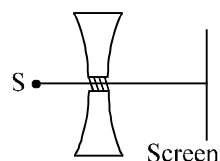


The two parts are separated slightly. The gap is filled by opaque material.

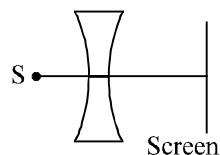
- (C) Interference pattern can be obtained if

screen is suitably positioned.

(S)



The two parts are separated slightly. The gap is filled by opaque material.



Small portion of each part near pole is removed. The remaining parts are joined.

SECTION-I

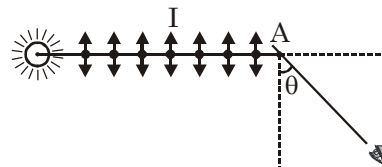
Single Correct Answer Type

12 Q. [3 M (-1)]

1. Unpolarized light of intensity I scattered from point A. Intensity of light perceived by given observer is :-

(A) I

(B) $\frac{I}{2} \cos^2 \theta$



(C) $\frac{I}{2} + \frac{I}{2} \sin^2 \theta$

(D) $\frac{I}{2} + \frac{I}{2} \cos^2 \theta$

2. An electron microscope is operated at 40 kV. The ratio of resolving power of this microscope and another one which uses yellow light of wavelength 6×10^{-7} m, is :-

(A) 9.78×10^6

(B) 9.78×10^4

(C) 9.78×10^{-4}

(D) 9.78×10^{-6}

3. Consider Fraunhofer diffraction pattern obtained with a single slit illuminated at normal incidence. At the angular position θ of the first diffraction minimum, the phase difference (in radians) between the wavelets from the opposite edges of the slit is :-

(A) π

(B) 2π

(C) $\pi/4$

(D) $\pi/2$

4. A small mirror of mass m is suspended by a light thread of length ℓ . The angle through which the thread will be deflected when a short pulse of laser of energy E falls normally on the mirror is :- (Assume mirror is a perfect reflector)

(A) $\frac{2E}{mc\sqrt{2g\ell}}$

(B) $\frac{mc\sqrt{g\ell}}{2E}$

(C) $\frac{mc\sqrt{2g\ell}}{2E}$

(D) $\frac{2E}{mc\sqrt{g\ell}}$

5. A beam of light has two wavelengths $3100 \text{ \AA}$ and $4133 \text{ \AA}$ with a total intensity of 12.8 W/m^2 equally distributed between the two wavelengths. The beam falls normally on an area of a clean metallic surface of work function 3.1 eV . Assume that there is no loss of energy by reflection and that each energetically capable photon ejects one electron. How many electrons will emit per second from the face area ?

(A) 2×10^{19}

(B) 10^{19}

(C) 10^{18}

(D) 2×10^{18}

6. If the surface of a metal is successfully exposed to radiation of $\lambda_1 = 350 \text{ nm}$ and $\lambda_2 = 450 \text{ nm}$ the maximum velocity of photoelectrons will differ by a factor 2. The work function of this metal is

(A) $2.84 \times 10^{-19} \text{ J}$

(B) $1.6 \times 10^{-19} \text{ J}$

(C) $3.9 \times 10^{-19} \text{ J}$

(D) $2.4 \times 10^{-19} \text{ J}$

7. In an experiment on photoelectric effect, the wavelength of the incident radiation is λ . The wavelength of the incident radiation is reduced to $\frac{1}{3}$ rd of the initial value and the maximum kinetic energy of the photoelectron is observed to be n times the previous value. The threshold wavelength for the metal plate is :-

(A) $\left(\frac{n-1}{n-3}\right)\lambda$

(B) $\left(\frac{n}{n-3}\right)\lambda$

(C) $\frac{(n+1)\lambda}{(n-3)}$

(D) $\frac{\lambda}{n}$

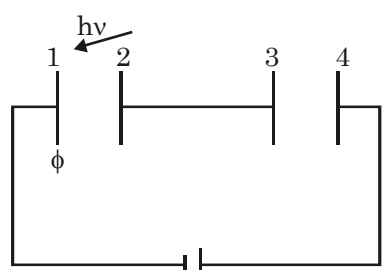
8. When 1 cm thick surface is illuminated with light of wavelength λ , the stopping potential is v . When the same surface is illuminated by light of wavelength 2λ , the stopping potential is $v/3$. Threshold wavelength for metallic surface is :-

(A) $\frac{4\lambda}{3}$

(B) 4λ

(C) $\frac{8\lambda}{3}$

(D) 6λ

9. If light of wavelength of maximum intensity emitted from surface at temperature T_1 is used to cause photoelectric emission from a metallic surface, the maximum kinetic energy of the emitted electron is 6 eV, which is 3 times the work function of the metallic surface. If light of wavelength of maximum intensity emitted from a surface at temperature T_2 ($T_2 = 2T_1$) is used, the maximum kinetic energy of the photoelectrons emitted is :-
 (A) 2 eV (B) 4 eV (C) 14 eV (D) 18 eV
10. Two identical capacitors are arranged as shown. Work function of plate 1 is ϕ eV. The emf of battery is $\frac{\phi}{e}$. If energy of incident photon is $h\nu$ then maximum kinetic energy of e^- reaching plate 2 is :-
 (A) $h\nu - \phi/2$ (B) $h\nu - \phi$
 (C) $h\nu - 2\phi$ (D) $h\nu$
- 
11. The work function of a certain metal is $\frac{hc}{\lambda_0}$. When a monochromatic light of wavelength $\lambda < \lambda_0$ is incident such that the plate gains a total power P . If the efficiency of photoelectric emission is $\eta\%$ and all the emitted photoelectrons are captured by a hollow conducting sphere of radius R already charged to potential V , then neglecting any interaction between plate and the sphere, expression of potential of the sphere at time t is :-
 (A) $V + \frac{100\eta\lambda Pet}{4\pi\epsilon_0 RhC}$ (B) $V - \frac{\eta\lambda Pet}{400\pi\epsilon_0 RhC}$ (C) V (D) $\frac{\lambda Pet}{4\pi\epsilon_0 RhC}$
12. A proton and an alpha particle are accelerated under the same potential difference. The ratio of de-Broglie wavelengths of the proton and the alpha particle is :
 (A) $\sqrt{8}$ (B) $\frac{1}{\sqrt{8}}$ (C) 1 (D) 2

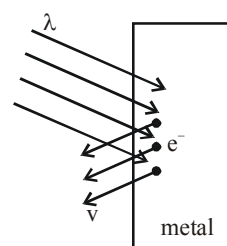
Multiple Correct Answer Type

3 Q. [4 M (-1)]

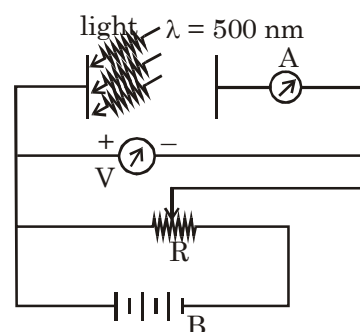
13. When a point light source of power W emitting monochromatic light of wavelength λ is kept at a distance 'a' from a photo-sensitive surface of wave function ϕ and area S then
 (A) Number of photons striking the surface per unit time is $\frac{W\lambda S}{4\pi hca^2}$
 (B) The stopping potential needed to stop the most energetic emitted photoelectrons is $\frac{1}{e\lambda}(hc - \lambda\phi)$
 (C) Photo emission occurs only if λ lies in the range $0 < \lambda \leq \frac{hc}{\phi}$
 (D) The maximum kinetic energy of the emitted photo electron is $\frac{1}{\lambda}(hc - \lambda\phi)$
14. When light of sufficient small wavelength is incident on a metal, electrons having different K.E. are emitted depending on loss of energy due to collision. The maximum K.E. of electron can be,

$$\text{K.E.}_{\text{max}} = \frac{hC}{\lambda} - \frac{hC}{\lambda_0}$$
 where λ_0 is threshold wavelength which depends on the metal. If wavelength of incident light is $\frac{\lambda_0}{4}$ and mass of electron is m then :

- (A) Minimum de-broglie wavelength of emitted electron is $\sqrt{\frac{6h\lambda_0}{mC}}$
 (B) Minimum de-broglie wavelength of emitted electron is $\sqrt{\frac{h\lambda_0}{6mC}}$
 (C) Maximum de-broglie wavelength of emitted electron tends to infinity
 (D) Maximum de-broglie wavelength of emitted electron tends to zero



15. A clean surface of a metal in a vacuum tube is irradiated with a monochromatic light of wavelength 500 nm and the resulting photoelectric current is measured by the ammeter A, as shown in the circuit. It is found that if the voltmeter (V) reading is greater than 0.8 V, the ammeter shows zero reading. Now, with a second source of light, the ammeter reading vanishes when the voltmeter reading is changed to 1.2 V. Then, [Battery B with rheostat R makes a potential divider circuit] [Given : $h = 6.6 \times 10^{-34}$ Js $c = 3 \times 10^8$ ms⁻¹, $e = 1.6 \times 10^{-19}$ C.]



- (A) the frequency of second source is 6.97×10^{14} Hz.
 (B) the kinetic energy of photoelectrons due to first source is less than or equal to 0.8 eV.
 (C) the work function of emitting surface is 1.68 eV.
 (D) the kinetic energy of photoelectrons due to second source is less than or equal to 1.2 eV.

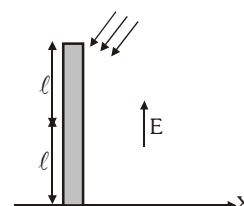
Linked Comprehension Type (1 Para × 3Q.)

[3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 16 to 18

A metal plate of length 2ℓ is placed as shown in figure. A light of frequency $5\nu_0$ [ν_0 is threshold frequency] is incident uniformly on upper half of the plate. Efficiency of photo-electric effect is 50% and all electrons comes out horizontally with KE = 1/4 of KE_{max}. Then



[Consider gravity free space & electric field $E = \frac{4h\nu_0}{e\ell}$ in vertically upward direction].

16. Kinetic energy of the emitted electron will be
 (A) $4h\nu_0$ (B) $2h\nu_0$ (C) $h\nu_0$ (D) $5h\nu_0$
 17. The length of the portion on the horizontal ground on which electron will strike the ground
 (A) $(\sqrt{2} - 1)\ell$ (B) $(\sqrt{2} + 1)\ell$ (C) $(\sqrt{3} - 1)\ell$ (D) $(\sqrt{3} + 1)\ell$
 18. At any instant charge density in the region where electrons fall
 (A) increases with x (B) decreases with x
 (C) is uniform (D) first increases & then decreases

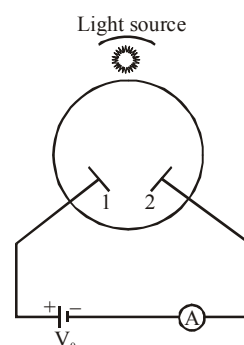
SECTION-II

Numerical Answer Type Question

1Q.[3(0)]

(upto second decimal place)

1. The figure below shows a vacuum tube containing electrodes made of different metals, 1 and 2 whose work functions are ϕ_1 and ϕ_2 . The electrodes are illuminated simultaneously. The maximum kinetic energy of photoelectrons reaching plate 2 is 2 eV and maximum kinetic energy of photoelectrons reaching plate 1 is 6 eV. Assume that photoelectron emitted from either plate do not interact with each other. $\phi_1 = 3$ eV and $\phi_2 = 1.4$ eV. Find wavelength (in nm) of the electromagnetic wave used.



SECTION-IV

Matrix Match Type (4 × 5)

1Q. [8 M (for each entry +2(0))]

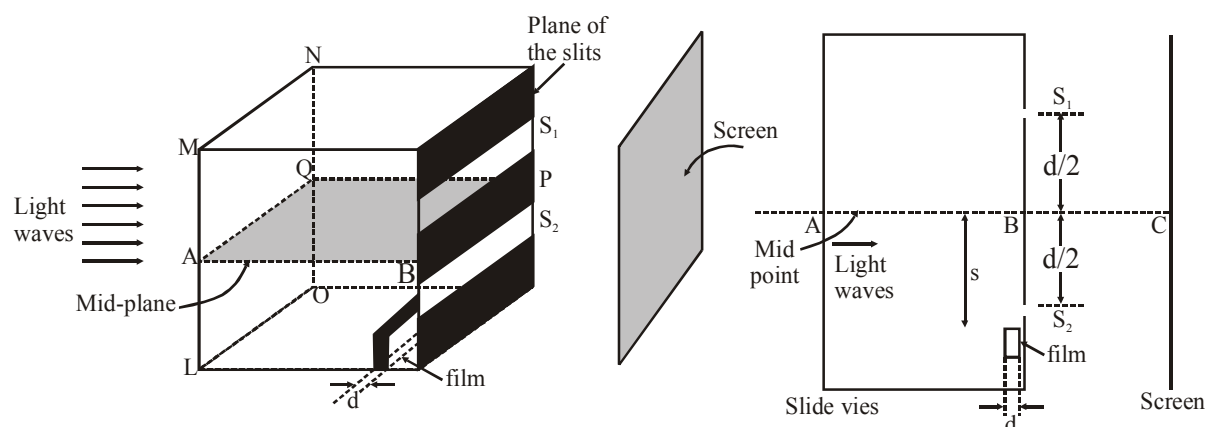
1. The diagram shows a cuboidal container with a non-viscous liquid of nonuniform density but a uniform refractive index μ_0 . The height of container is 0.4 SI units and the density of liquid varies linearly according to the relation :

$$\rho = \rho_0(1 - 4y)$$

where ρ_0 is a constant and y is the vertical distance measured from the midplane (ABPQ) of the container. (y is positive when measured upward, and negative when measured downward).

There are two horizontal slits, covered with perfectly transparent films, in one of the vertical faces as shown in the figure. A thin transparent strip of material having superficial dimensions equal to the slits and thickness equal to d (volume V_0) is released from rest from an initial depths below the mid-plane (ABPQ) (i.e., from $y = -s$) close to the slits. The strip has a constant density ρ_0 , uniform refractive index $\mu(> \mu_0)$ and it moves due to buoyancy keeping its plane parallel to the plane of the slits ; always remaining very close to the plane of the slits.

Monochromatic plane wavefronts (of light) parallel to the plane of the slits enter into the liquid through the face opposite the double-slit. The intensity of light is observed on the screen placed outside the container parallel to the plane of the slits.



Column-I

- (A) $s = 2d$ and $0 < t < 1$ sec
(B) $s = 3d$ and $0 < t < 2$ sec
(C) $s = d/4$ and $0 < t < 6$ sec
(D) $s = d$ and $0 < t < 3/4$ sec

Column-II

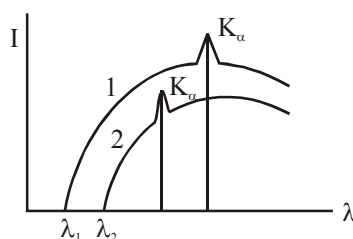
- (P) Central bright fringe is always at C
(Q) Central bright fringe is below C four times
(R) Central bright fringe is above C twice
(S) Central bright fringe is above C four times
(T) Central bright fringe is below C twice

SECTION-I

Single Correct Answer Type

7 Q. [3 M (–1)]

- The additional energy that should be given to an electron to reduce its de-Broglie wavelength from 1 nm to 0.5 nm is :
(A) 2 times the initial kinetic energy (B) 3 times the initial kinetic energy
(C) 0.5 times the initial kinetic energy (D) 4 times the initial kinetic energy
- Two particles A and B have de-Broglie's wavelengths $30\text{\AA}$ and $20\text{\AA}$, combined to form a particle C. Momentum is conserved in this process. The possible de-Broglie's wavelength of C is :-
(A) $10\text{\AA}$ (B) $20\text{\AA}$ (C) $65\text{\AA}$ (D) $80\text{\AA}$
- A perfectly absorbing, black, solid sphere with constant density and radius R, hovers stationary above the sun. This is because the gravitational attraction of the sun is balanced by the pressure due to the sun's light. Light pressure P is given by the intensity I of the absorbed light divided by the speed of light $c = 3 \times 10^8 \text{ m/s}$. $P = I/c$. Assume the sun is far enough away that it closely approximates a point source of light. The distance from the centre of the sun at which the sphere hovers is:
(A) proportional to R (B) proportional to $1/R$
(C) proportional to $1/R^2$ (D) independent of R
- If electron beam in a Coolidge tube is accelerated by 100 volt and targeted on high atomic number ($z > 30$) material. What will be spectrum of energy emitted?
(A) Superimposed continuous and characteristic energy spectrum of X-rays
(B) Only continuous spectrum in mainly ultraviolet and visible region
(C) Only characteristic energy spectrum of X-rays
(D) Only characteristic spectrum in visible and ultraviolet region.
- When two different materials A and B having atomic number Z_1 and Z_2 are used as the target in Coolidge X-ray tube at different operating voltage V_1 and V_2 respectively their spectrums are found as below. The correct relation is :

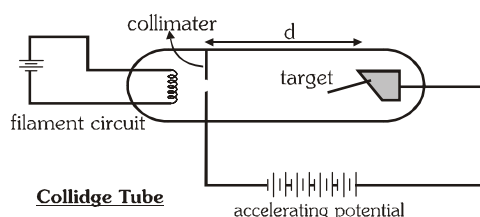


- $V_1 > V_2$ and $Z_1 > Z_2$ (B) $V_1 < V_2$ and $Z_1 < Z_2$
(C) $V_1 < V_2$ and $Z_1 > Z_2$ (D) $V_1 > V_2$ and $Z_1 < Z_2$
- The k_α wavelength of an element with atomic number z is λ_z . The k_α wavelength of an element with atomic number $2z$ is λ_{2z} . How are λ_z and λ_{2z} related ?
(A) $\lambda_z > 4\lambda_{2z}$ (B) $\lambda_z = 4\lambda_{2z}$
(C) $\lambda_z < 4\lambda_{2z}$ (D) Depending on z , λ_z could be greater than or less than $4\lambda_{2z}$
 - The half lives of radioisotopes P^{32} and P^{33} are 20 days and 30 days respectively. These radioisotopes are mixed in the ratio of 4 : 1 of their atoms. If the initial activity of the mixed sample is 7 mCi, then the activity of the mixed isotopes after 60 days.
(A) 2.5 mCi (B) 1 mCi (C) 3.2 mCi (D) 3 mCi

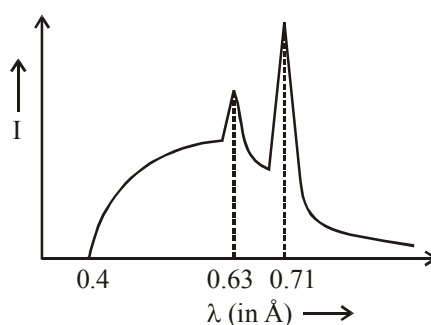
Multiple Correct Answer Type

6 Q. [4 M (–1)]

8. In an experiment suppose we want to increase the intensity to desired result



- (A) decreasing the distance d between filament (electron gun) & target metal
 (B) increasing the current in filament circuit
 (C) increasing the size of collimator
 (D) increasing the accelerating potential
9. In K-capture process,
 (A) a neutrino is emitted from the nucleus.
 (B) an X-ray photon is emitted having wavelength same as the X-ray photon emitted by parent nucleus.
 (C) an X-ray photon is emitted having wavelength same as the X-ray photon emitted by daughter nucleus.
 (D) proton number Z is reduced by one.
10. Figure shows intensity versus wavelength graph of X-rays coming from Coolidge-tube with molybdenum as anticathode. The two peaks shown in graph corresponds to K_α & K_β X-rays



- (A) Wavelength of L_α X-rays from Coolidge tube will be (approximately) 5.60 Å
 (B) Voltage applied across Coolidge tube is (approximately) 16 kV
 (C) If the potential difference across the Coolidge tube is increased such that the cut off wavelength of x-ray becomes 0.2 Å . Then the value of $(\lambda_{K_\alpha} - \lambda_{K_\beta})$ in this condition is 0.08 Å
 (D) If cut off wavelength is increased to 0.65 Å , intensity of K_α & K_β must increase.
11. Which of the following statements about X-rays is/are **TRUE**?
 (A) $E(K_\alpha) + E(L_\beta) = E(K_\beta) + E(M_\alpha) = E(K_\gamma)$
 (B) For the harder X-rays, the intensity is higher than soft X-rays
 (C) The continuous and the characteristic X-rays of same wavelengths differ only in the method of creation
 (D) The cut-off wavelength $\lambda_{\min}$ depends only on the accelerating voltage applied between the target and the filament.

12. In the α -decay of a U-238 nucleus the energy released in the decay is Q. The U-238 nucleus was initially stationary. Which of the following is (are) true?
 (A) Ratio of K.E. of α -particle and Thorium nucleus is 117 : 2
 (B) Ratio of K.E. of Thorium nucleus and α -particle is 1 : 234
 (C) Momentum of α -particle is $\sqrt{\frac{234Qm_{\alpha}}{119}}$
 (D) Recoil velocity of Thorium nucleus is $\sqrt{\frac{234Q}{119 \times 117m_{Th}}}$
13. A radioactive sample has a half life of 40 seconds. When its activity is measured 80 seconds after the beginning, it is found to be 6.932×10^{18} dps. During this time total energy released is 6×10^8 joule ($\ln 2 = 0.6932$) :-
 (A) The initial number of atoms in the sample is 1.6×10^{20}
 (B) The initial number of atoms in the sample is 1.6×10^{21}
 (C) Energy released per fission is 5×10^{-13} J
 (D) Energy released per fission is $\frac{5}{3} \times 10^{-13}$ J

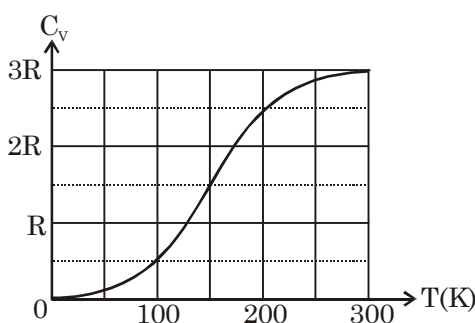
Linked Comprehension Type
(Single Correct Answer Type)

(2 Para $\times$ 2Q.) [3 M (-1)]

Paragraph for Question no. 14 and 15

THERMAL ACCELERATION

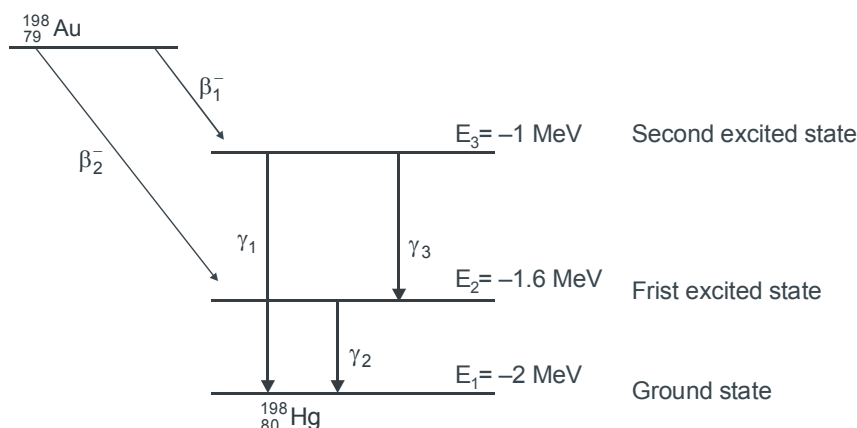
Consider a cube of side length $a = 1\text{cm}$, made of aluminium (density $\rho = 2.7\text{g/cm}^3$, molar mass $M_A = 23\text{g/mol}$). The heat capacity of one mole of aluminium is given as a function of temperature in the graph below. The speed of light $c = 3 \times 10^8\text{ m/s}$, universal gas constant $R = 8.31\text{J/(mole}\cdot\text{K)}$. The initial temperature of the cube is $T_0 = 300\text{K}$.



14. What is the total internal thermal (approximately) energy of such a cube at initial temperature T_0 :
 (A) 439 J (B) 878 J (C) 329.25 J (D) 109.75 J
15. Now, the cube has 5 faces painted in white (reflect all wavelengths) and one face painted in black (absorbs all these waves). The cube is surrounded by vacuum at very low temperature (near absolute zero); there is no gravity field. Initially, the cube is at rest; as it cools down due to heat radiation, it starts slowly moving. Find its terminal speed (assuming that radiation emitted by a cube face is normal to surface)
 (A) 2.16 mm/s (B) 0.54 mm/s (C) 1.08 mm/s (D) 1.62 mm/s

Paragraph for Questions 16 and 17

Gold Nucleus (${}_{79}\text{Au}^{198}$) can decay into mercury nucleus (${}_{80}\text{Hg}^{198}$) by two decay schemes shown in figure. (i) It can emit a β particle (β_1) and come to ground state by either emitting one γ ray (γ_1) or emitting two γ rays (γ_3 & γ_2) (ii) It can emit one β particle (β_2) and come to ground state by emitting γ_2 ray. Atomic masses : ${}^{198}\text{Au} = 197.9682$ amu, ${}^{198}\text{Hg} = 197.9662$ amu, $1 \text{ amu} = 930 \text{ MeV}/c^2$. The energy levels of the nucleus are shown in figure.



16. What is the maximum kinetic energy of emitted β_2 particles
(A) 1.44 MeV (B) 0.59 MeV (C) 1.86 MeV (D) 1.46 MeV
17. What is the maximum kinetic energy of emitted β_1 particle is -
(A) 1.28 MeV (B) 0.77 MeV (C) 1.86 MeV (D) 0.86 MeV

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

1. If the series limit of the Balmer series for hydrogen is $2700 \text{ \AA}$. Calculate the atomic number (Z) of the element which gives X-ray wavelength of K_α line as $1.0 \text{ \AA}$. If $Z = 10p + 1$ then find p.
2. When the nucleus of ${}_{7}\text{N}^{14}$ is bombard with an α - particle of sufficient kinetic energy, it gets transformed into ${}_{8}\text{O}^{17}$. If the masses of ${}_{7}\text{N}^{14}$, ${}_{2}\text{He}^4$, ${}_{1}\text{H}^1$ and ${}_{8}\text{O}^{17}$ are respectively 14.00307u , 4.00260u , 1.00783u and 16.99913u and $1\text{u} = 931.5 \text{ MeV}$, if the minimum kinetic energy that must be possessed by the α particle

is $xyz \times 10^{-2} \text{ MeV}$ then find value of $\frac{x+y+z}{2}$.

SECTION-IV

Matrix Match Type (4 × 5)

1 Q. [8 M (for each entry +2(0))]

1. With respect to photoelectric experiment, match the entries of Column I with the entries of Column II.

Column I

- (A) If ν (frequency) is increased keeping I (intensity) and ϕ (work function) constant
- (B) If I is increased keeping ν and ϕ constant
- (C) If the distance between anode and cathode increases.
- (D) If ϕ is decreased keeping ν and I constant

Column II

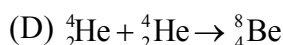
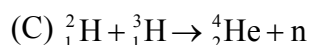
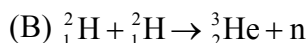
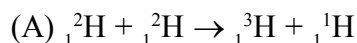
- (P) Stopping potential increases
- (Q) Saturation photocurrent increases
- (R) Maximum KE of the photoelectrons increase
- (S) Stopping potential remains the same

SECTION-I

Single Correct Answer Type

5 Q. [3 M (-1)]

1. Which of the following reaction(s) is/are an endoergic reaction :-



Mass in amu. Given $m({}_1^1\text{H}) = 1.0073$, $m(n) = 1.0087$, $m({}_1^2\text{H}) = 2.0141$, $m({}_1^3\text{H}) = 3.0160$,
 $m({}_2^3\text{He}) = 3.0160$, $m({}_2^4\text{He}) = 4.0026$ and $m({}_4^8\text{Be}) = 8.0053$

2. Consider the nuclear reaction $X^{200} \rightarrow A^{110} + B^{90}$ if the binding energy per nucleon for X, A and B is 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, what is the energy released ?

(A) 200 MeV

(B) 160 MeV

(C) 110 MeV

(D) 90 MeV

3. The following deuterium reactions and corresponding reaction energies are found to occur :



The notation ${}^{14}\text{N}(d, p){}^{15}\text{N}$ represents the reaction ${}^{14}\text{N} + d \rightarrow {}^{15}\text{N} + p$, ${}^4\text{He} = 4.0026 \text{ amu}$,
 ${}^2\text{He} = 2.014 \text{ amu}$, ${}^1\text{H} = 1.0078 \text{ amu}$, $n = 1.0087 \text{ amu}$ (1 amu = 931 MeV)

The Q values of the reaction ${}^{11}\text{B}(\alpha, n){}^{14}\text{N}$ is :-

(A) 0.5 eV

(B) 0.5 MeV

(C) 0.05 MeV

(D) 0.05 eV

4. Probability of decay of a particular nucleus of substance Z in next 1 hour is $\frac{1}{3}$. Then probability of decay of a nucleus of same substance :

(A) In next $\frac{1}{2}$ hours is $\frac{1}{\sqrt{3}}$

(B) In next 2 hours is $\frac{5}{9}$

(C) In next 3 hours is $\frac{1}{3}$

(D) In next 1.5 hour is $\frac{1}{3}$

5. If the β -rays spectrum is represented by $n \propto (E_{\max} - E)^2 \times \sqrt{E}$ where n is the number of β particles emitted with energy E and $E_{\max}$ is energy released in β decay, then maximum number of neutrinos would come out with energy equal to (approximately) :-

(A) $\frac{E_{\max}}{5}$

(B) $\frac{4E_{\max}}{5}$

(C) $E_{\max}$

(D) 0

Multiple Correct Answer Type

6 Q. [4 M (-1)]

6. Nuclei A decays to B with decay constant λ_1 and B decays to C with decay constant λ_2 . Initially at $t = 0$, number of nuclei of A and B are $2N_0$ and N_0 respectively. At $t = t_0$, number of nuclei of B stop decaying. If at this instant number of nuclei of B are $\frac{3N_0}{2}$.

(A) the value of t_0 is $\frac{1}{\lambda_1} \ln \frac{4\lambda_1}{3\lambda_2}$

(B) the value of t_0 is $\frac{1}{\lambda_2} \ln \frac{4\lambda_1}{3\lambda_2}$

(C) the value of N_A at t_0 is $\frac{3N_0}{2} \frac{\lambda_2}{\lambda_1}$

(D) the value of N_A at t_0 is $\frac{2N_0}{3} \frac{\lambda_2}{\lambda_1}$

7. Nuclei of a radioactive element X are being produced at a constant rate K and this element decays to a stable nucleus Y with a decay constant λ and half life $T_{1/2}$. At time $t = 0$, there are N_0 nuclei of the element X. :-

- (A) The number N_X of nuclei of X at time $t = T_{1/2}$ is $\frac{K + \lambda N_0}{2\lambda}$
- (B) The number N_Y of nucleus Y at time t is $Kt + \left(\frac{K - \lambda N_0}{\lambda}\right)e^{-\lambda t} - \frac{K - \lambda N_0}{\lambda}$
- (C) The number N_Y of nuclei of Y at $t = T_{1/2}$ is $K \frac{\ln 2}{\lambda} - \frac{1}{2} \left(\frac{K - \lambda N_0}{\lambda}\right)$
- (D) The number N_Y of nuclei of Y at $t = T_{1/2}$ is $K \frac{\ln 2}{\lambda} + \frac{1}{2} \left(\frac{K + \lambda N_0}{\lambda}\right)$

8. Choose the **CORRECT** option(s) :-

- (A) When two light nuclei fuse together, the rest mass of resulting nucleus is more than the sum of the rest masses of light nuclei.
- (B) When a heavy nucleus disintegrates, the binding energy per nucleon of resulting nuclei is higher.
- (C) In a typical fission reaction, approximately 200 MeV energy is released.
- (D) In a typical fusion reaction, approximately 1 GeV energy is released.

9. A stationary nucleus ${}^{226}_{88}\text{Ra}$ (ground state) decays into the nucleus ${}^{222}_{86}\text{Rn}$ (ground state) by emitting an α -particle. (Given that $m({}^{226}_{88}\text{Ra}) = 226.02540 \text{ u}$, $m({}^4_2\text{He}) = 4.00260 \text{ u}$, $m({}^{222}_{86}\text{Rn}) = 222.01750 \text{ u}$.) and Take $1 \text{ amu} = 931 \text{ MeV}/c^2$.

- (A) The Q-value for the α -decay of $({}^{226}_{88}\text{Ra})$ is (approximately) 4.93 MeV
- (B) The Q-value for the α -decay of $({}^{226}_{88}\text{Ra})$ is (approximately) 9.8 MeV
- (C) The K.E. of the emitted α -particle in the decay of ${}^{226}_{88}\text{Ra}$ is (approximately) 4.85 MeV
- (D) The K.E. of the emitted α -particle in the decay of ${}^{226}_{88}\text{Ra}$ is (approximately) 9.7 MeV

10. A nucleus has a radius of $7.2 \times 10^{-15} \text{ m}$. When an α -decay takes place from this nucleus, ratio of number of neutrons and number of protons in the daughter nucleus becomes $\frac{65}{41}$:-

- (A) Parent nucleus is ${}_{84}\text{Po}$. (B) Daughter nucleus is ${}_{82}\text{Pb}$.
- (C) Mass number of daughter nucleus is 216. (D) Mass number of parent nucleus is 216.

11. Choose the **CORRECT** statement(s) :

- (A) The stability of nucleus is directly proportional to total binding energy.
- (B) As the accelerating potential in Coolidge apparatus increases, difference between $\lambda_{\text{cut-off}}$ and wavelength K_α characteristic X-ray increases due to change in both $\lambda_{\text{cut-off}}$ & $\lambda_{K\alpha}$.
- (C) A neutron with kinetic energy of 24.2 eV collides with stationary H-atom in ground state then collision may be elastic, inelastic & perfectly inelastic.
- (D) Presence of K_α in characteristic X-ray ensures the availability of all types of characteristic X-rays.

Linked Comprehension Type (Single Correct Answer Type)

(2 Para × 2Q.) [3 M (-1)]

Paragraph for Question no. 12 and 13

Interaction within a nucleus is so very complex that no one has so far tried to do the quantum mechanical analysis of each and every nucleus. The complexity of interaction is so vast that no formula exists that is able to predict the binding energy of a given nucleus.

Based on the interactions understood so far, semi-empirically derived formula for binding energy per nucleon is given by : $E_b(A, Z) = a_1 - a_s A^{-1/3} - a_c Z^2 A^{-4/3} - a_A (A - 2Z)^2 A^{-2}$

Where,

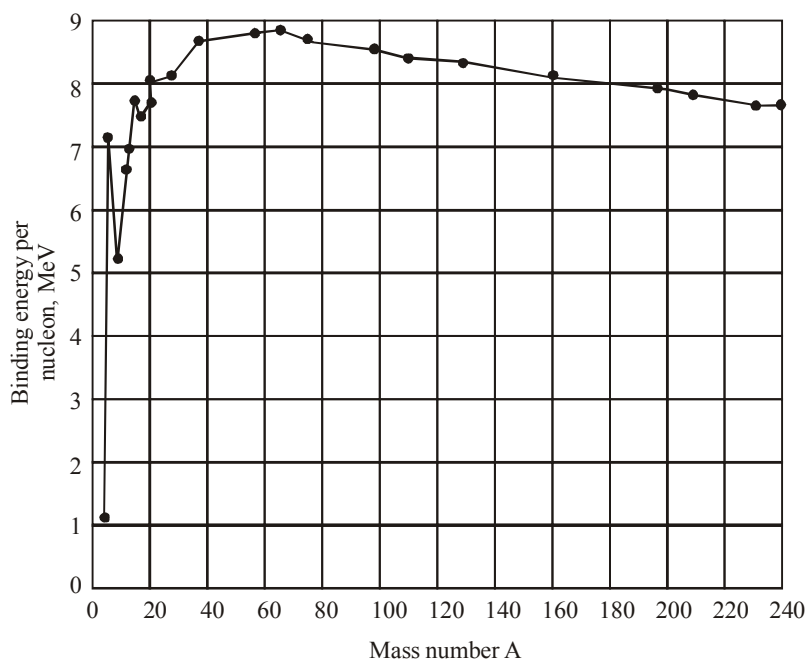
$a_1 = 15.5$ MeV : called as *interior bonding coefficient*

$a_s = 17.2$ MeV : called as *surface correction coefficient*

$a_c = (2048/3000)$ MeV : called as *coulomb repulsion coefficient*

$a_A = 24$ MeV : called as *asymmetry coefficient*

The constants given in above equation correctly predicts the measured values of the binding energy per nucleon for the nuclei with $A > 20$ to within 1% error in comparison to the experimental results shown below. But the same equation fails miserably when $A < 20$.



12. The measured binding energy per nucleon for N_7^{15} is $E_b = 7.700$ MeV, while that for O_8^{15} is 7.464 MeV. Why is average nucleon, in the oxygen nucleus, less tightly bound than in the nitrogen nucleus?
 (A) It has fewer nucleons overall. (B) It has more nucleons on the surface.
 (C) Electrostatic repulsion is greater. (D) The nucleus is asymmetric.
13. Find approximately the binding energy per nucleon for Zn_{30}^{64} :-
 (A) 1.51 MeV (B) 6.54 MeV (C) 8.71 MeV (D) 12.05 MeV

Paragraph for Question no. 14 and 15

Few atomic masses are given

${}_{92}^{238}\text{U} = 238.05079\text{u}$, ${}_2^4\text{He} = 4.00260\text{u}$, ${}_{90}^{234}\text{Th} = 234.04363\text{u}$, ${}_1^1\text{H} = 1.007834$, ${}_{91}^{237}\text{Pa} = 237.065121\text{u}$.

Answer the following questions on the basis of above data.

14. Calculate the energy released during the α -decay of ${}_{92}^{238}\text{U}$.
 (A) $Q = 4.25$ MeV (B) $Q = 8.5$ MeV (C) $Q = 3.25$ MeV (D) None of these
15. Spontaneous emission of a proton from ${}_{92}^{238}\text{U}$ is
 (A) possible (B) impossible (C) can't say (D) none of these

SECTION-II

Numerical Answer Type Question

1Q.[3(0)]

(upto second decimal place)

- The ratio of the components in a mixture, which consists of two elements, is to be determined. The atomic number of the elements are big, their atomic mass numbers are the same and the amount of the sample is only 58 mg. We know that both elements β decay when they are bombarded by neutrons. They behave similarly when absorbing neutrons. The half-life of element A is half an hour and the half-life of element B is an hour. Right after the neutron irradiation the β emission is measured. At this time 80 particles are measured in 2 seconds, and after an hour only 29 particles are measured during 2 seconds. Determine the mass of the element A (in mg) in the sample. Assume that spontaneous decay statistical law is obeyed by nuclei of both the elements.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

2 Q. [4 M (0)]

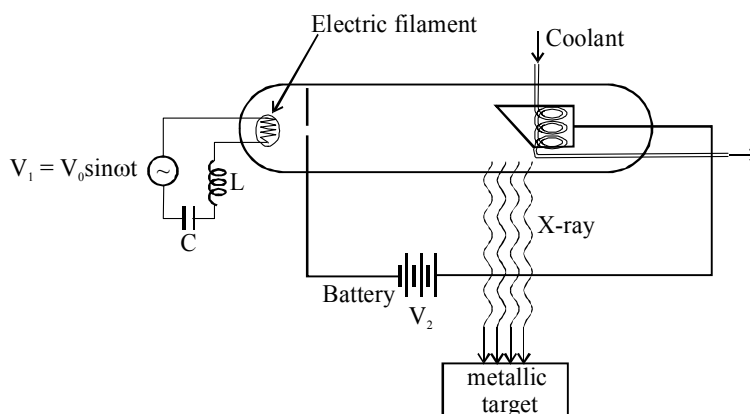
- A shell of radius 1m is coated with a thin layer of β^- active material. It's initial charge is zero and initial number of active atoms is $\frac{4}{3} \times 10^{12}$. If half life of decay is 1 hr and all the electron are emitted with an energy of 1.44 keV, find the time (in hr.) after which charge on the sphere becomes constant. Neglect the time taken by electron to return back.
- A nucleus with $Z = 72$ emits the following in a sequence : $\alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$. The Z of the resulting nucleus is 10N. Then fill the value of N in OMR sheet.

SECTION-IV

Matrix Match Type (4×5)

2 Q. [8 M (for each entry +2(0))]

- Figure shows a Coolidge tube being used to produce X-ray. Column-I shows change in various parameters while column-II shows effects occurring due to change in the parameters. Match them correctly. Initially filament circuit is in resonance condition and M series X-ray was obtained. The emitted X-rays were incident on a metallic target, such that photo electrons were emitted.



- | | |
|---|---|
| (A) On increasing frequency (ω) of AC source | (P) Photo current from metallic target increases |
| (B) On increasing pressure of gas inside the tube keeping temperature of gas constant | (Q) Photo current from metallic target decreases |
| (C) On increasing peak voltage (V_0) of AC | (R) Penetrating power of X-ray increases |
| (D) On increasing emf (V_2) of battery. | (S) Spectrum has more number of characteristic X-ray lines. |
| | (T) Temperature of coolant at output would be higher. |

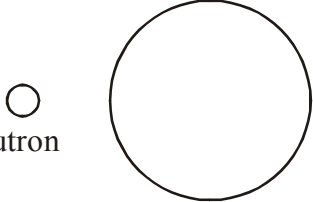
2. Column-I shows a projectile and a target. Left particles are projectile and right is target. Column-II shows subsequent event. All the collisions are head on. Mass of projectile is m and mass of target is M .

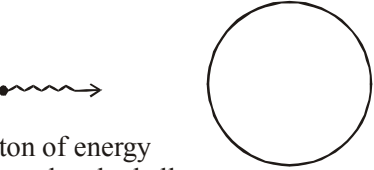
Column-I

Column-II

(A)  (P) nuclear reaction may take place.
Electron accelerated through 20 V H-atom

(B)  (Q) If energy is released it is of the order of MeV or greater
Electron positron pair or small nuclei of equal mass

(C)  (R) Mass may be converted to energy
Neutron U^{235} Nuclei

(D)  (S) Photon may be emitted
Photon of energy greater than k-shell binding energy. atom of high atomic mass
(T) Electron emission may take place

SECTION-I

Single Correct Answer Type

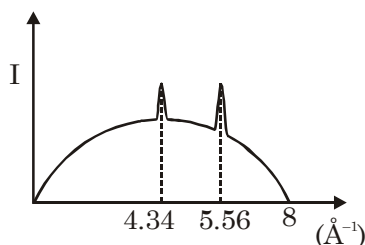
4 Q. [3 M (-1)]

- An environment study showed that the chernobyl disaster released 6.93 MCi of ^{137}Cs into the environment. If half life of ^{137}Cs is 30 years then the released mass of ^{137}Cs is-
(A) 112.5 kg (B) 56.3 kg (C) 79.6 kg (D) 7.9 kg
- A radioactive sample decays by two modes by α decay and by β -decay. 66.6% of times it decays by α -decay and 33.3% of times, it decays by β -decay. If half life of sample is 60 years then what will be half life of sample, if it decays only by α - decay.
(A) 30 years (B) 90 years (C) 120 years (D) 180 years
- A radioactive sample decays by β -emission. In first two seconds 'n' β -particles are emitted and in next 2 seconds, '0.25n' β -particles are emitted. The half life of radioactive nuclei is
(A) 2 sec (B) 4 sec (C) 1 sec (D) None of these
- Mass spectrometric analysis of potassium and argon atoms in a Moon rock sample shows that the ratio of the number of (stable) ^{40}Ar atoms present to the number of (radioactive) ^{40}K atoms is 10.3. Assume that all the argon atoms were produced by the decay of potassium atoms, with a half-life of 1.25×10^9 yr. How old is the rock [Given : $\ln(11.3) = 2.42$, $\ln 2 = 0.693$]
(A) 2.95×10^{11} yr (B) 2.95×10^9 yr (C) 4.37×10^9 yr (D) 4.37×10^{11} yr

Multiple Correct Answer Type

6 Q. [4 M (-1)]

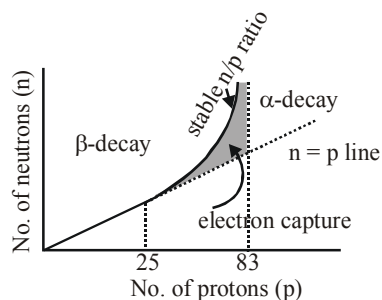
- A radioactive sample decays by three modes simultaneously. Half lives corresponding to these modes are in G.P. and half life of sample is 10 years. When the sample decays by the mode having largest half life it takes 70 years for the sample to become half. If sample decays exclusively by other modes possible values of half life is :
(A) 28 years (B) 35 years (C) 14 years (D) 17.5 years
- The intensity versus $\frac{1}{\lambda}$ graph for x rays is shown here. Two lines of characteristic x-ray spectra (k_α & k_β) are also shown.



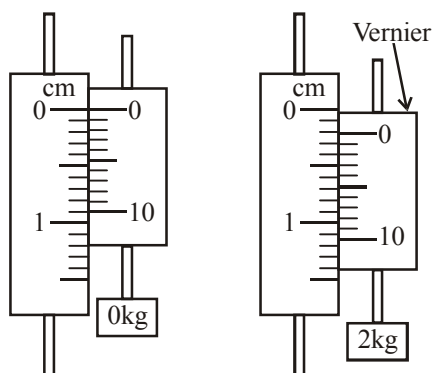
- The accelerating potential difference across the x-ray tube is 99.2 kV
- The atomic member of target is $Z = 74$.
- $\frac{1}{\lambda} = 5.56 (\text{\AA})^{-1}$ corresponds to K_α x ray & $\frac{1}{\lambda} = 4.34 (\text{\AA})^{-1}$ corresponds to K_β x ray
- $\frac{1}{\lambda} = 4.34 (\text{\AA})^{-1}$ corresponds to K_α x ray & $\frac{1}{\lambda} = 5.56 (\text{\AA})^{-1}$ corresponds to K_β x ray.

7. If all the known isotopes of the elements are plotted on a graph of number of neutrons (n) versus number of protons (p), it is observed that all isotopes lying outside of a "stable" n/p ratio region are radioactive as shown in figure.

The graph exhibits its straight line behavior with unit slope up to $p = 25$. Above $p = 25$ those isotopes with an n/p ratio lying below the stable region usually undergo electron capture while those with n/p ratios lying above the stable region usually undergo beta-decay. Very heavy isotopes $p > 83$ are unstable because of their relatively large nuclei and they undergo alpha decay.



- (A) The radioisotope of magnesium with mass number 27 and atomic number 12 may undergo Beta decay.
- (B) The radioisotope of magnesium with mass number 27 and atomic number 12 may undergo Alpha decay.
- (C) For a hypothetical isotope of an element ${}^{178}_{88}\text{X}$ we may expect α -decay
- (D) For a hypothetical isotope of an element ${}^{178}_{88}\text{X}$ we may expect β^+ -decay
8. In an experiment to determine the Young's modulus of a metal, two identical wires are suspended using a vernier arrangement as shown in figure. The left wire is held taut with a fixed mass. Different loads can be applied to the right wire and stretching of the wire can be measured by using the vernier scale attached to the wire on right. The test wire has a length of 2.2 m between support and vernier and a uniform diameter of 0.42 mm. Take $g = 10\text{ms}^{-2}$ and give your answer in appropriate significant digits.



- (A) The strain produced in the wire is 0.0010
- (B) The Young's modulus of the material is $1.4 \times 10^{11} \text{ Nm}^{-2}$
- (C) If the wire is loaded statically, then the increase in internal energy of the wire is 0.022 J
- (D) If the wire is loaded statically, then the increase in internal energy of the wire is 0.22 J

9. Select the **CORRECT** statement(s) :
- (A) Accuracy of measurement by screw gauge can be increased by making more divisions on circular scale
 (B) If N division on main scale is equal to $(N + m)$ divisions on vernier scale, then accuracy of measurement can be increased by taking m smallest
 (C) Zero errors can be minimised by taking large number of readings.
 (D) Zero error in a given instrument always have a fixed value.
10. To measure speed of sound in air, we use a resonance column apparatus. The tuning fork has a frequency of 1200 Hz. Resonance is obtained when height of air column is 6.8 cm, 20.6 cm and 34.8 cm.
 (A) The velocity of sound is 331 m/s (B) The end correction can be 0.2 cm
 (C) The error in wavelength 0.4 cm (D) The error in velocity of sound is 4.8 m/s

Linked Comprehension Type

(2 Para × 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 11 to 13

Polonium (P_0^{210}) emits ${}_2\alpha^4$ particles and is converted into lead (${}_{82}Pb^{206}$). This reaction is used for producing electric power in a space mission: P_0^{210} has half life of 138.6 days. (Given masses of the nuclei) $P_0^{210} = 209.98264$ amu, $Pb^{206} = 205.97440$, ${}_2\alpha^4 = 4.00260$ amu)

11. Assuming an efficiency of 10% of the thermoelectric machine, how much P_0^{210} is required to produce 1.2×10^7 J of electric energy per day at the end of 693 days?
 (A) 1.0 gm (B) 2.0 gm (C) 3.0 gm (D) 0.5 gm
12. What is the initial amount of P_0^{210} ?
 (A) 16 gm (B) 8 gm (C) 32 gm (D) 64 gm
13. Find the initial activity of the material.
 (A) 4.57×10^{21} per day (B) 3.57×10^{21} per day
 (C) 2.28×10^{21} per day (D) 1.785×10^{21} per day

Paragraph for Question 14 to 16

The major errors in the experiment to determine specific resistance of the material of a wire using meter bridge are due to the heating effect, end correction introduced due to shift of the zero of the scale at A and B, stray resistances in gaps of meter bridge and due to non-uniformity of wire.

14. The major errors mentioned above are
 (A) systematic in nature (B) random in nature
 (C) neither systematic nor random (D) both systematic & random
15. End correction α and β can be estimated by including known resistances R_1 and R_2 in the two gaps and finding the null point using equations
 दोनों रिक्त स्थानों में ज्ञात प्रतिरोधों R_1 व R_2 को लेते हुए अन्त्य संशोधन α व β को अनुमानित कर निम्न से कौनसी समीकरणों का उपयोग कर संतुलन बिन्दु ज्ञात किया जा सकता है?

$$(A) \frac{R_1}{R_2} = \frac{\ell_1 - \alpha}{(100 - \ell_1 - \beta)}, \frac{R_2}{R_1} = \frac{\ell_2 - \alpha}{(100 - \ell_2 - \beta)} \quad (B) \frac{R_1}{R_2} = \frac{\ell_1 + \alpha}{(100 - \ell_1 + \beta)}, \frac{R_2}{R_1} = \frac{\ell_2 + \alpha}{(100 - \ell_2 + \beta)}$$

$$(C) \frac{R_1}{R_2} = \frac{\ell_1 - \alpha}{(100 - \ell_1 + \beta)}, \frac{R_2}{R_1} = \frac{\ell_2 + \alpha}{(100 - \ell_2 + \beta)} \quad (D) \frac{R_1}{R_2} = \frac{\ell_1 + \alpha}{(100 - \ell_1 + \beta)}, \frac{R_2}{R_1} = \frac{\ell_2 - \alpha}{(100 - \ell_2 - \beta)}$$

16. Error due to non-uniformity of meter bridge wire can be reduced by interchanging the resistances in the gaps of the meter bridge.

Error in specific resistance is given by $\frac{d\rho}{\rho} = \frac{dR}{R} + 2\frac{dr}{r} + \frac{dL}{L}$. Here $\frac{dR}{R}$ depends on

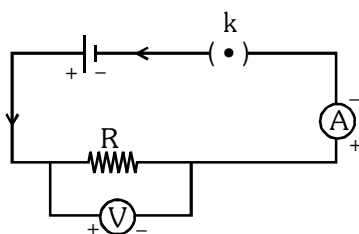
- (A) error in the measurement of resistance in R.B.
(B) error in the measurement of unknown resistance.
(C) error in the measurement of balancing length.
(D) All of the above.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9)

3 Q. [4 M (0)]

- A gas containing hydrogen like ions with atomic number Z , emits photons in transition $(n + 2) \rightarrow n$, where $n = Z$. These photons fall on a metallic plate and eject electrons having minimum de-Broglie wavelength λ of $5\text{\AA}$. Find Z . (the work function of metal is 4.2 eV)
- Mr. Vivek Taparia performed an experiment to verify Ohm's law. He connected following circuit to measure voltage and current.



Here R is the unknown resistance, V the voltmeter, A the ammeter and K is the key. The value of R from following readings is given by $\alpha \times 10^3\text{ k}\Omega$, then find the value of $\alpha + \beta$.

V(volt)	1	2	3	4	5
I(mA)	1.40	2.83	5.68	7.11	8.54

- Suppose that we have a vernier callipers which does not expand or contract on heating. Such a device is very useful in engineering applications where the dimensions are to be found at different temperatures. One such vernier callipers is used to measure the diameter of a rod at different temperatures as shown in table below. What will be the vernier scale reading (division coinciding) at a temperature of 50°C . 10 vernier scale divisions coincide with 9 main scale divisions and 1 main scale division = 1 mm.

Temperature T in $^\circ\text{C}$	Main scale reading	Vernier scale reading
0	2.4 cm	4th division coinciding
20	2.4 cm	8th division coinciding

SECTION-IV

Matrix Match Type (4×5)

1 Q. [8 M (for each entry +2(0))]

- Column I shows various effects occurring/involved to isolated samples undergoing various processes shown in Column II. Match them correctly

- Column I**
- (A) Remaining sample gets charged
(B) Electromagnetic force plays a substantial role
(C) Mass is getting converted into energy
(D) A chargeless particle carrying momentum is emitted

- Column II**
- (P) Thermionic emission
(Sample is electrically heated)
(Q) K-capture process accompanied by emission of characteristic X-ray
(R) Photoelectric effect
(sample is exposed to light having sufficient energy)
(S) β^- decay
(T) α -decay